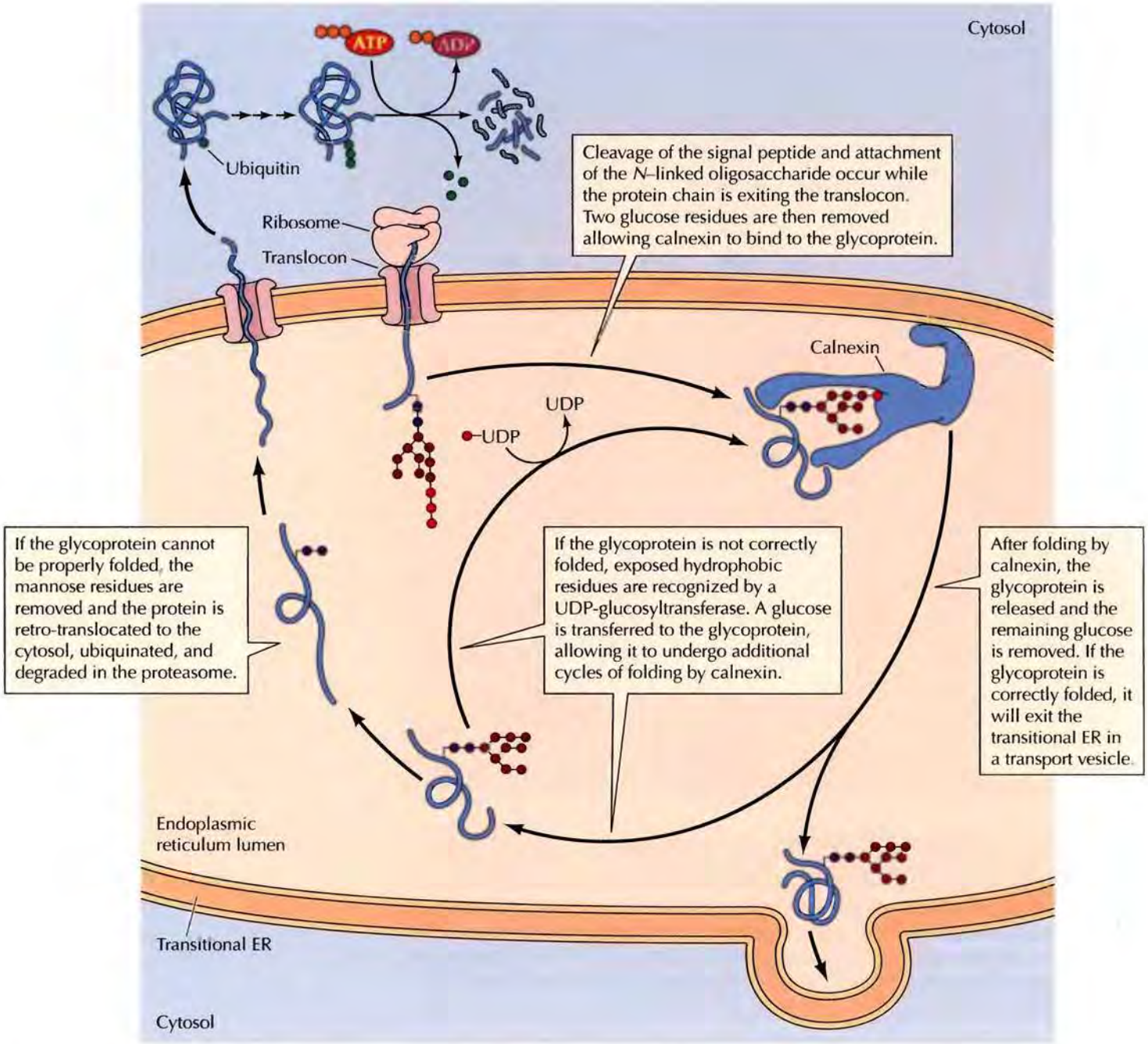


FIGURE 10.18 Glycoprotein folding by calnexin

unfolded proteins accumulates, as may result from a variety of types of cellular stress, signaling via BiP initiates a process known as the **unfolded protein response** (Figure 10.19). The levels of BiP in the ER lumen are normally sufficient not only to function in protein import and folding but also to bind signaling molecules, keeping them in an inactive state. However, if an excess of unfolded proteins accumulates, they compete for the available BiP. This releases the molecules that signal the unfolded protein response, which includes general inhibition of protein synthesis, increased expression of chaperones (such as calnexin, calreticulin, protein disulfide isomerase, and BiP itself), and an increase in the activity of the proteasome to enhance degradation of misfolded proteins (see Figure 10.18).

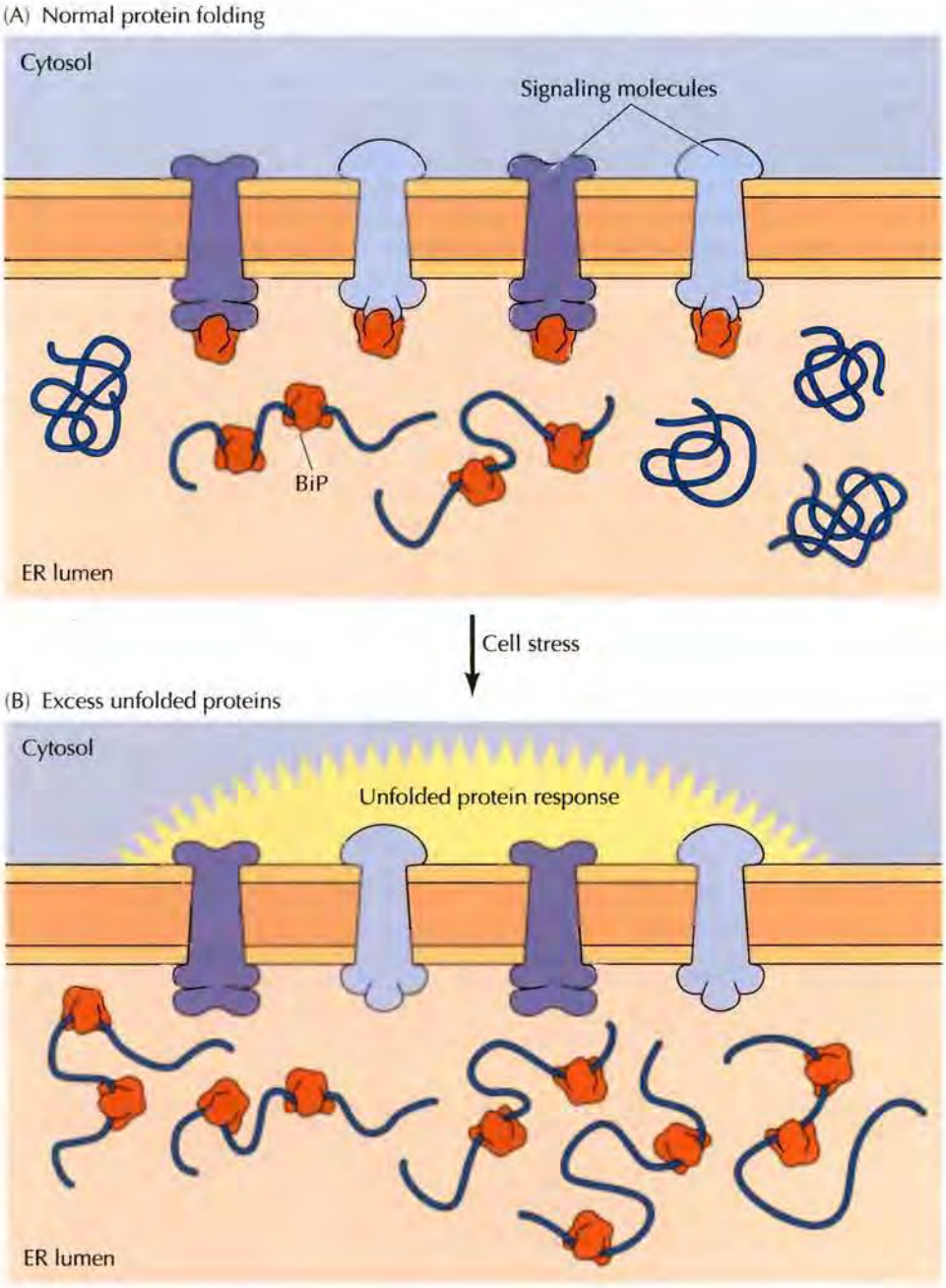


FIGURE 10.19 Unfolded protein response The chaperone protein, BiP, participates in the folding of proteins in the ER lumen. (A) In an unstressed cell there is sufficient BiP available to both fold newly synthesized proteins and keep several types of ER membrane signal molecules inactive. (B) Cellular stress, such as heat, chemical insult, or viral infection interferes with protein folding, so unfolded proteins accumulate in the ER. BiP has a higher affinity for unfolded proteins than for the ER membrane signal molecules, so the latter are released to become active and initiate the unfolded protein response.

The Smooth ER and Lipid Synthesis

In addition to its activities in the processing of secreted and membrane proteins, the ER is the major site at which membrane lipids are synthesized in eukaryotic cells. Because they are extremely hydrophobic, membrane lipids are synthesized in association with already existing cellular membranes rather than in the aqueous environment of the cytosol. Although some lipids are synthesized in association with other membranes, most are synthesized in the ER. They are then transported from the ER to their ultimate destinations either in vesicles or by carrier proteins, as will be discussed later in this chapter and in Chapter 13.

The membranes of eukaryotic cells are composed of three main types of lipids: phospholipids, glycolipids, and cholesterol. Most of the phospho-

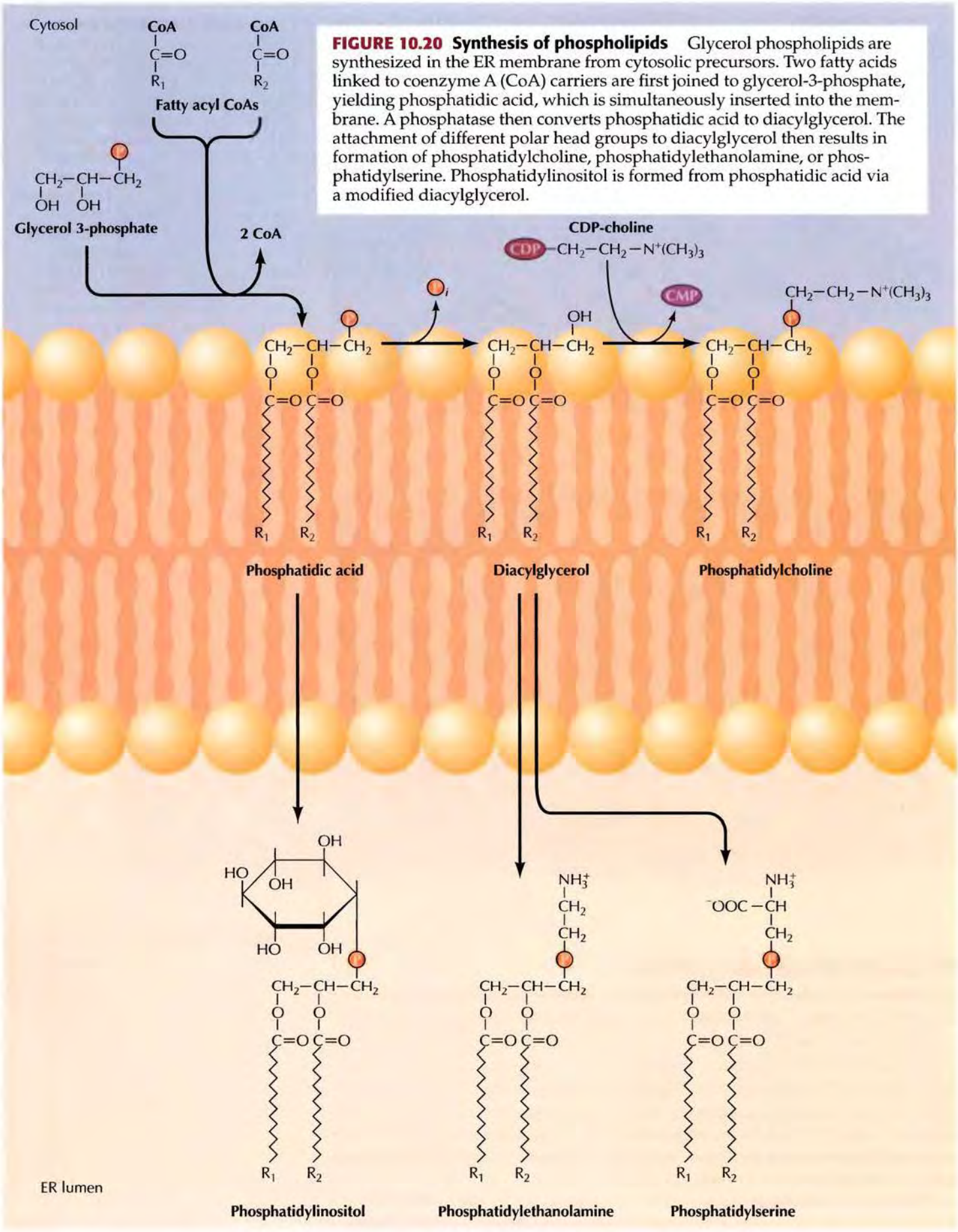


FIGURE 10.20 Synthesis of phospholipids Glycerol phospholipids are synthesized in the ER membrane from cytosolic precursors. Two fatty acids linked to coenzyme A (CoA) carriers are first joined to glycerol-3-phosphate, yielding phosphatidic acid, which is simultaneously inserted into the membrane. A phosphatase then converts phosphatidic acid to diacylglycerol. The attachment of different polar head groups to diacylglycerol then results in formation of phosphatidylcholine, phosphatidylethanolamine, or phosphatidylserine. Phosphatidylinositol is formed from phosphatidic acid via a modified diacylglycerol.

lipids, which are the basic structural components of the membrane, are derived from glycerol. They are synthesized on the cytosolic side of the ER membrane from water-soluble cytosolic precursors (Figure 10.20). Fatty acids are first transferred from coenzyme A carriers to glycerol-3-phosphate by membrane-bound enzymes, and the resulting phospholipid (phosphatidic acid) is inserted into the membrane. Enzymes on the cytosolic face of the ER membrane then either modify phosphatidic acid or directly catalyze the addition of different polar head groups, resulting in formation of phosphatidylcholine, phosphatidylserine, phosphatidylethanolamine, or phosphatidylinositol.

The synthesis of these phospholipids on the cytosolic side of the ER membrane allows the hydrophobic fatty acid chains to remain buried in the membrane while membrane-bound enzymes catalyze their reactions with water-soluble precursors (e.g., CDP-choline) in the cytosol. Because of this topography, however, new phospholipids are added only to the cytosolic half of the ER membrane (Figure 10.21). To maintain a stable membrane some of these newly synthesized phospholipids must therefore be transferred to the other (lumenal) half of the ER bilayer. This transfer does not occur spon-

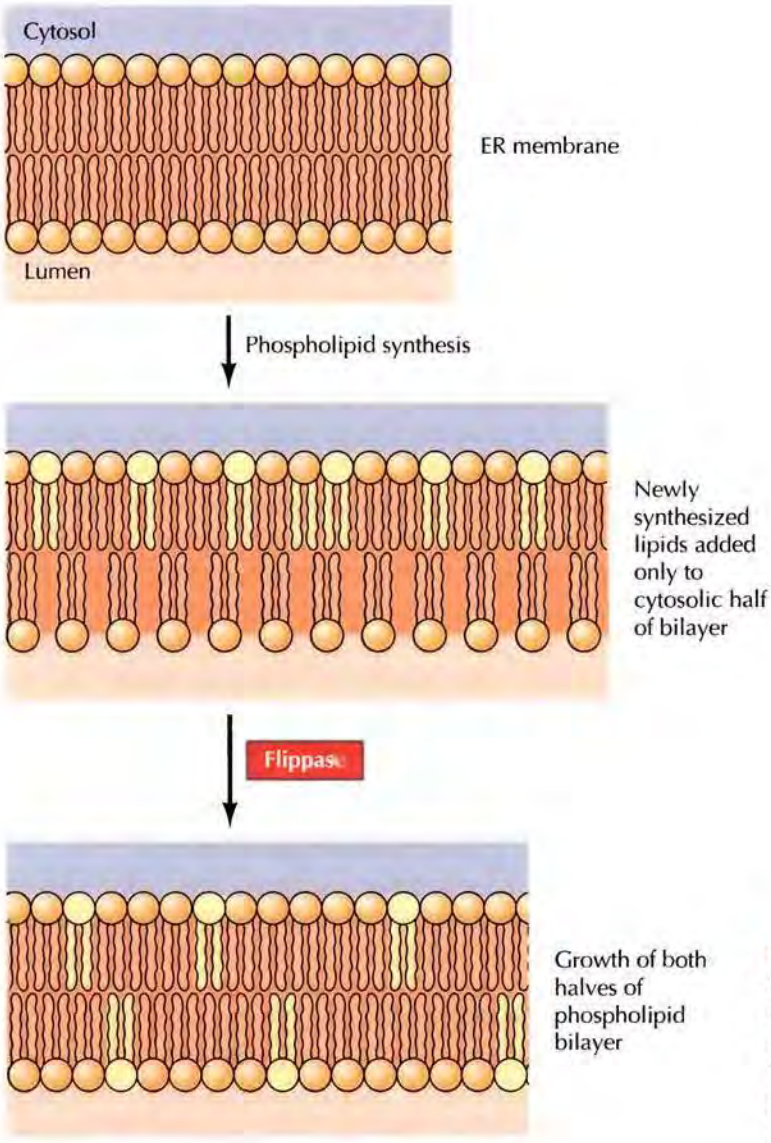


FIGURE 10.21 Translocation of phospholipids across the ER membrane Because phospholipids are synthesized on the cytosolic side of the ER membrane, they are added only to the cytosolic half of the bilayer. They are then translocated across the membrane by phospholipid flippases, resulting in even growth of both halves of the phospholipid bilayer.

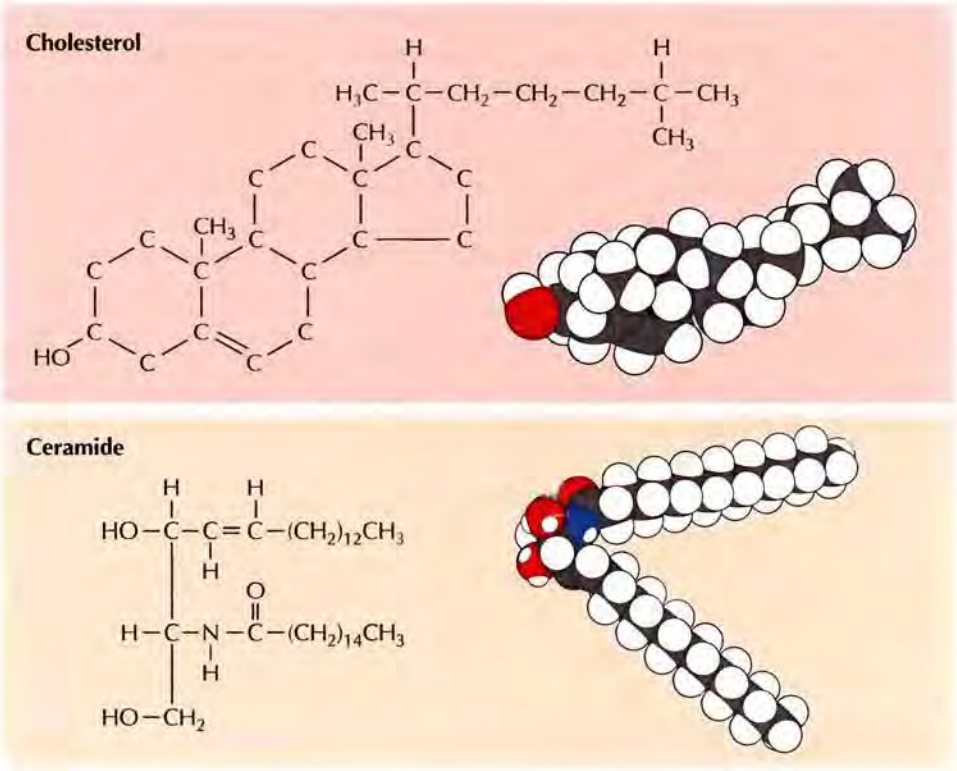


FIGURE 10.22 Structure of cholesterol and ceramide The hydrogens attached to the ring carbons of cholesterol are not shown.

taneously because it requires the passage of a polar head group through the membrane. Instead, membrane proteins called **flippases** catalyze the rapid translocation of phospholipids across the ER membrane resulting in even growth of both halves of the bilayer. At least some flippases are phospholipid-specific and ATP-dependent.

In addition to its role in synthesis of the glycerol phospholipids, the ER also serves as the major site of synthesis of two other membrane lipids: cholesterol and ceramide (Figure 10.22). As discussed later, ceramide is converted to either glycolipids or sphingomyelin (the only membrane phospholipid not derived from glycerol) in the Golgi apparatus. The ER is thus responsible for synthesis of either the final products or the precursors of all the major lipids of eukaryotic membranes. Cholesterol and sphingomyelin are important components of lipid rafts as discussed in Chapter 13.

Smooth ER is abundant in cell types that are particularly active in lipid metabolism. For example, steroid hormones are synthesized (from cholesterol) in the ER, so large amounts of smooth ER are found in steroid-producing cells, such as those in the testis and ovary. In addition, smooth ER is abundant in the liver where it contains enzymes that metabolize various lipid-soluble compounds. These detoxifying enzymes inactivate a number of potentially harmful drugs (e.g., phenobarbital) by converting them to water-soluble compounds that can be eliminated from the body in the urine. The smooth ER is thus involved in multiple aspects of the metabolism of lipids and lipid-soluble compounds.

Export of Proteins and Lipids from the ER

Both proteins and phospholipids travel along the secretory pathway in transport vesicles, which bud from the membrane of one organelle and then fuse with the membrane of another. Thus molecules are exported from the

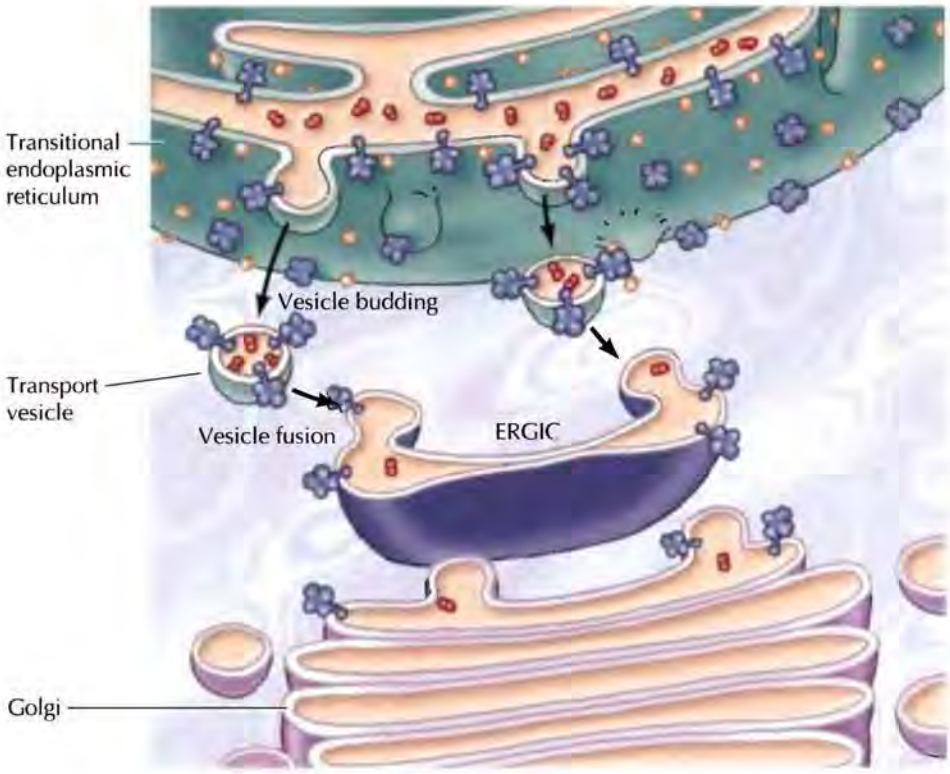


FIGURE 10.23 Vesicular transport from the ER to the Golgi Proteins and lipids are carried from the ER to the Golgi in transport vesicles that bud from the membrane of the transitional ER, fuse to form the vesicles and tubules of the ER-Golgi intermediate compartment (ERGIC), and are then carried to the Golgi. Lumenal ER proteins are taken up by the vesicles and released into the lumen of the Golgi. Membrane proteins maintain the same orientation in the Golgi as in the ER.

ER in vesicles that bud from the transitional ER and carry their cargo through the ER-Golgi intermediate compartment and then to the Golgi apparatus (Figure 10.23). Subsequent steps in the secretory pathway involve vesicular transport between different compartments of the Golgi and from the Golgi to lysosomes or to the plasma membrane. In most cases, proteins within the lumen of one organelle are packaged into the budding transport vesicle and then released into the lumen of the recipient organelle following vesicle fusion. Membrane proteins and lipids are transported similarly, and it is significant that their topological orientation is maintained as they travel from one membrane-enclosed organelle to another. For example, the domains of a protein exposed on the cytosolic side of the ER membrane will also be exposed on the cytosolic side of the Golgi and plasma membranes, whereas protein domains exposed on the lumenal side of the ER membrane will be exposed on the lumenal side of the Golgi and on the exterior of the cell (see Figure 10.11).

Most proteins that enter the transitional ER move through the ER-Golgi intermediate compartment and on to the Golgi. These proteins are marked by sequences that signal either their export from or retention within the ER (Figure 10.24). Many transmembrane proteins possess di-acidic or di-hydrophobic amino acid sequences in their cytosolic domains that function as ER export signals. Both GPI-anchored proteins (which are marked for export by their GPI-anchors) and lumenal secretory proteins appear to be recognized and sequestered by these transmembrane receptor proteins. Very few ER export signals have been detected on lumenal secretory proteins and their recognition may depend on the shape of the correctly folded protein. It is also possible that there is a default pathway where otherwise unmarked proteins in the ER lumen move to the Golgi and beyond.

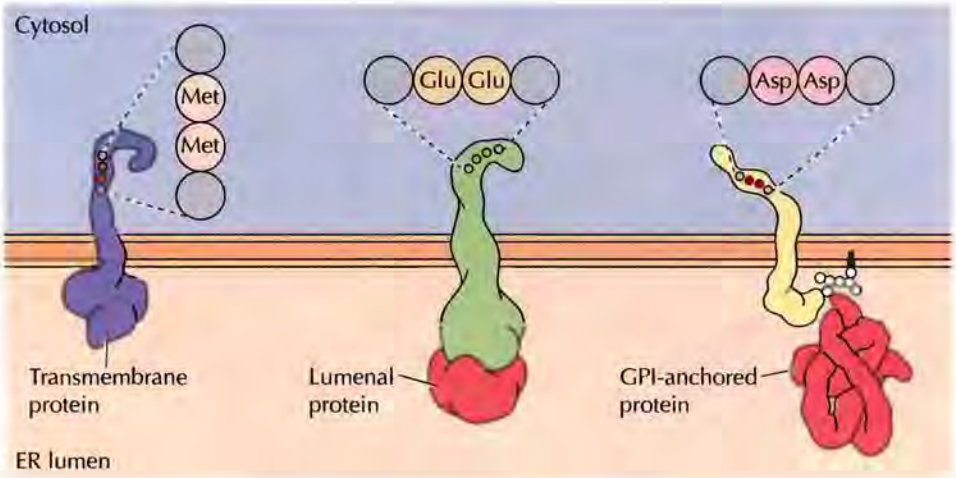


FIGURE 10.24 ER export signals
Three types of secretory proteins are recruited into budding vesicles by cytosolic adaptor proteins. Transmembrane proteins are recognized by di-acidic (e.g., Asp-Asp or Glu-Glu) or di-hydrophobic (e.g., Met-Met) signal sequences in their cytosolic segments. Some of these are also receptors for lumenal and GPI-anchored membrane secretory proteins. The transmembrane receptors recognize the GPI-anchors and signal sequences or signal patches characteristic of the folded lumenal proteins.

If proteins that function within the ER (including BiP, signal peptidase, protein disulfide isomerase, and other enzymes discussed earlier) are allowed to proceed along the secretory pathway, they will be lost to the cell. Thus many such proteins have a targeting sequence Lys-Asp-Glu-Leu (KDEL, in the single-letter code) at their carboxy terminus that directs their retrieval back to the ER. If this sequence is deleted from a protein that normally functions in the ER (e.g., BiP or protein disulfide isomerase), the mutated protein is instead transported to the Golgi and secreted from the cell. Conversely, addition of the KDEL sequence to the carboxy terminus of proteins that are normally secreted blocks their secretion. Some ER transmembrane proteins are similarly marked by short C-terminal sequences that contain two lysine residues (KKXX sequences).

Interestingly, the KDEL and KKXX signals do not prevent ER proteins from being packaged into vesicles and carried to the Golgi. Instead, these signals cause these ER resident proteins to be selectively retrieved from the ER-Golgi intermediate compartment or the Golgi complex and returned to the ER via a recycling pathway (Figure 10.25). Proteins bearing the KDEL and KKXX sequences bind to specific recycling receptors in the membranes of these compartments and are then selectively transported back to the ER. The KDEL and KKXX sequences are the best-characterized retention/retrieval signals but there may be others. Other proteins are retrieved because they specifically bind to KDEL-bearing proteins such as BiP. Thus continued movement along the secretory pathway or retrieval back from the Golgi to the ER is the first branch point encountered by proteins being sorted to their correct destinations in the secretory pathway. Similar branch points arise at each subsequent stage of transport, such as retention in the Golgi versus export to lysosomes or the plasma membrane. In each case, specific localization signals target proteins to their correct intracellular destinations.

The Golgi Apparatus

The **Golgi apparatus**, or **Golgi complex**, functions as a factory in which proteins received from the ER are further processed and sorted for transport to their eventual destinations: endosomes, lysosomes, the plasma

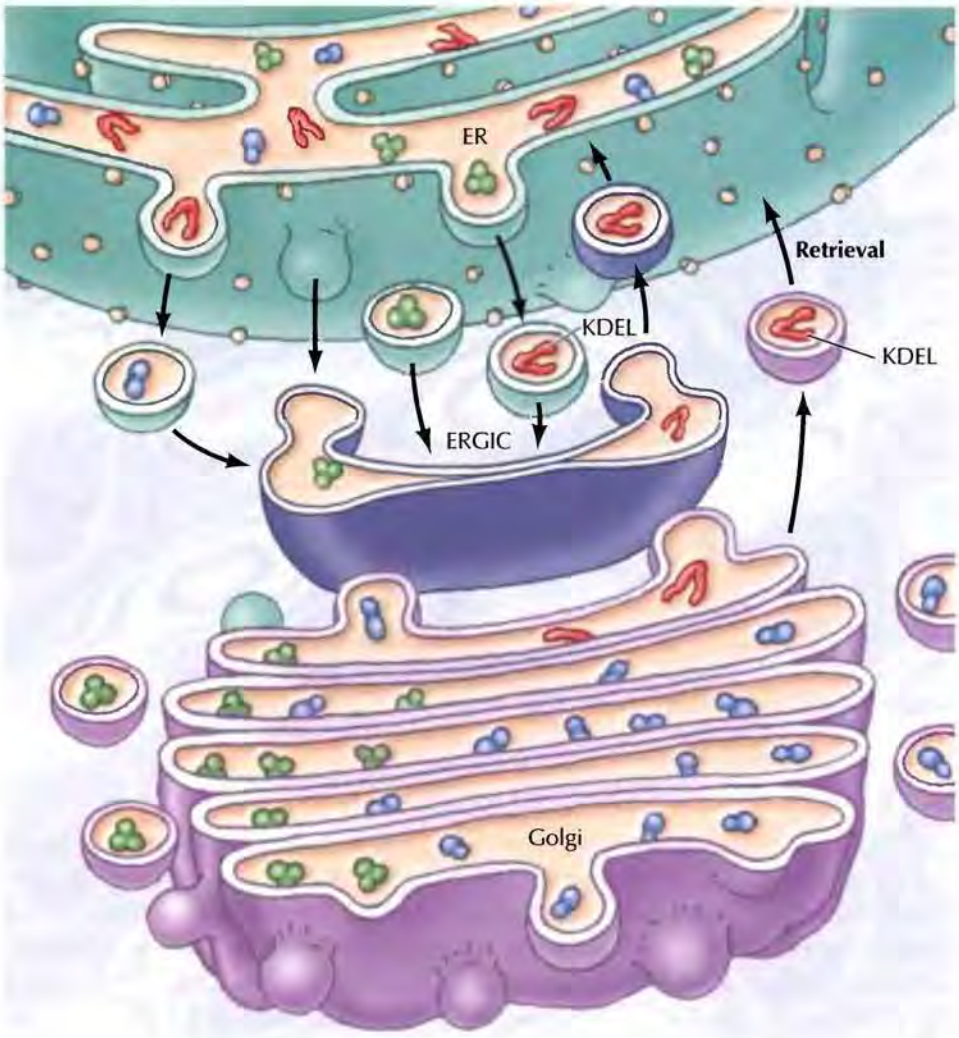


FIGURE 10.25 Retrieval of resident ER proteins Proteins destined to remain in the lumen of the ER are marked by the sequence Lys-Asp-Glu-Leu (KDEL) at their carboxy terminus. These proteins are exported from the ER to the Golgi, but they are recognized by a receptor in the ERGIC or the Golgi apparatus and selectively returned to the ER.

membrane, or secretion. In addition, as noted earlier, most glycolipids and sphingomyelin are synthesized within the Golgi. In plant cells, the Golgi apparatus further serves as the site at which the complex polysaccharides of the cell wall are synthesized. The Golgi apparatus is thus involved in processing the broad range of cellular constituents that travel along the secretory pathway.

Organization of the Golgi

In most cells, the Golgi is composed of flattened membrane-enclosed sacs (cisternae) and associated vesicles (Figure 10.26). A striking feature of the Golgi apparatus is its distinct polarity in both structure and function. Proteins from the ER enter at its *cis* face (entry face), which is convex and usually oriented toward the nucleus. They are then transported through the Golgi and exit from its concave *trans* face (exit face). The importance of this dynamic process for the structure of the Golgi is shown by the disappearance of the Golgi as an organized structure if vesicle transport from the ER is blocked. As they pass through the Golgi, proteins are modified and sorted for transport to their eventual destinations within the cell.

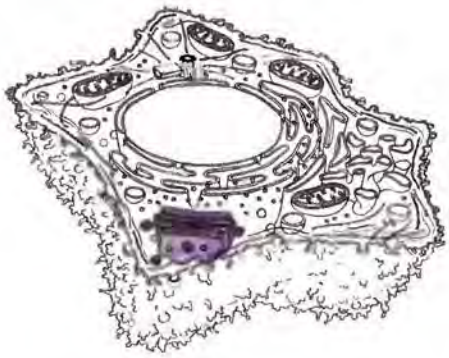


FIGURE 10.26 Electron micrograph of a Golgi apparatus The Golgi apparatus consists of a stack of flattened cisternae and associated vesicles. Proteins and lipids from the ER enter the Golgi apparatus at its *cis* face and exit near its *trans* face. (Courtesy of Dr. L. Andrew Staehelin, University of Colorado at Boulder.)

Distinct processing and sorting events appear to take place in an ordered sequence within different regions of the Golgi complex, so the Golgi is usually considered to consist of multiple discrete compartments. Although the number of such compartments has not been established, the Golgi is most commonly viewed as consisting of four functionally distinct regions: the ***cis* Golgi network**, the **Golgi stack** (which is divided into the *medial* and *trans* subcompartments), and the ***trans* Golgi network** (Figure 10.27). Proteins from the ER are transported to the ER-Golgi intermediate compartment and then enter the Golgi apparatus at the *cis* Golgi network. They then progress to the *medial* and *trans* compartments of the Golgi stack within which most metabolic activities of the Golgi apparatus take place. The modified proteins, lipids, and polysaccharides then move to the *trans* Golgi network, which acts as a sorting and distribution center, directing molecular traffic to endosomes, lysosomes, the plasma membrane, or the cell exterior.

Although the Golgi apparatus was first described over 100 years ago, the mechanism by which proteins move through the Golgi apparatus has still not been established and is an area of controversy among cell biologists. One possibility is that transport vesicles carry proteins between the cisternae of the Golgi compartments. However, there is considerable experimental support for proteins being carried through compartments of the Golgi within the Golgi cisternae, which gradually mature and progressively move through the Golgi in the *cis* to *trans* direction. It is likely that transport occurs by both of these processes. Recent work suggests that the dynamic structure of the Golgi is maintained by interactions of proteins in the cisternae membranes with the cytoskeleton.

Protein Glycosylation within the Golgi

Protein processing within the Golgi involves the modification and synthesis of the carbohydrate portions of glycoproteins. One of the major aspects of this processing is the modification of the *N*-linked oligosaccharides that

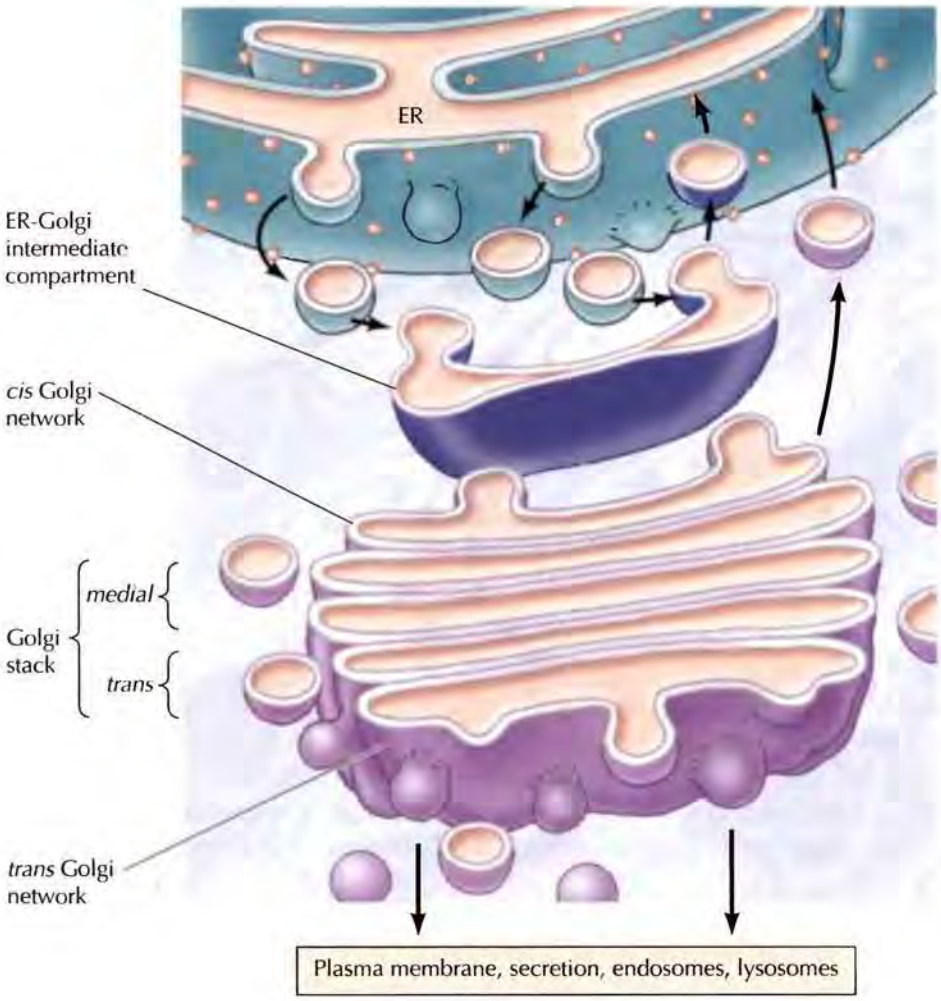


FIGURE 10.27 Regions of the Golgi apparatus Vesicles from the ER fuse to form the ER-Golgi intermediate compartment, and proteins from the ER are then transported to the *cis* Golgi network. Resident ER proteins are returned from the ER-Golgi intermediate compartment and the *cis* Golgi network via the recycling pathway. The *medial* and *trans* compartments of the Golgi stack correspond to the cisternae in the middle of the Golgi complex and are the sites of most protein modifications. Proteins are then carried to the *trans* Golgi network where they are sorted for transport to the plasma membrane, secretion, endosomes, or lysosomes.

10.1 WEBSITE ANIMATION

Organization of the Golgi The Golgi apparatus is composed of flattened membrane-enclosed sacs that receive proteins from the ER, process them, and sort them to their eventual destinations.

were added to proteins in the ER. As discussed earlier in this chapter, proteins are modified within the ER by the addition of an oligosaccharide consisting of 14 sugar residues (see Figure 10.16). Three glucose residues are removed while the polypeptides are still in the ER. Following transport to the Golgi apparatus, the *N*-linked oligosaccharides of these glycoproteins are subject to extensive further modifications.

N-linked oligosaccharides are processed within the Golgi apparatus in an ordered sequence of reactions (Figure 10.28). In most cases, the first modification of proteins destined for secretion or for the plasma membrane is the removal of four mannose residues. This is followed by the sequential addition of an *N*-acetylglucosamine, the removal of two more mannoses, and the addition of a fucose and two more *N*-acetylglucosamines. Finally, three galactose and three sialic acid residues are added. As noted in Chapter 8, different glycoproteins are modified to different extents during their passage through the Golgi, depending on both the structure of the protein and on the amount of processing enzymes that are present within the Golgi complexes of different types of cells. Consequently, proteins can emerge from the Golgi with a variety of different *N*-linked oligosaccharides. The enzymes that carry out the addition of sugar residues, **glycosyltransferases**, and those that remove them, **glycosidases**, are well-characterized, but the basis of their localization to specific cisternae of the Golgi is not known.

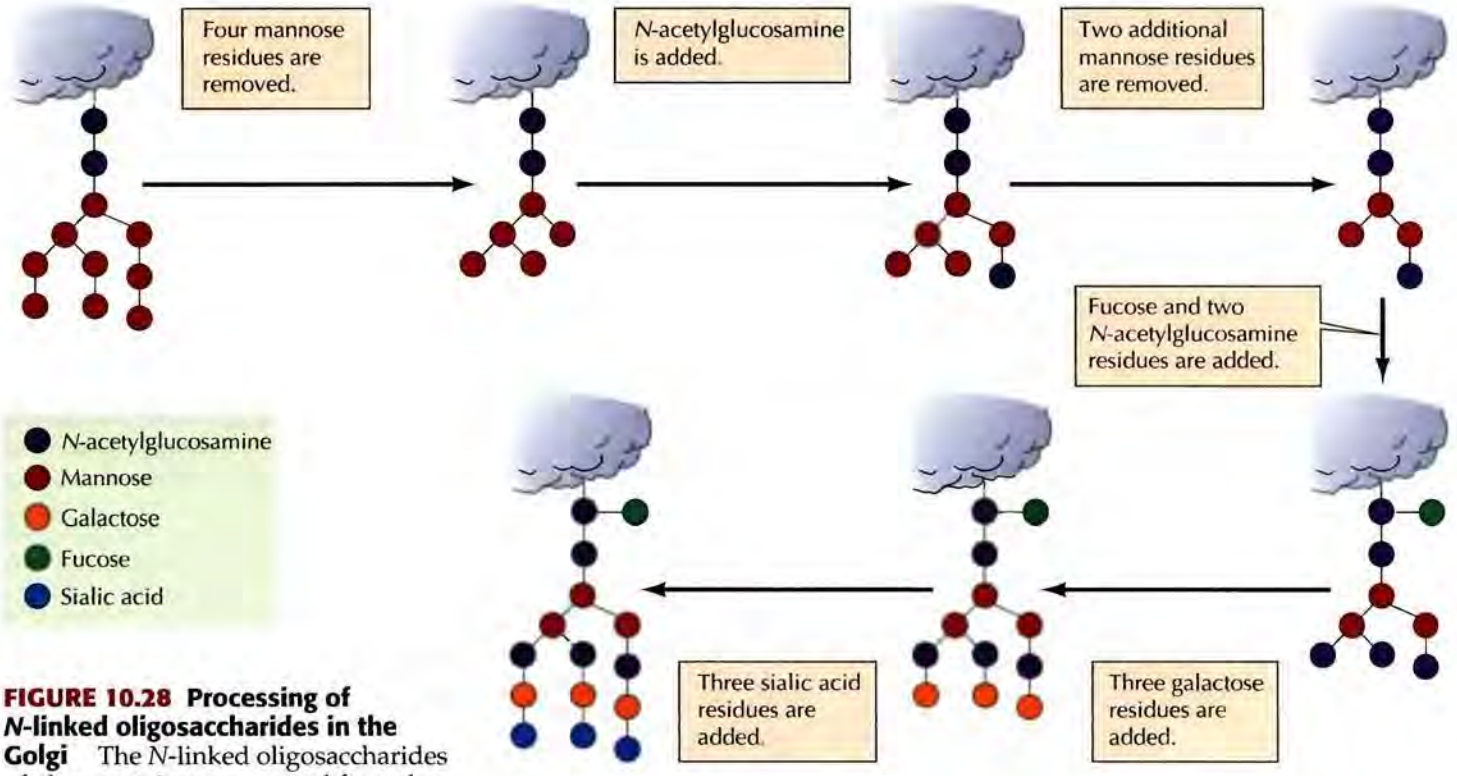


FIGURE 10.28 Processing of *N*-linked oligosaccharides in the Golgi The *N*-linked oligosaccharides of glycoproteins transported from the ER are further modified by an ordered sequence of reactions in the Golgi.

The processing of the *N*-linked oligosaccharide of lysosomal proteins differs from that of secreted and plasma membrane proteins. Rather than the initial removal of mannose residues, proteins destined for incorporation into lysosomes are modified by mannose phosphorylation. In the first step of this reaction, *N*-acetylglucosamine phosphates are added to specific mannose residues, probably while the protein is still in the *cis* Golgi network (Figure 10.29). This is followed by removal of the *N*-acetylglucosamine group, leaving **mannose-6-phosphate** residues on the *N*-linked oligosaccharide. Because of this modification these residues are not removed during further processing. Instead, these phosphorylated mannose residues are specifically recognized by a mannose-6-phosphate receptor in the *trans* Golgi network, which directs the transport of these proteins to endosomes and on to lysosomes.

The phosphorylation of mannose residues is thus a critical step in sorting lysosomal proteins to their correct intracellular destinations. The specificity of this process resides in the enzyme that catalyzes the first step in the reaction sequence—the selective addition of *N*-acetylglucosamine phosphates to lysosomal proteins. This enzyme recognizes a structural determinant that is present on lysosomal proteins but not on proteins destined for the plasma membrane or secretion. This recognition determinant is not a simple sequence of amino acids; rather, it is formed in the folded protein by the juxtaposition of amino acid sequences from different regions of the polypeptide chain. In contrast to the signal sequences that direct protein translocation to the ER, the recognition determinant that leads to mannose phosphorylation, and thus ultimately targets proteins to lysosomes, depends on the three-dimensional conformation of the folded protein. Such determinants are called **signal patches**, in contrast to the linear targeting signals discussed earlier in this chapter.

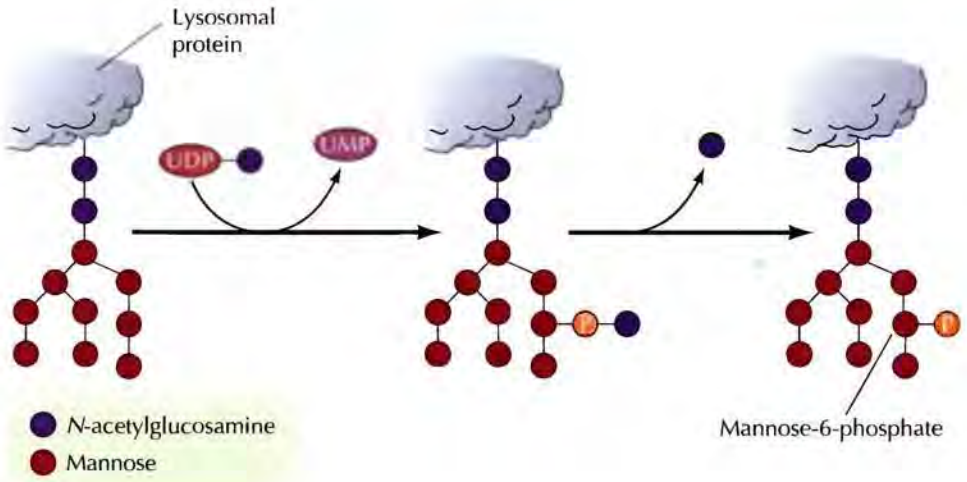


FIGURE 10.29 Targeting of lysosomal proteins by phosphorylation of mannose residues Proteins destined for incorporation into lysosomes are specifically recognized and modified by the addition of phosphate groups to the number 6 position of mannose residues. In the first step of the reaction, *N*-acetylglucosamine phosphates are transferred to mannose residues from UDP*N*-acetylglucosamine. The *N*-acetylglucosamine groups are then removed, leaving mannose-6-phosphates.

Proteins can also be modified by the addition of carbohydrates to the side chains of acceptor serine and threonine residues within specific sequences of amino acids (*O*-linked glycosylation) (see Figure 8.31). These modifications take place in the Golgi apparatus by the sequential addition of single sugar residues. The serine or threonine is usually linked directly to *N*-acetylgalactosamine to which other sugars can then be added. In some cases these sugars are further modified by the addition of sulfate groups.

Lipid and Polysaccharide Metabolism in the Golgi

In addition to its activities in processing and sorting glycoproteins, the Golgi apparatus functions in lipid metabolism—in particular, in the synthesis of glycolipids and sphingomyelin. As discussed earlier, the glycerol phospholipids, cholesterol, and ceramide are synthesized in the ER. Sphingomyelin and glycolipids are then synthesized from ceramide in the Golgi apparatus (Figure 10.30). Sphingomyelin (the only nonglycerol phospholipid in cell membranes) is synthesized by the transfer of a phosphorylcholine group from phosphatidylcholine to ceramide. Alternatively, the addition of carbohydrates to ceramide can yield a variety of different glycolipids.

Sphingomyelin is synthesized on the lumenal surface of the Golgi, but glucose is added to ceramide on the cytosolic side. Glucosylceramide then apparently

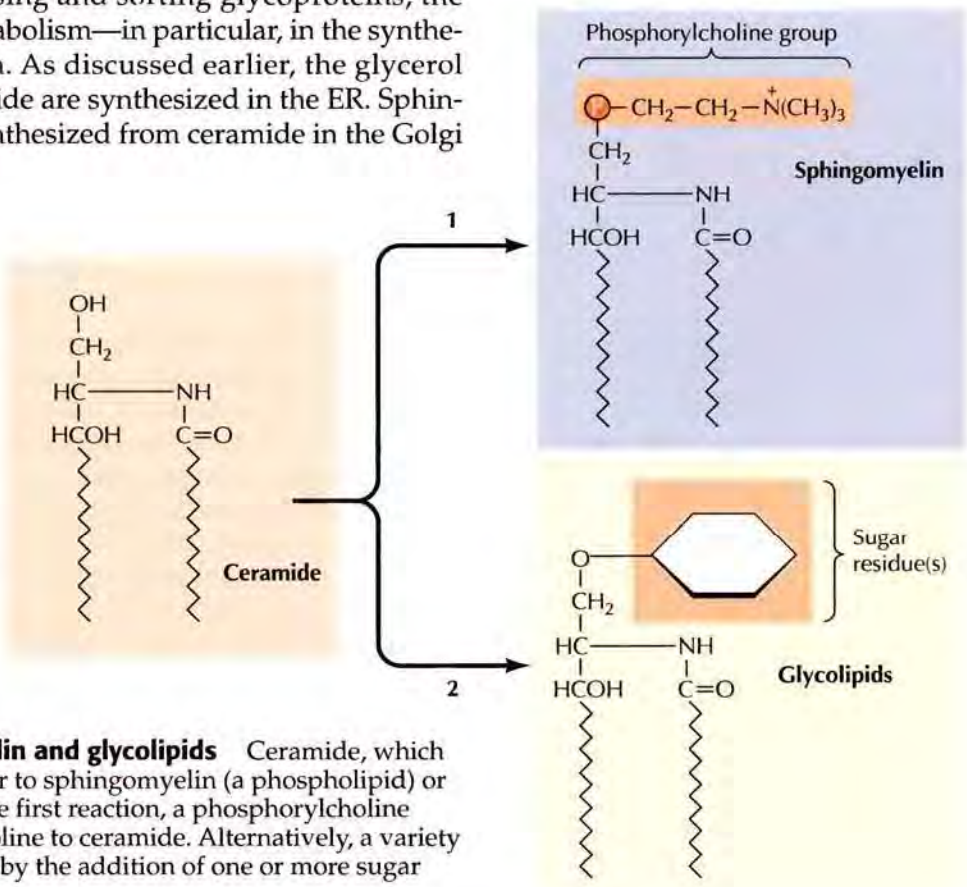


FIGURE 10.30 Synthesis of sphingomyelin and glycolipids Ceramide, which is synthesized in the ER, is converted either to sphingomyelin (a phospholipid) or to glycolipids in the Golgi apparatus. In the first reaction, a phosphorylcholine group is transferred from phosphatidylcholine to ceramide. Alternatively, a variety of different glycolipids can be synthesized by the addition of one or more sugar residues (e.g., glucose).

flips, however, and additional carbohydrates are added on the lumenal side of the membrane. Glycolipids are not able to translocate across the Golgi membrane, so they are found only in the lumenal half of the Golgi bilayer as is most sphingomyelin. Following vesicular transport they are correspondingly localized to the exterior half of the plasma membrane, with their polar head groups exposed on the cell surface. As will be discussed in Chapter 13, the oligosaccharide portions of glycolipids are important surface markers in cell-cell recognition.

In plant cells, the Golgi apparatus has the additional task of serving as the site where complex polysaccharides of the cell wall are synthesized. As discussed further in Chapter 14, the plant cell wall is composed of three major types of polysaccharides. Cellulose, the predominant constituent, is a simple linear polymer of glucose residues. It is synthesized at the cell surface by enzymes in the plasma membrane. The other cell wall polysaccharides (hemicelluloses and pectins), however, are complex branched chain molecules that are synthesized in the Golgi apparatus and then transported in vesicles to the cell surface. The synthesis of these cell wall polysaccharides is a major cellular function, and as much as 80% of the metabolic activity of the Golgi apparatus in plant cells may be devoted to polysaccharide synthesis.

Protein Sorting and Export from the Golgi Apparatus

Proteins as well as lipids and polysaccharides are transported from the Golgi apparatus to their final destinations through the secretory pathway. This involves the sorting of proteins into different kinds of transport vesicles, which bud from the *trans* Golgi network and deliver their contents to the appropriate cellular locations (Figure 10.31). Some proteins are carried from the Golgi to the plasma membrane by a constitutive secretory pathway, which accounts for the incorporation of new proteins and lipids into the plasma membrane as well as for the continuous secretion of proteins from the cell. Other proteins are transported to the cell surface by a distinct pathway of regulated secretion or are specifically targeted to other intracellular destinations, such as lysosomes in animal cells or vacuoles in yeast.

Proteins that function within the Golgi apparatus must be retained within that organelle rather than being transported along the secretory pathway. In contrast to the ER, all of the proteins known to be retained within the Golgi complex are associated with the Golgi membrane rather than being soluble proteins within the lumen. The signals responsible for retention of some proteins within the Golgi have been localized to their transmembrane domains, which prevent the protein from being packaged in the transport vesicles that leave the *trans* Golgi network. In addition, like the KKXX sequences of resident ER membrane proteins, signals in the cytoplasmic tails of some Golgi proteins mediate the retrieval of these proteins from subsequent compartments along the secretory pathway.

The constitutive secretory pathway, which operates in all cells, leads to continual unregulated protein secretion. However, some cells also possess a distinct regulated secretory pathway in which specific proteins are secreted in response to environmental signals. Examples of regulated secretion include the release of hormones from endocrine cells, the release of neurotransmitters from neurons, and the release of digestive enzymes from the pancreatic acinar cells discussed at the beginning of this chapter (see Figure 10.2). Proteins are sorted into the regulated secretory pathway in the *trans* Golgi network where they are packaged into specialized secretory vesicles. These immature secretory vesicles, which are larger than transport vesicles,

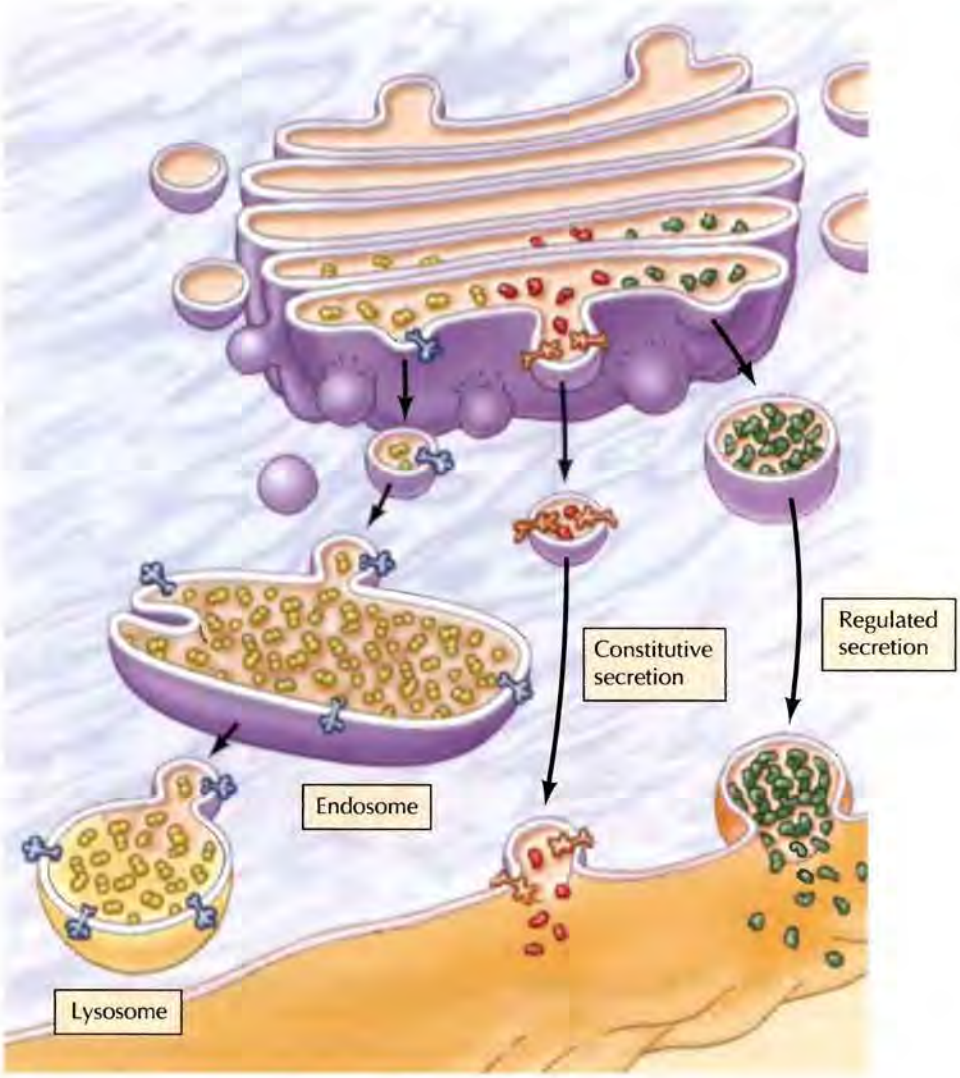


FIGURE 10.31 Transport from the Golgi apparatus Proteins are sorted in the *trans* Golgi network and transported in vesicles to their final destinations. In the absence of specific targeting signals, proteins are carried to the plasma membrane by constitutive secretion. Alternatively, proteins can be diverted from the constitutive secretion pathway and targeted to other destinations, such as endosomes and lysosomes or regulated secretion from the cells.

further process their protein contents and often fuse with each other. They also store their contents until specific signals direct their fusion with the plasma membrane. For example, the digestive enzymes produced by pancreatic acinar cells are stored in mature secretory vesicles until the presence of food in the stomach and small intestine triggers their secretion. The sorting of proteins into the regulated secretory pathway appears to involve the recognition of signal patches shared by multiple proteins that enter this pathway. These proteins selectively aggregate in cisternae of the *trans* Golgi network and are then released by budding as immature secretory vesicles. Proteins destined for very different locations, such as the plasma membrane versus lysosomes, are known to aggregate in different *trans* Golgi cisternae.

A further complication in the transport of proteins to the plasma membrane arises in many epithelial cells, which are polarized when they are organized into tissues. The plasma membrane of such cells is divided into two separate regions, the **apical domain** and the **basolateral domain**, that contain specific proteins related to their particular functions. For example, the apical membrane of intestinal epithelial cells faces the lumen of the intestine and is specialized for the efficient absorption of nutrients; the

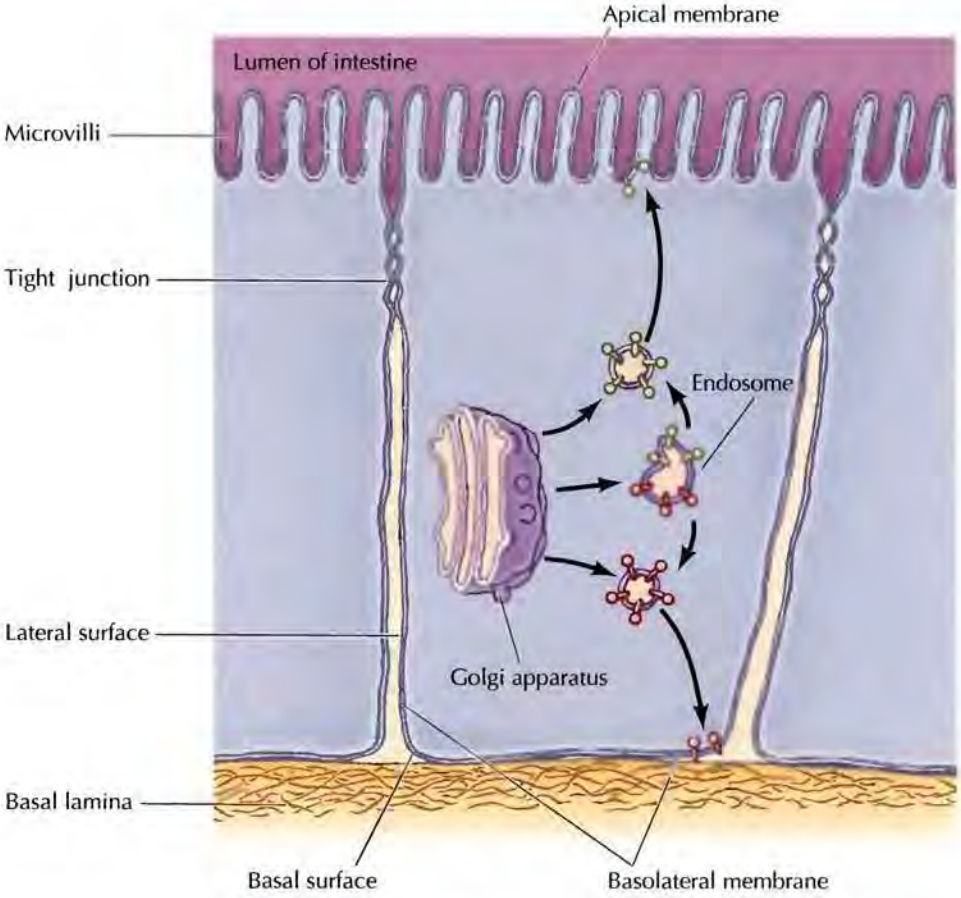


FIGURE 10.32 Transport to the plasma membrane of polarized cells The plasma membranes of polarized epithelial cells are divided into apical and basolateral domains. In this example (intestinal epithelium), the apical surface of the cell faces the lumen of the intestine, the lateral surfaces are in contact with neighboring cells, and the basal surface rests on a sheet of extracellular matrix (the basal lamina). The apical membrane is characterized by the presence of microvilli, which facilitate the absorption of nutrients by increasing surface area. Specific proteins are targeted to the apical or basolateral membranes either in the *trans* Golgi network or in an endosome. Tight junctions between neighboring cells maintain the identity of the apical and basolateral membranes by preventing the diffusion of proteins between these domains.

remainder of the cell is covered by the basolateral membrane (Figure 10.32). Distinct domains of the plasma membrane are present not only in epithelial cells but also in other cell types. Thus the constitutive secretory pathway must selectively transport proteins from the *trans* Golgi network to these distinct domains of the plasma membrane. This is accomplished by the selective packaging of proteins into at least two types of constitutive secretory vesicles targeted specifically for either the apical or basolateral plasma membrane domains of the cell. This may occur in the *trans* Golgi network or in an endosome.

The best-characterized pathway of protein sorting in the Golgi is the selective transport of proteins to lysosomes. As already discussed, lumenal lysosomal proteins are marked by mannose-6-phosphates that are formed by modification of their *N*-linked oligosaccharides shortly after entry into the Golgi apparatus. A specific receptor in the membrane of the *trans* Golgi network then recognizes these mannose-6-phosphate residues. The resulting complexes of receptor plus lysosomal enzyme are packaged into transport vesicles destined for lysosomes. Lysosomal membrane proteins are targeted by sequences in their cytoplasmic tails rather than by mannose-6-phosphates.

In yeasts and plant cells, both of which lack lysosomes, proteins are transported from the Golgi apparatus to an additional destination: the **vacuole** (Figure 10.33). Vacuoles assume the functions of lysosomes in these cells as well as perform a variety of other tasks, such as the storage of nutrients and the maintenance of turgor pressure and osmotic balance. In contrast to lysosomal targeting, proteins are directed to vacuoles by short peptide sequences instead of carbohydrate markers.

FIGURE 10.33 A plant cell vacuole The large central vacuole functions as a lysosome in addition to storing nutrients and maintaining osmotic balance. (E. H. Newcombe/Biological Photo Service.)

The Mechanism of Vesicular Transport

As is evident from the preceding sections of this chapter, transport vesicles play a central role in the traffic of molecules between different membrane-enclosed compartments of the secretory pathway. As discussed in Chapter 13, vesicles are similarly involved in the transport of materials taken up at the cell surface. Vesicular transport is thus a major cellular activity, responsible for molecular traffic between a variety of specific membrane-enclosed compartments. The selectivity of such transport is therefore key to maintaining the functional organization of the cell. For example, lysosomal enzymes must be transported specifically from the Golgi apparatus to lysosomes—not to the plasma membrane or to the ER. Some of the signals that target proteins to specific organelles, such as lysosomes, were discussed earlier in this chapter. These proteins are transported within vesicles, so the specificity of transport is based on the selective packaging of the intended cargo into vesicles that recognize and fuse only with the appropriate target membrane. Because of the central importance of vesicular transport to the organization of eukaryotic cells, understanding the molecular mechanisms that control vesicle packaging, budding, and fusion is a major area of research in cell biology.

Experimental Approaches to Understanding Vesicular Transport

Progress toward elucidating the molecular mechanisms of vesicular transport has been made by three distinct experimental approaches: (1) isolation of yeast mutants that are defective in protein transport and sorting; (2) reconstitution of vesicular transport in cell-free systems; and (3) biochemical analysis of synaptic vesicles, which are responsible for the regulated

secretion of neurotransmitters by neurons. Each of these experimental systems has distinct advantages for understanding particular aspects of the transport process. Most important, however, is the fact that the results from all three of these avenues of investigation have converged, indicating that similar molecular mechanisms regulate secretion in cells as different as yeasts and mammalian neurons.

As in other areas of cell biology, yeasts have proven to be advantageous in studying the secretory pathway because they are readily amenable to genetic analysis. In particular, Randy Schekman and his colleagues have pioneered the isolation of yeast mutants defective in vesicular transport. These include mutants that are defective at various stages of protein secretion (*sec* mutants), mutants that are unable to transport proteins to the vacuole, and mutants that are unable to retain resident ER proteins. The isolation of such mutants in yeasts led directly to the molecular cloning and analysis of the corresponding genes, thereby identifying a number of proteins involved in various steps of the secretory pathway. For example, the role of Sec61 as a major component of the protein translocation channel in the endoplasmic reticulum was discussed earlier in this chapter.

Biochemical studies of vesicular transport using reconstituted systems have complemented these genetic studies and have enabled the direct isolation of transport proteins from mammalian cells. The first cell-free transport system was developed by James Rothman and colleagues, who analyzed protein transport between compartments of the Golgi apparatus. Similar reconstituted systems have been developed to analyze transport between other compartments, including transport from the ER to the Golgi and transport from the Golgi to secretory vesicles, vacuoles, and the plasma membrane. The development of these *in vitro* systems has enabled biochemical studies of the transport process and functional analysis of proteins identified by mutations in yeasts, as well as direct isolation of some of the proteins involved in vesicle budding and fusion.

Insights into the molecular mechanisms of vesicular transport have come from studies of synaptic transmission in neurons, which represents a highly specialized form of regulated secretion. A synapse is the junction of a neuron with another cell, which may be either another neuron or an effector, such as a muscle cell. Information is transmitted across the synapse by chemical neurotransmitters, such as acetylcholine, which are stored within the neuron in **synaptic vesicles**. Stimulation of the transmitting neuron triggers the fusion of these synaptic vesicles with the plasma membrane causing neurotransmitters to be released into the synapse and stimulating the postsynaptic neuron or effector cell. Synaptic vesicles are extremely abundant in the brain, allowing them to be purified in large amounts for biochemical analysis. Some of the proteins isolated from synaptic vesicles are closely related to proteins that have been shown to play critical roles in vesicular transport by yeast genetics and reconstitution experiments, so biochemical analysis of these proteins has provided important insights into the molecular mechanisms of vesicle fusion.

Recent studies using GFP fusion proteins have allowed transport vesicles carrying specific proteins to be visualized by immunofluorescence as they move through the secretory pathway. In these experiments, cells are transfected with cDNA constructs encoding secretory proteins tagged with green fluorescent protein (GFP) (see Figure 1.27). The progress of the GFP-labeled proteins through the secretory pathway can then be followed in living cells, allowing characterization of many aspects of the dynamics and molecular interactions involved in vesicular transport.

Cargo Selection, Coat Proteins, and Vesicle Budding

Most transport vesicles that carry secretory proteins from the ER to the Golgi and from the Golgi to other targets are coated with cytosolic coat proteins and thus are called coated vesicles. Initially, the secretory proteins are sorted from proteins targeted for other destinations and from proteins that need to remain behind (see Figure 10.24). The coats assemble as the secretory protein-containing vesicle buds off the donor membrane and are generally removed from the vesicle in the cytosol before it reaches its target (Figure 10.34). Some of the remaining proteins allow the vesicles to travel along microtubules to their targets by interacting with specific tubulin-based molecular motors as described in Chapter 12. At the target membrane, the vesicles dock and fuse with the membrane, emptying their lumenal cargo and inserting their membrane proteins into the target membrane.

The formation of coated vesicles is regulated by small GTP-binding proteins related to Ras and Ran. Two families of GTP-binding proteins play roles in transport vesicle budding: ADP-ribosylation factors (ARFs 1–3 & Sar1) and a large family of Rab proteins. These regulate adaptor proteins that interact directly with a vesicle coat protein. The binding of GTP-binding proteins and adaptor proteins establishes a "platform" on the membrane for a specific process, such as assembly and budding of a transport vesicle directed from the transitional ER to the Golgi or from the *trans* Golgi network to endosomes and lysosomes. Individual proteins in the complex (coat proteins, adaptor proteins, and GTP-binding proteins) may participate in assembly of transport vesicles directed elsewhere, or in vesicle fusion (described later in the chapter), but each protein complex is apparently unique to a particular budding, transport, or fusion pathway.

Three kinds of coated vesicles have been characterized: two types of **COP-coated vesicles, COPI** and **COPII** (COP indicates coat protein), and

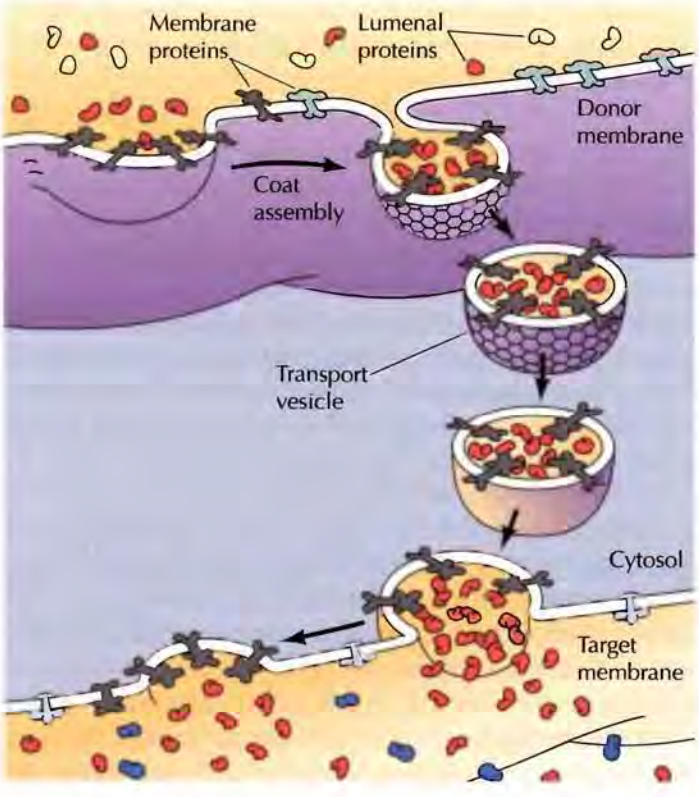


FIGURE 10.34 Formation and fusion of a transport vesicle Membrane and lumenal secretory proteins are collected into selected regions of a donor membrane where the formation of a cytosolic coat results in the budding off of a transport vesicle. During transport the coat is disassembled, and the transport vesicle docks and fuses with the target membrane.

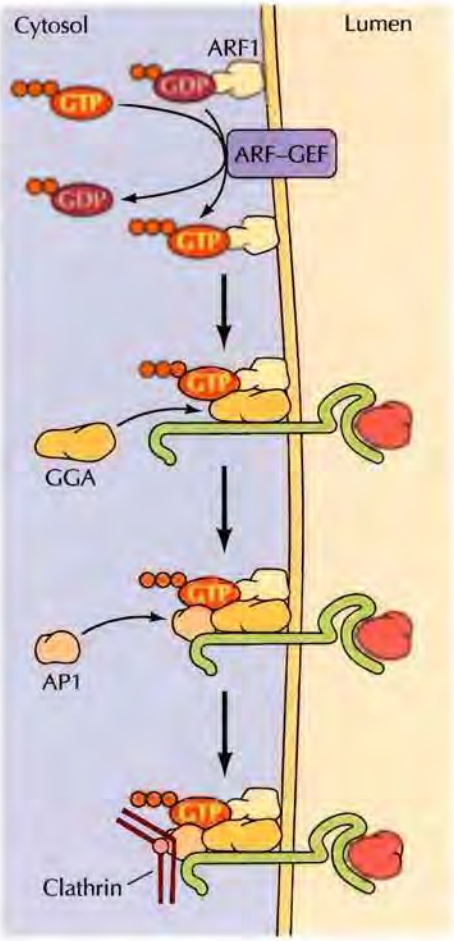


FIGURE 10.35 Initiation of a clathrin-coated vesicle by ARF1 The small GTP-binding protein, ARF1, can initiate the formation of a clathrin-coated vesicle on the *trans* Golgi membrane. After being delivered to the membrane, ARF/GDP is activated to ARF/GTP by a specific ARF-guanine nucleotide exchange factor (ARF-GEF). ARF/GTP recruits a GGA adaptor protein to the membrane and this protein recruits a transmembrane receptor with its bound lumenal cargo by interacting with the cytosolic tail of the receptor. GGA then recruits a second adaptor protein, AP1, which serves as a binding site for assembly of a clathrin coat.

clathrin-coated vesicles. COPII-coated vesicles carry secretory proteins from the ER to the ER-Golgi intermediate compartment or Golgi apparatus, budding from the transitional ER and carrying their cargo forward along the secretory pathway. COPI-coated vesicles bud from the ER-Golgi intermediate compartment, or the Golgi apparatus. COPI is the coat protein on vesicles moving from one Golgi cisternae to another during secretion and on the retrieval vesicles that return resident ER proteins marked by the KDEL or KKXX retrieval signals back to the ER from the ER-Golgi intermediate compartment or the *cis* Golgi network.

Clathrin-coated vesicles are the best understood and are responsible for the uptake of extracellular molecules from the plasma membrane by endocytosis (see Chapter 13) as well as the transport of molecules from the *trans* Golgi network to endosomes, lysosomes, or the plasma membrane. The formation of clathrin-coated vesicles on the *trans* Golgi network requires **clathrin**, the GTP-binding protein, ARF1, and at least two types of adaptor proteins (Figure 10.35). First, ARF/GDP binds to proteins in the Golgi membrane. An ARF-guanine nucleotide exchange factor in the membrane then stimulates the exchange of the GDP for GTP. ARF/GTP initiates the budding process by recruiting adaptor proteins, which then serve as binding sites for both transmembrane receptors and for clathrin. Clathrin actually plays a structural role in vesicle budding by assembling into a basketlike lattice structure that distorts the membrane and initiates the bud. This is illustrated for vesicles targeted to the lysosome in Figure 10.36. During transport, the GTP bound to ARF1 is hydrolyzed to GDP and the ARF/GDP is released from the membrane to recycle. Loss of ARF1 and the action of uncoating enzymes weakens the cooperative binding of the clathrin coat complex, allowing chaperone proteins in the cytoplasm to dissociate most of the coat from the vesicle membrane (see Figure 10.34).

While COPI- and COPII-coated vesicles have limited targets, clathrin-coated vesicles exit the *trans* Golgi for different destinations: endosomes, lysosomes, or different plasma membrane domains. Since these targets require specific cargoes, different adaptor proteins play a role in the assembly of vesicles for different destinations.

Vesicle Fusion

The fusion of a transport vesicle with its target involves two types of events. First, the transport vesicle must recognize the correct target membrane; for example, a vesicle carrying lysosomal enzymes has to deliver its cargo only to lysosomes. Second, the vesicle and target membranes must fuse, delivering the contents of the vesicle to the target organelle. Research over the last several years has led to the development of several models of vesicle fusion in which specific recognition between a vesicle and its target (tethering) is

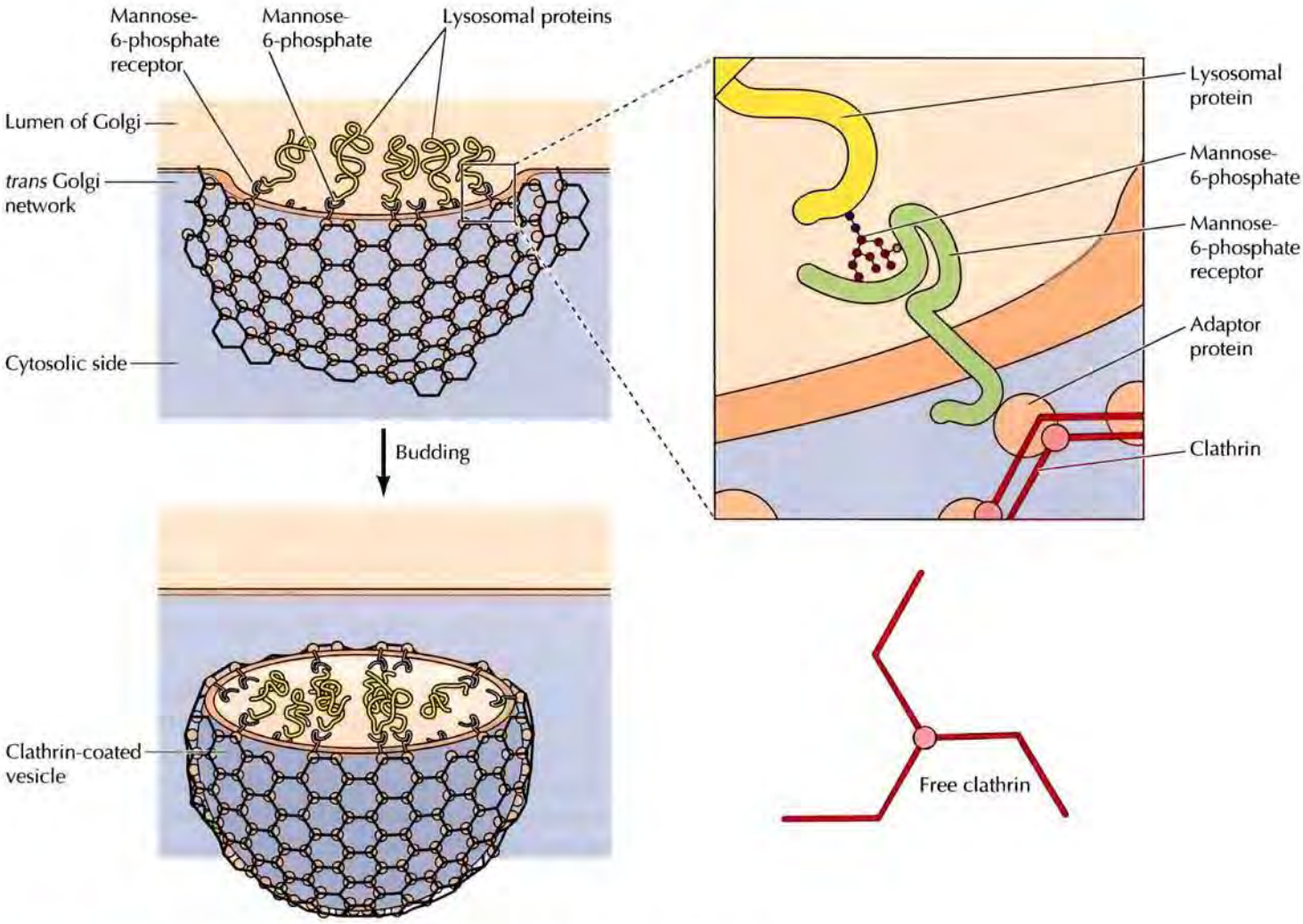


FIGURE 10.36 Incorporation of lysosomal proteins into clathrin-coated vesicles Proteins targeted for lysosomes are marked by mannose-6-phosphates, which bind to mannose-6-phosphate receptors in the *trans* Golgi network. The mannose-6-phosphate receptors span the Golgi membrane and serve as binding sites for cytosolic adaptor proteins, which in turn bind clathrin. Clathrins consist of three protein chains that associate with each other to form a basketlike lattice that distorts the membrane and drives vesicle budding.

mediated by interactions between proteins on the vesicle and the target membranes, followed by fusion between the phospholipid bilayers.

Proteins involved in vesicle fusion were initially identified in James Rothman's laboratory by biochemical analysis of reconstituted vesicular transport systems from mammalian cells. Analysis of the proteins involved in vesicle fusion in these systems led Rothman and his colleagues to propose a general model, called the **SNARE hypothesis**, in which vesicle fusion is mediated by interactions between specific pairs of transmembrane proteins, called SNAREs, on the vesicle and target membranes (v-SNAREs and t-SNAREs, respectively). According to the hypothesis, the formation of complexes between v-SNAREs on the vesicle and t-SNAREs on the target membranes leads to membrane fusion. This hypothesis was supported by the identification of SNAREs that were present on synaptic vesicles and by the finding of yeast secretion mutants that appeared to encode SNAREs

TABLE 10.1 Rab GTP-Binding Proteins and Their Sites of Action

Transport step	Rab proteins involved
Exocytosis	
Transitional ER to Golgi	Rab1, Rab1b, Rab2
Golgi back to ER	Rab6, Rab6b
Intra-Golgi	Rab1, Rab6, Rab6b
trans Golgi network to plasma membrane	Rab11a, Rab11b
Endocytosis	
Plasma membrane to early endosome	Rab5a, Rab5b, Rab5c
Early endosome to plasma membrane	Rab4, Rab15, Rab18
Early endosome to late endosome	Rab7
Special roles	
Exocytosis of secretory granules	Rab8b
Late endosome to *trans* Golgi network	Rab9, Rab11a, Rab11b
trans Golgi network to basolateral membrane	Rab8a
trans Golgi network to apical membrane	Rab21

Examples of the more than 60 mammalian Rab proteins whose locations and presumptive functions are known.

■ **Griscelli syndrome is a rare disease caused by mutations in the gene encoding Rab27a. Rab27a appears to play a key role in the transport of pigment-containing vesicles (melanosomes) to the skin and hair and in the exocytosis of vesicles in T lymphocytes. Patients with Griscelli syndrome exhibit partial albinism (lack of pigment) and are immunodeficient.**

required for a variety of vesicle transport events. For example, transport from the ER to the Golgi in yeast requires specific SNAREs that are located on both the vesicle and target membranes.

Recent research has confirmed that SNAREs are required for vesicle fusion with a target membrane and that SNARE-SNARE pairing provides the energy to bring the two bilayers sufficiently close to destabilize them and result in fusion. However, the docking, tethering, and fusion of transport vesicles to specific target membranes appears to be mediated by a sequentially assembled protein complex much like that which led to transport vesicle budding. Members of the **Rab** family of small GTP-binding proteins play key roles in this docking of transport vesicles. Rab proteins, like the ARF family, participate in many of the vesicle budding and fusion reactions during vesicular transport. More than 60 different Rab proteins have been identified and shown to function in specific vesicle transport processes (Table 10.1). They function in many steps of vesicle trafficking, including interacting with SNAREs to regulate and facilitate the formation of SNARE/SNARE complexes.

Individual Rab proteins or combinations of Rab proteins mark different organelles and transport vesicles, so their localization on the correct membrane is key to establishing the specificity of vesicular transport (Figure 10.37). The Rab proteins are carried through the cytosol in their GDP-bound form by GDP-dissociation inhibitors (GDIs). At a membrane, they are

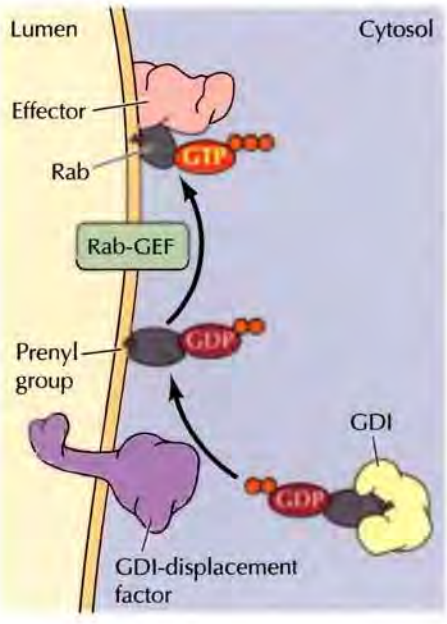


FIGURE 10.37 Delivery of Rab to a membrane The small GTP-binding protein Rab is modified by the addition of a prenyl group, which allows it to insert into a membrane (see Figure 8.33). Rab is carried in the cytosol bound to a GDP–dissociation inhibitor (GDI) keeping it in the Rab/GDP state. At a membrane, a non-specific GDI-displacement factor can remove the Rab/GDP from the GDI and insert it into the membrane. If a specific Rab-guanine nucleotide exchange factor (Rab-GEF) is present, the GDP on Rab will be exchanged for GTP and the active Rab/GTP can interact with effector proteins. If the correct Rab-guanine nucleotide exchange factor is not present, the Rab/GDP will be removed by a GDI and carried to another membrane.

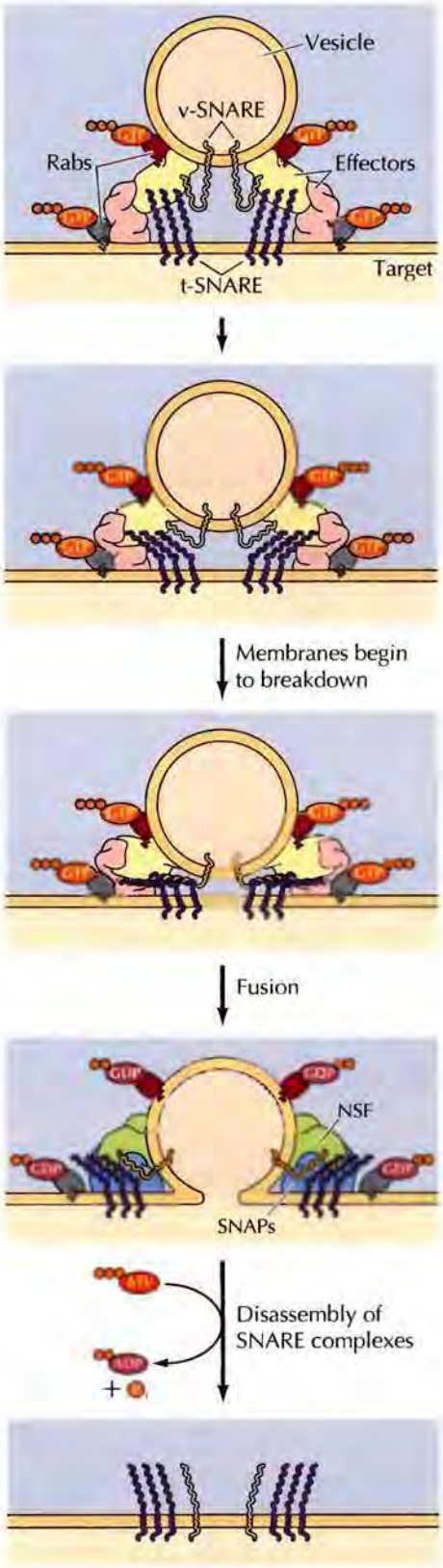


FIGURE 10.38 Vesicle fusion Vesicle fusion is initiated by Rab/GTP. Specific Rab proteins on the vesicle and target membranes bind effector proteins to tether the vesicle to the target membrane. Tethering allows the v-SNAREs and t-SNAREs to interact. The coiled-coil domains of the SNAREs zip together, providing energy to bring the vesicle and target membranes into close proximity. This close proximity of the membranes destabilizes the lipid bilayers, and the vesicle and target membranes fuse. The changes in protein-protein interactions recruit NSF and SNAPs to the SNARE complex, and they disassemble the complex using energy from the hydrolysis of ATP.

removed from GDIs by GDI-displacement factors. Specific guanine nucleotide exchange factors then convert Rab/GDP to the active Rab/GTP state. Individual guanine nucleotide exchange factors are localized to specific membranes and act on specific members of the Rab family, so they are, in part, responsible for formation of active Rab/GTP complexes at the correct membrane sites. In the absence of the appropriate exchange factor, Rab proteins remain as Rab/GDP and are removed from the membrane. Thus full activation of a Rab protein requires that it both bind GTP and be associated with a membrane.

To initiate transport vesicle fusion, Rab/GTP on the transport vesicle interacts with effector proteins and v-SNAREs to assemble a pre-fusion complex (Figure 10.38). A different Rab protein on the target membrane similarly organizes other effector proteins and t-SNAREs. When the transport vesicle encounters this target membrane, the effector proteins link the membranes by protein-protein interactions. This tethering of the vesicle to the target membrane stimulates Rab/GTP hydrolysis and allows the v-SNAREs to contact the t-SNAREs. All SNARE proteins have a long central coiled-coil domain like that found in nuclear lamins (see Figure 9.4). As in the lamins, this domain binds strongly to other coiled-coil domains and, in effect, zips the SNAREs together, bringing the two membranes into nearly direct contact. The simplest hypothesis is that this creates instability in the lipid bilayers and they fuse. Following membrane fusion, the NSF/SNAP complex disassembles the SNARE complex, allowing the SNAREs to be reused for subsequent rounds of vesicle transport. As the energy of SNARE-SNARE interaction drives the fusion of the membranes, energy from hydrolysis of ATP is required to separate the SNAREs.

Membrane fusion is a general process that occurs whenever a transport vesicle fuses with a target membrane. However, specific types of fusion may involve specialized sites on the plasma membrane. One of these is exocytosis, the fusion of a transport vesicle with the plasma membrane, resulting in secretion of the vesicle contents. Many types of exocytosis occur at specific protein complexes, called **exocysts,** on the plasma membrane. This eight protein complex was first discovered to be required for secretion in the yeast *Saccharomyces cerevisiae*, but it also plays an important role in secretion in polarized mammalian cells. The structure of exocysts is not well understood but their assembly appears to require sequential interactions among eight exocyst proteins localized on both the transport vesicle and the target membrane site (Figure 10.39). Interaction of these proteins results in efficient targeting of the transport vesicle to a specific location on the plasma membrane. Several small GTP-binding proteins are also associated with exocysts; some like the Rab proteins are involved in vesicle docking and fusion but others may play a role in localizing exocysts to apical or basolateral membranes or to axons or dendrites.

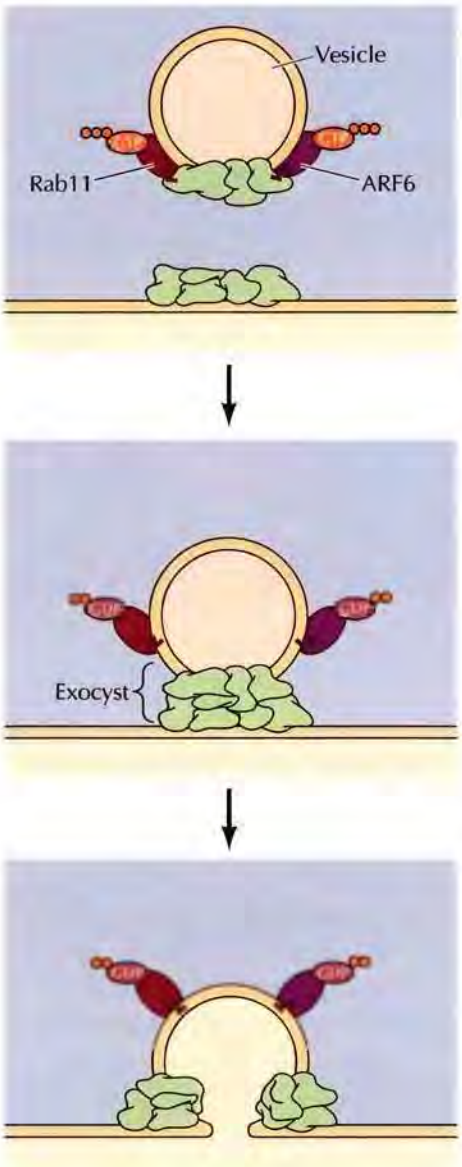


FIGURE 10.39 Exocyst assembly and vesicle targeting Exocysts are complexes of eight different proteins formed during exocytosis from proteins present on both transport vesicles and specific regions of the plasma membrane. Tethering and docking at exocysts results in normal SNARE-mediated vesicle fusion. Small GTP-binding proteins including Rab11 and ARF6 regulate assembly of the exocyst complex on the transport vesicle and coordinate its movement to the target site.

Lysosomes

Lysosomes are membrane-enclosed organelles that contain an array of enzymes capable of breaking down all types of biological polymers—proteins, nucleic acids, carbohydrates, and lipids. Lysosomes function as the digestive system of the cell, serving both to degrade material taken up from outside the cell and to digest obsolete components of the cell itself. In their simplest form, lysosomes are visualized as dense spherical vacuoles, but they can display considerable variation in size and shape as a result of differences in the materials that have been taken up for digestion (Figure 10.40). Lysosomes thus represent morphologically diverse organelles defined by the common function of degrading intracellular material.

Lysosomal Acid Hydrolases

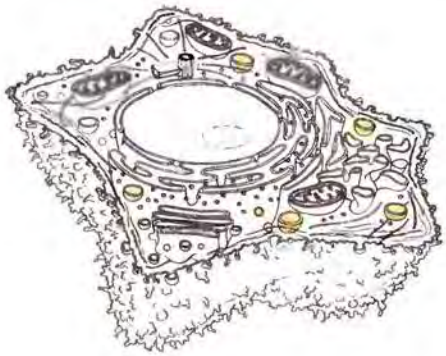
Lysosomes contain about 50 different degradative enzymes that can hydrolyze proteins, DNA, RNA, polysaccharides, and lipids. Mutations in the genes that encode these enzymes are responsible for more than 30 different human genetic diseases, which are called **lysosomal storage diseases** because undegraded material accumulates within the lysosomes of affected individuals. Most of these diseases result from deficiencies in single lysosomal enzymes. For example, Gaucher disease (the most common of these disorders) results from a mutation in the gene that encodes a lysosomal enzyme required for the breakdown of glycolipids. An intriguing exception is I-cell disease, which is caused by a deficiency in the enzyme that catalyzes the first step in the tagging of lysosomal enzymes with mannose-6-phosphate in the Golgi apparatus (see Figure 10.29). The result is a general failure of lysosomal enzymes to be incorporated into lysosomes.

Most lysosomal enzymes are acid hydrolases, which are active at the acidic pH (about 5) that is maintained within lysosomes but not at the neutral pH (about 7.2) characteristic of the rest of the cytoplasm (Figure 10.41). The requirement of these lysosomal hydrolases for acidic pH provides double protection against uncontrolled digestion of the contents of the cytosol; even if the lysosomal membrane were to break down, the released acid hydrolases would be inactive at the neutral pH of the cytosol. To maintain their acidic internal pH, lysosomes must actively concentrate H^+ ions (protons). This is accomplished by a proton pump in the lysosomal membrane, which actively transports protons into the lysosome from the cytosol. This pumping requires expenditure of energy in the form of ATP hydrolysis in order to maintain approximately a hundredfold higher H^+ concentration inside the lysosome.

Endocytosis and Lysosome Formation

One of the major functions of lysosomes is the digestion of material taken up from outside the cell by **endocytosis**, which is discussed in detail in Chapter 13. However, the role of lysosomes in the digestion of material taken up by endocytosis relates not only to the function of lysosomes but

■ **The antimalarial drug chloroquine is uncharged and able to cross membranes at physiological pH. At acidic pH, chloroquine is positively charged and accumulates within the parasite's digestive vacuoles (analogous to lysosomes). The high concentration of chloroquine within the digestive vacuoles is thought to be responsible for its activity against the parasite.**



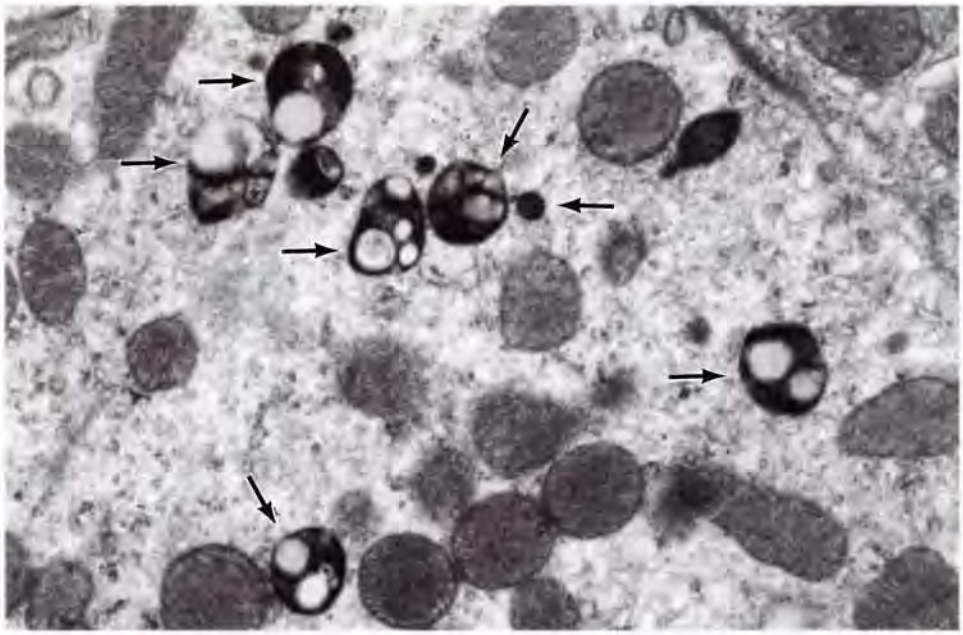


FIGURE 10.40 Electron micrograph of lysosomes and mitochondria in a mammalian cell Lysosomes are indicated by arrows. (K. G. Murti/Visuals Unlimited.)

also to their formation. In particular, lysosomes are formed when transport vesicles from the *trans* Golgi network fuse with endosomes, which contain molecules taken up by endocytosis at the plasma membrane.

The formation of **endosomes** and lysosomes thus represents an intersection between the secretory pathway through which lysosomal proteins are processed, and the endocytic pathway through which extracellular molecules are taken up at the cell surface (Figure 10.42). Material from outside the cell is taken up in clathrin-coated endocytic vesicles, which bud from the plasma membrane and then fuse with early endosomes. Membrane components are then recycled to the plasma membrane (discussed in detail in Chapter 13) and the early endosomes gradually mature into late endosomes, which are the precursors to lysosomes. One of the important changes during endosome maturation is the lowering of the internal pH to about 5.5, which plays a key role in the delivery of lysosomal acid hydrolases from the *trans* Golgi network.

As discussed earlier, acid hydrolases are targeted to lysosomes by mannose-6-phosphate residues, which are recognized by mannose-6-phosphate receptors in the *trans* Golgi network and packaged into clathrin-coated vesicles. Following removal of the clathrin coat, these transport vesicles fuse with late endosomes, and the acidic internal pH causes the hydrolases to dissociate from the mannose-6-phosphate receptor (see Figure 10.42). The hydrolases are thus released into the lumen of the endosome, while the receptors remain in the membrane and are eventually recycled to the Golgi. Late endosomes then mature into lysosomes as they acquire a full complement of acid hydrolases, which digest the molecules originally taken up by endocytosis.

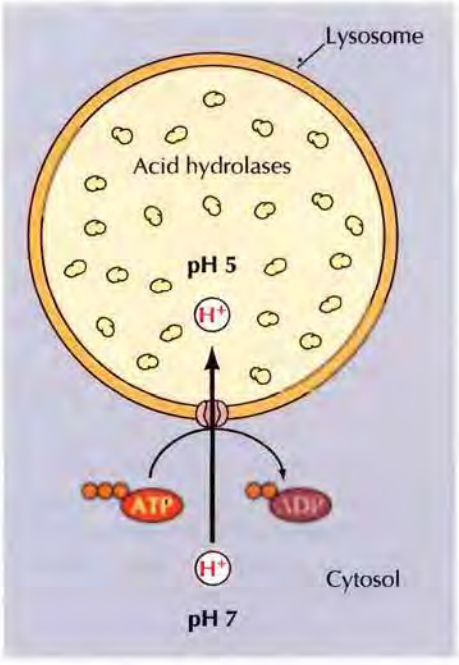
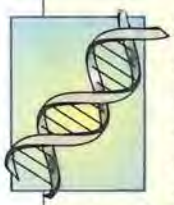


FIGURE 10.41 Organization of the lysosome Lysosomes contain a variety of acid hydrolases that are active at the acidic pH maintained within the lysosome but not at the neutral pH of the cytosol. The acidic internal pH of lysosomes results from the action of a proton pump in the lysosomal membrane, which imports protons from the cytosol coupled to ATP hydrolysis.

MOLECULAR MEDICINE

Gaucher Disease



The Disease

Gaucher disease is the most common of the lysosomal storage diseases, which are caused by a failure of lysosomes to degrade substances that they normally break down. The resulting accumulation of nondegraded compounds leads to an increase in the size and number of lysosomes within the cell, eventually resulting in cellular malfunction and pathological consequences to affected organs. Gaucher disease is found primarily in the Jewish population, where it has a frequency of about 1 in 2500 individuals. There are three types of Gaucher disease, which differ in severity and nervous system involvement. In the most common form of the disease (type I), the nervous system is not involved; the disease is manifest as spleen and liver enlargement and development of bone lesions. Many patients with this form of the disease have no serious symptoms, and their life span is unaffected. The more severe forms of the disease (types II and III) are much rarer and found in both Jewish and non-Jewish populations. The most devastating is type II disease in which extensive neurological involvement is evident in infancy, and patients die early in life. Type III disease, intermediate in severity between types I and II, is characterized by the onset of neurological symptoms (including dementia and spasticity) by about age ten.

Molecular and Cellular Basis

Gaucher disease is caused by a deficiency of the lysosomal enzyme glucocerebrosidase, which catalyzes the hydrolysis of glucocerebroside to glucose and ceramide (see figure). This enzyme deficiency was demonstrated in 1965, and the responsible gene was cloned in 1985. Since then more than 30 different mutations responsible for Gaucher disease have been identified. Interestingly, the severity of the disease can be largely predicted from the nature of these mutations. For example, patients with a mutation leading to the relatively conservative amino acid substitution of serine for asparagine have type I disease; patients with a mutation leading to substitution of proline for leucine have more severe enzyme deficiencies and develop either type II or III disease.

Except for the very rare type II and III forms of the disease, the only cells affected in Gaucher disease are macrophages. Because their function is to eliminate aged and damaged cells by phagocytosis, macrophages continually ingest large amounts of lipids, which are normally degraded in lysosomes. Deficiencies of glucocerebrosidase are therefore particularly evident in macrophages of both the spleen and the liver, consistent with these organ abnormalities being the primary manifestation in most cases of Gaucher disease.

Prevention and Treatment

Gaucher disease is a prime example of a disease that can be treated by enzyme replacement therapy in which exogenous administration of an enzyme is used to correct an enzyme defect. This approach to treatment of lysosomal storage diseases was suggested by Christian de Duve in the 1960s, based on the idea that exogenously administered enzymes might be taken up by endocytosis and transported to lysosomes. In type I Gaucher disease, this approach is particularly attractive because the single target cell is the macrophage. In the 1970s it was discovered that macrophages express cell surface receptors that bind mannose residues on extracellular glycoproteins and then internalize these proteins by endocytosis. This finding suggested that exogenously administered glucocerebrosidase could be specifically targeted to macrophages by modifications that would expose mannose residues. Enzyme prepared from human placenta was appropriately modified, and clinical studies have clearly demonstrated its effectiveness in the treatment of Gaucher disease.

Unfortunately, enzyme replacement therapy for Gaucher disease is expensive. The expense of this treatment puts it far beyond the resources of individual patients and has raised a number of societal issues concerning the cost of expensive drugs for the treatment of rare disorders.

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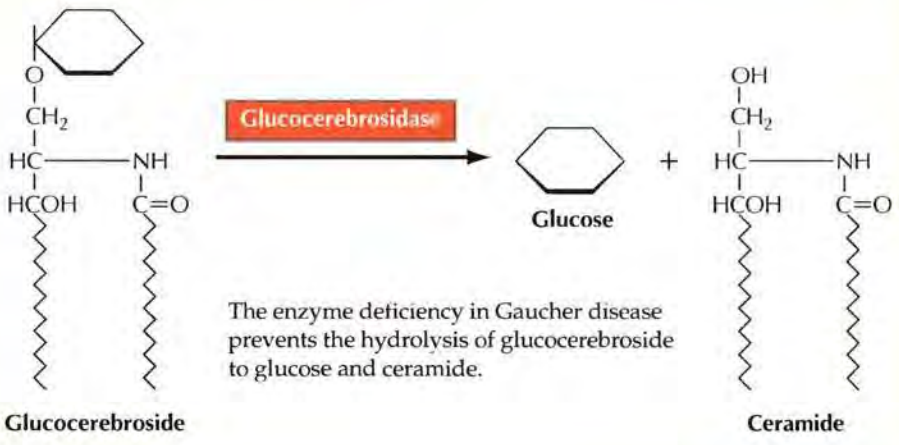

The enzyme deficiency in Gaucher disease prevents the hydrolysis of glucocerebroside to glucose and ceramide.

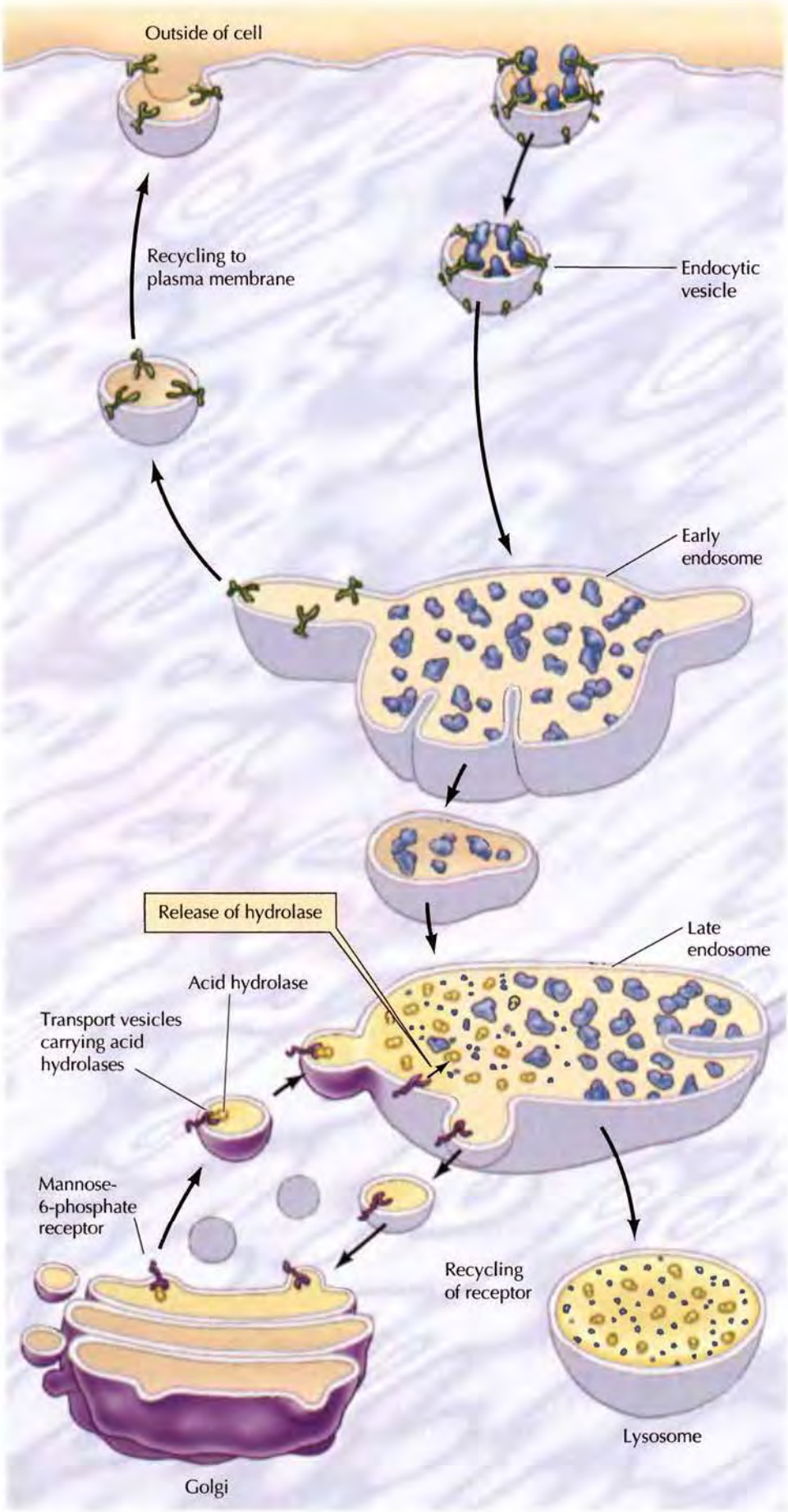

FIGURE 10.42 Endocytosis and lysosome formation Molecules are taken up from outside the cell in endocytic vesicles, which fuse with early endosomes. Membrane components are recycled as the early endosomes mature into late endosomes. Transport vesicles carrying acid hydrolases from the Golgi apparatus then fuse with late endosomes, which mature into lysosomes as they acquire a full complement of lysosomal enzymes. The acid hydrolases dissociate from the mannose-6-phosphate receptors when the transport vesicles fuse with late endosomes, and the receptors are recycled to the Golgi apparatus.

13.3 WEBSITE ANIMATION

Endocytosis In endocytosis, the plasma membrane pinches off small vesicles (pinocytosis) or large vesicles (phagocytosis) to take in extracellular materials.

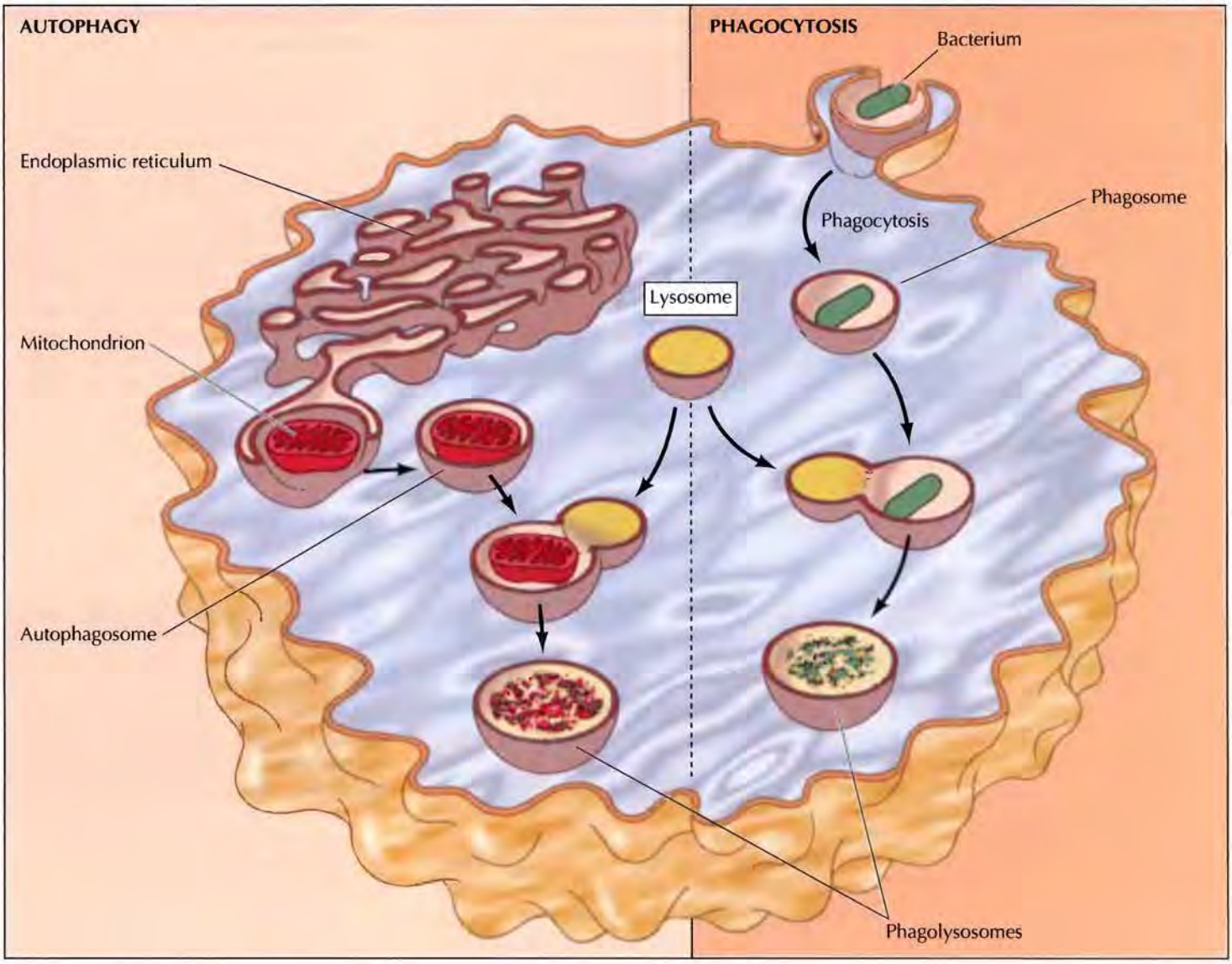

FIGURE 10.43 Lysosomes in phagocytosis and autophagy In phagocytosis, large particles (such as bacteria) are taken up into phagocytic vacuoles or phagosomes. In autophagy, internal organelles (such as mitochondria) are enclosed by membrane fragments from the ER, forming autophagosomes. Both phagosomes and autophagosomes fuse with lysosomes to form large phagolysosomes in which their contents are digested.

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

Phagocytosis and Autophagy

In addition to degrading molecules taken up by endocytosis, lysosomes digest material derived from two other routes: phagocytosis and autophagy (Figure 10.43). In **phagocytosis**, specialized cells, such as macrophages, take up and degrade large particles, including bacteria, cell debris, and aged cells that need to be eliminated from the body. Such large particles are taken up in phagocytic vacuoles (**phagosomes**), which then fuse with lysosomes, resulting in digestion of their contents. The lysosomes formed in this way (**phagolysosomes**) can be quite large and heterogeneous since their size and shape is determined by the content of material that is being digested.

Lysosomes are also responsible for **autophagy**, the gradual turnover of the cell's own components (discussed in Chapter 8). In contrast to phagocytosis, autophagy is a function of all cells and is critical at certain stages of embryonic development when it allows rapid reallocation of the cell's resources. The first step of autophagy appears to be the enclosure of an organelle (e.g., a mitochondrion) in membrane derived from the ER. The resulting vesicle (an **autophagosome**) then fuses with a lysosome, and its contents are digested (see Figure 10.43).

SUMMARY

THE ENDOPLASMIC RETICULUM

The Endoplasmic Reticulum and Protein Secretion: The endoplasmic reticulum is the first branch point in protein sorting. In mammalian cells, proteins destined for secretion, lysosomes, or the plasma membrane are translated on membrane-bound ribosomes and transferred into the rough ER as their translation proceeds.

endoplasmic reticulum (ER), rough ER, transitional ER, smooth ER, secretory vesicle, secretory pathway

Targeting Proteins to the Endoplasmic Reticulum: Proteins can be targeted to the ER either while their translation is still in progress or following completion of translation in the cytosol. In mammalian cells, most proteins are translocated into the ER while they are being translated on membrane-bound ribosomes. Ribosomes engaged in the synthesis of secreted proteins are targeted to the endoplasmic reticulum by signal sequences at the amino terminus of the polypeptide chain. Growing polypeptide chains are then translocated into the ER through protein channels and released into the ER lumen by cleavage of the signal sequence.

signal sequence, microsome, signal recognition particle (SRP), srpRNA, SRP receptor, translocon, signal peptidase

Insertion of Proteins into the ER Membrane: Integral membrane proteins of the plasma membrane or the membranes of the ER, Golgi apparatus, and lysosomes are initially inserted into the membrane of the ER. Rather than being translocated into the ER lumen, these proteins are anchored by membrane-spanning α helices that stop the transfer of the growing polypeptide chain across the membrane.

Protein Folding and Processing in the ER: Polypeptide chains are folded into their correct three-dimensional conformations within the ER. The ER is also the site of *N*-linked glycosylation and addition of GPI anchors.

protein disulfide isomerase, glycosylphosphatidylinositol (GPI) anchor

Quality Control in the ER: Many secretory proteins are not folded correctly the first time. Chaperones detect incorrectly folded proteins and recycle them through the folding pathway. Those proteins that cannot be correctly folded are diverted from the secretory pathway and marked for degradation.

unfolded protein response

The Smooth ER and Lipid Synthesis: The ER is the major site of lipid synthesis in eukaryotic cells, and the smooth ER is abundant in cells that are active in lipid metabolism and detoxification of lipidsoluble drugs.

flippase

Export of Proteins and Lipids from the ER: Proteins and lipids are transported in vesicles from the ER to the Golgi apparatus. Resident ER proteins are marked by sequences that signal their return from the Golgi to the ER by a recycling pathway. Other targeting sequences mediate the selective packaging of exported proteins into vesicles that transport them to the Golgi.

THE GOLGI APPARATUS

Organization of the Golgi: The Golgi apparatus functions in protein processing and sorting as well as in the synthesis of lipids and polysaccharides. Proteins are transported from the endoplasmic reticulum to the *cis* Golgi network. From there they are transported to the Golgi stack, which represents the site of most metabolic activities of the Golgi apparatus. Modified proteins are transported from the Golgi stack to the *trans* Golgi network, where they are sorted and packaged in vesicles for transport to endosomes, lysosomes, the plasma membrane, or the exterior of the cell.

Golgi apparatus, Golgi complex, *cis* Golgi network, Golgi stack, *trans* Golgi network

KEY TERMS

KEY TERMS

mannose-6-phosphate, glycosyltransferase, glycosidase, signal patch

apical domain, basolateral domain, vacuole

synaptic vesicle

COP-coated vesicle, COPI, COPII, clathrin-coated vesicle, clathrin

SNARE hypothesis, Rab, exocyst

lysosome, lysosomal storage disease

endocytosis, endosome

phagocytosis, phagosome, phagolysosome, autophagy, autophagosome

SUMMARY

Protein Glycosylation within the Golgi: The *N*-linked oligosaccharides added to proteins in the ER are modified within the Golgi. Those proteins destined for lysosomes are specifically phosphorylated on mannose residues, and mannose-6-phosphate serves as a targeting signal that directs their transport to lysosomes from the *trans* Golgi network. *O*-linked glycosylation also takes place within the Golgi.

Lipid and Polysaccharide Metabolism in the Golgi: The Golgi apparatus is the site of synthesis of glycolipids, sphingomyelin, and the complex polysaccharides of plant cell walls.

Protein Sorting and Export from the Golgi Apparatus: Proteins are sorted in the *trans* Golgi network for packaging into transport vesicles targeted for secretion, the plasma membrane, lysosomes, or yeast and plant vacuoles. In polarized cells, proteins are specifically targeted to the apical and basolateral domains of the plasma membrane.

THE MECHANISM OF VESICULAR TRANSPORT

Experimental Approaches to Understanding Vesicular Transport: The mechanism of vesicular transport has been elucidated through studies of yeast mutants, reconstituted cell-free systems, and synaptic vesicles.

Cargo Selection, Coat Proteins, and Vesicle Budding: The cytoplasmic surfaces of vesicles are coated with proteins that drive vesicle budding and select the specific molecules to be transported.

Vesicle Fusion: Vesicle binding to the correct target membrane is mediated by interactions between pairs of transmembrane proteins, which leads to membrane fusion. Some types of fusion with the plasma membrane (exocytosis) occur at specific multiprotein complexes called exocysts.

LYSOSOMES

Lysosomal Acid Hydrolases: Lysosomes contain an array of acid hydrolases that degrade proteins, nucleic acids, polysaccharides, and lipids. These enzymes function specifically at the acidic pH maintained within lysosomes.

Endocytosis and Lysosome Formation: Extracellular molecules taken up by endocytosis are transported to endosomes, which mature to lysosomes as lysosomal acid hydrolases are delivered from the Golgi.

Phagocytosis and Autophagy: Lysosomes are responsible for the degradation of large particles taken up by phagocytosis and for the gradual digestion of the cell's own components by autophagy.

Questions

1. What was the original experimental evidence for the secretory pathway from rough ER → Golgi apparatus → secretory vesicles → secreted protein?

2. How did *in vitro* translation of mRNAs provide evidence for the existence of a signal sequence that targets secretory proteins to the rough endoplasmic reticulum?

3. Compare and contrast cotranslational and posttranslational translocation of polypeptide chains into the endoplasmic reticulum.

4. Sec61 is a critical component of the protein channel through the ER membrane. In Sec61 mutant yeast, what is the fate of proteins that are normally localized to the Golgi apparatus?

5. Why are the carbohydrate groups of glycoproteins always exposed on the surface of the cell?

6. What would be the effect of mutating the KDEL sequence of a resident ER protein like BiP? Would this effect be similar or different from that of mutating the KDEL receptor protein?

7. How is a lysosomal protein targeted to a lysosome? What effect would the addition of a lysosome-targeting signal patch have on the subcellular localization of a protein that is normally cytosolic? How would it affect localization of a protein that is normally secreted?

8. What is the predicted fate of lysosomal acid hydrolases in I-cell disease in which cells are deficient in the enzyme required for formation of mannose-6-phosphate residues?

9. What processes result in glycolipids and sphingomyelin being found in the outer—but not the inner—half of the plasma membrane bilayer?

10. What experimental evidence demonstrated vesicular transport of proteins between Golgi cisternae?

11. A patient comes to your clinic with an accumulation of glucocerebrosides in macrophage lysosomes. What is your diagnosis, and what therapy would you suggest if price is not a limiting factor?

12. Lysosomes contain powerful hydrolytic enzymes, which are transported there from the site of their synthesis in the ER via the Golgi apparatus. Why don't these enzymes damage the constituents of these organelles?

13. What is the source of energy for fusion between target and vesicle membranes?

14. Why does activation of Rab proteins require association with a membrane?

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CHAPTER 11

Bioenergetics and Metabolism
Mitochondria, Chloroplasts, and Peroxisomes

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IN ADDITION TO BEING INVOLVED IN PROTEIN SORTING AND TRANSPORT, cytoplasmic organelles provide specialized compartments in which a variety of metabolic activities take place. The generation of metabolic energy is a major activity of all cells, and two cytoplasmic organelles are specifically devoted to energy metabolism and the production of ATP. Mitochondria are responsible for generating most of the useful energy derived from the breakdown of lipids and carbohydrates, and chloroplasts use energy captured from sunlight to generate both ATP and the reducing power needed to synthesize carbohydrates from CO_2 and H_2O. The third organelle discussed in this chapter, the peroxisome, contains enzymes involved in a variety of different metabolic pathways, including the breakdown of fatty acids and the metabolism of a byproduct of photosynthesis.

Mitochondria, chloroplasts, and peroxisomes differ from the organelles discussed in the preceding chapter not only in their functions but also in their mechanism of assembly. Rather than being synthesized on membrane-bound ribosomes and translocated into the endoplasmic reticulum, most proteins destined for peroxisomes, mitochondria, and chloroplasts are synthesized on free ribosomes in the cytosol and imported into their target organelles as completed polypeptide chains. Mitochondria and chloroplasts also contain their own genomes, which include some genes that are transcribed and translated within the organelle. Protein sorting to the cytoplasmic organelles discussed in this chapter is thus distinct from the pathways of vesicular transport that connect the endoplasmic reticulum, Golgi apparatus, lysosomes, and plasma membrane.

Mitochondria

Mitochondria play a critical role in the generation of metabolic energy in eukaryotic cells. As reviewed in Chapter 3, they are responsible for most of the useful energy derived from the breakdown of carbohydrates and fatty acids, which is converted to ATP by the process of oxidative phosphorylation. Most mitochondrial proteins are translated on free cytosolic ribosomes and imported into the organelle by specific targeting signals. In addition, mitochondria are unique among the cytoplasmic organelles already discussed in that they contain their own DNA, which encodes tRNAs, rRNAs, and some mitochondrial proteins. The assembly of mitochondria thus involves proteins encoded by their own genomes and translated within the organelle, as well as proteins encoded by the nuclear genome and imported from the cytosol.

Organization and Function of Mitochondria

Mitochondria are surrounded by a double-membrane system, consisting of inner and outer mitochondrial membranes separated by an intermembrane space (Figure 11.1). The inner membrane forms numerous folds (**cristae**), which extend into the interior (or **matrix**) of the organelle. Each of these components plays distinct functional roles, with the matrix and inner membrane representing the major working compartments of mitochondria.

The matrix contains the mitochondrial genetic system as well as the enzymes responsible for the central reactions of oxidative metabolism (Figure 11.2). As discussed in Chapter 3, the oxidative breakdown of glucose and fatty acids is the principal source of metabolic energy in animal cells. The initial stages of glucose metabolism (glycolysis) occur in the cytosol, where glucose is converted to pyruvate (see Figure 3.11). Pyruvate is then transported into the mitochondria, where its complete oxidation to CO_2 yields the bulk of usable energy (ATP) obtained from glucose metabolism. This involves the initial oxidation of pyruvate to acetyl CoA, which is then broken down to CO_2 via the citric acid cycle (see Figures 3.12 and 3.13). The oxidation of fatty acids also yields acetyl CoA (see Figure 3.15), which is similarly metabolized by the citric acid cycle in mitochondria. The enzymes of the citric acid cycle

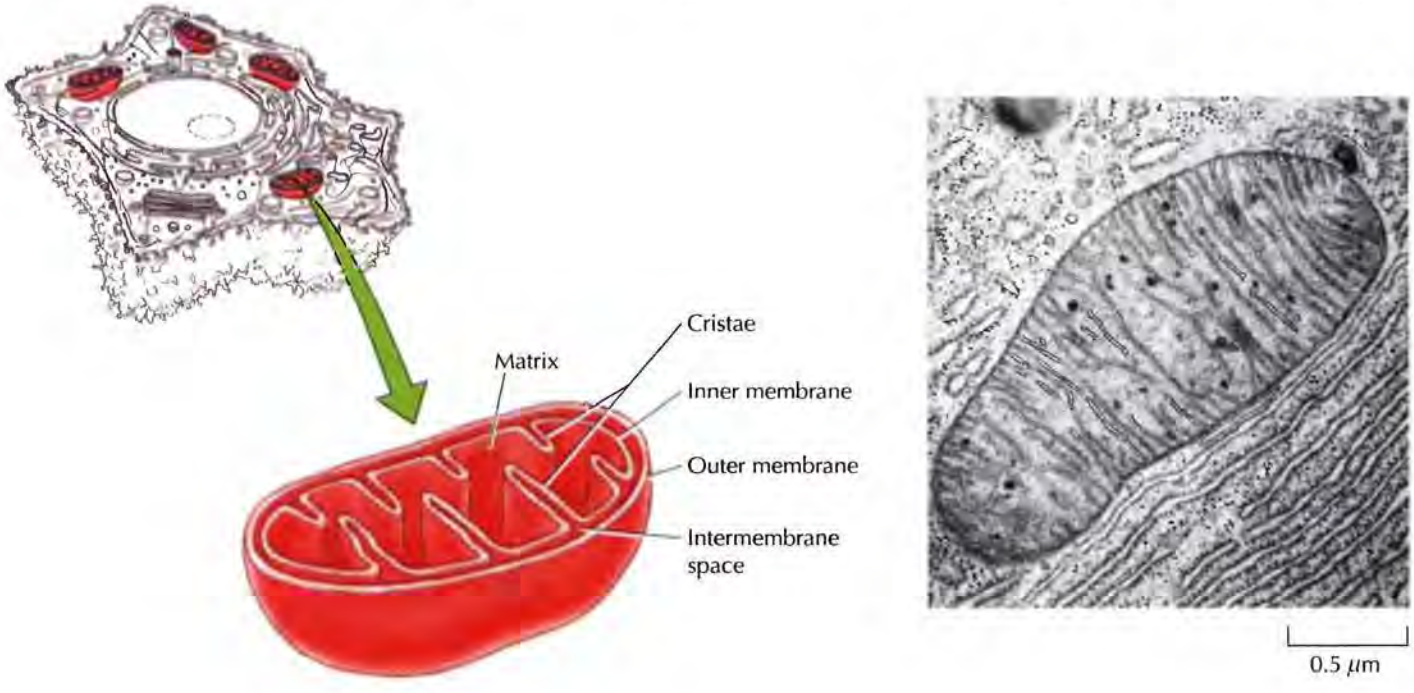

FIGURE 11.1 Structure of a mitochondrion Mitochondria are bounded by a double-membrane system, consisting of inner and outer membranes. Folds of the inner membrane (cristae) extend into the matrix. (Micrograph by K. R. Porter/Photo Researchers, Inc.)

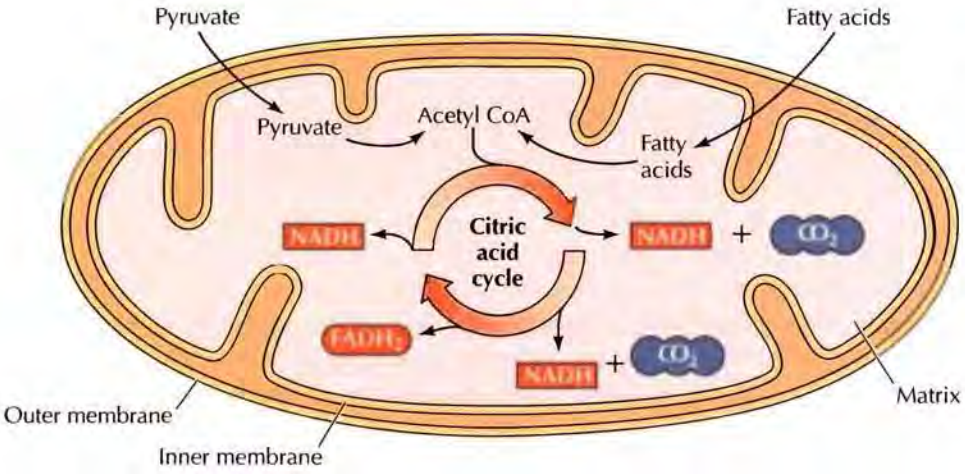

FIGURE 11.2 Metabolism in the matrix of mitochondria Pyruvate and fatty acids are imported from the cytosol and converted to acetyl CoA in the mitochondrial matrix. Acetyl CoA is then oxidized to CO_2 via the citric acid cycle, the central pathway of oxidative metabolism.

(located in the matrix of mitochondria) thus are central players in the oxidative breakdown of both carbohydrates and fatty acids.

The oxidation of acetyl CoA to CO_2 is coupled to the reduction of NAD^+ and FAD to NADH and $FADH_2$, respectively. Most of the energy derived from oxidative metabolism is then produced by the process of oxidative phosphorylation (discussed in detail in the next section), which takes place in the inner mitochondrial membrane. The high-energy electrons from NADH and $FADH_2$ are transferred through a series of carriers in the membrane to molecular oxygen. The energy derived from these electron transfer reactions is converted to potential energy stored in a proton gradient across the membrane, which is then used to drive ATP synthesis. The inner mitochondrial membrane thus represents the principal site of ATP generation, and this critical role is reflected in its structure. First, its surface area is substantially increased by its folding into cristae. In addition, the inner mitochondrial membrane contains an unusually high percentage (greater than 70%) of proteins, which are involved in oxidative phosphorylation as well as in the transport of metabolites (e.g., pyruvate and fatty acids) between the cytosol and mitochondria. Otherwise, the inner membrane is impermeable to most ions and small molecules—a property critical to maintaining the proton gradient that drives oxidative phosphorylation.

In contrast to the inner membrane, the outer mitochondrial membrane is highly permeable to small molecules. This is because it contains proteins called **porins**, which form channels that allow the free diffusion of molecules smaller than about 1000 daltons. The composition of the intermembrane space is therefore similar to the cytosol with respect to ions and small molecules. Consequently, the inner mitochondrial membrane is the functional barrier to the passage of small molecules between the cytosol and the matrix, and it maintains the proton gradient that drives oxidative phosphorylation.

The Genetic System of Mitochondria

Mitochondria contain their own genetic system, which is separate and distinct from the nuclear genome of the cell. As reviewed in Chapter 1, mitochondria are thought to have evolved from bacteria that developed a symbiotic relationship in which they lived within larger cells (**endosymbiosis**). The genomes of living organisms that are most similar to the mitochondrial genome are those of free living α-proteobacteria, which code for 6700–8300 proteins. An intracellular parasite like the α-proteobacterium *Rickettsia*

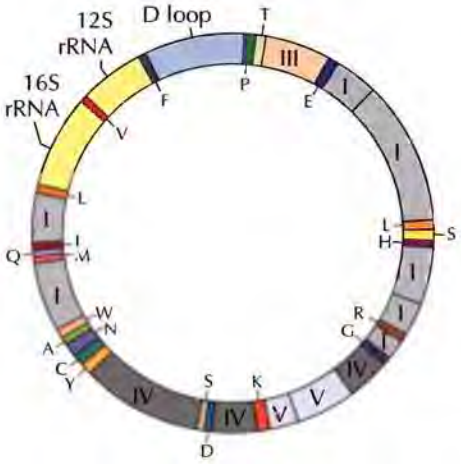

FIGURE 11.3 The human mitochondrial genome The genome contains 13 protein-coding sequences, which are designated as components of respiratory complexes I, III, IV, or V. In addition, the genome contains genes for 12S and 16S rRNAs and for 22 tRNAs, which are designated by the one-letter code for the corresponding amino acid. The region of the genome designated "D loop" contains an origin of DNA replication and transcriptional promoter sequences.

prowazekii has a smaller genome, about 830 protein-coding genes. Like mitochondria, *Rickettsia prowazekii* is able to reproduce only within eukaryotic cells, but unlike mitochondria it still transcribes and translates most of its own genes.

Mitochondrial genomes are usually circular DNA molecules like those of bacteria, which are present in multiple copies per organelle. They vary considerably in size between different species. The genomes of human and most other animal mitochondria are only about 16 kb, but substantially larger mitochondrial genomes are found in yeasts (approximately 80 kb) and plants (more than 200 kb). However, these larger mitochondrial genomes are composed predominantly of noncoding sequences and do not appear to contain significantly more genetic information. For example, the largest sequenced mitochondrial genome is that of the plant *Arabidopsis thaliana*. Although *Arabidopsis* mitochondrial DNA is approximately 367 kb, it encodes only 31 proteins: just more than twice the number encoded by human mitochondrial DNA. The largest number of mitochondrial genes has been found in mitochondrial DNA of the protozoan *Reclinomonas americana*, which is 69 kb and contains 67 genes. The smallest mitochondrial genome is that of the protist *Plasmodium falciparum*, which is 6 kb and codes for only 3 proteins. In contrast, the genomes of the free-living α-proteobacteria are 7–10 Mb. Most present-day mitochondrial genomes encode only a small number of proteins that are essential components of the oxidative phosphorylation system. In addition, mitochondrial genomes encode all of the ribosomal RNAs and most of the transfer RNAs needed for translation of these protein-coding sequences within mitochondria. Other mitochondrial proteins are encoded by nuclear genes, which are thought to have been transferred to the nucleus from the ancestral mitochondrial genome.

The human mitochondrial genome encodes 13 proteins involved in electron transport and oxidative phosphorylation (Figure 11.3). In addition, human mitochondrial DNA encodes 16S and 12S rRNAs and 22 tRNAs, which are required for translation of the proteins encoded by the organelle genome. The two rRNAs are the only RNA components of animal and yeast mitochondrial ribosomes, in contrast to the three rRNAs of bacterial ribosomes (23S, 16S, and 5S). Plant mitochondrial DNAs, however, also encode a third rRNA: 5S. The mitochondria of plants and protozoans also differ in importing and utilizing tRNAs encoded by the nuclear as well as the mitochondrial genome, whereas in animal mitochondria, all the tRNAs are encoded by the organelle genome.

■ **Since almost all mitochondria are inherited from the mother, it is possible to trace the human maternal lineage back to our most recent common female ancestor: Mitochondrial Eve.**

The small number of tRNAs encoded by the mitochondrial genome highlights an important feature of the mitochondrial genetic system—the use of a slightly different genetic code, which is distinct from the "universal" genetic code used by both prokaryotic and eukaryotic cells (Table 11.1). As discussed in Chapter 4, there are 64 possible triplet codons, of which 61 encode the 20 different amino acids incorporated into proteins (see Table 4.1). Many tRNAs in both prokaryotic and eukaryotic cells are able to recog-

nize more than a single codon in mRNA because of "wobble," which allows some mispairing between the tRNA anticodon and the third position of certain complementary codons (see Figure 8.3). However, at least 30 different tRNAs are required to translate the universal code according to the wobble rules. Yet human mitochondrial DNA encodes only 22 tRNA species, and these are the only tRNAs used for translation of mitochondrial mRNAs. This is accomplished by an extreme form of wobble in which U in the anticodon of the tRNA can pair with any of the four bases in the third codon position of mRNA, allowing four codons to be recognized by a single tRNA. In addition, some codons specify different amino acids in mitochondria than in the universal code.

Like the DNA of nuclear genomes, mitochondrial DNA can be altered by mutations, which are frequently deleterious to the organelle. Since almost all the mitochondria of fertilized eggs are contributed by the oocyte rather than by the sperm, germ-line mutations in mitochondrial DNA are transmitted to the next generation by the mother. Such mutations have been associated with a number of diseases. Mutations in one mitochondrial tRNA gene are associated with metabolic syndrome, the human condition associated with obesity and diabetes. In addition, Leber's hereditary optic neuropathy, a disease that leads to blindness, can be caused by mutations in mitochondrial genes that encode components of the electron transport chain. In addition, the progressive accumulation of mutations in mitochondrial DNA during the lifetime of individuals has been suggested to contribute to the process of aging.

In contrast to the mitochondrial genome, the proteins present in mitochondria are much less-well understood. Mammalian mitochondria are thought to contain between 1000–2000 different proteins, representing ~5% of the proteins encoded by mammalian genomes. However, nearly half of all mitochondrial proteins remain unidentified. Protein analysis has been difficult because mitochondria from different tissues contain different proteins. For example, only half the proteins in human mitochondria are found in all tissues. Some of these differences can be traced to tissue-specific functions of the mitochondria, such as complex steroid synthesis in adrenal cells or heme biosynthesis in bone marrow; others likely play roles in poorly understood processes such as control of mitochondrial shape or number, and mitochondrial inheritance.

TABLE 11.1 Differences between the Universal and Mitochondrial Genetic Codes

Codon	Universal code	Human mitochondrial code
UGA	Stop	Trp
AGA	Arg	Stop
AGG	Arg	Stop
AUA	Ile	Met

Other codons vary from the universal code in yeast and plant mitochondria.

Protein Import and Mitochondrial Assembly

In contrast to the RNA components of the mitochondrial translation apparatus (rRNAs and tRNAs), most mitochondrial genomes do not encode the proteins required for DNA replication, transcription, or translation. Instead, the genes that encode proteins required for the replication and expression of mitochondrial DNA are contained in the nucleus. In addition, the nucleus contains the genes that encode most of the mitochondrial proteins required for oxidative phosphorylation and all of the enzymes involved in mitochondrial metabolism (e.g., enzymes of the citric acid cycle). Some of these genes were transferred to the nucleus from the original prokaryotic ancestor of mitochondria.

Approximately 1000 proteins encoded by nuclear genes (more than 95% of mitochondrial proteins) are synthesized on free cytosolic ribosomes and imported into mitochondria as completed polypeptide chains. Because of the double-membrane structure of mitochondria, the import of proteins is considerably more complicated than the transfer of a polypeptide across a single phospholipid bilayer. Proteins targeted to the matrix have to cross

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Diseases of Mitochondria: Leber's Hereditary Optic Neuropathy

The Disease

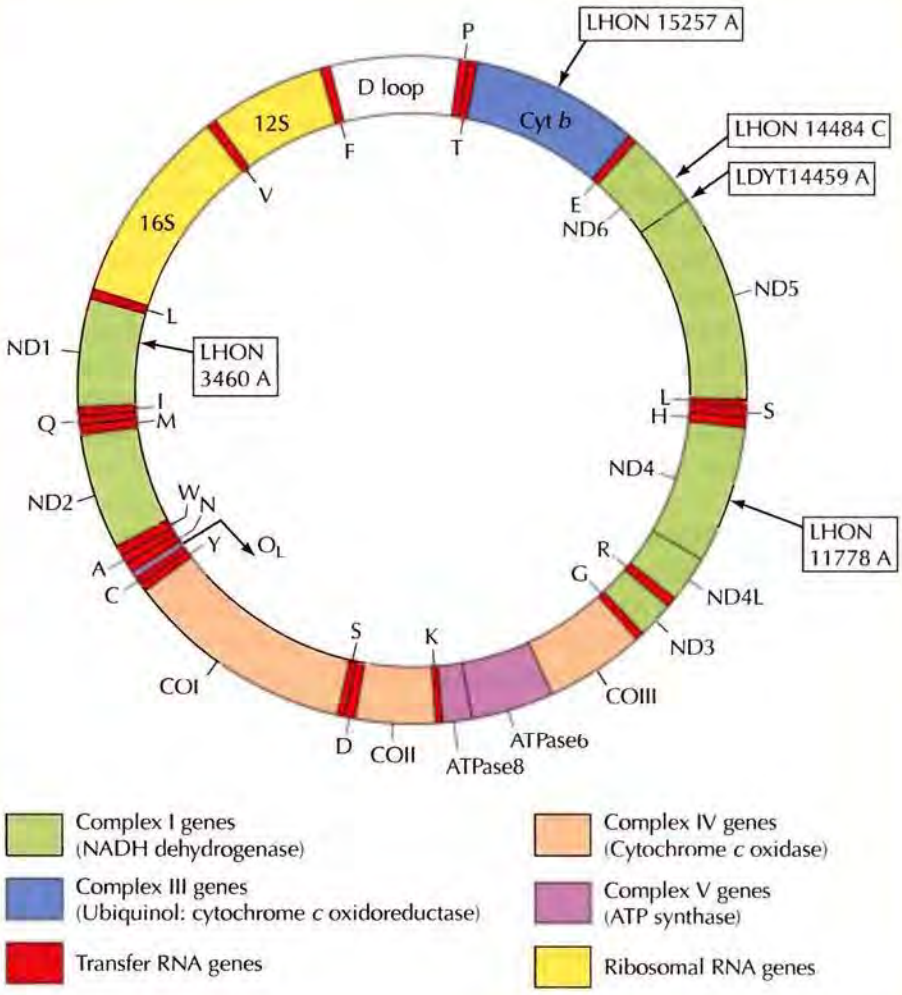
Leber's hereditary optic neuropathy (LHON) is a rare inherited disease that results in blindness because of degeneration of the optic nerve. Vision loss usually occurs between the ages of 15 and 35, and is generally the only manifestation of the disease. Not all individuals who inherit the genetic defects responsible for LHON develop the disease, and females are affected less frequently than males. This propensity to affect males might suggest that LHON is an X-linked disease. This is not the case, however, because males never transmit LHON to their offspring. Instead, the inheritance of LHON is entirely by maternal transmission. This characteristic is consistent with cytoplasmic rather than nuclear inheritance of LHON, since the cytoplasm of fertilized eggs is derived almost entirely from the oocyte.

Molecular and Cellular Basis

In 1988 Douglas Wallace and his colleagues identified a mutation in the mitochondrial DNA of LHON patients. This mutation (at base pair 11778) affects one of the subunits of complex I of the electron transport chain (NADH dehydrogenase), resulting in the substitution of a histidine for an arginine. The 11778 mutation accounts for approximately half of all cases of LHON. Three other mutations of mitochondrial DNA have also been identified as primary causes of LHON. Two of these mutations affect other subunits of complex I, while the third affects cytochrome *b*, which is a component of complex III (see figure). Together, these four mutations account for more than 80% of LHON cases. A fifth mutation (at base pair 14459), affecting a complex I subunit, can cause either LHON or muscular disorders.

The mutations causing LHON reduce the capacity of mitochondria to carry out oxidative phosphorylation and generate ATP. This has the greatest effect on those tissues that are most dependent on oxidative phosphorylation, so defects in components of mitochondria can lead to clinical manifestations in specific organs, rather than to systemic disease. The central nervous system (including the brain and optic nerve) is most highly dependent on oxidative metabolism, consistent with blindness being the primary clinical manifestation resulting from the mitochondrial DNA mutations responsible for LHON.

As already noted, inheritance of LHON mutations does not always lead to development of the disease; only about 10% of females and 50% of males possessing a mutation suffer vision loss. One factor that may contribute to this low incidence of disease among carriers of LHON mutations is that each cell contains thousands of copies of mitochondrial DNA, which can be present in mixtures of mutant and normal mitochondria. These mitochondria are randomly distributed to daughter cells at cell division, so the population of mitochondria can change as cells divide, leading to the formation of cells containing either greater or lesser proportions of mutant



LHON mutations in mitochondrial DNA.

MOLECULAR MEDICINE

organelles. Importantly, however, many individuals who bear predominantly mutant mitochondrial DNAs still fail to develop the disease. Thus additional genetic or environmental factors, which have yet to be identified, appear to play a significant role in the development of LHON.

Prevention and Treatment

The identification of mitochondrial DNA mutations responsible for LHON allows molecular diagnosis of the disease, which can be important in establishing a definitive diagnosis of patients without a family history. However, the detection of mutations in mitochondrial DNA is of little value for screening members of affected families or for family planning. This contrasts to the utility of detecting inherited mutations of nuclear genes, where molecular analysis can determine whether a family member or embryo has inherited a mutant or wild-type allele. In LHON, however, mutant mitochondria are present in large numbers and are maternally transmitted to all offspring. As noted above, not all such offspring develop the disease, but this cannot be predicted by genetic analysis.

The finding that LHON is caused by mutations of mitochondrial DNA suggests the potential of new therapies. One approach is metabolic therapy intended to enhance oxidative phosphorylation by administration of substrates or cofactors in the electron transport pathway, such as succinate or coenzyme Q. Another possibility that has been considered for treatment of LHON is gene therapy designed to relocate a normal gene allele to the nucleus. An appropriate targeting signal would be added to direct the gene product to mitochondria, where it could substitute for the defective mitochondrial-encoded protein.

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both the outer and inner mitochondrial membranes, while other proteins need to be sorted to distinct compartments within the organelle (e.g., the intermembrane space).

The import of proteins to the matrix is the best-understood aspect of mitochondrial protein sorting (Figure 11.4). Most proteins are targeted to mitochondria by amino-terminal sequences of 20 to 35 amino acids (called **presequences**) that are removed by proteolytic cleavage following their import into the organelle. The presequences of mitochondrial proteins, first characterized by Gottfried Schatz, contain multiple positively charged amino acid residues, usually in an amphipathic α helix. The first step in protein import is the binding of these presequences to receptors on the surface of mitochondria. These receptors are part of a protein complex that directs translocation across the outer membrane (the *t*ranslocase of the *o*uter *m*embrane or **Tom complex**). Individual Tom proteins are designated according to their molecular weights, so the receptors are called Tom20, Tom22, and Tom5. From these receptors, proteins are transferred to the Tom40 pore protein and translocated across the outer membrane. The proteins are then transferred to a second protein complex in the inner membrane (one of two different *t*ranslocases of the *i*nner *m*embrane or **Tim complexes**). Proteins with presequences cross the inner membrane through the Tim23 complex. Continuing protein translocation requires the electrochemical potential established across the inner mitochondrial membrane during electron transport. As discussed in the next section of this chapter, the transfer of high-energy electrons from NADH and $FADH_2$ to molecular oxygen is coupled to the transfer of protons from the mitochondrial matrix to the

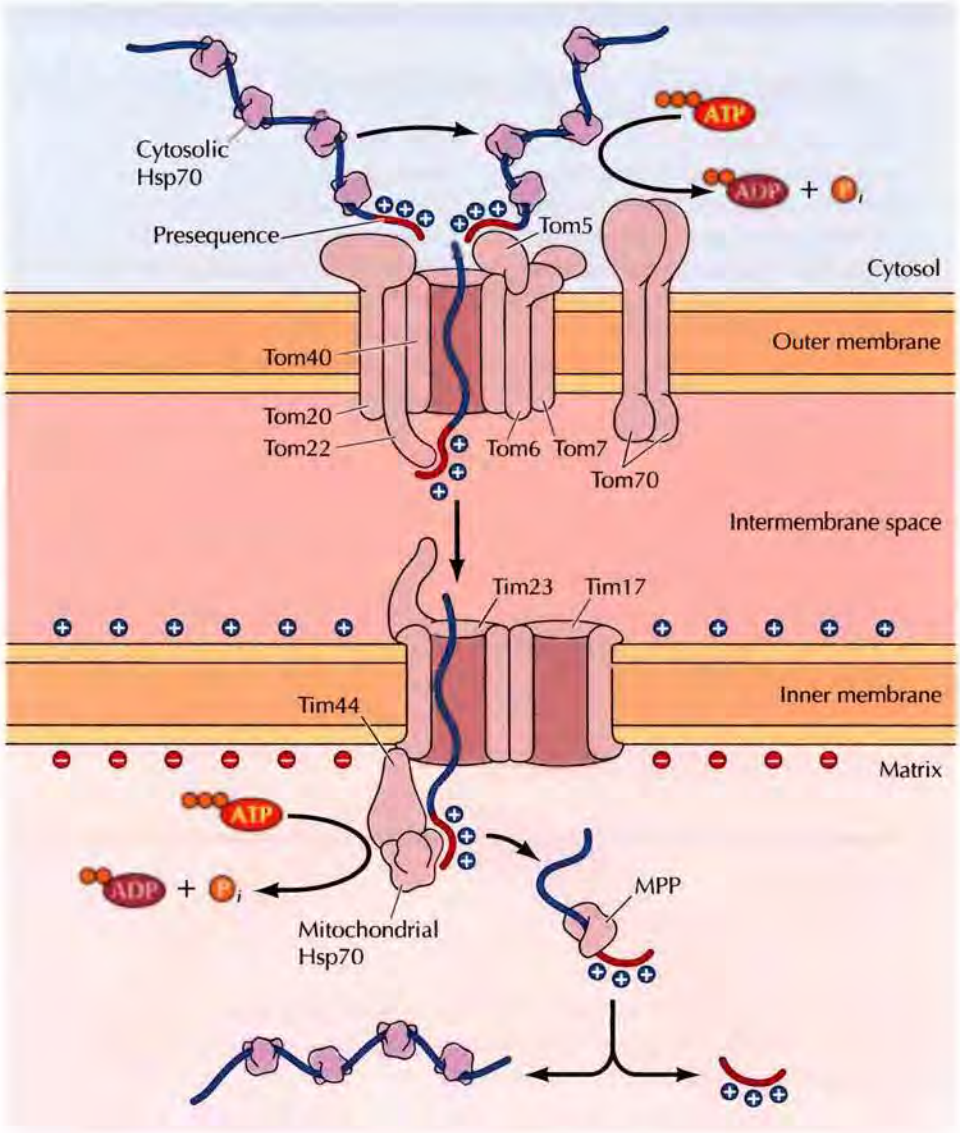


FIGURE 11.4 Import of mitochondrial matrix proteins Proteins are targeted to the Tom complex in the mitochondrial outer membrane by amino-terminal presequences containing positively charged amino acids. The presequence first binds to Tom20 and is transferred to Tom5 and then to the import pore, Tom40. Following passage through the outer membrane, the presequence binds the intermembrane domain of Tom22 and is passed to the Tim23 complex in the inner membrane. In the matrix, an Hsp70 chaperone associated with Tim44 acts as a ratchet, using ATP hydrolysis to translocate the protein across the inner membrane. Most proteins destined for the mitochondrial matrix have their presequences removed by the mitochondrial matrix processing peptidase (MPP) and associate with soluble Hsp70 proteins that assist in their folding.

intermembrane space. Since protons are charged particles, this transfer establishes an electric potential across the inner membrane, with the matrix being negative. During protein import, this electric potential drives translocation of the positively charged presequence.

To be translocated across the mitochondrial membrane, proteins must be at least partially unfolded. Consequently, protein import into mitochondria requires molecular chaperones in addition to the membrane proteins involved in translocation (see Figure 11.4). On the cytosolic side, members of the Hsp70 family of chaperones both maintain proteins in a partially unfolded state and present them to the translocase so that they can be inserted into the mitochondrial membrane. As they cross the inner membrane, the unfolded polypeptide chains are bound by another Hsp70 chaperone, which is associated with the Tim23 complex and acts as a ratchet that uses repeated ATP hydrolysis to drive protein import (Figure 11.5). In most cases, the presequence is then cleaved by the **matrix processing peptidase (MPP)** and the polypeptide chain is bound by other matrix Hsp70 chaperones that facilitate its folding. Some polypeptides are then transferred to a

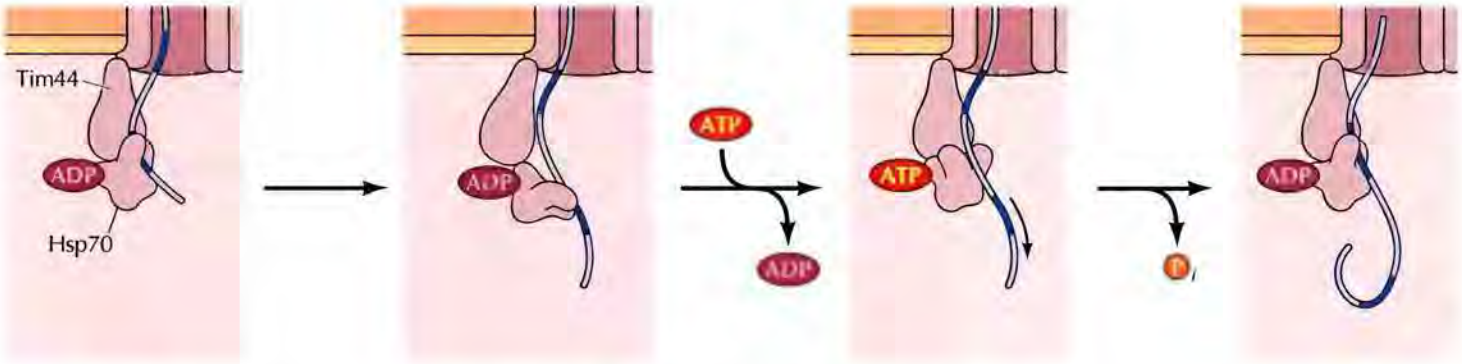


FIGURE 11.5 Binding cycle of an Hsp70 chaperone Hsp70 chaperones provide the force for translocating proteins across many membranes. To do this they exploit their ability to reversibly bind short hydrophobic sequences in polypeptides and, when bound to another structure such as Tim44, can act as a ratchet. In the ADP-bound state, the binding pocket for the substrate protein is closed and the hydrophobic sequence in the polypeptide is tightly bound. The displacement of ADP by ATP opens the pocket, allowing Hsp70 to release the polypeptide. Recognition of a second hydrophobic sequence results in hydrolysis of ATP by Hsp70, which then closes the pocket and tightly binds the polypeptide.

chaperone of the Hsp60 family (a chaperonin, see Figure 8.23), within which additional protein folding takes place. These interactions of polypeptide chains with molecular chaperones depend on ATP, so protein import requires ATP both outside and inside the mitochondria as well as the electric potential across the inner membrane.

As noted above, some mitochondrial proteins are targeted to the outer membrane, inner membrane, or intermembrane space rather than to the matrix, so additional mechanisms are needed to direct these proteins to the correct submitochondrial compartment. Many of the proteins in the inner membrane are small molecule transporters, which are multi-pass transmembrane proteins that exchange nucleotides and ions between the mitochondria and the cytosol. These proteins do not contain presequences but instead have multiple internal mitochondrial import signals. Consequently, they are not recognized by Tom20. Instead, these inner membrane proteins, in association with an Hsp90 chaperone, are recognized by a distinct receptor on the mitochondrial outer membrane (Tom70), and then translocated across the membrane through the Tom40 channel (Figure 11.6). In the intermembrane space, the proteins are recognized by mobile components of a distinct Tim complex, that of Tim22. These small Tim proteins (called "tiny Tim" proteins) serve as both chaperones and shuttle proteins that escort proteins to Tim22. The proteins are then partially translocated through Tim22, before internal stop-transfer signals cause them to exit the Tim22 pore laterally and insert into the inner membrane.

Some proteins destined for the outer or inner membranes or the intermembrane space have both a presequence and internal signal sequences. Since they contain presequences, these proteins are recognized by the Tom20 receptor on the outer membrane and translocated through the Tom40 channel (Figure 11.7). Some outer membrane proteins exit the Tom40 channel laterally, although the porin proteins and Tom40 itself pass through the Tom complex and interact with a second protein complex to insert into

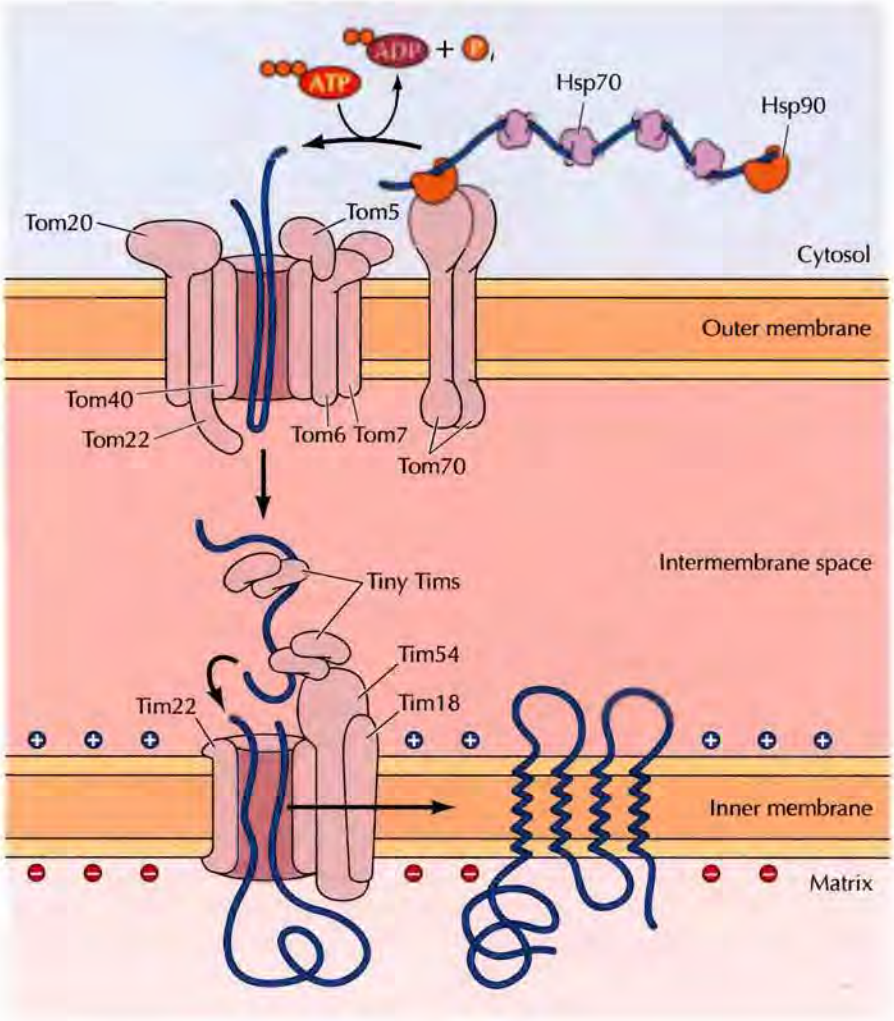


FIGURE 11.6 Import of small molecule transport proteins into the mitochondrial inner membrane These multiple-pass transmembrane proteins have internal signal sequences, rather than N-terminal presequences. The internal signal sequences in association with Hsp90 chaperones interact with the Tom70 receptors, from which the transmembrane protein is transferred to the Tom40 channel. In the intermembrane space, the protein is bound by small mobile Tim proteins, the "tiny Tim" proteins that guide it to the Tim22 complex in the inner membrane. The tiny Tim proteins transfer the protein to Tim54 and then to the Tim22 import pore. Internal stop-transfer sequences halt translocation, and the protein is transferred laterally into the mitochondrial inner membrane.

the outer membrane from the intermembrane space. Other proteins with complex signal sequences pass through the outer membrane but remain in the intermembrane space instead of entering Tim23. Other proteins destined for the intermembrane space, as well as some inner membrane proteins, are first transported across the inner membrane into the mitochondrial matrix through the Tim23 complex. They are then targeted for further transport by a second sorting signal that is uncovered by the removal of the presequence in the matrix. This second sorting signal targets them to another translocase, Oxa1, where they are either passed into the intermembrane space or arrested in transit by internal stop-transfer signals and inserted into the inner membrane. Oxa1 is also the translocase for those intermembrane and inner membrane proteins that are encoded by the mitochondrial genome and are synthesized on mitochondrial ribosomes in the matrix.

Not only the proteins but also the phospholipids of mitochondrial membranes are imported from the cytosol. In animal cells, phosphatidylcholine and phosphatidylethanolamine are synthesized in the ER and carried to mitochondria by **phospholipid transfer proteins**, which extract single phospholipid molecules from the membrane of the ER. The lipid can then be transported through the aqueous environment of the cytosol, buried in a hydrophobic binding site of the protein, and released when the complex

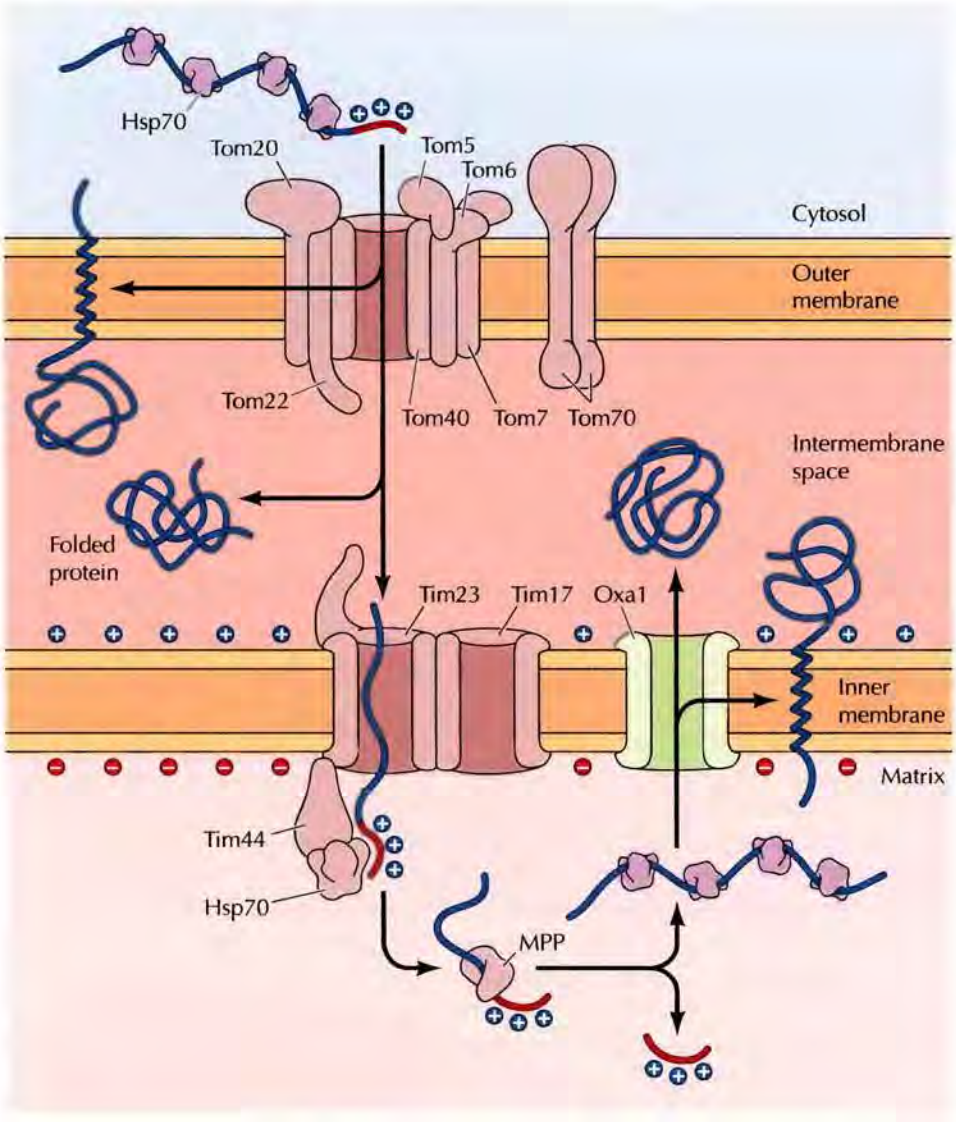


FIGURE 11.7 Sorting of proteins containing presequences to different mitochondrial compartments Mitochondrial proteins with N-terminal presequences can be imported to the outer membrane, inner membrane, or intermembrane space. The presequences of these proteins are recognized by the Tom20 receptor and transferred to Tom40. Proteins destined for the outer membrane halt translocation in the Tom40 complex and pass laterally into the membrane. Some proteins destined for the intermembrane space are translocated through Tom40 but remain in the intermembrane space rather than interacting with the Tim23 complex. Other proteins are transferred through Tim23 into the mitochondrial matrix. Removal of the presequence within the matrix then exposes a second sorting signal that targets these proteins back into either the inner membrane or the intermembrane space through the Oxa1 translocation pore.

reaches a new membrane, such as that of mitochondria. The mitochondria then synthesize phosphatidylserine from phosphatidylethanolamine, in addition to catalyzing the synthesis of the unusual phospholipid **cardiolipin**, which contains four fatty acid chains (Figure 11.8).

The Mechanism of Oxidative Phosphorylation

Most of the usable energy obtained from the breakdown of carbohydrates or fats is derived by oxidative phosphorylation, which takes place within mitochondria. For example, the breakdown of glucose by glycolysis and the citric acid cycle yields a total of four molecules of ATP, ten molecules of NADH, and two molecules of $FADH_2$ (see Chapter 3). Electrons from NADH and $FADH_2$ are then transferred to molecular oxygen, which is coupled to the formation of an additional 32 to 34 ATP molecules by oxidative phosphorylation. Electron transport and oxidative phosphorylation are critical activities of protein complexes in the inner mitochondrial membrane which ultimately serve as the major sources of cellular energy.

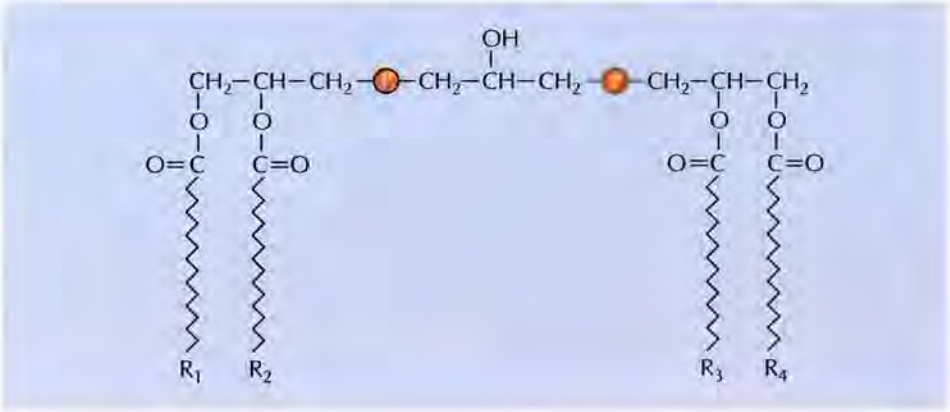


FIGURE 11.8 Structure of cardiolipin Cardiolipin is an unusual "double" phospholipid, containing four fatty acid chains, that is found primarily in the inner mitochondrial membrane.

The Electron Transport Chain

During **oxidative phosphorylation**, electrons derived from NADH and $FADH_2$ combine with O_2, and the energy released from these oxidation/reduction reactions is used to drive the synthesis of ATP from ADP. The transfer of electrons from NADH to O_2 is a very energy-yielding reaction, with $\Delta G^{o\prime} = -52.5$ kcal/mol for each pair of electrons transferred. To be harvested in usable form this energy must be produced gradually by the passage of electrons through a series of carriers, which constitute the **electron transport chain**. These carriers are organized into four complexes in the inner mitochondrial membrane. A fifth protein complex then serves to couple the energy-yielding reactions of electron transport to ATP synthesis.

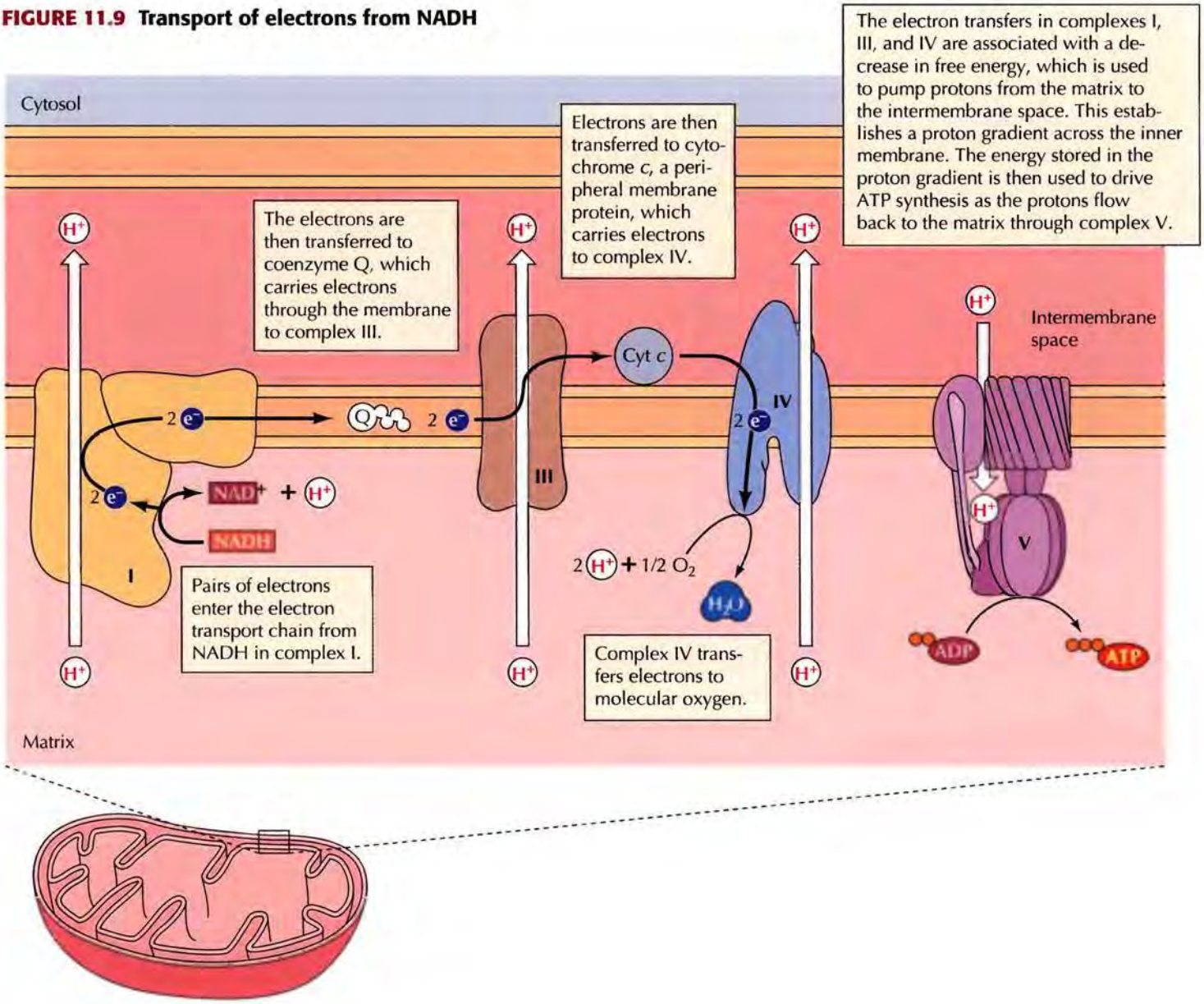
■ **Rotenone, an inhibitor of electron transfer from complex I to coenzyme Q, is used as a broad-spectrum insecticide.**

Electrons from NADH enter the electron transport chain in complex I, which consists of nearly 40 polypeptide chains (Figure 11.9). These electrons are initially transferred from NADH to flavin mononucleotide and then, through an iron-sulfur carrier, to coenzyme Q—an energy-yielding process $\Delta G^{o\prime} = -16.6$ kcal/mol. **Coenzyme Q** (also called **ubiquinone**) is a small, lipid-soluble molecule that carries electrons from complex I through the membrane to complex III, which consists of about ten polypeptides. In complex III, electrons are transferred from cytochrome *b* to cytochrome *c*—an energy-yielding reaction with $\Delta G^{o\prime} = -10.1$ kcal/mol. **Cytochrome *c***, a peripheral membrane protein bound to the outer face of the inner membrane then carries electrons to complex IV (**cytochrome oxidase**), where they are finally transferred to O_2 ($\Delta G^{o\prime} = -25.8$ kcal/mol).

A distinct protein complex (complex II), which consists of four polypeptides, receives electrons from the citric acid cycle intermediate, succinate (Figure 11.10). These electrons are transferred to $FADH_2$, rather than to NADH, and then to coenzyme Q. From coenzyme Q, electrons are transferred to complex III and then to complex IV as already described. In contrast to the transfer of electrons from NADH to coenzyme Q at complex I, the transfer of electrons from $FADH_2$ to coenzyme Q is not associated with a significant decrease in free energy and, therefore, is not coupled to ATP synthesis. Consequently, the passage of electrons derived from $FADH_2$ through the electron transport chain yields free energy only at complexes III and IV.

The free energy derived from the passage of electrons through complexes I, III, and IV is harvested by being coupled to the synthesis of ATP. Importantly, the mechanism by which the energy derived from these electron transport reactions is coupled to ATP synthesis, is fundamentally different from the synthesis of ATP during glycolysis or the citric acid cycle. In

FIGURE 11.9 Transport of electrons from NADH



the latter cases, a high-energy phosphate is transferred directly to ADP from the other substrate of an energy-yielding reaction. For example, in the final reaction of glycolysis, the high-energy phosphate of phosphoenolpyruvate is transferred to ADP, yielding pyruvate plus ATP (see Figure 3.11). Such direct transfer of high-energy phosphate groups does not occur during electron transport. Instead, the energy derived from electron transport is coupled to the generation of a proton gradient across the inner mitochondrial membrane. The potential energy stored in this gradient is then harvested by a fifth protein complex, which couples the energetically favorable flow of protons back across the membrane to the synthesis of ATP.

■ **In addition to its role in electron transport, cytochrome *c* is a key regulator of programmed cell death in mammalian cells (discussed in Chapter 16).**

Chemiosmotic Coupling

The mechanism of coupling electron transport to ATP generation, **chemiosmotic coupling**, is a striking example of the relationship between structure and function in cell biology. The hypothesis of chemiosmotic coupling was first proposed in 1961 by Peter Mitchell, who suggested that ATP is gener-

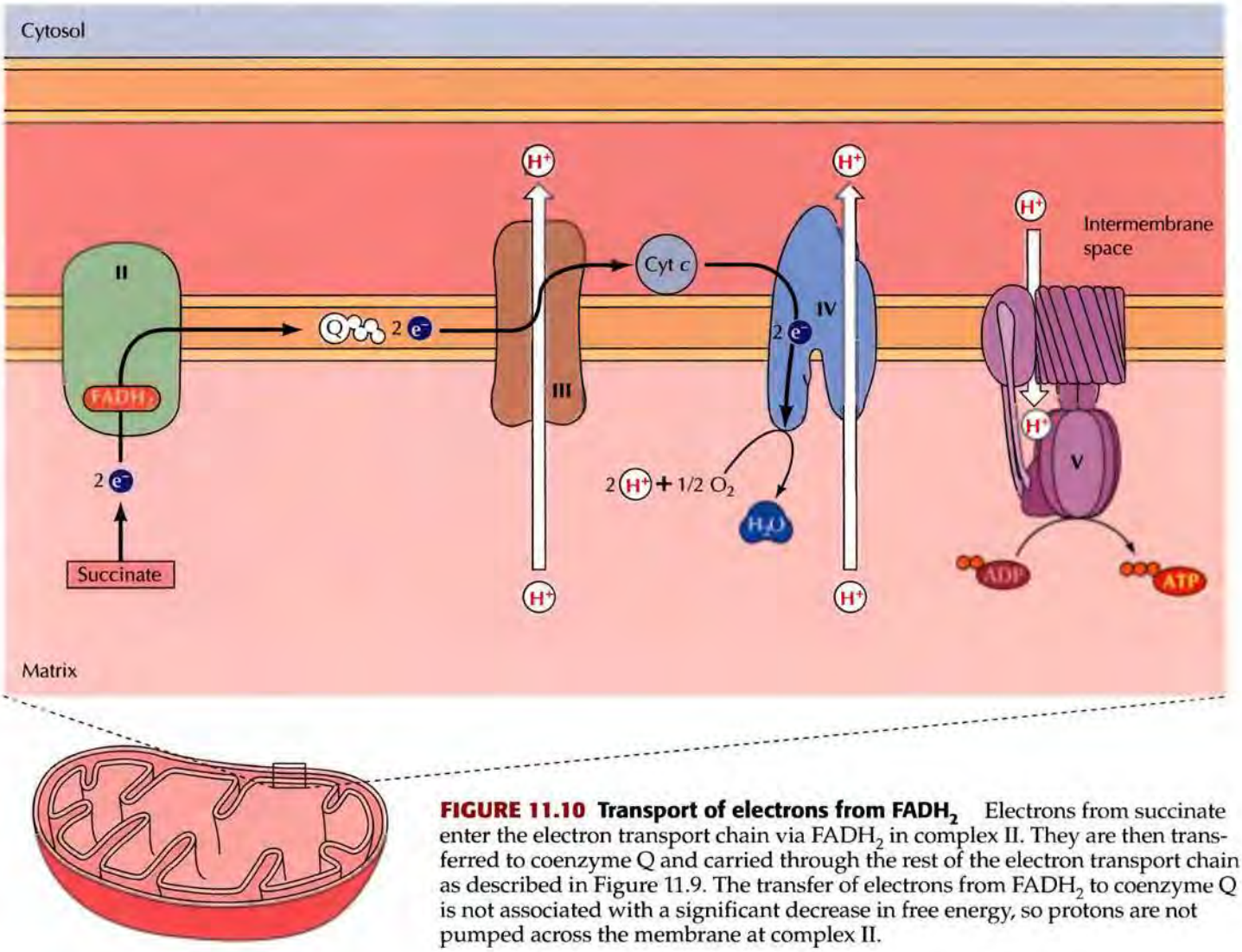


FIGURE 11.10 Transport of electrons from $FADH_2$ Electrons from succinate enter the electron transport chain via $FADH_2$ in complex II. They are then transferred to coenzyme Q and carried through the rest of the electron transport chain as described in Figure 11.9. The transfer of electrons from $FADH_2$ to coenzyme Q is not associated with a significant decrease in free energy, so protons are not pumped across the membrane at complex II.

ated by the use of energy stored in the form of proton gradients across biological membranes, rather than by direct chemical transfer of high-energy groups. Biochemists were initially highly skeptical of this proposal, and the chemiosmotic hypothesis took more than a decade to win general acceptance in the scientific community. Overwhelming evidence eventually accumulated in its favor, however, and chemiosmotic coupling is now recognized as a general mechanism of ATP generation, operating not only in mitochondria but also in chloroplasts and in bacteria, where ATP is generated via a proton gradient across the plasma membrane.

Electron transport through complexes I, III, and IV is coupled to the transport of protons out of the interior of the mitochondrion (see Figure 11.9). Thus the energy-yielding reactions of electron transport are coupled to the transfer of protons from the matrix to the intermembrane space, which establishes a proton gradient across the inner membrane. Complexes I and IV appear to act as proton pumps that transfer protons across the membrane as a result of conformational changes induced by electron transport. In complex III, protons are carried across the membrane by coenzyme Q, which accepts protons from the matrix at complexes I or II and releases them into the intermembrane space at complex III. Complexes I and III each transfer four protons across the membrane per pair of electrons. In complex

IV, two protons per pair of electrons are pumped across the membrane and another two protons per pair of electrons are combined with O_2 to form H_2O within the matrix. Thus the equivalent of four protons per pair of electrons is transported out of the mitochondrial matrix at each of these three complexes. This transfer of protons from the matrix to the intermembrane space plays the critical role of converting the energy derived from the oxidation/reduction reactions of electron transport to the potential energy stored in a proton gradient.

Because protons are electrically charged particles, the potential energy stored in the proton gradient is electric as well as chemical in nature. The electric component corresponds to the voltage difference across the inner mitochondrial membrane, with the matrix of the mitochondrion being negative and the intermembrane space being positive. The corresponding free energy is given by the equation

$$\Delta G = -F\Delta V$$

where F is the Faraday constant and ΔV is the membrane potential. The additional free energy corresponding to the difference in proton concentration across the membrane is given by the equation

$$\Delta G = RT \ln \frac{[H^+]_i}{[H^+]_o}$$

where $[H^+]_i$ and $[H^+]_o$ refer, respectively, to the proton concentrations inside and outside the mitochondria.

In metabolically active cells, protons are typically pumped out of the matrix such that the proton gradient across the inner membrane corresponds to about one pH unit, or a tenfold lower concentration of protons within mitochondria (Figure 11.11). The pH of the mitochondrial matrix is

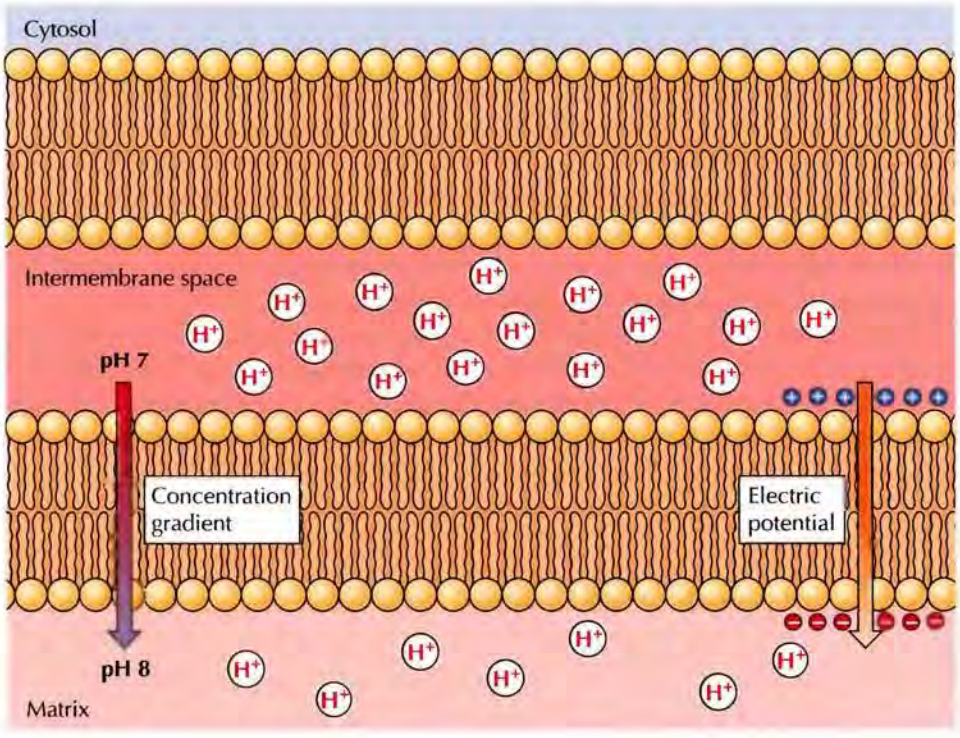


FIGURE 11.11 The electrochemical nature of the proton gradient Since protons are positively charged, the proton gradient established across the inner mitochondrial membrane has both chemical and electric components. The chemical component is the proton concentration, or pH, gradient, which corresponds to about a tenfold higher concentration of protons on the cytosolic side of the inner mitochondrial membrane (a difference of one pH unit). In addition, there is an electric potential across the membrane, resulting from the net increase in positive charge on the cytosolic side.

KEY EXPERIMENT

The Chemiosmotic Theory

Coupling of Phosphorylation to Electron and Hydrogen Transfer by a Chemiosmotic Type of Mechanism

Peter Mitchell

University of Edinburgh, Edinburgh, Scotland

Nature, 1961, Volume 191, pages 144–148

Peter Mitchell

UPI/Corbis-Bettmann

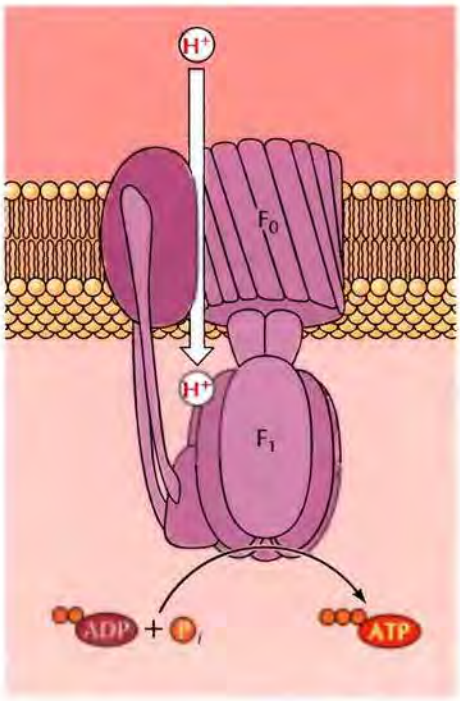
The Context

By the 1950s it had been clearly established that oxidative phosphorylation involved the stepwise transfer of electrons through a series of carriers to molecular oxygen. But how the energy derived from these electron transfer reactions was converted to ATP remained a mystery. The natural assumption was that ADP was converted to ATP by direct transfer of high-energy phosphate groups from some other intermediate, as was known to occur during glycolysis. Thus it was postulated that high-energy intermediates were produced as a result of electron transfer reactions and that these intermediates then drove ATP synthesis by phosphate group transfer.

The search for these postulated high-energy intermediates became a central goal of research during the 1950s and 1960s. But despite many false leads, no such intermediates were to be found. Moreover, several features of oxidative phosphorylation were difficult to reconcile with the orthodox hypothesis that ATP synthesis was driven by simple phosphate group transfer. In particular, phosphorylation was closely associated with membranes and was inhibited by a variety of compounds that disrupted membrane structure. These considerations led Peter Mitchell to propose a fundamentally different mechanism of energy coupling in which ATP synthesis was driven by an electrochemical gradient across a membrane rather than by the elusive high-energy intermediates sought by other investigators.

The Experiments

The fundamental proposal of the chemiosmotic hypothesis was that the "intermediate" that coupled electron transport to ATP synthesis was a proton electrochemical gradient across the membrane. Mitchell postulated that such a gradient was produced by electron transport and that the flow of protons back across the membrane in the energetically favorable direction

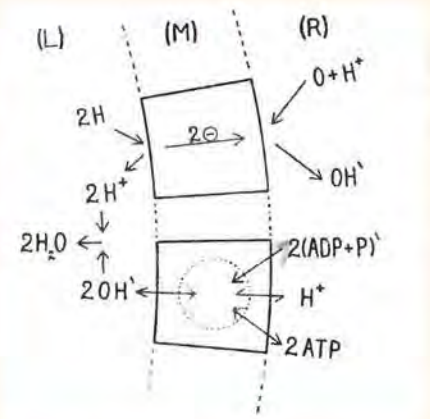


therefore about 8, compared to the neutral pH (approximately 7) of the cytosol and intermembrane space. This gradient also generates an electric potential of approximately 0.14 V across the membrane, with the matrix being negative. Both the pH gradient and the electric potential drive protons back into the matrix from the cytosol, so they combine to form an **electrochemical gradient** across the inner mitochondrial membrane, corresponding to a ΔG of approximately –5 kcal/mol per proton.

Because the phospholipid bilayer is impermeable to ions, protons are able to cross the membrane only through a protein channel. This restriction allows the energy in the electrochemical gradient to be harnessed and converted to ATP as a result of the action of the fifth complex involved in oxidative phosphorylation, complex V, or **ATP synthase** (see Figure 11.9).

FIGURE 11.12 Structure of ATP synthase The mitochondrial ATP synthase (complex V) consists of two multisubunit components, F_0 and F_1, which are linked by a slender stalk. F_0 spans the lipid bilayer, forming a channel through which protons can cross the membrane. F_1 harvests the free energy derived from proton movement down the electrochemical gradient by catalyzing the synthesis of ATP.

KEY EXPERIMENT



Mitchell's representation of chemiosmotic coupling between an electron transport system (top) and an ATP-generating system (bottom) in a membrane (M) enclosing aqueous phase L within aqueous phase R.

was then coupled to ATP synthesis (see figure).

The hypothesis of chemiosmotic coupling clearly explained the lack of success in identifying a chemical high-energy intermediate, as well as the fact that intact membranes were required to synthesize ATP. Yet it was a radical concept that went against the biochemical dogma of the time. In a concluding paragraph of this 1961 paper, Mitchell took a philosophical view of his revolutionary proposal:

> *In the exact sciences, cause and effect are no more than events linked in sequence. Biochemists now generally accept the idea that metabolism is the cause of membrane transport. The underlying thesis of the hypothesis put forward here is that if the processes that we call metabolism and transport represent events in a sequence, not only can metabolism be the cause of transport, but also transport can be the cause of metabolism.*

The Impact

Mitchell's hypothesis was greeted with skepticism and remained the subject of acrimonious debate for more than a decade. However, the wealth of supporting evidence obtained by Mitchell and his colleagues, as well as by other investigators, eventually led to general acceptance of the chemiosmotic hypothesis—which became known instead as the chemiosmotic theory. It is now accepted not only as the basis for the generation of ATP during oxidative phosphorylation and photosynthesis in bacteria, mitochondria, and chloroplasts but also for the energy-requiring transport of a variety of molecules across cell membranes.

Mitchell's work was recognized with a Nobel Prize in 1978. The lecture he delivered on that occasion began as follows:

> *Although I had hoped that the chemiosmotic rationale of vectorial metabolism and biological energy transfer might one day come to be generally accepted, it would have been presumptuous of me to expect it to happen. Was it not Max Planck who remarked that a new scientific idea does not triumph by convincing its opponents, but rather because its opponents eventually die? The fact that what began as the chemiosmotic hypothesis has now been acclaimed as the chemiosmotic theory ... has therefore both astonished and delighted me, particularly because those who were formerly my most capable opponents are still in the prime of their scientific lives.*

ATP synthase is organized into two structurally distinct components, F_0 and F_1, which are linked by a slender stalk (Figure 11.12). The F_0 portion is an electrically driven motor that spans the inner membrane and provides a channel through which protons are able to flow back from the intermembrane space to the matrix. The energetically favorable return of protons to the matrix is coupled to ATP synthesis by the F_1 subunit, which catalyzes the synthesis of ATP from ADP and phosphate ions (P_i). Detailed structural studies have established the mechanism of ATP synthase action, which involves mechanical coupling between the F_0 and F_1 subunits. In particular, the flow of protons through F_0 drives the rotation of a part of F_1, which acts as a rotary motor to drive ATP synthesis.

It appears that the flow of four protons back across the membrane through F_0 is required to drive the synthesis of one molecule of ATP by F_1, consistent with the proton transfers at complexes I, III, and IV each contributing sufficient free energy to the proton gradient to drive the synthesis of one ATP molecule. The oxidation of one molecule of NADH thus leads to the synthesis of three molecules of ATP, whereas the oxidation of $FADH_2$, which enters the electron transport chain at complex II, yields only two ATP molecules.

■ **Another example of an electrochemical rotary motor is the bacterial flagellum.**

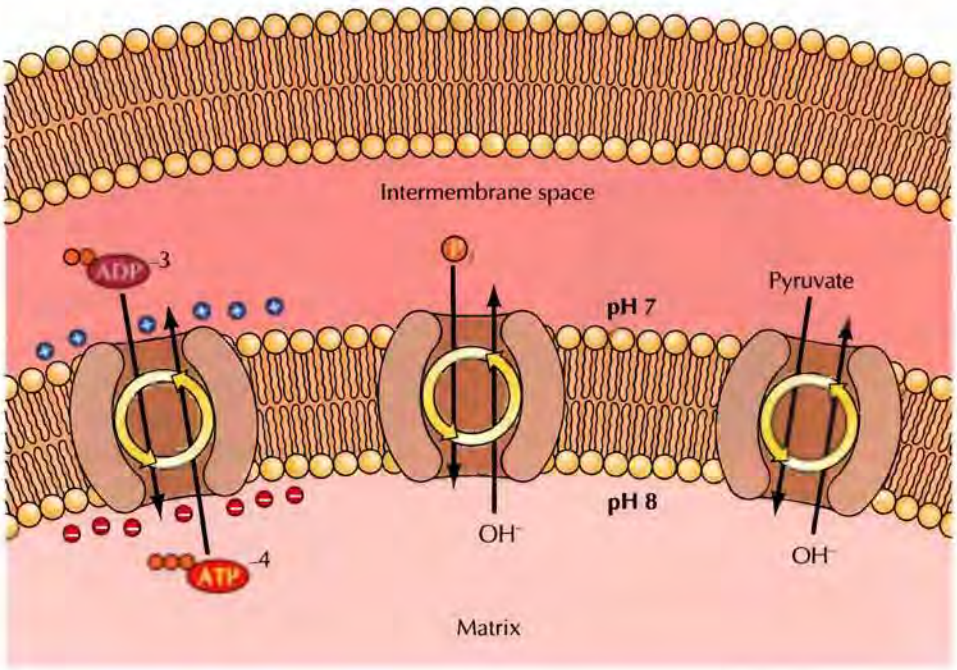


FIGURE 11.13 Transport of metabolites across the mitochondrial inner membrane The transport of small molecules across the inner membrane of mitochondria is mediated by membrane-spanning proteins and driven by the electrochemical gradient. For example, ATP is exported from mitochondria to the cytosol by a transporter that exchanges it for ADP. The voltage component of the electrochemical gradient drives this exchange: ATP carries a greater negative charge (–4) than ADP (–3), so ATP is exported from the mitochondrial matrix to the cytosol while ADP is imported into mitochondria. In contrast, the transport of phosphate (P_i) and pyruvate is coupled to an exchange for hydroxyl ions (OH^-); in this case, the pH component of the electrochemical gradient drives the export of hydroxyl ions, coupled to the transport of P_i and pyruvate into mitochondria.

Transport of Metabolites across the Inner Membrane

In addition to driving the synthesis of ATP, the potential energy stored in the electrochemical gradient drives the transport of small molecules into and out of mitochondria. For example, the ATP synthesized within mitochondria has to be exported to the cytosol, while ADP and P_i need to be imported from the cytosol for ATP synthesis to continue. The electrochemical gradient generated by proton pumping provides energy required for the transport of these molecules and other metabolites that need to be concentrated within mitochondria (Figure 11.13).

The transport of ATP and ADP across the inner membrane is mediated by an integral membrane protein, the adenine nucleotide translocator, which transports one molecule of ADP into the mitochondrion in exchange for one molecule of ATP transferred from the mitochondrion to the cytosol. Because ATP carries more negative charge than ADP (–4 compared to –3), this exchange is driven by the voltage component of the electrochemical gradient. Since the proton gradient establishes a positive charge on the cytosolic side of the membrane, the export of ATP in exchange for ADP is energetically favorable.

The synthesis of ATP within the mitochondrion requires phosphate ions (P_i) as well as ADP, so P_i must also be imported from the cytosol. This is mediated by another membrane transport protein, which imports phosphate ($H_2PO_4^-$) and exports hydroxyl ions (OH^-). This exchange is electrically neutral because both phosphate and hydroxyl ions have a charge of –1. However, the exchange is driven by the proton concentration gradient; the higher pH within mitochondria corresponds to a higher concentration of hydroxyl ions, favoring their translocation to the cytosolic side of the membrane.

Energy from the electrochemical gradient is similarly used to drive the transport of other metabolites into mitochondria. For example, the import of pyruvate from the cytosol (where it is produced by glycolysis) is medi-

■ **All newborn mammals (and certain adult mammals) contain a specialized tissue called brown fat. The mitochondria in cells of brown fat contain an uncoupling protein called thermogenin that utilizes the proton gradient to generate heat. Brown fat is very important for thermoregulation in neonates and in hibernating mammals.**

ated by a transport protein that exchanges pyruvate for hydroxyl ions. Other intermediates of the citric acid cycle are able to shuttle between mitochondria and the cytosol by similar exchange mechanisms.

Chloroplasts and Other Plastids

Chloroplasts, the organelles responsible for photosynthesis, are in many respects similar to mitochondria. Both chloroplasts and mitochondria function to generate metabolic energy, both have evolved by endosymbiosis, both contain their own genetic systems, and both replicate by division. However, chloroplasts are larger and more complex than mitochondria, and they perform several critical tasks in addition to the generation of ATP. Most importantly, chloroplasts are responsible for the photosynthetic conversion of CO_2 to carbohydrates. In addition, chloroplasts synthesize amino acids, fatty acids, and the lipid components of their own membranes. The reduction of nitrite (NO_2^-) to ammonia (NH_3), an essential step in the incorporation of nitrogen into organic compounds, also occurs in chloroplasts. Moreover, chloroplasts are only one of several types of related organelles (plastids) that play a variety of roles in plant cells.

The Structure and Function of Chloroplasts

Plant chloroplasts are large organelles (5 to 10 μm long) that, like mitochondria, are bounded by a double membrane called the chloroplast envelope (Figure 11.14). In addition to the inner and outer membranes of the envelope, chloroplasts have a third internal membrane system, called the thylakoid membrane. The **thylakoid membrane** forms a network of flattened discs called thylakoids, which are frequently arranged in stacks called grana. Because of this three-membrane structure, the internal organization of chloroplasts is more complex than that of mitochondria. In particular, their three membranes divide chloroplasts into three distinct internal compartments: (1) the intermembrane space between the two membranes of the chloroplast envelope; (2) the **stroma**, which lies inside the envelope but outside the thylakoid membrane; and (3) the thylakoid lumen.

Despite this greater complexity, the membranes of chloroplasts have clear functional similarities with those of mitochondria—as expected, given the role of both organelles in the chemiosmotic generation of ATP. The outer membrane of the chloroplast envelope, like that of mitochondria, contains porins and is therefore freely permeable to small molecules. In contrast, the inner membrane is impermeable to ions and metabolites, which are therefore able to enter chloroplasts only via specific membrane transporters. These properties of the inner and outer membranes of the chloroplast envelope are similar to the inner and outer membranes of mitochondria: In both cases the inner membrane restricts the passage of molecules between the cytosol and the interior of the organelle. The chloroplast stroma is also equivalent in function to the mitochondrial matrix: It contains the chloroplast genetic system and a variety of metabolic enzymes, including those responsible for the critical conversion of CO_2 to carbohydrates during photosynthesis.

The major difference between chloroplasts and mitochondria, in terms of both structure and function, is the thylakoid membrane. This membrane is of central importance in chloroplasts, where it fills the role of the inner mitochondrial membrane in electron transport and the chemiosmotic gen-

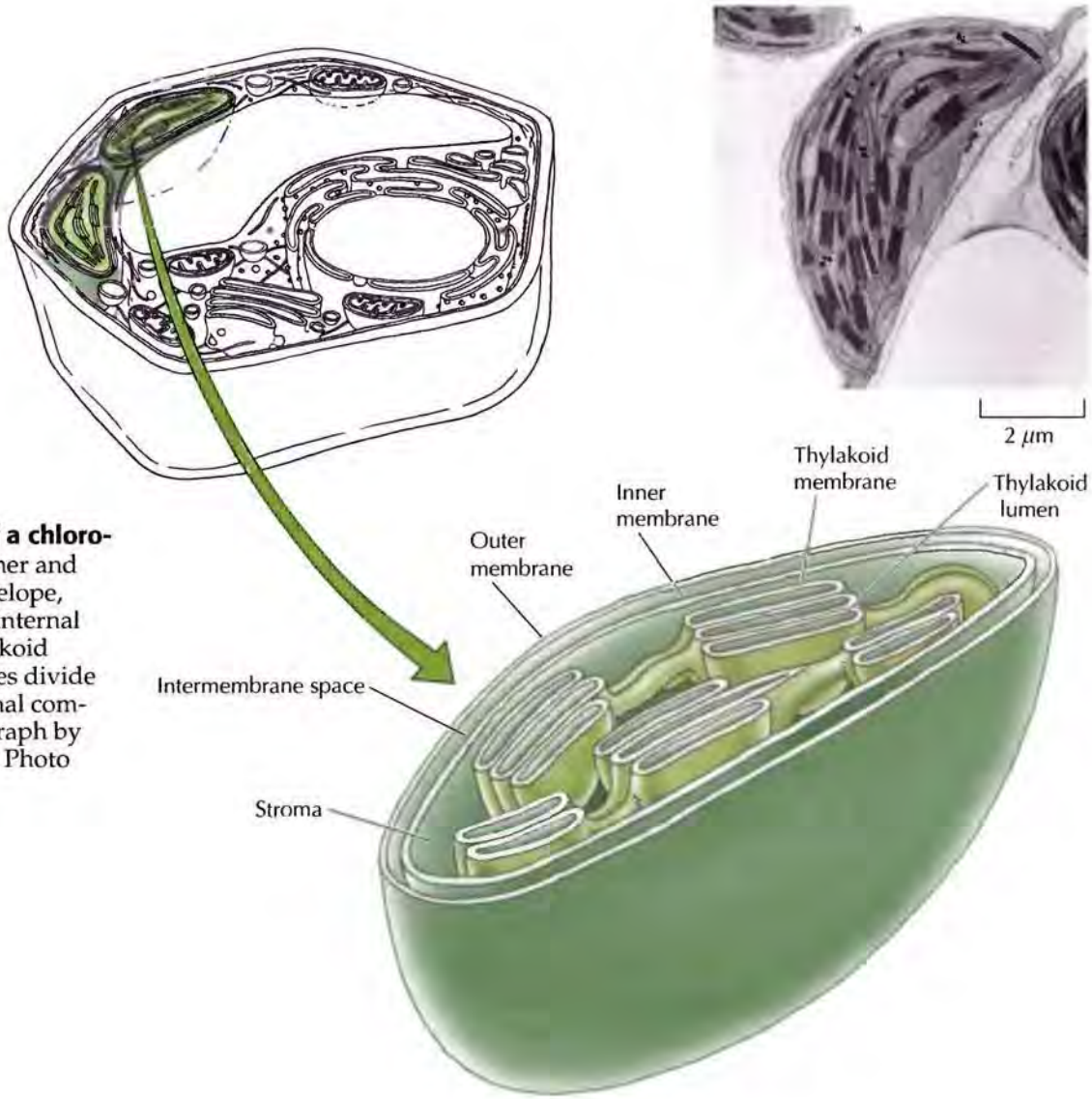


FIGURE 11.14 Structure of a chloroplast In addition to the inner and outer membranes of the envelope, chloroplasts contain a third internal membrane system: the thylakoid membrane. These membranes divide chloroplasts into three internal compartments. (Electron micrograph by E. H. Newcombe/Biological Photo Service.)

eration of ATP (Figure 11.15). The inner membrane of the chloroplast envelope (which is not folded into cristae) does not function in either electron transport or photosynthesis. Instead, the chloroplast electron transport system is located in the thylakoid membrane, and protons are pumped across this membrane from the stroma to the thylakoid lumen. The resulting electrochemical gradient then drives ATP synthesis as protons cross back into the stroma. In terms of its role in generation of metabolic energy, the thylakoid membrane of chloroplasts is thus equivalent to the inner membrane of mitochondria.

The Chloroplast Genome

Like mitochondria, chloroplasts contain their own genetic system, reflecting their evolutionary origins from photosynthetic bacteria. The 6–9 Mb genomes of present day free living photosynthetic cyanobacteria code for between 5400 and 7200 proteins. Like those of mitochondria, the genomes of chloroplasts consist of circular DNA molecules present in multiple copies per organelle. However, chloroplast genomes are larger and more complex than those of mitochondria, ranging from 120 to 160 kb and containing approximately 150 genes.

The chloroplast genomes of several plants have been completely sequenced, leading to the identification of many of the genes contained in

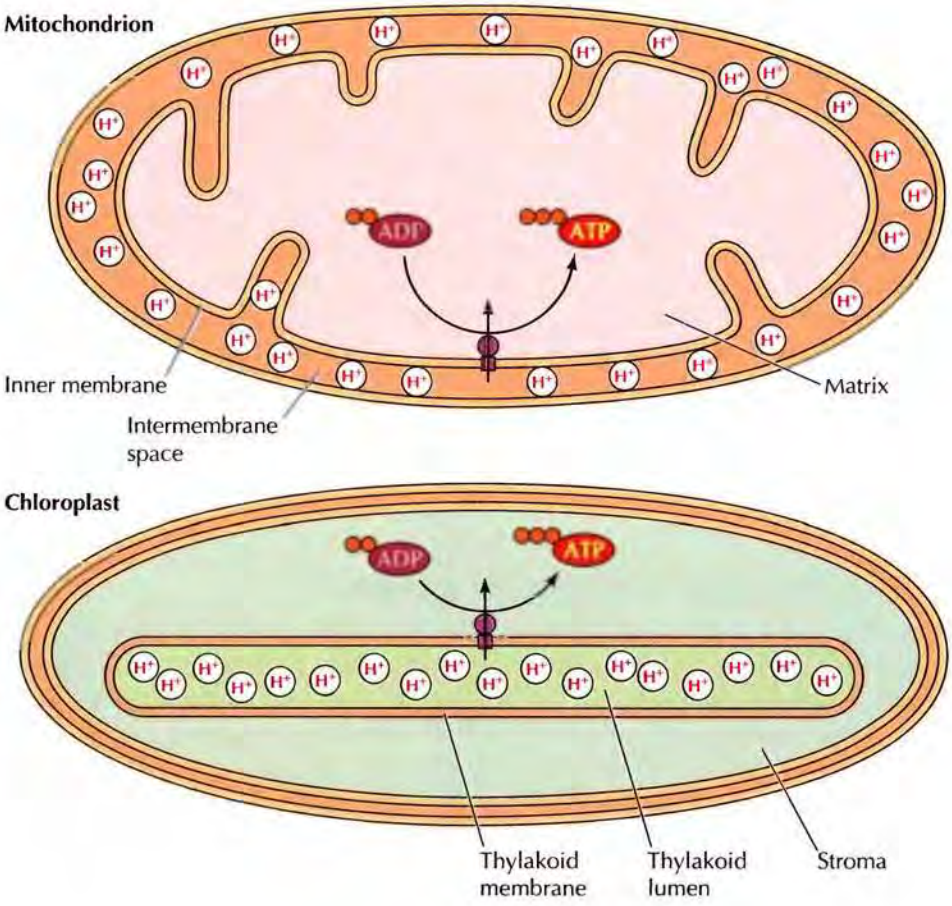


FIGURE 11.15 Chemiosmotic generation of ATP in chloroplasts and mitochondria In mitochondria, electron transport generates a proton gradient across the inner membrane, which is then used to drive ATP synthesis in the matrix. In chloroplasts, the proton gradient is generated across the thylakoid membrane and used to drive ATP synthesis in the stroma.

the organelle DNAs. These chloroplast genes encode both RNAs and proteins involved in gene expression, as well as a variety of proteins that function in photosynthesis (Table 11.2). Both the ribosomal and transfer RNAs used for translation of chloroplast mRNAs are encoded by the organelle genome. These include four rRNAs (23S, 16S, 5S, and 4.5S) and 30 tRNA species. In contrast to the smaller number of tRNAs encoded by the mitochondrial genome, the chloroplast tRNAs are sufficient to translate all the mRNA codons according to the universal genetic code. In addition to these

TABLE 11.2 Genes Encoded by Chloroplast DNA

Function	Number of genes
Genes for the genetic apparatus	
rRNAs (23S, 16S, 5S, 4.5S)	4
tRNAs	30
Ribosomal proteins	21
RNA polymerase subunits	4
Genes for photosynthesis	
Photosystem I	5
Photosystem II	12
Cytochrome *bf* complex	4
ATP synthase	6
Ribulose bisphosphate carboxylase	1

Sequence analysis indicates that chloroplast genomes contain about 30 genes in addition to those listed here. Some of these encode proteins involved in respiration, but most remain to be identified.

RNA components of the translation system, the chloroplast genome encodes about 20 ribosomal proteins, which represent approximately a third of the proteins of chloroplast ribosomes. Some subunits of RNA polymerase are also encoded by chloroplasts, although additional RNA polymerase subunits and other factors needed for chloroplast gene expression are encoded in the nucleus.

The chloroplast genome also encodes approximately 30 proteins that are involved in photosynthesis, including components of photosystems I and II, of the cytochrome *bf* complex, and of ATP synthase. In addition, one of the subunits of ribulose bisphosphate carboxylase (rubisco) is encoded by chloroplast DNA. Rubisco is the critical enzyme that catalyzes the addition of CO_2 to ribulose-1,5-bisphosphate during the Calvin cycle (see Figure 3.18). Not only is it the major protein component of the chloroplast stroma, but it is also thought to be the single most abundant protein on Earth, so it is noteworthy that one of its subunits is encoded by the chloroplast genome.

Import and Sorting of Chloroplast Proteins

Although chloroplasts encode more of their own proteins than mitochondria, about 3500 or 95% of chloroplast proteins are still encoded by nuclear genes. As with mitochondria, these proteins are synthesized on cytosolic ribosomes and then imported into chloroplasts as completed polypeptide chains. They must then be sorted to their appropriate location within chloroplasts—an even more complicated task than protein sorting in mitochondria, since chloroplasts contain three separate membranes that divide them into three distinct internal compartments.

Proteins are targeted for import into chloroplasts by N-terminal sequences of 30 to 100 amino acids, called **transit peptides**, which direct protein translocation across the two membranes of the chloroplast envelope and are then removed by proteolytic cleavage (Figure 11.16). A **guidance complex** initially recognizes the transit peptides and directs them to the translocase of the chloroplast outer member (the **Toc complex**), where they bind to the Toc159 and Toc34 receptors. In contrast to the presequences for mitochondrial import, transit peptides are not positively charged and the chloroplast inner membrane does not have a strong electric potential. Protein import into chloroplasts requires Hsp70 molecules to keep the preprotein in an unfolded state. In addition, Hsp70 molecules are attached to the Toc complex where they drive protein import by hydrolysis of ATP (see Figure 11.5). At least one Toc protein, Toc 34, binds GTP, and hydrolysis of GTP may provide an additional source of energy for translocation.

After preproteins are transported through the Toc complex, they are transferred to the translocase of the inner membrane (the **Tic complex**) and transported across the inner membrane into the stroma. Like the Toc, Tom, and Tim complexes, Tic is a multiprotein complex with one or more protein channels. However, little is known about the specific proteins, perhaps because there is more than one type of Tic complex. A chaperone of the Hsp100 family (another family of chaperones in addition to those that were discussed in Chapter 8) is associated with the stromal side of the Tic complex. This chloroplast Hsp100 acts to draw the preprotein through the inner membrane. In the stroma, the transit peptide is cleaved by a **stromal processing peptidase (SPP)**, and the protein associates with stromal Hsp70 chaperones. As in the mitochondrial matrix, some proteins that remain in the stroma complete their folding within an Hsp60 chaperonin.

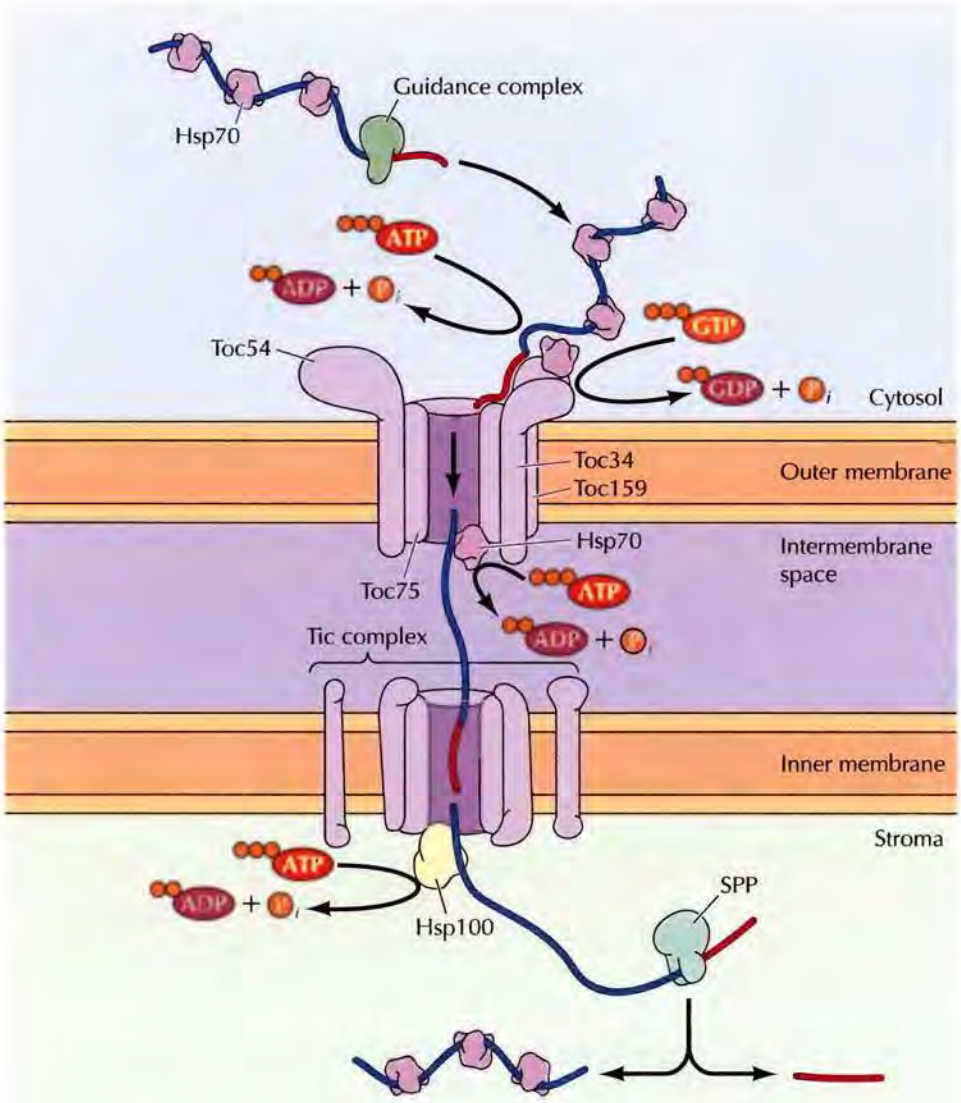


FIGURE 11.16 Import of proteins into the chloroplast stroma
Proteins with N-terminal transit peptides are recognized by a guidance complex that targets them to the Toc complex in the chloroplast outer membrane. The transit peptide binds Toc159 and Toc34, which are associated with Hsp70, before being passed to the Toc75 import pore. Passage through the outer membrane also requires ATP hydrolysis by Hsp70 in the intermembrane space, and possibly the hydrolysis of GTP by Toc34. Once through the chloroplast outer membrane, the transit peptide is passed to the Tic complex in the inner membrane. The preprotein is drawn through the Tic complex by the action of an Hsp100. In the stroma, the transit peptide is removed by the chloroplast stromal processing peptidase (SPP) and the protein interacts with Hsp70.

Little is known about how proteins are targeted to the outer or inner membranes of the chloroplast. Proteins to be incorporated into the thylakoid membrane or lumen are transported to their destination in two steps. They are first imported into the stroma, as already described, and then most are targeted for translocation across the thylakoid membrane by a second signal sequence, which is exposed following cleavage of the transit peptide. Proteins are translocated into the thylakoid lumen by at least three different pathways (Figure 11.17). In the Sec pathway, the thylakoid signal sequence is recognized by the SecA protein and translocated in an ATP-dependent manner through the Sec translocon. The second pathway (the TAT pathway) uses a twin-arginine signal sequence and depends on the proton gradient across the thylakoid membrane to translocate fully folded proteins. A third pathway (the SRP pathway) is used for thylakoid membrane proteins, which are recognized by a stromal signal recognition particle (SRP). In addition, some proteins may insert directly into the thylakoid membrane.

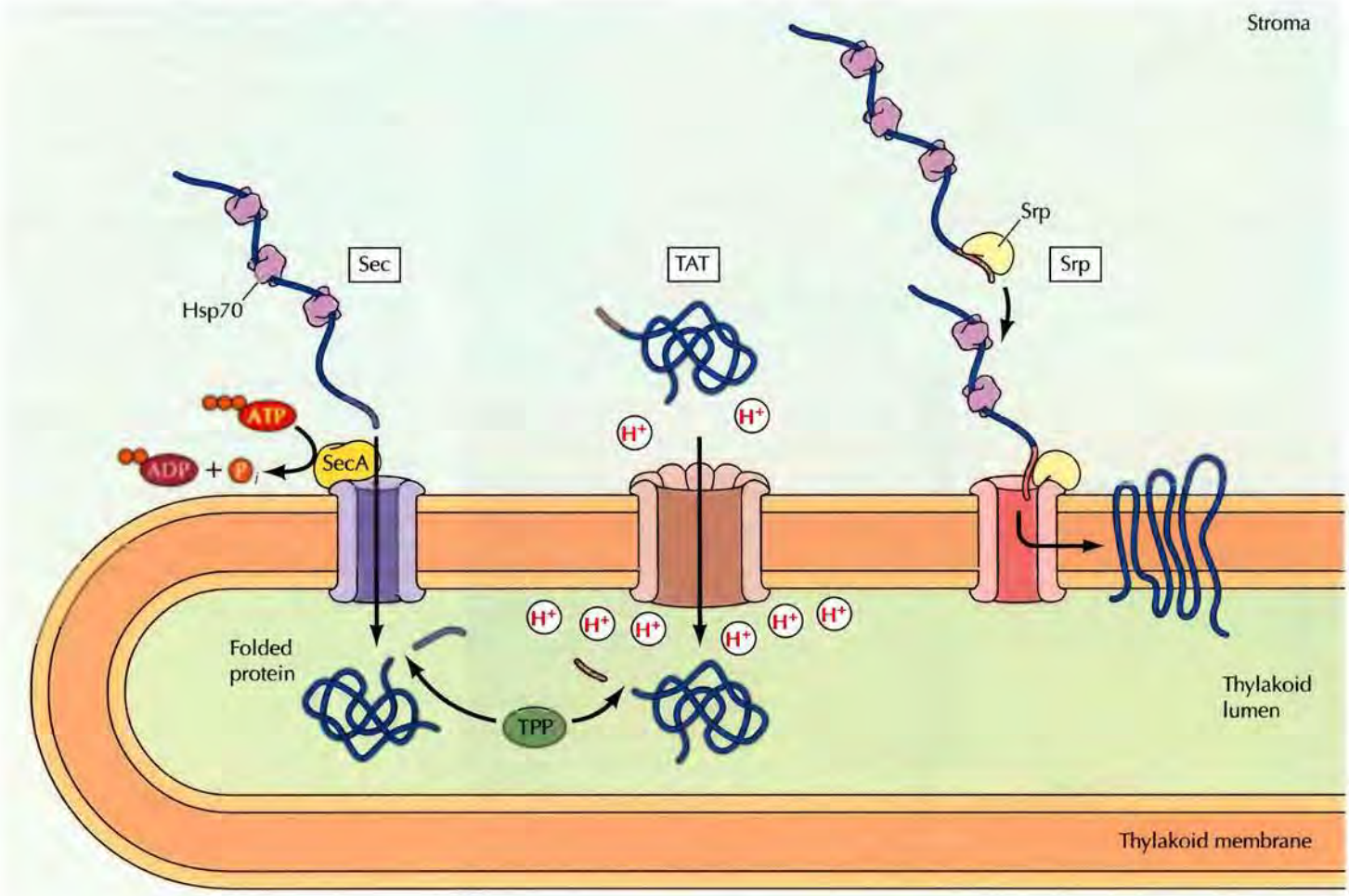


FIGURE 11.17 Import of proteins into the thylakoid lumen or membrane
Three characterized pathways transfer proteins from the stroma to the thylakoid lumen or membrane. In the Sec pathway, the SecA protein recognizes a thylakoid signal sequence and targets the protein to the Sec translocon, using energy derived from ATP hydrolysis to transfer the protein to the lumen. In the lumen, the thylakoid signal sequence is cleaved by a thylakoid processing protease (TPP). In the twin-arginine-translocation (TAT) pathway, a cleavable thylakoid transfer sequence containing two arginines near the amino terminus targets fully folded proteins directly to a novel translocon, which inserts them into the thylakoid lumen. Both assembly of the translocon and translocation of the protein are dependent on the proton gradient across the thylakoid membrane. In the Srp pathway, transmembrane proteins are recognized by the chloroplast signal recognition particle (Srp) and inserted into the thylakoid membrane.

Other Plastids

Chloroplasts are only one, albeit the most prominent, member of a larger family of plant organelles called **plastids**. All plastids contain the same genome as chloroplasts, but they differ in both structure and function. Chloroplasts are specialized for photosynthesis and are unique in that they contain the internal thylakoid membrane system. Other plastids, which are involved in different aspects of plant cell metabolism (synthesis of amino acids, fatty acids and lipids, plant hormones, nucleotides, vitamins, and sec-

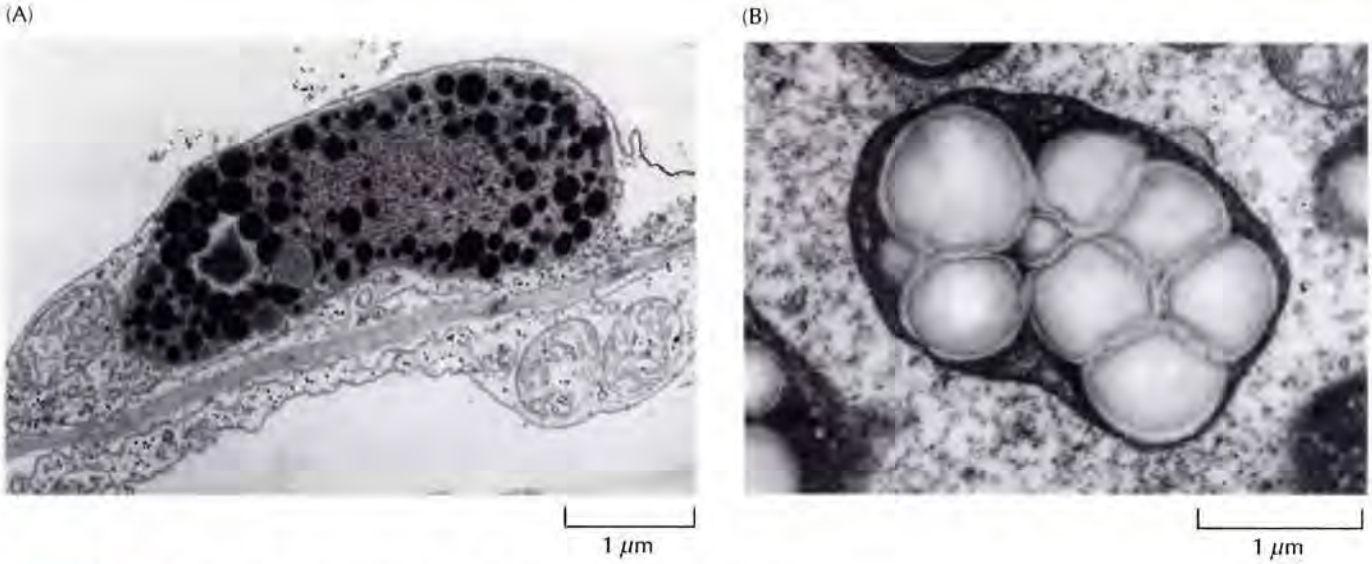


FIGURE 11.18 Electron micrographs of chromoplasts and amyloplasts
(A) Chromoplasts contain lipid droplets in which carotenoids are stored. (B) Amyloplasts contain large starch granules. (A, Biophoto Associates/Photo Researchers, Inc.; B, Dr. Jeremy Burgess/Photo Researchers, Inc.)

ondary metabolites), are bounded by the two membranes of the plastid envelope but lack both the thylakoid membranes and other components of the photosynthetic apparatus.

The different types of plastids are frequently classified according to the kinds of pigments they contain. Chloroplasts are so named because they contain chlorophyll. **Chromoplasts** (Figure 11.18A) lack chlorophyll but contain carotenoids; they are responsible for the yellow, orange, and red colors of some flowers and fruits, although their precise function in cell metabolism is not clear. **Leucoplasts** are nonpigmented plastids, which store a variety of energy sources in nonphotosynthetic tissues. **Amyloplasts** (Figure 11.18B) and **elaioplasts** are examples of leucoplasts that store starch and lipids, respectively.

All plastids, including chloroplasts, develop from **proplastids**, small (0.5 to 1 μm in diameter) undifferentiated organelles present in the rapidly dividing cells of plant roots and shoots. Proplastids then develop into the various types of mature plastids according to the needs of differentiated cells. In addition, mature plastids are able to change from one type to another. Chromoplasts develop from chloroplasts, for example, during the ripening of fruit (e.g., tomatoes). During this process, chlorophyll and the thylakoid membranes break down, while new types of carotenoids are synthesized.

An interesting feature of plastids is that their development is controlled both by environmental signals and by intrinsic programs of cell differentiation. In the photosynthetic cells of leaves, for example, proplastids develop into chloroplasts (Figure 11.19). During this process, the thylakoid membrane is formed by vesicles budding from the inner membrane of the plastid envelope and the various components of the photosynthetic apparatus are synthesized and assembled. However, chloroplasts develop only in the presence of light. If plants are kept in the dark, the development of proplastids in leaves is arrested at an intermediate stage (called **etioplasts**) in

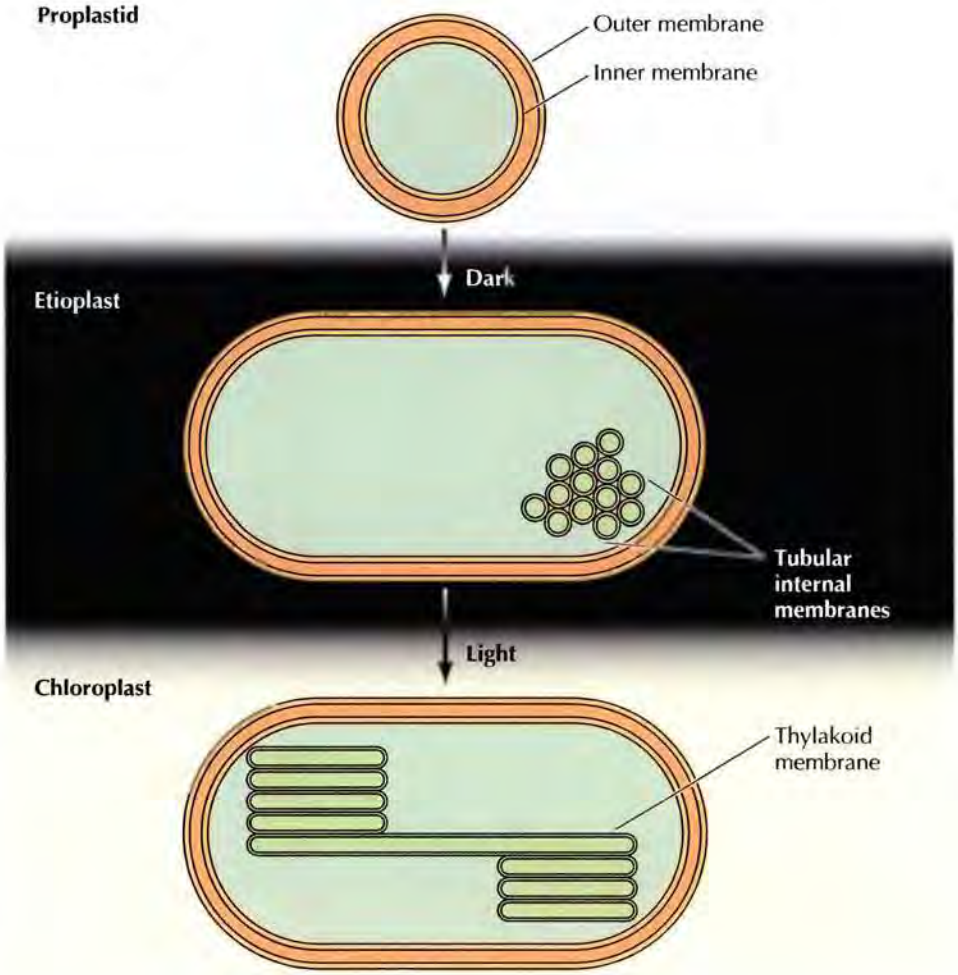


FIGURE 11.19 Development of chloroplasts Chloroplasts develop from proplastids in the photosynthetic cells of leaves. Proplastids contain only the inner and outer envelope membranes; the thylakoid membrane is formed by vesicle budding from the inner membrane during chloroplast development. If the plant is kept in the dark, chloroplast development is arrested at an intermediate stage (etioplasts). Etioplasts lack chlorophyll and contain semicrystalline arrays of membrane tubules. In the presence of light, they continue their development to chloroplasts.

11.1 WEBSITE ANIMATION

From Proplastid to Chloroplast A proplastid with inner and outer membranes develops into a mature chloroplast with a third membrane system—the thylakoid membranes.

FIGURE 11.20 Electron micrograph of an etioplast (John N. A. Lott/Biological Photo Service.)

which a semicrystalline array of tubular internal membranes has formed but chlorophyll has not been synthesized (Figure 11.20). If dark-grown plants are then exposed to light, the etioplasts continue their development to chloroplasts. It is noteworthy that this dual control of plastid development involves the coordinated expression of genes within both the plastid and nuclear genomes. The mechanisms responsible for such coordinated gene expression are largely unknown, and their elucidation represents a challenging problem in plant molecular biology.

Photosynthesis

During photosynthesis, energy from sunlight is harvested and used to drive the synthesis of glucose from CO_2 and H_2O. By converting the energy of sunlight to a usable form of potential chemical energy, photosynthesis is the ultimate source of metabolic energy for all biological systems. Photosynthesis takes place in two distinct stages. In the light reactions, energy from sunlight drives the synthesis of ATP and NADPH, coupled to the formation of O_2 from H_2O. In the dark reactions (so named because they do not require sunlight) the ATP and NADPH produced by the light reactions drive glucose synthesis. In eukaryotic cells, both the light and dark reactions of photosynthesis occur within chloroplasts—the light reactions in the thylakoid

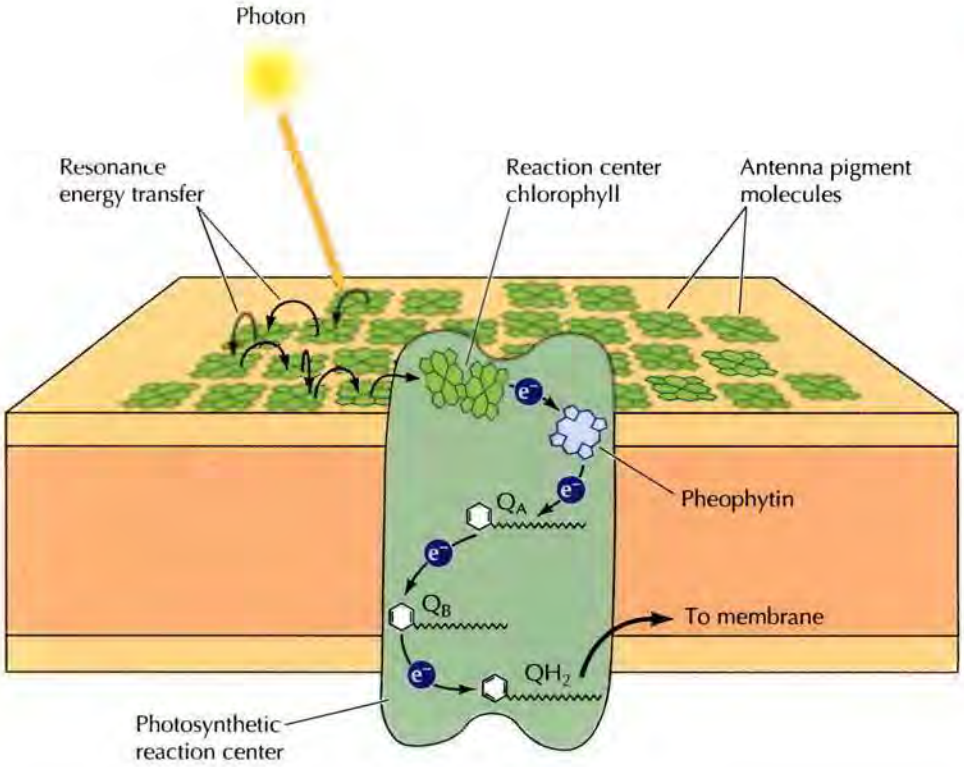


FIGURE 11.21 Organization of a photocenter Each photocenter consists of hundreds of antenna pigment molecules, which absorb photons and transfer energy to a reaction center chlorophyll. The reaction center chlorophyll then transfers its excited electron to an acceptor in the electron transport chain. The reaction center illustrated is that of photosystem II in which electrons are transferred from the reaction center chlorophyll to pheophytin and then to quinones (Q_A, Q_B, and QH_2).

membrane and the dark reactions within the stroma. This section discusses the light reactions of photosynthesis, which are related to oxidative phosphorylation in mitochondria. The dark reactions were discussed in detail in Chapter 3.

Electron Flow through Photosystems I and II

Sunlight is absorbed by photosynthetic pigments, the most abundant of which in plants are the **chlorophylls**. Absorption of light excites an electron to a higher energy state, thus converting the energy of sunlight to potential chemical energy. The photosynthetic pigments are organized into **photocenters** in the thylakoid membrane, each of which contains hundreds of pigment molecules (Figure 11.21). The many pigment molecules in each photocenter act as antennae to absorb light and transfer the energy of their excited electrons to a chlorophyll molecule that serves as a reaction center. The reaction center chlorophyll then transfers its high-energy electron to an acceptor molecule in an electron transport chain. High-energy electrons are then transferred through a series of membrane carriers, coupled to the synthesis of ATP and NADPH.

The proteins involved in the light reactions of photosynthesis are organized into multiprotein complexes in photosynthetic membranes (Figure 11.22). The earliest characterized photosynthetic reaction center was that of the bacterium *Rhodopseudomonas viridis*, the structure of which was determined by Johann Deisenhofer, Hartmut Michel, Robert Huber, and their colleagues in 1985 (Figure 11.23). The reaction center consists of three trans-

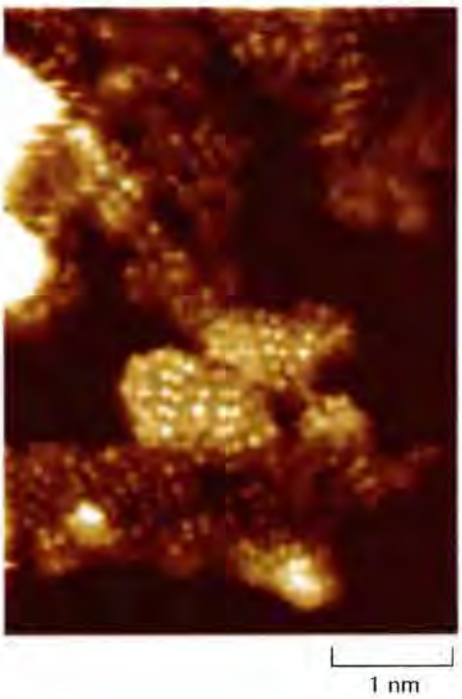


FIGURE 11.22 Photosynthetic complexes Atomic force micrograph of photosynthetic membrane fragments from the purple bacterium *Rhodobacter sphaeroides* showing the high density of photosynthetic complexes. (From S. Bahatyrova et al., 2004. *Nature* 430: 1059.)

FIGURE 11.23 Structure of a photosynthetic reaction center The reaction center of *R. viridis* consists of three transmembrane proteins (purple, blue, and beige) and a *c*-type cytochrome (green). Chlorophylls and other prosthetic groups are colored yellow. (Courtesy of Johann Deisenhofer, University of Texas Medical Center and The Nobel Foundation, 1989.)

■ **A variety of compounds that bind photosystem II and inhibit electron transfer are used as herbicides for the control of weeds.**

membrane polypeptides, bound to a *c*-type cytochrome on the exterior side of the membrane. Energy from sunlight is captured by a pair of chlorophyll molecules known as the special pair. Electrons are then transferred from the special pair to another pair of chlorophylls and from there to other prosthetic groups (pheophytins and quinones). From there the electrons are transferred to a cytochrome *bc* complex in which electron transport is coupled to the generation of a proton gradient. The electrons are then transferred to the reaction center cytochrome and finally returned to the chlorophyll special pair. The reaction center thus converts the energy of sunlight to high-energy electrons, the potential energy of which is converted to a proton gradient by the cytochrome *bc* complex.

The proteins involved in the light reactions of photosynthesis in plants are organized into five multiprotein complexes in the thylakoid membrane (Figure 11.24). Two of these complexes are photosystems (**photosystems I** and **II**) in which light is absorbed and transferred to reaction center chlorophylls. High-energy electrons are then transferred through a series of carriers in both photosystems and in a third protein complex, the **cytochrome *bf* complex**. As in mitochondria, these electron transfers are coupled to the transfer of protons into the thylakoid lumen, thereby establishing a proton gradient across the thylakoid membrane. The energy stored in this proton gradient is then harvested by a fourth protein complex in the thylakoid membrane, ATP synthase, which (like the mitochondrial enzyme) couples proton flow back across the membrane to the synthesis of ATP.

One important difference between electron transport in chloroplasts and that in mitochondria is that the energy derived from sunlight during photosynthesis not only is converted to ATP but also is used to generate the NADPH required for subsequent conversion of CO_2 to carbohydrates. This is accomplished by the use of two different photosystems in the light reactions of photosynthesis, one to generate ATP and the other to generate NADPH. Electrons are transferred sequentially between the two photosystems, with photosystem I acting to generate NADPH and photosystem II acting to generate ATP.

The pathway of electron flow starts at photosystem II, which is homologous to the photosynthetic reaction center of *R. viridis*, already described. However, at photosystem II the energy derived from absorption of photons is used to split water molecules to molecular oxygen and protons (see Figure 11.24). This reaction takes place within the thylakoid lumen, so the release of protons from H_2O establishes a proton gradient across the thylakoid membrane. The high-energy electrons derived from this process are transferred through a series of carriers to plastoquinone, a lipid-soluble carrier similar to coenzyme Q (ubiquinone) of mitochondria. Plastoquinone carries electrons from photosystem II to the cytochrome *bf* complex, within which electrons are transferred to plastocyanin and additional protons are pumped into the thylakoid lumen. Electron transport through photosystem II is thus coupled to establishment of a proton gradient, which drives the chemiosmotic synthesis of ATP.

From photosystem II, electrons are carried by plastocyanin (a peripheral membrane protein) to photosystem I, where the absorption of additional photons again generates high-energy electrons. Photosystem I, however, does not act as a proton pump; instead, it uses these high-energy electrons to reduce $NADP^+$ to NADPH. The reaction center chlorophyll of photosystem I transfers its excited electrons through a series of carriers to ferredoxin, a small protein on the stromal side of the thylakoid membrane, which then complexes with the enzyme **NADP reductase** and transfers electrons from

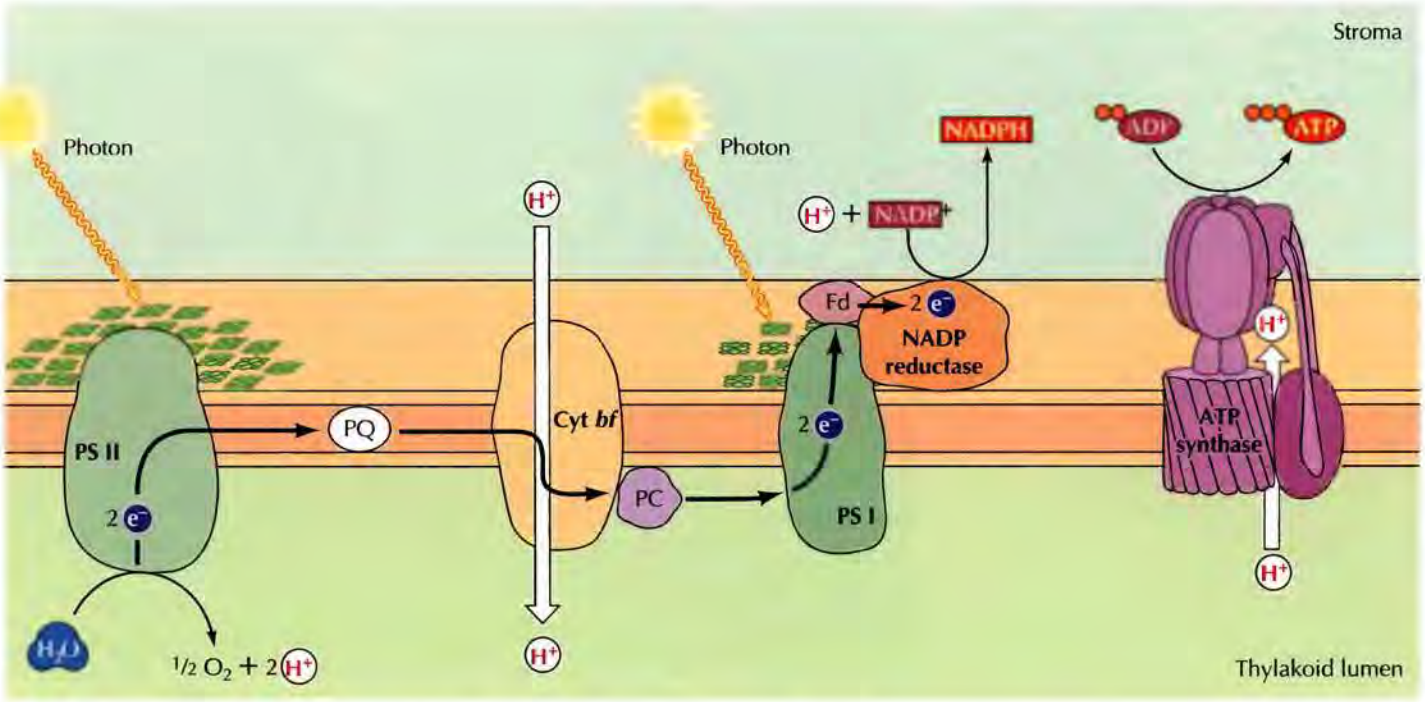


FIGURE 11.24 Electron transport and ATP synthesis during photosynthesis
Four protein complexes in the thylakoid membrane function in electron transport and the synthesis of ATP and NADPH. Photons are absorbed by complexes of pigment molecules associated with photosystems I and II (PS I and PS II). At photosystem II, energy derived from photon absorption is used to split a water molecule within the thylakoid lumen. Electrons are then carried by plastoquinone (PQ) to the cytochrome *bf* complex, where they are transferred to a lower energy state and protons are pumped into the thylakoid lumen. Electrons are then transferred to photosystem I by plastocyanin (PC). At photosystem I, energy derived from light absorption again generates high-energy electrons, which are transferred to ferredoxin (Fd) and used to reduce $NADP^+$ to NADPH in the stroma. ATP synthase then uses the energy stored in the proton gradient to convert ADP to ATP.

11.2 WEBSITE ANIMATION
The Light Reactions
During the light reactions of photosynthesis, energy absorbed from sunlight drives the synthesis of ATP and NADPH, coupled to the oxidation of H_2O to O_2.

ferrodoxin to $NADP^+$, generating NADPH. The passage of electrons through photosystems I and II thus generates both ATP and NADPH, which are used by the Calvin cycle enzymes in the chloroplast stroma to convert CO_2 to carbohydrates (see Figure 3.18).

Cyclic Electron Flow

A second electron transport pathway, called **cyclic electron flow**, produces ATP without the synthesis of NADPH, thereby supplying additional ATP for other metabolic processes. In cyclic electron flow, light energy harvested at photosystem I is used for ATP synthesis rather than NADPH synthesis (Figure 11.25). Instead of being transferred to $NADP^+$, high-energy electrons from photosystem I are transferred to the cytochrome *bf* complex. Electron transfer through the cytochrome *bf* complex is then coupled, as in photosystem II, to the establishment of a proton gradient across the thylakoid membrane. Plastocyanin then returns these electrons to photosystem I in a lower energy state, completing a cycle of electron transport in which light energy harvested at photosystem I is used to pump protons at the cytochrome *bf* complex. Electron transfer from photosystem I can thus generate either ATP or NADPH, depending on the metabolic needs of the cell.

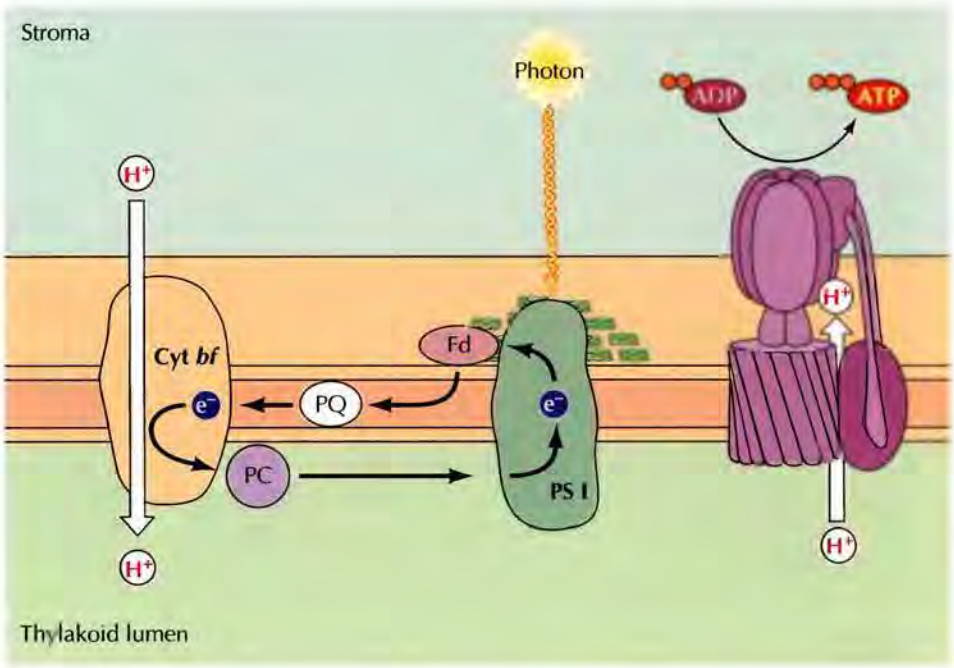


FIGURE 11.25 The pathway of cyclic electron flow Light energy absorbed at photosystem I (PS I) is used for ATP synthesis rather than NADPH synthesis. High-energy electrons generated by photon absorption are transferred to the cytochrome *bf* complex rather than to $NADP^+$. At the cytochrome *bf* complex, electrons are transferred to a lower energy state and protons are pumped into the thylakoid lumen. The electrons are then returned to photosystem I by plastocyanin (PC).

ATP Synthesis

The ATP synthase of the thylakoid membrane is similar to the mitochondrial enzyme. However, the energy stored in the proton gradient across the thylakoid membrane, in contrast to the inner mitochondrial membrane, is almost entirely chemical in nature. This is because the thylakoid membrane, although impermeable to protons, differs from the inner mitochondrial membrane in being permeable to other ions, particularly Mg^{2+} and Cl^-. The free passage of these ions neutralizes the voltage component of the proton gradient, so the energy derived from photosynthesis is conserved mainly as the difference in proton concentration (pH) across the thylakoid membrane. However, because the thylakoid lumen is a closed compartment, this difference in proton concentration can be quite large, corresponding to a differential of more than three pH units between the stroma and the thylakoid lumen. Because of the magnitude of this pH differential, the total free energy stored across the thylakoid membrane is similar to that stored across the inner mitochondrial membrane.

For each pair of electrons transported, two protons are transferred across the thylakoid membrane at photosystem II and two to four protons at the cytochrome *bf* complex. Since four protons are needed to drive the synthesis of one molecule of ATP, passage of each pair of electrons through photosystems I and II by noncyclic electron flow yields between 1 and 1.5 ATP molecules. Cyclic electron flow has a lower yield, corresponding to between 0.5 and 1 ATP molecules per pair of electrons.

Peroxisomes

Peroxisomes are small, single membrane-enclosed organelles (Figure 11.26) that contain enzymes involved in a variety of metabolic reactions, including several aspects of energy metabolism. Most human cells contain about 500 peroxisomes. Peroxisomes do not have their own genomes and all their pro-

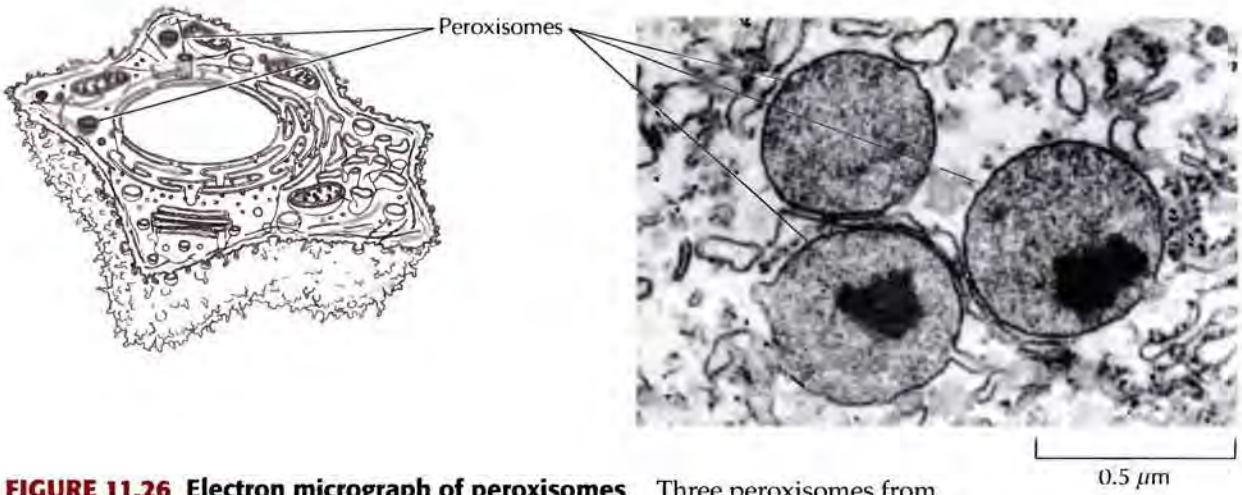


FIGURE 11.26 Electron micrograph of peroxisomes Three peroxisomes from rat liver are shown. Two contain dense regions, which are paracrystalline arrays of the enzyme urate oxidase. (Don Fawcett/Photo Researchers, Inc.)

teins, **peroxins** (Pex1, Pex2, etc.), are synthesized from the nuclear genome. Most peroxins are synthesized on free ribosomes and then imported into peroxisomes as completed polypeptide chains. Like mitochondria and chloroplasts, peroxisomes can replicate by division. However, unlike those organelles, peroxisomes can also be rapidly regenerated even if entirely lost to the cell. While many mitochondrial and plastid proteins resemble those of prokaryotes, reflecting their endosymbiotic origin, the peroxins resemble typical eukaryotic proteins.

Functions of Peroxisomes

Peroxisomes contain at least 50 different enzymes, which are involved in a variety of biochemical pathways in different types of cells. Peroxisomes originally were defined as organelles that carry out oxidation reactions leading to the production of hydrogen peroxide. Because hydrogen peroxide is harmful to the cell, peroxisomes also contain the enzyme **catalase**, which decomposes hydrogen peroxide either by converting it to water or by using it to oxidize another organic compound. A variety of substrates are broken down by such oxidative reactions in peroxisomes, including uric acid, amino acids, purines, methanol, and fatty acids. The oxidation of fatty acids (Figure 11.27) is a particularly important example, since it provides a major source of metabolic energy. In animal cells, fatty acids are oxidized in

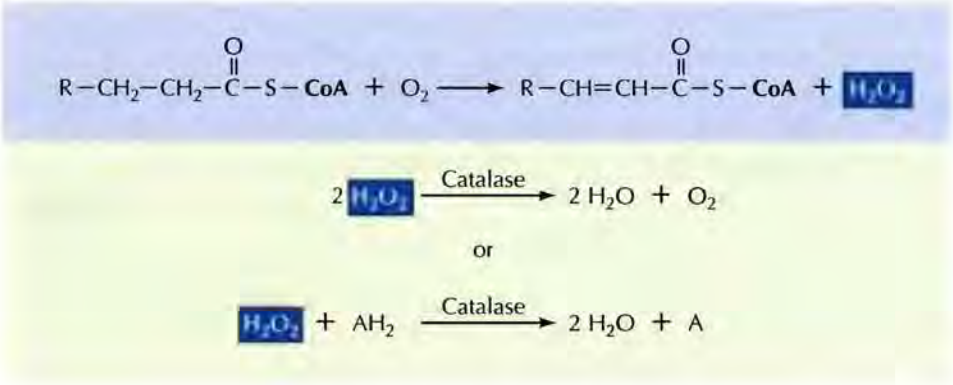


FIGURE 11.27 Fatty acid oxidation in peroxisomes The oxidation of a fatty acid is accompanied by the production of hydrogen peroxide (H_2O_2) from oxygen. The hydrogen peroxide is decomposed by catalase, either by conversion to water or by oxidation of another organic compound (designated AH_2).

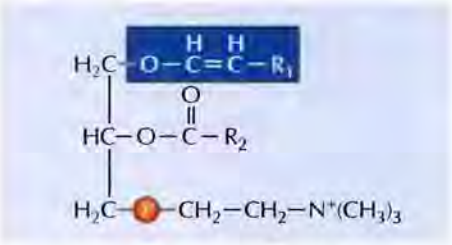


FIGURE 11.28 Structure of a plasmalogen The plasmalogen shown is analogous to phosphatidylcholine. However, one of the fatty acid chains is joined to glycerol by an ether, rather than an ester, bond.

both peroxisomes and mitochondria, but in yeasts and plants, fatty acid oxidation is restricted to peroxisomes.

In addition to providing a compartment for oxidation reactions, peroxisomes are involved in biosynthesis of lipids and the amino acid, lysine. In animal cells, cholesterol and dolichol are synthesized in peroxisomes as well as in the ER. In the liver, peroxisomes are also involved in the synthesis of bile acids, which are derived from cholesterol. In addition, peroxisomes contain enzymes required for the synthesis of **plasmalogens**—a family of phospholipids in which one of the hydrocarbon chains is joined to glycerol by an ether bond rather than an ester bond (Figure 11.28). Plasmalogens are important membrane components in some tissues, particularly heart and brain, although they are absent in others. Peroxisomes carry out different biochemical reactions in different tissues. However, it is currently unknown whether there are subpopulations of peroxisomes that specialize in one or a limited number of processes within a cell.

Peroxisomes play two particularly important roles in plants. First, peroxisomes in seeds are responsible for the conversion of stored fatty acids to carbohydrates, which is critical to providing energy and raw materials for growth of the germinating plant. This occurs via a series of reactions termed the **glyoxylate cycle**, which is a variant of the citric acid cycle (Figure 11.29). The peroxisomes in which this takes place are sometimes called **glyoxysomes**.

Second, peroxisomes in leaves are involved in **photorespiration**, which serves to metabolize a side product formed during photosynthesis (Figure 11.30). CO_2 is converted to carbohydrates during photosynthesis via a series

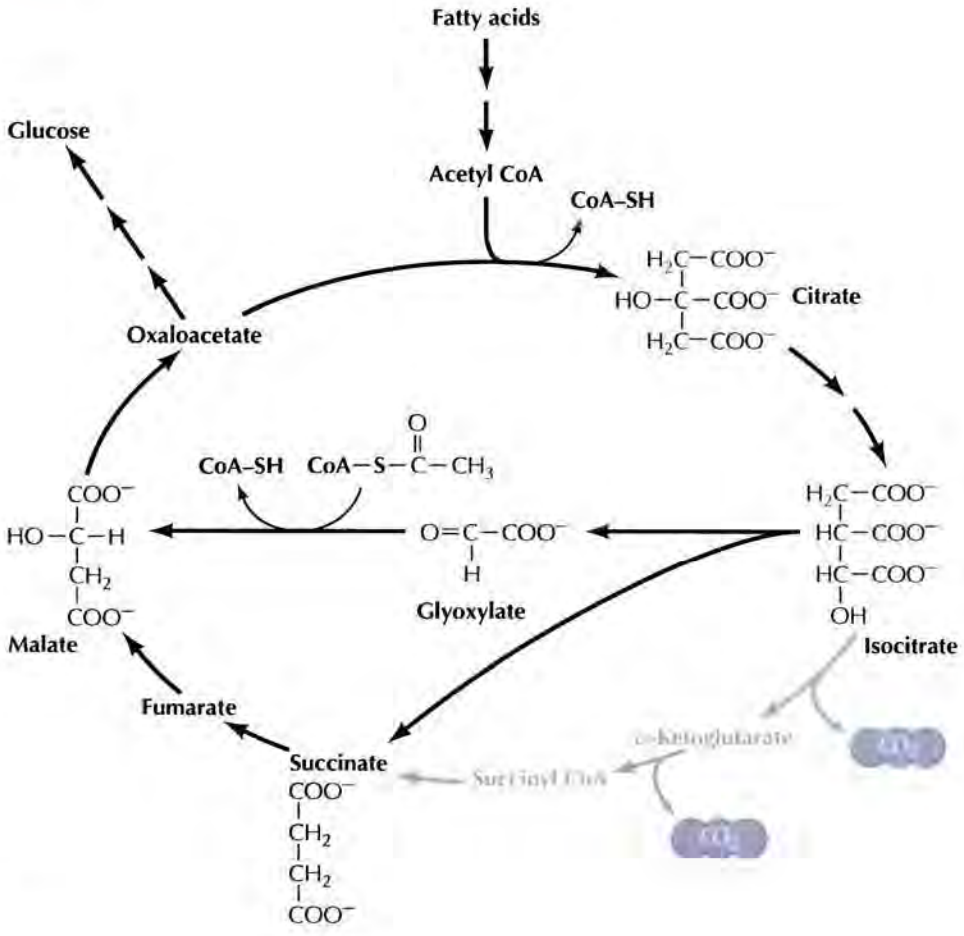


FIGURE 11.29 The glyoxylate cycle Plants are capable of synthesizing carbohydrates from fatty acids via the glyoxylate cycle, which is a variant of the citric acid cycle (see Figure 3.13). As in the citric acid cycle, acetyl CoA combines with oxaloacetate to form citrate, which is converted to isocitrate. However, instead of being degraded to CO_2 and α-ketoglutarate (gray arrows), isocitrate is converted to succinate and glyoxylate. Glyoxylate then reacts with another molecule of acetyl CoA to yield malate, which is converted to oxaloacetate and used for glucose synthesis.

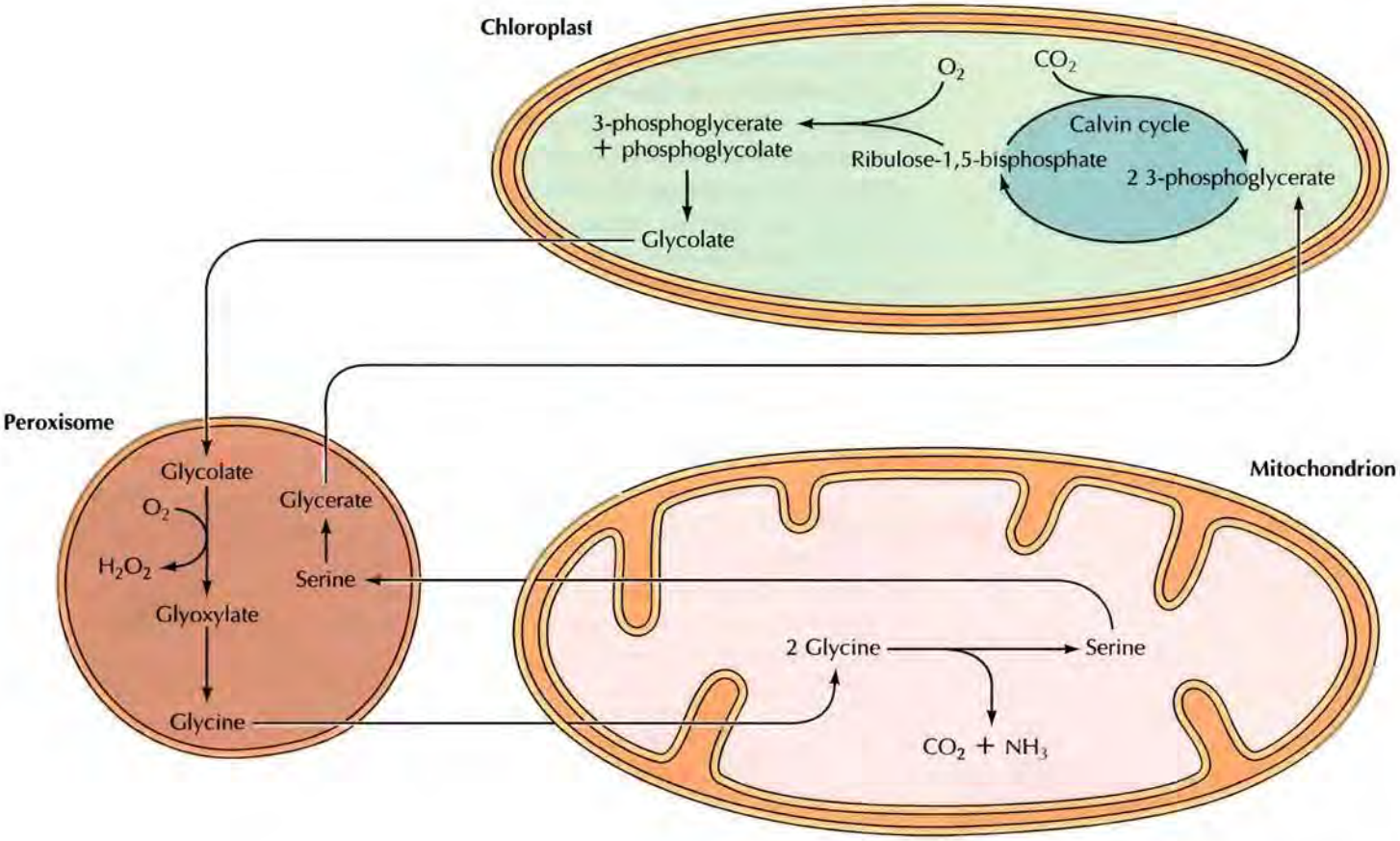


FIGURE 11.30 Role of peroxisomes in photorespiration During photosynthesis CO_2 is converted to carbohydrates by the Calvin cycle, which initiates with the addition of CO_2 to the five-carbon sugar ribulose-1,5-bisphosphate. However, the enzyme involved sometimes catalyzes the addition of O_2 instead, resulting in production of the two-carbon compound phosphoglycolate. Phosphoglycolate is converted to glycolate, which is then transferred to peroxisomes, where it is oxidized and converted to glycine. Glycine is then transferred to mitochondria and converted to serine. The serine is returned to peroxisomes and converted to glycerate, which is transferred back to chloroplasts.

of reactions called the Calvin cycle (see Figure 3.18). The first step is the addition of CO_2 to the five-carbon sugar ribulose-1,5-bisphosphate, yielding two molecules of 3-phosphoglycerate (three carbons each). However, the enzyme involved (ribulose bisphosphate carboxylase or rubisco) sometimes catalyzes the addition of O_2 instead of CO_2, producing one molecule of 3-phosphoglycerate and one molecule of phosphoglycolate (two carbons). This is a side reaction, and phosphoglycolate is not a useful metabolite. It is first converted to glycolate and then transferred to peroxisomes, where it is oxidized and converted to glycine. Glycine is then transferred to mitochondria, where two molecules of glycine are converted to one molecule of serine, with the loss of CO_2 and NH_3. The serine is then returned to peroxisomes, where it is converted to glycerate. Finally, the glycerate is transferred back to chloroplasts, where it reenters the Calvin cycle. Photorespiration does not appear to be beneficial for the plant, since it is essentially the reverse of photosynthesis—O_2 is consumed and CO_2 is released without any gain of ATP. However, the occasional utilization of O_2 in place of CO_2 appears to be an inherent property of rubisco, so photorespiration is a general accompaniment of photosynthesis. Peroxisomes thus play an important role by allowing most of the carbon in glycolate to be recovered and utilized.

Peroxisome Assembly

Although most peroxins are synthesized on free cytosolic ribosomes and then imported to peroxisomes, recent experiments indicate that peroxisome assembly begins on the rough ER, where two peroxins, Pex3 and Pex19, ini-

tially localize (Figure 11.31). Pex3 is an integral transmembrane protein while Pex19 is a farnesylated protein found largely in the cytosol. Pex3 recruits Pex19 to the ER membrane, where their interaction causes Pex3/Pex19-containing vesicles to bud off the ER. These vesicles may then fuse either with pre-existing peroxisomes or with one another to form new peroxisomes.

Pex3, Pex19, and other peroxisomal membrane proteins then act as receptors for import of the other peroxins, which are translated on free cytosolic ribosomes and then transported into peroxisomes as completed and folded polypeptides. They are targeted to the interior of peroxisomes by at least two pathways, which are conserved from yeasts to humans. Most peroxins are targeted by the simple amino acid sequence Ser-Lys-Leu

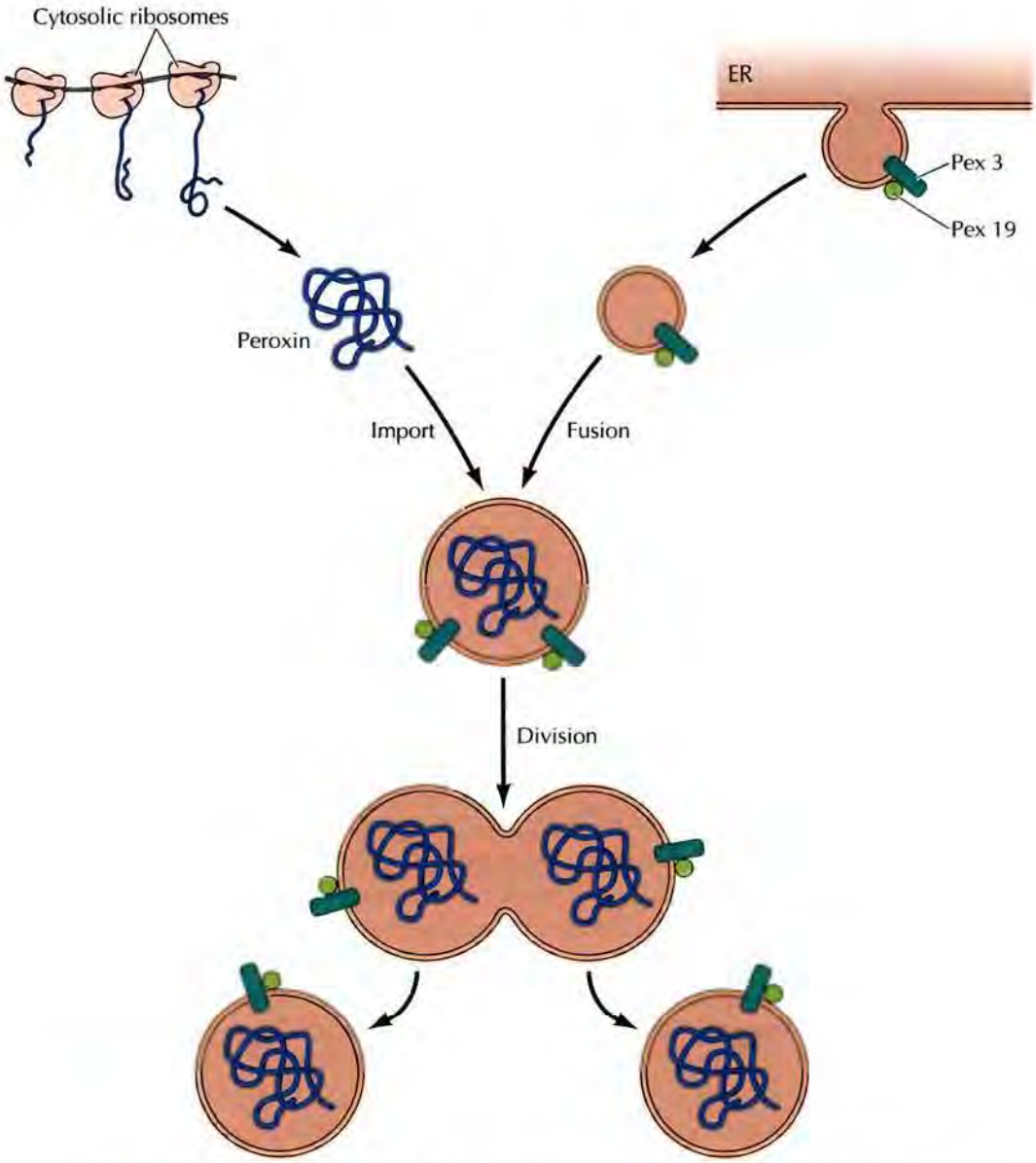


FIGURE 11.31 Assembly of peroxisomes Initiation of peroxisome assembly begins in the rough ER when the transmembrane protein peroxin 3 (Pex3) recruits the soluble farnesylated protein peroxin 19 (Pex19) and initiates budding of a nascent peroxisome. These nascent peroxisomes fuse either with each other or with existing peroxisomes. Additional peroxins are synthesized on free cytosolic ribosomes and imported as completed polypeptide chains to form functional peroxisomes, which grow larger and divide.

at their carboxy terminus (peroxisome targeting signal 1, or PTS1). A small number of proteins are targeted by a sequence of nine amino acids at their amino terminus (peroxisome targeting signal 2, or PTS2). PTS1 and PTS2 are recognized by distinct cytosolic receptors and then passed through a poorly understood channel in the peroxisomal membrane into the matrix. The receptors are subsequently retrieved from the peroxisome and recycled. Unlike the translocation of polypeptide chains across the membranes of the endoplasmic reticulum, mitochondria, and chloroplasts, the targeting signals are usually not cleaved during the import of proteins into peroxisomes and the mechanism of translocation is not known.

Protein import, together with the continuing addition of lipids from the rough ER, results in peroxisome growth, and new peroxisomes can be formed by division of old ones. In addition, peroxisomes undergo a complex maturation process that involves the import of different classes of proteins from the cytosol at different times. As a result, the enzyme content, and therefore the metabolic activities, of peroxisomes may change as they mature.

Interestingly, some components of peroxisome import pathways have been identified not only as mutants of yeasts but also as mutations associated with serious human diseases involving disorders of peroxisomes. In some such diseases, only a single peroxisomal enzyme is deficient. However, in other diseases resulting from defects in peroxisome function, multiple peroxisomal enzymes fail to be imported to peroxisomes and instead are localized in the cytosol. The latter group of diseases results from deficiencies in the PTS1 or PTS2 pathways responsible for peroxisomal protein import. The prototypical example is Zellweger syndrome, which is lethal within the first ten years of life. Zellweger syndrome can result from mutations in at least ten different genes affecting peroxisomal protein import, one of which has been identified as the gene encoding the receptor for the peroxisome targeting signal PTS1.

COMPANION WEBSITE

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SUMMARY

MITOCHONDRIA

Organization and Function of Mitochondria: Mitochondria, which play a critical role in the generation of metabolic energy, are surrounded by a double-membrane system. The matrix contains the enzymes of the citric acid cycle; the inner membrane contains protein complexes involved in electron transport and oxidative phosphorylation. In contrast to the inner membrane, the outer membrane is freely permeable to small molecules.

The Genetic System of Mitochondria: Mitochondria contain their own genomes, which encode rRNAs, tRNAs, and some of the proteins that are involved in oxidative phosphorylation.

Protein Import and Mitochondrial Assembly: Most mitochondrial proteins are encoded by the nuclear genome. These proteins are translated on free ribosomes and imported into mitochondria as completed polypeptide chains. Positively charged presequences target proteins for import to the mitochondrial matrix. Phospholipids are carried to mitochondria from the endoplasmic reticulum by phospholipid transfer proteins.

KEY TERMS

mitochondria, crista, matrix, porin

endosymbiosis

presequence, Tom complex, Tim complex, matrix processing peptidase (MPP), phospholipid transfer protein, cardiolipin

KEY TERMS

SUMMARY

THE MECHANISM OF OXIDATIVE PHOSPHORYLATION

oxidative phosphorylation, electron transport chain, coenzyme Q, ubiquinone, cytochrome *c*, cytochrome oxidase

The Electron Transport Chain: Most of the energy derived from oxidative metabolism comes from the transfer of electrons from NADH and $FADH_2$ to O_2. In order to harvest this energy in usable form, electrons are transferred through a series of carriers organized into four protein complexes in the inner mitochondrial membrane.

chemiosmotic coupling, electrochemical gradient, ATP synthase

Chemiosmotic Coupling: The energy-yielding reactions of electron transport are coupled to the generation of a proton gradient across the inner mitochondrial membrane. The potential energy stored in this gradient is harvested by a fifth protein complex, ATP synthase, which couples ATP synthesis to the energetically favorable return of protons to the mitochondrial matrix.

Transport of Metabolites across the Inner Membrane: In addition to driving ATP synthesis, potential energy stored in the proton gradient drives the transport of ATP, ADP, and other metabolites into and out of mitochondria.

CHLOROPLASTS AND OTHER PLASTIDS

chloroplast, thylakoid membrane, stroma

The Structure and Function of Chloroplasts: Chloroplasts are large organelles that function in photosynthesis and a variety of other metabolic activities. Like mitochondria, chloroplasts are bounded by a double-membrane envelope. In addition, chloroplasts have an internal thylakoid membrane, which is the site of electron transport and the chemiosmotic generation of ATP.

The Chloroplast Genome: Chloroplast genomes contain more than 100 genes, including genes encoding rRNAs, tRNAs, some ribosomal proteins, and some proteins involved in photosynthesis.

transit peptide, guidance complex, Toc complex, Tic complex, stromal processing peptidase (SPP)

Import and Sorting of Chloroplast Proteins: Most chloroplast proteins are synthesized on free ribosomes in the cytosol and targeted for import to chloroplasts by amino-terminal transit peptides. Most proteins incorporated into the thylakoid lumen are first imported into the chloroplast stroma and then targeted for transport across the thylakoid membrane by several different pathways.

plastid, chromoplast, leucoplast, amyloplast, elaioplast, proplastid, etioplast

Other Plastids: The chloroplast is only one member of a family of related organelles, all of which contain the same genome. Other plastids serve to store energy sources, such as starch and lipids, and function in other aspects of plant metabolism.

PHOTOSYNTHESIS

chlorophyll, photocenter, photosystem I, photosystem II, cytochrome *bf* complex, NADP reductase

Electron Flow through Photosystems I and II: During photosynthesis, energy from sunlight is harvested and converted to usable forms of potential chemical energy. Absorption of light by chlorophylls excites electrons to a higher energy state. These high-energy electrons are then transferred through a series of carriers organized into two photosystems and the cytochrome *bf* complex in the thylakoid membrane. The sequential flow of electrons through both photosystems is coupled to the syn-

SUMMARY

thesis of ATP at photosystem II and the reduction of $NADP^+$ to NADPH at photosystem I. Both ATP and NADPH are then used in the synthesis of carbohydrates from CO_2, which takes place in the chloroplast stroma.

Cyclic Electron Flow: The alternative pathway of cyclic electron flow allows light energy harvested at photosystem I to be converted to ATP, rather than NADPH.

ATP Synthesis: The chemiosmotic synthesis of ATP is driven by a proton gradient across the thylakoid membrane.

PEROXISOMES

Functions of Peroxisomes: Peroxisomes are small organelles, bounded by a single membrane, that contain enzymes involved in a variety of metabolic reactions, including fatty acid oxidation, the glyoxylate cycle, and photorespiration.

Peroxisome Assembly: Peroxisome assembly begins on the ER with the formation of specific vesicles. However, most peroxisomal proteins are synthesized on free ribosomes in the cytosol and imported to peroxisomes as complete polypeptide chains. At least two types of signals target proteins to the interior of peroxisomes, but the mechanism of protein import is not well understood.

KEY TERMS

cyclic electron flow

peroxisome, peroxin, catalase, plasmalogen, glyoxylate cycle, glyoxysome, photorespiration

Questions

1. What two properties of the mitochondrial inner membrane allow it to have unusually high metabolic activity?

2. According to standard "wobble" rules, protein synthesis requires a minimum of 30 different tRNAs. How do human mitochondria manage to translate mRNAs into proteins with only 22 tRNA species?

3. Assume that the electric potential across the inner mitochondrial membrane is dissipated, so the electrochemical gradient is composed solely of a proton concentration gradient corresponding to one pH unit. Calculate the free energy stored in this gradient. For your calculation, use $R = 1.98 \times 10^{-3}$ kcal/mol/deg, T = 298°K (25°C), and note that $\ln(x) = 2.3 \log_{10}(x)$.

4. What roles do molecular chaperones play in mitochondrial protein import?

5. Cytochrome *b2* is synthesized in the cytosol and has a second, hydrophobic signal sequence behind the usual positively charged mitochondrial presequence. Suggest a pathway that will get it to its final location in the mitochondrial intermembrane space.

6. What are the roles of coenzyme Q and cytochome *c* in the electron transport chain?

7. ATP synthase consists of two multisubunit complexes, F_0 and F_1. Where is each one located in mitochondria and in chloroplasts, and what is the function of each?

8. Why do the transit peptides of chloroplast proteins, in contrast to the presequences of mitochondrial proteins, not need to be positively charged?

9. How many high-energy electrons are required to drive the synthesis of one molecule of glucose during photosynthesis, coupled to the formation of six molecules of O_2? How many molecules of ATP and NADPH are generated by passage of these electrons through photosystems I and II?

10. What fraction of the carbon atoms converted to glycolate during photorespiration is salvaged by peroxisomes?

11. How are proteins targeted to peroxisomes?

12. Why is the energy stored across the thylakoid membrane almost entirely chemical?

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Peroxisomes

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CHAPTER 12

The Cytoskeleton and Cell Movement

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THE MEMBRANE-ENCLOSED ORGANELLES discussed in the preceding chapters constitute one level of the organizational substructure of eukaryotic cells. A further level of organization is provided by the cytoskeleton, which consists of a network of protein filaments extending throughout the cytoplasm of all eukaryotic cells. The cytoskeleton provides a structural framework for the cell, serving as a scaffold that determines cell shape, the positions of organelles, and the general organization of the cytoplasm. In addition to playing this structural role, the cytoskeleton is responsible for cell movements. These include not only the movements of entire cells, but also the internal transport of organelles and other structures (such as mitotic chromosomes) through the cytoplasm. Importantly, the cytoskeleton is much less rigid and permanent than its name implies. Rather, it is a dynamic structure that is continually reorganized as cells move and change shape—for example, during cell division.

The cytoskeleton is composed of three principal types of protein filaments: actin filaments, intermediate filaments, and microtubules, which are held together and linked to subcellular organelles and the plasma membrane by a variety of accessory proteins. This chapter discusses the structure and organization of each of these three major components of the cytoskeleton as well as their roles in cell motility, organelle transport, cell division, and other types of cell movements.

Structure and Organization of Actin Filaments

The major cytoskeletal protein of most cells is **actin**, which polymerizes to form actin filaments—thin, flexible fibers approximately 7 nm in diameter and up to several micrometers in length (Figure 12.1). Within the cell, actin filaments (also called **microfilaments**) are organized into higher-order structures, forming bundles or three-dimensional networks with the prop-

FIGURE 12.1 Actin filaments
Electron micrograph of actin filaments. (Courtesy of Roger Craig, University of Massachusetts Medical Center.)

12.1 WEBSITE ANIMATION
Assembly of an Actin Filament Actin monomers polymerize to form actin filaments, a process that is reversible and is regulated within the cell by actin-binding proteins.

erties of semisolid gels. The assembly and disassembly of actin filaments, their cross-linking into bundles and networks, and their association with other cell structures (such as the plasma membrane) are regulated by a variety of actin-binding proteins, which are critical components of the actin cytoskeleton. Actin filaments are particularly abundant beneath the plasma membrane where they form a network that provides mechanical support, determines cell shape, and allows movement of the cell surface, thereby enabling cells to migrate, engulf particles, and divide.

Assembly and Disassembly of Actin Filaments

Actin was first isolated from muscle cells, in which it constitutes approximately 20% of total cell protein, in 1942. Although actin was initially thought to be uniquely involved in muscle contraction, it is now known to be an extremely abundant protein (typically 5 to 10% of total protein) in all types of eukaryotic cells. Yeasts have only a single actin gene, but higher eukaryotes have several distinct types of actin, which are encoded by different members of the actin gene family. Mammals, for example, have six distinct actin genes: Four are expressed in different types of muscle and two are expressed in nonmuscle cells. All of the actins, however, are very similar in amino acid sequence and have been highly conserved throughout the evolution of eukaryotes. Yeast actin, for example, is 90% identical in amino acid sequence to the actins of mammalian cells. The prokaryotic ancestor of actin is a protein called MreB, which gives rod-shaped bacteria their structure.

The three-dimensional structures of both individual actin molecules and actin filaments were determined in 1990 by Kenneth Holmes, Wolfgang Kabsch, and their colleagues. Individual actin molecules are globular proteins of 375 amino acids (43 kd). Each actin monomer (**globular [G] actin**) has tight binding sites that mediate head-to-tail interactions with two other actin monomers, so actin monomers polymerize to form filaments (**filamentous [F] actin**) (Figure 12.2). Each monomer is rotated by 166° in the filaments, which therefore have the appearance of a double-stranded helix. Because all the actin monomers are oriented in the same direction, actin filaments have a distinct polarity and their ends (called barbed or plus ends, and pointed or minus ends) are distinguishable from one another. This polarity of actin filaments is important both in their assembly and in establishing a specific direction of myosin movement relative to actin, as discussed later in the chapter.

While actin monomers and filaments are tightly controlled within cells, much of their behavior is a property of the actin monomers and filaments themselves. In solutions of low ionic strength *in vitro*, actin filaments depolymerize to monomers. Actin then polymerizes spontaneously if the ionic strength is increased to physiological levels. The first step in actin polymerization (called nucleation) is the formation of a small aggregate consisting of three actin monomers. Actin filaments are then able to grow by the reversible addition of monomers to both ends, but one end (the barbed end) elongates five to ten times faster than the pointed end. The actin monomers also bind ATP, which is hydrolyzed to ADP following filament assembly. Although ATP is not required for polymerization, actin monomers to which ATP is bound polymerize more readily than those to which ADP is bound. As discussed below, ATP binding and hydrolysis play a key role in regulating the assembly and dynamic behavior of actin filaments.

Because actin polymerization is reversible, filaments can depolymerize by the dissociation of actin subunits, allowing actin filaments to be broken down when necessary (Figure 12.3). *In vitro*, an equilibrium exists between

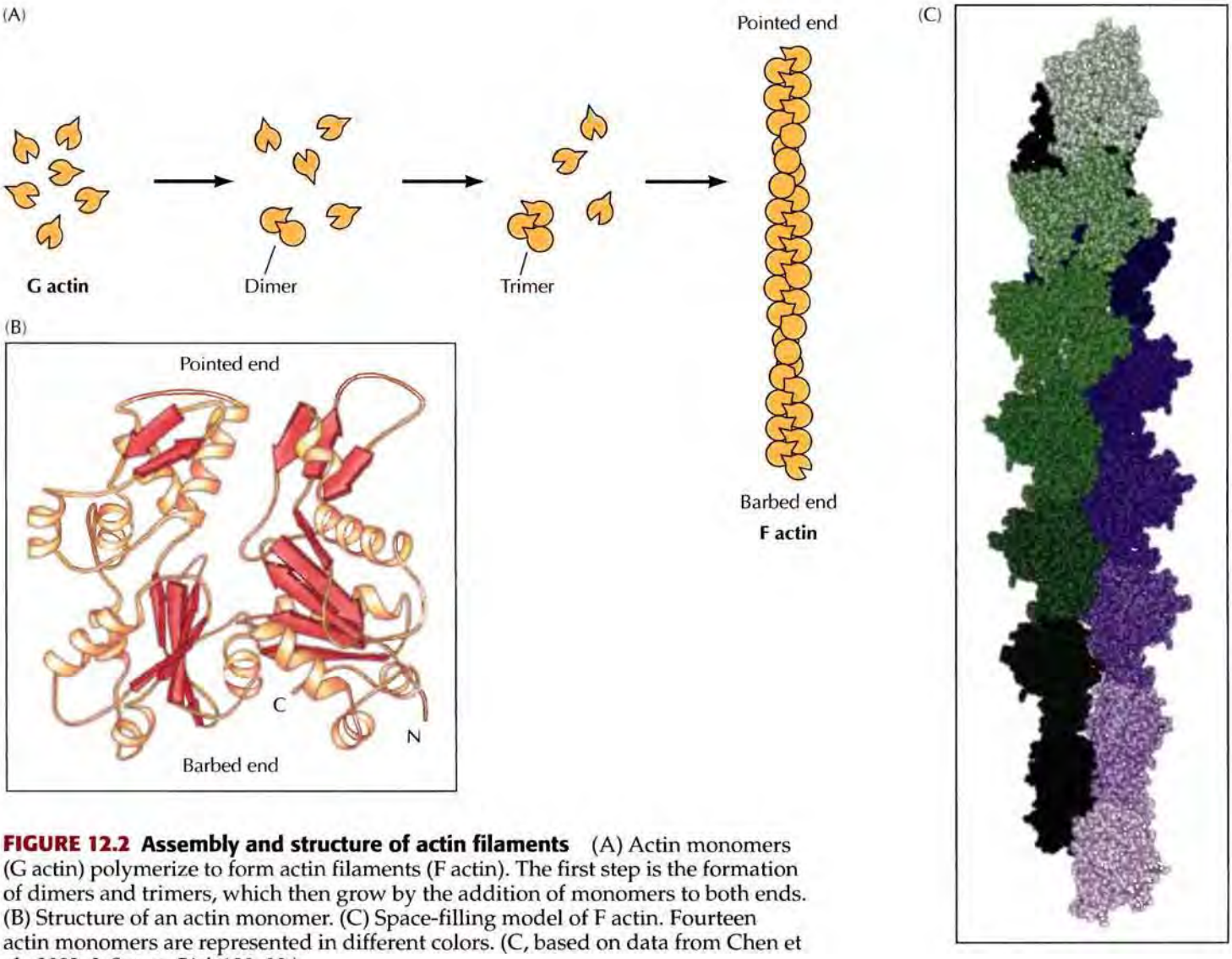


FIGURE 12.2 Assembly and structure of actin filaments (A) Actin monomers (G actin) polymerize to form actin filaments (F actin). The first step is the formation of dimers and trimers, which then grow by the addition of monomers to both ends. (B) Structure of an actin monomer. (C) Space-filling model of F actin. Fourteen actin monomers are represented in different colors. (C, based on data from Chen et al., 2002. *J. Struct. Biol.* 138: 92.)

actin monomers and filaments. The rate at which actin monomers are incorporated into filaments is proportional to their concentration, so there is a critical concentration of actin monomers at which the rate of their polymerization into filaments equals the rate of dissociation. At this critical concentration, monomers and filaments are in apparent equilibrium.

As noted earlier, the two ends of an actin filament grow at different rates, with monomers being added to the fast-growing barbed end five to ten times faster than to the slow-growing pointed end. Because ATP-actin dissociates less readily than ADP-actin, this results in a difference in the criti-

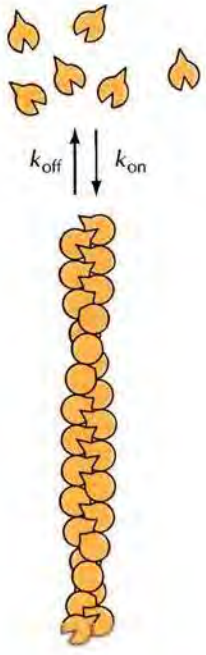


FIGURE 12.3 Reversible polymerization of actin monomers Actin polymerization is a reversible process in which monomers both associate with and dissociate from the ends of actin filaments. The rate of subunit dissociation (k_{off}) is independent of monomer concentration, while the rate of subunit association is proportional to the concentration of free monomers and given by $C \times k_{on}$ (C = concentration of free monomers). An apparent equilibrium is reached at the critical concentration of monomers (C_c), where $k_{off} = C_c \times k_{on}$.

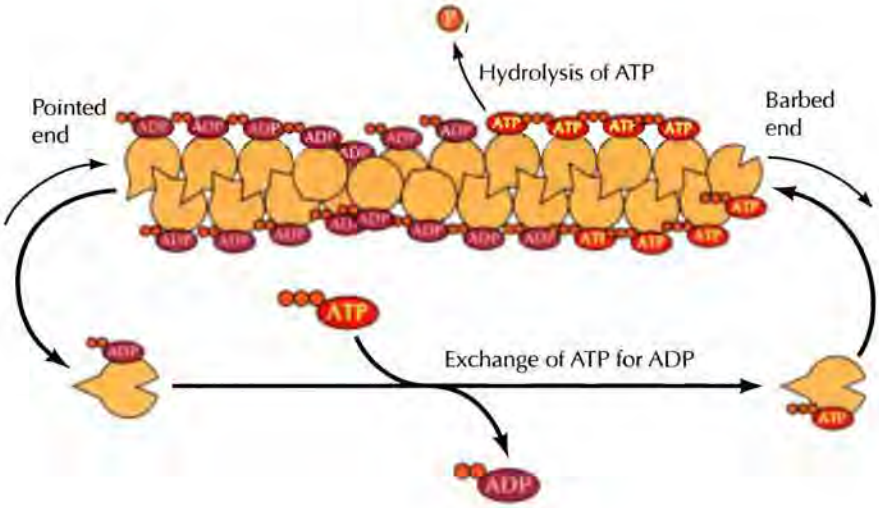


FIGURE 12.4 Treadmilling and the role of ATP in microfilament polymerization The pointed ends of actin filaments grow less rapidly than the barbed ends. This difference in growth rate is reflected in a difference in the critical concentration for addition of monomers to the two ends of the filament. Actin bound to ATP associates with the rapidly growing barbed ends, and the ATP bound to actin is then hydrolyzed to ADP. Because ADP-actin dissociates from filaments more readily than ATP-actin, the critical concentration of actin monomers is higher for addition to the pointed end than to the barbed end of actin filaments. Treadmilling takes place at monomer concentrations intermediate between the critical concentrations for the barbed and pointed ends. Under these conditions, there is a net dissociation of monomers (bound to ADP) from the pointed end, balanced by the addition of monomers (bound to ATP) to the barbed end.

cal concentration of monomers needed for polymerization at the two ends. This difference can result in the phenomenon known as **treadmilling**, which illustrates the dynamic behavior of actin filaments (Figure 12.4). For the *in vitro* system to be at an overall steady state, the concentration of free actin monomers must be intermediate between the critical concentrations required for polymerization at the barbed and pointed ends of the filaments. Under these conditions, there is a net loss of monomers from the pointed end, which is balanced by a net addition to the barbed end. Treadmilling requires ATP, with ATP-actin polymerizing at the barbed end of filaments while ADP-actin dissociates from the pointed end. As discussed later in this chapter, treadmilling is important in the formation of cell processes and in cell movement.

It is noteworthy that several drugs useful in cell biology act by binding to actin and affecting its polymerization. For example, the **cytochalasins** bind to the barbed ends of actin filaments and block their elongation. This results in changes in cell shape as well as inhibition of some types of cell movements (e.g., cell division following mitosis), indicating that actin polymerization is required for these processes. Another drug, **phalloidin**, binds tightly to actin filaments and prevents their dissociation into individual actin molecules. Phalloidin labeled with a fluorescent dye is frequently used to visualize actin filaments by fluorescence microscopy.

■ **Phalloidin is derived from the deathcap mushroom (*Amanita phalloides*). This mushroom also produces α-amanitin, a specific inhibitor of RNA polymerase II. The toxins produced by the deathcap and related mushrooms primarily attack the liver and kidney and can cause death if left untreated.**

Within the cell, the concentration of actin filaments and monomers is far from equilibrium. In parts of the cell, the turnover of actin filaments can be 100 times faster than it is *in vitro* while in other parts of the cell, filaments can be stabilized against treadmilling. The assembly and disassembly of actin filaments within the cell is regulated by a diverse group of **actin-binding proteins** (Table 12.1). These proteins can regulate the formation and stability of the actin cytoskeleton at several different levels (Figure 12.5). Some of these proteins bind to actin filaments, either stabilizing them or cross-linking them to one another. Other actin-binding proteins stabilize filaments by capping their ends and preventing the dissociation of actin monomers. Conversely, some proteins act to disassemble actin filaments either by severing them or stimulating their depolymerization. Finally, some actin-binding proteins bind monomeric actin and control its assembly into filaments by regulating the exchange of ATP for ADP.

The initial and rate-limiting step in the formation of actin filaments is nucleation, which requires monomers to interact correctly. Two types of

TABLE 12.1 Actin-binding Proteins

Cellular Role	Representative Proteins
Filament initiation and polymerization	Arp2/3, formin
Filament stabilization	Nebulin, tropomyosin
Filament cross-linking	α-actinin, filamin, fimbrin, villin
End-capping	CapZ, tropomodulin
Filament severing/depolymerization	ADF/cofilin, gelsolin, thymosin
Monomer binding	Profilin, twinfilin
Actin filament linkage to other proteins	α-catenin, dystrophin, spectrin, talin, vinculin

proteins, **formin** and the **Arp2/3 complex** (*a*ctin-*r*elated *p*rotein), determine where filaments are formed within the cell by nucleating actin filaments. Formins are a family of large (140–200 kd) barbed-end tracking proteins that both nucleate the initial actin monomers and then move along the growing filament, adding new monomers to the barbed end (Figure 12.6). It is thought that formins nucleate long unbranched actin filaments (see Figure 12.1) that make up stress fibers and the thin filaments of muscle cells (discussed later in this chapter). Many of these filaments are relatively stable due to filament-stabilizing proteins, such as members of the **tropomyosin** family. Tropomyosins are 30–36 kd fibrous proteins that bind lengthwise along the groove of actin filaments.

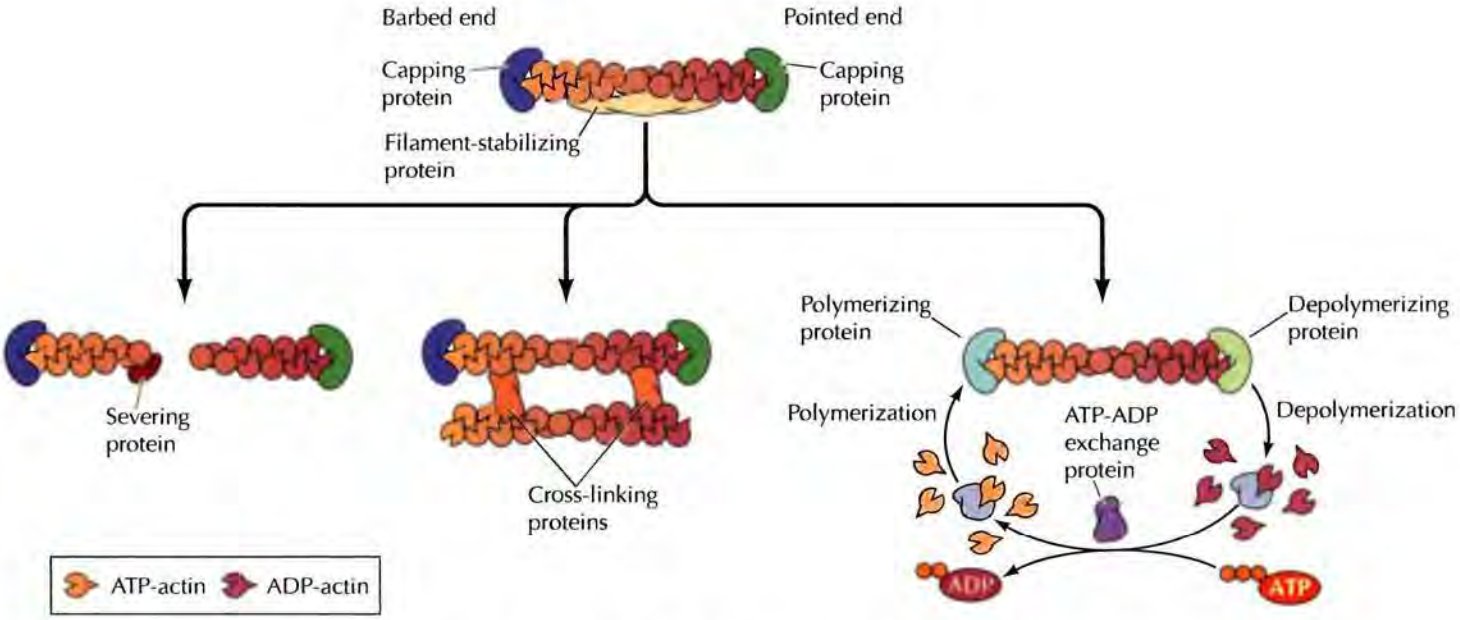


FIGURE 12.5 Regulation of actin filament organization by actin-binding proteins Actin-binding proteins play several roles in the dynamic behavior of actin filaments. Actin filaments can be stabilized by filament-stabilizing proteins that bind along their length. Both the barbed and pointed ends can also be capped and the filaments themselves can be cross-linked. Intact filaments can also be split by filament-severing proteins. The equilibrium between actin monomers and filaments can be regulated by filament-depolymerizing proteins, filament-polymerizing proteins, or proteins that modulate the exchange of ATP for ADP on actin monomers.

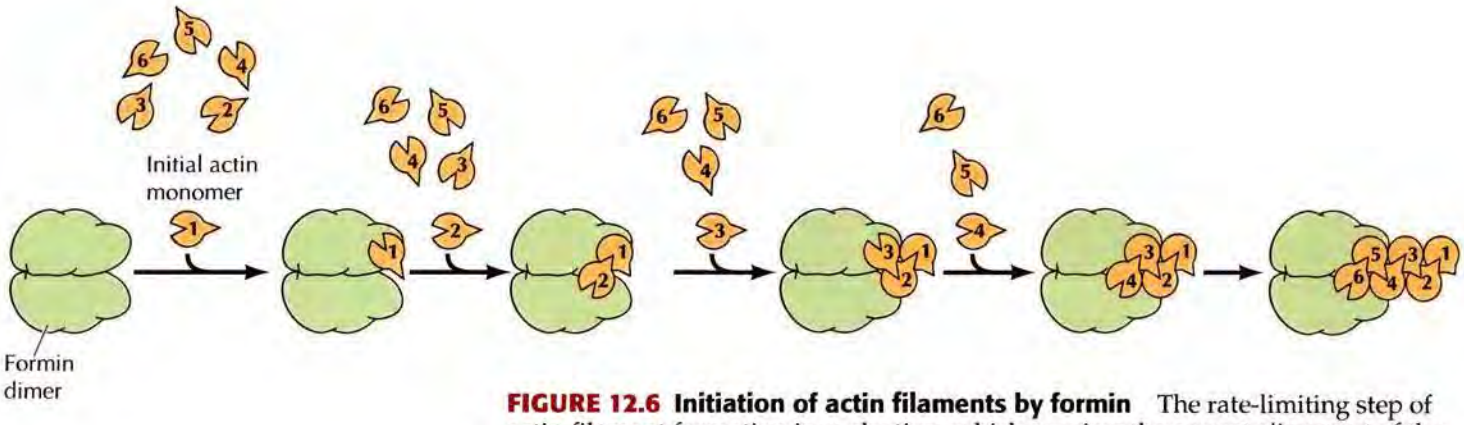


FIGURE 12.6 Initiation of actin filaments by formin The rate-limiting step of actin filament formation is nucleation, which requires the correct alignment of the first three actin monomers to allow subsequent polymerization. Within the cell, nucleation is facilitated by the actin-binding protein, formin. Each subunit of a formin dimer binds an actin monomer. The actin monomers are held in the correct configuration to allow binding of the third monomer, followed by rapid polymerization during which formin continues to track the barbed end.

In contrast, at the leading edge of cell processes or moving cells, actin filaments both actively treadmill and branch extensively. The densely packed and highly branched actin filaments in these areas are nucleated by the Arp2/3 complex, which binds actin/ATP near the barbed ends of microfilaments (Figure 12.7). The Arp2/3 complex contains seven proteins, two of which are similar to actin. The complex by itself has very low activity but is activated by several proteins that bind and activate it. Once activated, the Arp2/3 complex binds to the side of an existing actin filament near the barbed end and forms a new branch.

Another type of actin binding protein remodels or modifies existing filaments. One family of proteins responsible for actin filament remodeling

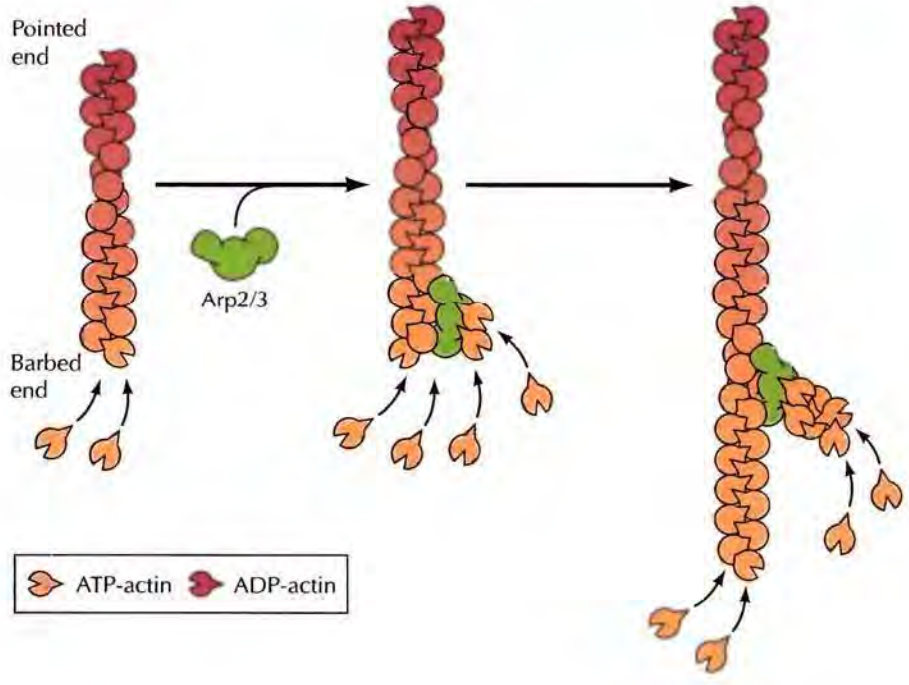


FIGURE 12.7 Initiation of actin filament branches The Arp2/3 complex binds to actin filaments near their barbed ends and initiates the formation of branches.

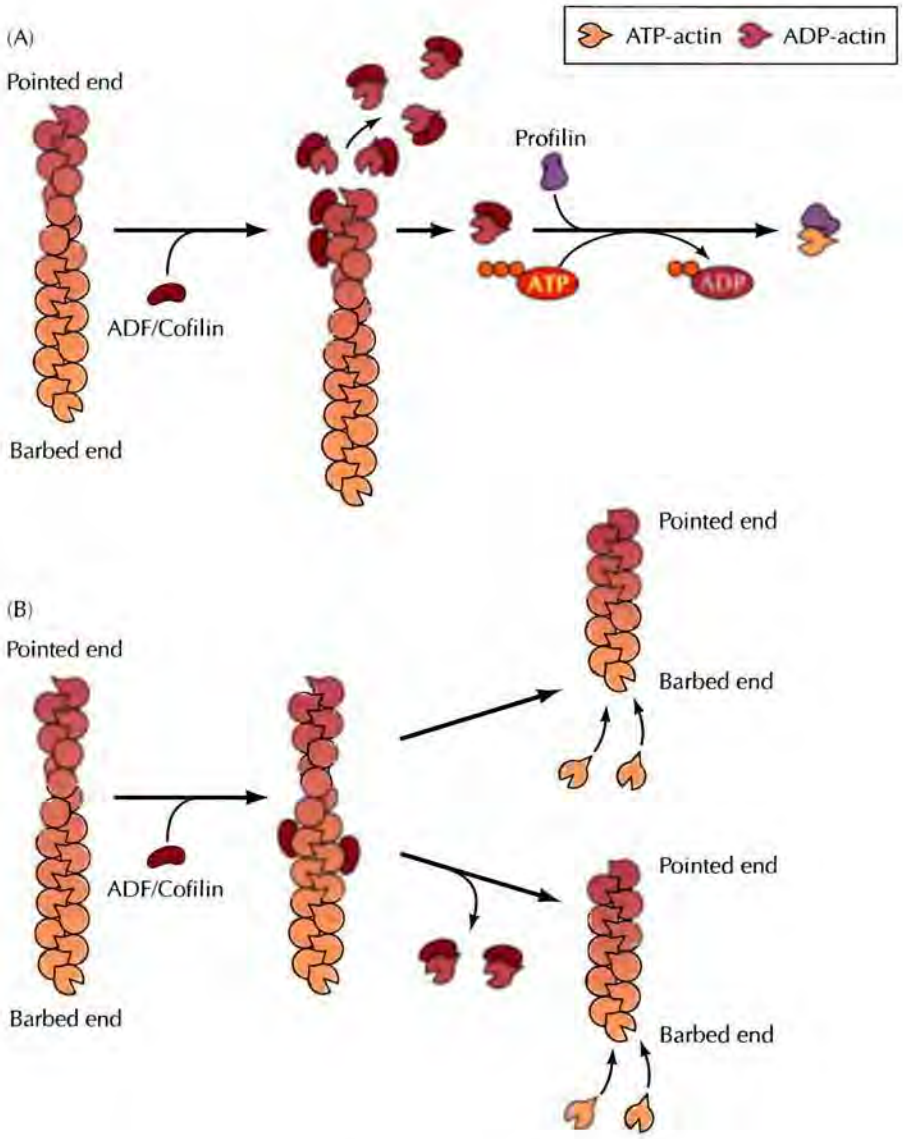


FIGURE 12.8 Effects of ADF/cofilin and profilin on actin filaments ADF/cofilin has two different activities. (A) It is an *a*ctin *d*epolymerization *f*actor, binding to actin filaments and increasing the rate of dissociation of actin monomers from the pointed end. ADF/cofilin remains bound to the actin/ADP monomers, preventing their reassembly into filaments. Profilin can stimulate the exchange of bound ADP for ATP, resulting in the formation of actin/ATP monomers that can be repolymerized into filaments. (B) ADF/cofilin can also bind to and sever actin filaments, creating new plus ends.

within the cell is the **ADF/cofilin** (*a*ctin *d*epolymerizing *f*actor) family (Figure 12.8). These proteins bind to actin filaments and enhance the rate of dissociation of actin/ADP monomers from the pointed end. ADF/cofilin can also sever actin filaments. ADF/cofilin preferentially binds to ADP-actin, so it remains bound to actin monomers following filament disassembly and sequesters them in the ADP-bound form, preventing their reincorporation into filaments. However, another actin-binding protein, **profilin**, can reverse this effect of cofilin and stimulate the incorporation of actin monomers into filaments. Profilin acts by stimulating the exchange of bound ADP for ATP, resulting in the formation of actin/ATP monomers, which dissociate from cofilin and are then available for assembly into filaments.

As might be expected, the activities of these proteins are controlled by a variety of cell signaling mechanisms (discussed in Chapter 15), allowing actin polymerization to be appropriately regulated in response to environmental signals. ADF/cofilin, profilin, formin, and the Arp2/3 complex (as well as other actin-binding proteins) can thus act together to promote the rapid turnover of actin filaments and remodeling of the actin cytoskeleton that is required for a variety of cell movements and changes in cell shape, as discussed later in this chapter. This is a major undertaking, and in some cell

types, microfilament assembly and disassembly are responsible for half the hydrolysis and turnover of ATP in the cell.

Several of the actin-related proteins (Arp4–8) and actin itself occur in the nucleus. The actin-related proteins are involved in chromatin remodeling in both plants and animals and may participate in the assembly of the nucleus after cell division. The role of nuclear actin remains controversial.

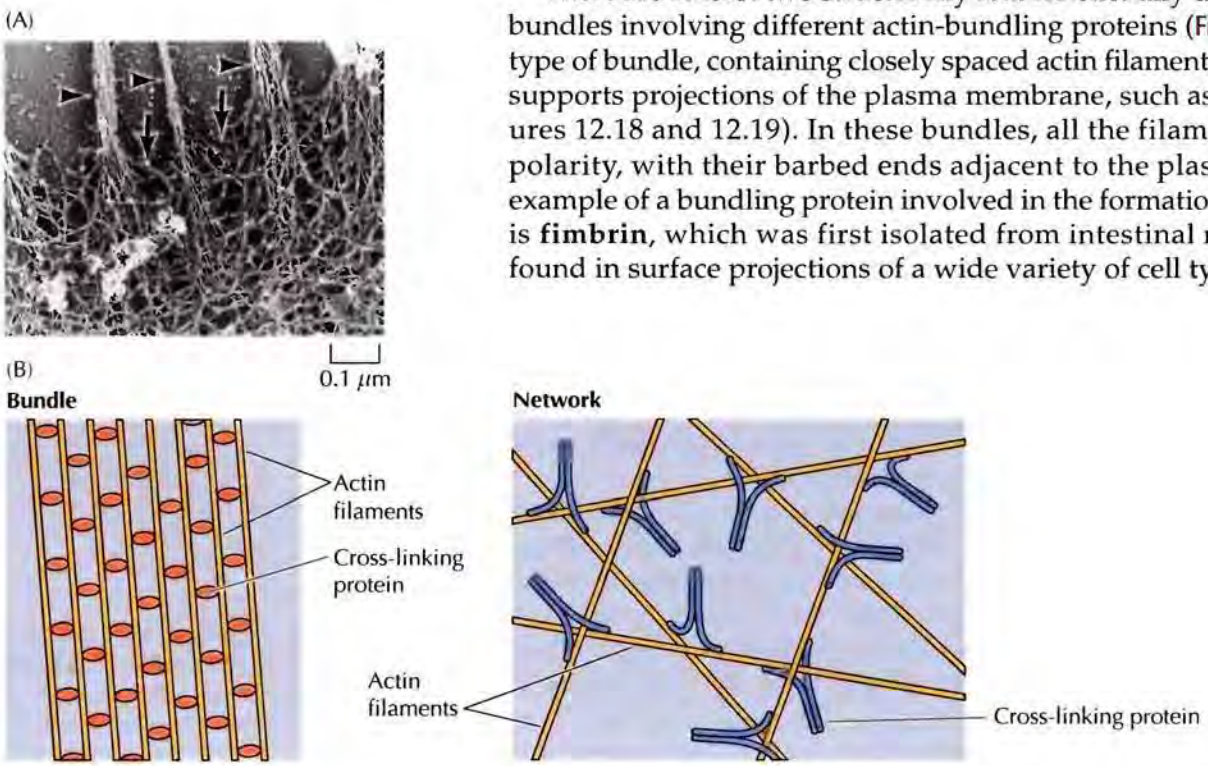
Organization of Actin Filaments

Individual actin filaments are assembled into two general types of structures called **actin bundles** and **actin networks**, which play different roles in the cell (Figure 12.9). In bundles, the actin filaments are cross-linked into closely packed parallel arrays. In networks, the actin filaments are cross-linked in orthogonal arrays that form three-dimensional meshworks with the properties of semisolid gels. As mentioned previously, the formation of these structures is governed by a variety of actin-binding proteins that cross-link actin filaments in distinct patterns (see Table 12.1).

All of the actin-binding proteins involved in cross-linking contain at least two domains that bind actin, allowing them to bind and cross-link two different actin filaments. The nature of the association between these filaments is then determined by the size and shape of the cross-linking proteins (see Figure 12.9). The proteins that cross-link actin filaments into bundles (called **actin-bundling proteins**) usually are small rigid proteins that force the filaments to align closely with one another. In contrast, the proteins that organize actin filaments into networks tend to be large flexible proteins that can cross-link perpendicular filaments. Actin cross-linking proteins are modular proteins consisting of related structural units. In particular, the actin-binding domains of many of these proteins are similar in structure. They are separated by regulatory domains and spacer sequences that vary in length and flexibility, and it is these differences in the spacer sequences that are responsible for the distinct cross-linking properties of different actin-binding proteins.

There are at least two structurally and functionally distinct types of actin bundles involving different actin-bundling proteins (Figure 12.10). The first type of bundle, containing closely spaced actin filaments aligned in parallel, supports projections of the plasma membrane, such as microvilli (see Figures 12.18 and 12.19). In these bundles, all the filaments have the same polarity, with their barbed ends adjacent to the plasma membrane. An example of a bundling protein involved in the formation of these structures is **fimbrin**, which was first isolated from intestinal microvilli and later found in surface projections of a wide variety of cell types. Fimbrin is a 68

FIGURE 12.9 Actin bundles and networks (A) Electron micrograph of actin bundles (arrowheads) projecting from the actin network (arrows) underlying the plasma membrane of a macrophage. The bundles support cell surface projections called filopodia (see Figure 12.20). (B) Schematic organization of bundles and networks. Actin filaments in bundles are cross-linked into parallel arrays by small proteins that align the filaments closely with one another. In contrast, networks are formed by large flexible proteins that cross-link orthogonal filaments. (A, courtesy of John H. Hartwig, Brigham & Women's Hospital.)



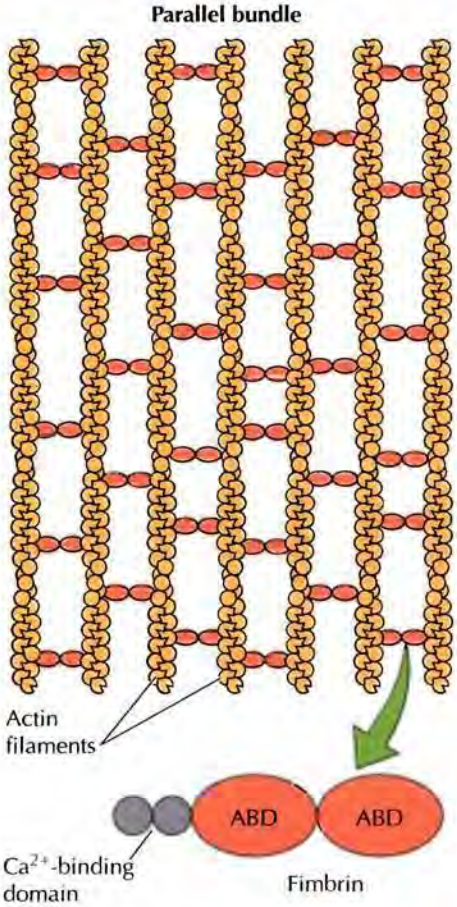


FIGURE 12.10 Actin-bundling proteins Actin filaments are associated into two types of bundles by different actin-bundling proteins. Fimbrin has two adjacent actin-binding domains (ABD) and cross-links actin filaments into closely packed parallel bundles in which the filaments are approximately 14 nm apart. In contrast, the two separated actin-binding domains of α-actinin dimers cross-link filaments into more loosely spaced contractile bundles in which the filaments are separated by 40 nm. Both fimbrin and α-actinin contain two related Ca^{2+}-binding domains, and α-actinin contains four repeated α-helical spacer domains.

kd protein containing two adjacent actin-binding domains. It binds to actin filaments as a monomer, holding two parallel filaments close together.

The second type of actin bundle is composed of filaments that are more widely spaced, allowing the bundle to contract. The more open structure of these bundles (which are called **contractile bundles**) reflects the properties of the cross-linking protein, **α-actinin**. In contrast to fimbrin, α-actinin binds to actin as a dimer, each subunit of which is a 102 kd protein containing a single actin-binding site. Filaments cross-linked by α-actinin are consequently separated by a greater distance than those cross-linked by fimbrin (40 nm apart instead of 14 nm). The increased spacing between filaments allows the motor protein myosin to interact with the actin filaments in these bundles, which (as discussed later) enables the bundles to contract.

The actin filaments in networks are held together by large actin-binding proteins, such as **filamin** (Figure 12.11). Filamin binds actin as a dimer of two 280 kd subunits. The actin-binding domains and dimerization domains are at opposite ends of each subunit, so the filamin dimer is a flexible V-shaped molecule with actin-binding domains at the ends of each arm. As a result, filamin forms cross-links between orthogonal actin filaments, creating a loose three-dimensional meshwork. As discussed in the next section, such networks of actin filaments underlie the plasma membrane and support the surface of the cell.

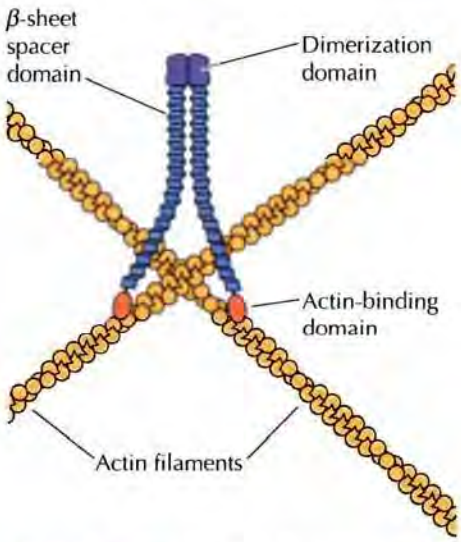


FIGURE 12.11 Actin networks and filamin Filamin is a dimer of two large (280 kd) subunits, forming a flexible V-shaped molecule that cross-links actin filaments into orthogonal networks. The carboxy-terminal dimerization domain is separated from the amino-terminal actin-binding domain by repeated β-sheet spacer domains.

FIGURE 12.12 Morphology of red blood cells Scanning electron micrograph of red blood cells illustrating their biconcave shape. (Omikron/Photo Researchers, Inc.)

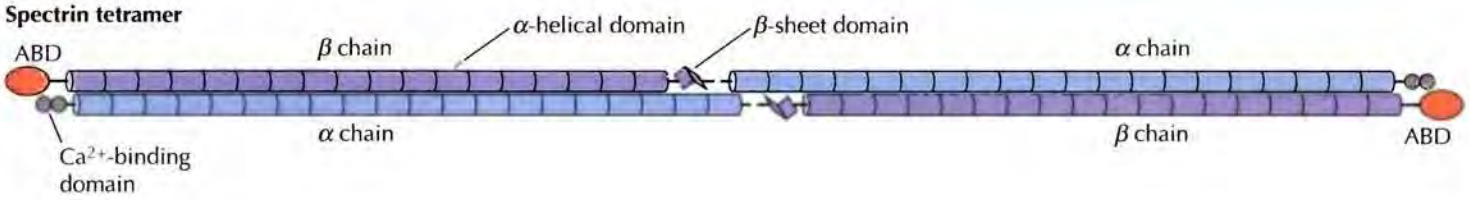
Association of Actin Filaments with the Plasma Membrane

Actin filaments are highly concentrated at the periphery of the cell where they form a three-dimensional network beneath the plasma membrane (see Figure 12.9). This network of actin filaments and associated actin-binding proteins (called the **cell cortex**) determines cell shape and is involved in a variety of cell surface activities, including movement. The association of the actin cytoskeleton with the plasma membrane is thus central to cell structure and function.

Red blood cells (erythrocytes) have proven particularly useful for studies of both the plasma membrane (discussed in the next chapter) and the cortical cytoskeleton. The principal advantage of red blood cells for these studies is that they contain no nucleus or internal organelles, so their plasma membrane and associated proteins can be easily isolated without contamination by the various internal membranes that are abundant in other cell types. In addition, human erythrocytes lack other cytoskeletal components (microtubules and intermediate filaments), so the cortical actin cytoskeleton is the principal determinant of their distinctive shape as biconcave discs (Figure 12.12).

The major protein that provides the structural basis for the cortical cytoskeleton in erythrocytes is the actin-binding protein **spectrin** (Figure 12.13). Spectrin is a member of the large calponin family of actin-binding proteins, which includes α-actinin, filamin, and fimbrin. Erythrocyte spec-

FIGURE 12.13 Structure of spectrin Spectrin is a tetramer consisting of two α and two β chains. Each β chain has a single actin-binding domain (ABD) at its amino terminus. Both α and β chains contain multiple repeats of α-helical spacer domains, which separate the two actin-binding domains of the tetramer. The α chain has two Ca^{2+} binding domains at its carboxy terminus.



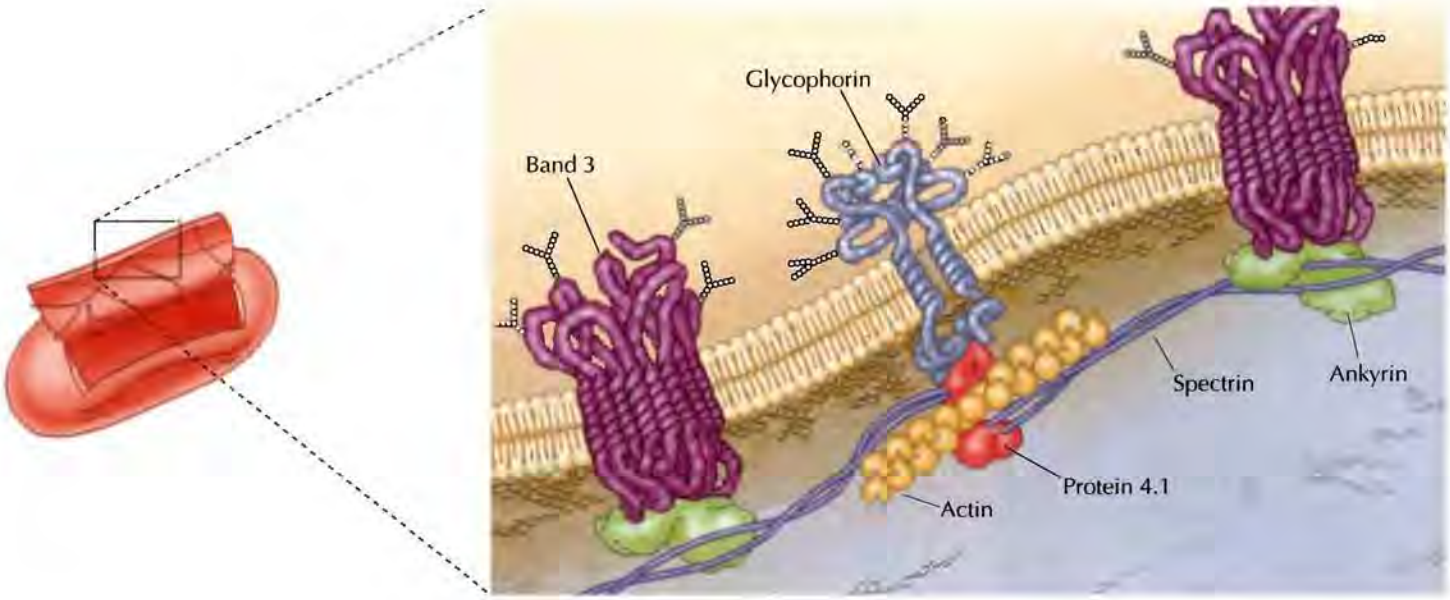


FIGURE 12.14 Association of the erythrocyte cortical cytoskeleton with the plasma membrane
The plasma membrane is associated with a network of spectrin tetramers cross-linked by short actin filaments in association with protein 4.1. The spectrin-actin network is linked to the membrane by ankyrin, which binds to both spectrin and the abundant transmembrane protein, band 3. An additional link is provided by the binding of protein 4.1 to glycophorin.

trin is a tetramer consisting of two distinct polypeptide chains called α and β, with molecular weights of 240 and 220 kd, respectively. The β chain has a single actin-binding domain at its amino terminus. The α and β chains associate laterally to form dimers, which then join head-to-head to form tetramers with two actin-binding domains separated by approximately 200 nm. The ends of the spectrin tetramers then associate with short actin filaments, resulting in the spectrin-actin network that forms the cortical cytoskeleton of red blood cells (Figure 12.14). The major link between the spectrin-actin network and the plasma membrane is provided by a protein called **ankyrin**, which binds both to spectrin and to the cytoplasmic domain of an abundant transmembrane protein called band 3. An additional link between the spectrin-actin network and the plasma membrane is provided by protein 4.1, which binds to spectrin-actin junctions as well as recognizing the cytoplasmic domain of glycophorin (another abundant transmembrane protein).

Other types of cells contain linkages between the cortical cytoskeleton and the plasma membrane that are similar to those observed in red blood cells. Proteins related to spectrin (nonerythroid spectrin is also called **fodrin**), ankyrin, and protein 4.1 are expressed in a wide range of cell types where they fulfill functions analogous to those described for erythrocytes. For example, a family of proteins related to protein 4.1 (**ERM proteins**) link actin filaments to the plasma membranes of many different kinds of cells, and another member of the calponin family, filamin (see Figure 12.11), constitutes a major link between actin filaments and the plasma membrane of blood platelets. An additional member of the calponin family, **dystrophin**, is of particular interest because it is the product of the gene responsible for two types of muscular dystrophy (Duchenne's and Becker's). These X-linked inherited diseases result in progressive degeneration of skeletal muscle, and patients with the more severe form of the disease (Duchenne's muscular dystrophy) usually die in their teens or early twenties. Molecular cloning of the gene responsible for this disorder revealed that it encodes a large protein (427 kd) that is either absent or abnormal in patients with Duchenne's or Becker's muscular dystrophy, respectively. The sequence of dystrophin fur-

FIGURE 12.15 Stress fibers and focal adhesions Fluorescence microscopy of a human fibroblast in which actin filaments have been stained with a fluorescent dye. Stress fibers are revealed as bundles of actin filaments anchored at sites of cell attachment to the culture dish surface (focal adhesions). (Don Fawcett/Photo Researchers, Inc.)

ther indicated that it has a single actin-binding domain at its amino terminus and a membrane-binding domain at its carboxy terminus. Like spectrin, dystrophin links actin filaments to transmembrane proteins of the muscle cell plasma membrane. These transmembrane proteins in turn link the cytoskeleton to the extracellular matrix, which plays an important role in maintaining cell stability during muscle contraction.

In contrast to the uniform surface of red blood cells, most cells have specialized regions of the plasma membrane that form contacts with adjacent cells, the extracellular matrix (discussed in Chapter 14), or other substrata (such as the surface of a culture dish). These regions can also serve as attachment sites for bundles of actin filaments that anchor the cytoskeleton to areas of cell contact. These attachments of actin filaments are particularly evident in fibroblasts maintained in tissue culture (Figure 12.15). Such cultured fibroblasts secrete extracellular matrix proteins that stick to the surface of the culture dish. The fibroblasts then attach to this extracellular matrix on the culture dish via the binding of transmembrane proteins (called **integrins**). The sites of attachment are discrete regions (called **focal adhesions**) that also serve as attachment sites for large bundles of actin filaments called **stress fibers**.

Stress fibers are contractile bundles of actin filaments, cross-linked by α-actinin and stabilized by tropomysosin, which anchor the cell and exert tension against the substratum. They are attached to the plasma membrane at focal adhesions via interactions with integrin. These complex associations are mediated by several other proteins, including **talin** and **vinculin** (Figure 12.16). For example, both talin and α-actinin bind to the cytoplasmic domains of integrins. Talin also binds to vinculin and both proteins also bind actin. Other proteins found at focal adhesions also participate in the attachment of actin filaments, and a combination of these interactions is responsible for the linkage of actin filaments to the plasma membrane.

The actin cytoskeleton is similarly anchored to regions of cell-cell contact called **adherens junctions** (Figure 12.17). In sheets of epithelial cells, these junctions form a continuous beltlike structure (called an **adhesion belt**) around each cell in which an underlying contractile bundle of actin filaments is linked to the plasma membrane. Contact between cells at adherens

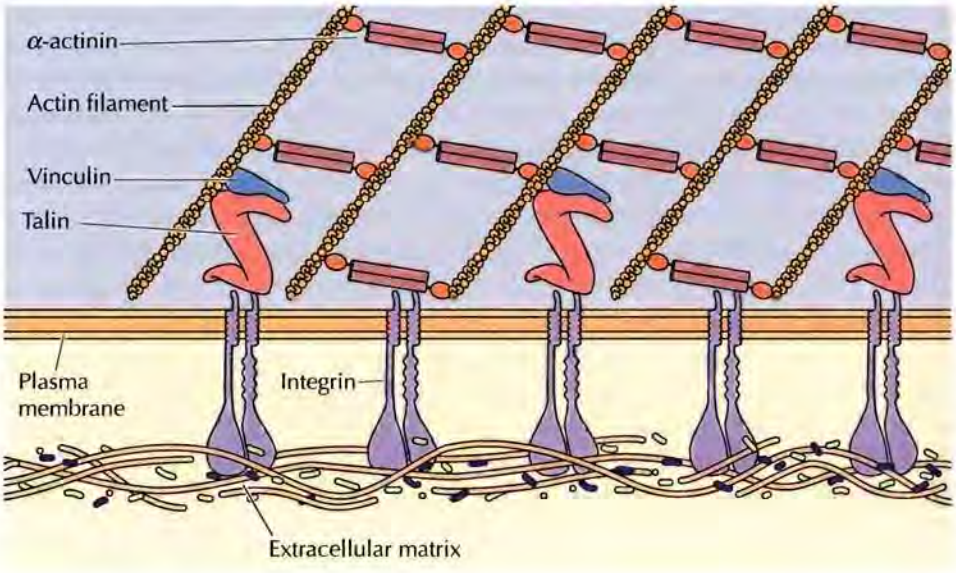


FIGURE 12.16 Attachment of stress fibers to the plasma membrane at focal adhesions Focal adhesions are mediated by the binding of integrins to proteins of the extracellular matrix. Stress fibers (bundles of actin filaments cross-linked by α-actinin) are then bound to the cytoplasmic domain of integrins by complex associations involving a number of proteins. Two possible associations are illustrated: 1) talin binds to both integrin and vinculin, and both talin and vinculin bind to actin, and 2) integrin binds to α-actinin. A number of other proteins (not shown) are also present at focal adhesions and may be involved in anchoring stress fibers to the plasma membrane.

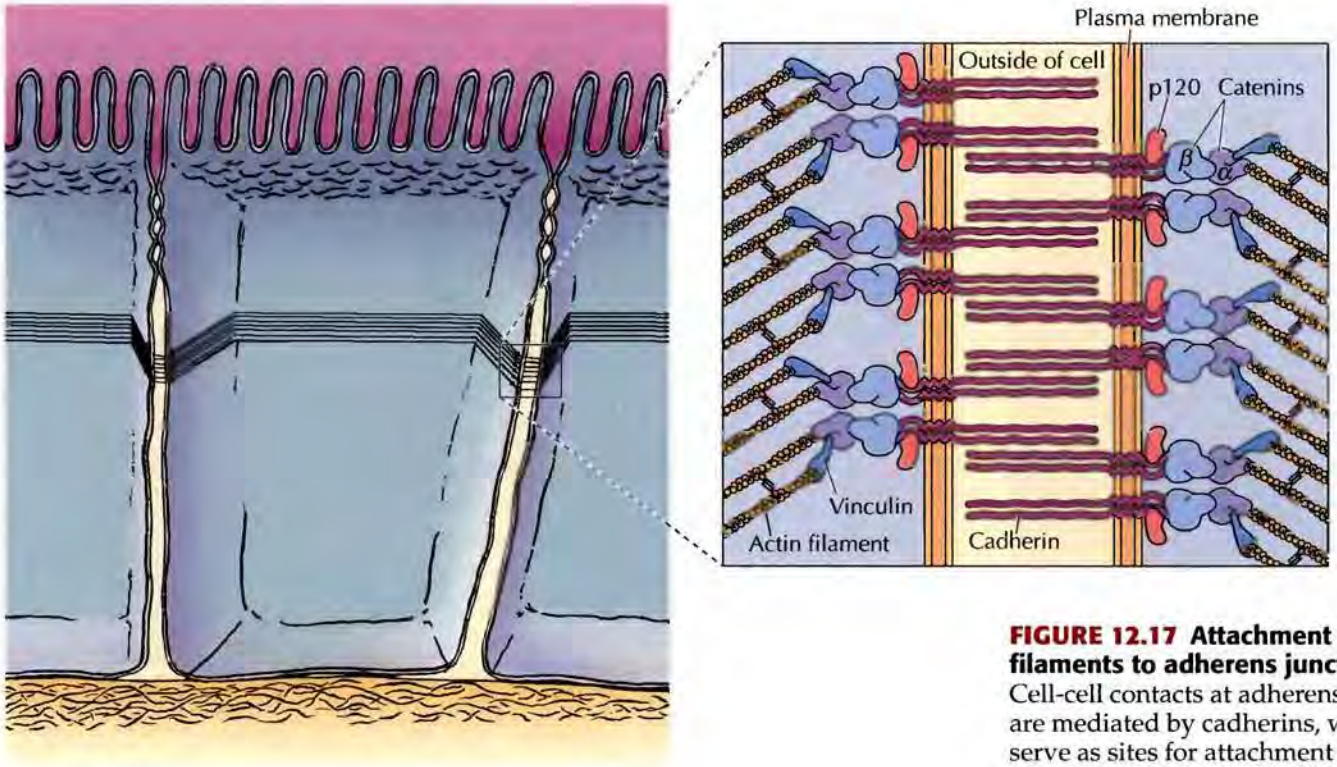


FIGURE 12.17 Attachment of actin filaments to adherens junctions
Cell-cell contacts at adherens junctions are mediated by cadherins, which serve as sites for attachment of actin filaments. In sheets of epithelial cells, these junctions form a continuous belt of actin filaments around each cell. The transmembrane cadherins bind β-catenin at their cytoplasmic domains. β-catenin interacts with α-catenin, which binds both actin filaments and vinculin. p120 regulates the stability of the junction.

junctions is mediated by transmembrane proteins called **cadherins**, which are discussed further in Chapter 14. The cadherins form a complex with cytoplasmic proteins called **catenins**, which associate with actin filaments.

Protrusions of the Cell Surface

The surfaces of most cells have a variety of protrusions or extensions that are involved in cell movement, phagocytosis, or specialized functions, such as absorption of nutrients. Most of these cell surface extensions are based on actin filaments, which are organized into either relatively permanent or rapidly rearranging bundles or networks.

The best-characterized of these actin-based cell surface protrusions are **microvilli**, fingerlike extensions of the plasma membrane that are particularly abundant on the surfaces of cells involved in absorption, such as the epithelial cells lining the intestine (Figure 12.18). The microvilli of these cells form a layer on the apical surface (called a **brush border**) that consists of approximately a thousand microvilli per cell and increases the exposed surface area available for absorption by ten to twentyfold. In addition to their role in absorption, specialized forms of microvilli, the **stereocilia** of auditory hair cells, are responsible for hearing by detecting sound vibrations.

Their abundance and ease of isolation have facilitated detailed structural analysis of intestinal microvilli, which contain closely packed parallel bundles of 20 to 30 actin filaments (Figure 12.19). The filaments in these bundles are cross-linked in part by fimbrin, an actin-bundling protein (discussed earlier) that is present in surface projections of a variety of cell types. However, the major actin-bundling protein in intestinal microvilli is **villin**, a 95 kd protein present in microvilli of only a few specialized types of cells, such as those lining the intestine and kidney tubules. Along their length, the actin bundles of microvilli are attached to the plasma membrane by lateral arms consisting of the calcium-binding protein calmodulin in association

■ **In addition to its structural role at adherens junctions, β-catenin is a critical component of the Wnt signaling pathway where it functions as a transcriptional activator (see Chapter 15).**

FIGURE 12.18 Electron micrograph of microvilli The microvilli of intestinal epithelial cells are fingerlike projections of the plasma membrane. They are supported by actin bundles anchored in a dense region of the cortex called the terminal web. (Courtesy of Nobutaka Hirokawa.)

with myosin I, which may be involved in movement of the plasma membrane along the actin bundle of the microvillus. At their base, the actin bundles are anchored in a spectrin-rich region of the actin cortex called the terminal web, which cross-links and stabilizes the microvilli.

In contrast to microvilli many surface protrusions are transient structures that form in response to environmental stimuli. Several types of these structures extend from the leading edge of a moving cell and are involved in cell locomotion (Figure 12.20). **Pseudopodia** are extensions of moderate width, based on actin filaments cross-linked into a three-dimensional network, that are responsible for phagocytosis and for the movement of amoebas across a surface. **Lamellipodia** are broad, sheetlike extensions at the leading edge of fibroblasts, which similarly contain a network of actin filaments. Many cells also extend **microspikes** or **filopodia**, thin projections of the plasma membrane supported by actin bundles. The formation and retraction of these structures is based on the regulated assembly and disassembly of actin filaments, as discussed later in this chapter.

Actin, Myosin, and Cell Movement

Actin filaments, often in association with myosin, are responsible for many types of cell movements. **Myosin** is the prototype of a **molecular motor**—a protein that converts chemical energy in the form of ATP to mechanical energy, thus generating force and movement. The most striking variety of such movement is muscle contraction, which has provided the model for understanding actin-myosin interactions and the motor activity of myosin molecules. However, interactions of actin and myosin are responsible not only for muscle contraction but also for a variety of

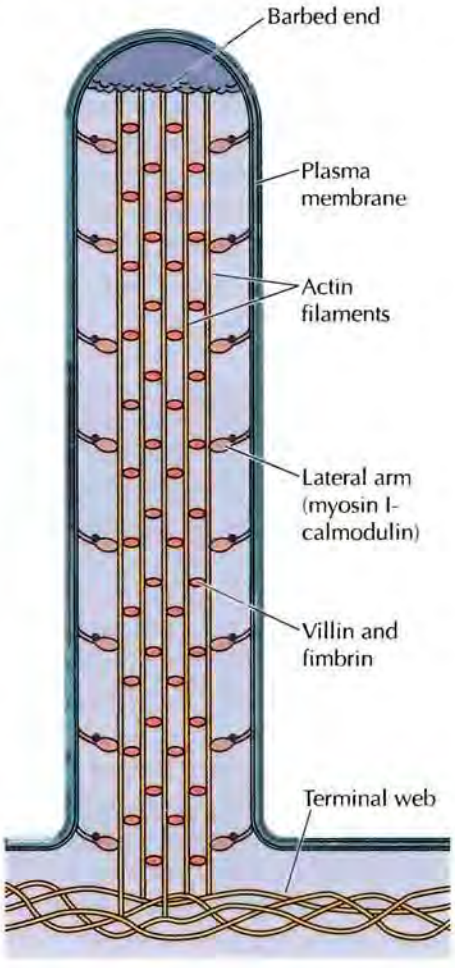


FIGURE 12.19 Organization of microvilli The core actin filaments of microvilli are cross-linked into closely packed bundles by fimbrin and villin. They are attached to the plasma membrane along their length by lateral arms, consisting of myosin I and calmodulin. The barbed ends of the actin filaments are embedded in a cap of unidentified proteins at the tip of the microvillus.

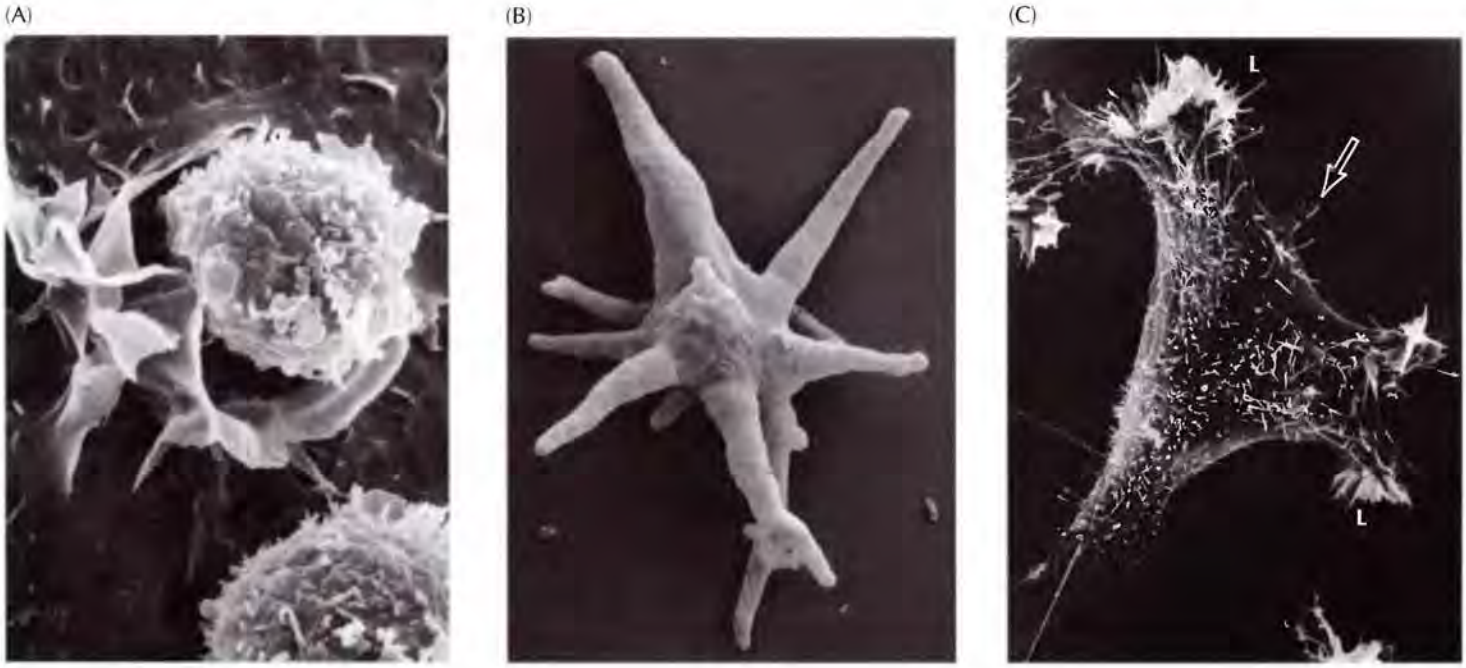


FIGURE 12.20 Examples of cell surface projections involved in phagocytosis and movement (A) Scanning electron micrograph showing pseudopodia of a macrophage engulfing a tumor cell during phagocytosis. (B) An amoeba with several extended pseudopodia. (C) A tissue culture cell illustrating lamellipodia (L) and filopodia (arrow). (A, K. Wassermann/Visuals Unlimited; B, Stanley Flegler/Visuals Unlimited; C, Don Fawcett/Photo Researchers, Inc.)

movements of nonmuscle cells, including cell division, so these interactions play a central role in cell biology.

Muscle Contraction

Muscle cells are highly specialized for a single task—contraction—and it is this specialization in structure and function that has made muscle the prototype for studying movement at the cellular and molecular levels. There are three distinct types of muscle cells in vertebrates: skeletal muscle, which is responsible for all voluntary movements; cardiac muscle, which pumps blood from the heart; and smooth muscle, which is responsible for involuntary movements of organs such as the stomach, intestine, uterus, and blood vessels. In both skeletal and cardiac muscle, the contractile elements of the cytoskeleton are present in highly organized arrays that give rise to characteristic patterns of cross-striations. It is the characterization of these structures in skeletal muscle that has led to our current understanding of muscle contraction and other actin-based cell movements at the molecular level.

Skeletal muscles are bundles of **muscle fibers**, which are single large cells (approximately 50 μm in diameter and up to several centimeters in length) formed by the fusion of many individual cells during development (Figure 12.21). Most of the cytoplasm consists of **myofibrils**, which are cylindrical bundles of two types of filaments: thick filaments of myosin (about 15 nm in diameter) and thin filaments of actin (about 7 nm in diameter). Each myofibril is organized as a chain of contractile units called **sarcomeres**, which are responsible for the striated appearance of skeletal and cardiac muscle.

■ **Molecules that regulate the activity of smooth muscle cells are important drugs. For example, albuterol, a drug that relaxes smooth muscles, is used in the treatment of asthma and to prevent premature labor in pregnant women. In contrast, oxytocin, a hormone that stimulates smooth muscle contraction, is administered to induce labor.**

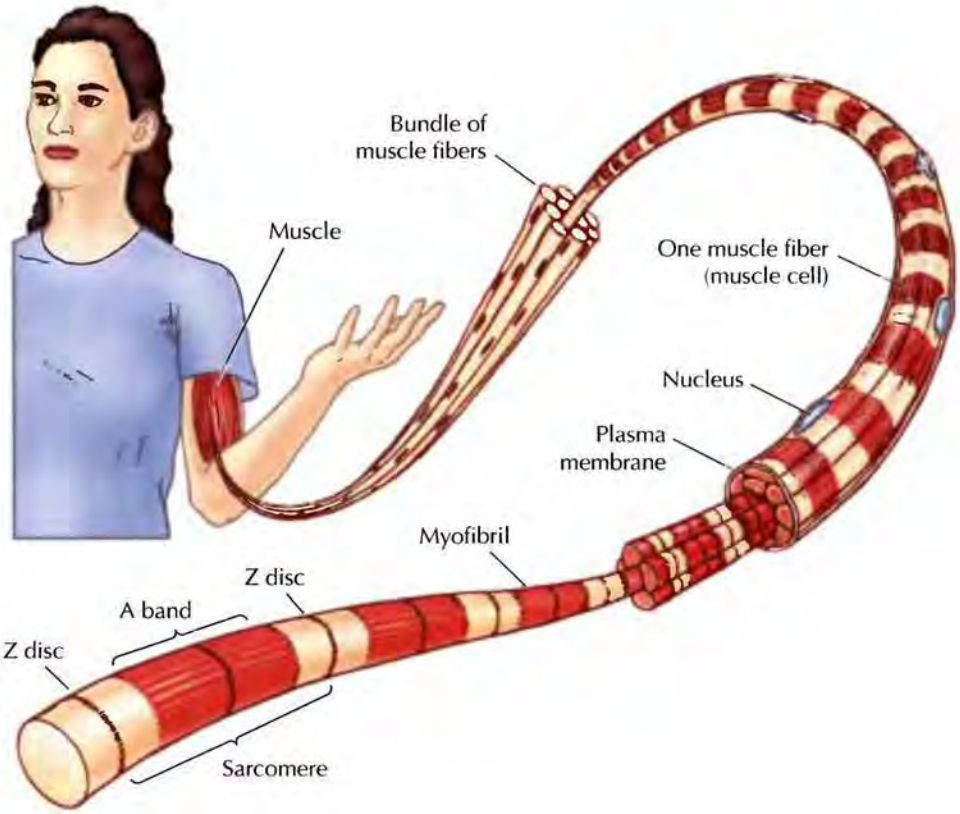


FIGURE 12.21 Structure of muscle cells Muscles are composed of bundles of single large cells (called muscle fibers) that form by cell fusion and contain multiple nuclei. Each muscle fiber contains many myofibrils, which are bundles of actin and myosin filaments organized into a chain of repeating units called sarcomeres.

The sarcomeres (which are approximately 2.3 μm long) consist of several distinct regions discernible by electron microscopy, which provided critical insights into the mechanism of muscle contraction (Figure 12.22). The ends of each sarcomere are defined by the Z disc. Within each sarcomere, dark bands (called A bands because they are *a*nisotropic when viewed with

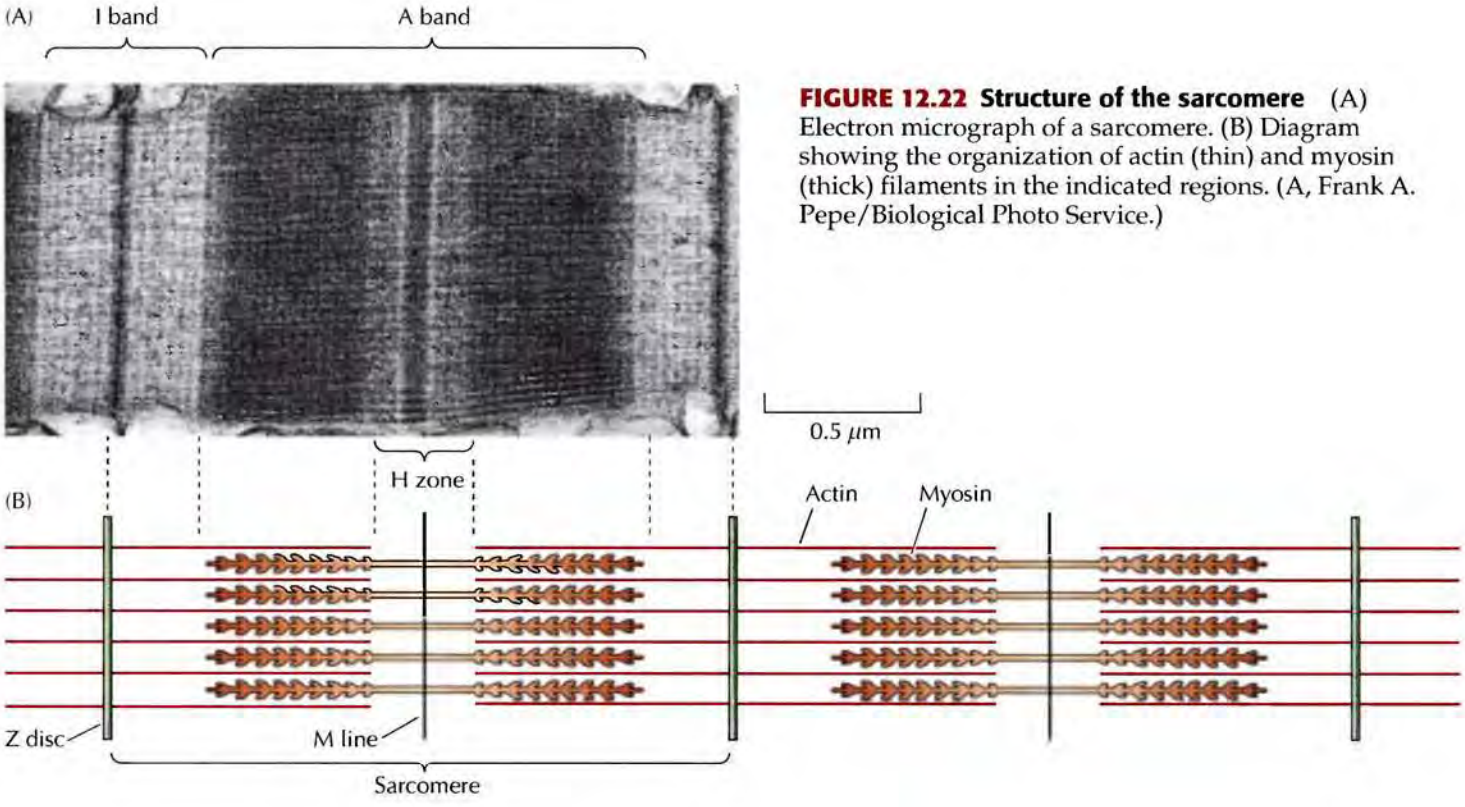


FIGURE 12.22 Structure of the sarcomere (A) Electron micrograph of a sarcomere. (B) Diagram showing the organization of actin (thin) and myosin (thick) filaments in the indicated regions. (A, Frank A. Pepe/Biological Photo Service.)

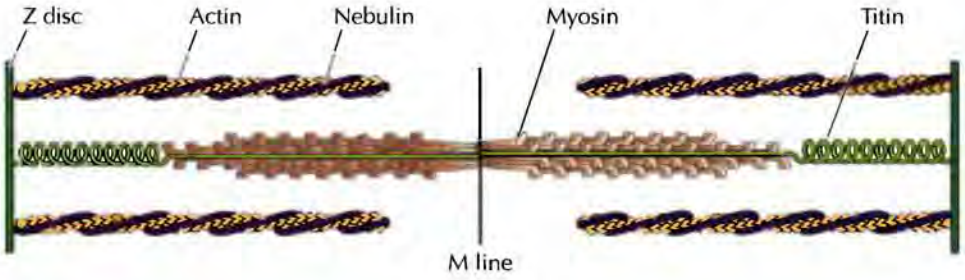


FIGURE 12.23 Titin and nebulin Molecules of titin extend from the Z disc to the M line and act as springs to keep myosin filaments centered in the sarcomere. Molecules of nebulin extend from the Z disc and are thought to determine the length of associated actin filaments.

polarized light) alternate with light bands (called I bands for *i*sotropic). These bands correspond to the presence or absence of myosin filaments. The I bands contain only thin (actin) filaments, whereas the A bands contain thick (myosin) filaments. The myosin and actin filaments overlap in peripheral regions of the A band, whereas a middle region (called the H zone) contains only myosin. The actin filaments are attached at their barbed ends to the Z disc, which includes the cross-linking protein α-actinin. The myosin filaments are anchored at the M line in the middle of the sarcomere.

Two additional proteins (**titin** and **nebulin**) also contribute to sarcomere structure and stability (Figure 12.23). Titin is an extremely large protein (3000 kd), and single titin molecules extend from the M line to the Z disc. These long molecules of titin are thought to act like springs that keep the myosin filaments centered in the sarcomere and maintain the resting tension that allows a muscle to snap back if overextended. Nebulin filaments are associated with actin and are thought to regulate the assembly of actin filaments by acting as rulers that determine their length.

The basis for understanding muscle contraction is the **sliding filament model**, first proposed in 1954 both by Andrew Huxley and Ralph Niedergerke and by Hugh Huxley and Jean Hanson (Figure 12.24). During muscle contraction each sarcomere shortens, bringing the Z discs closer together. There is no change in the width of the A band, but both the I bands and the H zone almost completely disappear. These changes are explained by the

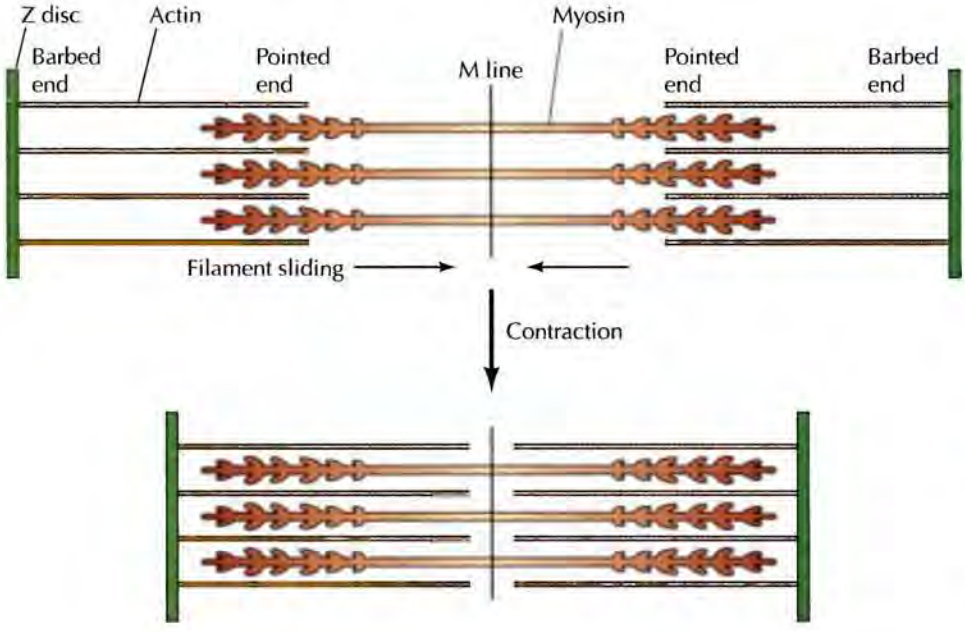


FIGURE 12.24 Sliding-filament model of muscle contraction The actin filaments slide past the myosin filaments toward the middle of the sarcomere. The result is shortening of the sarcomere without any change in filament length.

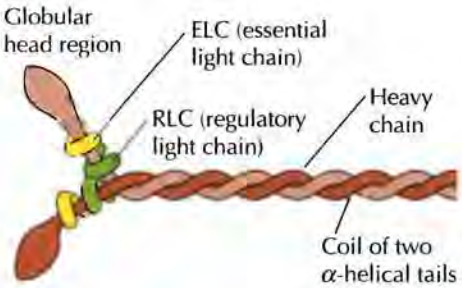


FIGURE 12.25 Myosin II The myosin II molecule consists of two heavy chains and two pairs of light chains (called the essential and regulatory light chains). The heavy chains have globular head regions and long α-helical tails, which coil around each other to form dimers.

actin and myosin filaments sliding past one another so that the actin filaments move into the A band and H zone. Muscle contraction thus results from an interaction between the actin and myosin filaments that generates their movement relative to one another. The molecular basis for this interaction is the binding of myosin to actin filaments, allowing myosin to function as a motor that drives filament sliding.

The type of myosin present in muscle (**myosin II**) is a very large protein (about 500 kd) consisting of two identical heavy chains (about 200 kd each) and two pairs of light chains (about 20 kd each) (Figure 12.25). Each heavy chain consists of a globular head region and a long α-helical tail. The α-helical tails of two heavy chains twist around each other in a coiled-coil structure to form a dimer, and two light chains associate with the neck of each head region to form the complete myosin II molecule.

The thick filaments of muscle consist of several hundred myosin molecules associated in a parallel staggered array by interactions between their tails (Figure 12.26). The globular heads of myosin bind actin, forming cross-bridges between the thick and thin filaments. It is important to note that the orientation of myosin molecules in the thick filaments reverses at the M line of the sarcomere. The polarity of actin filaments (which are attached to Z discs at their barbed ends) similarly reverses at the M line, so the relative orientation of myosin and actin filaments is the same on both halves of the sarcomere. As discussed later, the motor activity of myosin moves its head groups along the actin filament in the direction of the barbed end. This movement slides the actin filaments from both sides of the sarcomere toward the M line, shortening the sarcomere and resulting in muscle contraction.

In addition to binding actin the myosin heads bind and hydrolyze ATP, which provides the energy to drive filament sliding. This translation of chemical energy to movement is mediated by changes in the shape of myosin resulting from ATP binding. The generally accepted model (the swinging-cross-bridge model) is that ATP hydrolysis drives repeated cycles of interaction between myosin heads and actin. During each cycle, conformational changes in myosin result in the movement of myosin heads along actin filaments.

Although the molecular mechanisms are still not fully understood, a plausible working model for myosin function has been derived both from

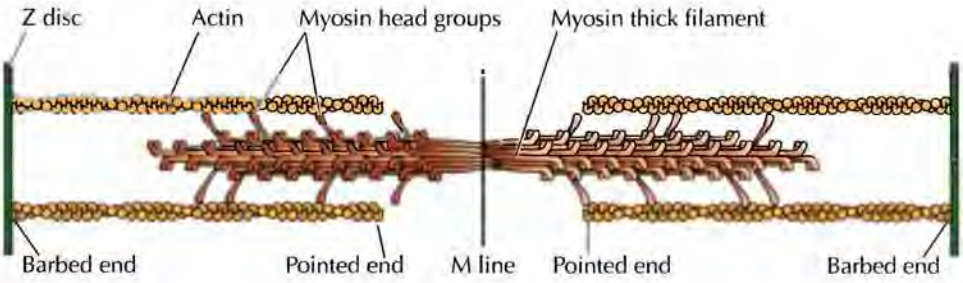


FIGURE 12.26 Organization of myosin thick filaments Thick filaments are formed by the association of several hundred myosin II molecules in a staggered array. The globular heads of myosin bind actin, forming cross-bridges between the myosin and actin filaments. The orientation of both actin and myosin filaments reverses at the M line, so their relative polarity is the same on both sides of the sarcomere.

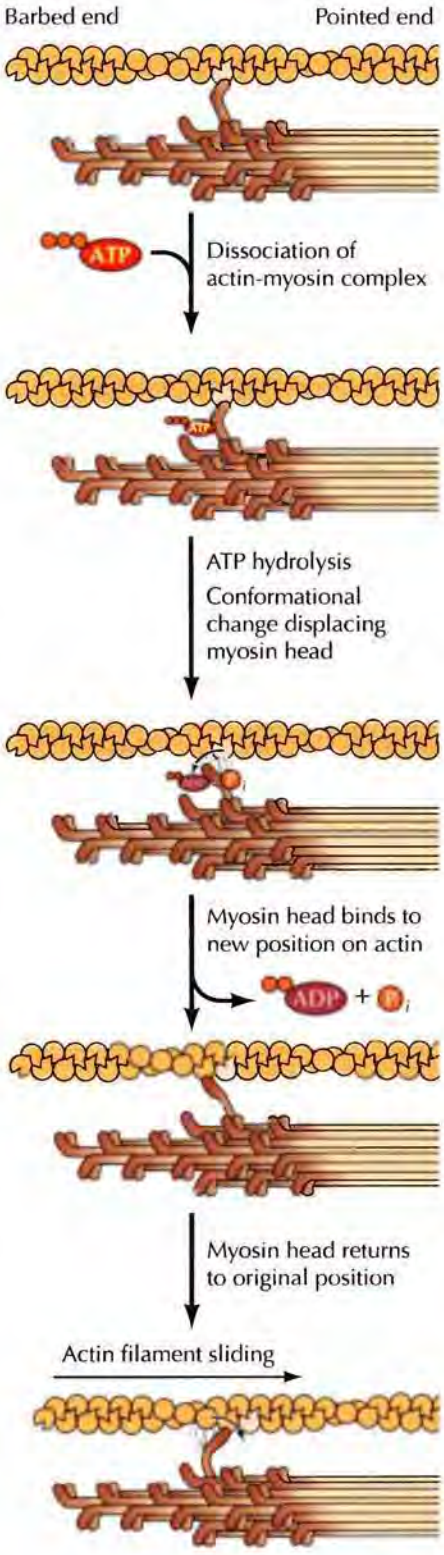


FIGURE 12.27 Model for myosin action The binding of ATP dissociates myosin from actin. ATP hydrolysis then induces a conformational change that displaces the myosin head group. This is followed by binding of the myosin head to a new position on the actin filament and release of ADP and P_i. The return of the myosin head to its original conformation drives actin filament sliding.

in vitro studies of myosin movement along actin filaments (a system developed by James Spudich and Michael Sheetz) and from determination of the three-dimensional structure of myosin by Ivan Rayment and his colleagues (Figure 12.27). The cycle starts with myosin (in the absence of ATP) tightly bound to actin. ATP binding dissociates the myosin-actin complex and the hydrolysis of ATP then induces a conformational change in myosin. This change affects the neck region of myosin that binds the light chains (see Figure 12.25), which acts as a lever arm to displace the myosin head by about 5 nm. The products of hydrolysis (ADP and P_i) remain bound to the myosin head, which is said to be in the "cocked" position. The myosin head then rebinds at a new position on the actin filament, resulting in the release of ADP and P_i and triggering the "power stroke" in which the myosin head returns to its initial conformation, thereby sliding the actin filament toward the M line of the sarcomere.

The contraction of skeletal muscle is triggered by nerve impulses, which stimulate the release of Ca^{2+} from the **sarcoplasmic reticulum**—a specialized network of internal membranes (similar to the endoplasmic reticulum) that stores high concentrations of Ca^{2+} ions. The release of Ca^{2+} from the sarcoplasmic reticulum increases the concentration of Ca^{2+} in the cytosol from approximately 10^{-7} to 10^{-5} *M*. The increased Ca^{2+} concentration signals muscle contraction via the action of two actin filament binding proteins: tropomyosin and **troponin** (Figure 12.28). In striated muscle each tropomyosin molecule is bound to troponin, which is a complex of three polypeptides: troponin C (Ca^{2+}-binding), troponin I (inhibitory), and troponin T (tropomyosin-binding). When the concentration of Ca^{2+} is low, the complex of the troponins with tropomyosin blocks the interactions of almost all actins with the myosin head groups, so the muscle does not contract. At high concentrations, Ca^{2+} binding to troponin C shifts the position of the complex, allowing access of the myosin head groups to an increasing number of actins and allowing contraction to proceed.

Contractile Assemblies of Actin and Myosin in Nonmuscle Cells

Contractile assemblies of actin and myosin resembling small-scale versions of muscle fibers are present also in nonmuscle cells. As in muscle, the actin filaments in these contractile assemblies are interdigitated with bipolar filaments of myosin II, consisting of 15 to 20 myosin II molecules, which produce contraction by sliding the actin filaments relative to one another (Figure 12.29). The actin filaments in contractile bundles in nonmuscle cells are also associated with tropomyosin, which facilitates their interaction with myosin II.

Two examples of contractile assemblies in nonmuscle cells, stress fibers, and adhesion belts, were discussed earlier with respect to attachment of the actin cytoskeleton to regions of cell-substratum and cell-cell contacts (see Figures 12.16 and 12.17). The contraction of stress fibers produces tension across the cell, allowing the cell to pull on a substratum (e.g., the extracellular matrix) to which it is anchored. The contraction of adhesion belts alters

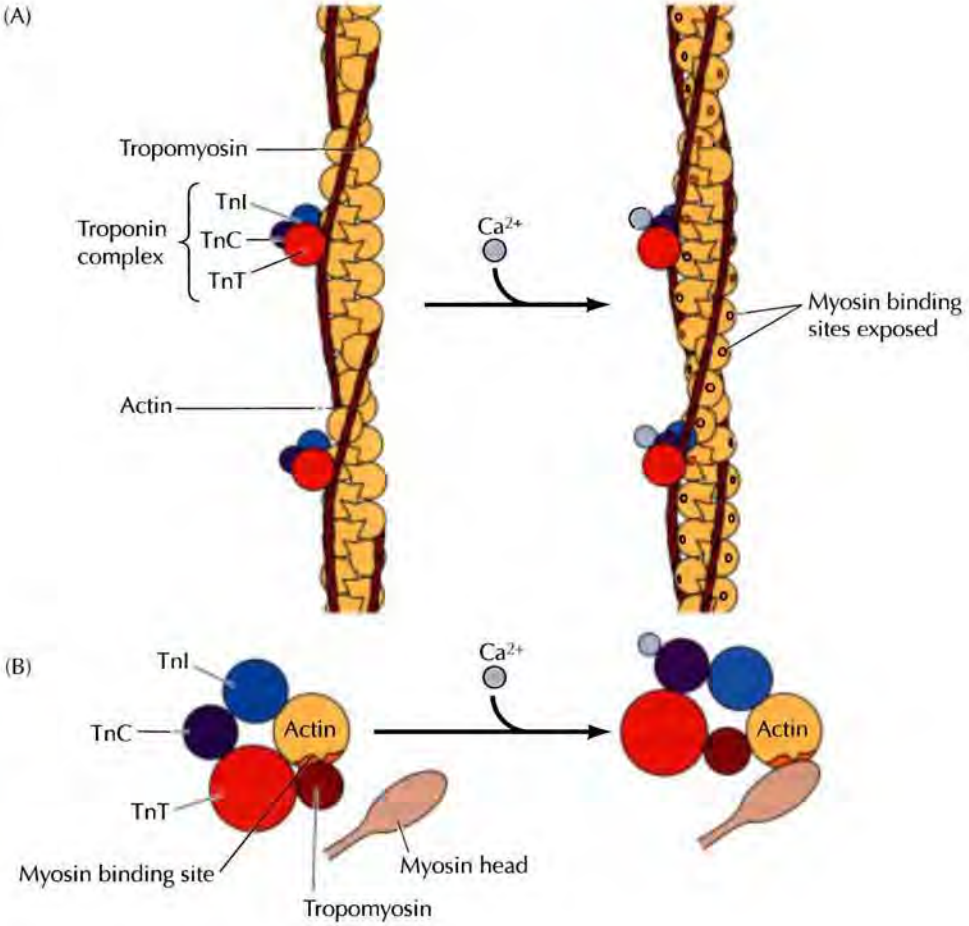


FIGURE 12.28 Association of tropomyosin and troponins with actin filaments (A) Tropomyosin binds lengthwise along actin filaments, and in striated muscle, is associated with a complex of three troponins: troponin I (TnI), troponin C (TnC), and troponin T (TnT). In the absence of Ca^{2+}, the tropomyosin-troponin complex blocks the binding of myosin to actin. Binding of Ca^{2+} to TnC shifts the complex, relieving this inhibition and allowing contraction to proceed. (B) Cross-sectional view.

12.2 WEBSITE ANIMATION

A Thin Filament The thin filaments of skeletal muscle consist of actin filaments decorated with tropomyosin-troponin complexes covering the myosin binding sites.

the shape of epithelial cell sheets: a process that is particularly important during embryonic development when sheets of epithelial cells fold into structures such as tubes.

The most dramatic example of actin-myosin contraction in nonmuscle cells, however, is provided by **cytokinesis**—the division of a cell into two cells following mitosis (Figure 12.30). Toward the end of mitosis in yeast and animal cells, a **contractile ring** consisting of actin filaments and myosin II assembles just underneath the plasma membrane. Its contraction pulls the plasma membrane progressively inward, constricting the center of the cell and pinching it in two. Interestingly, the thickness of the contractile ring remains constant as it contracts, implying that actin filaments disassemble as contraction proceeds. The ring then disperses completely following cell

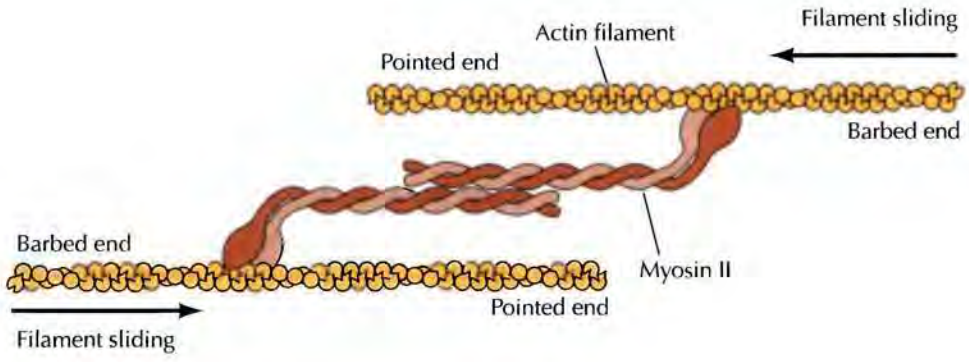


FIGURE 12.29 Contractile assemblies in nonmuscle cells Bipolar filaments of myosin II produce contraction by sliding actin filaments in opposite directions.

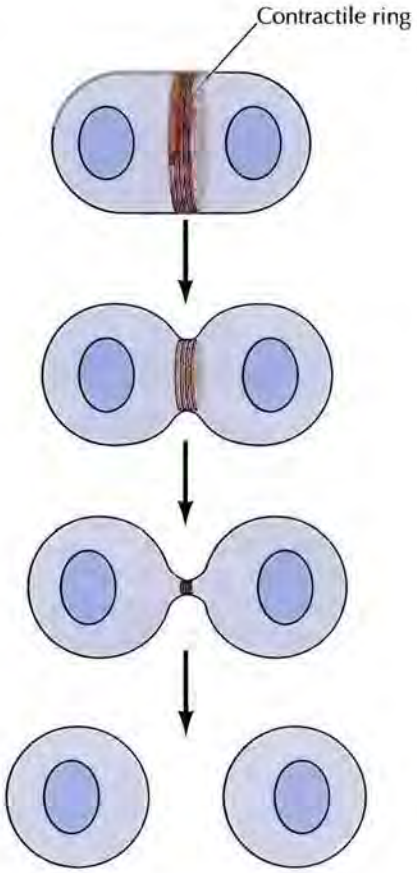


FIGURE 12.30 Cytokinesis Following completion of mitosis (nuclear division), a contractile ring consisting of actin filaments and myosin II divides the cell in two.

division. This mechanism appears to be evolutionarily ancient, as the bacterial MreB protein also functions in cell division.

The regulation of actin-myosin contraction in striated muscle, discussed earlier, is mediated by the binding of Ca^{2+} to troponin. In nonmuscle cells and in smooth muscle, however, contraction is regulated primarily by phosphorylation of one of the myosin light chains called the regulatory light chain (Figure 12.31). Phosphorylation of the regulatory light chain in these cells has at least two effects: It promotes the assembly of myosin into filaments, and it increases myosin catalytic activity enabling contraction to proceed. The enzyme that catalyzes this phosphorylation, called **myosin light-chain kinase**, is itself regulated by association with the Ca^{2+}-binding protein **calmodulin**. Increases in cytosolic Ca^{2+} promote the binding of calmodulin to the kinase resulting in phosphorylation of the myosin regulatory light chain. Increases in cytosolic Ca^{2+} are thus responsible, albeit indirectly, for activating myosin in smooth muscle and nonmuscle cells, as well as in striated muscle.

Nonmuscle Myosins

In addition to myosin II (the two-headed myosins found in muscle cells), several other types of myosin are found in nonmuscle cells. These myosins do not have tails able to form coiled-coils, so they do not form filaments and are not involved in contraction. However, they are important in a variety of cell movements, such as the transport of membrane vesicles and organelles along actin filaments, phagocytosis, and extension of pseudopods in amoebae (see Figure 12.20).

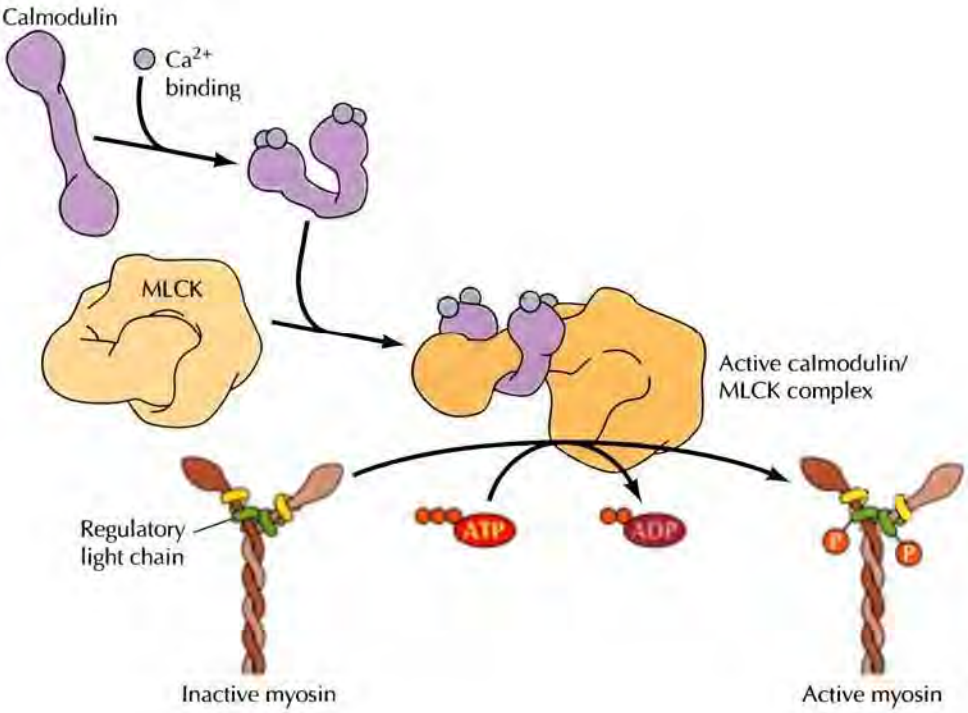


FIGURE 12.31 Regulation of myosin by phosphorylation Ca^{2+} binds to calmodulin, which in turn binds to myosin light-chain kinase (MLCK). The active calmodulin-MLCK complex then phosphorylates the myosin II regulatory light chain converting myosin from an inactive to an active state.

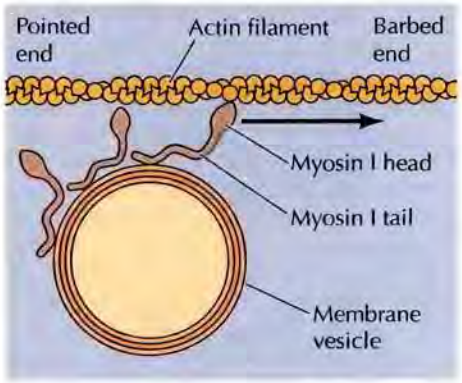


FIGURE 12.32 Myosin I Myosin I contains a head group similar to myosin II, but it has a comparatively short tail and does not form dimers or filaments. Although it cannot induce contraction, myosin I can move along actin filaments (toward the barbed end), carrying a variety of cargoes (such as membrane vesicles) attached to its tail.

One family of nonmuscle myosins are members of the **myosin I** family (Figure 12.32). The myosin I proteins contain a globular head group that acts as a molecular motor, like that of myosin II, but myosin I proteins are much smaller molecules (about 110 kd in mammalian cells) that lack the long tail of myosin II. Rather than forming dimers, their tails bind to other structures, such as membrane vesicles or organelles. The movement of myosin I along an actin filament can then transport its attached cargo. One function of myosin I, discussed earlier, is to form the lateral arms that link actin bundles to the plasma membrane of intestinal microvilli (see Figure 12.19). In these structures, the motor activity of myosin I may move the plasma membrane along the actin bundles toward the tip of the microvillus. Additional functions of myosin I may include the transport of vesicles and organelles along actin filaments and the movement of the plasma membrane during phagocytosis and pseudopod extension.

In addition to myosins I and II at least 12 other classes of nonmuscle myosins (III through XIV) have been identified. Some of these nonmuscle myosins are one-headed like myosin I, whereas others like myosin V (Figure 12.33) are two-headed like myosin II. The functions of most of these nonmuscle myosins remain to be determined, but some have been shown to play important roles in cargo transport and organelle movement (myosins V and VI) and in sensory functions such as vision (myosin III) and hearing (myosins VI and VII). Myosin VI is apparently unique among myosins in that it moves toward the pointed ends of actin filaments. Finally, some of

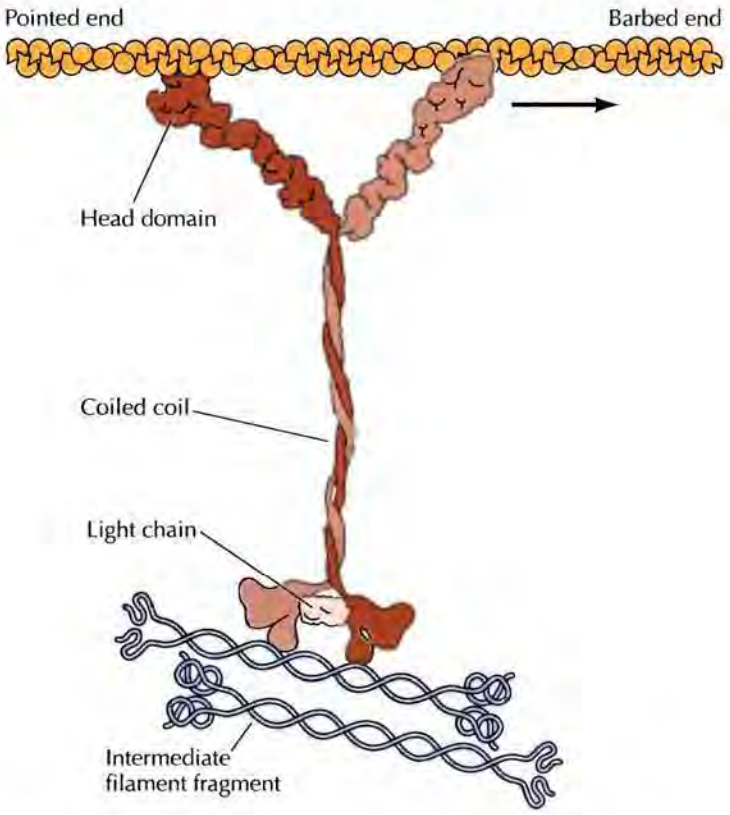


FIGURE 12.33 Myosin V Myosin V is a two-headed myosin like myosin II. It transports organelles and other cargo (for example, intermediate filaments) toward the barbed ends of actin filaments. The model shown is based on X-ray crystallography data (R. D. Vale, 2003. *Cell* 112: 467).

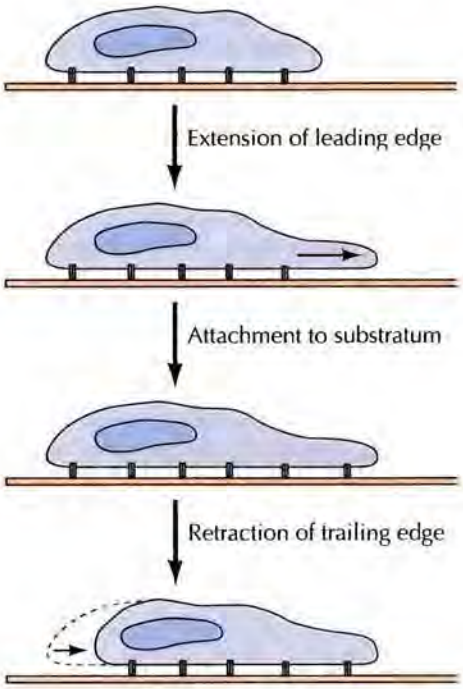


FIGURE 12.34 Cell migration The movement of cells across a surface can be viewed as three stages of coordinated movements: (1) extension of the leading edge, (2) attachment of the leading edge to the substratum, and (3) retraction of the rear of the cell into the cell body.

the other myosins do not move cargo but participate in actin filament reorganization or anchor actin filaments to the plasma membrane.

Formation of Protrusions and Cell Movement

The movement of cells across a surface represents a basic form of cell locomotion employed by a wide variety of different kinds of cells. Examples include the crawling of amoebas, the migration of embryonic cells during development, the invasion of tissues by white blood cells to fight infection, the migration of cells involved in wound healing, and the spread of cancer cells during the metastasis of malignant tumors. Similar types of movement are also responsible for phagocytosis and for the extension of nerve cell processes during development of the nervous system. All of these movements are based on local specializations and extensions of the plasma membrane driven by the dynamic properties of the actin cytoskeleton.

Cell movement or the extension of long cellular processes involves a coordinated cycle of movements, which can be viewed in several stages (Figure 12.34). First, cells must develop an initial polarity via specialization of the plasma membrane or the cell cortex. Second, protrusions such as pseudopodia, lamellipodia, or filopodia (see Figure 12.20) must be extended to establish a leading edge of the cell. These extensions must then attach to the substratum across which the cell is moving. Finally, during cell migration the trailing edge of the cell must dissociate from the substratum and retract into the cell body. A variety of experiments indicate that extension of the leading edge involves the branching and polymerization of actin filaments. For example, inhibition of actin polymerization (e.g., by treatment with cytochalasin) blocks the formation of cell surface protrusions. As described below, the process that underlies the extension of cell protrusions is the force of actin filament polymerization against the cell membrane at the leading edge.

■ **Certain pathogenic bacteria subvert the normal dynamics of the actin cytoskeleton to move through the host cell cytoplasm and infect neighboring cells.**

In most cases, cells move in response to signals from other cells or the environment. For example, in wound healing, cells at the edge of a cut extend lamellipodia and move across the extracellular matrix or the underlying cells to cover the wound. The signals that stimulate cell movement activate receptors in a small area of the cell membrane, leading to the recruitment of membrane proteins and the formation of specialized lipids in that area. These proteins and lipids in turn recruit actin binding proteins, including the Arp2/3 complex, its activator the **WASP/Scar complex,** and barbed-end tracking proteins that connect the growing actin filaments to the plasma membrane (Figure 12.35). WASP/Scar activates the Arp2/3 complex, which then initiates actin filament branches near the barbed ends, thereby increasing the number of growing barbed ends that are able to push against the cell membrane. At high local concentrations of ATP-actin, the growth of the barbed ends of actin filaments is energetically favorable and can generate considerable force. While a single filament would not exert enough force to extend the cell membrane, many filaments can easily do so. As the barbed ends of the actin filaments at the leading edge branch and grow, the pointed ends of the filaments are actively disassembled by

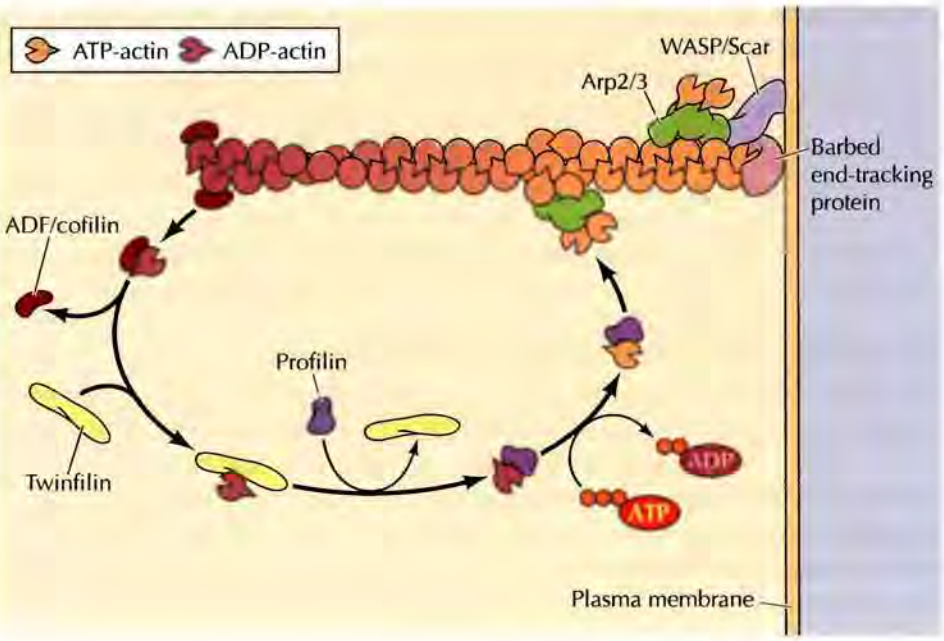


FIGURE 12.35 Actin filament branching at the leading edge The Arp2/3 complex, WASP/Scar proteins, and barbed end-tracking proteins are recruited to a small area of the plasma membrane at the leading edge of a cell. WASP/Scar activates the Arp2/3 complex to initiate actin filament branches near the barbed ends, which are connected to the plasma membrane by barbed end-tracking proteins. At the pointed ends of the filaments, monomers of ADP-actin are removed by ADF/cofilin. The ADP-actin monomers are carried to the growing barbed ends by twinfilin and reactivated through ADP/ATP exchange by profilin.

ADF/cofilin. The ADP-actin monomers are reactivated through ADP/ATP exchange by profilin and carried to the growing barbed ends by **twinfilin**. As the new microfilaments extend into the growing cell process, reorganized microtubules and the new microfilaments provide pathways for the delivery of lipid vesicles and proteins needed for continued extension. In neurons, myosin V is required to provide the new membrane components for extension of filopodia. The regulation of this entire process involves small GTP-binding proteins of the Rho family, which are discussed in Chapter 15.

Cell attachment to the surface requires rebuilding cell-substratum or cell-cell adhesions. For slow-moving cells, such as epithelial cells or fibroblasts, attachment involves the formation of focal adhesions (see Figure 12.16). Among the cargo proteins delivered to the growing cell protrusion by actin filaments and microtubules are actin-bundling proteins (which are needed for the formation of actin bundles and stress fibers immediately behind the leading edge) as well as focal adhesion proteins, such as talin and vinculin. Intermediate filament proteins are also transported toward the leading edge where they are used for the reorganization of the intermediate filament meshwork.

Reconstruction of focal adhesions occurs in two steps: the appearance of small focal complexes containing a few microfilaments attached to integrins and the growth of those focal complexes into mature focal contacts (illustrated in Figure 12.16). Vinculin and talin are activated by contact with the cell membrane lipids. They activate integrins to bind to the extracellular matrix as well as connecting the integrins to actin filaments. The appearance of mature focal contacts also requires the development of tension between the cell and the substratum, which is generated by myosin motors acting on actin bundles or stress fibers. Cells moving more rapidly, such as amoebas or white blood cells, form more diffuse contacts with the substratum, the molecular composition of which is not known.

The final stage of cell migration—retraction of the trailing edge—involves the action of small GTP-binding proteins of the ARF family (see Chapter 10), which break down existing focal adhesions at the trailing edge

of the cell. The myosin II-mediated sliding of microfilaments in actin bundles and stress fibers connected to the new focal adhesions formed at the leading edge then generates the force necessary to pull the trailing edge of the cell forward.

Intermediate Filaments

Intermediate filaments have diameters between 8 and 11 nm, which is intermediate between the diameters of the two other principal elements of the cytoskeleton, actin filaments (about 7 nm) and microtubules (about 25 nm). In contrast to actin filaments and microtubules the intermediate filaments are not directly involved in cell movements. Instead, they appear to play basically a structural role by providing mechanical strength to cells and tissues.

Intermediate Filament Proteins

Whereas actin filaments and microtubules are polymers of single types of proteins (actin and tubulin, respectively), intermediate filaments are composed of a variety of proteins that are expressed in different types of cells. More than 65 different intermediate filament proteins have been identified and classified into six groups based on similarities between their amino acid sequences (Table 12.2). Types I and II consist of two groups of **keratins**, each consisting of about 15 different proteins, which are expressed in epithelial cells. Each type of epithelial cell synthesizes at least one type I (acidic) and one type II (neutral/basic) keratin, which copolymerize to form filaments. Some type I and type II keratins (called hard keratins) are used for production of structures such as hair, nails, and horns. The other type I and type II keratins (soft keratins) are abundant in the cytoplasm of epithelial cells, with different keratins being expressed in various differentiated cell types.

TABLE 12.2 Intermediate Filament Proteins

Type	Protein	Size (kd)	Site of expression
I	Acidic keratins (~15 proteins)	40–60	Epithelial cells
II	Neutral or basic keratins (~15 proteins)	50–70	Epithelial cells
III	Vimentin	54	Fibroblasts, white blood cells, and other cell types
	Desmin	53	Muscle cells
	Glial fibrillary acidic protein	51	Glial cells
	Peripherin	57	Peripheral neurons
IV	Neurofilament proteins		
	NF-L	67	Neurons
	NF-M	150	Neurons
	NF-H	200	Neurons
	α-internexin	66	Neurons
V	Nuclear lamins	60–75	Nuclear lamina of all cell types
VI	Nestin	200	Stem cells, especially of the central nervous system

Nestins are sometimes classified as type IV rather than type VI intermediate filaments

The type III intermediate filament proteins include **vimentin**, which is found in several different kinds of cells, including fibroblasts, smooth muscle cells, and white blood cells. Unlike actin filaments, vimentin forms a network extending out from the nucleus toward the cell periphery. Another type III protein, **desmin**, is specifically expressed in muscle cells where it connects the Z discs of individual contractile elements. A third type III intermediate filament protein is specifically expressed in glial cells, and a fourth is expressed in neurons of the peripheral nervous system.

The type IV intermediate filament proteins include the three **neurofilament (NF) proteins** (designated NF-L, NF-M, and NF-H for light, medium, and heavy, respectively). These proteins form the major intermediate filaments of many types of mature neurons. They are particularly abundant in the axons of motor neurons and are thought to play a critical role in supporting these long, thin processes, which can extend more than a meter in length. Another type IV protein (α-internexin) is expressed at an earlier stage of neuron development prior to expression of the neurofilament proteins. The type V intermediate filament proteins are the nuclear lamins, which are found in most eukaryotic cells. Rather than being part of the cytoskeleton, the nuclear lamins are components of the nucleus where they assemble to form an orthogonal meshwork underlying the nuclear membrane and extending more diffusely into the nucleus (see Figure 9.5). Nestins (the type VI intermediate filaments) are expressed during embryonic development in several types of stem cells. They differ from other intermediate filaments in that they only polymerize if other intermediate filaments are present in the cell; nestins are sometimes classified as another type IV rather a type VI intermediate filament.

Despite considerable diversity in size and amino acid sequence, the various intermediate filament proteins share a common structural organization (Figure 12.36). All of the intermediate filament proteins have a central α-helical rod domain of approximately 310 amino acids (350 amino acids in the nuclear lamins). This central rod domain is flanked by amino- and carboxy-terminal domains, which vary among the different intermediate filament proteins in size, sequence, and secondary structure. As discussed next, the α-helical rod domain plays a central role in filament assembly, while the variable head and tail domains presumably determine the specific functions of the different intermediate filament proteins.

Assembly of Intermediate Filaments

The first stage of filament assembly is the formation of dimers in which the central rod domains of two polypeptide chains are wound around each other in a coiled-coil structure similar to that formed by myosin II heavy chains (Figure 12.37). The dimers of cytoskeletal intermediate filaments then

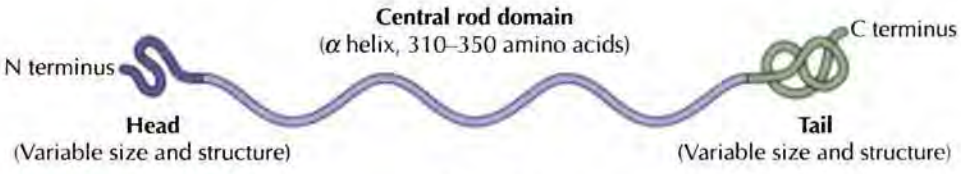


FIGURE 12.36 Structure of intermediate filament proteins Intermediate filament proteins contain a central α-helical rod domain of approximately 310 amino acids (350 amino acids in the nuclear lamins). The N-terminal head and C-terminal tail domains vary in size and shape.

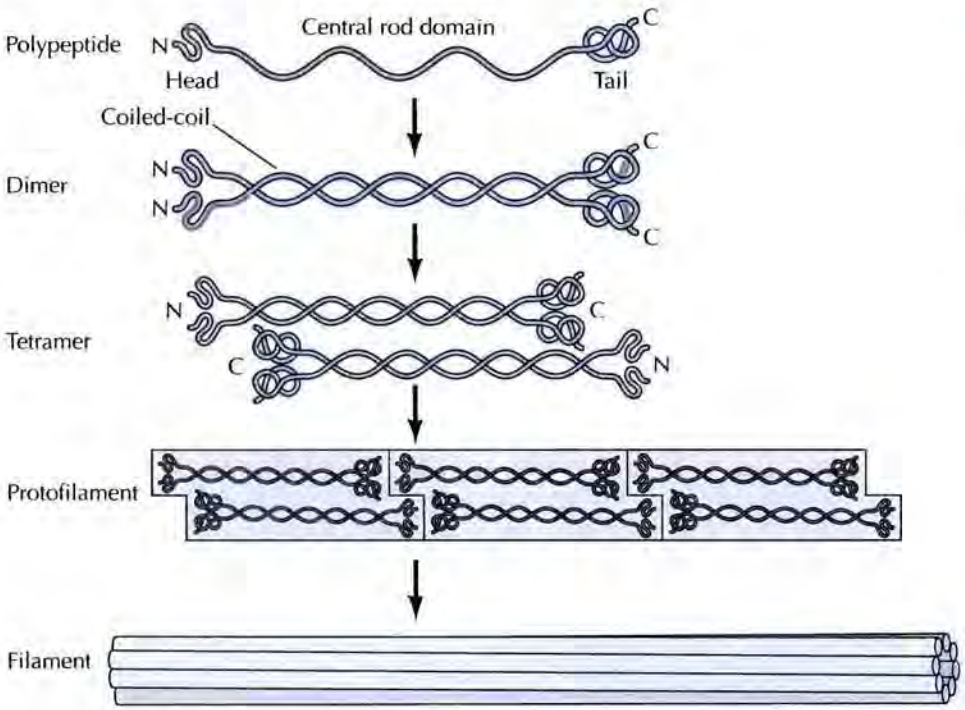


FIGURE 12.37 Assembly of intermediate filaments The central rod domains of two polypeptides wind around each other in a coiled-coil structure to form dimers. Dimers then associate in a staggered antiparallel fashion to form tetramers. Tetramers associate end-to-end to form protofilaments and laterally to form filaments. Each filament contains approximately eight protofilaments wound around each other in a ropelike structure.

associate in a staggered antiparallel fashion to form tetramers, which can assemble end-to-end to form protofilaments. A common step is the interaction of approximately eight protofilaments wound around each other in a ropelike structure. Because they are assembled from antiparallel tetramers, both ends of intermediate filaments are equivalent. Consequently, in contrast to actin filaments and microtubules, intermediate filaments are apolar—they do not have distinct plus and minus ends.

Filament assembly requires interactions between specific types of intermediate filament proteins. For example, keratin filaments are always assembled from heterodimers containing one type I and one type II polypeptide. In contrast, the type III proteins can assemble into filaments containing only a single polypeptide (e.g., vimentin) or consisting of two different type III proteins (e.g., vimentin plus desmin). The type III proteins do not, however, form copolymers with the keratins. Among the type IV proteins, α-internexin can assemble into filaments by itself, whereas the three neurofilament proteins copolymerize to form heteropolymers.

Intermediate filaments are generally more stable than actin filaments or microtubules and do not exhibit the dynamic behavior associated with these other elements of the cytoskeleton (e.g., the treadmilling of actin filaments illustrated in Figure 12.4). However, intermediate filament proteins are frequently modified by phosphorylation, which can regulate their assembly and disassembly within the cell. One example is phosphorylation of the nuclear lamins (see Chapter 16), which results in disassembly of the nuclear lamina and breakdown of the nuclear envelope during mitosis. Cytoplasmic intermediate filaments, such as vimentin, are also phosphorylated, which can lead to their disassembly and reorganization in dividing or migrating cells.

Intracellular Organization of Intermediate Filaments

Intermediate filaments form an elaborate network in the cytoplasm of most cells, extending from a ring surrounding the nucleus to the plasma mem-

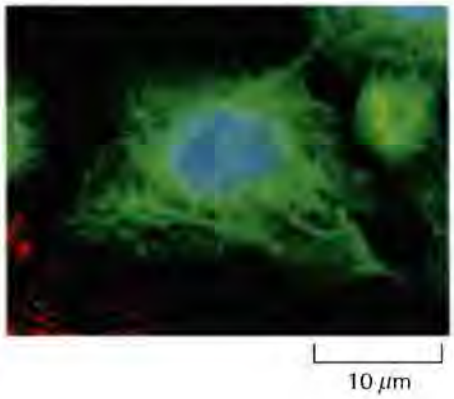


FIGURE 12.38 Intracellular organization of keratin filaments Micrograph of epithelial cells stained with fluorescent antibodies to keratin (green). Nuclei are stained blue. The keratin filaments extend from a ring surrounding the nucleus to the plasma membrane. (Nancy Kedersha/Immunogen/Photo Researchers, Inc.)

brane (Figure 12.38). Both keratin and vimentin filaments attach to the nuclear envelope, apparently serving to position and anchor the nucleus within the cell. In addition, intermediate filaments can associate not only with the plasma membrane but also with the other elements of the cytoskeleton, actin filaments and microtubules. Intermediate filaments thus provide a scaffold that integrates the components of the cytoskeleton and organizes the internal structure of the cell.

The keratin filaments of epithelial cells are tightly anchored to the plasma membrane at two areas of specialized cell contacts: **desmosomes** and **hemidesmosomes** (Figure 12.39). Desmosomes are junctions between adjacent cells at which cell-cell contacts are mediated by transmembrane proteins related to the cadherins. On their cytoplasmic side, desmosomes are associated with a characteristic dense plaque of intracellular proteins to which keratin filaments are attached. These attachments are mediated by desmoplakin, a member of a family of proteins called **plakins** that bind intermediate filaments and link them to other cellular structures. Hemidesmosomes are morphologically similar junctions between epithelial cells and underlying connective tissue at which keratin filaments are linked by different members of the plakin family (e.g., plectin) to integrins. Desmosomes and hemidesmosomes thus anchor intermediate filaments to regions of cell-cell and cell-substratum contact, respectively, similar to the attachment of the actin cytoskeleton to the plasma membrane at adherens junctions and focal adhesions. It is important to note that the keratin filaments anchored to both sides of desmosomes serve as a mechanical link between adjacent cells in an epithelial layer, thereby providing mechanical stability to the entire tissue.

In addition to linking intermediate filaments to cell junctions, some plakins link intermediate filaments to other elements of the cytoskeleton. Plectin, for example, binds actin filaments and microtubules in addition to intermediate filaments, so it can provide bridges between these cytoskeletal components (Figure 12.40). These bridges to intermediate filaments are thought to brace and stabilize actin filaments and microtubules, thereby increasing the mechanical stability of the cell.

Two types of intermediate filaments—desmin and the neurofilaments—play specialized roles in muscle and nerve cells, respectively. Desmin connects the individual actin-myosin assemblies of muscle cells both to one another and to the plasma membrane, thereby linking the actions of individual contractile elements. Neurofilaments are the major intermediate filaments in most mature neurons. They are particularly abundant in the long axons of motor neurons where they appear to be anchored to actin filaments and microtubules by neuronal members of the plakin family. Neuro-

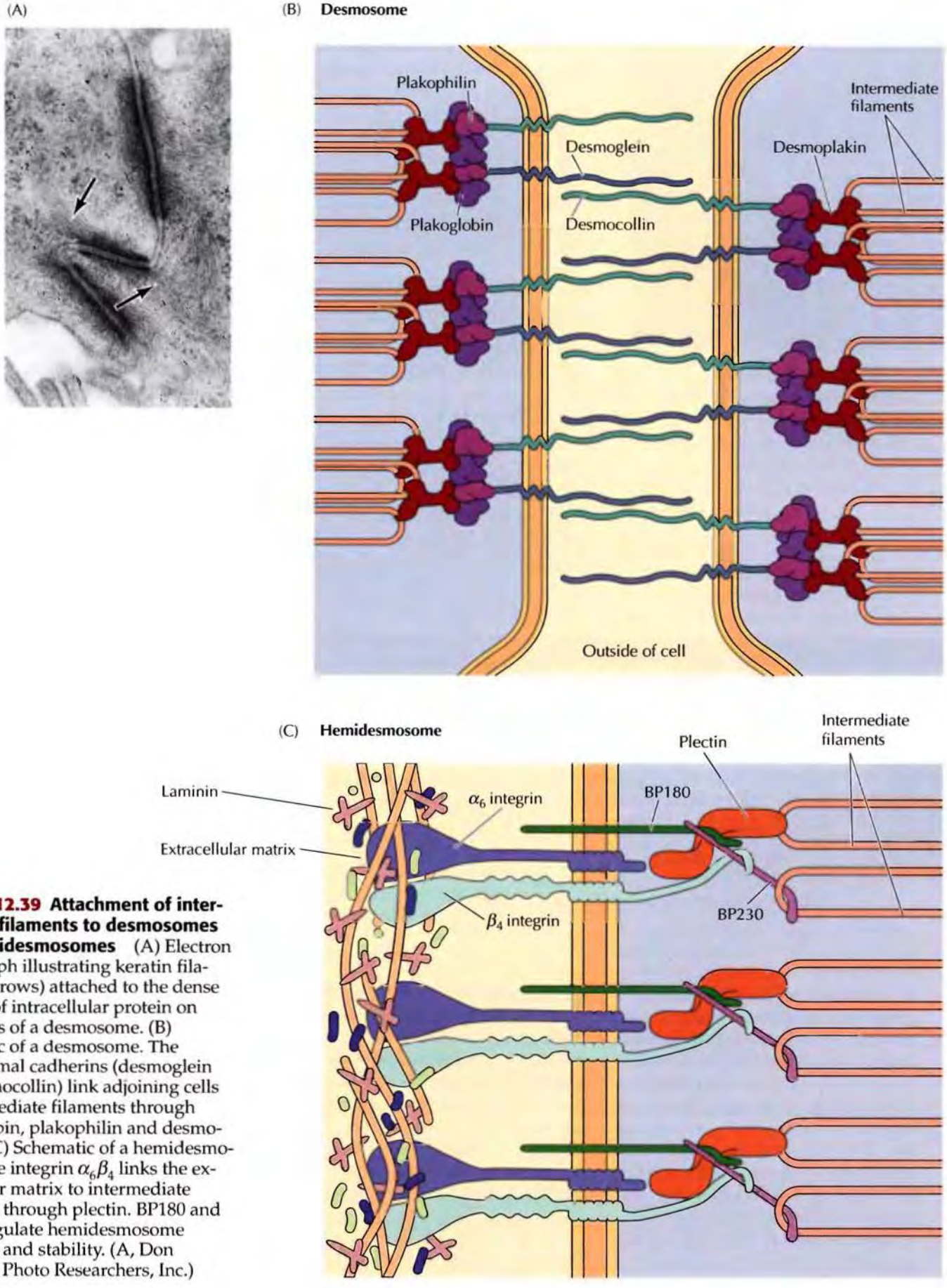


FIGURE 12.39 Attachment of intermediate filaments to desmosomes and hemidesmosomes (A) Electron micrograph illustrating keratin filaments (arrows) attached to the dense plaques of intracellular protein on both sides of a desmosome. (B) Schematic of a desmosome. The desmosomal cadherins (desmoglein and desmocollin) link adjoining cells to intermediate filaments through plakoglobin, plakophilin and desmoplakin. (C) Schematic of a hemidesmosome. The integrin $\alpha_6\beta_4$ links the extracellular matrix to intermediate filaments through plectin. BP180 and BP230 regulate hemidesmosome assembly and stability. (A, Don Fawcett/ Photo Researchers, Inc.)

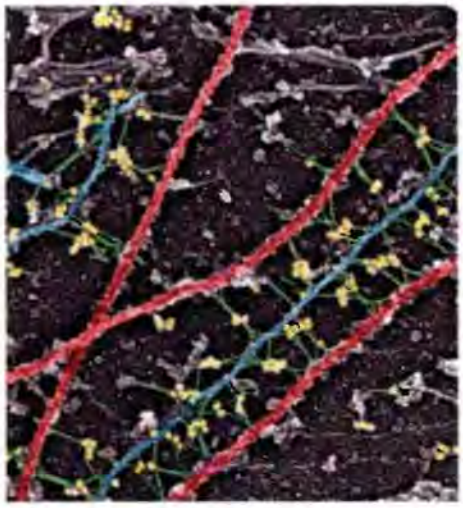


FIGURE 12.40 Electron micrograph of plectin bridges between intermediate filaments and microtubules Micrograph of a fibroblast stained with antibody against plectin. The micrograph has been artificially colored to show plectin (green), antibodies against plectin (yellow), intermediate filaments (blue), and microtubules (red). (Courtesy of Tatyana Svitkina and Gary Borisy, University of Wisconsin, Madison.)

filaments are thought to play an important role in providing mechanical support and stabilizing other elements of the cytoskeleton in these long, thin extensions of nerve cells.

Functions of Keratins and Neurofilaments: Diseases of the Skin and Nervous System

Although intermediate filaments have long been thought to provide structural support to the cell, direct evidence for their function has only recently been obtained. Some cells in culture make no intermediate filament proteins, indicating that these proteins are not required for the growth of cells *in vitro*. Similarly, injection of cultured cells with antibody against vimentin disrupts intermediate filament networks without affecting cell growth or movement. Therefore, it has been thought that intermediate filaments are most needed to strengthen the cytoskeleton of cells in the tissues of multicellular organisms where they are subjected to a variety of mechanical stresses that do not affect cells in the isolated environment of a culture dish.

KEY EXPERIMENT

Expression of Mutant Keratin Causes Abnormal Skin Development

Mutant Keratin Expression in Transgenic Mice Causes Marked Abnormalities Resembling a Human Genetic Skin Disease

Robert Vassar, Pierre A. Coulombe, Linda Degenstein, Kathryn Albers and Elaine Fuchs

University of Chicago

Cell, 1991, Volume 64, pages 365–380

Elaine Fuchs

The Context

By 1991 intermediate filaments in epithelial cells were well known and the developmental appearance of different forms of type I and type II keratins in the skin was being studied. What remained unknown for any intermediate filaments was their function. While all vertebrate cells contain intermediate filaments, cell lines that lack them survive in culture and continue to proliferate. Thus whatever function intermediate filaments had, they were important not for cells in culture but perhaps for cells within the tissues of multicellular organisms.

Fuchs and coworkers knew that during early development the epithelium of the skin expresses keratin 5 (type I) and keratin 14 (type II). Because keratins polymerize as heterodimers, it was thought that expression of an abnormal protein might interfere with the formation of normal intermediate filaments. Fuchs and her colleagues initially tested this possibility in cultured skin cells and demonstrated that expression of a truncated keratin 14 interfered with keratin filament formation. This suggested that similar expression of the mutant keratin in transgenic mice might cause a defect in the intermediate filament network in the skin cells of an embryo. If this occurred, it would provide a test of the role of intermediate filaments in an intact tissue.

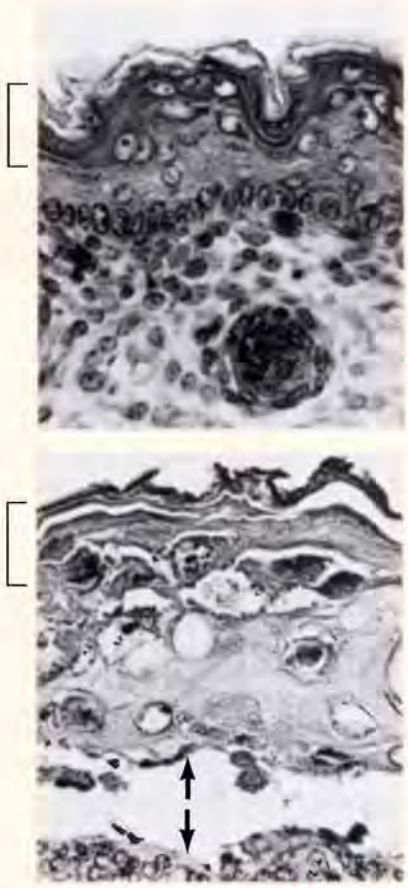
KEY EXPERIMENT

In the experiments described here, Fuchs and her colleagues demonstrated that expression of a mutant keratin in transgenic mice not only disrupted the intermediate filament network of skin cells but also led to severe defects in the organization and tissue stability of the skin. These experiments thus provided the first demonstration of a physiological role for intermediate filaments.

The Experiments

For their earlier experiments in cultured cells, Fuchs and colleagues had constructed a mutant keratin 14 gene in which a truncated protein missing 30% of the central α-helical domain and all of the carboxy-terminal tail was expressed from the normal keratin promoter. To investigate the role of keratin 14 in early mouse development, they introduced the plasmid encoding this mutant keratin 14 into fertilized mouse eggs, which were transferred to foster mothers and allowed to develop into offspring. All of the offspring were analyzed for keratin 14, and some were found to be transgenic and express the mutant keratin 14 protein.

Most of the transgenic animals died within 24 hours of birth. Those that survived longer showed severe skin abnormalities, including blisters due to epidermal cell lysis following mild mechanical trauma, such as rubbing of the skin. Analysis of stained tissue sections of the skin from transgenic animals demonstrated severe disorganization of the epidermis in the most affected animals (see figure) and patches of disorganized tissue in others. Patchy expression is characteristic of a mosaic animal where some tissue develops from normal embryo cells and some from cells carrying the transgene. By analyzing for the mutant protein, they found that the areas of disorganized tissue correlated with expression of the mutant keratin 14. In addition, there was a clear correlation between the amount of disorganized epidermis and susceptibility to skin damage and death during the trauma of birth.



Skin from a normal and a transgenic mouse. (Top) Skin from a normal mouse showing the highly organized outer layers (brackets) with no intervening spaces between the intact underlying tissue. (Bottom) Skin from a transgenic mouse showing severe disruption of the outer layers, which contain spaces due to abrasion by mechanical trauma and are separated from the underlying tissue (arrows).

Fuchs and her colleagues further noted that the pattern of tissue disorganization in the transgenic mice resembled that seen in a group of human skin diseases called epidermolysis bullosa simplex. Thus they compared sections of the transgenic mouse tissue with sections obtained from the skin of a human patient and found very similar patterns of tissue disruption. From this, Fuchs and coworkers concluded that defects in keratins or intermediate filament-related proteins might be a cause of human genetic diseases of the skin.

The Impact

The skin abnormalities of these transgenic mice provided the first direct support for the presumed role of keratins in providing mechanical strength to epithelial cells in tissues. It is now known that the intermediate filament cytoskeleton is critical for the structure of tissues such as skin, intestine, heart, and skeletal muscle that are subject to mechanical stress. In contrast, single-celled eukaryotes such as yeast and many small invertebrates survive perfectly well without intermediate filaments, often modifying the actin or tubulin cytoskeleton to serve a structural role. The results of Fuchs and her colleagues also suggested a basis for several human diseases. In fact, there are now more than 17 different keratins known to be defective in human disease; these include keratin 5 and keratin 14, both of which were initially studied in the transgenic mice.

Experimental evidence for such an *in vivo* role of intermediate filaments was first provided in 1991 by studies in the laboratory of Elaine Fuchs. These investigators used transgenic mice to investigate the *in vivo* effects of expressing a keratin deletion mutant encoding a truncated polypeptide that disrupted the formation of normal keratin filaments (Figure 12.41). This

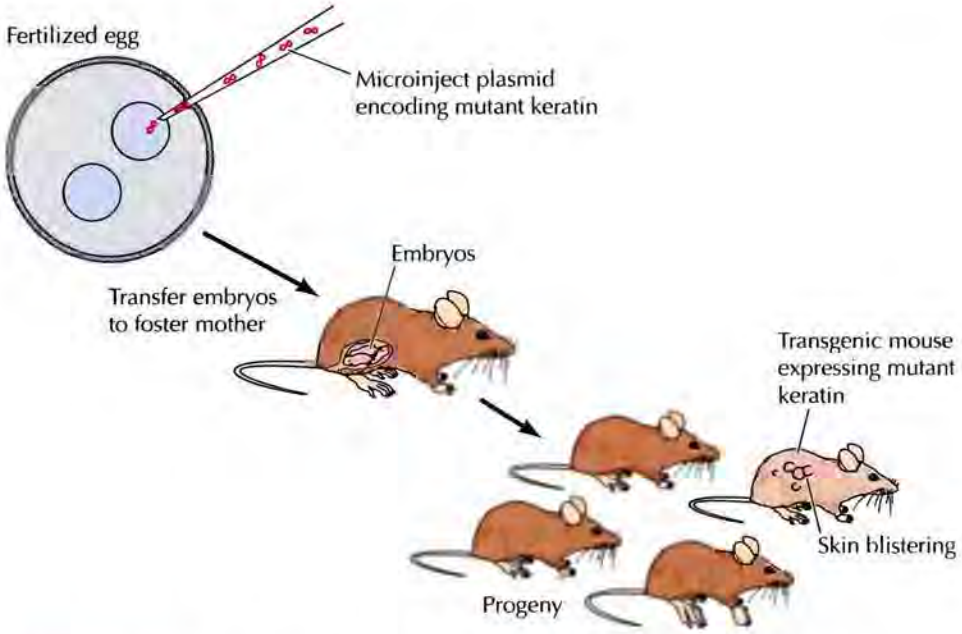


FIGURE 12.41 Experimental demonstration of keratin function A plasmid encoding a mutant keratin that interferes with the normal assembly of keratin filaments was microinjected into one pronucleus of a fertilized egg. Microinjected embryos were then transferred to a foster mother, and some of the offspring were found to have incorporated the mutant keratin gene into their genome. Expression of the mutant gene in these transgenic mice disrupted the keratin cytoskeleton of cells of the epidermis, resulting in severe skin blistering due to cell lysis following mild mechanical stress.

■ **The physical properties of intermediate filaments are well suited to their role in providing structural support. They are not normally very rigid but harden and resist breakage when subjected to high stress.**

mutant keratin gene was introduced into transgenic mice where it was expressed in basal cells of the epidermis and disrupted formation of a normal keratin cytoskeleton. This resulted in the development of severe skin abnormalities, including blisters due to epidermal cell lysis following mild mechanical trauma, such as rubbing of the skin. The skin abnormalities of these transgenic mice thus provided direct support for the presumed role of keratins in providing mechanical strength to epithelial cells in tissues. Subsequently, the same result has been shown in mice in which the gene for the same keratin was inactivated by homologous recombination.

These experiments also pointed to the molecular basis of a human genetic disease, epidermolysis bullosa simplex (EBS). Like the transgenic mice expressing mutant keratin genes, patients with this disease develop skin blisters resulting from cell lysis after minor trauma. This similarity prompted studies of the keratin genes in EBS patients, leading to the demonstration that EBS is caused by keratin gene mutations that interfere with the normal assembly of keratin filaments. Thus both experimental studies in transgenic mice and molecular analysis of a human genetic disease have demonstrated the role of keratins in allowing skin cells to withstand mechanical stress. Continuing studies have shown that mutations in other keratins are responsible for several other inherited skin diseases, which are similarly characterized by abnormal fragility of epidermal cells.

Other studies in transgenic mice have implicated abnormalities of neurofilaments in diseases of motor neurons, particularly amyotrophic lateral sclerosis (ALS). ALS, known as Lou Gehrig's disease (and the disease afflicting the renowned physicist Stephen Hawking), results from progres-

sive loss of motor neurons, which in turn leads to muscle atrophy, paralysis, and eventual death. ALS and other types of motor neuron disease are characterized by the accumulation and abnormal assembly of neurofilaments, suggesting that neurofilament abnormalities might contribute to these pathologies. Consistent with this possibility, overexpression of NF-L or NF-H in transgenic mice has been found to result in the development of a condition similar to ALS. Although the mechanism involved remains to be understood, these experiments clearly suggest the involvement of neurofilaments in the pathogenesis of motor neuron disease.

Microtubules

Microtubules, the third principal component of the cytoskeleton, are rigid hollow rods approximately 25 nm in diameter. Like actin filaments, microtubules are dynamic structures that undergo continual assembly and disassembly within the cell. They function both to determine cell shape and in a variety of cell movements, including some forms of cell locomotion, the intracellular transport of organelles, and the separation of chromosomes during mitosis.

Structure and Dynamic Organization of Microtubules

In contrast to intermediate filaments, which are composed of a variety of different fibrous proteins, microtubules are composed of a single type of globular protein called **tubulin**. The building blocks of microtubules are tubulin dimers consisting of two closely related 55 kd polypeptides: α-tubulin and β-tubulin. Like actin, both α- and β-tubulin are encoded by small families of related genes. In addition, a third type of tubulin (γ-tubulin) is concentrated in the centrosome where it plays a critical role in initiating microtubule assembly (discussed shortly). The evolutionary ancestor of all plant and animal tubulins appears to be a protein similar to the prokaryote protein, FtsZ.

Like actin, tubulin polymerization can be studied *in vitro*. Tubulin dimers polymerize to form microtubules, which generally consist of 13 linear protofilaments assembled around a hollow core (Figure 12.42). The protofilaments, which are composed of head-to-tail arrays of tubulin dimers, are arranged in parallel. Consequently, microtubules (like actin filaments) are polar structures with two distinct ends: a fast-growing plus end and a slow-growing minus end. This polarity is an important consideration in determining the direction of movement along microtubules, just as the polarity of actin filaments defines the direction of myosin movement.

Tubulin dimers can depolymerize as well as polymerize, and microtubules can undergo rapid cycles of assembly and disassembly. Both α- and β-tubulin bind GTP, which functions analogously to the ATP bound to actin to regulate polymerization. In particular, the GTP bound to β-tubulin (though not that bound to α-tubulin) is hydrolyzed to GDP during or shortly after polymerization. This GTP hydrolysis weakens the binding affinity of tubulin for adjacent molecules, thereby favoring depolymeriza-

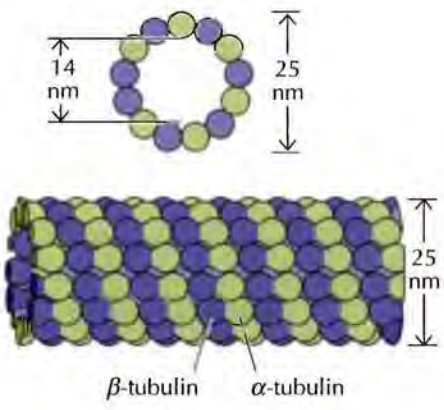


FIGURE 12.42 Structure of microtubules Dimers of α- and β-tubulin polymerize to form microtubules, which are composed of 13 protofilaments assembled around a hollow core.

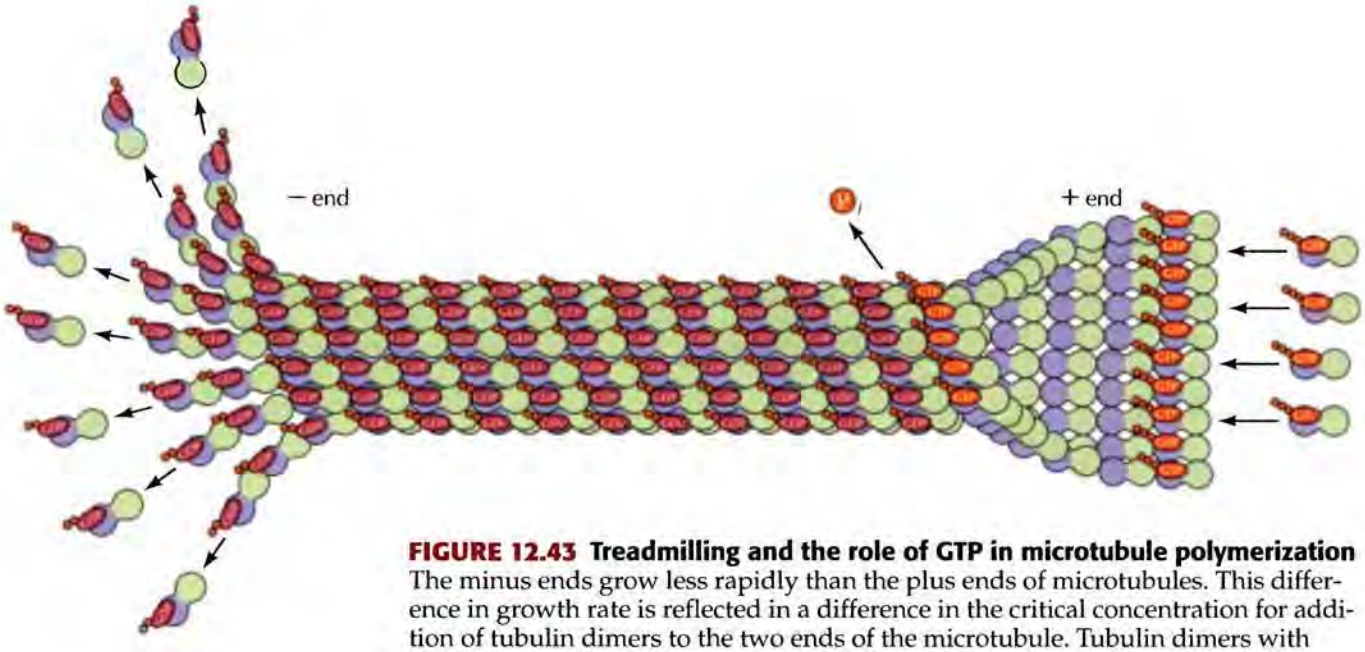


FIGURE 12.43 Treadmilling and the role of GTP in microtubule polymerization The minus ends grow less rapidly than the plus ends of microtubules. This difference in growth rate is reflected in a difference in the critical concentration for addition of tubulin dimers to the two ends of the microtubule. Tubulin dimers with GTP bound to β-tubulin associate with the rapidly growing plus ends in a flat sheet, which then zips up into the mature microtubule just behind the region of growth. Shortly after polymerization the GTP bound to β-tubulin is hydrolyzed to GDP, and since GDP-bound tubulin is less stable in the microtubule, the dimers at the minus end begin to peel off. Treadmilling takes place at tubulin dimer concentrations intermediate between the critical concentrations for the plus and minus ends. Under these conditions there is a net dissociation of dimers (bound to GDP) from the minus end, balanced by the addition of dimers (bound to GTP) to the plus end.

tion and resulting in the dynamic behavior of microtubules. Like actin filaments, microtubules undergo treadmilling (Figure 12.43), a dynamic behavior in which tubulin molecules bound to GDP are continually lost from the minus end and replaced by the addition of tubulin molecules bound to GTP to the plus end of the same microtubule. In microtubules, rapid GTP hydrolysis also results in the behavior known as **dynamic instability** in which individual microtubules alternate between cycles of growth and shrinkage (Figure 12.44). Whether a microtubule grows or shrinks is determined in part by the rate of tubulin addition relative to the rate of GTP hydrolysis. As long as new GTP-bound tubulin molecules are added more rapidly than GTP is hydrolyzed, the microtubule retains a GTP cap at its plus end and microtubule growth continues. However, if the rate of polymerization slows, the GTP bound to tubulin at the plus end of the microtubule will be hydrolyzed to GDP. If this occurs, the GDP-bound tubulin will dissociate, resulting in rapid depolymerization and shrinkage of the microtubule.

Dynamic instability, described by Tim Mitchison and Marc Kirschner in 1984, allows for the continual and rapid turnover of many microtubules within the cell, some of which have half-lives of only several minutes. As discussed later, this rapid turnover of microtubules is particularly critical for the remodeling of the cytoskeleton that occurs during mitosis. Because of the central role of microtubules in mitosis, drugs that affect microtubule assembly are useful not only as experimental tools in cell biology but also in the treatment of cancer. **Colchicine** and **colcemid** are examples of com-

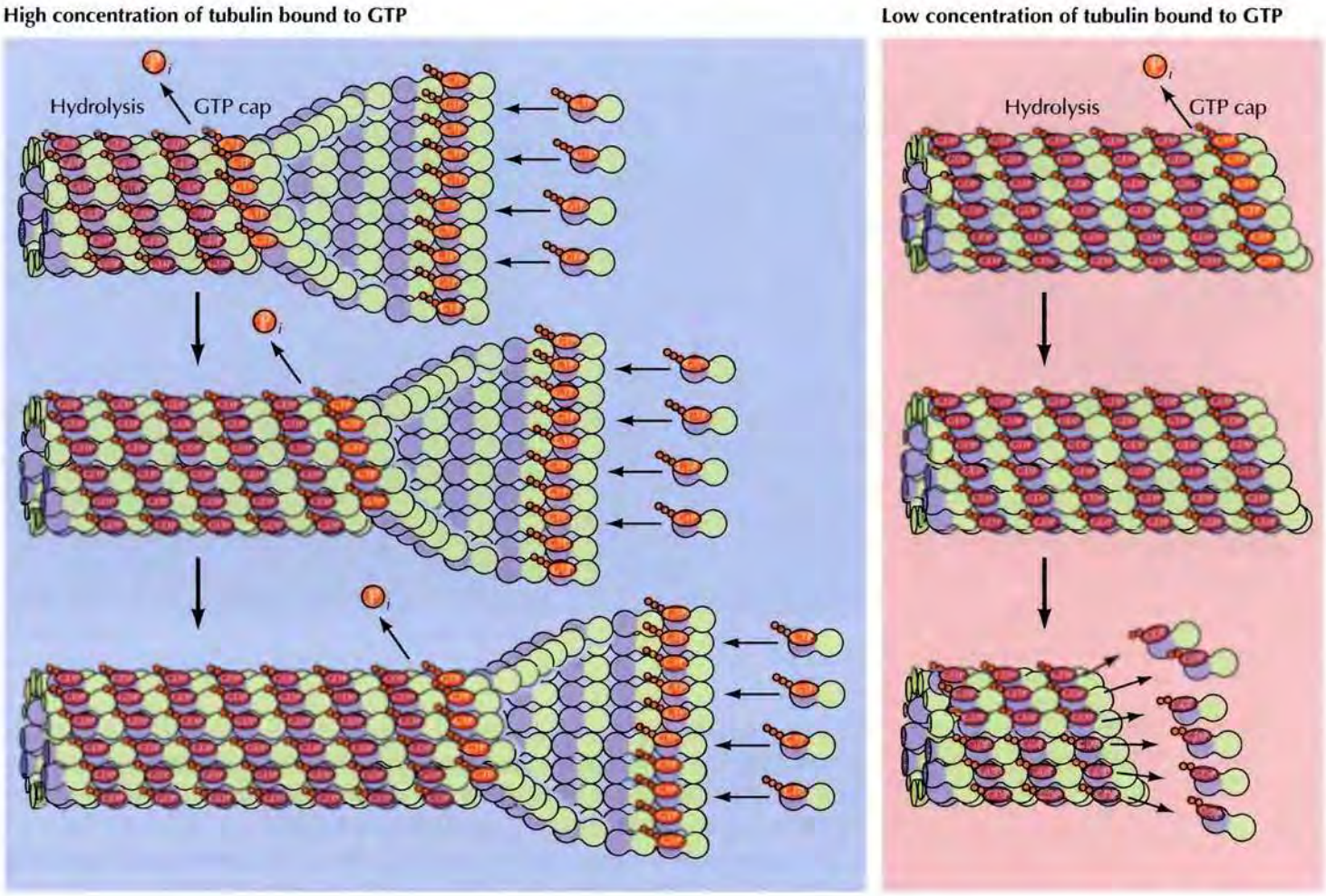


FIGURE 12.44 Dynamic instability of microtubules Dynamic instability results from the hydrolysis of GTP bound to β-tubulin during or shortly after polymerization, which reduces its binding affinity for adjacent molecules. Growth of microtubules continues as long as there is a high concentration of tubulin bound to GTP. New GTP-bound tubulin molecules are then added more rapidly than GTP is hydrolyzed, so a GTP cap is retained at the growing end. However, if GTP is hydrolyzed more rapidly than new subunits are added, the presence of GDP-bound tubulin at the plus end of the microtubule leads to disassembly and shrinkage.

12.3 WEBSITE ANIMATION

Microtubule Assembly

Tubulin dimers can polymerize as well as depolymerize to assemble or break down microtubules.

monly used experimental drugs that bind tubulin and inhibit microtubule polymerization, which in turn blocks mitosis. Two related drugs (**vincristine** and **vinblastine**) are used in cancer chemotherapy because they selectively inhibit rapidly dividing cells. Another useful drug, **taxol**, stabilizes microtubules rather than inhibiting their assembly. Such stabilization also blocks cell division, and taxol is used as an anticancer agent as well as an experimental tool.

■ **Taxol was originally isolated from the bark of Pacific yew trees. The Pacific yew is one of the slowest growing trees and several trees were needed to extract enough taxol for the treatment of one patient, limiting the availability of taxol. Fortunately, taxol and its derivatives are now produced economically via a semisynthetic process wherein a compound similar to taxol is extracted from related trees and modified chemically.**

Assembly of Microtubules

In animal cells, most microtubules extend outward from the **centrosome** (first described by Theodor Boveri in 1888), which is located adjacent to the nucleus near the center of interphase (nondividing) cells (Figure 12.45). During mitosis, microtubules similarly extend outward from duplicated centrosomes to form the mitotic spindle, which is responsible for the separation

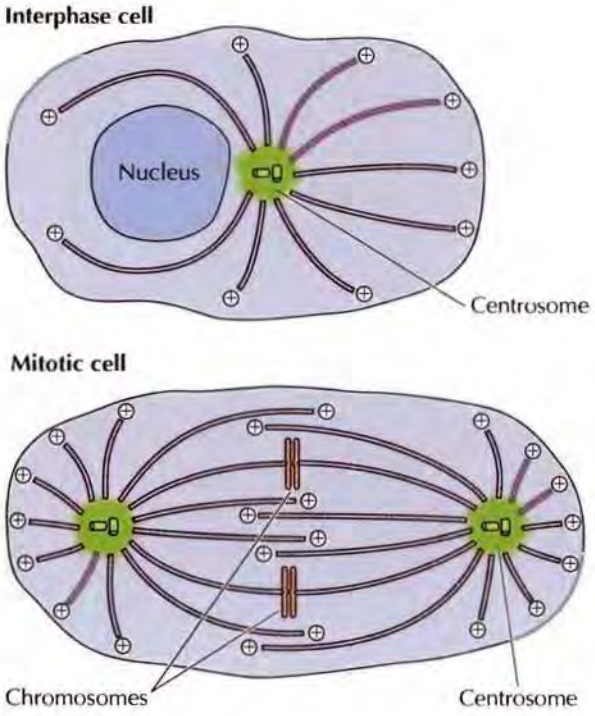


FIGURE 12.45 Intracellular organization of microtubules The minus ends of microtubules are anchored in the centrosome. In interphase cells, the centrosome is located near the nucleus and microtubules extend outward to the cell periphery. During mitosis, duplicated centrosomes separate and microtubules reorganize to form the mitotic spindle.

and distribution of chromosomes to daughter cells. The centrosome thus plays a key role in determining the intracellular organization of microtubules. Plant cells do not have an organized centrosome; instead, microtubules in most plants extend outward from the nucleus.

The centrosome is now known to function as a microtubule-organizing center in which the minus ends of the microtubules are anchored. It serves as the initiation site for the assembly of microtubules, which then grow outward from the centrosome toward the periphery of the cell. This can be clearly visualized in cells that have been treated with colcemid to disassemble their microtubules (Figure 12.46). When the drug is removed, the cells

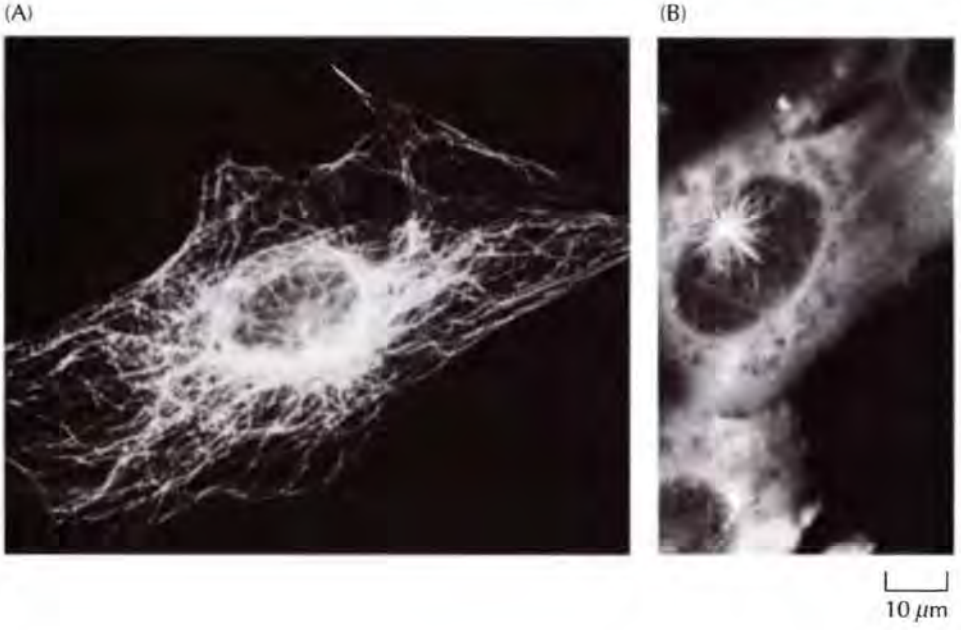


FIGURE 12.46 Growth of microtubules from the centrosome Microtubules in mouse fibroblasts are visualized by immunofluorescence microscopy using an antibody against tubulin. (A) The distribution of microtubules in a normal interphase cell. (B) This cell was treated with colcemid for one hour to disassemble microtubules. The drug was then removed and the cell allowed to recover for 30 minutes, allowing the visualization of new microtubules growing out of the centrosome. (From M. Osborn and K. Weber, 1976. *Proc. Natl. Acad. Sci. U.S.A.* 73: 867.)

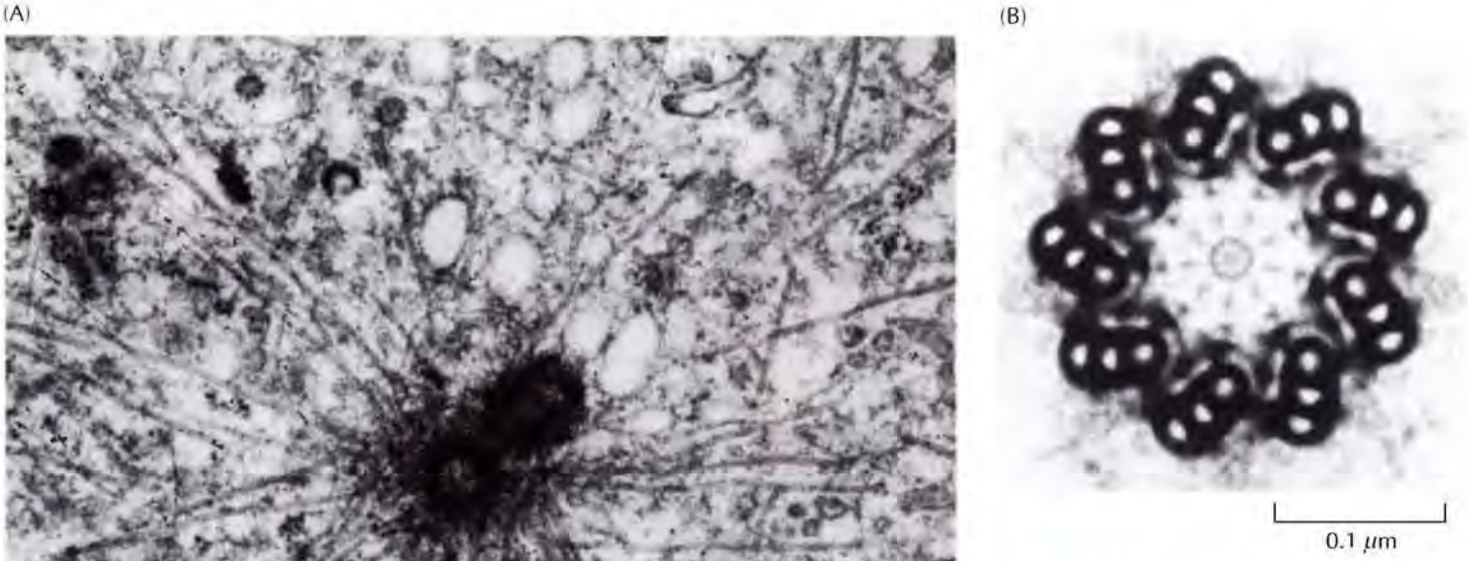


FIGURE 12.47 Structure of centrosomes (A) Electron micrograph of a centrosome showing microtubules radiating from the pericentriolar material that surrounds a pair of centrioles. (B) Transverse section of a centriole illustrating its nine triplets of microtubules. (A, © Cytographics; B, Don Fawcett, Photo Researchers, Inc.)

recover and new microtubules can be seen growing outward from the centrosome. Thus the initiation of microtubule growth at the centrosome establishes the polarity of microtubules within the cell. Note that microtubules grow by the addition of tubulin to their plus ends, which extend outward from the centrosome toward the cell periphery, so the role of the centrosome is to *initiate* microtubule growth. The key protein in the centrosome is γ-tubulin, a minor species of tubulin first identified in fungi, which nucleates assembly of microtubules. γ-tubulin is associated with eight or more other proteins in a ring-shaped structure called the **γ-tubulin ring complex**. This is thought to act as a seed for rapid microtubule growth, bypassing the rate-limiting nucleation step. Once assembled, microtubules can be released from the microtubule organizing center to organize elsewhere in the cell. This is especially evident in polarized epithelial cells or nerve cells as well as in plant root cells during growth of root hairs. Cells can initiate microtubules in the absence of a centrosome because γ-tubulin is also found in the cytosol, and in plant cells γ-tubulin is concentrated at the periphery of the nucleus.

The centrosomes of most animal cells contain a pair of **centrioles**, oriented perpendicular to each other and surrounded by amorphous **pericentriolar material** (Figure 12.47). The centrioles are cylindrical structures based on nine triplets of microtubules, similar to the basal bodies of cilia and flagella (discussed later in the chapter). Centrioles are necessary to form basal bodies, cilia, and flagella, and play a poorly understood role in coordinating the cell cycle. However, they are not necessary for the microtubule-organizing functions of the centrosome and are not found in plant cells, many unicellular eukaryotes, and most meiotic animal cells (such as mouse eggs). It is the pericentriolar material, not the centrioles, that initiates microtubule assembly. However, removal of the centrioles from animal cells results in dispersion of the centrosome contents and a decline in microtubule turnover.

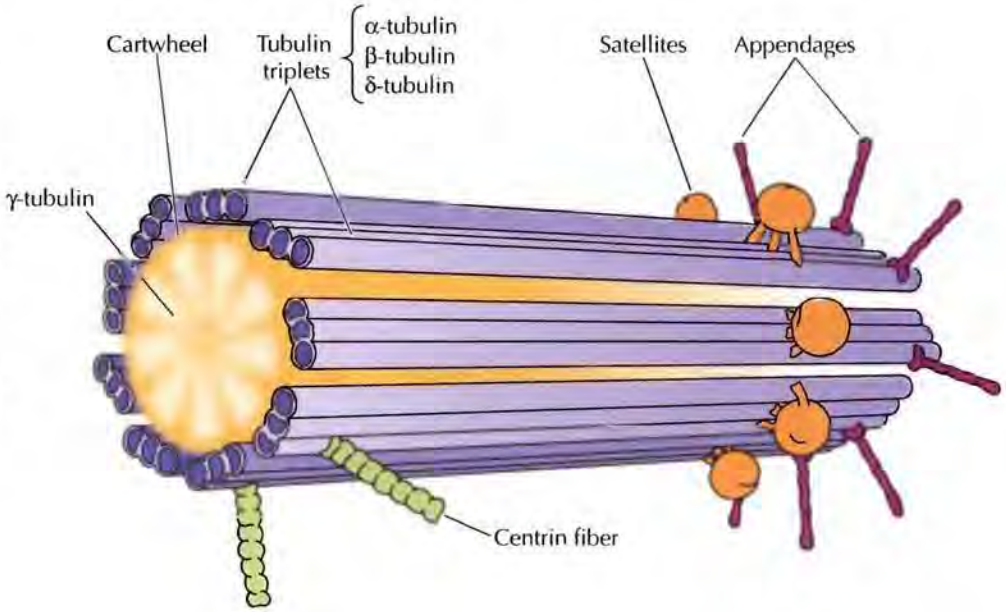


FIGURE 12.48 Structure of a centriole Centrioles are highly polar structures with a cartwheel protein structure at one end and several types of processes extending from the other end. These processes (called satellites and appendages) are thought to interact with specific proteins in the centrosome matrix. Other centrosome matrix proteins such as γ-tubulin are associated with the lumen of the centriole. The triplet microtubules contain highly modified α- and β-tubulins and the unique δ-tubulin. Centrin fibers extend out from the triplet microtubules and connect to the other centriole. (Modified from W. F. Marshall, 2001. *Curr. Biol.* 11: R487.)

Centrioles are complex structures with a clear polarity, a cartwheel-like protein structure at one end, and numerous extensions into the centrosome (Figure 12.48). The extensions are thought to help organize the centrosome matrix. Centrioles also contain several unique proteins, including δ-tubulin, which is part of the characteristic microtubule triplet. During interphase the two centrioles are connected by one or more fibers that contain centrin (a Ca^{2+}-binding protein related to calmodulin).

Organization of Microtubules within Cells

Because of their inherent dynamic instability, most microtubules would be frequently disassembled within the cell. This dynamic behavior can, however, be modified by the interactions of microtubules with **microtubule-associated proteins (MAPs)**. As with microfilaments, some of these proteins stabilize microtubules by capping their ends, while others act to disassemble microtubules, either by severing them or by increasing the rate of tubulin depolymerization at their ends. Several MAPs are plus-end tracking proteins, which bind to tubulin/GTP and act to track growing microtubules toward specific cellular locations. Such interactions allow the cell to stabilize microtubules in particular locations and provide an important mechanism for determining cell shape and polarity.

Many different MAPs have been identified and they vary depending on the type of cell. Among these are MAP-1, MAP-2, and tau (isolated from neuronal cells), and MAP-4, which is present in all non-neuronal vertebrate cell types. The tau protein has been extensively studied because it is the main component of the characteristic lesions found in the brains of Alzheimer's patients. The activity of many MAPs is regulated by phosphorylation, allowing the cell to control microtubule stability.

A good example of the role of stable microtubules in determining cell polarity is provided by nerve cells, which consist of two distinct types of processes (axons and dendrites) extending from a cell body (Figure 12.49). Both axons and dendrites are supported by stable microtubules, together with the neurofilaments discussed in the preceding section of this chapter. However, the microtubules in axons and dendrites are organized differently and associated with distinct MAPs. In axons, the microtubules are all ori-

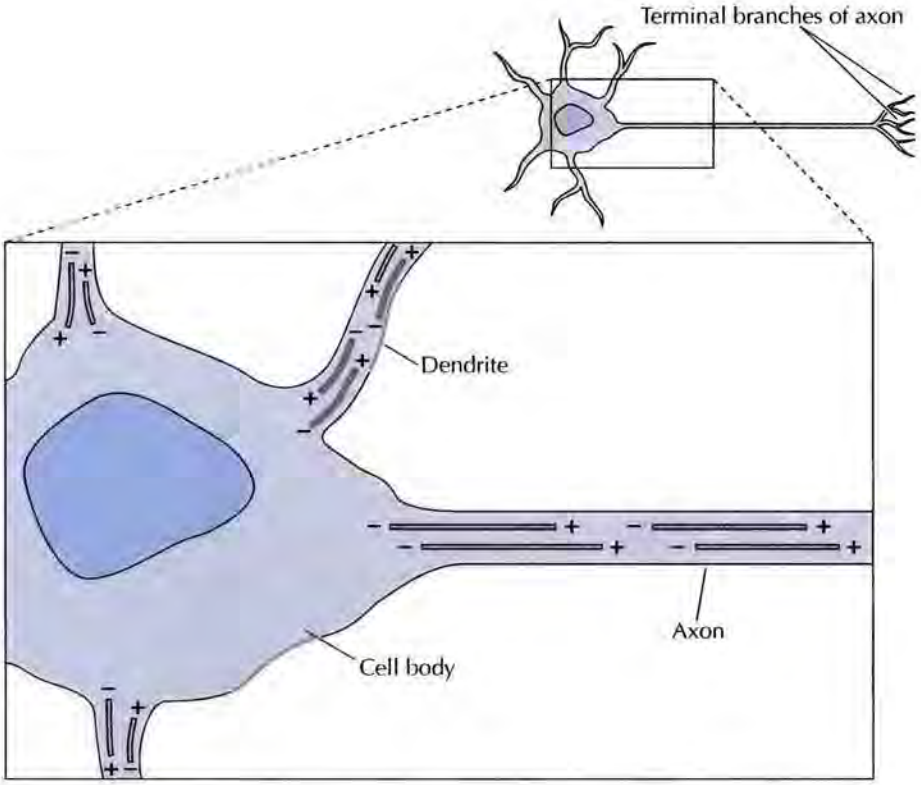


FIGURE 12.49 Organization of microtubules in nerve cells Two distinct types of processes extend from the cell body of nerve cells (neurons). Dendrites are short processes that receive stimuli from other nerve cells. The single long axon then carries impulses from the cell body to other cells, which may be either other neurons or an effector cell, such as a muscle. Stable microtubules in both axons and dendrites terminate in the cytoplasm rather than being anchored in the centrosome. In dendrites, microtubules are oriented in both directions, with their plus ends pointing both toward and away from the cell body. In contrast, all of the axon microtubules are oriented with their plus ends pointing toward the tip of the axon.

ented with their plus ends away from the cell body, similar to the general orientation of microtubules in other cell types. The minus ends of most of the microtubules in axons, however, are not anchored in the centrosome; instead, both the plus and minus ends of these microtubules terminate in the cytoplasm of the axon. In dendrites, the microtubules are oriented in both directions; some plus ends point toward the cell body and some point toward the cell periphery. These distinct microtubule arrangements are paralleled by differences in MAPs: Axons contain tau proteins, but no MAP-2, whereas dendrites contain MAP-2, but no tau proteins, and it appears that these differences in MAP-2 and tau distribution are responsible for the distinct organization of stable microtubules in axons and dendrites.

Microtubule Motors and Movement

Microtubules are responsible for a variety of cell movements, including the intracellular transport and positioning of membrane vesicles and organelles, the separation of chromosomes at mitosis, and the beating of cilia and flagella. As discussed for actin filaments earlier in this chapter, movement along microtubules is based on the action of motor proteins that utilize energy derived from ATP hydrolysis to produce force and movement. Members of two large families of motor proteins—the **kinesins** and the **dyneins**—are responsible for powering the variety of movements in which microtubules participate.

Identification of Microtubule Motor Proteins

Kinesin and dynein, the prototypes of microtubule motor proteins, move along microtubules in opposite directions—most kinesins toward the plus end and dyneins toward the minus end (Figure 12.50). The first of these

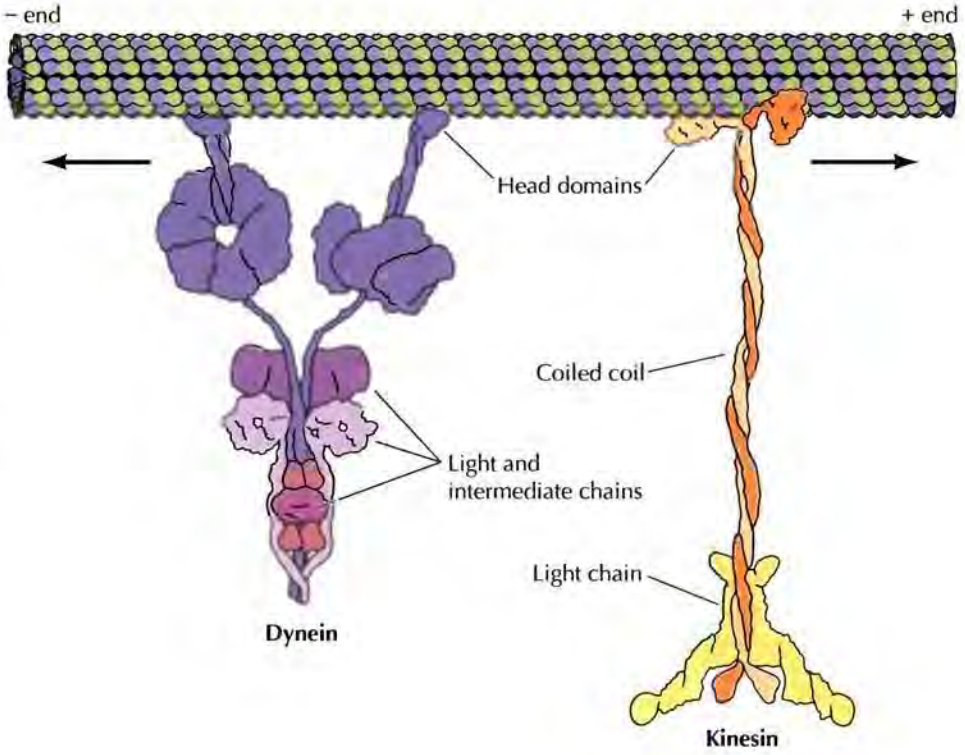


FIGURE 12.50 Microtubule motor proteins Kinesin and dynein move in opposite directions along microtubules, toward the plus and minus ends, respectively. Kinesin consists of two heavy chains (wound around each other in a coiled-coil structure) and two light chains. The globular head domains of the heavy chains bind microtubules and are the motor domains of the molecule. Dynein consists of two or three heavy chains (two are shown here) in association with multiple light and intermediate chains. The globular head domains of the heavy chains are the motor domains. The models shown are based on X-ray crystallography data (After R. D. Vale, 2003. *Cell* 112: 467).

12.4 WEBSITE ANIMATION

Kinesin Kinesin is a motor protein that moves vesicles and organelles toward the plus ends of microtubules.

microtubule motor proteins to be identified was dynein, which was isolated by Ian Gibbons in 1965. The purification of this form of dynein (called **axonemal dynein**) was facilitated because it is a highly abundant protein in cilia, just as the abundance of myosin facilitated its isolation from muscle cells. The identification of other microtubule-based motors, however, was more problematic because the proteins responsible for processes such as chromosome movement and organelle transport are present at comparatively low concentrations in the cytoplasm. Isolation of these proteins therefore depended on the development of new experimental methods to detect the activity of molecular motors in cell-free systems.

The development of *in vitro* assays for cytoplasmic motor proteins was based on the use of **video-enhanced microscopy** (developed by Robert Allen and Shinya Inoué in the early 1980s) to study the movement of membrane vesicles and organelles along microtubules in squid axons. In this method, a video camera is used to increase the contrast of images obtained with the light microscope, substantially improving the detection of small objects and allowing the movement of organelles to be followed in living cells. Using this approach, Allen, Scott Brady, and Ray Lasek demonstrated that organelle movements also took place in a cell-free system in which the plasma membrane had been removed and a cytoplasmic extract had been spread on a glass slide. These observations led to the development of an *in vitro* reconstructed system, which provided an assay capable of detecting cellular proteins responsible for organelle movement. In 1985 Brady, as well as Ronald Vale, Thomas Reese, and Michael Sheetz, capitalized on these developments to identify kinesin (now know as "conventional" kinesin, or kinesin I) as a novel microtubule motor protein present in both squid axons and bovine brain.

Further studies demonstrated that kinesin I translocates along microtubules in only a single direction—toward the plus end. Because the plus ends of microtubules in axons are all oriented away from the cell body (see

Figure 12.49), the movement of kinesin I in this direction transports vesicles and organelles away from the cell body toward the tip of the axon. Within intact axons, however, vesicles and organelles also had been observed to move back toward the cell body, implying that a different motor protein might be responsible for movement along microtubules in the opposite direction—toward the minus end. Consistent with this prediction, further experiments showed that a protein previously identified as the microtubule-associated protein MAP-1C was in fact a motor protein that moved along microtubules in the minus end direction. Subsequent analysis demonstrated that MAP-1C is related to the dynein isolated from cilia (axonemal dynein), so MAP-1C is now referred to as **cytoplasmic dynein**.

Kinesin I is a molecule of approximately 380 kd consisting of two heavy chains (120 kd each) and two light chains (64 kd each) (see Figure 12.50). The heavy chains have long α-helical regions that wind around each other in a coiled-coil structure. The amino-terminal globular head domains of the heavy chains are the motor domains of the molecule. They bind to both microtubules and ATP, the hydrolysis of which provides the energy required for movement. Although the motor domain of kinesin (approximately 340 amino acids) is much smaller than that of myosin (about 850 amino acids), X-ray crystallography indicates that kinesin and myosin (see Figure 12.33) evolved from a common ancestor and are structurally similar. The tail portion of the kinesin molecule consists of the light chains in association with the carboxy-terminal domains of the heavy chains. This portion of kinesin is responsible for binding to other cell components (such as membrane vesicles and organelles) that are transported along microtubules by the action of kinesin motors.

Cytoplasmic dynein is an extremely large molecule (up to 2000 kd), which consists of two or three heavy chains (each about 500 kd) complexed with a variable number of light and intermediate chains, which range from 14 to 120 kd (see Figure 12.50). As in kinesin, the heavy chains form globular ATP-binding motor domains that are responsible for movement along microtubules. The basal portion of the molecule, including the light and intermediate chains, is thought to bind to other subcellular structures, such as organelles and vesicles. In many situations cytoplasmic dynein acts along with a protein called **dynactin** to move cargoes over long distances along the microtubules.

Like the myosins, both kinesin and dynein define families of related motor proteins. Following the initial isolation of kinesin in 1985, a variety of kinesin-related proteins have been identified. Eighteen different kinesins are encoded in the genome of *C. elegans*, and there are forty-five kinesins known in humans. Most of these, like kinesin I, move along microtubules in the plus end direction (see Figure 12.50). Other kinesins move in the opposite direction toward the minus end, and three kinesins do not move along microtubules at all but use their tubulin-bind properties to anchor microtubules to other structures (discussed later). There is a direct correlation between where in the molecule the motor domain is located and the direction of movement along the microtubule; plus end-directed kinesins have N-terminal motor domains, minus end-directed kinesins have C-terminal motor domains, and those that do not move along microtubules have motor domains in the middle of the heavy chain (middle motor kinesins). Different members of the kinesin family vary in the sequences of their carboxy-terminal tails and are responsible for the movements of different types of "cargo" along microtubules. There are at least two types of cytoplasmic dyneins and several types of axonemal dyneins. While all members of the

KEY EXPERIMENT

The Isolation of Kinesin

Identification of a Novel Force-Generating Protein, Kinesin, Involved in Microtubule-Based Motility

Ronald D. Vale, Thomas S. Reese, and Michael P. Sheetz

National Institute of Neurological and Communicative Disorders and Stroke, Marine Biological Laboratory, Woods Hole, MA; University of Connecticut Health Center, Farmington, CT; Stanford University School of Medicine, Stanford, CA

Cell, Volume 42, 1985, pages 39–50

R. D. Vale

T. S. Reese

M. P. Sheetz

The Context

The transport and positioning of cytoplasmic organelles is key to the organization of eukaryotic cells, so understanding the mechanisms responsible for vesicle and organelle transport is a fundamental question in cell biology. In 1982 Robert Allen, Scott Brady, Ray Lasek, and their colleagues used video-enhanced microscopy to visualize the movement of organelles along cytoplasmic filaments in squid giant axons, both *in vivo* and in a cell-free system. These filaments were then identified as microtubules by electron microscopy, but the motor proteins responsible for organelle movement were unknown. The only microtubule motor identified at the time was axonemal dynein, which was present only in cilia and flagella. In 1985 Ronald Vale, Thomas Reese, and Michael Sheetz described the isolation of a novel motor protein, kinesin, which was responsible for the movement of organelles along microtubules. Similar experiments were reported at the same time by Scott Brady (*Nature*, 1985, 317: 73–75).

The Experiments

Two experimental strategies were key to the isolation of kinesin. The first, based on the work of Allen and his colleagues, was the use of an *in vitro* system in which motor protein activity could be detected. Vale, Reese, and Sheetz used a system in which proteins in the cytoplasm of axons were found to power the movement of microtubules across the surface of glass coverslips. This cell-free system provided a sensitive and rapid functional assay for the activity of kinesin as a molecular motor.

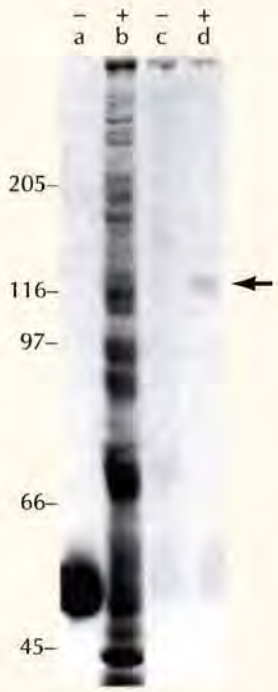
The second important approach used in the isolation of kinesin was to take advantage of its binding to microtubules. Lasek and Brady (*Nature*, 1985, 316: 645–647) had made the key observation that *in vitro* movements of microtubules require ATP and are inhibited by an ATP analog (adenylyl imidodiphosphate, AMP-PNP) that cannot be hydrolyzed and therefore does not provide a usable source of energy. In the presence of AMP-PNP, organelles remained attached to microtubules, suggesting that the motor protein responsible for organelle movement might also remain bound to microtubules under these conditions.

dynein family move toward the minus ends of microtubules, different cytoplasmic dyneins may transport different cargoes.

Cargo Transport and Intracellular Organization

One of the major roles of microtubules is to transport macromolecules, membrane vesicles, and organelles through the cytoplasm of eukaryotic cells. As already discussed, such cytoplasmic organelle transport is particularly evident in nerve cell axons, which may extend more than a meter in length. Ribosomes are present only in the cell body and dendrites, so proteins, membrane vesicles, and organelles (e.g., mitochondria) must be transported from the cell body to the axon. Via video-enhanced microscopy, the

KEY EXPERIMENT



Binding of a motor protein to microtubules in the presence of AMP-PNP. Protein samples were analyzed by electrophoresis through a polyacrylamide gel, stained, and photographed. Molecular weights of marker proteins are indicated in kilodaltons at the left. Lane a represents purified microtubules in which only tubulin (55 kd) is detected. Lane b is a soluble extract of squid axon cytoplasm containing many different polypeptides. This soluble extract was incubated with microtubules either without (lane c) or with (lane d) AMP-PNP. Microtubules were then recovered and incubated with ATP to release bound proteins, which were subjected to electrophoresis. Note that a 110 kd polypeptide was specifically bound to microtubules in the presence of AMP-PNP and then released by ATP. The proteins bound in the presence of AMP-PNP and released by ATP were also found to induce microtubule movement (indicated by "+" above the gel).

On the basis of these findings, Vale and colleagues incubated microtubules with cytoplasmic proteins from squid axon in the presence of AMP-PNP. The microtubules were then recovered and incubated with ATP to release proteins that were specifically bound in the presence of the nonhydrolyzable ATP analog. This experiment identified a 110 kd polypeptide that was bound to microtubules in the presence of AMP-PNP and released by subsequent incubation with ATP (see figure). Furthermore, the proteins bound to microtubules and then released by ATP were also shown to support the movement of microtubules *in vitro*. Binding to microtubules in the presence of AMP-PNP thus provided an efficient approach to isolation of the motor protein, which was shown by further biochemical studies to contain the 110 kd polypeptide complexed to polypeptides of 60 to 70 kd. By a similar approach, a related protein was also purified from bovine brain. The authors concluded that these proteins "represent a novel class of motility proteins that are structurally as well as enzymatically distinct from dynein, and we propose to call these translocators kinesin (from the Greek *kinein*, to move)."

The Impact

The movement of vesicles and organelles along microtubules is fundamental to the organization of eukaryotic cells, so the motor proteins responsible for these movements play critical roles in cell biology. Subsequent experiments using *in vitro* assays similar to those described here established that kinesin moves along microtubules in the plus-end direction, whereas cytoplasmic dynein is responsible for the transport of vesicles and organelles toward microtubule minus ends. Moreover, a large family of kinesin-related proteins has since been identified. In addition to their roles in vesicle and organelle transport, members of these families of motor proteins are responsible for the separation and distribution of chromosomes during mitosis. The identification of kinesin thus opened the door to understanding a variety of microtubule-based movements that are critical to the structure and function of eukaryotic cells.

transport of membrane vesicles and organelles in both directions can be visualized along axon microtubules where kinesin and dynein carry their cargoes to and from the tips of the axons, respectively. For example, secretory vesicles containing neurotransmitters are carried from the Golgi apparatus to the terminal branches of the axon by kinesin. In the reverse direction, cytoplasmic dynein transports endocytic vesicles from the axon back to the cell body.

Microtubules similarly transport macromolecules, membrane vesicles, and organelles in other types of cells. Because microtubules are usually oriented with their minus end anchored in the centrosome and their plus end extending toward the cell periphery, different members of the kinesin and

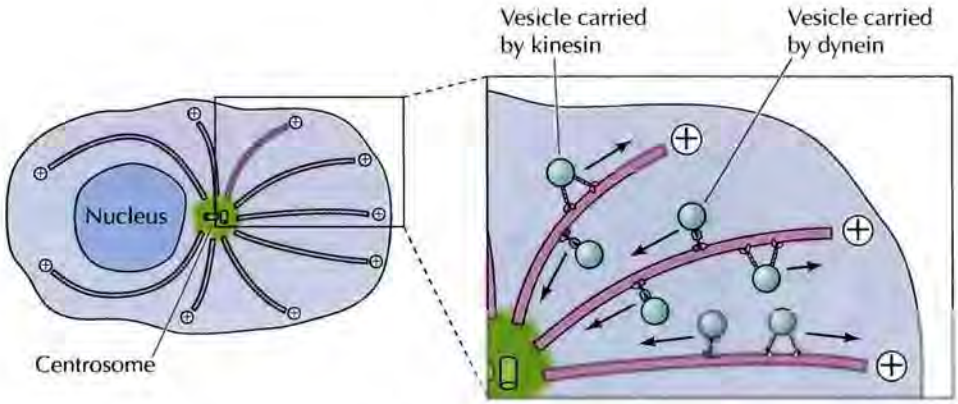


FIGURE 12.51 Transport of vesicles along microtubules Kinesin and other plus end-directed members of the kinesin family transport vesicles and organelles in the direction of microtubule plus ends, which extend toward the cell periphery. In contrast, dynein and minus end-directed members of the kinesin family carry their cargo in the direction of microtubule minus ends, which are anchored in the center of the cell.

dynein families are thought to transport cargo in opposite directions through the cytoplasm (Figure 12.51). Kinesin I and other plus end-directed members of the kinesin family carry their cargo toward the cell periphery, whereas cytoplasmic dyneins and minus end-directed members of the kinesin family transport materials toward the center of the cell. Cargo selection can be very specific. Kinesin II, a plus end-directed motor, transports selected mRNAs toward the cell cortex in *Xenopus* oocytes, a process that appears to be general to all vertebrates. Kinesin I similarly transports actin mRNA in fibroblasts, and dynein moves specific mRNAs toward one side of the *Drosophila* embryo. In addition to transporting membrane vesicles in the endocytic and secretory pathways, microtubules and associated motor proteins position membrane-enclosed organelles (such as the endoplasmic reticulum, Golgi apparatus, lysosomes, and mitochondria) within the cell. For example, the endoplasmic reticulum extends to the periphery of the cell in association with microtubules (Figure 12.52). Drugs that depolymerize microtubules cause the endoplasmic reticulum to retract toward the cell center, indicating that association with microtubules is required to maintain the endoplasmic reticulum in its extended state. This positioning of the endoplasmic reticulum appears to involve the action of kinesin I (or possibly multiple members of the kinesin family), which pulls the endoplasmic reticulum along microtubules in the plus-end direction, toward the cell periphery. Similarly, kinesin appears to play a key role in the positioning of lysosomes away from the center of the cell, and three different members of the kinesin family have been implicated in the movements of mitochondria.

Conversely, cytoplasmic dynein is thought to play a role in positioning the Golgi apparatus. The Golgi apparatus is located in the center of the cell near the centrosome. If microtubules are disrupted, either by a drug or when the cell enters mitosis, the Golgi breaks up into small vesicles that disperse throughout the cytoplasm. When the microtubules re-form, the Golgi apparatus also reassembles, with the Golgi vesicles apparently being transported to the center of the cell (toward the minus end of the microtubules) by cytoplasmic dynein. Movement along microtubules is thus responsible

FIGURE 12.52 Association of the endoplasmic reticulum with microtubules Fluorescence microscopy of the endoplasmic reticulum (A) and microtubules (B) in an epithelial cell. The endoplasmic reticulum is stained with a fluorescent dye, and microtubules are stained with an antibody against tubulin. Note the close correlation between the endoplasmic reticulum and microtubules at the periphery of the cell. (From M. Terasaki, L. B. Chen and K. Fujiwara, 1986. *J. Cell Biol.* 103: 1557.)

not only for vesicle transport but also for establishing the positions of membrane-enclosed organelles within the cytoplasm of eukaryotic cells.

Cilia and Flagella

Cilia and **flagella** are microtubule-based projections of the plasma membrane that are responsible for movement of a variety of eukaryotic cells. Many bacteria also have flagella, but these prokaryotic flagella are quite different from those of eukaryotes. Bacterial flagella (which are not discussed further here) are protein filaments projecting from the cell surface, rather than projections of the plasma membrane supported by microtubules.

Eukaryotic cilia and flagella are very similar structures, each with a diameter of approximately 0.25 μm (Figure 12.53). Many cells are covered by numerous cilia, which are about 10 μm in length. Cilia beat in a coordinated back-and-forth motion, which either moves the cell through fluid or moves fluid over the surface of the cell. For example, the cilia of some protozoans (such as *Paramecium*) are responsible both for cell motility and for sweeping food organisms over the cell surface and into the oral cavity. In animals, an important function of cilia is to move fluid or mucus over the surface of epithelial cell sheets. A good example is provided by the ciliated cells lining

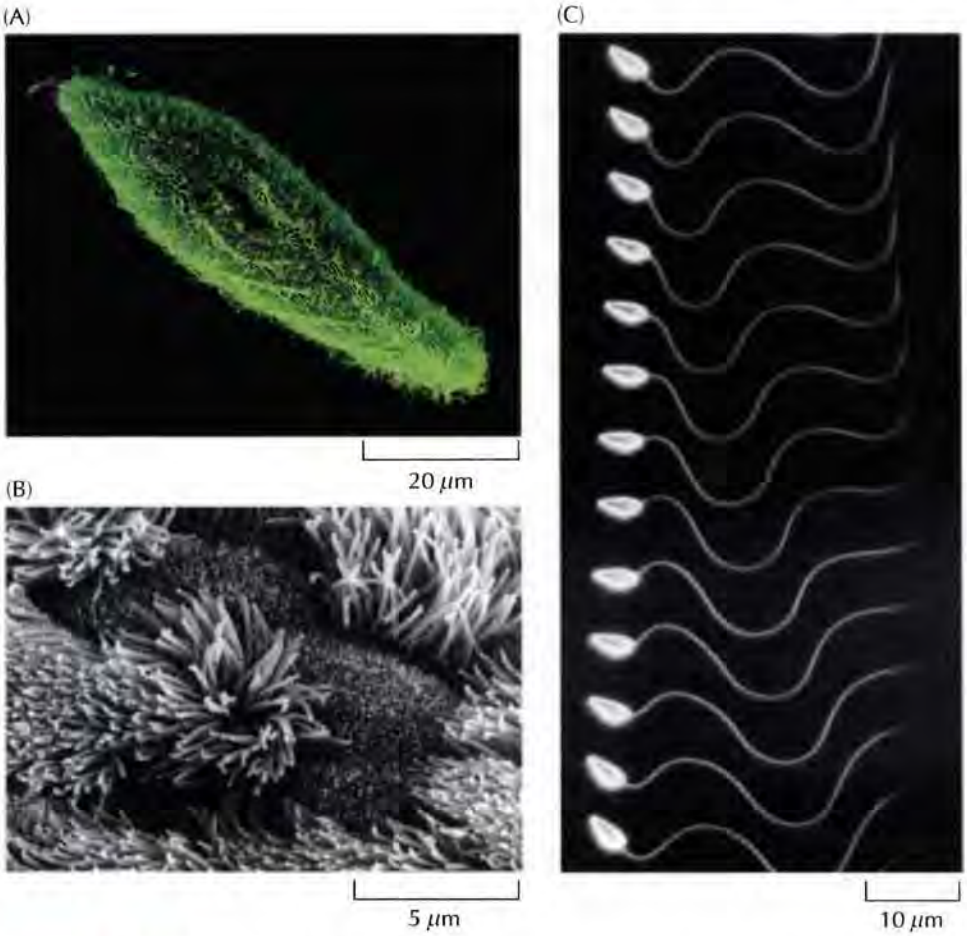


FIGURE 12.53 Examples of cilia and flagella (A) Scanning electron micrograph showing numerous cilia covering the surface of *Paramecium*. (B) Scanning electron micrograph of ciliated epithelial cells lining the surface of a trachea. (C) Multiple-flash photograph (500 flashes per second) showing the wavelike movement of a sea urchin sperm flagellum. (A, Karl Aufderheide/Visuals Unlimited; B, Fred E. Hossler/Visuals Unlimited; and C, C. J. Brokaw, California Institute of Technology.)

the respiratory tract, which clear mucus and dust from the respiratory passages. Flagella differ from cilia in their length (they can be as long as 200 μm) and in their wavelike pattern of beating. Cells usually have only one or two flagella, which are responsible for the locomotion of a variety of protozoans and of sperm.

The fundamental structure of both cilia and flagella is the **axoneme**, which is composed of microtubules and their associated proteins (Figure 12.54). The microtubules are arranged in a characteristic "9 + 2" pattern in which a central pair of microtubules is surrounded by nine outer microtubule doublets. The two fused microtubules of each outer doublet are distinct: One (called the A tubule) is a complete microtubule consisting of 13 protofilaments; the other (the B tubule) is incomplete, containing only 10 or 11 protofilaments fused to the A tubule. The outer microtubule doublets are connected to the central pair by radial spokes and to each other by links of a protein called **nexin**. In addition, two arms of dynein are attached to each A tubule, and it is the motor activity of these axonemal dyneins that drives the beating of cilia and flagella.

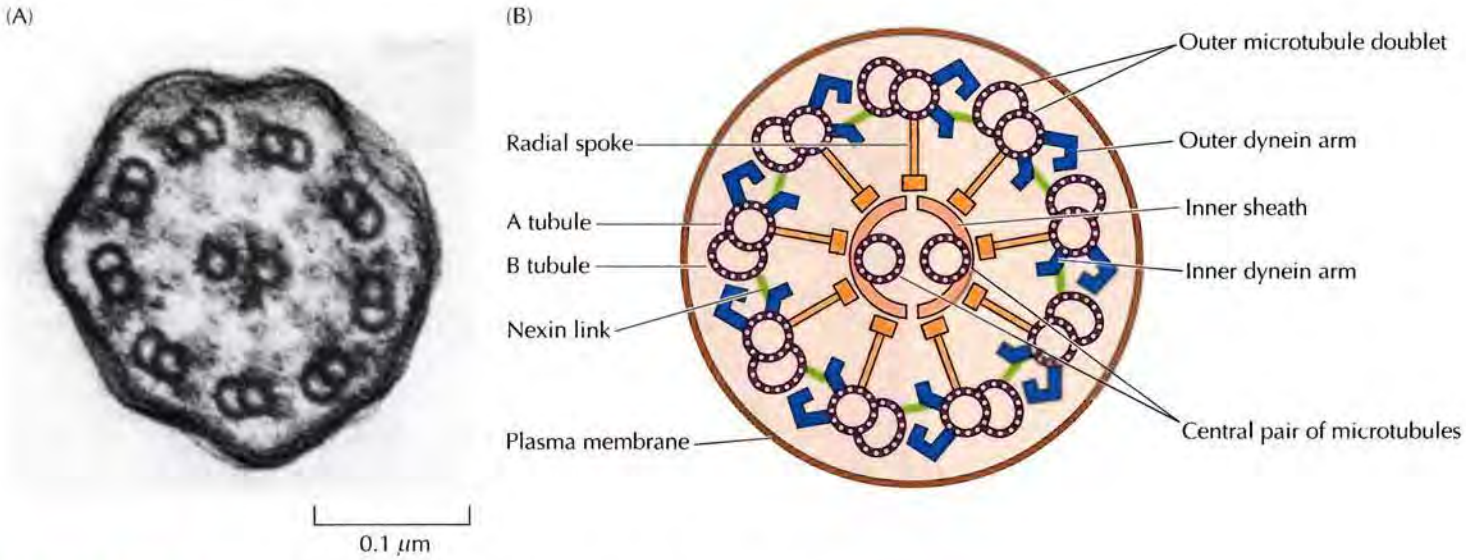
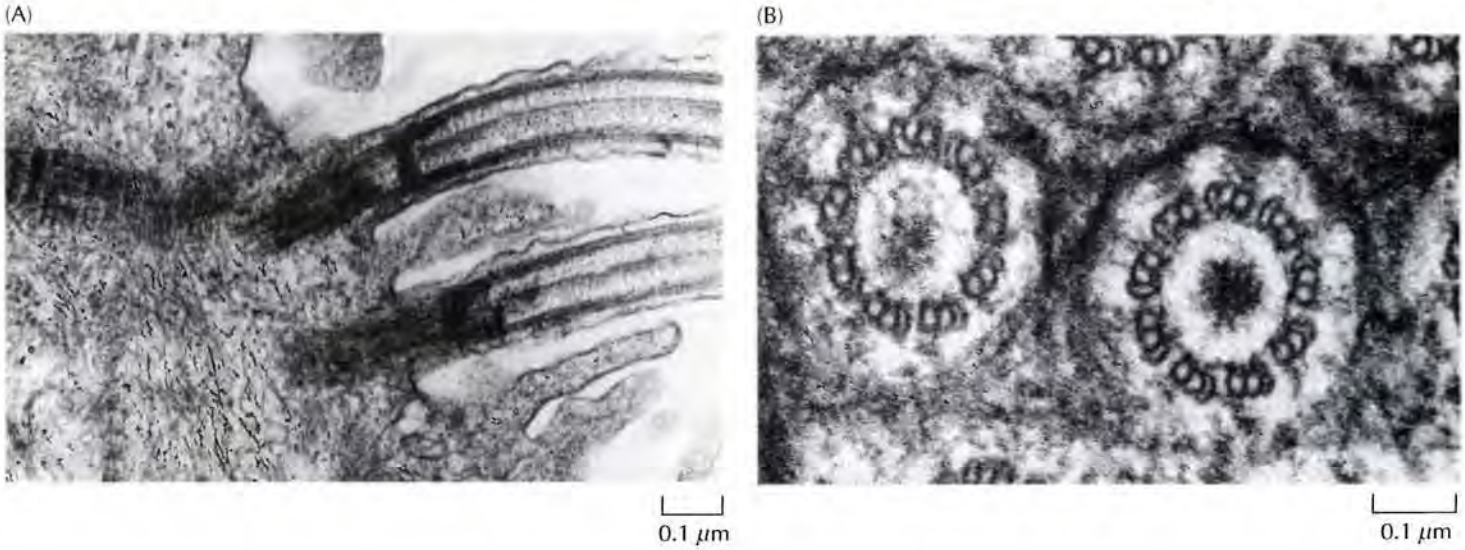


FIGURE 12.54 Structure of the axoneme of cilia and flagella (A) Computer-enhanced electron micrograph of a cross-section of the axoneme of a rat sperm flagellum. (B) Schematic cross-section of an axoneme. The nine outer doublets consist of one complete (A) microtubule and one incomplete (B) microtubule, containing only 10 or 11 protofilaments. The outer doublets are joined to each other by nexin links and to the central pair of microtubules by radial spokes. Each outer microtubule doublet is associated with inner and outer dynein arms. (A, K. G. Murti/Visuals Unlimited.)

The minus ends of the microtubules of cilia and flagella are anchored in a **basal body**, which is similar in structure to a centriole and contains nine triplets of microtubules (Figure 12.55). Centrioles were discussed earlier as components of the centrosome in which their function is complex and poorly understood. Basal bodies, however, play a clear role in organization of the axoneme microtubules. Namely, each of the outer microtubule doublets of the axoneme is initiated by extension of two of the microtubules present in the triplets of the basal body. Basal bodies thus serve to initiate

FIGURE 12.55 Electron micrographs of basal bodies (A) A longitudinal view of cilia anchored in basal bodies. (B) A cross-section of basal bodies. Each basal body consists of nine triplets of microtubules. (A, Conly L. Reider/Biological Photo Service; B, W. L. Dentler, Biological Photo Service.)



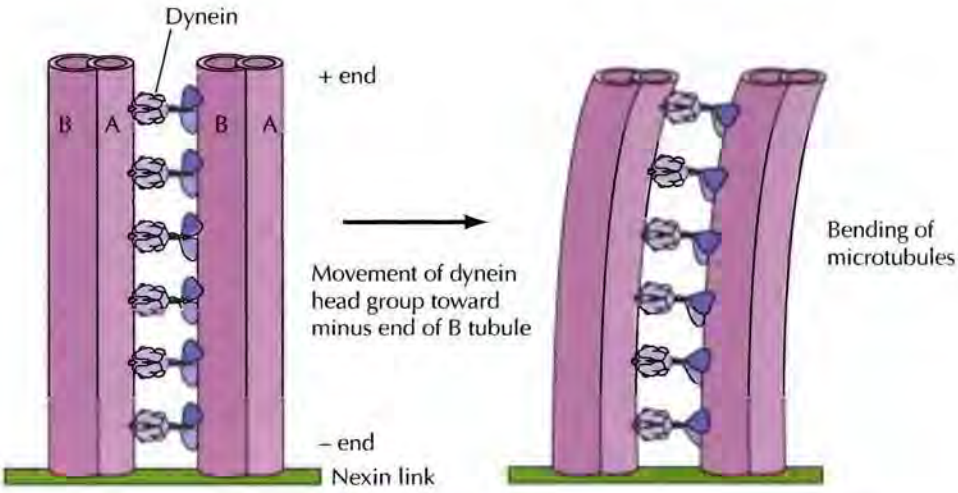


FIGURE 12.56 Movement of microtubules in cilia and flagella The bases of dynein arms are attached to A tubules, and the motor head groups interact with the B tubules of adjacent doublets. Movement of the dynein head groups in the minus end direction (toward the base of the cilium) then causes the A tubule of one doublet to slide toward the base of the adjacent B tubule. Because both microtubule doublets are connected by nexin links, this sliding movement forces them to bend.

the growth of axonemal microtubules as well as anchoring cilia and flagella to the surface of the cell.

The movements of cilia and flagella result from the sliding of outer microtubule doublets relative to one another, powered by the motor activity of the axonemal dyneins (Figure 12.56). The dynein bases bind to the A tubules while the dynein head groups bind to the B tubules of adjacent doublets. Movement of the dynein head groups in the minus end direction then causes the A tubule of one doublet to slide toward the basal end of the adjacent B tubule. Because the microtubule doublets in an axoneme are connected by nexin links, the sliding of one doublet along another causes them to bend, forming the basis of the beating movements of cilia and flagella. It is apparent, however, that the activities of dynein molecules in different regions of the axoneme must be carefully regulated to produce the coordinated beating of cilia and the wavelike oscillations of flagella—a process about which little is currently understood.

Reorganization of Microtubules during Mitosis

As noted earlier, microtubules completely reorganize during mitosis, providing a dramatic example of the importance of their dynamic nature. The microtubule array present in interphase cells disassembles and the free tubulin subunits are reassembled to form the **mitotic spindle**, which is responsible for the separation of daughter chromosomes (Figure 12.57). This restructuring of the microtubule cytoskeleton is directed by duplication of the centrosome to form two separate microtubule-organizing centers at opposite poles of the mitotic spindle.

The centrioles and other components of the centrosome are duplicated in interphase cells, but they remain together on one side of the nucleus until the beginning of mitosis (Figure 12.58). The two centrosomes then separate and move to opposite sides of the nucleus, forming the two poles of the mitotic spindle. As the cell enters mitosis, the dynamics of microtubule

FIGURE 12.57 Electron micrograph of the mitotic spindle The spindle microtubules are attached to condensed chromosomes at metaphase. (From C. L. Rieder and S. S. Bowser, 1985. *J. Histochem. Cytochem.* 33: 165/Biological Photo Service.)

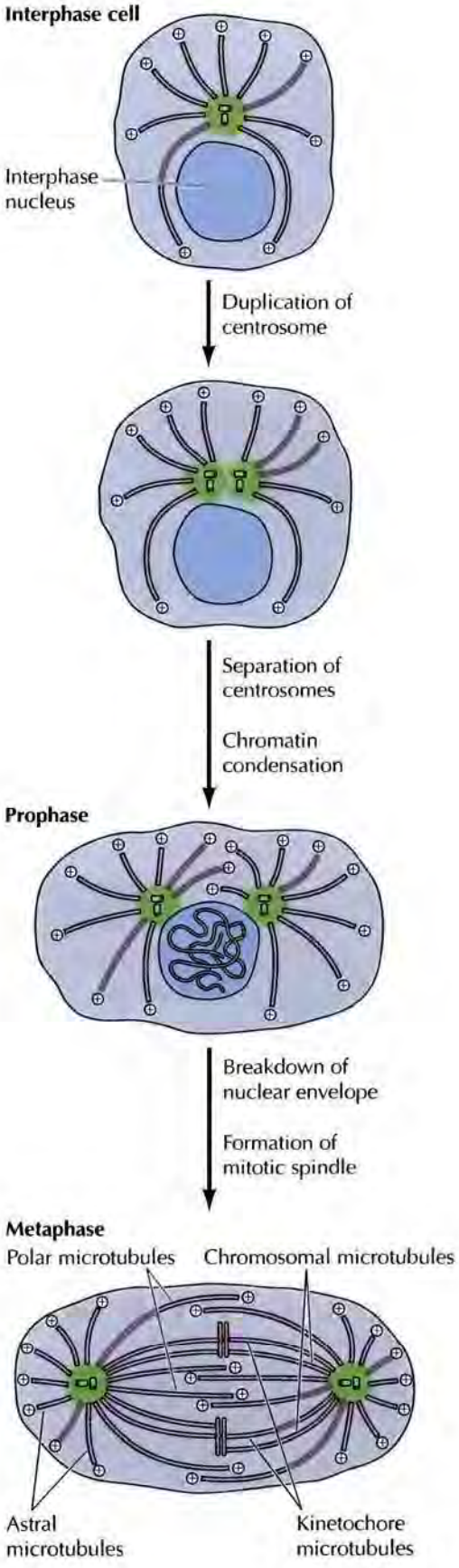


FIGURE 12.58 Formation of the mitotic spindle The centrioles and centrosomes duplicate during interphase. During prophase of mitosis the duplicated centrosomes separate and move to opposite sides of the nucleus. The nuclear envelope then disassembles, and microtubules reorganize to form the mitotic spindle. Kinetochore microtubules are attached to the kinetochores of condensed chromosomes and chromosomal microtubules are attached to their ends. Polar microtubules overlap with each other in the center of the cell, and astral microtubules extend outward to the cell periphery. At metaphase, the condensed chromosomes are aligned at the center of the spindle.

assembly and disassembly also change dramatically. First, the rate of microtubule disassembly increases about tenfold, resulting in overall depolymerization and shrinkage of microtubules. At the same time, the number of microtubules emanating from the centrosome increases by five- to tenfold. In combination, these changes result in disassembly of the interphase microtubules and the outgrowth of large numbers of short microtubules from the centrosomes.

As first proposed by Marc Kirschner and Tim Mitchison in 1986, formation of the mitotic spindle involves the selective stabilization of some of the microtubules radiating from the centrosomes. These microtubules are of four types, three of which make up the mitotic spindle. **Kinetochore microtubules** attach to the condensed chromosomes of mitotic cells at their centromeres, which are associated with specific proteins to form the kinetochore (see Figure 5.19). Attachment to the kinetochore stabilizes these microtubules, which as discussed below play a critical role in separation of the mitotic chromosomes. Also emanating from the centrosomes are the **chromosomal microtubules,** which connect to the ends of the chromosomes via chromokinesin. The third type of microtubules found in the mitotic spindle (**polar microtubules**) is not attached to chromosomes. Instead, the polar microtubules are stabilized by overlapping with each other in the center of the cell. **Astral microtubules** extend outward from the centrosomes to the cell periphery and have freely exposed plus ends. As discussed later, both the polar and astral microtubules contribute to chromosome movement by pushing the spindle poles apart.

As mitosis proceeds, the condensed chromosomes first align on the metaphase plate and then separate, with the two chromatids of each chromosome being pulled to opposite poles of the spindle. Chromosome movement is mediated by motor proteins associated with the spindle microtubules, as will be discussed shortly. In the final stage of mitosis, nuclear envelopes re-form, the chromosomes decondense, and cytokinesis takes place. Each daughter cell then contains one centrosome, which nucleates the formation of a new network of interphase microtubules.

Chromosome Movement

After the two centrosomes move to opposite sides of the cell at the beginning of mitosis, the duplicated chromosomes attach to kinetochore and chromosomal microtubules and align on the metaphase plate, equidistant from the two spindle poles. This alignment of chromosomes is mediated by rapid growth of the kinetochore microtubules and capture of the kinetochores by plus-end tracking proteins. In addi-

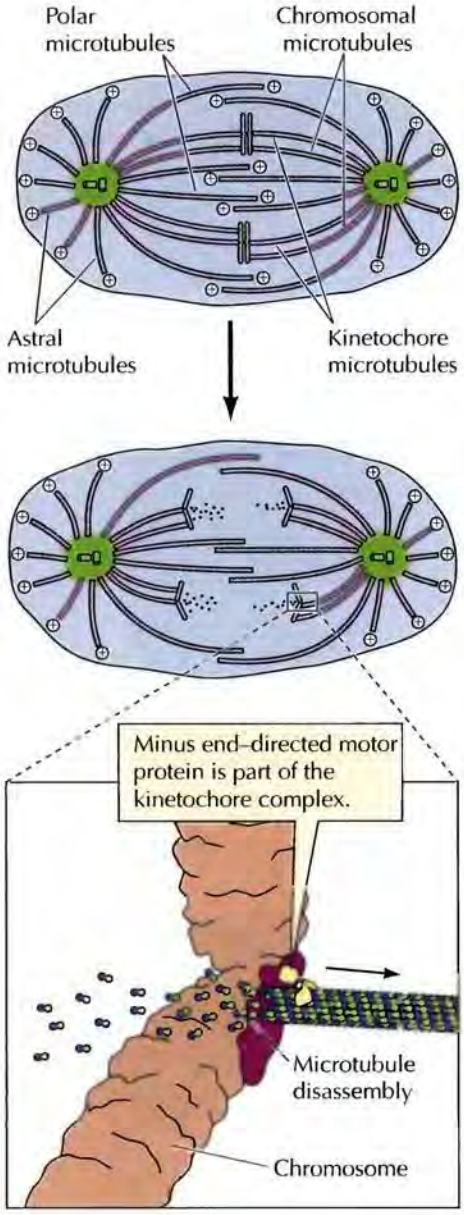


FIGURE 12.59 Anaphase A chromosome movement Chromosomes move toward the spindle poles along the kinetochore microtubules. Chromosome movement is driven by minus end-directed motor proteins associated with the kinetochore. The action of these motor proteins is coupled to disassembly and shortening of both the kinetochore and chromosomal microtubules.

tion, the chromosome ends are pushed toward the metaphase plate by chromokinesin moving along the chromosomal microtubules. Once all of the chromosomes have aligned on the metaphase plate, the links between the sister chromatids are severed and anaphase begins. During anaphase of mitosis, the sister chromatids separate and move to opposite poles of the spindle. Chromosome movement proceeds by two distinct mechanisms, referred to as **anaphase A** and **anaphase B**, which involve different types of spindle microtubules.

Anaphase A consists of the movement of chromosomes toward the spindle poles along the kinetochore microtubules, which shorten as chromosome movement proceeds (Figure 12.59). Movement of chromosomes along the spindle microtubules in the minus end direction (toward the centrosomes) is driven by kinetochore-associated motor proteins. Cytoplasmic

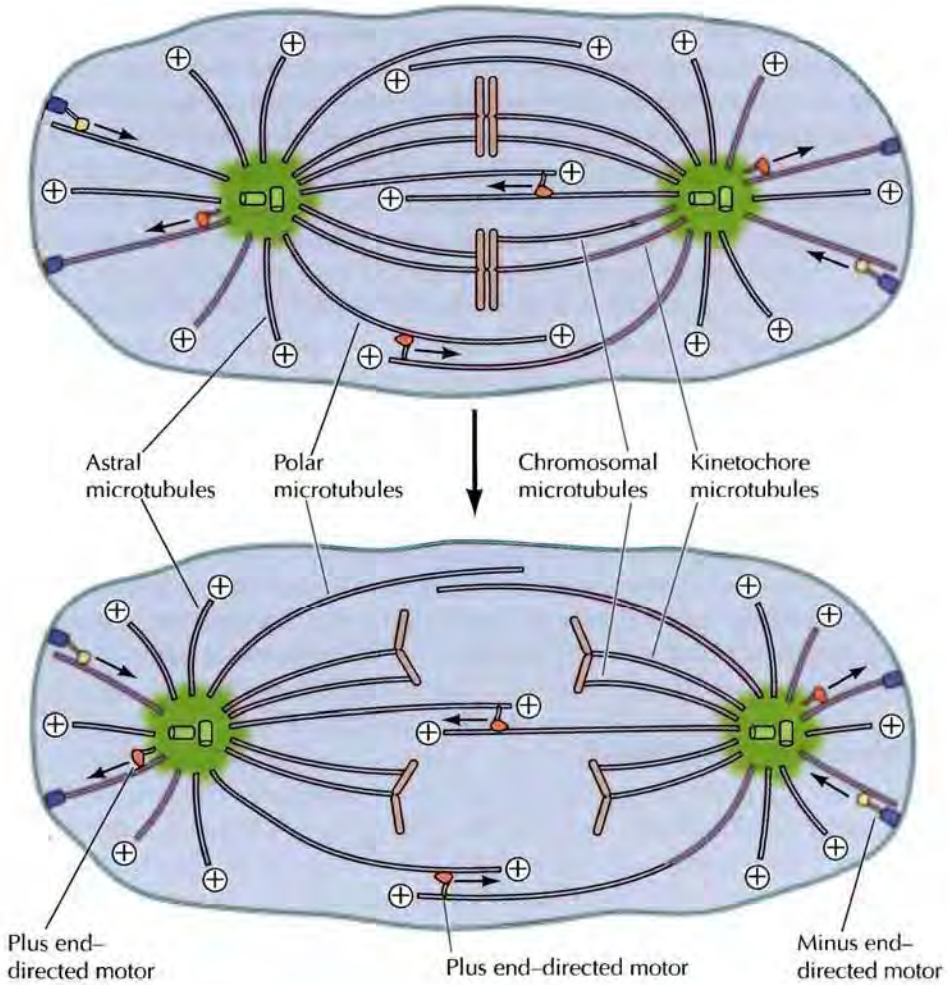


FIGURE 12.60 Spindle pole separation in anaphase B The separation of spindle poles results from two types of movement. First, overlapping polar microtubules slide past each other to push the spindle poles apart, probably as a result of the action of plus end-directed motor proteins. Second, the spindle poles are pulled apart by the astral microtubules. The driving force appears to be a minus end-directed motor anchored to the cell cortex.

dynein is associated with kinetochores and may play a role in poleward chromosome movement, as may minus end-directed members of the kinesin family. The action of these kinetochore motor proteins is coupled to disassembly and shortening of both the kinetochore and chromosomal microtubules, which is mediated by middle motor kinesins that act as microtubule-depolymerizing enzymes.

Anaphase B refers to the separation of the spindle poles themselves (Figure 12.60). Spindle-pole separation is accompanied by elongation of the polar microtubules and is similar to the initial separation of duplicated centrosomes to form the spindle poles at the beginning of mitosis (see Figure 12.58). During anaphase B the overlapping polar microtubules slide against one another, pushing the spindle poles apart. This type of movement has been found to result from the action of several plus end–directed kinesins, which cross-link polar microtubules and move them toward the plus end of their overlapping microtubule away from the opposite spindle pole. In addition, the spindle poles are pulled apart by the astral microtubules. This type of movement involves the action of cytoplasmic dynein anchored to the cell cortex. The movement of this anchored dynein along astral microtubules in the minus end direction pulls the spindle poles apart toward the periphery of the cell. There is a simultaneous depolymerization of the astral microtubules by middle motor kinesins, leading to separation of the spindle poles and their movement to the periphery of the cell, prior to the formation of two daughter cells at the end of mitosis.

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

SUMMARY

STRUCTURE AND ORGANIZATION OF ACTIN FILAMENTS

Assembly and Disassembly of Actin Filaments: Actin filaments are formed by the head-to-tail polymerization of actin monomers into a helix. A variety of actin-binding proteins regulate the assembly and disassembly of actin filaments within the cell.

Organization of Actin Filaments: Actin filaments are cross-linked by actin-binding proteins to form bundles or three-dimensional networks.

Association of Actin Filaments with the Plasma Membrane: A network of actin filaments and other cytoskeletal proteins underlies the plasma membrane and determines cell shape. Actin bundles also attach to the plasma membrane and anchor the cell at regions of cell-cell and cell-substratum contact.

Protrusions of the Cell Surface: Actin filaments support permanent protrusions of the cell surface, such as microvilli, as well as transient extensions that are responsible for phagocytosis and cell locomotion.

KEY TERMS

actin, microfilament, globular [G] actin, filamentous [F] actin, treadmilling, cytochalasin, phalloidin, actin-binding protein, formin, Arp2/3 complex, tropomyosin, ADF/cofilin, profilin

actin bundle, actin network, actin-bundling protein, fimbrin, contractile bundle, α-actinin, filamin

cell cortex, spectrin, ankyrin, fodrin, ERM proteins, dystrophin, integrin, focal adhesion, stress fiber, talin, vinculin, adherens junction, adhesion belt, cadherin, catenin

microvillus, brush border, stereocilium, villin, pseudopodium, lamellipodium, microspike, filopodium

KEY TERMS

myosin, molecular motor, muscle fiber, myofibril, sarcomere, titin, nebulin, sliding filament model, myosin II, sarcoplasmic reticulum, troponin

cytokinesis, contractile ring, myosin light-chain kinase, calmodulin

myosin I

WASP/Scar complex, twinfilin

intermediate filament, keratin, vimentin, desmin, neurofilament (NF) protein

desmosome, hemidesmosome, plakin

SUMMARY

ACTIN, MYOSIN, AND CELL MOVEMENT

Muscle Contraction: Studies of muscle established the role of myosin as a motor protein that uses the energy derived from ATP hydrolysis to generate force and movement. Muscle contraction results from the sliding of actin and myosin filaments past each other. ATP hydrolysis drives repeated cycles of interaction between myosin and actin during which conformational changes result in movement of the myosin head group along actin filaments.

Contractile Assemblies of Actin and Myosin in Nonmuscle Cells: Assemblies of actin and myosin II are responsible for a variety of movements of nonmuscle cells, including cytokinesis.

Nonmuscle Myosins: Other types of myosin that do not function in contraction serve to transport membrane vesicles and organelles along actin filaments.

Formation of Protrusions and Cell Movement: Extension of cell protrusions is mediated by the growth of multiple actin filament branches at the leading edge of the cell. Cell movement is a complex process in which adhesions form at the ends of the new cell protrusions, the cell body is brought forward by the action of myosin II along stress fibers, and the trailing edge retracts into the cell body.

INTERMEDIATE FILAMENTS

Intermediate Filament Proteins: Intermediate filaments are polymers of more than 50 different proteins that are expressed in various types of cells. They are not involved in cell movement but provide mechanical support to cells and tissues.

Assembly of Intermediate Filaments: Intermediate filaments are formed from dimers of two polypeptide chains wound around each other in a coiled-coil structure. The dimers then associate to form tetramers, which assemble into protofilaments. Intermediate filaments are formed from protofilaments wound around one another in a ropelike structure.

Intracellular Organization of Intermediate Filaments: Intermediate filaments form a network extending from a ring surrounding the nucleus, to the plasma membrane of most cell types. In epithelial cells, intermediate filaments are anchored to the plasma membrane at regions of specialized cell contacts (desmosomes and hemidesmosomes). Intermediate filaments also play specialized roles in muscle and nerve cells.

Functions of Keratins and Neurofilaments: Diseases of the Skin and Nervous System: The importance of intermediate filaments in providing mechanical strength to cells in tissues has been demonstrated by the introduction of mutant keratin genes into transgenic mice. Similar keratin gene mutations are responsible for some human skin diseases, and abnormalities of neurofilaments have been implicated in the development of motor neuron disease.

SUMMARY

MICROTUBULES

Structure and Dynamic Organization of Microtubules: Microtubules are formed by the reversible polymerization of tubulin. They display dynamic instability and undergo continual cycles of assembly and disassembly as a result of GTP hydrolysis following tubulin polymerization.

microtubule, tubulin, dynamic instability, colchicine, colcemid, vincristine, vinblastine, taxol

Assembly of Microtubules: The microtubules in most cells extend outward from a microtubule-organizing center, or centrosome, located near the center of the cell. In animal cells, the centrosome usually contains a pair of centrioles surrounded by pericentriolar material. The growth of microtubules is initiated in the pericentriolar material, which then serves to anchor their minus ends.

centrosome, γ-tubulin ring complex, centriole, pericentriolar material

Organization of Microtubules within Cells: Selective stabilization of microtubules by microtubule-associated proteins can determine cell shape and polarity, such as the extension of nerve cell axons and dendrites.

microtubule-associated protein (MAP)

MICROTUBULE MOTORS AND MOVEMENT

Identification of Microtubule Motor Proteins: Two families of motor proteins, the kinesins and the dyneins, are responsible for movement along microtubules. Kinesin and most kinesin-related proteins move in the plus-end direction, whereas the dyneins and some members of the kinesin family move toward microtubule minus ends.

kinesin, dynein, axonemal dynein, video-enhanced microscopy, cytoplasmic dynein, dynactin

Cargo Transport and Intracellular Organization: Movement along microtubules transports macromolecules, membrane vesicles, and organelles through the cytoplasm, as well as positioning cytoplasmic organelles within the cell.

Cilia and Flagella: Cilia and flagella are microtubule-based extensions of the plasma membrane. Their movements result from the sliding of microtubules driven by the action of dynein motors.

cilium, flagellum, axoneme, nexin, basal body

Reorganization of Microtubules during Mitosis: Microtubules reorganize at the beginning of mitosis to form the mitotic spindle, which is responsible for chromosome separation.

mitotic spindle, kinetochore microtubule, chromosomal microtubule, polar microtubule, astral microtubule

Chromosome Movement: The duplicated chromosomes align on the metaphase plate. During anaphase of mitosis, daughter chromosomes separate and move to opposite poles of the mitotic spindle. Chromosome separation results from several types of movements in which different classes of spindle microtubules and motor proteins participate.

anaphase A, anaphase B

Questions

1. Why do actin filaments have a distinct polarity? Why is the polarity of actin filaments important in muscle contraction?

2. What is treadmilling, and at what concentration of monomers does it occur?

3. How would cytochalasin and phalloidin affect treadmilling actin filaments?

4. How do ADF/cofilin, profilin, and the Arp2/3 complex regulate actin filament assembly and turnover?

5. Which bands or zones of a muscle sarcomere change length during contraction? Why doesn't the A band change length?

6. How does Ca^{2+} regulate the contraction of smooth muscle cells?

7. How would expression of siRNAs targeted against vimentin affect the growth of fibroblasts in culture?

8. Why are intermediate filaments apolar, even though they are assembled from monomers that have distinct ends?

9. What key observation helped Vale and colleagues devise a strategy for the isolation of kinesin?

10. How would the removal of nexin affect the beating of cilia?

11. You are studying the transport of secretory vesicles containing insulin along microtubules in cultured pancreatic cells. How would treatment with colcemid affect the transport of these vesicles?

12. What is the cellular function of γ-tubulin?

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CHAPTER 13

The Plasma Membrane

- ***Structure of the Plasma Membrane 529***
- ***Transport of Small Molecules 540***
- ***Endocytosis 556***
- ***KEY EXPERIMENT:*** The LDL Receptor 559
- ***MOLECULAR MEDICINE:*** Cystic Fibrosis 554

ALL CELLS—BOTH PROKARYOTIC AND EUKARYOTIC—are surrounded by a plasma membrane, which defines the boundary of the cell and separates its internal contents from the environment. By serving as a selective barrier to the passage of molecules, the plasma membrane determines the composition of the cytoplasm. This ultimately defines the very identity of the cell, so the plasma membrane is one of the most fundamental structures of cellular evolution. Indeed, as discussed in Chapter 1, the first cell is thought to have arisen by the enclosure of self-replicating RNA in a membrane of phospholipids.

The plasma membranes of present-day cells are composed of both lipids and proteins. The basic structure of the plasma membrane is the phospholipid bilayer, which is impermeable to most water-soluble molecules. The passage of ions and most biological molecules across the plasma membrane is therefore mediated by proteins, which are responsible for the selective traffic of molecules into and out of the cell. Other proteins of the plasma membrane control the interactions between cells of multicellular organisms and serve as sensors through which the cell receives signals from its environment. The plasma membrane thus plays a dual role: It both isolates the cytoplasm and mediates interactions between the cell and its environment.

Structure of the Plasma Membrane

Like all other cellular membranes, the plasma membrane consists of both lipids and proteins. The fundamental structure of the membrane is the phospholipid bilayer, which forms a stable barrier between two aqueous compartments. In the case of the plasma membrane, these compartments are the inside and the outside of the cell. Proteins embedded within the phospholipid bilayer carry out the specific functions of the plasma membrane, including selective transport of molecules and cell-cell recognition.

The Phospholipid Bilayer

The plasma membrane is the most thoroughly studied of all cell membranes, and it is largely through investigations of the plasma membrane that our current concepts of membrane structure have evolved. The plasma membranes of mammalian red blood cells (erythrocytes) have been particularly useful as a model for studies of membrane structure. Mammalian red blood cells do not contain nuclei or internal membranes, so they represent a source from which pure plasma membranes can be easily isolated for biochemical analysis. Indeed, studies of the red blood cell plasma membrane provided the first evidence that biological membranes consist of lipid bilayers. In 1925, two Dutch scientists (Edwin Gorter and F. Grendel) extracted the membrane lipids from a known number of red blood cells corresponding to a known surface area of plasma membrane. They then determined the surface area occupied by a monolayer of the extracted lipid spread out at an air-water interface. The surface area of the lipid monolayer turned out to be twice that occupied by the erythrocyte plasma membranes, leading to the conclusion that the membranes consisted of lipid bilayers rather than monolayers.

The bilayer structure of the erythrocyte plasma membrane is clearly evident in high-magnification electron micrographs (Figure 13.1). The plasma membrane appears as two dense lines separated by an intervening space—a morphology frequently referred to as a "railroad track" appearance. This image results from the binding of the electron-dense heavy metals used as stains in transmission electron microscopy (see Chapter 1) to the polar head groups of the phospholipids, which therefore appear as dark lines. These dense lines are separated by the lightly stained interior portion of the membrane, which contains the hydrophobic fatty acid chains.

As discussed in Chapter 2, the plasma membranes of animal cells contain four major phospholipids (**phosphatidylcholine**, **phosphatidylethanolamine**, **phosphatidylserine**, and **sphingomyelin**), which together account for more than half of the lipid in most membranes. These phospholipids are asymmetrically distributed between the two halves of the membrane bilayer (Figure 13.2). The outer leaflet of the plasma membrane consists mainly of phosphatidylcholine and sphingomyelin, whereas phosphatidylethanolamine and phosphatidylserine are the predominant phos-

FIGURE 13.1 Bilayer structure of the plasma membrane Electron micrograph of a human red blood cell. Note the "railroad track" appearance of the plasma membrane. (Courtesy of J. David Robertson, Duke University Medical Center.)

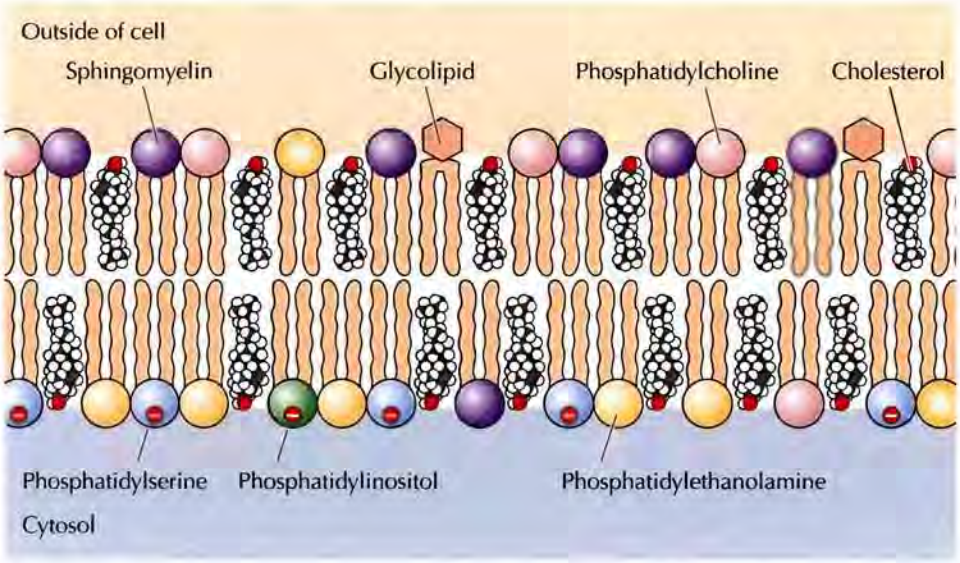


FIGURE 13.2 Lipid components of the plasma membrane The outer leaflet consists predominantly of phosphatidylcholine, sphingomyelin, and glycolipids, whereas the inner leaflet contains phosphatidylethanolamine, phosphatidylserine, and phosphatidylinositol. Cholesterol is distributed in both leaflets. The net negative charge of the head groups of phosphatidylserine and phosphatidylinositol is indicated. The structures of phospholipids, glycolipids, and cholesterol are shown in Figures 2.7, 2.8, and 2.9, respectively.

pholipids of the inner leaflet. The head groups of both phosphatidylserine and phosphatidylinositol are negatively charged, so their predominance in the inner leaflet results in a net negative charge on the cytosolic face of the plasma membrane. A fifth phospholipid, **phosphatidylinositol**, is also localized to the inner half of the plasma membrane. Although phosphatidylinositol is a quantitatively minor membrane component, it plays an important role in cell signaling, as discussed in Chapter 15.

In addition to the phospholipids, the plasma membranes of animal cells contain **glycolipids** and **cholesterol**. The glycolipids are found exclusively in the outer leaflet of the plasma membrane, with their carbohydrate portions exposed on the cell surface. They are relatively minor membrane components, constituting only about 2% of the lipids of most plasma membranes. Cholesterol, on the other hand, is a major membrane constituent of animal cells, being present in about the same molar amounts as the phospholipids.

Two general features of phospholipid bilayers are critical to membrane function. First, the structure of phospholipids is responsible for the basic function of membranes as barriers between two aqueous compartments. Because the interior of the phospholipid bilayer is occupied by hydrophobic fatty acid chains, the membrane is impermeable to water-soluble molecules, including ions and most biological molecules. Second, bilayers of the naturally occurring phospholipids are viscous fluids, not solids. The fatty acids of most natural phospholipids have one or more double bonds, which introduce kinks into the hydrocarbon chains and make them difficult to pack together. The long hydrocarbon chains of the fatty acids therefore move freely in the interior of the membrane, so the membrane itself is soft and flexible. In addition, both phospholipids and proteins are free to diffuse laterally within the membrane—a property that is critical for many membrane functions.

Because of its rigid ring structure, cholesterol plays a distinct role in membrane structure. Cholesterol will not form a membrane by itself but inserts into a bilayer of phospholipids with its polar hydroxyl group close to the phospholipid head groups (see Figure 13.2). Depending on the temperature, cholesterol has distinct effects on membrane fluidity. At high temperatures, cholesterol

interferes with the movement of the phospholipid fatty acid chains, making the outer part of the membrane less fluid and reducing its permeability to small molecules. At low temperatures, however, cholesterol has the opposite effect: By interfering with interactions between fatty acid chains, cholesterol prevents membranes from freezing and maintains membrane fluidity. Although cholesterol is not present in bacteria, it is an essential component of animal cell plasma membranes. Plant cells also lack cholesterol, but they contain related compounds (sterols) that fulfill a similar function.

Rather than diffusing freely in the plasma membrane, cholesterol and the sphingolipids (sphingomyelin and glycolipids) form discrete membrane domains known as **lipid rafts**. These clusters of sphingolipids and cholesterol move laterally within the plasma membrane and associate with specific membrane proteins. Although the functions of lipid rafts remain to be fully understood, they play important roles in processes such as cell movement and the uptake of extracellular molecules by endocytosis (as discussed later in this chapter) as well as in cell signaling (discussed in Chapter 15).

Membrane Proteins

While lipids are the fundamental structural elements of membranes, proteins are responsible for carrying out specific membrane functions. Most plasma membranes consist of approximately 50% lipid and 50% protein by weight, with the carbohydrate portions of glycolipids and glycoproteins constituting 5 to 10% of the membrane mass. Since proteins are much larger than lipids, this percentage corresponds to about one protein molecule per every 50 to 100 molecules of lipid. In 1972 Jonathan Singer and Garth Nicolson proposed the **fluid mosaic model** of membrane structure, which is now generally accepted as the basic paradigm for the organization of all biological membranes. In this model, membranes are viewed as two-dimensional fluids in which proteins are inserted into lipid bilayers (Figure 13.3).

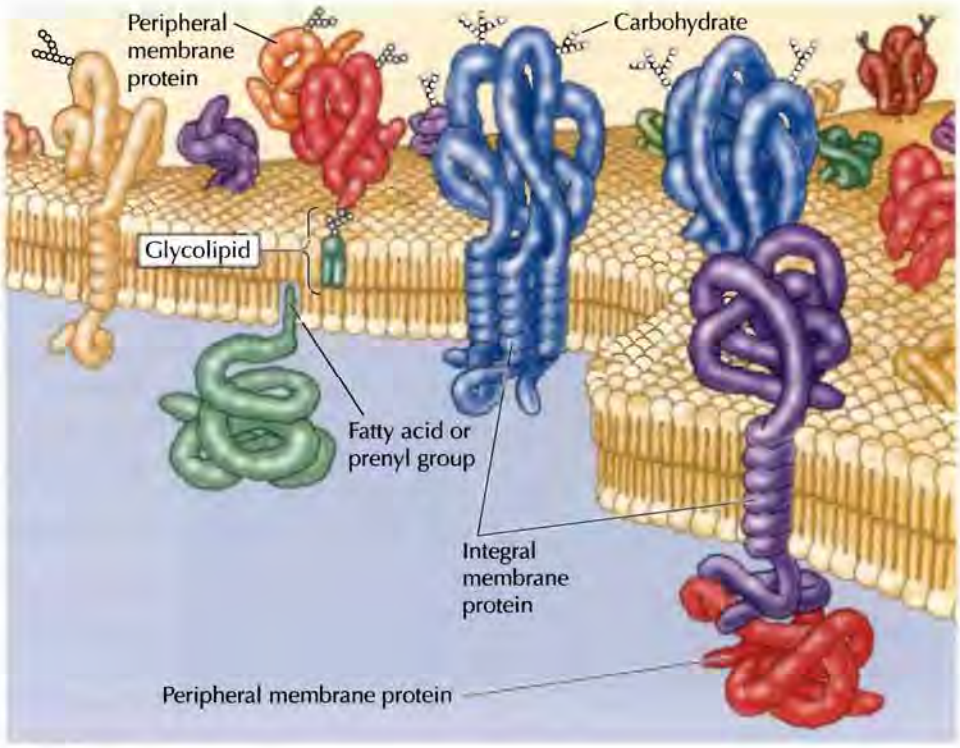


FIGURE 13.3 Fluid mosaic model of the plasma membrane Integral membrane proteins are inserted into the lipid bilayer, whereas peripheral proteins are bound to the membrane indirectly by protein-protein interactions. Most integral membrane proteins are transmembrane proteins with portions exposed on both sides of the lipid bilayer. The extracellular portions of these proteins are usually glycosylated, as are the peripheral membrane proteins bound to the external face of the membrane.

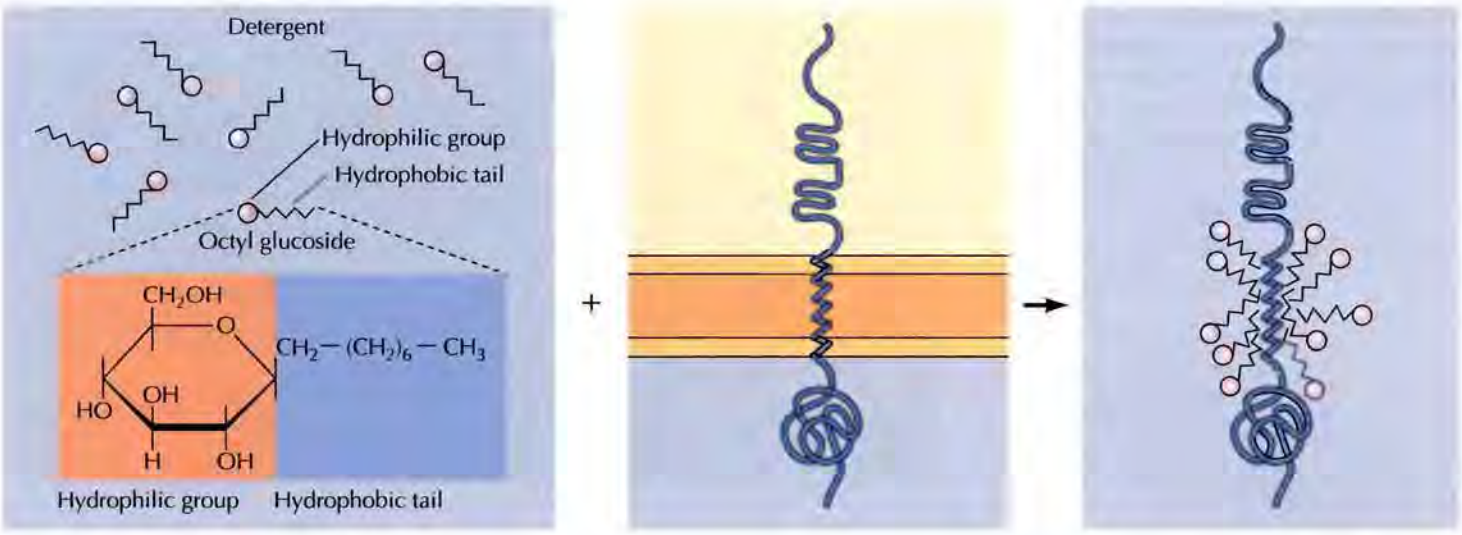


FIGURE 13.4 Solubilization of integral membrane proteins by detergents Detergents (e.g., octyl glucoside) are amphipathic molecules containing hydrophilic head groups and hydrophobic tails. The hydrophobic tails bind to the hydrophobic regions of integral membrane proteins, forming detergent-protein complexes that are soluble in aqueous solution.

Singer and Nicolson distinguished two classes of membrane-associated proteins, which they called **peripheral** and **integral membrane proteins**. Peripheral membrane proteins were operationally defined as proteins that dissociate from the membrane following treatments with polar reagents, such as solutions of extreme pH or high salt concentration that do not disrupt the phospholipid bilayer. Once dissociated from the membrane, peripheral membrane proteins are soluble in aqueous buffers. These proteins are not inserted into the hydrophobic interior of the lipid bilayer. Instead, they are indirectly associated with membranes through protein-protein interactions. These interactions frequently involve ionic bonds, which are disrupted by extreme pH or high salt.

In contrast to the peripheral membrane proteins, integral membrane proteins can be released only by treatments that disrupt the phospholipid bilayer. Portions of these integral membrane proteins are inserted into the lipid bilayer, so they can be dissociated only by reagents that disrupt hydrophobic interactions. The most commonly used reagents for solubilization of integral membrane proteins are detergents, which are small amphipathic molecules containing both hydrophobic and hydrophilic groups (Figure 13.4). The hydrophobic portions of detergents displace the membrane lipids and bind to the hydrophobic portions of integral membrane proteins. Because the other end of the detergent molecule is hydrophilic, the detergent-protein complexes are soluble in aqueous solutions.

Many integral proteins are **transmembrane proteins**, which span the lipid bilayer with portions exposed on both sides of the membrane. These proteins can be visualized in electron micrographs of plasma membranes prepared by the freeze-fracture technique (see Figure 1.36). In these specimens, the membrane is split and separated into its two leaflets. Transmembrane proteins are then apparent as particles on the internal faces of the membrane (Figure 13.5).

The membrane-spanning portions of transmembrane proteins are usually α helices of 20 to 25 hydrophobic amino acids that are inserted into the membrane of the endoplasmic reticulum during synthesis of the polypeptide chain (see Figures 10.12–10.14). These proteins are then transported in membrane vesicles from the endoplasmic reticulum to the Golgi apparatus and from there to the plasma membrane. Carbohydrate groups are added

FIGURE 13.5 Freeze-fracture electron micrograph of human red blood cell membranes The particles in the membrane are transmembrane proteins. (Harold H. Edwards/Visuals Unlimited.)

to the polypeptide chains in both the endoplasmic reticulum and Golgi apparatus, so most transmembrane proteins of the plasma membrane are glycoproteins with their oligosaccharides exposed on the surface of the cell.

Studies of red blood cells have provided good examples of both peripheral and integral proteins associated with the plasma membrane. The membranes of human erythrocytes contain about a dozen major proteins, which were originally identified by gel electrophoresis of membrane preparations. Most of these are peripheral membrane proteins that have been identified as components of the cortical cytoskeleton, which underlies the plasma membrane and determines cell shape (see Chapter 12). For example, the most abundant peripheral membrane protein of red blood cells is spectrin, which is the major cytoskeletal protein of erythrocytes. Other peripheral membrane proteins of red blood cells include actin, ankyrin, and band 4.1. Ankyrin serves as the principal link between the plasma membrane and the cytoskeleton by binding to both spectrin and the integral membrane protein band 3 (see Figure 12.14). An additional link between the membrane and the cytoskeleton is provided by band 4.1, which binds to the junctions of spectrin and actin, as well as to glycophorin (the other major integral membrane protein of erythrocytes).

The two major integral membrane proteins of red blood cells—glycophorin and band 3—provide well-studied examples of transmembrane protein structure (Figure 13.6). Glycophorin is a small glycoprotein of 131 amino acids, with a molecular weight of about 30,000, half of which is protein and half carbohydrate. Glycophorin crosses the membrane with a single membrane-spanning α helix of 23 amino acids, with its glycosylated amino-terminal portion exposed on the cell surface. Although glycophorin was one of the first transmembrane proteins to be characterized, its precise function remains unknown. In contrast, the function of the other major transmembrane protein of red blood cells is well understood. This protein, originally known as band 3, is the anion transporter responsible for the passage of bicarbonate (HCO_3^-) and chloride (Cl^-) ions across the red blood cell membrane. The band 3 polypeptide chain is 929 amino acids and is thought to have 14 membrane-spanning α-helical regions. Within the membrane, dimers of band 3 form globular structures containing internal channels through which ions are able to travel across the lipid bilayer.

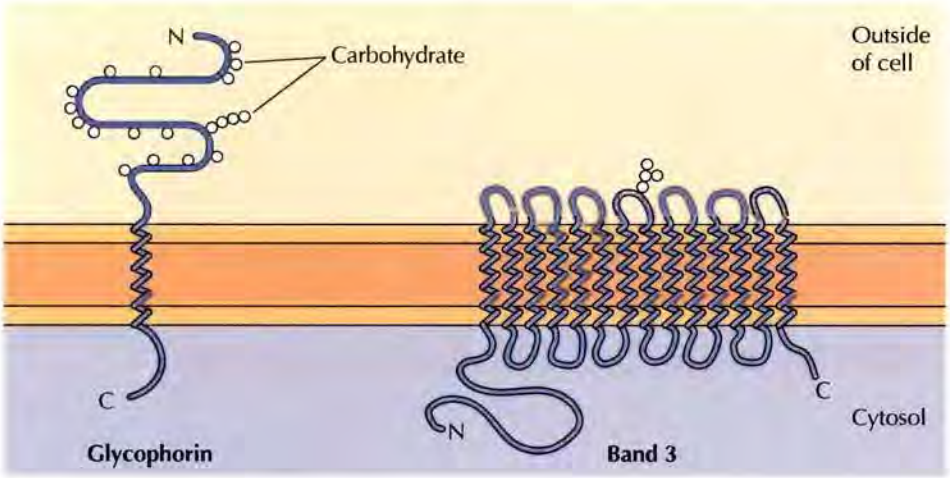


FIGURE 13.6 Integral membrane proteins of red blood cells Glycophorin (131 amino acids) contains a single transmembrane α helix. It is heavily glyocosylated, with oligosaccharides attached to 16 sites on the extracellular portion of the polypeptide chain. Band 3 (929 amino acids) has multiple transmembrane α helices and is thought to cross the membrane 14 times.

Because of their amphipathic character, transmembrane proteins have proved difficult to crystallize, as required for three-dimensional structural analysis by X-ray diffraction. The first transmembrane protein to be analyzed by X-ray crystallography was the photosynthetic reaction center of the bacterium *Rhodopseudomonas viridis* whose structure was reported in 1985 (Figure 13.7). The reaction center contains three transmembrane proteins designated L, M, and H (light, medium, and heavy) according to their apparent sizes indicated by gel electrophoresis. The L and M subunits each have five membrane-spanning α helices. The H subunit has only a single transmembrane α helix, with the bulk of the polypeptide chain on the cytosolic side of the membrane. The fourth subunit of the reaction center is a cytochrome, which is a peripheral membrane protein bound to the complex by protein-protein interactions.

Although most transmembrane proteins span the membrane by α-helical regions, this is not always the case. A well-characterized exception is provided by the **porins**—a class of proteins that form channels in the outer membranes of some bacteria. Many bacteria, including *E. coli*, have a dual membrane system in which the plasma membrane (or inner membrane) is surrounded by the cell wall and a distinct outer membrane (Figure 13.8). In contrast to the plasma membrane, the outer membrane is highly permeable to ions and small polar molecules (in the case of *E. coli*, with molecular weights up to 600). This permeability results from the porins, which form aqueous channels through the lipid bilayer. As discussed in Chapter 11, proteins related to the bacterial porins are also found in the outer membranes of mitochondria and chloroplasts.

Structural analysis shows that the porins do not contain hydrophobic β-helical regions. Instead, they cross the membrane as β barrels in which 8-22 β sheets fold up into a barrel-like structure enclosing an aqueous pore. The side chains of polar amino acids line the pore, whereas side chains of hydrophobic amino acids interact with the interior of the membrane. Some

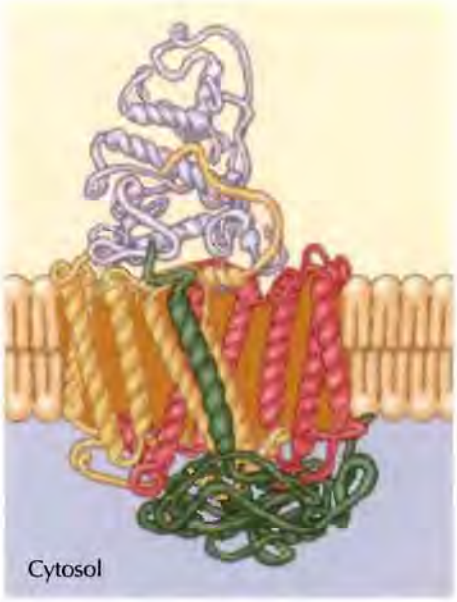


FIGURE 13.7 A bacterial photosynthetic reaction center The reaction center consists of three transmembrane proteins designated L (red), M (yellow), and H (green). The L and M subunits each have five transmembrane α helices, whereas the H subunit has only one. The fourth subunit of the reaction center is a cytochrome (white), which is a peripheral membrane protein.

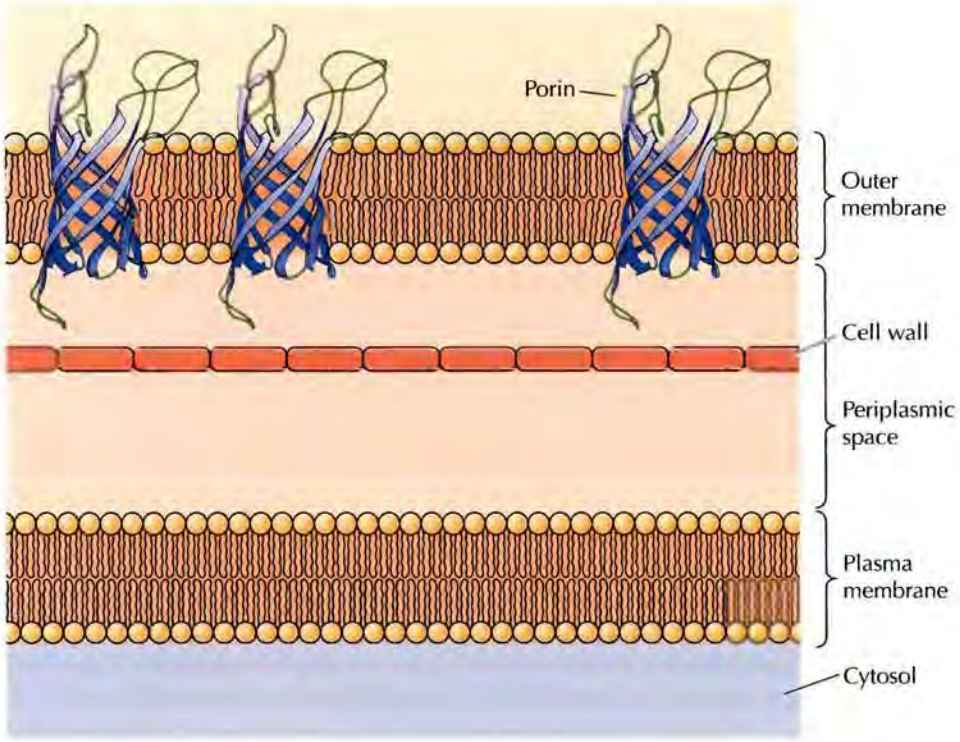


FIGURE 13.8 Bacterial outer membranes The plasma membrane of some bacteria is surrounded by a cell wall and a distinct outer membrane. The outer membrane contains porins, which form aqueous channels allowing the free passage of ions and small molecules. Porins cross the membrane as β-barrels.

porins exist in the membrane as monomers while others associate to form stable multimers, containing multiple channels through which polar molecules can diffuse across the membrane.

In contrast to transmembrane proteins, a variety of proteins (many of which behave as integral membrane proteins) are anchored in the plasma membrane by covalently attached lipids or glycolipids (Figure 13.9). Members of one class of these proteins are inserted into the outer leaflet of the plasma membrane by **glycosylphosphatidylinositol (GPI) anchors**. GPI anchors are added to certain proteins that have been transferred into the endoplasmic reticulum and are anchored in the membrane by a C-terminal transmembrane region (see Figure 10.17). The transmembrane region is cleaved as the GPI anchor is added, so these proteins remain attached to the membrane only by the glycolipid. Since the polypeptide chains of GPI-anchored proteins are transferred into the endoplasmic reticulum, they are glycosylated and exposed on the surface of the cell following transport to the plasma membrane.

Other proteins are anchored in the inner leaflet of the plasma membrane by covalently attached lipids. Rather than being processed through the secretory pathway, these proteins are synthesized on free cytosolic ribosomes and then modified by the addition of lipids. These modifications include the addition of myristic acid (a 14-carbon fatty acid) to the amino terminus of the polypeptide chain, the addition of prenyl groups (15 or 20 carbons) to the side chains of carboxy-terminal cysteine residues, and the addition of palmitic acid (16 carbons) to the side chains of cysteine residues (see Figures 8.32–8.34). In some cases, these proteins (many of which behave as peripheral membrane proteins) are targeted to the plasma membrane by positively charged regions of the polypeptide chain as well as by the attached lipids. These positively charged protein domains may interact with the negatively charged head groups of phosphatidylserine on the

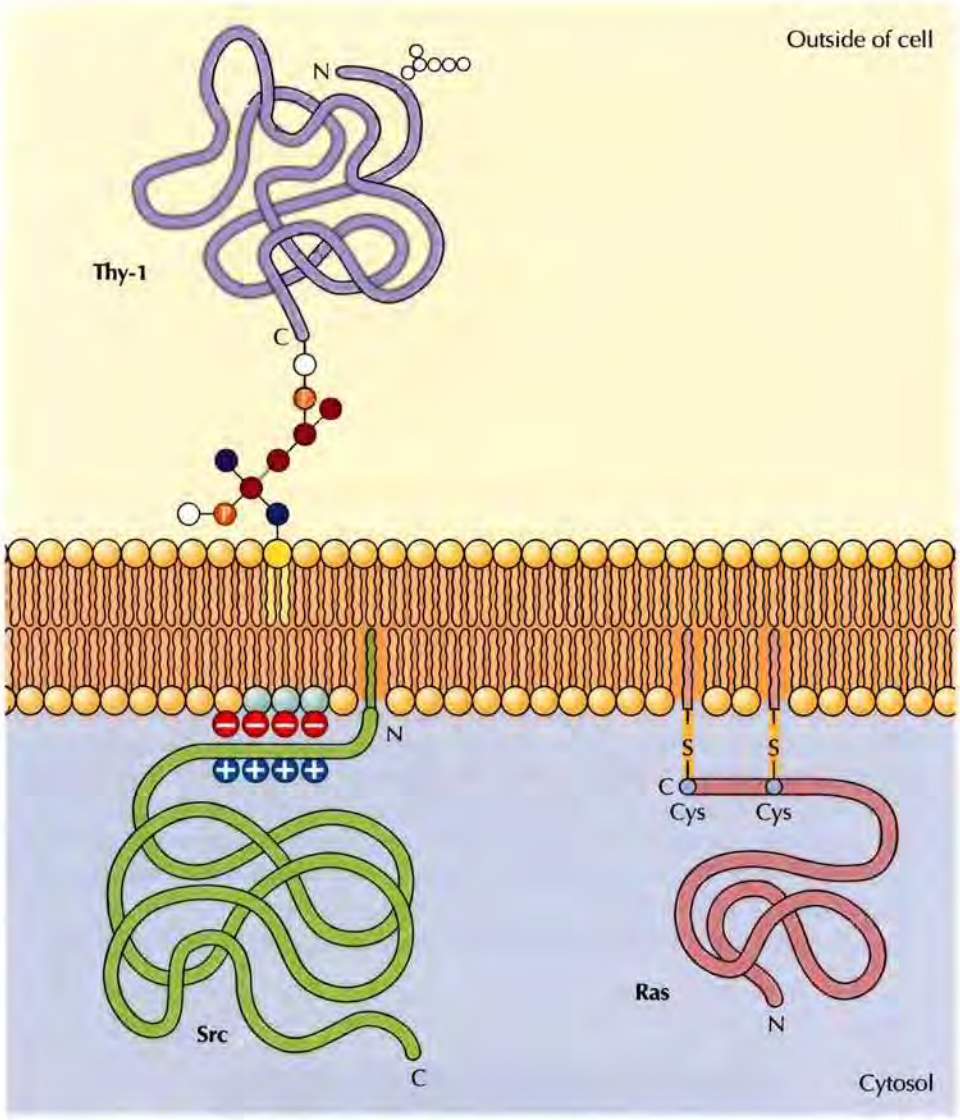


FIGURE 13.9 Examples of proteins anchored in the plasma membrane by lipids and glycolipids Some proteins (e.g., the lymphocyte protein Thy-1) are anchored in the outer leaflet of the plasma membrane by GPI anchors added to their C terminus in the endoplasmic reticulum. These proteins are glycosylated and exposed on the cell surface. Other proteins are anchored in the inner leaflet of the plasma membrane following their translation on free cytosolic ribosomes. The Ras protein illustrated is anchored by a prenyl group attached to the side chain of a C-terminal cysteine and by a palmitoyl group attached to a cysteine located five amino acids upstream. The Src protein is anchored by a myristoyl group attached to its N terminus. A positively charged region of Src also plays a role in membrane association, perhaps by interacting with the negatively charged head groups of phosphatidylserine. The structures of these lipid and glycolipid groups are illustrated in Figures 8.32 through 8.34.

cytosolic face of the plasma membrane. It is noteworthy that many of the proteins anchored in the inner leaflet of the plasma membrane (including the Src and Ras proteins illustrated in Figure 13.9) play important roles in the transmission of signals from cell surface receptors to intracellular targets, as discussed in Chapter 15.

Mobility of Membrane Proteins

Membrane proteins and phospholipids are unable to move back and forth between the inner and outer leaflets of the membrane at an appreciable rate. However, because they are inserted into a fluid lipid bilayer, both proteins and lipids are able to diffuse laterally through the membrane. This lateral movement was first shown directly in an experiment reported by Larry Frye and Michael Edidin in 1970, which provided support for the fluid mosaic model. Frye and Edidin fused human and mouse cells in culture to produce human-mouse cell hybrids (Figure 13.10). They then analyzed the distribution of proteins in the membranes of these hybrid cells using antibodies that specifically recognize proteins of human and mouse origin. These antibodies were labeled with different fluorescent dyes, so the

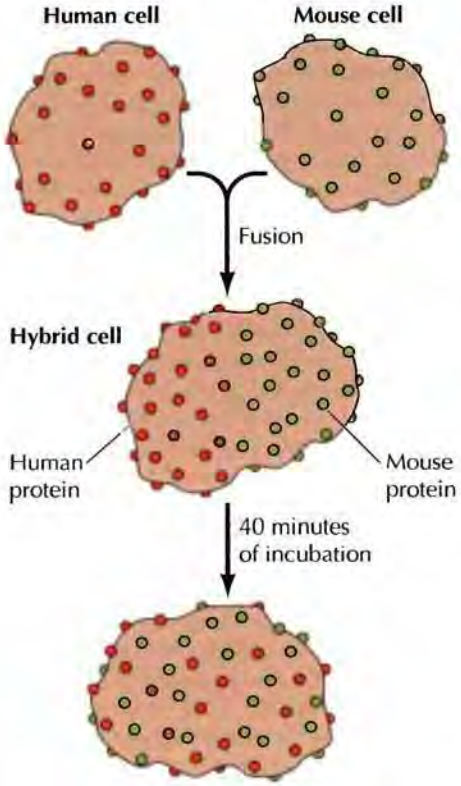


FIGURE 13.10 Mobility of membrane proteins Human and mouse cells were fused to produce hybrid cells. The distribution of cell surface proteins was then analyzed using anti-human and anti-mouse antibodies labeled with different fluorescent dyes (red and green, respectively). The human and mouse proteins were detected in different halves of the hybrid cells immediately after fusion but had intermingled over the cell surface following 40 minutes of incubation.

human and mouse proteins could be distinguished by fluorescence microscopy. Immediately after fusion, human and mouse proteins were localized to different halves of the hybrid cells. However, after a brief period of incubation at 37°C the human and mouse proteins were completely intermixed over the cell surface, indicating that they moved freely through the plasma membrane.

However, not all proteins are able to diffuse freely through the membrane. In some cases, the mobility of membrane proteins is restricted by their association with the cytoskeleton. For example, a fraction of band 3 in the red blood cell membrane is immobilized as a result of its association with ankyrin and spectrin. In other cases, the mobility of membrane proteins may be restricted by their associations with other membrane proteins, with proteins on the surface of adjacent cells, or with the extracellular matrix.

In contrast to blood cells, epithelial cells are polarized when they are organized into tissues, with different parts of the cell responsible for performing distinct functions. Consequently, the plasma membranes of many epithelial cells are divided into distinct **apical** and **basolateral domains** that differ in function and protein composition (Figure 13.11). For example, epithelial cells of the small intestine function to absorb nutrients from the

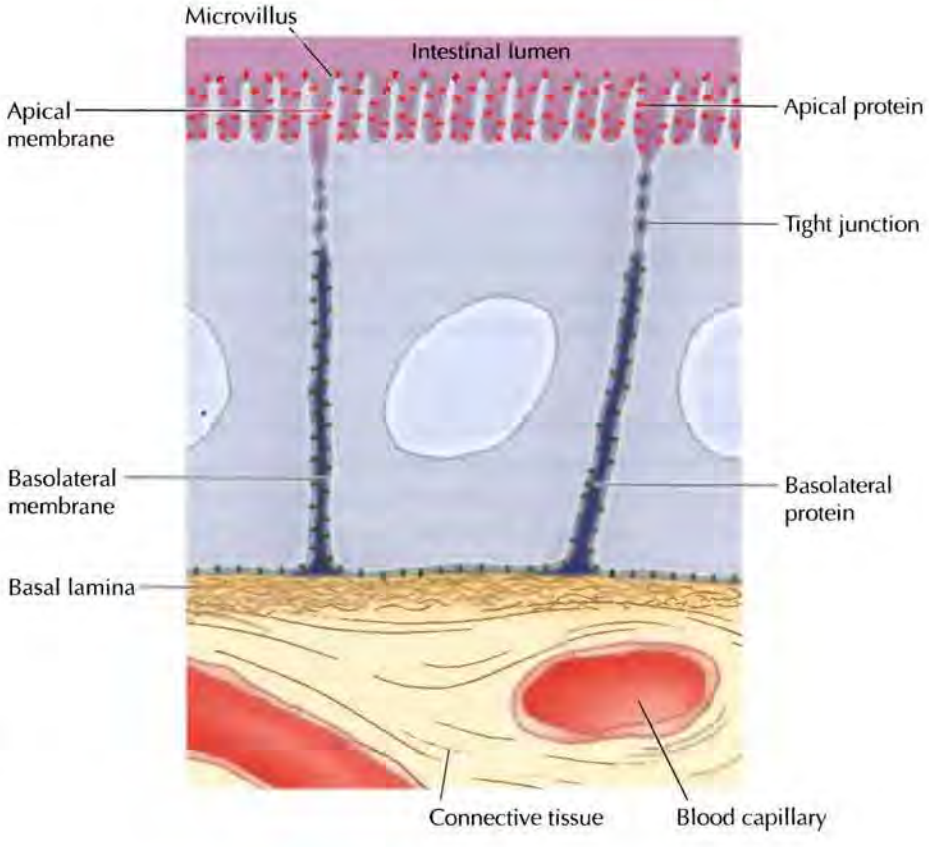


FIGURE 13.11 A polarized intestinal epithelial cell The apical surface of the cell contains microvilli and is specialized for absorption of nutrients from the intestinal lumen. The basolateral surface is specialized for the transfer of absorbed nutrients to the underlying connective tissue, which contains blood capillaries. Tight junctions separate the apical and basolateral domains of the plasma membrane. Membrane proteins are free to diffuse within each domain but are not able to cross from one domain to the other.

digestive tract. The apical surface of these cells, which faces the intestinal lumen, is therefore covered by microvilli, which increase its surface area and facilitate nutrient absorption. The basolateral surface, which faces underlying connective tissue and the blood supply, is specialized to mediate the transfer of absorbed nutrients into the circulation. In order to maintain these distinct functions, the mobility of plasma membrane proteins must be restricted to the appropriate domains of the cell surface. At least part of the mechanism by which this occurs involves the formation of tight junctions (which are discussed in Chapter 14) between adjacent cells of the epithelium. These junctions not only seal the space between cells but also serve as barriers to the movement of membrane lipids and proteins. As a result, proteins are able to diffuse within either the apical or basolateral domains of the plasma membrane but are not able to cross from one domain to the other.

Lipid composition can also perturb the free diffusion of membrane proteins. The melting temperatures of the sphingolipids (sphingomyelin and glycolipids) differ from those of the phospholipids derived from glycerol. Sphingomyelin and glycolipids tend to cluster in small semisolid patches termed lipid rafts, which are also enriched in cholesterol because of the packing affinity of cholesterol and the sphingolipids (Figure 13.12). Rafts are also enriched in GPI-anchored proteins, and several proteins found in the bulk phospholipid bilayer, such as GTP-binding proteins and several protein kinases, move in and out of rafts. The transient presence of these proteins in rafts allows the clustering necessary for processes such as endocytosis, extension of cell processes, and receptor-mediated signaling.

FIGURE 13.12 Structure of lipid rafts Lipid rafts are organized by the interactions of sphingomyelin, glycolipids, and cholesterol. GPI-anchored proteins are preferentially found in lipid rafts, and several other types of membrane proteins are present transiently in rafts to mediate cell signaling or endocytosis.

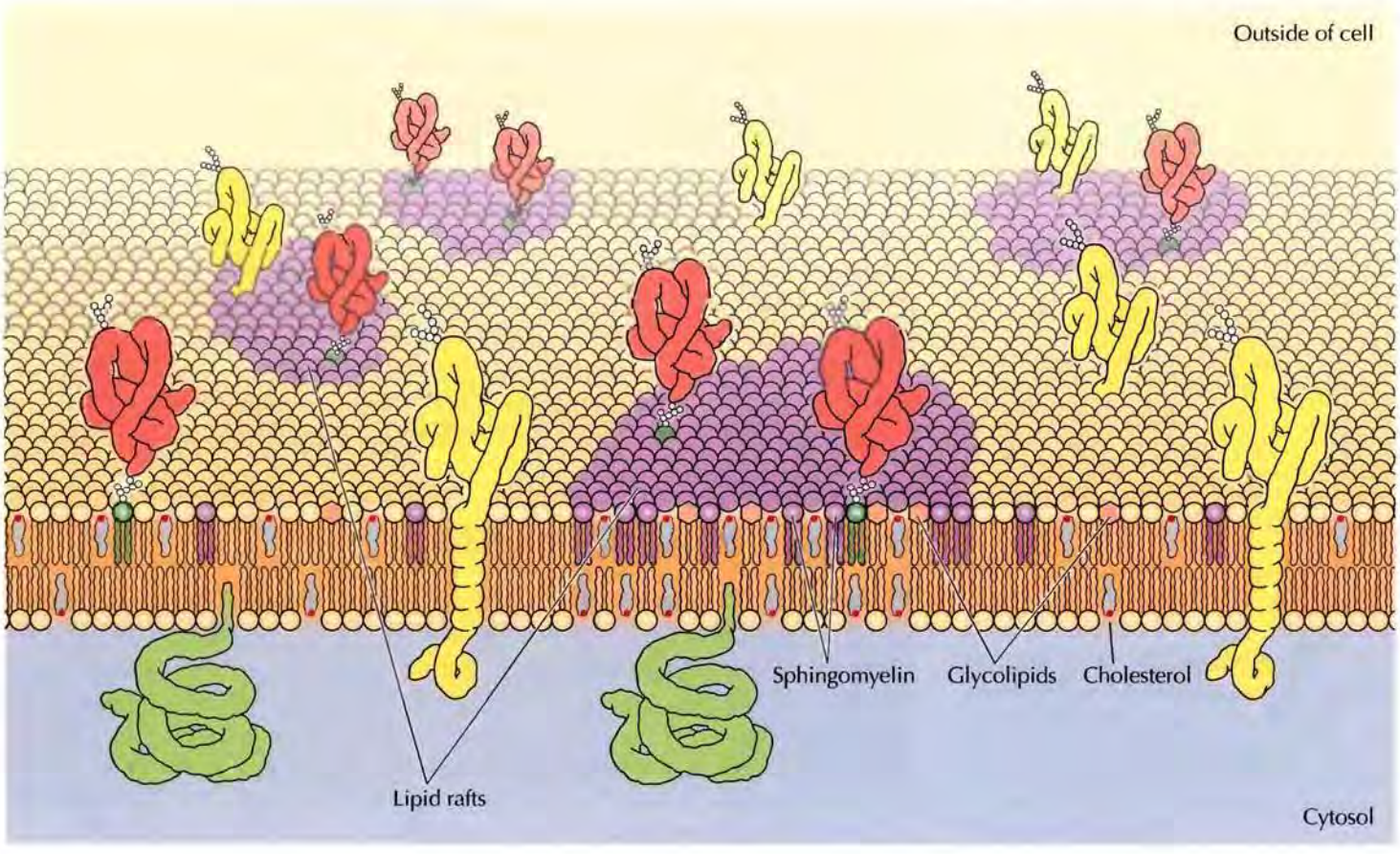


FIGURE 13.13 The glycocalyx An electron micrograph of intestinal epithelium illustrating the glycocalyx (arrows). (Don Fawcett/Visuals Unlimited.)

Lipid rafts can also be stabilized by interactions with the cytoskeleton through peripheral membrane proteins. One such peripheral protein is caveolin, which is associated with a subclass of lipid rafts called caveolae. As discussed later in this chapter, caveolae function in endocytosis.

The Glycocalyx

As already discussed, the extracellular portions of plasma membrane proteins are generally glycosylated. Likewise, the carbohydrate portions of glycolipids are exposed on the outer face of the plasma membrane. Consequently, the surface of the cell is covered by a carbohydrate coat known as the **glycocalyx**, which is formed by the oligosaccharides of glycolipids and transmembrane glycoproteins (Figure 13.13).

Part of the role of the glycocalyx is to protect the cell surface. In addition, the oligosaccharides of the glycocalyx serve as markers for a variety of cell-cell interactions. A well-studied example of these interactions is the adhesion of white blood cells (leukocytes) to the endothelial cells that line blood vessels—a process that allows the leukocytes to leave the circulatory system and mediate the inflammatory response in injured tissues. The initial step in adhesion between leukocytes and endothelial cells is mediated by a family of transmembrane proteins called **selectins**, which recognize specific carbohydrates on the cell surface (Figure 13.14). Two members of the selectin family (E-selectin and P-selectin) expressed by endothelial cells and platelets bind to specific oligosaccharides expressed on the surface of leukocytes. A different selectin (L-selectin) is expressed by leukocytes and recognizes an oligosaccharide on the surface of endothelial cells. The oligosaccharides exposed on the cell surface thus provide a set of markers that help identify the distinct cell types of multicellular organisms.

Transport of Small Molecules

The internal composition of the cell is maintained because the plasma membrane is selectively permeable to small molecules. Most biological molecules are unable to diffuse through the phospholipid bilayer, so the plasma membrane forms a barrier that blocks the free exchange of molecules between the cytoplasm and the external environment of the cell. Specific transport proteins (carrier proteins and channel proteins) then mediate the selective passage of small molecules across the membrane, allowing the cell to control the composition of its cytoplasm.

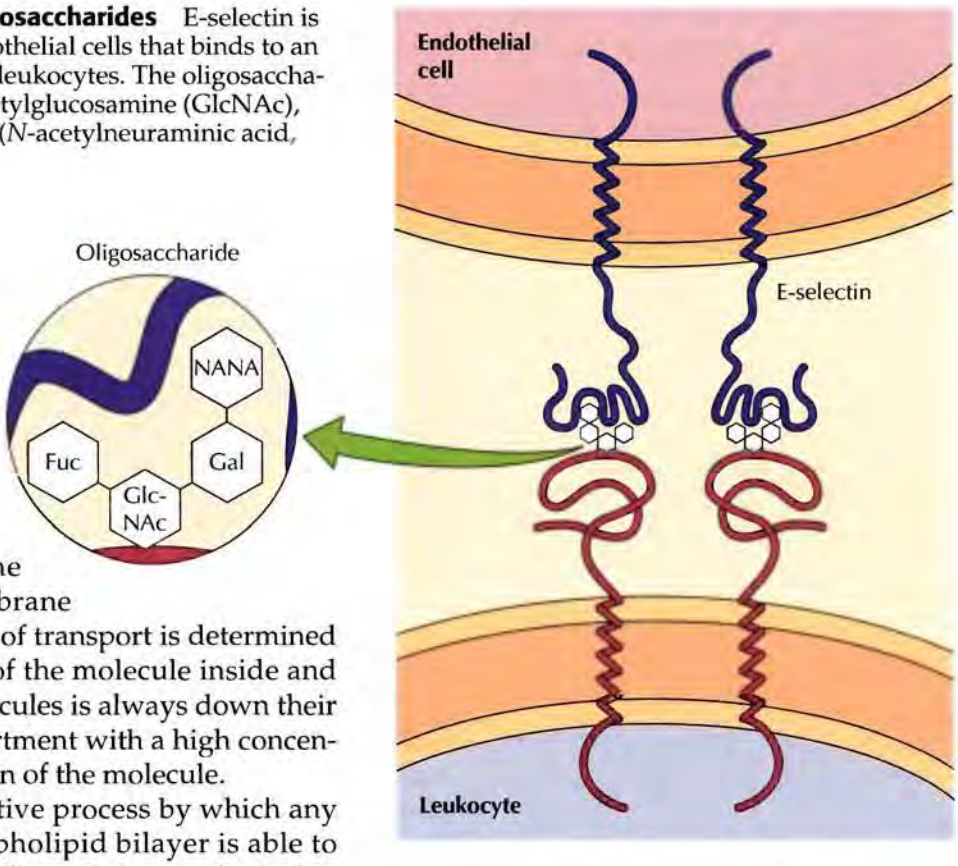


FIGURE 13.14 Binding of selectins to oligosaccharides E-selectin is a transmembrane protein expressed by endothelial cells that binds to an oligosaccharide expressed on the surface of leukocytes. The oligosaccharide recognized by E-selectin contains *N*-acetylglucosamine (GlcNAc), fucose (Fuc), galactose (Gal), and sialic acid (*N*-acetylneuraminic acid, NANA).

Passive Diffusion

The simplest mechanism by which molecules can cross the plasma membrane is **passive diffusion**. During passive diffusion, a molecule simply dissolves in the phospholipid bilayer, diffuses across it, and then dissolves in the aqueous solution at the other side of the membrane. No membrane proteins are involved and the direction of transport is determined simply by the relative concentrations of the molecule inside and outside of the cell. The net flow of molecules is always down their concentration gradient—from a compartment with a high concentration to one with a lower concentration of the molecule.

Passive diffusion is thus a nonselective process by which any molecule able to dissolve in the phospholipid bilayer is able to cross the plasma membrane and equilibrate between the inside and outside of the cell. Importantly, only small, relatively hydrophobic molecules are able to diffuse across a phospholipid bilayer at significant rates (Figure 13.15). Thus gases (such as O_2 and CO_2), hydrophobic molecules (such as benzene), and small polar but uncharged molecules (such as H_2O and ethanol) are able to passively diffuse across the plasma membrane. Other biological molecules, however, are unable to dissolve in the hydrophobic interior of the phospholipid bilayer. Consequently, larger

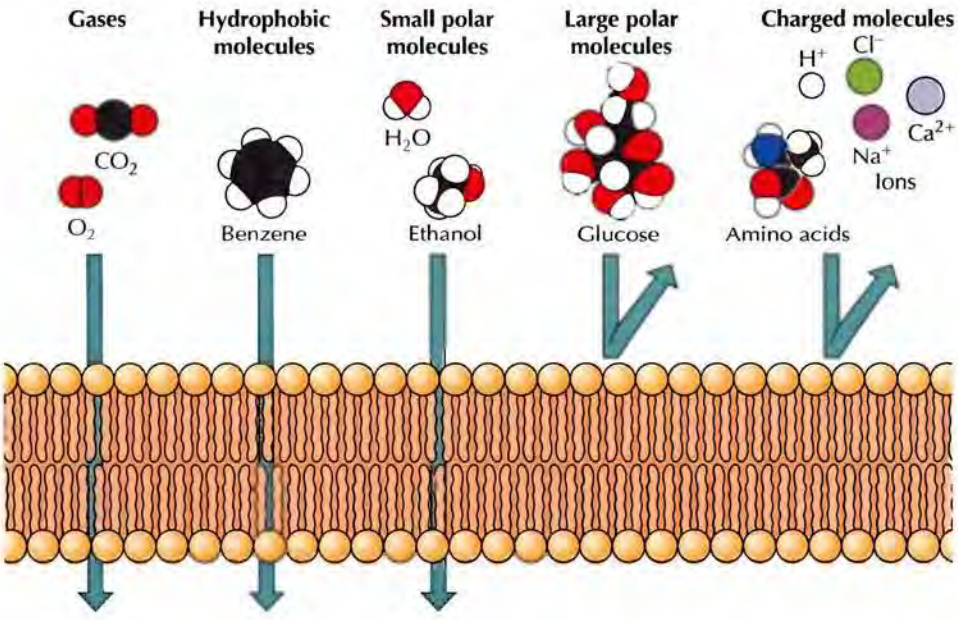


FIGURE 13.15 Permeability of phospholipid bilayers Gases, hydrophobic molecules, and small polar uncharged molecules can passively diffuse through phospholipid bilayers. Larger polar molecules and charged molecules cannot.

uncharged polar molecules such as glucose are unable to cross the plasma membrane by passive diffusion, as are charged molecules of any size (including small ions such as H^+, Na^+, K^+, and Cl^-). The passage of these molecules across the membrane instead requires the activity of specific transport and channel proteins, which therefore control the traffic of most biological molecules into and out of the cell.

Facilitated Diffusion and Carrier Proteins

Facilitated diffusion, like passive diffusion, involves the movement of molecules in the direction determined by their relative concentrations inside and outside of the cell. No external source of energy is provided, so molecules travel across the membrane in the direction determined by their concentration gradients and, in the case of charged molecules, by the electric potential across the membrane. However, facilitated diffusion differs from passive diffusion in that the transported molecules do not dissolve in the phospholipid bilayer. Instead, their passage is mediated by proteins that enable the transported molecules to cross the membrane without directly interacting with its hydrophobic interior. Facilitated diffusion therefore allows polar and charged molecules, such as carbohydrates, amino acids, nucleosides, and ions, to cross the plasma membrane.

Two classes of proteins that mediate facilitated diffusion have generally been distinguished: carrier proteins and channel proteins. **Carrier proteins** bind specific molecules to be transported on one side of the membrane. They then undergo conformational changes that allow the molecule to pass through the membrane and be released on the other side. In contrast, **channel proteins** form open pores through the membrane, allowing the free diffusion of any molecule of the appropriate size and charge.

Carrier proteins are responsible for the facilitated diffusion of sugars, amino acids, and nucleosides across the plasma membranes of most cells. The uptake of glucose, which serves as a primary source of metabolic energy, is one of the most important transport functions of the plasma membrane, and the glucose transporter provides a well-studied example of a carrier protein. The glucose transporter was initially identified as a 55-kd protein in human red blood cells in which it represents approximately 5% of total membrane protein. Subsequent isolation and sequence analysis of a cDNA clone revealed that the glucose transporter has 12 α-helical transmembrane segments—a structure typical of many carrier proteins (Figure 13.16). These transmembrane α-helices contain predominantly hydrophobic amino acids, but several also contain polar amino acid residues that are thought to form the glucose-binding site in the interior of the protein.

As with many membrane proteins, the three-dimensional structure of the glucose transporter is not known, so the molecular mechanism of glucose transport remains an open question. However, kinetic studies indicate that the glucose transporter functions by alternating between two conformational states (Figure 13.17). In the first conformation, a glucose-binding site faces the outside of the cell. The binding of glucose to this exterior site induces a conformational change in the transporter such that the glucose-binding site now faces the interior of the cell. Glucose can then be released into the cytosol, followed by the return of the transporter to its original conformation.

Most cells, including erythrocytes, are exposed to extracellular glucose concentrations that are higher than those inside the cell, so facilitated diffusion results in the net inward transport of glucose. Once glucose is taken up by these cells it is rapidly metabolized, so intracellular glucose concentra-

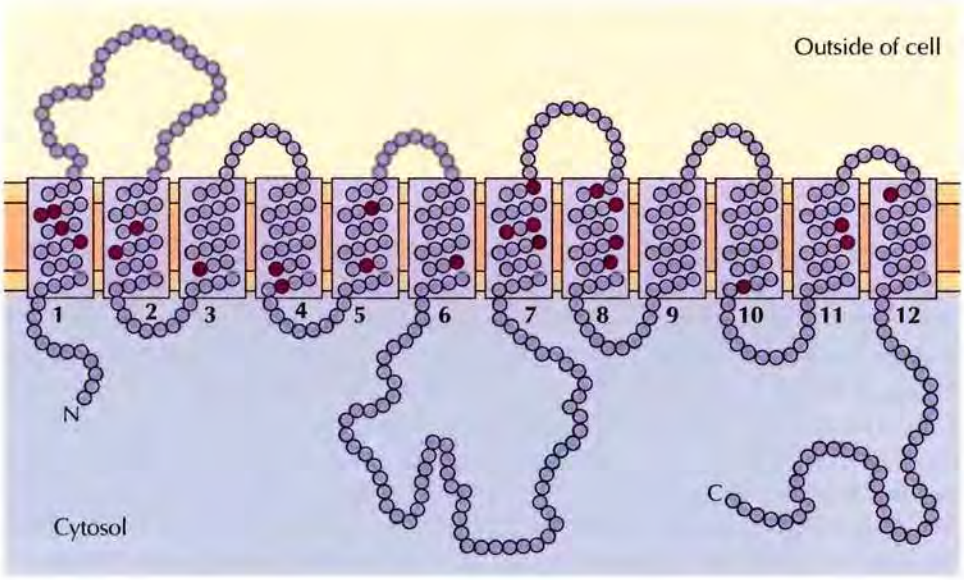


FIGURE 13.16 Structure of the glucose transporter The glucose transporter has 12 transmembrane α helices. Polar amino acid residues located within the phospholipid bilayer are indicated as dark purple circles. (Adapted from G. I. Bell, C. F. Burant, J. Takeda and G. W. Gould, 1993. *J. Biol. Chem.* 268: 19161.)

tions remain low and glucose continues to be transported into the cell from the extracellular fluids. Because the conformational changes of the glucose transporter are reversible, however, glucose can be transported in the opposite direction simply by reversing the steps in Figure 13.17. Such reverse flow occurs, for example, in liver cells in which glucose is synthesized and released into the circulation.

Ion Channels

In contrast to carrier proteins, channel proteins form open pores in the membrane, allowing small molecules of the appropriate size and charge to pass freely through the lipid bilayer. One group of channel proteins discussed earlier is the porins, which permit the free passage of ions and small

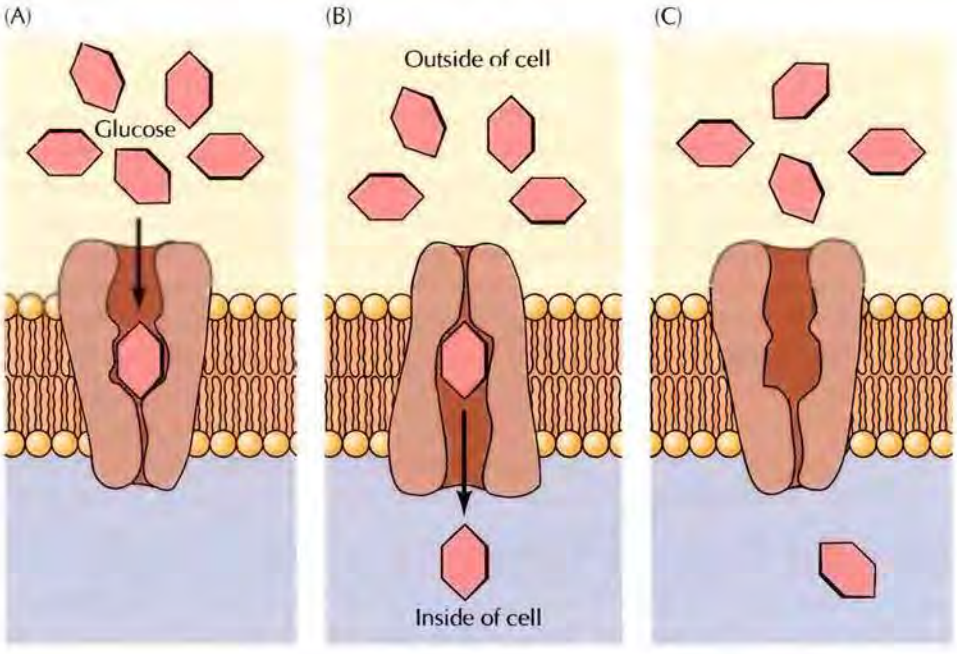
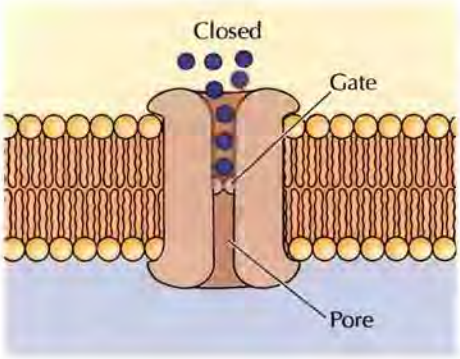


FIGURE 13.17 Model for the facilitated diffusion of glucose The glucose transporter alternates between two conformations in which a glucose-binding site is alternately exposed on the outside and the inside of the cell. In the first conformation shown (A), glucose binds to a site exposed on the outside of the plasma membrane. The transporter then undergoes a conformational change such that the glucose-binding site faces the inside of the cell and glucose is released into the cytosol (B). The transporter then returns to its original conformation (C).



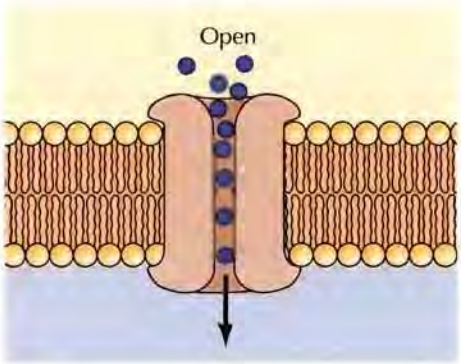


FIGURE 13.18 Model of an ion channel In the closed conformation, the flow of ions is blocked by a gate. Opening of the gate allows ions to flow rapidly through the channel. The channel contains a narrow pore that restricts passage to ions of the appropriate size and charge.

polar molecules through the outer membranes of bacteria (see Figure 13.8). Gap junctions, which are discussed in Chapter 14, contain channel proteins that permit the passage of molecules between connected cells. The plasma membranes of many cells also contain water channel proteins (aquaporins) through which water molecules are able to cross the membrane much more rapidly than they can diffuse through the phospholipid bilayer. The best-characterized channel proteins, however, are the **ion channels**, which mediate the passage of ions across plasma membranes. Although ion channels are present in the membranes of all cells, they have been especially well studied in nerve and muscle, where their regulated opening and closing is responsible for the transmission of electric signals.

Three properties of ion channels are central to their function (Figure 13.18). First, transport through channels is extremely rapid. More than a million ions per second flow through open channels—a flow rate approximately a thousand times greater than the rate of transport by carrier proteins. Second, ion channels are highly selective because narrow pores in the channel restrict passage to ions of the appropriate size and charge. Thus specific channel proteins allow the passage of Na^+, K^+, Ca^{2+}, and Cl^- across the membrane. Third, most ion channels are not permanently open. Instead, the opening of ion channels is regulated by "gates" that transiently open in response to specific stimuli. Some channels (called **ligand-gated channels**) open in response to the binding of neurotransmitters or other signaling molecules; others (**voltage-gated channels**) open in response to changes in electric potential across the plasma membrane.

The fundamental role of ion channels in the transmission of electric impulses was elucidated through a series of elegant experiments reported by Alan Hodgkin and Andrew Huxley in 1952. These investigators used the giant nerve cells of the squid as a model. The axons of these giant neurons have a diameter of about 1 mm, making it possible to insert electrodes and measure the changes in membrane potential that take place during the transmission of nerve impulses. Using this approach, Hodgkin and Huxley demonstrated that these changes in membrane potential result from the regulated opening and closing of Na^+ and K^+ channels in the plasma membrane. It subsequently became possible to study the activity of individual ion channels using the **patch clamp technique** developed by Erwin Neher and Bert Sakmann in 1976 (Figure 13.19). In this method, a micropipette with a tip diameter of about 1 μm is used to isolate a small patch of membrane, allowing the flow of ions through a single channel to be analyzed and greatly increasing the precision with which the activities of ion channels can be studied.

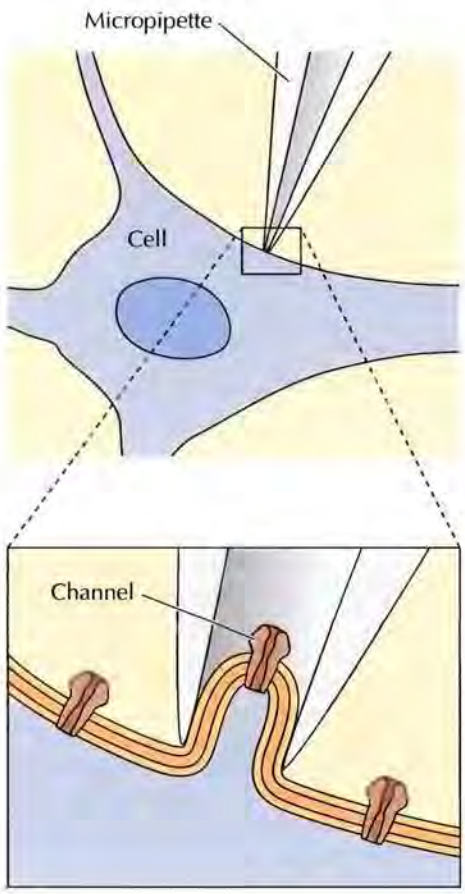


FIGURE 13.19 The patch clamp technique A small patch of membrane is isolated in the tip of a micropipette. Stimuli can then be applied from within the pipette, allowing the behavior of the trapped channel to be measured. (Adapted from E. Neher and B. Sakmann, 1992. *Sci. Am.* 266(3): 44.)

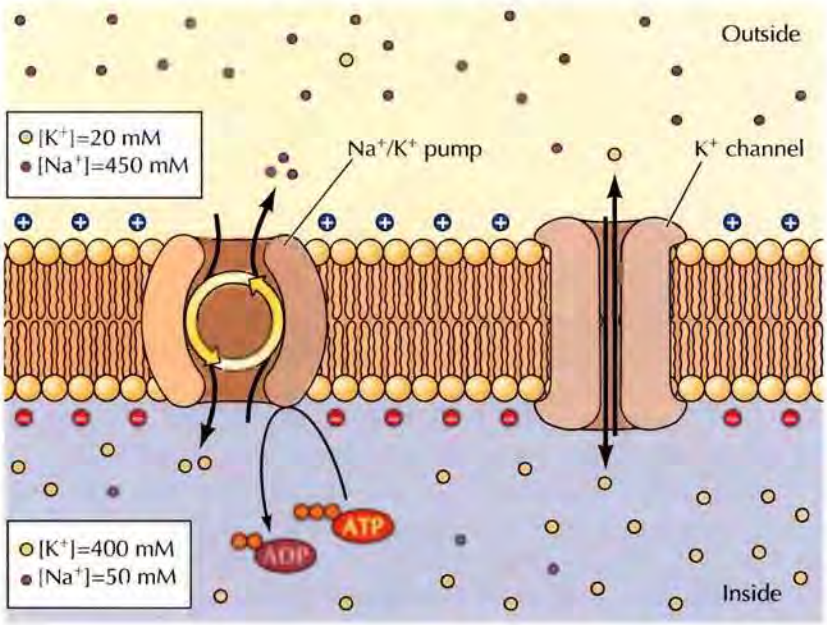


FIGURE 13.20 Ion gradients and resting membrane potential of the giant squid axon Only the concentrations of Na^+ and K^+ are shown because these are the ions that function in the transmission of nerve impulses. Na^+ is pumped out of the cell while K^+ is pumped in, so the concentration of Na^+ is higher outside than inside of the axon, whereas the concentration of K^+ is higher inside than out. The resting membrane is more permeable to K^+ than to Na^+ or other ions because it contains open K^+ channels. The flow of K^+ through these channels makes the major contribution to the resting membrane potential of –60 mV, which is therefore close to the K^+ equilibrium potential.

The flow of ions through membrane channels is dependent on the establishment of ion gradients across the plasma membrane. All cells, including nerve and muscle, contain ion pumps (discussed in the next section) that use energy derived from ATP hydrolysis to actively transport ions across the plasma membrane. As a result, the ionic composition of the cytoplasm is substantially different from that of extracellular fluids (Table 13.1). For example, Na^+ is actively pumped out of cells while K^+ is pumped in. Therefore, in the squid axon the concentration of Na^+ is about 10 times higher in extracellular fluids than inside the cell, whereas the concentration of K^+ is approximately 20 times higher in the cytosol than in the surrounding medium.

Because ions are electrically charged, their transport results in the establishment of an electric gradient across the plasma membrane. In resting squid axons there is an electric potential of about 60 mV across the plasma membrane, with the inside of the cell negative with respect to the outside (Figure 13.20). This electric potential arises both from ion pumps and from the flow of ions through channels that are open in the resting cell plasma membrane. The plasma membrane of resting squid axons contains open K^+ channels, so it is more permeable to K^+ than to Na^+ or other ions. Consequently, the flow of K^+ makes the largest contribution to the resting membrane potential.

As discussed in Chapter 11, the flow of ions across a membrane is driven by both the concentration and voltage components of an electrochemical gradient. For example, the twentyfold higher concentration of K^+ inside the squid axon as compared to the extracellular fluid drives the flow of K^+ out of the cell. However, because K^+ is positively charged this efflux of K^+ from the cell generates an electric potential across the membrane, with the inside of the cell becoming negatively charged. This membrane potential opposes the continuing flow of K^+ out of the cell, and the system approaches the equilibrium state in which the membrane potential balances the K^+ concentration gradient.

■ **The squid giant axon is approximately 100 times larger in diameter than typical mammalian axons. Its large diameter allows high-speed transduction of nerve impulses, which is vital for the rapid escape response of squids from predators.**

TABLE 13.1 Extracellular and Intracellular Ion Concentrations

	Concentration (m*M*)	
Ion	**Intracellular**	**Extracellular**
Squid axon		
K^+	400	20
Na^+	50	450
Cl^-	40–150	560
Ca^{2+}	0.0001	10
Mammalian cell		
K^+	140	5
Na^+	5–15	145
Cl^-	4	110
Ca^{2+}	0.0001	2.5–5

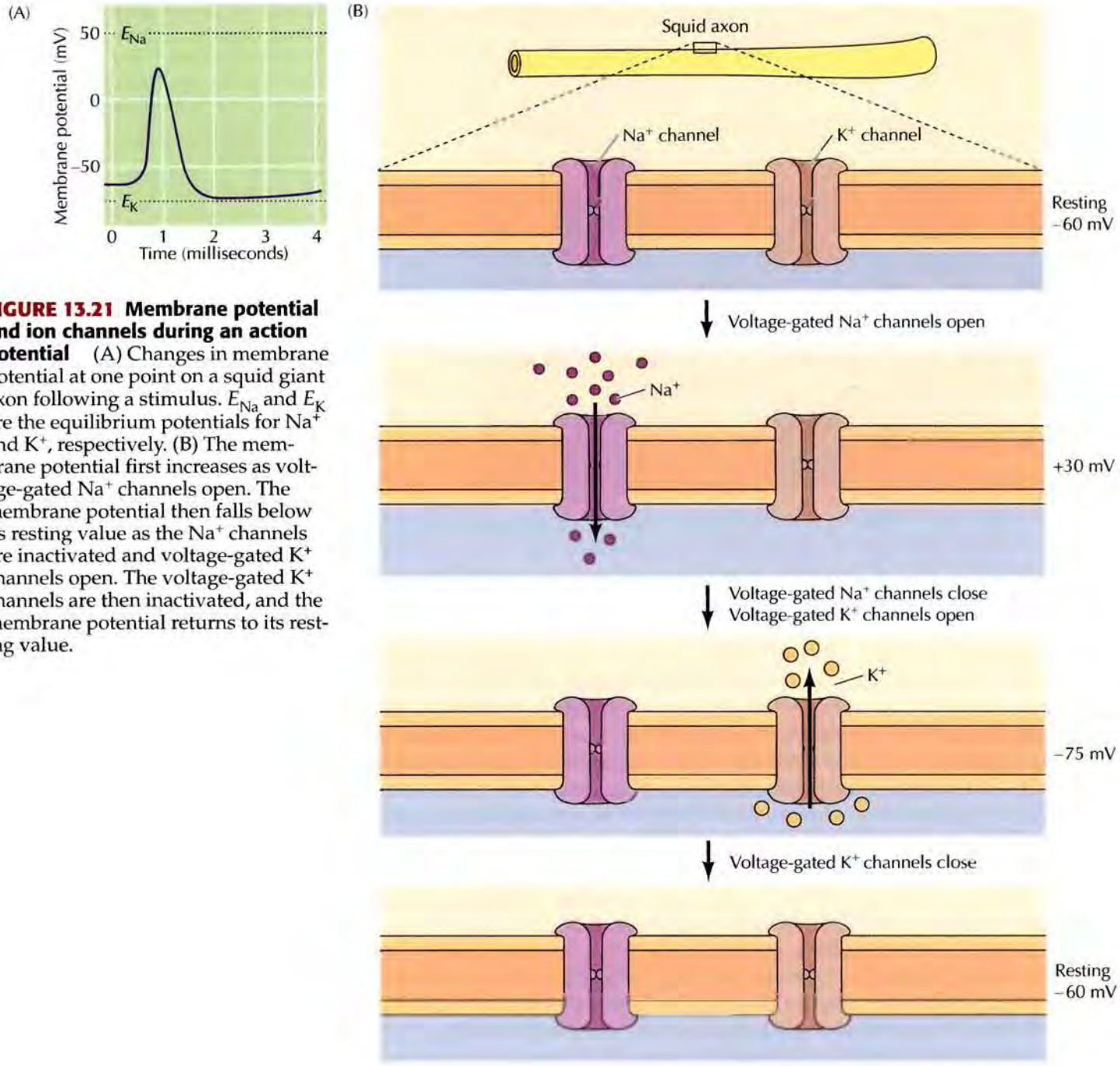


FIGURE 13.21 Membrane potential and ion channels during an action potential (A) Changes in membrane potential at one point on a squid giant axon following a stimulus. E_{Na} and E_K are the equilibrium potentials for Na^+ and K^+, respectively. (B) The membrane potential first increases as voltage-gated Na^+ channels open. The membrane potential then falls below its resting value as the Na^+ channels are inactivated and voltage-gated K^+ channels open. The voltage-gated K^+ channels are then inactivated, and the membrane potential returns to its resting value.

Quantitatively, the relationship between ion concentration and membrane potential is given by the **Nernst equation**:

$$V = \frac{RT}{zF} \ln \frac{C_o}{C_i}$$

where V is the equilibrium potential in volts, R is the gas constant, T is the absolute temperature, z is the charge of the ion, F is Faraday's constant, and C_o and C_i are the concentrations of the ion outside and inside of the cell, respectively. An equilibrium potential exists separately for each ion, and the membrane potential is determined by the flow of all the ions that cross the plasma membrane. However, because resting squid axons are more perme-

able to K^+ than to Na^+ or other ions (including Cl^-), the resting membrane potential (–60 mV) is close to the equilibrium potential determined by the intracellular and extracellular K^+ concentrations (–75 mV).

As nerve impulses (**action potentials**) travel along axons, the membrane depolarizes (Figure 13.21). The membrane potential changes from –60 mV to approximately +30 mV in less than a millisecond, after which it becomes negative again and returns to its resting value. These changes result from the rapid sequential opening and closing of voltage-gated Na^+ and K^+ channels. Relatively small initial changes in membrane potential (from –60 mV to about –40 mV) lead to the rapid but transient opening of Na^+ channels. This allows Na^+ to flow into the cell, driven by both its concentration gradient and the membrane potential. The sudden entry of Na^+ leads to a large change in membrane potential, which increases to nearly +30 mV, approaching the Na^+ equilibrium potential of approximately +50 mV. At this time, the Na^+ channels are inactivated and voltage-gated K^+ channels open, substantially increasing the permeability of the membrane to K^+. K^+ then flows rapidly out of the cell, driven by both the membrane potential and the K^+ concentration gradient, leading to a rapid decrease in membrane potential to about –75 mV (the K^+ equilibrium potential). This change in membrane potential inactivates the voltage-gated K^+ channels and the membrane potential returns to its resting level of –60 mV, determined by the flow of K^+ and other ions through the channels that remain open in unstimulated cells.

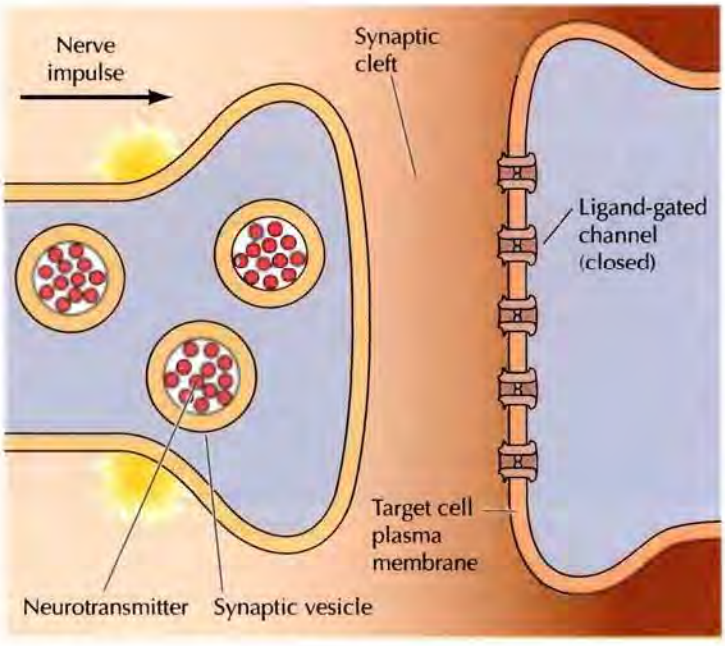
Depolarization of adjacent regions of the plasma membrane allows action potentials to travel down the length of nerve cell axons as electric signals, resulting in the rapid transmission of nerve impulses over long distances. For example, the axons of human motor neurons can be more than a meter long. The arrival of action potentials at the terminus of most neurons then signals the release of neurotransmitters, such as acetylcholine, which carry signals between cells at a synapse (Figure 13.22). Neurotransmitters released from presynaptic cells bind to receptors on the membranes of postsynaptic cells, where they act to open ligand-gated ion channels. One of the best-characterized of these channels is the nicotinic acetylcholine receptor of muscle

■ **Decreased concentrations of the neurotransmitter serotonin at neuronal synapses are thought to play a key role in clinical depression. Drugs like Prozac, which are used for the treatment of depression, are selective serotonin re-uptake inhibitors (SSRIs). As the name suggests, SSRIs selectively block transporters responsible for the re-uptake of serotonin by the presynaptic neuron, leading to increased concentrations of serotonin at the synapse.**

13.1 WEBSITE ANIMATION

A Chemical Synapse

The arrival of a nerve impulse at the terminus of a neuron triggers synaptic vesicles to fuse with the plasma membrane, thereby releasing neurotransmitter.



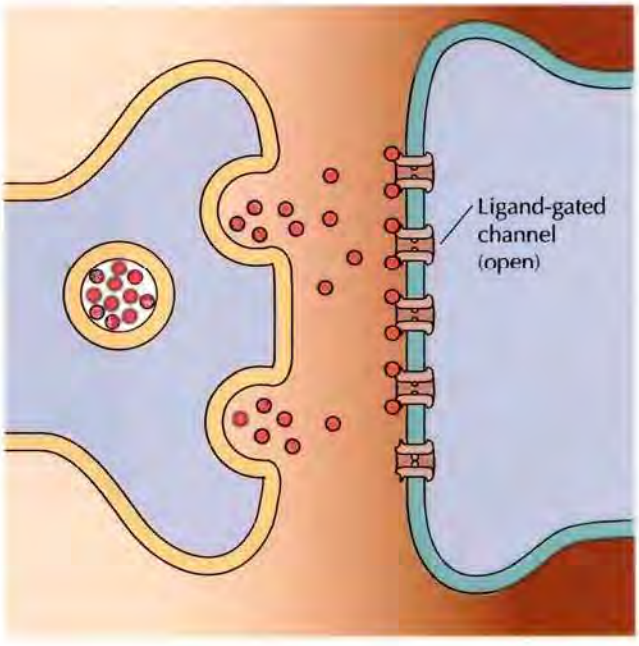


FIGURE 13.22 Signaling by neurotransmitter release at a synapse
The arrival of a nerve impulse at the terminus of the neuron signals the fusion of synaptic vesicles with the plasma membrane, resulting in the release of neurotransmitter from the presynaptic cell into the synaptic cleft. The neurotransmitter binds to receptors and opens ligand-gated ion channels in the target cell plasma membrane.

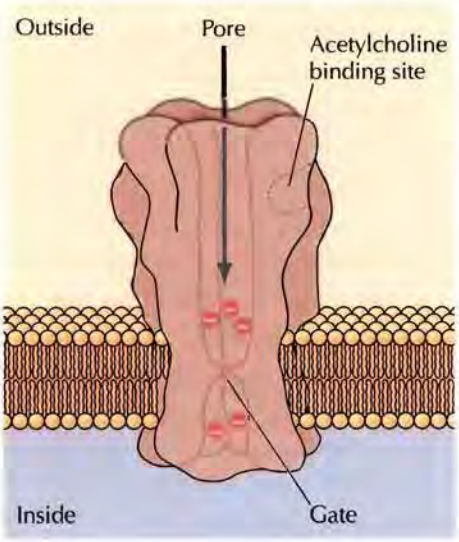


FIGURE 13.23 Model of the nicotinic acetylcholine receptor The receptor consists of five subunits arranged around a central pore. The binding of acetylcholine to a site in the extracellular region of the receptor induces allosteric changes that open the channel gate. The channel is lined by negatively charged amino acids that prevent the flow of negatively charged ions. (Adapted from N. Unwin, 1993. *Cell* 72/*Neuron* 10 (Suppl.): 31.)

cells. Binding of acetylcholine opens a channel that is permeable to both Na^+ and K^+. This permits the rapid influx of Na^+, which depolarizes the muscle cell membrane and triggers an action potential. The action potential then results in the opening of voltage-gated Ca^{2+} channels, leading to the increase in intracellular Ca^{2+} that signals contraction (see Figure 12.28).

This acetylcholine receptor, initially isolated from the electric organ of *Torpedo* rays in the 1970s, is the prototype of ligand-gated channels. The receptor consists of five subunits arranged as a cylinder in the membrane (Figure 13.23). In its closed state, the channel pore is thought to be blocked by the side chains of hydrophobic amino acids. The binding of acetylcholine induces a conformational change in the receptor such that these hydrophobic side chains shift out of the channel, opening a pore that allows the passage of positively charged ions, including Na^+ and K^+. However, the channel remains impermeable to negatively charged ions, such as Cl^-, because it is lined by negatively charged amino acids.

A greater degree of ion selectivity is displayed by the voltage-gated Na^+ and K^+ channels. Na^+ channels are more than ten times more permeable to Na^+ than to K^+, whereas K^+ channels are more than a thousand times more permeable to K^+ than to Na^+. The selectivity of the Na^+ channel can be explained, at least in part, on the basis of a narrow pore that acts as a size filter. The ionic radius of Na^+ (0.95 Å) is smaller than that of K^+ (1.33 Å), and it is thought that the Na^+ channel pore is narrow enough to interfere with the passage of K^+ or larger ions (Figure 13.24).

■ **Nicotinic acetylcholine receptors are so named because they respond to nicotine. Nicotine binds to the receptor and keeps the channel in its open state. Molecules that block the activity of nicotinic acetylcholine receptors are potent neurotoxins; these include several snake venom toxins and curare.**

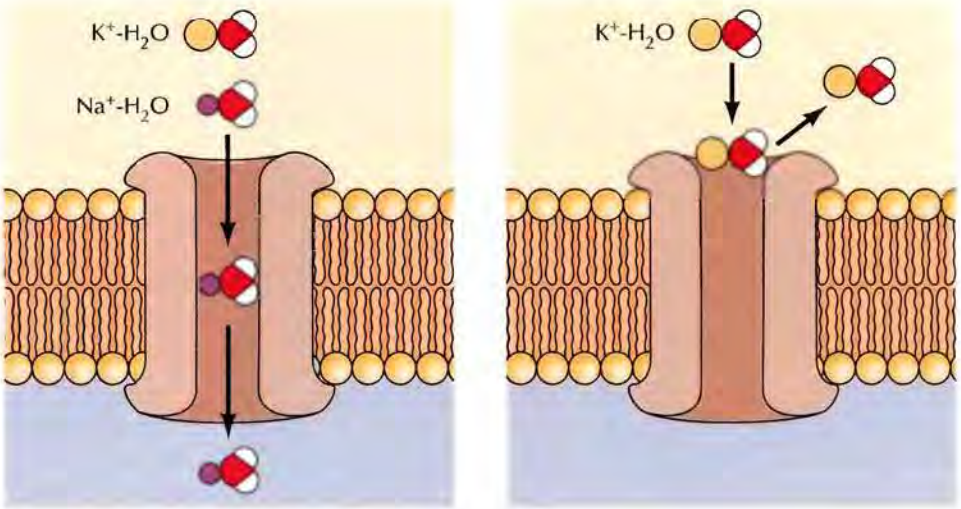


FIGURE 13.24 Ion selectivity of Na^+ channels A narrow pore permits the passage of Na^+ bound to a single water molecule but interferes with the passage of K^+ or larger ions.

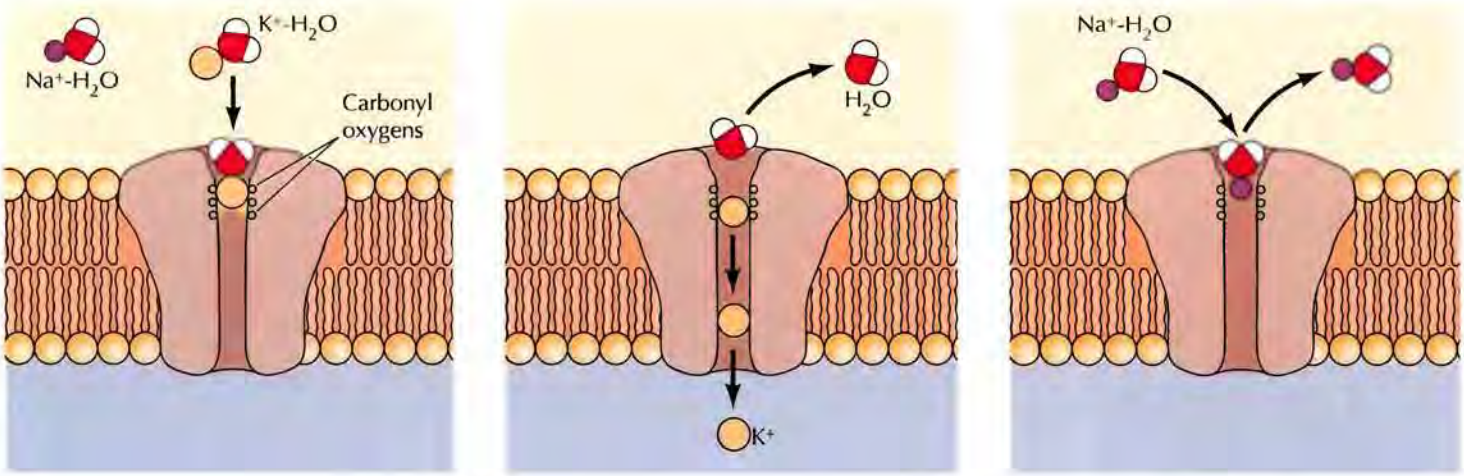


FIGURE 13.25 Selectivity of K^+ channels The K^+ channel contains a narrow selectivity filter lined with carbonyl oxygen (C=O) atoms. The pore is just wide enough to allow the passage of dehydrated K^+ from which all associated water molecules have been displaced as a result of interactions between K^+ and these carbonyl oxygens. Na^+ is too small to interact with the carbonyl oxygens of the selectivity filter, so it remains bound to water in a complex that is too large to pass through the channel pore.

K^+ channels also have narrow pores, which prevent the passage of larger ions. However, since Na^+ has a smaller ionic radius, this does not account for the selective permeability of these channels to K^+. Selectivity of the K^+ channel is based on a different mechanism, which was elucidated with the determination of the three-dimensional structure of a K^+ channel by X-ray crystallography in 1998 (Figure 13.25). The channel pore contains a narrow selectivity filter that is lined with carbonyl oxygen (C=O) atoms from the polypeptide backbone. When a K^+ ion enters the selectivity filter, interactions with these carbonyl oxygens displace the water molecules to which K^+ is bound, allowing dehydrated K^+ to pass through the pore. In contrast, a dehydrated Na^+ is too small to interact with these carbonyl oxygens in the selectivity filter, which is held rigidly open. Consequently, Na^+ remains bound to water molecules in a hydrated complex that is too large to pass through the channel.

Voltage-gated Na^+, K^+, and Ca^{2+} channels all belong to a large family of related proteins (Figure 13.26). For example, the genome sequence of *C. elegans* has revealed nearly 200 genes encoding ion channels, which presumably are needed to play diverse roles in cell signaling. K^+ channels consist of four identical subunits, each containing several transmembrane α helices. Na^+ and Ca^{2+} channels consist of a single polypeptide chain, but each polypeptide contains four repeated domains that correspond to the K^+ channel subunits. Voltage gating is mediated by α helix number 4, which contains multiple positively charged amino acids and is thought to move in response to changes in the membrane potential. However, the mechanism by which the movement of these charges opens and closes the channel remains controversial.

A wide variety of ion channels (including Ca^{2+} and Cl^- channels) respond to different neurotransmitters or open and close with different kinetics following membrane depolarization. The concerted actions of these multiple channels are responsible for the complexities of signaling in the nervous system. Moreover, as discussed in the next chapter, the roles of ion channels are not restricted to the electrically excitable cells of nerve and muscle; they also play critical roles in signaling in other cell types. The regulated opening and closing of ion channels thus provides cells with a sensitive and versatile mechanism for responding to a variety of environmental stimuli.

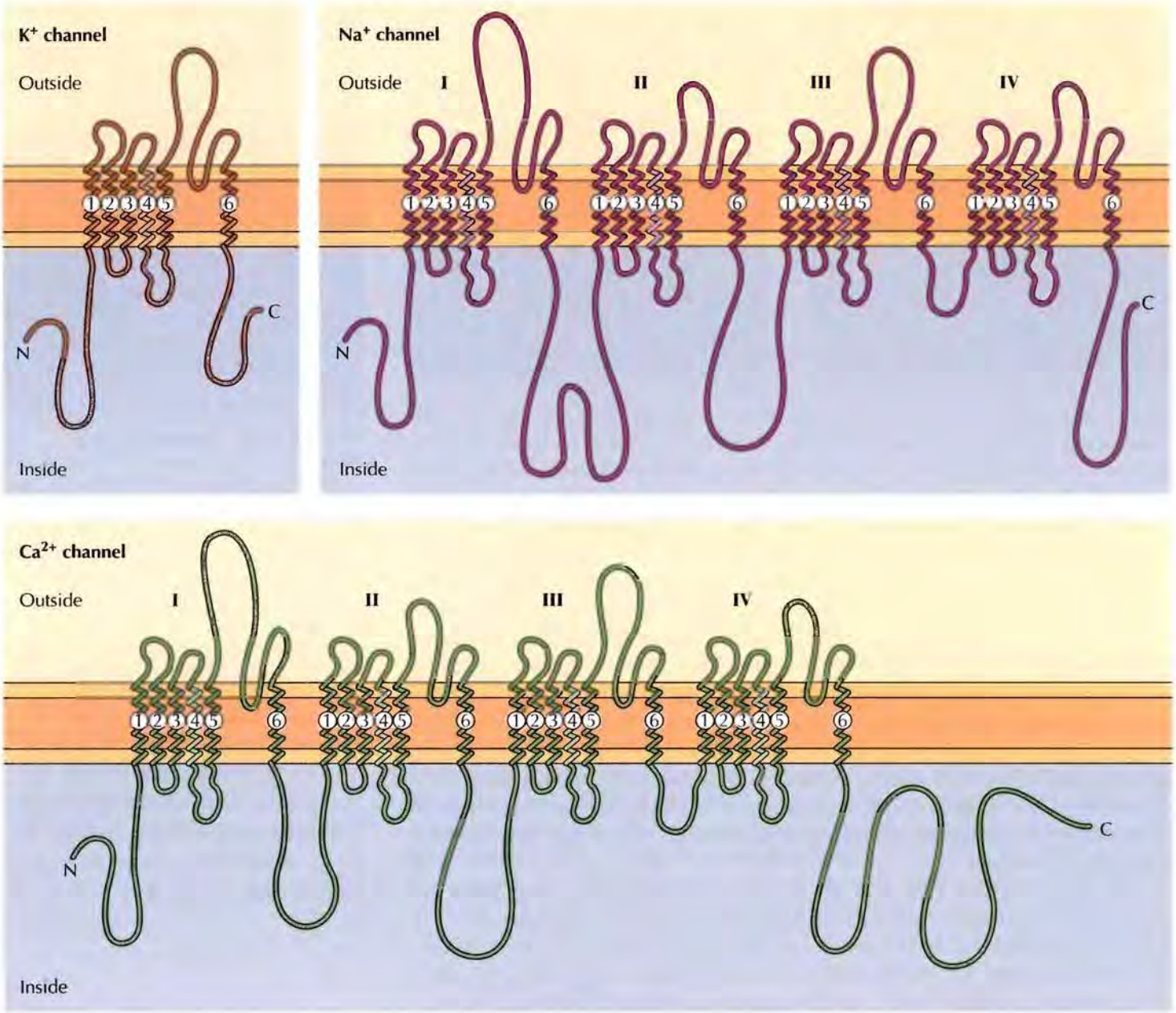


FIGURE 13.26 Structures of voltage-gated cation channels The K^+, Na^+, and Ca^{2+} channels belong to a family of related proteins. The K^+ channel is formed from the association of four identical subunits, one of which is shown. The Na^+ channel consists of a single polypeptide chain containing four repeated domains, each of which is similar to one K^+ channel subunit. The Ca^{2+} channel is similar to the Na^+ channel. Each subunit or domain contains six α helices. The α helix designated 4 contains multiple positively charged amino acids and acts as the voltage sensor that mediates channel opening in response to changes in membrane potential.

Active Transport Driven by ATP Hydrolysis

The net flow of molecules by facilitated diffusion, through either carrier proteins or channel proteins, is always energetically downhill in the direction determined by electrochemical gradients across the membrane. In many cases, however, the cell must transport molecules against their concentration gradients. In **active transport**, energy provided by another coupled reaction (such as the hydrolysis of ATP) is used to drive the uphill transport of molecules in the energetically unfavorable direction.

The **ion pumps** responsible for maintaining gradients of ions across the plasma membrane provide important examples of active transport driven directly by ATP hydrolysis. As discussed earlier (see Table 13.1), the concentration of Na^+ is approximately ten times higher outside than inside of cells, whereas the concentration of K^+ is higher inside than out. These ion gradients are maintained by the **Na^+-K^+ pump** (also called the **Na^+-K^+ ATPase**)

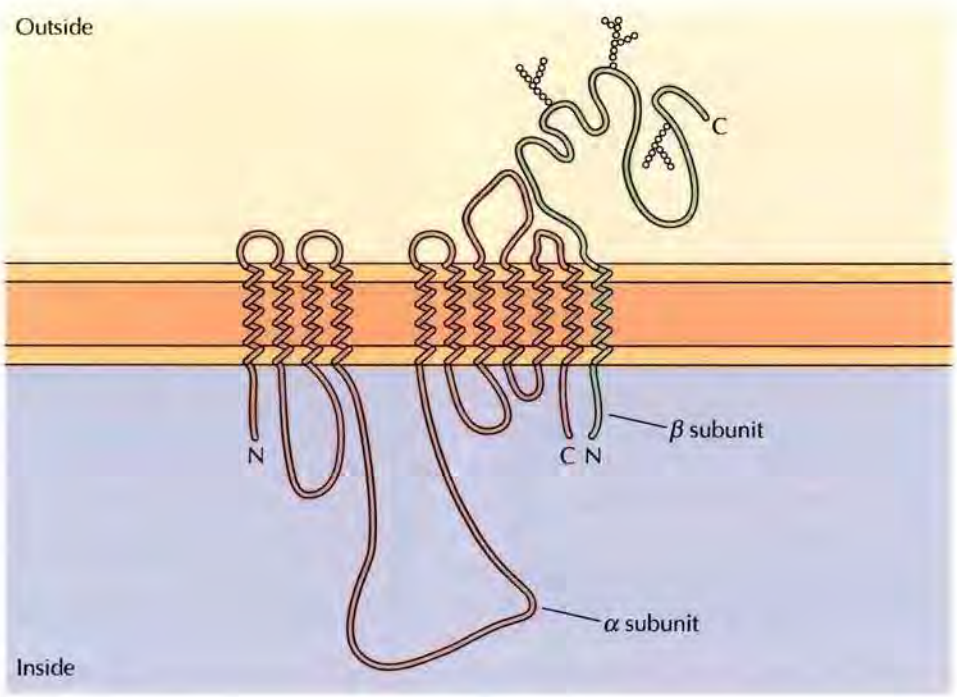


FIGURE 13.27 Structure of the Na^+-K^+ pump The Na^+-K^+ pump is a heterodimer with a 10 transmembrane domain α subunit and a single transmembrane domain β subunit. The β subunit is heavily glycosylated.

(Figure 13.27), which uses energy derived from ATP hydrolysis to transport Na^+ and K^+ against their electrochemical gradients. This process is a result of ATP-driven conformational changes in the pump (Figure 13.28). First, Na^+ ions bind to high-affinity sites inside the cell. This binding stimulates the hydrolysis of ATP and phosphorylation of the pump, inducing a conformational change that exposes the Na^+-binding sites to the outside of the cell and reduces their affinity for Na^+. Consequently, the bound Na^+ is released into the extracellular fluids. At the same time, high-affinity K^+-binding sites are exposed on the cell surface. The binding of extracellular K^+ to these sites then stimulates hydrolysis of the phosphate group bound to the pump, which induces a second conformational change, exposing the K^+-binding sites to the cytosol and lowering their binding affinity so that K^+ is released inside the cell. The pump has three binding sites for Na^+ and two for K^+, so each cycle transports three Na^+ and two K^+ ions across the plasma membrane at the expense of one molecule of ATP.

The importance of the Na^+-K^+ pump is indicated by the fact that it is estimated to consume nearly 25% of the ATP utilized by many animal cells. One critical role of the Na^+ and K^+ gradients established by the pump is the propagation of electric signals in nerve and muscle. As will be discussed shortly, the Na^+ gradient established by the pump is also utilized to drive the active transport of a variety of other molecules. Yet another important role of the Na^+-K^+ pump in most animal cells is to maintain osmotic balance and cell volume. The cytoplasm contains a high concentration of organic molecules, including macromolecules, amino acids, sugars, and nucleotides. In the absence of a counterbalance, this would drive the inward flow of water by osmosis, which if unchecked would result in swelling and eventual bursting of the cell. The required counterbalance is provided by the ion gradients established by the Na^+-K^+ pump (Figure 13.29). In particular, the pump establishes a higher concentration of Na^+ outside than inside the cell. As already discussed, the flow of K^+ through open channels further establishes an electric potential across the plasma

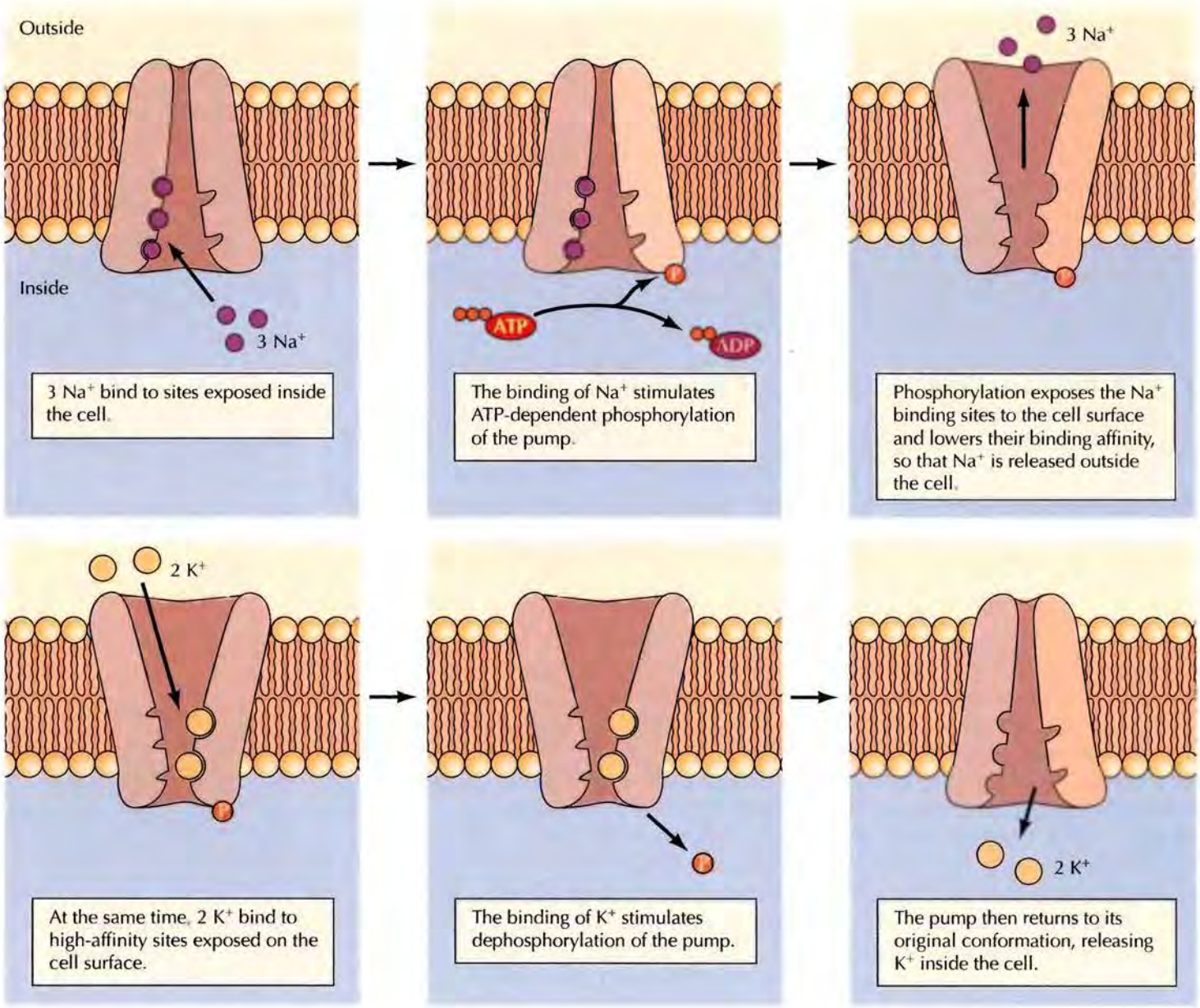


FIGURE 13.28 Model for operation of the Na^+-K^+ pump

13.2 WEBSITE ANIMATION

The Sodium-Potassium Pump The sodium and potassium ion gradients across the plasma membrane are maintained by the Na^+-K^+ pump, which hydrolyzes ATP to fuel the transport of these ions against their electrochemical gradients.

membrane. This membrane potential in turn drives Cl^- out of the cell, so the concentration of Cl^- (like that of Na^+) is about ten times higher in extracellular fluids than in the cytoplasm. These differences in ion concentrations balance the high concentrations of organic molecules inside cells, equalizing the osmotic pressure and preventing the net influx of water.

The active transport of Ca^{2+} across the plasma membrane is driven by a Ca^{2+} pump that is structurally related to the Na^+-K^+ pump and is similarly powered by ATP hydrolysis. The Ca^{2+} pump transports Ca^{2+} out of the cell, so intracellular Ca^{2+} concentrations are extremely low: approximately 0.1 μM in comparison to extracellular concentrations of about 1 mM. This low intracellular concentration of Ca^{2+} makes the cell sensitive to small increases in intracellular Ca^{2+} levels. Such transient and highly localized increases in intracellular Ca^{2+} play important roles in cell signaling as noted already with respect to muscle contraction (see Figure 12.28) and discussed further in Chapter 15.

Similar ion pumps in the plasma membranes of bacteria, yeasts, and plant cells are responsible for the active transport of H^+ out of the cell. In addition, H^+ is actively pumped out of cells lining the stomach, resulting in the acidity of gastric fluids. Structurally distinct pumps are responsible for the active transport of H^+ into lysosomes and endosomes (see Figure 10. 41). Yet a third type of H^+ pump is exemplified by the ATP synthases of mitochondria and chloroplasts: In these cases the pumps can be viewed as operating in reverse, with the movement of ions down the electrochemical gradient being used to drive ATP synthesis.

The largest family of membrane transporters consists of the **ABC transporters**, so called because they are characterized by highly conserved ATP-binding domains or *A*TP-*b*inding *c*assettes (Figure 13.30). More than a hundred family members have been identified in both prokaryotic and eukaryotic cells and all use energy derived from ATP hydrolysis to transport molecules in one direction. In bacteria, most ABC transporters bring a wide range of nutrients, including ions, sugars, and amino acids into the cell, while in eukaryotic cells they transport toxic substances out of the cell. The first eukaryotic ABC transporter was discovered as the product of a gene (called the multidrug resistance, or *mdr*, gene) that makes cancer cells resistant to a variety of drugs used in chemotherapy. Two MDR transporters have now been identified. They are normally expressed in a variety of cells, where they function to remove potentially toxic foreign compounds. For example, expression of an MDR transporter in capillary endothelial cells of the brain appears to play an important role in protecting the brain from toxic chemicals. Unfortunately, the MDR transporters are frequently expressed at high levels in cancer cells where they recognize a variety of drugs and pump them out of cells, thereby making the cancer cells resistant to a broad spectrum of chemotherapeutic agents and posing a major obstacle to successful cancer treatment.

Another medically important member of the ABC transporter family is the gene responsible for cystic fibrosis. Although it is a member of the ABC family, the product of this gene (called the cystic fibrosis transmembrane conductance regulator, or CFTR) functions as a Cl^- channel in epithelial cells, and defective Cl^- transport is characteristic of the disease. The CFTR Cl^- channel is also unusual in that it appears to require both ATP hydrolysis

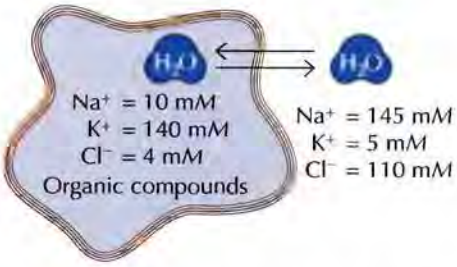


FIGURE 13.29 Ion gradients across the plasma membrane of a typical mammalian cell The concentrations of Na^+ and Cl^- are higher outside than inside the cell, whereas the concentration of K^+ is higher inside than out. The low concentrations of Na^+ and Cl^- balance the high intracellular concentration of organic compounds, equalizing the osmotic pressure and preventing the net influx of water.

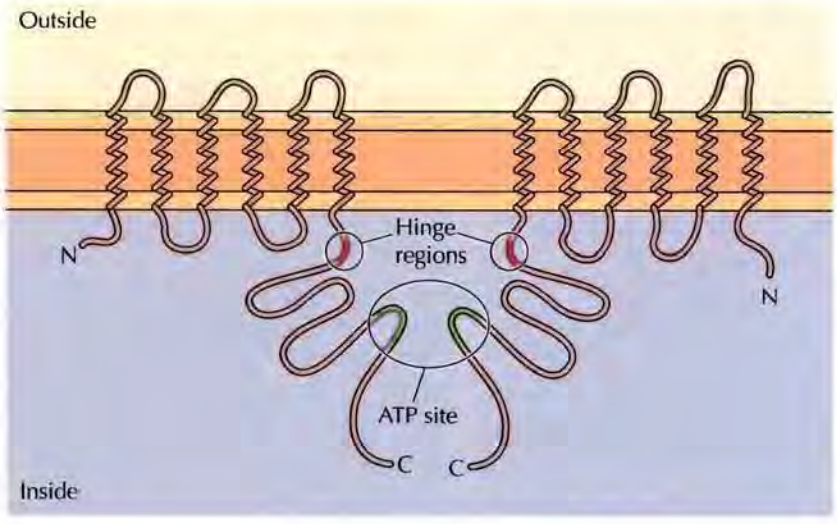


FIGURE 13.30 Structure of an ABC transporter Many ABC transporters are dimers, with each polypeptide consisting of six transmembrane domains connected via a hinge region to an ATP-binding cassette.

MOLECULAR MEDICINE

Cystic Fibrosis

The Disease

Cystic fibrosis is a recessive genetic disease affecting children and young adults. It is the most common lethal inherited disease of Caucasians, with approximately one in 2500 newborns affected, although it is rare in other races. The characteristic dysfunction in cystic fibrosis is the production of abnormally thick sticky mucus by several types of epithelial cells, including the cells lining the respiratory and gastrointestinal tracts. The primary clinical manifestation is respiratory disease resulting from obstruction of the pulmonary airways by thick plugs of mucus, followed by the development of recurrent bacterial infections. In most patients, the pancreas is also involved because the pancreatic ducts are blocked by mucus. Sweat glands also function abnormally, and the presence of excessive salt in sweat is diagnostic of cystic fibrosis.

Current management of the disease includes physical therapy to promote bronchial drainage, antibiotic administration, and pancreatic enzyme replacement. Although such treatment has extended the survival of affected individuals to about 30 years of age, cystic fibrosis is ultimately fatal, with lung disease being responsible for 95% of mortality.

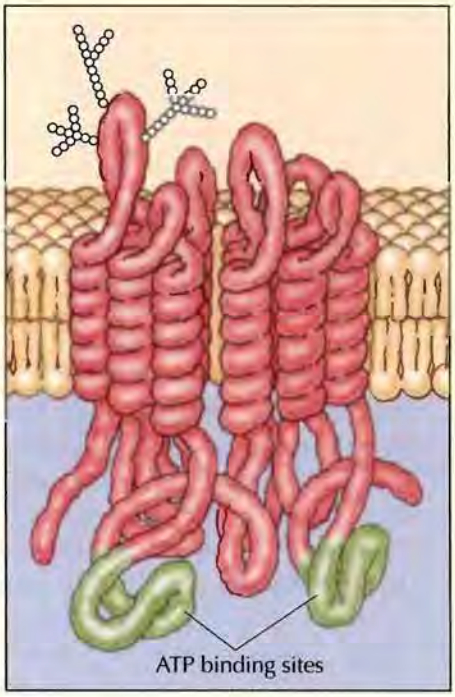
Molecular and Cellular Basis

The hallmark of cystic fibrosis is defective Cl^- transport in affected epithelia, including sweat ducts and the cells lining the respiratory tract. In 1984 it was demonstrated that Cl^- channels fail to function normally in epithelial cells from cystic fibrosis patients. The molecular basis of the disease was then elucidated in 1989 with the isolation of the cystic fibrosis gene as a molecular clone. The sequence of the gene revealed that it encodes a protein (called CFTR for *c*ystic *f*ibrosis *t*ransmembrane conductance *r*egulator) belonging to the ABC transporter family. A variety of subsequent studies then demonstrated that CFTR functions as a Cl^- channel and that the inherited mutations responsible for cystic fibrosis result directly in defective Cl^- transport. More than 70% of these are a single point mutation at phenylalanine 508 that disrupts folding or assembly of the protein.

Prevention and Treatment

As with other inherited diseases, isolation of the cystic fibrosis gene opens the possibility of genetic screening to identify individuals carrying mutant alleles. In some populations, the frequency of heterozygote carriers of mutant genes is as high as one in 25 individuals, suggesting the possibility of general population screening to identify couples at risk and provide genetic counseling. In addition, understanding the function of CFTR as a Cl^- channel has suggested new approaches to treatment. One possibility is the use of drugs that stimulate the opening of other Cl^- channels in affected epithelia. Alternatively, gene therapy provides the potential of replacing normal CFTR genes in the respiratory epithelium of cystic fibrosis patients.

The potential of gene therapy has been supported by experiments demonstrating that introduction of a normal CFTR gene into cultured cells of cystic fibrosis patients is sufficient to restore Cl^- channel function. The possible application of gene therapy to cystic fibrosis is also enhanced by the accessibility of the epithelial cells lining the airway to aerosol delivery systems. Studies with experimental animals have demonstrated that viral vectors can transmit CFTR cDNA to the respiratory epithelium, and the first human trial of gene therapy for cystic fibrosis was initiated in 1993. Trials to date have demonstrated that the CFTR can be safely delivered to and expressed in the bronchial epithelial cells of cystic fibrosis patients. So far, however, the efficiency of gene transfer has been low and expression of the transferred CFTR cDNA has persisted for less than a month. Thus the principle of successful gene transfer has been established, but significant obstacles in developing an effective gene therapy protocol remain to be overcome.



Model of the cystic fibrosis transmembrane conductance regulator (CFTR).

References

Collins, F. S. 1992. Cystic fibrosis: Molecular biology and therapeutic implications. *Science* 256: 774–779.

Driskell, R. R. and J. F. Engelhardt. 2003. Current status of gene therapy for inherited lung disease. *Ann. Rev. Physiol.* 65: 585–612.

Riordan, J. R. 1993. The cystic fibrosis transmembrane conductance regulator. *Ann. Rev. Physiol.* 55: 609–630.

and cAMP-dependent phosphorylation in order to open. Most cases of cystic fibrosis are the result of a single point mutation in the Cl^- conductance channel that interferes with both protein folding and Cl^- transport.

Active Transport Driven by Ion Gradients

The ion pumps and ABC transporters discussed in the previous section utilize energy derived directly from ATP hydrolysis to transport molecules against their electrochemical gradients. Other molecules are transported against their concentration gradients using energy derived not from ATP hydrolysis but from the coupled transport of a second molecule in the energetically favorable direction. The Na^+ gradient established by the Na^+-K^+ pump provides a source of energy that is frequently used to power the active transport of sugars, amino acids, and ions in mammalian cells. The H^+ gradients established by the H^+ pumps of bacteria, yeast, and plant cells play similar roles.

The epithelial cells lining the intestine provide a good example of active transport driven by the Na^+ gradient. These cells use active-transport systems in the apical domains of their plasma membranes to take up dietary sugars and amino acids from the lumen of the intestine. The uptake of glucose, for example, is carried out by a transporter that coordinately transports two Na^+ ions and one glucose molecule into the cell (Figure 13.31). The flow of Na^+ down its electrochemical gradient provides the energy required to take up dietary glucose and to accumulate high intracellular glucose concentrations. Glucose is then released into the underlying connective tissue (which contains blood capillaries) at the basolateral surface of the intestinal epithelium, where it is transported down its concentration gradient by facilitated diffusion (Figure 13.32). The uptake of glucose from the intestinal lumen and its release into the circulation thus provides a good example of the polarized function of epithelial cells, which results from the specific localization of active transport and facilitated diffusion carriers to the apical and basolateral domains of the plasma membrane, respectively.

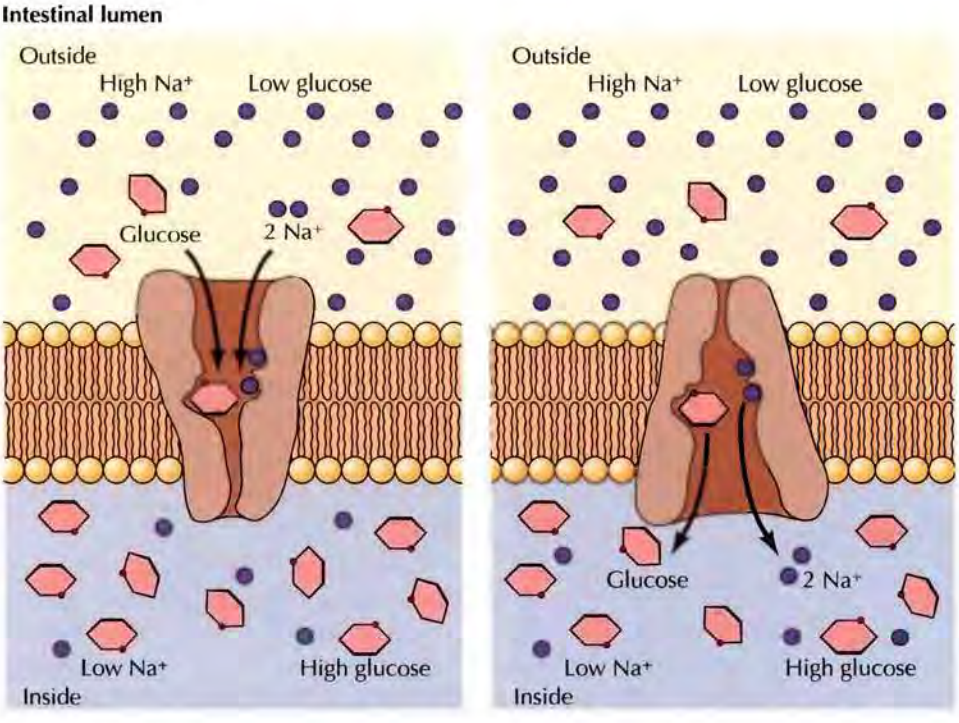


FIGURE 13.31 Active transport of glucose Active transport driven by the Na^+ gradient is responsible for the uptake of glucose from the intestinal lumen. The transporter coordinately binds and transports one glucose molecule and two Na^+ ions into the cell. The transport of Na^+ in the energetically favorable direction drives the uptake of glucose against its concentration gradient.

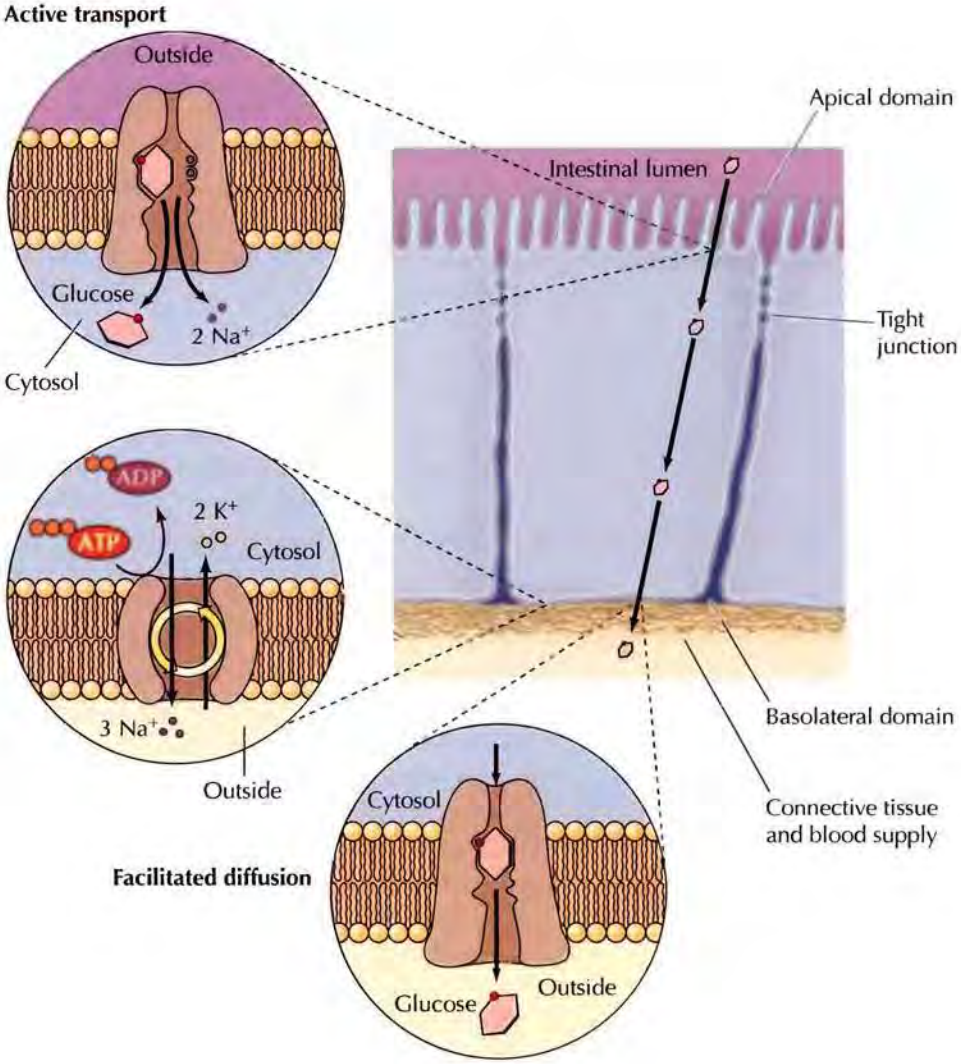


FIGURE 13.32 Glucose transport by intestinal epithelial cells A transporter in the apical domain of the plasma membrane is responsible for the active uptake of glucose (by cotransport with Na^+) from the intestinal lumen. As a result, dietary glucose is absorbed and concentrated within intestinal epithelial cells. Glucose is then transferred from these cells to the underlying connective tissue and blood supply by facilitated diffusion, mediated by a transporter in the basolateral domain of the plasma membrane. The system is driven by the Na^+-K^+ pump, which is also found in the basolateral domain. Note that the uptake of glucose from the digestive tract and its transfer to the circulation is dependent on the restricted localization of glucose transporters mediating active transport and facilitated diffusion to the apical and basolateral domains of the plasma membrane, respectively.

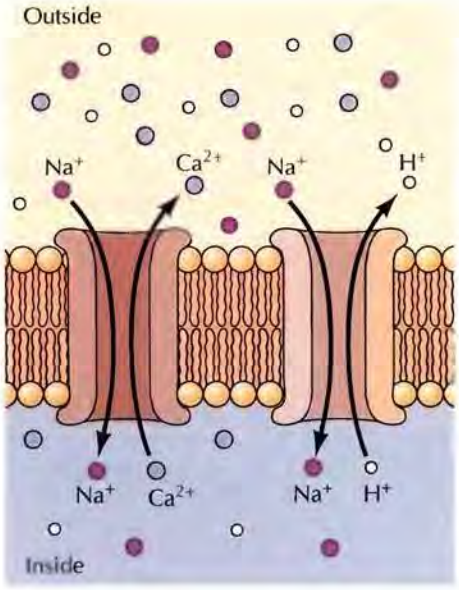


FIGURE 13.33 Examples of antiport Ca^{2+} and H^+ are exported from cells by antiporters, which couple their export to the energetically favorable import of Na^+.

The coordinated uptake of glucose and Na^+ is an example of **symport**, the transport of two molecules in the same direction. In contrast, the facilitated diffusion of glucose is an example of **uniport**, the transport of only a single molecule. Active transport can also take place by **antiport** in which two molecules are transported in opposite directions (Figure 13.33). For example, Ca^{2+} is exported from cells not only by the Ca^{2+} pump but also by a Na^+-Ca^{2+} antiporter that transports Na^+ into the cell and Ca^{2+} out. Another example is provided by the Na^+-H^+ exchange protein, which functions in the regulation of intracellular pH. The Na^+-H^+ antiporter couples the transport of Na^+ into the cell with the export of H^+, thereby removing excess H^+ produced by metabolic reactions and preventing acidification of the cytoplasm.

Endocytosis

The carrier and channel proteins discussed in the preceding section transport small molecules through the phospholipid bilayer. Eukaryotic cells are also able to take up macromolecules and particles from the surrounding medium by a distinct process called **endocytosis**. In endocytosis the mate-

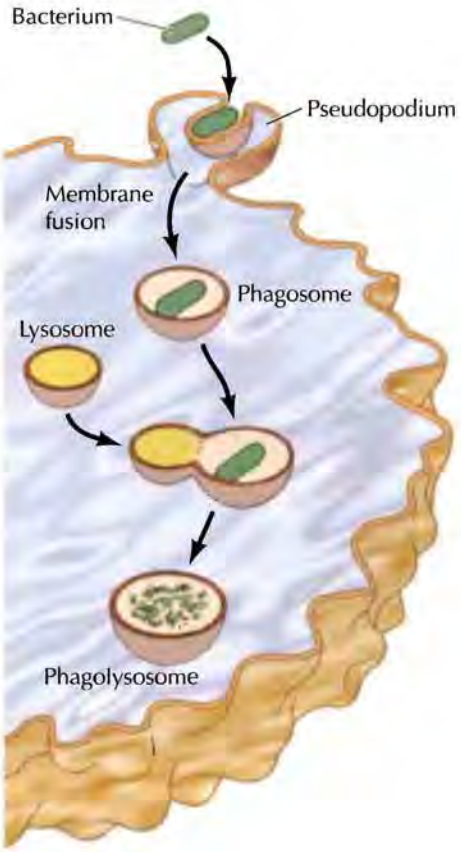


FIGURE 13.34 Phagocytosis Binding of a bacterium to the cell surface stimulates the extension of a pseudopodium, which eventually engulfs the bacterium. Fusion of the pseudopodium membranes then results in formation of a large intracellular vesicle (a phagosome). The phagosome fuses with lysosomes to form a phagolysosome within which the ingested bacterium is digested.

rial to be internalized is surrounded by an area of plasma membrane, which then buds off inside the cell to form a vesicle containing the ingested material. The term "endocytosis" was coined by Christian deDuve in 1963 to include both the ingestion of large particles (such as bacteria) and the uptake of fluids or macromolecules in small vesicles. The former of these activities is known as **phagocytosis** (cell eating) and occurs largely in specialized types of cells. **Pinocytosis** (cell drinking) is a property of all eukaryotic cells and takes place by several different mechanisms.

Phagocytosis

During phagocytosis cells engulf large particles such as bacteria, cell debris, or even intact cells (Figure 13.34). Binding of the particle to receptors on the surface of the phagocytic cell triggers the extension of pseudopodia—an actin-based movement of the cell surface. The pseudopodia eventually surround the particle and their membranes fuse to form a large intracellular vesicle (>0.25 μm in diameter) called a **phagosome**. The phagosomes then fuse with lysosomes, producing **phagolysosomes** in which the ingested material is digested by the action of lysosomal acid hydrolases (see Chapter 10). During maturation of the phagolysosome, some of the internalized membrane proteins are recycled to the plasma membrane as discussed in the next section for receptor-mediated endocytosis.

The ingestion of large particles by phagocytosis plays distinct roles in different kinds of cells (Figure 13.35). Many amoebas use phagocytosis to capture food particles, such as bacteria or other protozoans. In multicellular animals, the major roles of phagocytosis are to provide a defense against

13.3 WEBSITE ANIMATION

Endocytosis In endocytosis, the plasma membrane pinches off small vesicles (pinocytosis) or large vesicles (phagocytosis) to take in extracellular materials.

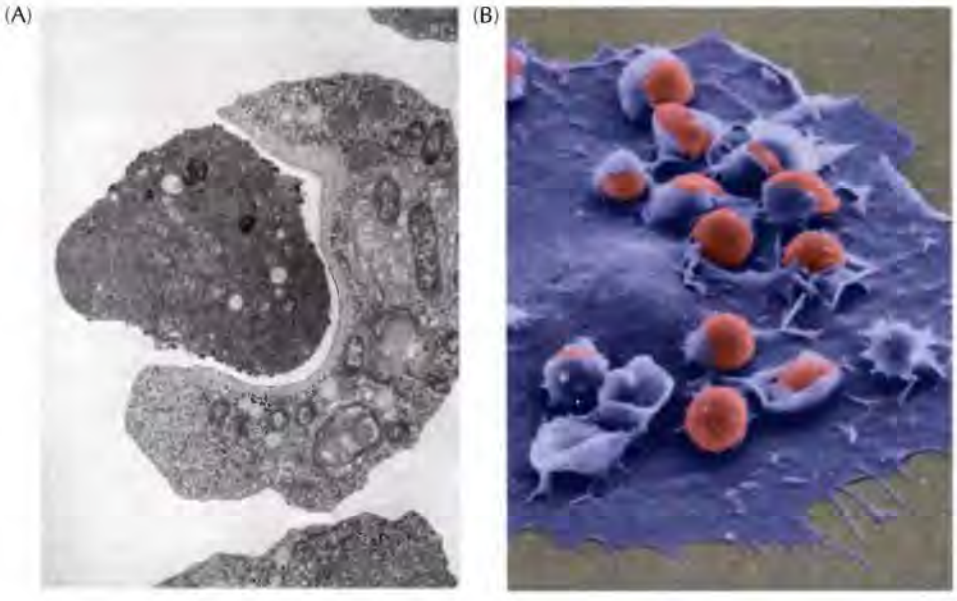


FIGURE 13.35 Examples of phagocytic cells (A) An amoeba engulfing another protist. (B) Macrophages ingesting red blood cells. False color has been added to the micrograph. (A, R. N. Band and H. S. Pankratz/Biological Photo Service; B, courtesy of Joel Swanson.)

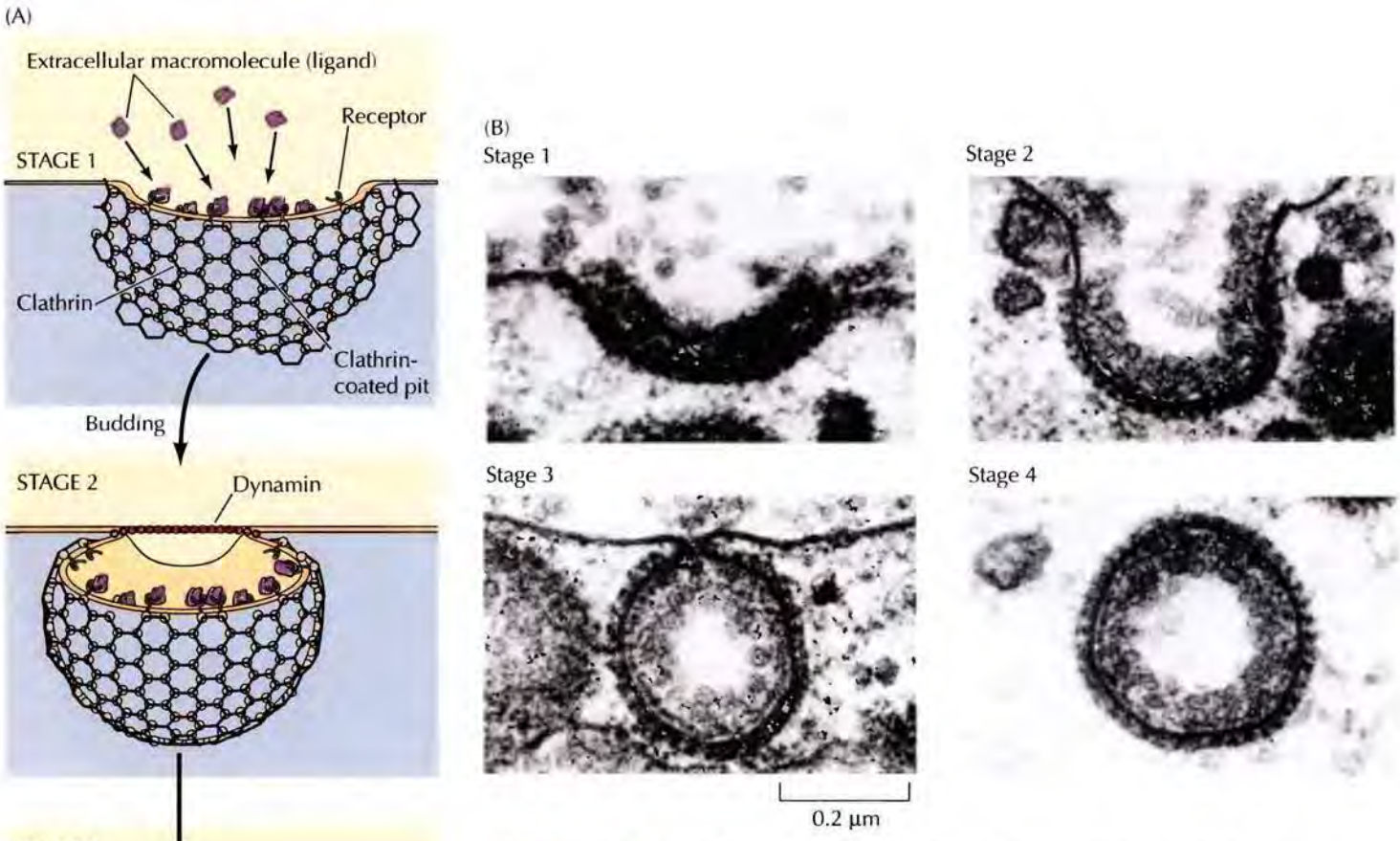


FIGURE 13.36 Clathrin-coated vesicle formation (A) Extracellular macromolecules (ligands) bind to cell surface receptors that are concentrated in clathrin-coated pits. With the assistance of the GTP-binding protein dynamin, these pits bud from the plasma membrane to form intracellular clathrin-coated vesicles. (B) Electron micrographs showing four stages in the formation of a clathrin-coated vesicle from a clathrin-coated pit. (B, M. M. Perry, 1979. *J. Cell Science* 34: 266.)

13.4 WEBSITE ANIMATION

Clathrin-Coated Pits and Vesicles Receptor molecules in the plasma membrane become bound to specific macromolecules from outside the cell, coalesce into clathrin-coated pits, and then pinch off in the form of clathrin-coated vesicles.

invading microorganisms and to eliminate aged or damaged cells from the body. In mammals, phagocytosis is the function of primarily two types of white blood cells—macrophages and neutrophils—which are frequently referred to as "professional phagocytes." Both macrophages and neutrophils play critical roles in the body's defense systems by eliminating microorganisms from infected tissues. In addition, macrophages eliminate aged or dead cells from tissues throughout the body. A striking example of the scope of this activity is provided by the macrophages of the human spleen and liver, which are responsible for the disposal of more than 10^{11} aged blood cells on a daily basis.

Receptor-Mediated Endocytosis

The best-characterized form of pinocytosis is **receptor-mediated endocytosis**, which provides a mechanism for the selective uptake of specific macromolecules (Figure 13.36). The macromolecules to be internalized first bind to specific cell surface receptors. Most of these receptors are concentrated in specialized regions of the plasma membrane called **clathrin-coated pits**. With assistance from the membrane-associated GTP-binding protein, **dynamin**, these pits bud from the membrane to form small **clathrin-coated vesicles** containing the receptors and their bound macromolecules (**ligands**). The clathrin-coated vesicles then fuse with early endosomes in which their contents are sorted for transport to lysosomes or recycling to the plasma membrane.

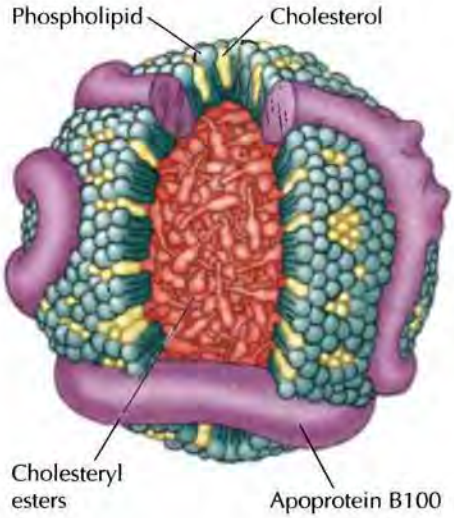


FIGURE 13.37 Structure of LDL Each particle of LDL contains approximately 1500 molecules of cholesteryl esters in an oily core. The core is surrounded by a coat containing 500 molecules of cholesterol, 800 molecules of phospholipid, and one molecule of apoprotein B100.

The uptake of cholesterol by mammalian cells has provided a key model for understanding receptor-mediated endocytosis at the molecular level. Cholesterol is transported through the bloodstream in the form of lipoprotein particles, the most common of which is called **low-density lipoprotein**, or **LDL** (Figure 13.37). Studies in the laboratories of Michael Brown and Joseph Goldstein demonstrated that the uptake of LDL by mammalian cells requires the binding of LDL to a specific cell surface receptor that is concentrated in clathrin-coated pits and internalized by endocytosis. As discussed in the next section, the receptor is then recycled to the plasma membrane while LDL is transported to lysosomes where cholesterol is released for use by the cell.

KEY EXPERIMENT

The LDL Receptor

Familial Hypercholesterolemia: Defective Binding of Lipoproteins to Cultured Fibroblasts Associated with Impaired Regulation of 3-Hydroxy-3-Methylglutaryl Coenzyme A Reductase Activity

Michael S. Brown and Joseph L. Goldstein

University of Texas Southwestern Medical School, Dallas

Proceedings of the National Academy of Science USA, 1974, Volume 71, pages 788–792.

Michael S. Brown

Joseph L. Goldstein

The Context

Familial hypercholesterolemia (FH) is a genetic disease in which patients have greatly elevated levels of serum cholesterol and suffer from heart attacks early in life. Michael Brown and Joseph Goldstein began their efforts to understand this disease in 1972 with the idea that cholesterol overproduction results from a defect in the control mechanisms that normally regulate cholesterol biosynthesis. Consistent with this hypothesis, they found that addition of LDL to the culture medium of normal human fibroblasts inhibits the activity of 3-hydroxy-3-methylglutaryl coenzyme A reductase (HMG-CoA reductase), the rate-limiting enzyme in the cholesterol biosynthetic pathway. In contrast, HMG-CoA reductase activity is unaffected by addition of LDL to the cells of FH patients, resulting in overproduction of cholesterol by FH cells.

Perhaps surprisingly, subsequent experiments indicated that this abnormality in HMG-CoA reductase regulation is not the result of a mutation in the HMG-CoA reductase gene. Instead, the abnormal regulation of HMG-CoA reductase appeared to be due to the inability of FH cells to extract cholesterol from LDL. In 1974 Brown and Goldstein demonstrated that the lesion in FH cells is a defect in LDL binding to a receptor on the cell surface. This identification of the LDL receptor led to a series of groundbreaking experiments in which Brown, Goldstein, and their colleagues delineated the pathway of receptor-mediated endocytosis.

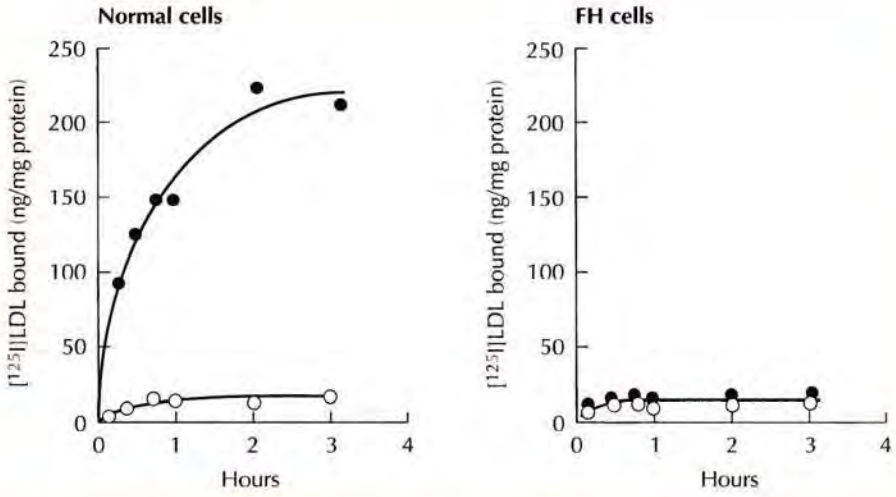
The Experiments

In their 1974 paper, Brown and Goldstein reported the results of experi-

Continued on next page

KEY EXPERIMENT

ments in which they investigated the binding of radiolabeled LDL to fibroblasts from either normal individuals or FH patients. Small amounts of radioactive LDL were added to the culture media, and the amount of radioactivity bound to the cells was determined after varying times of incubation (see figure). Increasing amounts of radioactive LDL bound to normal cells as a function of incubation time. Importantly, addition of excess unlabeled LDL reduced the binding of radioactive LDL, indicating that binding was due to a specific interaction of LDL with a limited number of sites on the cell surface. The specificity of the interaction was further demonstrated by the failure of excess amounts of other lipoproteins to interfere with LDL binding.



Time course of the binding of radioactive LDL to normal and FH cells. Cells were incubated with radioactive [^{125}I]LDL in the presence (open symbols) or absence (closed symbols) of excess unlabeled LDL. Cells were then harvested, and the amount of radioactive LDL bound was determined. Data are presented as nanograms of LDL bound per milligram of cell protein.

In contrast to these results with normal fibroblasts, the cells of FH patients failed to specifically bind radioactive LDL. It therefore appeared that normal fibroblasts possessed a specific LDL receptor that was absent or defective in FH cells. Brown and Goldstein concluded that the defect in LDL binding observed in FH cells "may represent the primary genetic lesion in this disorder," accounting for the inability of LDL to inhibit HMG-CoA reductase and the resultant overproduction of cholesterol. Additional experiments showed that the LDL bound to normal fibroblasts is associated with the membrane fraction of the cell, suggesting that the LDL receptor is a cell surface protein.

The Impact

Following this identification of the LDL receptor, Brown and Goldstein demonstrated that LDL bound to the cell surface is rapidly internalized and degraded to free cholesterol in lysosomes. In collaboration with Richard Anderson, they further established that the LDL receptor is internalized by endocytosis from coated pits. In addition, their early studies demonstrated that the LDL receptor is recycled to the plasma membrane after dissociation of its ligand inside the cell. Experiments that had been initiated with the goal of understanding the regulation of cholesterol biosynthesis thus led to the elucidation of a major pathway by which eukaryotic cells internalize specific macromolecules—a striking example of the way in which science and scientists can proceed in unanticipated but exciting new directions.

■ Statins are an important class of drugs, used in the treatment of hypercholesterolemia. They exert their effect by inhibiting the enzyme HMG-CoA reductase and blocking cholesterol biosynthesis.

The key insights into this process came from studies of patients with the inherited disease known as familial hypercholesterolemia. Patients with this disease have very high levels of serum cholesterol and suffer heart attacks early in life. Brown and Goldstein found that cells of these patients are unable to internalize LDL from extracellular fluids, resulting in the accumulation of high levels of cholesterol in the circulation. Further experiments demonstrated that cells of normal individuals possess a receptor for LDL, which is concentrated in clathrin-coated pits, and that familial hypercholesterolemia results from inherited mutations in the LDL receptor. These mutations are of two types. Cells from most patients with familial hypercholesterolemia simply fail to bind LDL, demonstrating that a specific cell surface receptor was required for LDL uptake. In addition, a few patients

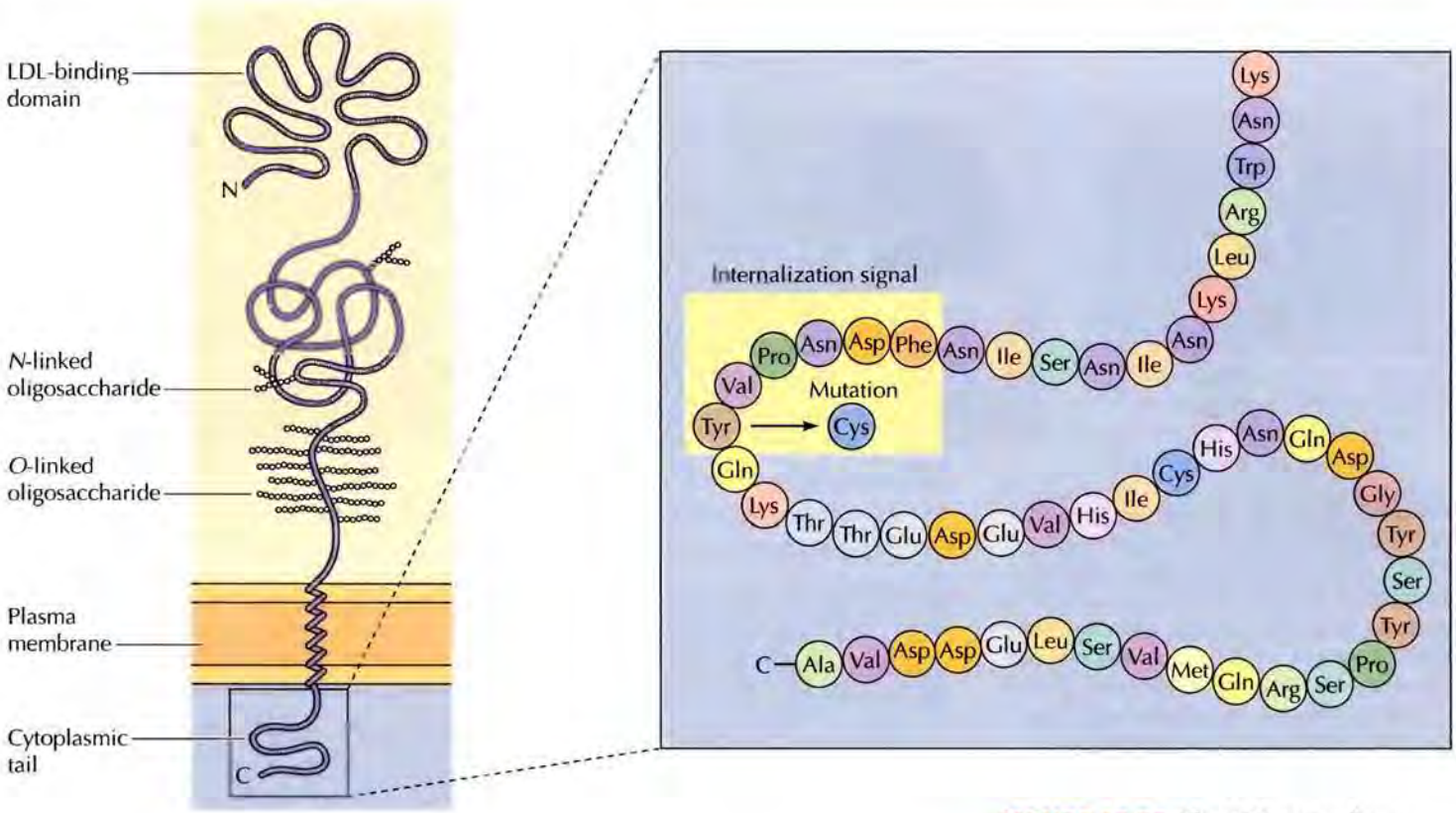


FIGURE 13.38 The LDL receptor
The LDL receptor includes 700 extracellular amino acids, a transmembrane α helix of 22 amino acids, and a cytoplasmic tail of 50 amino acids. The N-terminal 292 amino acids constitute the LDL-binding domain. Six amino acids within the cytoplasmic tail define the internalization signal, first recognized because the mutation of Tyr to Cys in a case of familial hypercholesterolemia prevented concentration of the receptor in coated pits.

were identified whose cells bound LDL but were unable to internalize it. The LDL receptors of these patients failed to concentrate in coated pits, providing direct evidence for the central role of coated pits in receptor-mediated endocytosis.

The mutations that prevent the LDL receptor from concentrating in coated pits lie within the cytoplasmic tail of the receptor and can be as subtle as the change of a single tyrosine to cysteine (Figure 13.38). Further studies have defined the internalization signal of the LDL receptor as a sequence of six amino acids, including the essential tyrosine. Similar internalization signals, frequently including tyrosine residues, are found in the cytoplasmic tails of other receptors taken up via clathrin-coated pits. These internalization signals bind to adaptor proteins, which in turn bind clathrin on the cytosolic side of the membrane, similar to the way in which clathrin-coated vesicles form during the transport of lysosomal hydrolases from the *trans* Golgi network (see Figure 10.35). Clathrin assembles into a basketlike structure that distorts the membrane, forming invaginated pits (Figure 13.39). The GTP-binding protein dynamin assembles into rings around the necks of these invaginated pits, eventually leading to the release of coated vesicles inside the cell.

Receptor-mediated endocytosis is a major activity of the plasma membranes of eukaryotic cells. More than 20 different receptors have been shown to be selectively internalized by this pathway. Extracellular fluids are also incorporated into coated vesicles as they bud from the plasma membrane, so receptor-mediated endocytosis results in the nonselective uptake of extracellular fluids and their contents (**fluid phase endocytosis**) in addition to the internalization of specific macromolecules. Clathrin-coated pits typically occupy 1 to 2% of the surface area of the plasma mem-

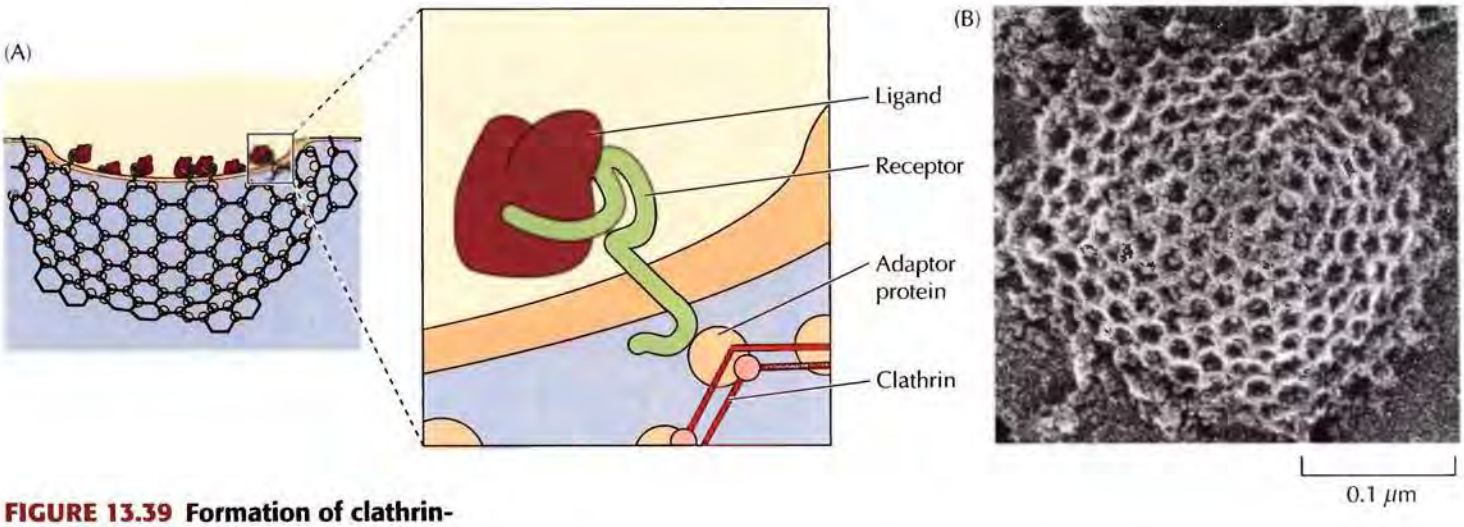


FIGURE 13.39 Formation of clathrin-coated pits (A) Adaptor proteins bind both to clathrin and to the internalization signals present in the cytoplasmic tails of receptors. (B) Electron micrograph of a clathrin-coated pit showing the basketlike structure of the clathrin network. (B, courtesy of John E. Heuser, Washington University School of Medicine.)

brane and are estimated to have a lifetime of 1 to 2 minutes. From these figures, one can calculate that receptor-mediated endocytosis results in the internalization of an area of cell surface equivalent to the entire plasma membrane approximately every 2 hours.

Cells also possess clathrin-independent pathways of endocytosis. One pathway of clathrin-independent endocytosis involves the uptake of molecules in **caveolae**, small invaginations of the plasma membrane (50 to 80 nm in diameter) that are organized by **caveolin** (Figure 13.40). Caveolins are a family of proteins that interact with cholesterol in lipid rafts, insert into the cell membrane, and interact with one another to form the structure of the caveolae. Caveolae are relatively stable structures and the regulation of their internalization is not well-understood. They carry out receptor-mediated endocytosis via specific transmembrane receptors, but the caveolae lipids and caveolin itself also serve as "receptors" for the uptake of specific molecules, including high-density lipoprotein (HDL). Recent studies have demonstrated that there are additional endocytosis pathways that are independent of both caveolin and clathrin. In addition, large vesicles (0.15 to 5.0 μm in diameter) can mediate the uptake of fluids in a process known as **macropinocytosis**. Thus while clathrin-dependent endocytosis clearly pro-

FIGURE 13.40 Caveolae Electron micrographs of caveolae. (A, courtesy of John E. Heuser, Washington University School of Medicine; B, courtesy of R. G. W. Anderson, University of Texas Southwestern Medical School, Dallas.)

vides a major pathway for the uptake of specific macromolecules, cells also use several clathrin-independent mechanisms.

Protein Trafficking in Endocytosis

Following their internalization, clathrin-coated vesicles rapidly shed their coats and fuse with early **endosomes,** which are vesicles with tubular extensions located at the periphery of the cell. The fusion of endocytic vesicles with endosomes is mediated by Rab GTP-binding proteins, their effectors, and complementary pairs of transmembrane proteins of the vesicle and target membranes (SNARE proteins) (see Figure 10.38). The early endosomes serve as a sorting compartment (they are sometimes called sorting endosomes) from which molecules taken up by endocytosis are either recycled to the plasma membrane or transported to lysosomes for degradation. In addition, the early endosomes of polarized cells can transfer endocytosed proteins between different domains of the plasma membrane—for example, between the apical and basolateral domains of epithelial cells.

An important feature of early endosomes is that they maintain an acidic internal pH (about 6.0 to 6.2) as the result of the action of a membrane H^+ pump. This acidic pH leads to the dissociation of many ligands from their receptors within the early endosome compartment. Following this uncoupling, the receptors and their ligands can be transported to different intracellular destinations. A classic example is provided by LDL, which dissociates from its receptor within early endosomes (Figure 13.41). The receptor is then returned to the plasma membrane via transport vesicles that bud from the tubular extensions of endosomes. In contrast, LDL is transported (along with other soluble contents of the endosome) to lysosomes, where its degradation releases cholesterol.

Recycling to the plasma membrane is the major fate of membrane proteins taken up by receptor-mediated endocytosis, with many receptors (like the LDL receptor) being returned to the plasma membrane following dissociation of their bound ligands in early endosomes. The recycling of these receptors results in the continuous internalization of their ligands. Each LDL receptor, for example, makes a round-trip from the plasma membrane to endosomes and back approximately every 10 minutes. The importance of the recycling pathway is further emphasized by the magnitude of membrane traffic resulting from endocytosis. As already noted, approximately 50% of the plasma membrane is internalized by receptor-mediated endocytosis

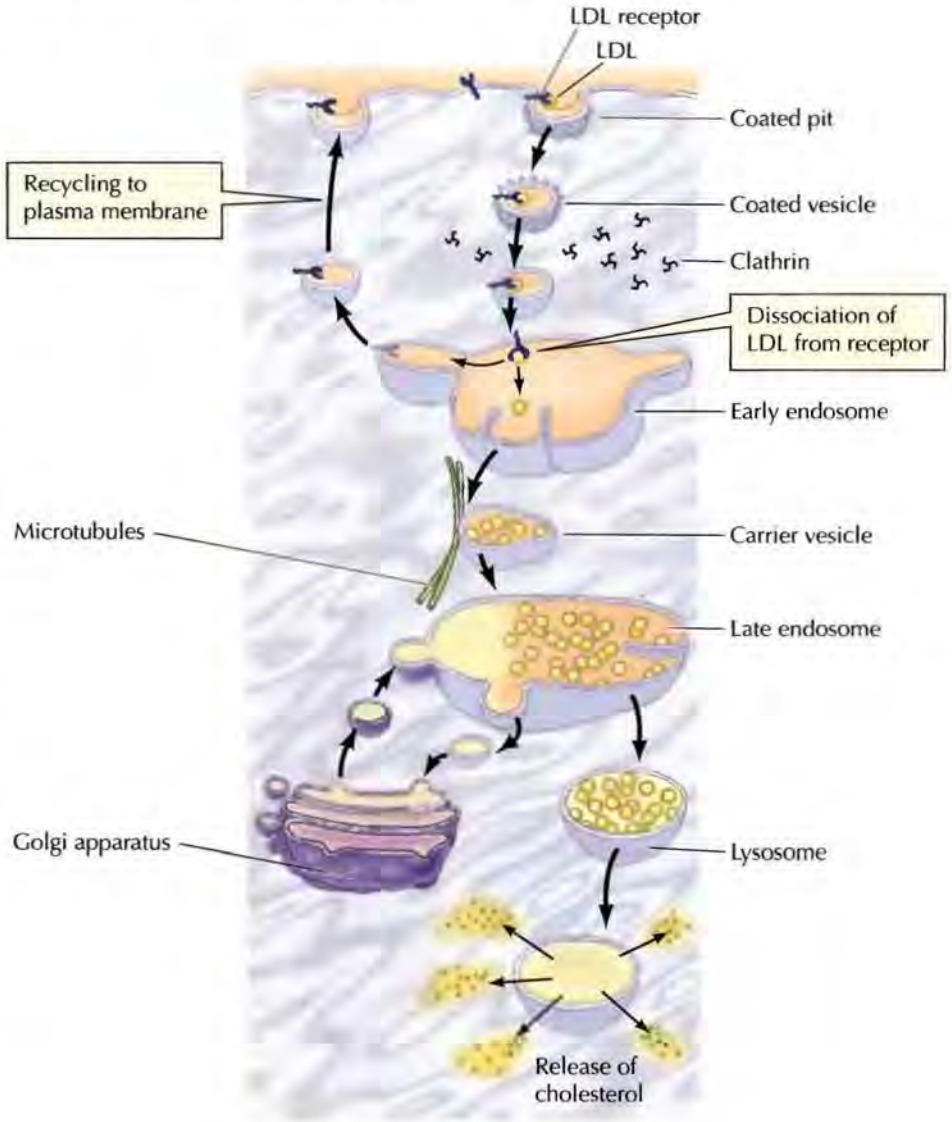


FIGURE 13.41 Sorting in early endosomes LDL bound to its receptor is internalized in clathrin-coated vesicles, which shed their coats and fuse with early endosomes. At the acidic pH of early endosomes, LDL dissociates from its receptor and the endocytosed materials are sorted for degradation in lysosomes or recycling to the plasma membrane. LDL is transported from early to late endosomes in large carrier vesicles that move along microtubules. Transport vesicles carrying lysosomal hydrolases from the Golgi apparatus then fuse with late endosomes, which mature to lysosomes where LDL is degraded and cholesterol is released. In contrast, the LDL receptor is recycled from early endosomes to the plasma membrane.

■ **Certain bacterial protein toxins, like cholera toxin, bind to a glycolipid found in lipid rafts and are taken up by endocytosis. Once inside endocytic vesicles the toxin commandeers the secretory pathway to gain entry into the cytoplasm, where it exerts its toxic effect.**

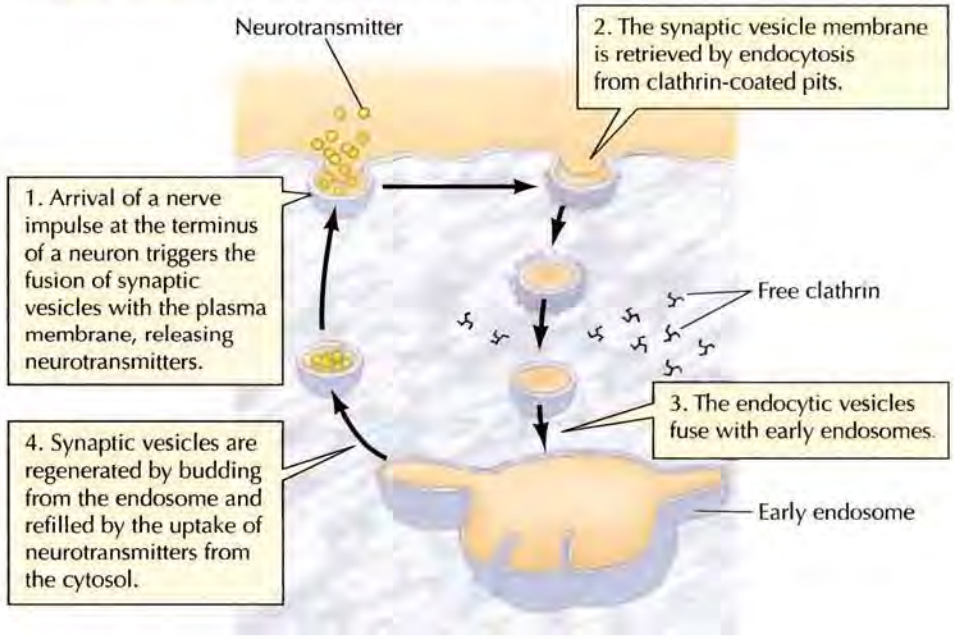
every hour and must therefore be replaced at an equivalent rate. Most of this replacement is the result of receptor recycling; only about 5% of the cell surface is newly synthesized per hour.

Ligands and membrane proteins destined for degradation in lysosomes are transported from early endosomes to late endosomes, which are located near the nucleus (see Figure 13.41). Transport from early to late endosomes is mediated by the movement of large endocytic carrier vesicles along microtubules. The late endosomes are more acidic than early endosomes (pH about 5.5 to 6.0) and, as discussed in Chapter 10, are able to fuse with transport vesicles carrying lysosomal hydrolases from the Golgi apparatus. Late endosomes then mature into lysosomes as they acquire a full complement of lysosomal enzymes and become still more acidic (pH about 5). Within lysosomes the endocytosed materials are degraded by the action of acid hydrolases.

Although many receptors (like the LDL receptor) are recycled to the plasma membrane, others follow different fates. Some are transported to lysosomes and degraded along with their ligands. For example, the cell surface receptors for several growth factors (discussed in Chapter 15) are internalized following growth factor binding and eventually degraded in lysosomes. The effect of this process is to remove the receptor-ligand complexes from the plasma membrane, thereby terminating the response of the cell to growth factor stimulation—a phenomenon known as **receptor down-regulation**.

A specialized kind of recycling from endosomes plays an important role in the transmission of nerve impulses across synapses (Figure 13.42). As discussed earlier in this chapter, the arrival of an action potential at the terminus of most neurons signals the fusion of synaptic vesicles with the plasma membrane, releasing the neurotransmitters that carry the signal to postsynaptic cells. The empty synaptic vesicles are then recovered from the

FIGURE 13.42 Recycling of synaptic vesicles



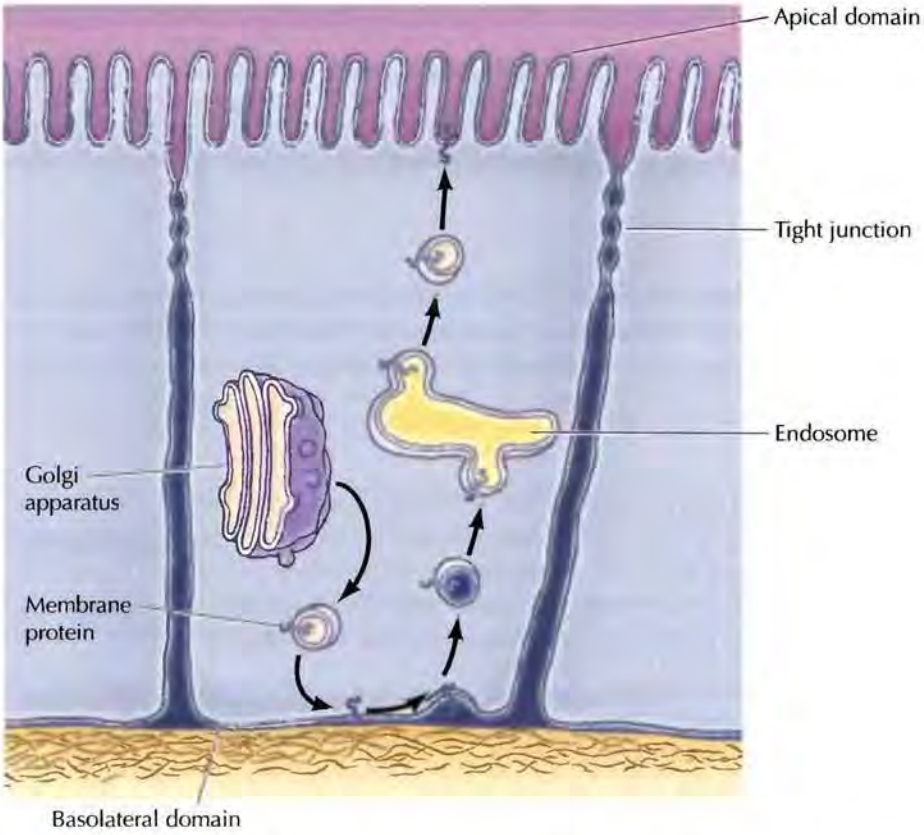


FIGURE 13.43 Protein sorting by transcytosis A protein destined for the apical domain of the plasma membrane is first transported from the Golgi apparatus to the basolateral domain. It is then endocytosed and selectively transported to the apical domain from early endosomes.

plasma membrane in clathrin-coated vesicles, which fuse with early endosomes. The synaptic vesicles are then regenerated directly by budding from endosomes. They accumulate a new supply of neurotransmitter and recycle to the plasma membrane, ready for the next cycle of synaptic transmission.

In polarized cells (e.g., epithelial cells) internalized receptors can also be transferred across the cell to the opposite domain of the plasma membrane—a process called **transcytosis**. For example, a receptor endocytosed from the basolateral domain of the plasma membrane can be sorted in early endosomes for transport to the apical membrane. In some cells, this is an important mechanism for sorting membrane proteins (Figure 13.43). Rather than being sorted for delivery to the apical or basolateral domains in the *trans* Golgi network (see Figure 10.31), proteins are initially delivered to the basolateral membrane. Proteins targeted for the apical membrane are then transferred to that site by transcytosis. In addition, transcytosis provides a mechanism for the transfer of extracellular macromolecules across epithelial cell sheets. For example, endothelial cells and many kinds of epithelial cells transport antibodies from the blood to a variety of secreted fluids, such as milk. The antibodies bind to receptors on the basolateral surface and are then transcytosed along with their receptors to the apical surface. The receptors are then cleaved, releasing the antibodies into extracellular secretions.

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

KEY TERMS

phosphatidylcholine, phosphatidylethanolamine, phosphatidylserine, sphingomyelin, phosphatidylinositol, glycolipid, cholesterol, lipid raft

fluid mosaic model, peripheral membrane protein, integral membrane protein, transmembrane protein, porin, glycosylphosphatidylinositol (GPI) anchor

apical domain, basolateral domain

glycocalyx, selectin

passive diffusion

facilitated diffusion, carrier protein, channel protein

ion channel, ligand-gated channel, voltage-gated channel, patch clamp technique, Nernst equation, action potential

active transport, ion pump, Na^+-K^+ pump, Na^+-K^+ ATPase, ABC transporter

symport, uniport, antiport

endocytosis, phagocytosis, pinocytosis, phagosome, phagolysosome

SUMMARY

STRUCTURE OF THE PLASMA MEMBRANE

The Phospholipid Bilayer: The fundamental structure of the plasma membrane is a phospholipid bilayer, which also contains glycolipids and cholesterol.

Membrane Proteins: Associated proteins are responsible for carrying out specific membrane functions. Membranes are viewed as fluid mosaics in which proteins are inserted into phospholipid bilayers.

Mobility of Membrane Proteins: Proteins are free to diffuse laterally through the phospholipid bilayer. However, the mobility of some proteins is restricted by their associations with other proteins or specific lipids. In addition, tight junctions prevent proteins from moving between distinct plasma membrane domains of epithelial cells.

The Glycocalyx: The cell surface is covered by a carbohydrate coat called the glycocalyx. Cell surface carbohydrates serve as markers for cell-cell recognition.

TRANSPORT OF SMALL MOLECULES

Passive Diffusion: Small hydrophobic molecules are able to cross the plasma membrane by diffusing through the phospholipid bilayer.

Facilitated Diffusion and Carrier Proteins: The passage of most biological molecules is mediated by carrier or channel proteins that allow polar and charged molecules to cross the plasma membrane without interacting with its hydrophobic interior.

Ion Channels: Ion channels mediate the rapid passage of selected ions across the plasma membrane. They are particularly well characterized in nerve and muscle cells, where they are responsible for the transmission of electric signals.

Active Transport Driven by ATP Hydrolysis: Energy derived from ATP hydrolysis can drive the transport of molecules against their electrochemical gradients.

Active Transport Driven by Ion Gradients: Ion gradients are frequently used as a source of energy to drive the active transport of other molecules.

ENDOCYTOSIS

Phagocytosis: Cells ingest large particles, such as bacteria and cell debris, by phagocytosis.

SUMMARY

Receptor-Mediated Endocytosis: The best-characterized form of endocytosis is receptor-mediated endocytosis, which provides a mechanism for the selective uptake of specific macromolecules.

Protein Trafficking in Endocytosis: Molecules taken up by endocytosis are transported to endosomes, where they are sorted for recycling to the plasma membrane or degradation in lysosomes.

KEY TERMS

receptor-mediated endocytosis, clathrin-coated pit, dynamin, clathrin-coated vesicle, ligand, low-density lipoprotein (LDL), fluid phase endocytosis, caveolae, caveolin, macropinocytosis

endosome, receptor down-regulation, transcytosis

Questions

1. How does cholesterol extend the functional temperature range of a lipid bilayer?

2. How are peripheral membrane proteins distinguished from integral membrane proteins?

3. How did the cell fusion experiments of Frye and Edidin support the fluid mosaic model of membrane structure? What results would they have obtained if they had incubated fused cells at 2° C?

4. What are lipid rafts and what are the cellular processes they are involved in?

5. What are the main functions of the glycocalyx?

6. The concentration of K^+ is about 20 times higher inside squid axons than in extracellular fluids, generating an equilibrium membrane potential of –75 mV. What would be the expected equilibrium membrane potential if the K^+ concentration were only 10 times higher inside than outside the cell? Why does the actual resting membrane potential (–60 mV) differ from the K^+ equilibrium potential of –75 mV? (Given: $R = 1.98 \times 10^{-3}$ kcal / mol / deg, T = 298 K, $\ln(x) = 2.3 \log_{10}(x)$, F = 23 kcal / V / mol)

7. Curare binds to nicotinic acetylcholine receptors and prevents them from opening. How would it affect the contraction of muscles?

8. How does the selectivity filter of K^+ channels differentiate between K^+ and Na^+ ions?

9. How can glucose be transported against its concentration gradient without the direct expenditure of ATP in intestinal epithelial cells?

10. How does the *mdr* gene confer drug resistance upon cancer cells?

11. How did Brown and Goldstein determine that LDL binds to a limited number of specific binding sites on the surface of normal cells?

12. What have studies on cells from children with familial hypercholesterolemia told us about the mechanisms of receptor-mediated endocytosis?

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CHAPTER 14

Cell Walls, the Extracellular Matrix, and Cell Interactions

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  Gap Junction Diseases 591

ALTHOUGH CELL BOUNDARIES ARE DEFINED BY THE PLASMA MEMBRANE, many cells are surrounded by an insoluble array of secreted macromolecules. Cells of bacteria, fungi, algae, and higher plants are surrounded by rigid cell walls, which are an integral part of the cell. Although animal cells are not surrounded by cell walls, most of the cells in animal tissues are embedded in an extracellular matrix consisting of secreted proteins and polysaccharides. The extracellular matrix not only provides structural support to cells and tissues, but also plays important roles in regulating cell behavior. Interactions of cells with the extracellular matrix anchor the cytoskeleton and regulate cell shape and movement. Likewise, direct interactions between cells are key to the organization of cells in the tissues of both plants and animals, as well as providing channels through which cells can communicate with their neighbors.

Cell Walls

The rigid cell walls that surround bacteria and many types of eukaryotic cells (fungi, algae, and higher plants) determine cell shape and prevent cells from swelling and bursting as a result of osmotic pressure. Despite their common functions, the cell walls of bacteria and eukaryotes are structurally very different. Bacterial cell walls consist of polysaccharides cross-linked by short peptides, which form a covalent shell around the entire cell. In contrast, the cell walls of eukaryotes are composed principally of polysaccharides embedded in a gel-like matrix. Rather than being fixed structures, plant cell walls can be modified both during development of the plant and in response to signals from the environment, so the cell walls of plants play critical roles in determining the organization of plant tissues and the structure of entire plants.

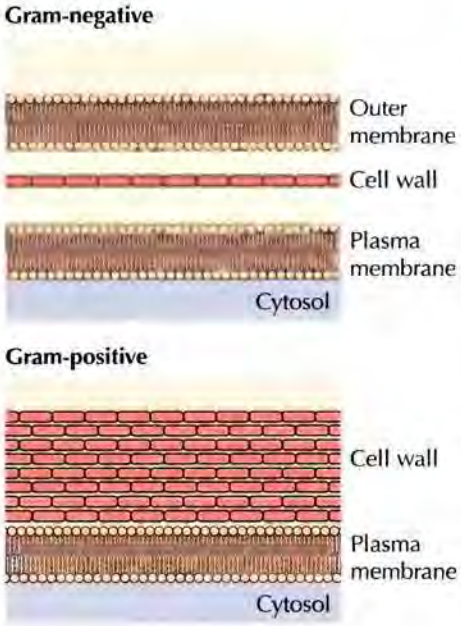


FIGURE 14.1 Bacterial cell walls The plasma membrane of Gram-negative bacteria is surrounded by a thin cell wall beneath the outer membrane. Gram-positive bacteria lack outer membranes and have thick cell walls.

■ **The outer membrane of Gram-negative bacteria contains a lipopolysaccharide (LPS) known as endotoxin. The release of LPS into the bloodstream leads to fever, lowered blood pressure, and inflammation.**

Bacterial Cell Walls

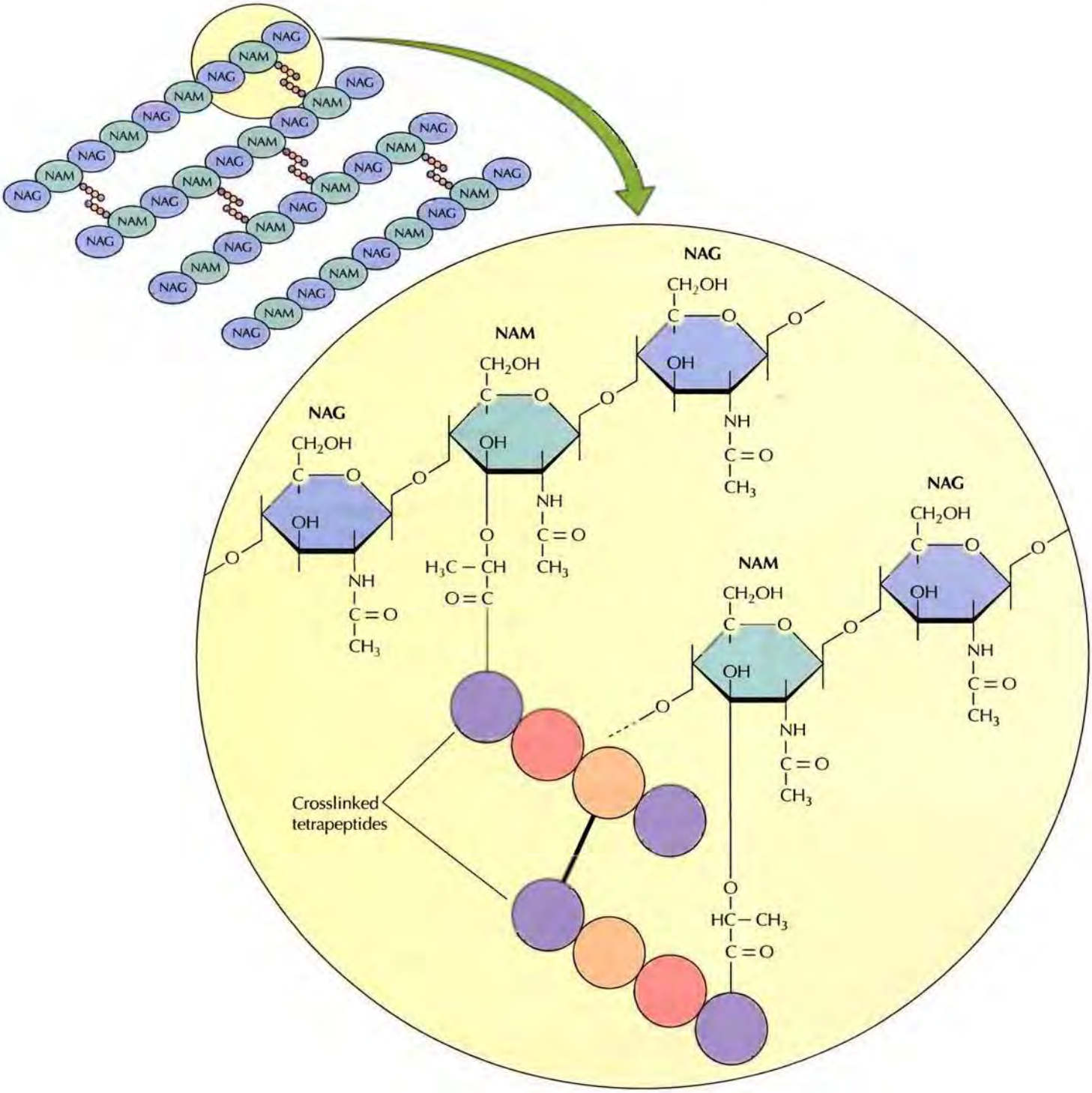
The rigid cell walls of bacteria determine the characteristic shapes of different kinds of bacterial cells. For example, some bacteria (such as *E. coli*) are rod-shaped, whereas others are spherical (e.g., *Pneumococcus* and *Staphylococcus*) or spiral-shaped (e.g., the spirochete *Treponema pallidum*, which causes syphilis). In addition, the structure of their cell walls divides bacteria into two broad classes that can be distinguished by a staining procedure known as the Gram stain, developed by Christian Gram in 1884 (Figure 14.1). As described in Chapter 13, Gram-negative bacteria (such as *E. coli*) have a dual membrane system in which the plasma membrane is surrounded by a permeable outer membrane (see Figure 13.8). These bacteria have thin cell walls located between their inner and outer membranes. In contrast, Gram-positive bacteria (such as the common human pathogen *Staphylococcus aureus*) have only a single plasma membrane, which is surrounded by a much thicker cell wall.

Despite these structural differences, the principal component of the cell walls of both Gram-positive and Gram-negative bacteria is a **peptidoglycan** (Figure 14.2), which consists of linear polysaccharide chains cross-linked by short peptides. Because of this cross-linked structure, the peptidoglycan forms a strong covalent shell around the entire bacterial cell. Interestingly, the unique structure of their cell walls also makes bacteria vulnerable to some antibiotics. Penicillin, for example, inhibits the enzyme responsible for forming cross-links between different strands of the peptidoglycan, thereby interfering with cell wall synthesis and blocking bacterial growth.

Eukaryotic Cell Walls

In contrast to bacteria, the cell walls of eukaryotes (including fungi, algae, and higher plants) are composed principally of polysaccharides (Figure 14.3). The basic structural polysaccharide of fungal cell walls is **chitin,** which also forms the shells of crabs and the exoskeletons of insects and other arthropods. Chitin is a linear polymer of *N*-acetylglucosamine residues. The cell walls of most algae and higher plants are composed principally of **cellulose**, which is the most abundant polymer on earth. Cellulose is a linear polymer of glucose residues, often containing more than 10,000 glucose monomers. The sugar residues in both chitin and cellulose are joined by β (1→4) linkages, which allow the polysaccharides to form long straight chains. Following transport across the plasma membrane to the extracellular space, single polysaccharide chains associate in parallel with one another to form microfibrils. Cellulose microfibrils can extend for many micrometers in length.

Within the plant cell wall, cellulose microfibrils are embedded in a matrix consisting of proteins and two other types of polysaccharides: **hemicelluloses** and **pectins** (Figure 14.4). Hemicelluloses are highly branched polysaccharides that are hydrogen-bonded to the surface of cellulose microfibrils (Figure 14.5). This stabilizes the cellulose microfibrils into a tough fiber, which is responsible for the mechanical strength of plant cell walls. The cellulose microfibrils are cross-linked by pectins, which are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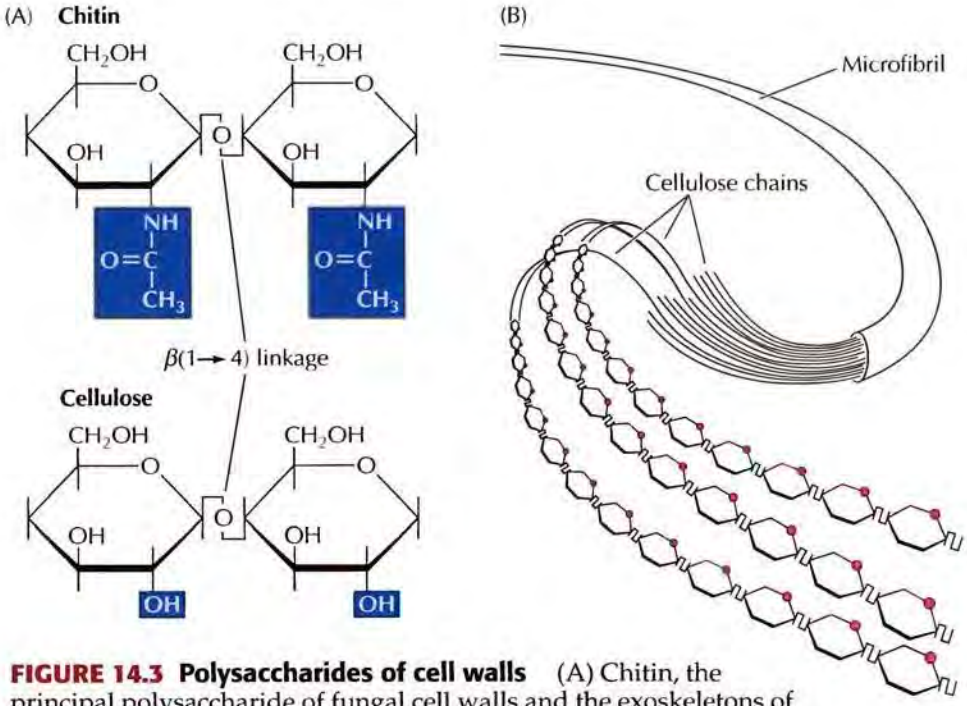


FIGURE 14.3 Polysaccharides of cell walls (A) Chitin, the principal polysaccharide of fungal cell walls and the exoskeletons of crabs and insects, is a linear polymer of *N*-acetylglucosamine residues, whereas cellulose is a linear polymer of glucose. As in peptidoglycan, the carbohydrate monomers are joined by β (1→4) linkages, allowing the polysaccharides to form long, straight chains. (B) Parallel chains of cellulose associate to form microfibrils.

Both the structure and function of cell walls change as plant cells develop. The walls of growing plant cells (called **primary cell walls**) are relatively thin and flexible, allowing the cell to expand in size. Once cells have ceased growth, they frequently lay down **secondary cell walls** between the plasma membrane and the primary cell wall (Figure 14.6). Such secondary cell walls, which are both thicker and more rigid than primary walls, are particularly important in cell types responsible for conducting water and providing mechanical strength to the plant.

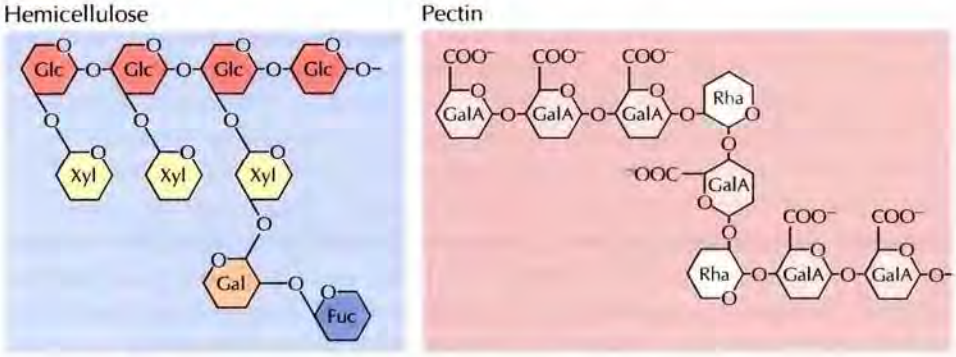


FIGURE 14.4 Structures of hemicellulose and pectin A representative hemicellulose (xyloglucan) consists of a backbone of glucose (Glc) residues with side chains of xylose (Xyl), galactose (Gal), and fucose (Fuc). The backbone of rhamnogalacturonan (a representative pectin) contains galacturonic acid (GalA) and rhamnose (Rha) residues to which numerous side chains are also attached

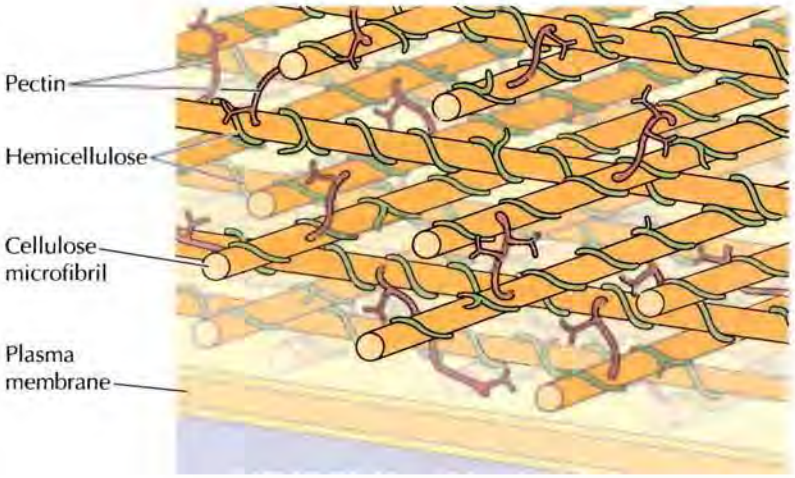
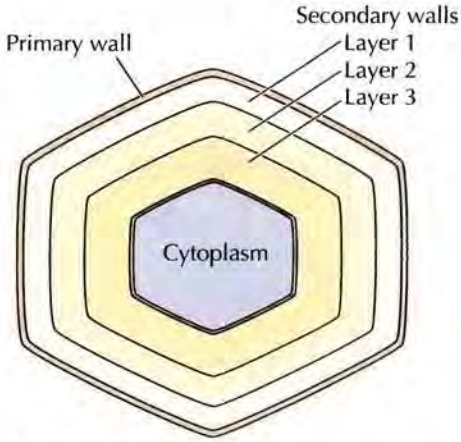


FIGURE 14.5 Model of a plant cell wall
Cellulose is organized into microfibrils that are oriented in layers. Hemicelluloses (green) are tightly associated with the surface of cellulose microfibrils, which are cross-linked by pectins (red).

Primary and secondary cell walls differ in composition as well as in thickness. Primary cell walls contain approximately equal amounts of cellulose, hemicelluloses, and pectins. In contrast, the more rigid secondary walls generally lack pectin and contain 50 to 80% cellulose. Many secondary walls are further strengthened by **lignin**, a complex polymer of phenolic residues that is responsible for much of the strength and density of wood. The orientation of cellulose microfibrils also differs in primary and secondary cell walls. The cellulose fibers of primary walls appear to be randomly arranged, whereas those of secondary walls are highly ordered (see Figure 14.6). Secondary walls are frequently laid down in layers in which the cellulose fibers differ in orientation, forming a laminated structure that greatly increases cell wall strength.

One of the critical functions of plant cell walls is to prevent cell swelling as a result of osmotic pressure. In contrast to animal cells, plant cells do not maintain an osmotic balance between their cytosol and extracellular fluids. Consequently, osmotic pressure continually drives the flow of water into the cell. This water influx is tolerated by plant cells because their rigid cell walls prevent swelling and bursting. Instead, an internal hydrostatic pressure (called **turgor pressure**) builds up within the cell, eventually equalizing the osmotic pressure and preventing the further influx of water.

Turgor pressure is responsible for much of the rigidity of plant tissues as is readily apparent from examination of a dehydrated, wilted plant. In addition, turgor pressure provides the basis for a form of cell growth that is unique to plants. In particular, plant cells frequently expand by taking up water without synthesizing new cytoplasmic components (Figure 14.7). Cell expansion by this mechanism is signaled by plant hormones (**auxins**) that activate proteins called expansins. The expansins act to weaken a region of the cell wall, allowing turgor pressure to drive the expansion of the cell in



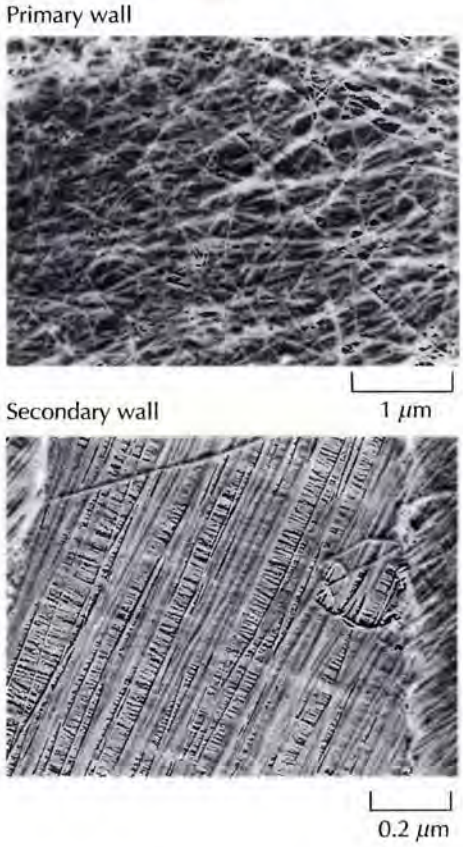


FIGURE 14.6 Primary and secondary cell walls Secondary cell walls are laid down between the primary cell wall and the plasma membrane. Secondary walls frequently consist of three layers, which differ in the orientation of their cellulose microfibrils. Electron micrographs show cellulose microfibrils in primary and secondary cell walls. (Primary wall, courtesy of F. C. Steward; secondary wall, Biophoto Associates/Photo Researchers, Inc.)

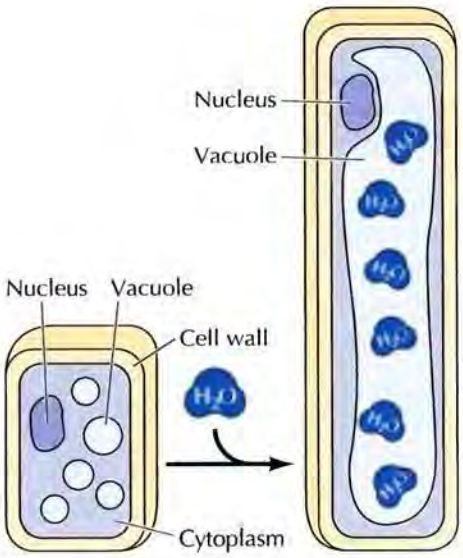


FIGURE 14.7 Expansion of plant cells Turgor pressure drives the expansion of plant cells by the uptake of water, which is accumulated in a large central vacuole.

that direction. As this occurs, the water that flows into the cell accumulates within a large central vacuole, so the cell expands without increasing the volume of its cytosol. Such expansion can result in a ten to one hundredfold increase in the size of plant cells during development.

As cells expand, new components of the cell wall are deposited outside the plasma membrane. While matrix components, including hemicelluloses and pectins, are synthesized in the Golgi apparatus and secreted, cellulose is synthesized by the plasma membrane enzyme complex **cellulose synthase** (Figure 14.8). Cellulose synthase is a transmembrane enzyme that synthesizes cellulose from UDP-glucose in the cytosol. The growing cellulose chain remains bound to the enzyme as it is synthesized and is translocated across the plasma membrane to the exterior of the cell through a pore created by multiple enzyme subunits. A similar mechanism is used to synthesize chitin and hyaluronan, a component of the extracellular matrix discussed later in this chapter.

In expanding cells, the newly synthesized cellulose microfibrils are deposited at right angles to the direction of cell elongation—an orientation that is thought to play an important role in determining the direction of further cell expansion (see Figure 14.8). Interestingly, the cellulose microfibrils in elongating cell walls are laid down in parallel to cortical microtubules underlying the plasma membrane. These microtubules appear to define the orientation of newly synthesized cellulose microfibrils, possibly by determining the direction of movement of the cellulose synthase complexes in the membrane. The cortical microtubules thus define the direction of cell wall growth, which in turn determines the direction of cell expansion and ultimately the shape of the entire plant.

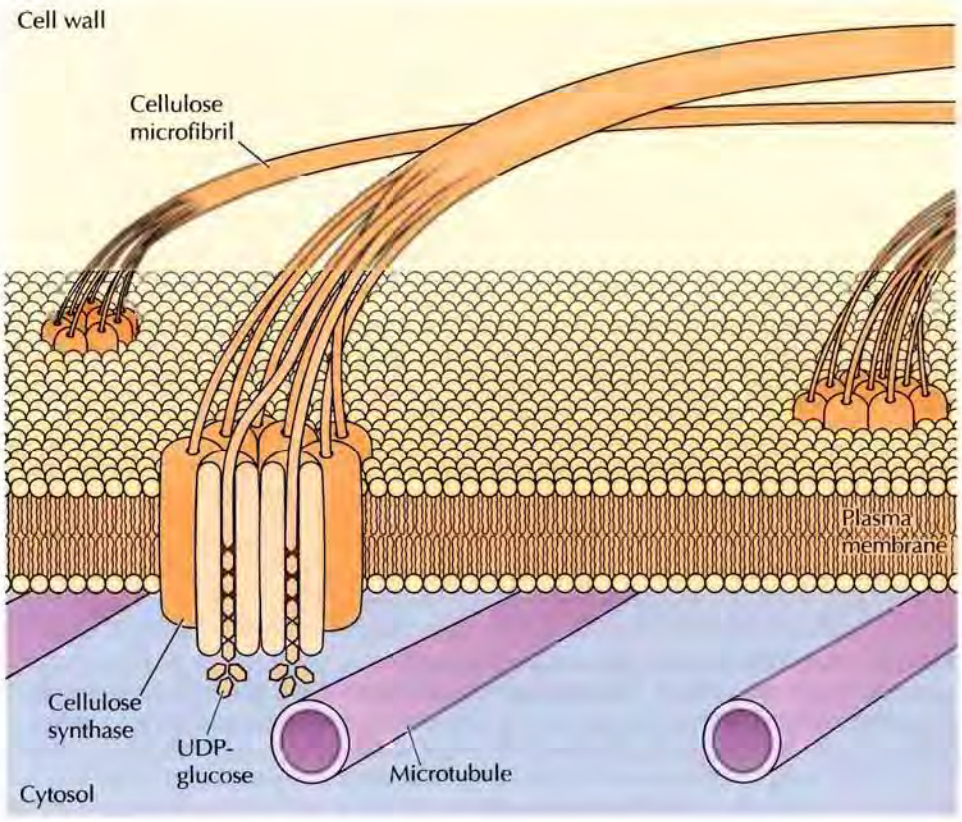


FIGURE 14.8 Cellulose synthesis during cell elongation Cellulose synthases are transmembrane enzymes that synthesize cellulose from UDP-glucose. UDP-glucose binds to the catalytic domain of cellulose synthase in the cytosol and the growing cellulose chain is translocated to the outside of the cell through a pore created by two subunits of the enzyme in the plasma membrane. Complexes of 8–10 cellulose synthase dimers track microtubules beneath the plasma membrane such that new cellulose microfibrils are laid down at right angles to the direction of cell elongation.

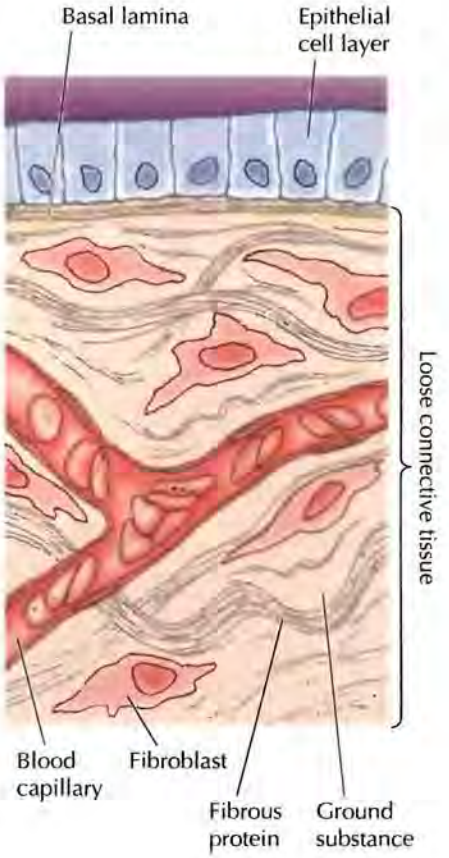


FIGURE 14.9 Examples of extracellular matrix Sheets of epithelial cells rest on a thin layer of extracellular matrix called a basal lamina. Beneath the basal lamina is loose connective tissue, which consists largely of extracellular matrix secreted by fibroblasts. The extracellular matrix contains fibrous structural proteins embedded in a gel-like polysaccharide ground substance.

Cell walls from different plant tissues, such as leaves, stems, roots and flowers, consist primarily of cellulose but differ in their matrix components and perhaps in the organization of their cellulose fibrils. There are at least ten different cellulose synthase enzymes and each plasma membrane enzyme complex contains two or more different forms, perhaps accounting for the varying organization of cell walls in different tissues.

The Extracellular Matrix and Cell-Matrix Interactions

Although animal cells are not surrounded by cell walls, most animal cells in tissues are embedded in an **extracellular matrix** that fills the spaces between cells and binds cells and tissues together. There are several types of extracellular matrices, which consist of a variety of secreted proteins and polysaccharides. One type of extracellular matrix is exemplified by the thin, sheetlike **basal laminae,** previously called basement membranes, upon which layers of epithelial cells rest (Figure 14.9). In addition to supporting sheets of epithelial cells, basal laminae surround muscle cells, adipose cells, and peripheral nerves. Extracellular matrix, however, is most abundant in connective tissues. For example, the loose connective tissue beneath epithelial cell layers consists predominantly of an extracellular matrix in which fibroblasts are distributed. Other types of connective tissue, such as bone, tendon, and cartilage, similarly consist largely of extracellular matrix, which is principally responsible for their structure and function. Several of the proteins and polysaccharides found in the extracellular matrix are also found closely associated with the plasma membrane. These can be seen on the plasma membrane of endothelial cells or the apical membrane of intestinal epithelial cells as a polysaccharide rich coat termed the glycocalyx (see Figure 13.13).

Matrix Structural Proteins

Extracellular matrices are composed of tough fibrous proteins embedded in a gel-like polysaccharide ground substance—a design basically similar to that of plant cell walls. In addition to fibrous structural proteins and polysaccharides, the extracellular matrix contains adhesion proteins that link components of the matrix both to one another and to attached cells. The differences between various types of extracellular matrices result from both quantitative variations in the types or amounts of these different constituents and from modifications in their organization. For example, tendons contain a high proportion of fibrous proteins, whereas cartilage contains a high concentration of polysaccharides that form a firm compression-resistant gel. In bone, the extracellular matrix is hardened by deposition of calcium phosphate crystals. The sheetlike structure of basal laminae results from a matrix composition that differs from that found in connective tissues.

The major structural protein of the extracellular matrix is **collagen,** which is the single most abundant protein in animal tissues. The collagens

Cancer cells secrete proteases that digest proteins of the extracellular matrix, allowing the cancer cells to invade surrounding tissue and metastasize to other parts of the body.

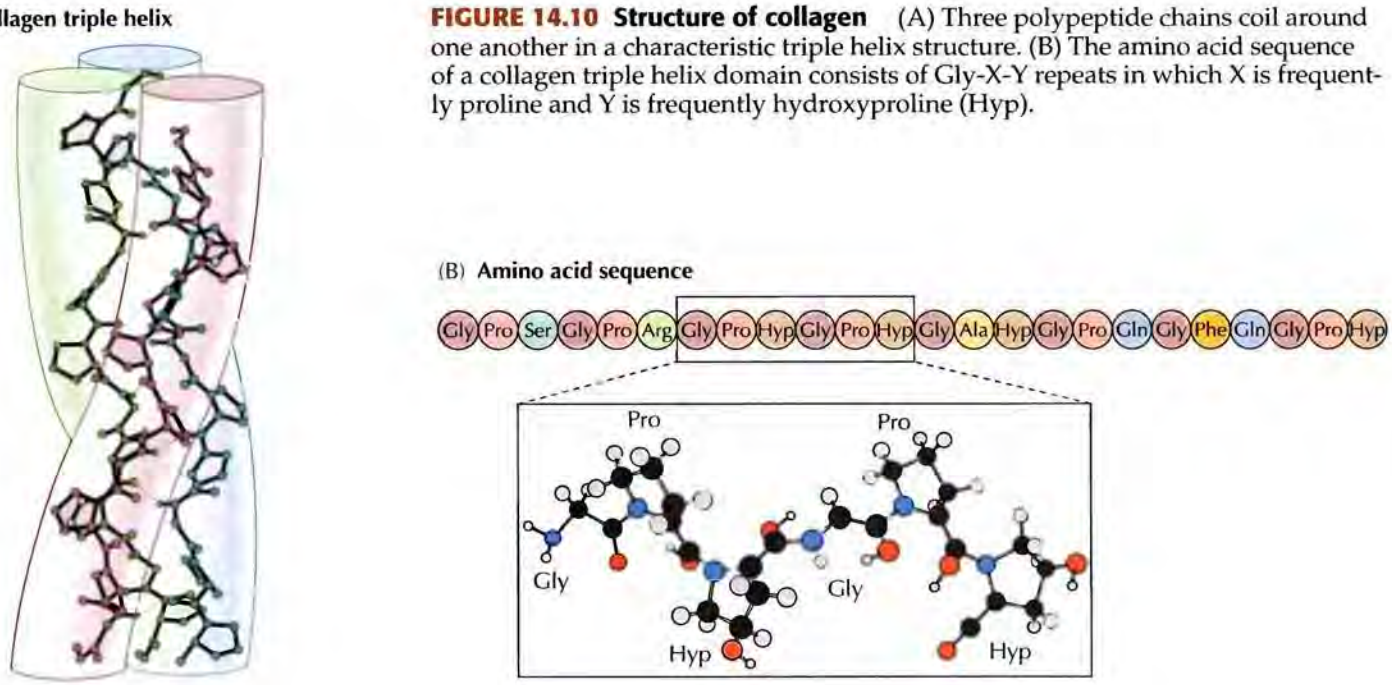


FIGURE 14.10 Structure of collagen (A) Three polypeptide chains coil around one another in a characteristic triple helix structure. (B) The amino acid sequence of a collagen triple helix domain consists of Gly-X-Y repeats in which X is frequently proline and Y is frequently hydroxyproline (Hyp).

■ **Prolyl hydroxylase, the enzyme that catalyzes the modification of proline to hydroxyproline residues in collagen, requires vitamin C for its activity. Vitamin C deficiency causes scurvy, a disorder characterized by skin lesions and blood vessel hemorrhages as a result of weakened connective tissue.**

are a large family of proteins containing at least 27 different members. They are characterized by the formation of triple helices in which three polypeptide chains are wound tightly around one another in a ropelike structure (Figure 14.10). The different collagen polypeptides can assemble into 42 different trimers. The triple helix domains of the collagens consist of repeats of the amino acid sequence Gly-X-Y. A glycine (the smallest amino acid, with a side chain consisting only of a hydrogen) is required in every third position in order for the polypeptide chains to pack together close enough to form the collagen triple helix. Proline is frequently found in the X position and hydroxyproline in the Y position; because of their ring structure these amino acids stabilize the helical conformations of the polypeptide chains.

The unusual amino acid hydroxyproline is formed within the endoplasmic reticulum by modification of proline residues that have already been incorporated into collagen polypeptide chains (Figure 14.11). Lysine residues in collagen are also frequently converted to hydroxylysines. The hydroxyl groups of these modified amino acids are thought to stabilize the collagen triple helix by forming hydrogen bonds between polypeptide chains. These amino acids are rarely found in other proteins although hydroxyproline is also common in some of the glycoproteins of plant cell walls.

The most abundant type of collagen (type I collagen) is one of the fibril-forming collagens that are the basic structural components of connective tissues (Table 14.1). The polypeptide chains of these collagens consist of approximately a thousand amino acids or 330 Gly-X-Y repeats. After being secreted from the cell these collagens assemble into **collagen fibrils** in which the triple helical molecules are associated in regular staggered arrays (Figure 14.12). These fibrils do not form within the cell because the fibril-

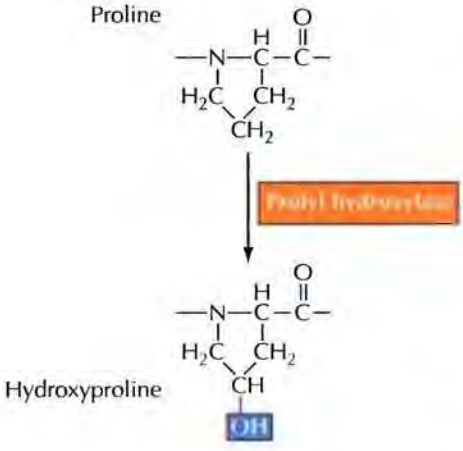


FIGURE 14.11 Formation of hydroxyproline Prolyl hydroxylase converts proline residues in collagen to hydroxyproline.

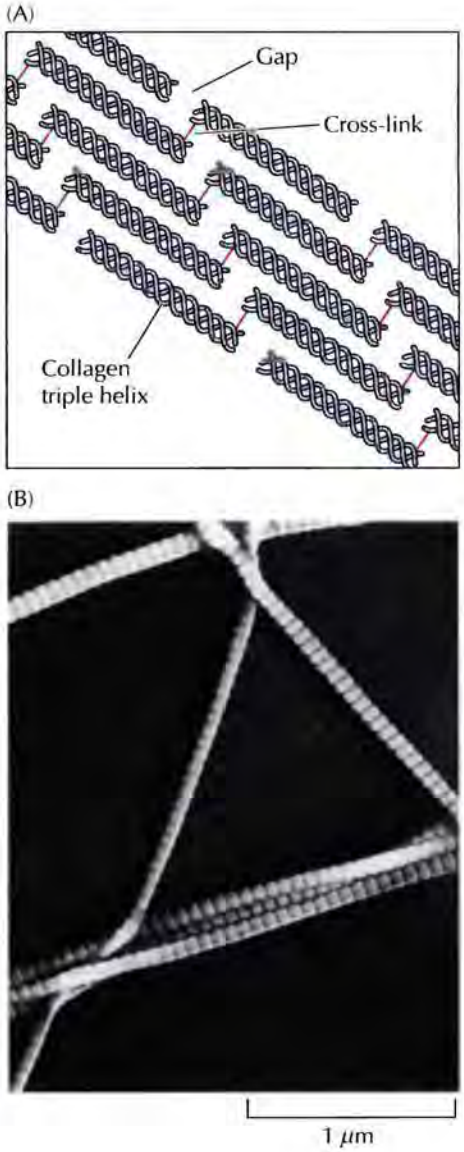


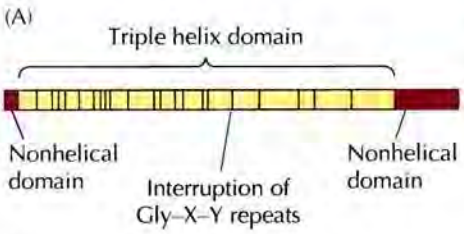
FIGURE 14.12 Collagen fibrils (A) Collagen molecules assemble in a regular staggered array to form fibrils. The molecules overlap by one-fourth of their length, and there is a short gap between the N terminus of one molecule and the C terminus of the next. The assembly is strengthened by covalent cross-links between side chains of lysine or hydroxylysine residues, primarily at the ends of the molecules. (B) Electron micrograph of collagen fibrils. The staggered arrangement of collagen molecules and the gaps between them are responsible for the characteristic cross-striations in the fibrils. (B, J. Gross, F. O. Sahmitt, and D. Fawcett/Visuals Unlimited.)

forming collagens are synthesized as soluble precursors (**procollagens**) that contain nonhelical segments at both ends of the polypeptide chain. Procollagen is cleaved to collagen after its secretion, so the assembly of collagen into fibrils takes place only outside the cell. The association of collagen molecules in fibrils is further strengthened by the formation of covalent cross-links between the side chains of lysine and hydroxylysine residues. Frequently, the fibrils further associate with one another to form collagen fibers, which can be several micrometers in diameter.

Several other types of collagen do not form fibrils but play distinct roles in various kinds of extracellular matrices (see Table 14.1). In addition to the fibril-forming collagens, connective tissues contain fibril-associated collagens, which bind to the surface of collagen fibrils and link them both to one another and to other matrix components. Basal laminae form from a different type of collagen (type IV collagen), which is a network-forming collagen (Figure 14.13). The Gly-X-Y repeats of these collagens are frequently inter-

TABLE 14.1 Representative Members of the Collagen Family

Collagen class	Types	Tissue distribution
Fibril-forming	I	Most connective tissues
	II	Cartilage and vitreous humor
	III	Extensible connective tissues (e.g., skin and lung)
	V	Tissues containing collagen I
	XI	Cartilage
	XXIV	Bone and cornea
	XXVII	Eye, ear, and lung
Fibril-associated	IX	Cartilage
	XII	Tissues containing collagen I
	XIV	Tissues containing collagen I
	XVI	Many tissues
	XIX	Many tissues
	XX	Cornea
	XXI	Many tissues
	XXII	Cell junctions
	XXVI	Testis and ovary
Network-forming	IV	Basal laminae
	VIII	Many tissues
	X	Cartilage
Anchoring fibrils	VII	Attachments of basal laminae to underlying connective tissue
Transmembrane	XVII	Skin hemidesmosomes
	XXV	Nerve cells



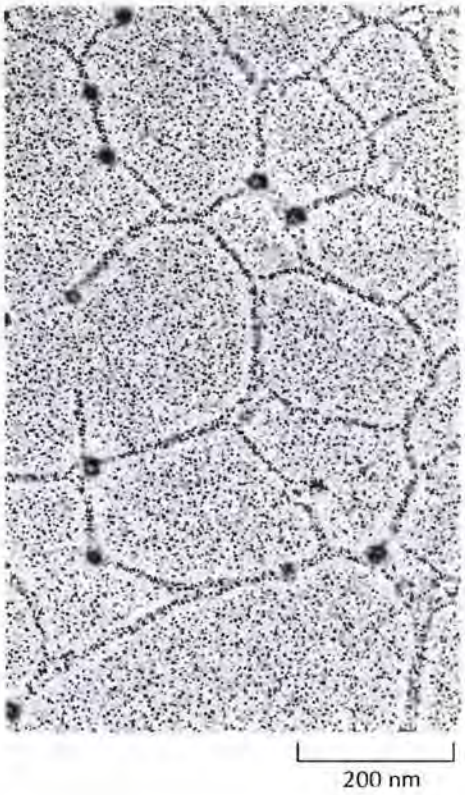


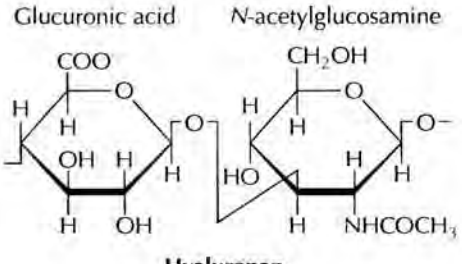
FIGURE 14.13 Type IV collagen (A) The Gly-X-Y repeat structure of type IV collagen (yellow) is interrupted by multiple nonhelical sequences (bars). (B) Electron micrograph of a type IV collagen network. (B, P. D. Yurchenco and J. C. Schittny, 1990. *FASEB J.* 4: 1577.)

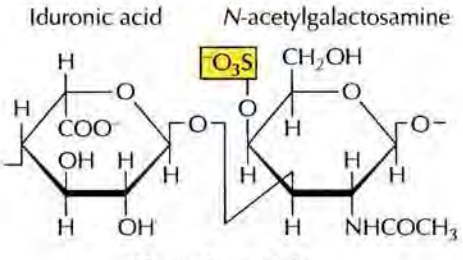
rupted by short nonhelical sequences. Because of these interruptions, the network-forming collagens are more flexible than the fibril-forming collagens. Consequently, they assemble into two-dimensional cross-linked networks instead of fibrils. Yet another type of collagen forms anchoring fibrils, which link some basal laminae to underlying connective tissues. Other types of collagen are transmembrane proteins that participate in cell-matrix interactions.

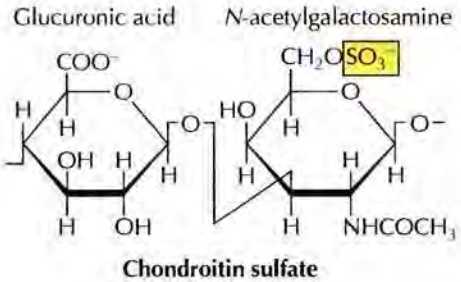
Connective tissues also contain **elastic fibers**, which are particularly abundant in organs that regularly stretch and then return to their original shape. The lungs, for example, stretch each time a breath is inhaled and return to their original shape with each exhalation. Elastic fibers are composed principally of a protein called **elastin**, which is cross-linked into a network by covalent bonds formed between the side chains of lysine residues (similar to those found in collagen). This network of cross-linked elastin chains behaves like a rubber band, stretching under tension and then snapping back when the tension is released.

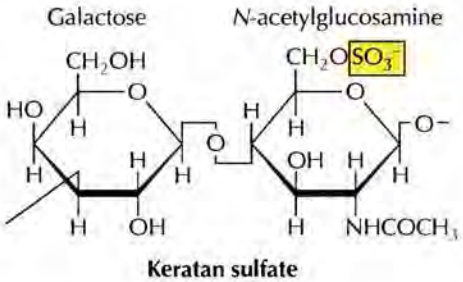
Matrix Polysaccharides

The fibrous structural proteins of the extracellular matrix are embedded in gels formed from polysaccharides called **glycosaminoglycans (GAGs)**, which consist of repeating units of disaccharides (Figure 14.14). One sugar of the disaccharide is either *N*-acetylglucosamine or *N*-acetylgalactosamine and the second is usually acidic (either glucuronic acid or iduronic acid). With the exception of hyaluronan, these sugars are modified by the addition of sulfate groups. Consequently, GAGs are highly negatively charged.









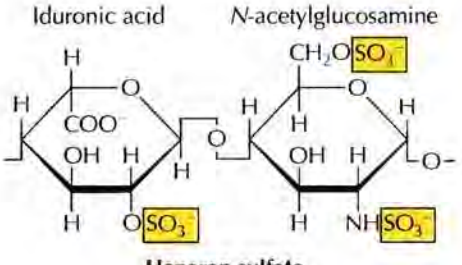


FIGURE 14.14 Major types of glycosaminoglycans Glycosaminoglycans consist of repeating disaccharide units. With the exception of hyaluronan, the sugars frequently contain sulfate.

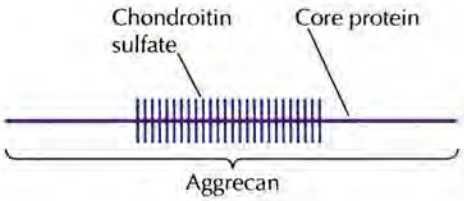
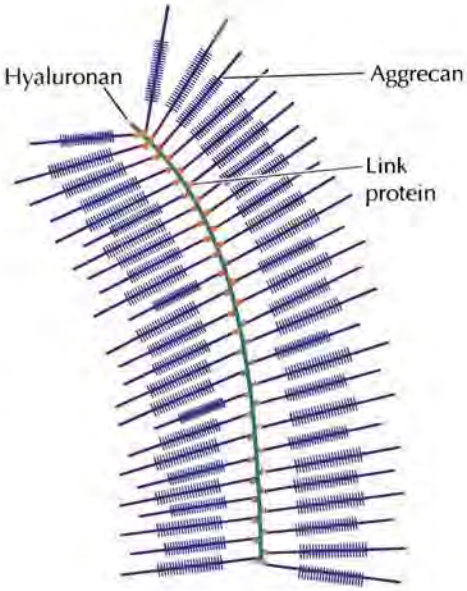


FIGURE 14.15 Complexes of aggrecan and hyaluronan Aggrecan is a large proteoglycan consisting of more than 100 chondroitin sulfate chains joined to a core protein. Multiple aggrecan molecules bind to long chains of hyaluronan forming large complexes in the extracellular matrix of cartilage. This association is stabilized by link proteins.

Like the pectins of plant cell walls, they bind positively charged ions and trap water molecules to form hydrated gels, thereby providing mechanical support to the extracellular matrix. The common sulfated GAGs are chondroitin sulfate, dermatan sulfate, heparan sulfate, and keratan sulfate.

Hyaluronan is the only GAG that occurs as a single long polysaccharide chain. Like cellulose and chitin, it is synthesized at the plasma membrane by a transmembrane hyaluronan synthase. All of the other GAGs are linked to proteins to form **proteoglycans**, which can consist of up to 95% polysaccharide by weight. Proteoglycans can contain as few as one or as many as more than a hundred GAG chains attached to serine residues of a core protein. A variety of core proteins (ranging from 10 to >500 kd) have been identified, so the proteoglycans are a diverse group of macromolecules. In addition to being components of the extracellular matrix, some proteoglycans (syndecans and glypicans) are cell surface proteins that function along with integrins in cell-cell and cell-matrix adhesion.



A number of proteoglycans interact with hyaluronan to form large complexes in the extracellular matrix. A well-characterized example is aggrecan, the major proteoglycan of cartilage (Figure 14.15). More than a hundred chains of chondroitin sulfate are attached to a core protein of about 250 kd, forming a proteoglycan of about 3000 kd. Multiple aggrecan molecules then associate with chains of hyaluronan, forming large aggregates (>100,000 kd) that become trapped in the collagen network. Proteoglycans also interact with both collagen and other matrix proteins to form gel-like networks in which the fibrous structural proteins of the extracellular matrix are embedded. For example, perlecan (the major heparan sulfate proteoglycan of basal laminae) binds to both type IV collagen and to the adhesion protein laminin, which is discussed shortly.

Matrix Adhesion Proteins

Matrix adhesion proteins, the final class of extracellular matrix constituents, are responsible for linking the components of the matrix to one another and to the surfaces of cells. They interact with collagen and proteoglycans to specify matrix organization and are the major binding sites for integrins.

The prototype of these molecules is **fibronectin**, which is the principal adhesion protein of connective tissues. Fibronectin is a dimeric glycoprotein consisting of two polypeptide chains, each containing nearly 2500 amino acids (Figure 14.16). Within the extracellular matrix, fibronectin is often cross-linked into fibrils. Fibronectin has binding sites for both collagen and GAGs so it cross-links these matrix components. A distinct site on the fibronectin molecule is recognized by cell surface receptors (such as integrins) and is

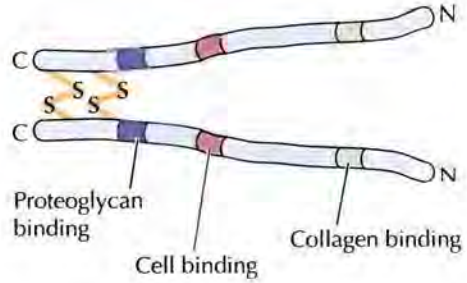


FIGURE 14.16 Structure of fibronectin Fibronectin is a dimer of similar polypeptide chains joined by disulfide bonds near the C terminus. Sites for binding to proteoglycans, cells, and collagen are indicated. The molecule also contains additional binding sites that are not shown.

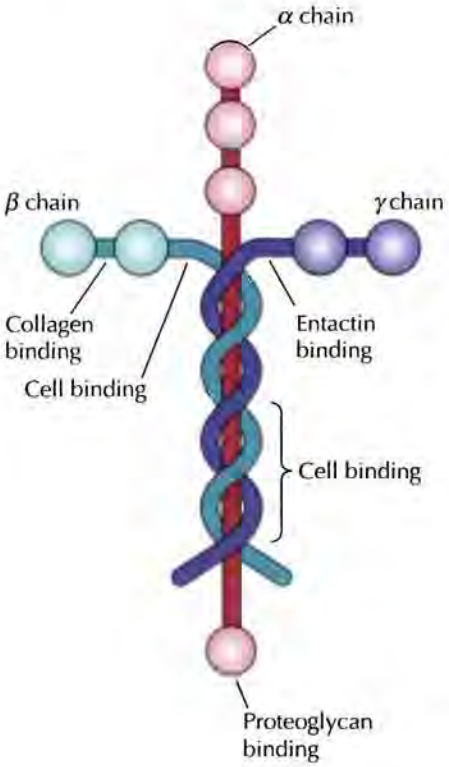


FIGURE 14.17 Structure of laminins Laminins consist of three polypeptide chains designated α, β, and γ. Some of the binding sites for entactin, type IV collagen, proteoglycans, and cell surface receptors are indicated.

thus responsible for the attachment of cells to the extracellular matrix. Fibronectin proteins vary greatly from tissue to tissue but all are derived by alternative splicing of the mRNA of a single gene.

Basal laminae contain distinct adhesion proteins of the **laminin** family (Figure 14.17). Laminins are heterotrimers of α, β, and γ subunits which are the products of five α genes, four β genes, and three γ genes. Like type IV collagen, laminins can self-assemble into meshlike polymers. Such laminin networks are the major structural components of the basal laminae synthesized in very early embryos, which do not contain collagen. The laminins also have binding sites for cell surface receptors such as integrins, type IV collagen, and the heparan sulfate proteoglycan, perlecan. In addition, laminins are tightly associated with another adhesion protein, called **entactin,** which also binds to type IV collagen. As a result of these multiple interactions, laminin, entactin, type IV collagen, and perlecan form cross-linked networks in the basal lamina.

Cell-Matrix Interactions

The major cell surface receptors responsible for the attachment of cells to the extracellular matrix are the **integrins**. The integrins are a family of transmembrane proteins consisting of two subunits, designated α and β (Figure 14.18). More than 24 different integrins formed from combinations of 18 known α subunits and 8 known β subunits have been identified. The integrins bind to short amino acid sequences present in multiple components of the extracellular matrix, including collagen, fibronectin, and laminin. The first such integrin-binding site to be characterized was the sequence Arg-Gly-Asp in fibronectin, which is recognized by several members of the integrin family. Other integrins, however, bind to distinct peptide sequences, including recognition sequences in collagens and laminin. Transmembrane proteoglycans on the surface of a variety of cells also bind to components of the extracellular matrix and modulate cell-matrix interactions.

In addition to attaching cells to the extracellular matrix the integrins serve as anchors for the cytoskeleton (Figure 14.19). The resulting linkage of the cytoskeleton to the extracellular matrix is responsible for the stability of cell-matrix junctions. Distinct interactions between integrins and the cytoskeleton are found at two types of cell-matrix junctions: **focal adhesions** and **hemidesmosomes**, both of which were discussed in Chapter 12. Focal adhesions attach a variety of cells, including fibroblasts, to the extracellular matrix. At focal adhesions, the cytoplasmic domains of the β subunits of integrins anchor the actin cytoskeleton by associating with bundles of actin filaments through actin-binding proteins such as α-actinin, talin, and vinculin. Hemidesmosomes mediate epithelial cell attachments at which a specific integrin (designated $\alpha_6\beta_4$) interacts through plectin and BP230 with intermediate filaments instead of with actin. This is mediated

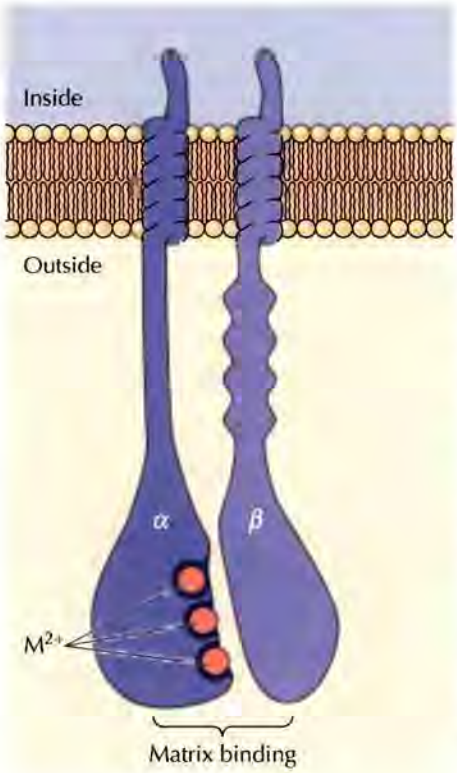


FIGURE 14.18 Structure of integrins The integrins are heterodimers of two transmembrane subunits, designated α and β. The α subunit binds divalent cations, designated M^{2+}. The matrix-binding region is composed of portions of both subunits.

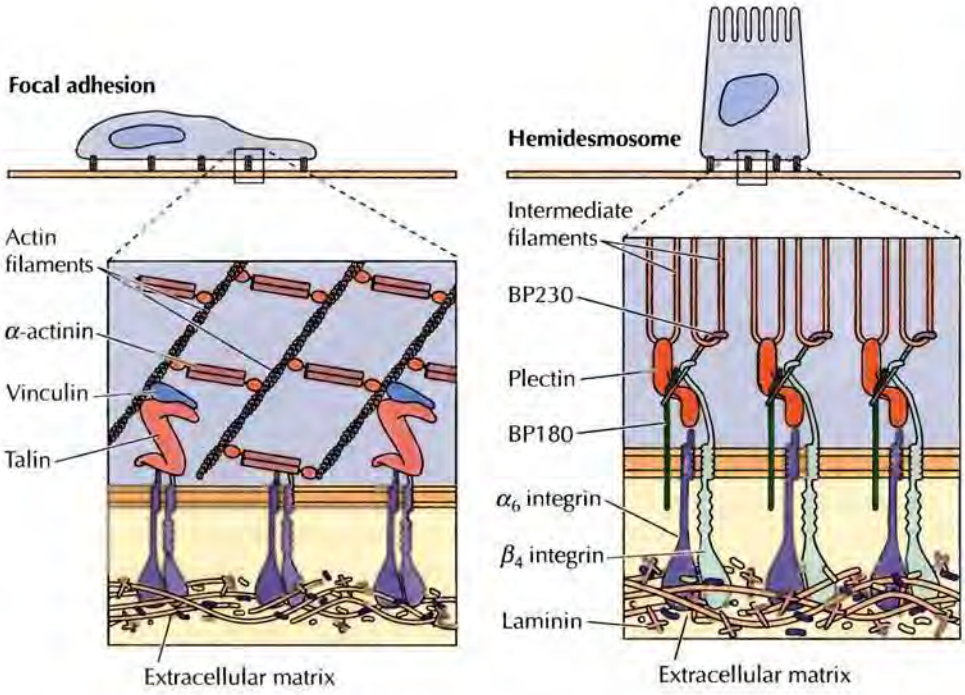


FIGURE 14.19 Cell-matrix junctions mediated by integrins Integrins mediate two types of stable junctions in which the cytoskeleton is linked to the extracellular matrix. In focal adhesions, bundles of actin filaments are anchored to the β subunits of most integrins via associations with a number of other proteins, including α-actinin, talin, and vinculin. In hemidesmosomes, $\alpha_6\beta_4$ integrin links the basal lamina to intermediate filaments via plectin and BP230. BP180 functions in hemidesmosome assembly and stability.

by the long cytoplasmic tail of the β_4 integrin subunit. The $\alpha_6\beta_4$ integrin binds to laminin, so hemidesmosomes anchor epithelial cells to the basal lamina. An additional protein, BP180, is important in hemidesmosome assembly and stability. Plectin and BP230 are members of the **plakin** family, which is a family of proteins involved in forming links to intermediate filaments at both hemidesmosomes and desmosomes (discussed in the next section of this chapter). BP180 shares sequence homology with the transmembrane collagens.

Cell-matrix interactions, like the cell-cell interactions discussed in the next section, develop by stepwise recruitment of specific junctional molecules to the cell membrane. Focal adhesions develop from a small cluster of integrins, termed **focal complexes**, by the sequential recruitment of talin, vinculin, and α-actinin. These form the initial connections to the actin cytoskeleton and recruit formin, which initiates actin bundle formation (see Figure 12.6), and myosin II, which leads to the development of tension at the point of adhesion. Tension allows the formation of a larger area of cell-matrix adhesion and the recruitment of signaling molecules (discussed in Chapter 15) to the cell junctions. Focal adhesions can be very stable interactions involved in tissue structure, or they can turn over rapidly as cells move (see Figure 12.34). During cell migration, the formation of new focal complexes at the leading edge of the cell results in the loss of tension at the old focal adhesions, leading to the inactivation of integrin binding to the extracellular matrix.

Recent progress in analyzing the structure of integrins has permitted an understanding of the regulated changes in integrin activity that underlie the rapid assembly and disassembly of focal adhesions during cell movement. The ability of integrins to reversibly bind matrix components is dependent on their ability to change conformation between active and inactive states (Figure 14.20). In the inactive state integrins are unable to bind to the matrix because the head groups containing the ligand-binding site are held close to the cell surface. Signals from the cytosol, via talin or vinculin binding, change the conformation of the cytosolic and extracellular

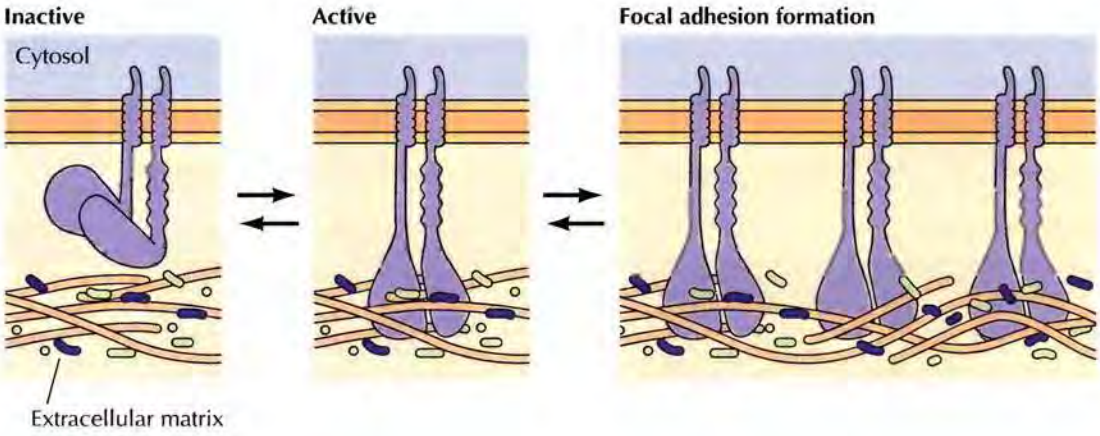


FIGURE 14.20 Integrin activation In the inactive state, integrin head groups are held close to the cell surface. Signals from the cytosol activate integrins, extending the head groups and enabling them to bind to the extracellular matrix. This leads to recruitment of additional integrins and formation of a focal adhesion. In migrating cells, the reverse set of conformational changes can dissociate integrins from the matrix.

domains of the integrins, extending the head groups into the matrix and allowing binding of the ligand. This in turn conveys a signal to the cytosol that allows the cell to respond to integrin binding and recruits additional integrins to the site of adhesion, leading to focal adhesion formation. Subsequent signaling from inside the cell can cause the reverse set of conformational changes and dissociate the integrin from its ligand, leading to disassembly of focal adhesions.

KEY EXPERIMENT

The Characterization of Integrin

Structure of Integrin, a Glycoprotein Involved in the Transmembrane Linkage between Fibronectin and Actin

John W. Tamkun, Douglas W. DeSimone, Deborah Fonda, Ramila S. Patel, Clayton Buck, Alan F. Horwitz and Richard O. Hynes

Massachusetts Institute of Technology, Cambridge, Massachusetts (JWT, DWD, DF, RSP and ROH), The Wistar Institute, Philadelphia, Pennsylvania (CB) and The University of Pennsylvania School of Medicine (AFH)

Cell, Volume 46, 1986, pages 271–282

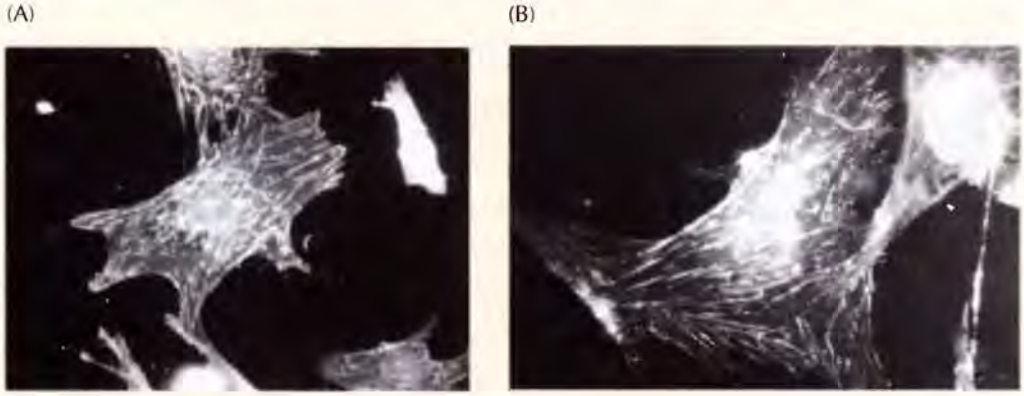
Richard O. Hynes

The Context

The molecular basis of cell adhesion to the extracellular matrix has been of interest to cell biologists since it was observed that adhesion to the matrix was reduced in cancer cells, potentially related to their abnormal growth and ability to metastasize or spread throughout the body. In the late 1970s and early 1980s work from several laboratories, including that of Richard Hynes, established that there was a physical link between the stress fibers of the actin cytoskeleton and fibronectin in the extracellular matrix. Several of the cytoskeletal proteins involved in this linkage, including vinculin and talin, had been identified, but the critical transmembrane proteins that linked cells to the extracellular matrix remained unknown.

Candidates for transmembrane proteins involved in cell adhesion to the extracellular matrix had been identified using antibodies prepared by several groups of scientists, including Alan Horwitz and Clayton Buck. These antibodies recognized a complex of 140 kd glycoproteins that appeared to be transmembrane proteins. Immunofluorescence and immunoelectron microscopy localized these glycoproteins to points of cell adhesion to the matrix. Additional studies showed that the 140 kd glycoproteins bound to fibronectin and were involved in cell adhesion. It thus appeared that these 140 kd glycoproteins were likely candidates for the transmembrane proteins responsible for cell-matrix adhesion. In the experiments described by Tamkun et al. these antibodies were used to isolate a molecular clone encoding one of these glycoproteins, thereby providing the first molecular characterization of integrin.

KEY EXPERIMENT



Immunofluorescence of chick embryo fibroblasts stained with the original antiserum to the 140 kd glycoprotein complex (A) or with antibodies purified against the protein expressed from the cDNA clones (B).

The Experiments

To isolate a molecular clone encoding one of the 140 kd glycoproteins Tamkun et al. prepared a cDNA library from mRNA of chick embryo fibroblasts. This cDNA library was prepared in a bacteriophage λ expression vector that directed high level transcription and translation of the eukaryotic cDNA inserts in *E. coli* (see Figure 4.21). The cDNA library contained approximately 100,000 independent cDNA inserts, sufficient to represent clones of all the mRNAs expressed in the chick embryo fibroblasts from which it was prepared. The library was then screened with antibodies against the 140 kd glycoproteins to identify those recombinant phages carrying the cDNAs of interest. Plaques produced by individual recombinant phages were transferred to nitrocellulose filters as would be done to screen a recombinant library by nucleic acid hybridization (see Figure 4.26). However, to screen the expression library, the filters were probed with an antibody to identify clones expressing the desired protein. By such antibody screening, Tamkun et al. successfully identified several recombinant clones expressing proteins that were recognized by antibodies against the 140 kd glycoproteins.

Their next challenge was to determine whether the cloned cDNAs actually encoded one or more of the 140 kd glycoproteins. To do this they used proteins expressed from the clones to purify antibodies that recognized the cloned proteins. In addition, they injected rabbits with the cloned proteins to raise new antibodies specifically directed against the proteins encoded by the cloned cDNAs. The resulting antibodies recognized one of the proteins of the 140 kd complex from chick embryo fibroblasts in immunoblot assays, establishing the relationship of the cloned cDNAs to this protein. In addition, antibodies purified against the cloned proteins stained cells similarly to the original antibody in immunofluorescence assays, yielding a staining pattern corresponding to the sites of stress fiber attachment to the extracellular matrix (see figure). The results of both immunoblotting and immunofluorescence thus indicated that the cloned cDNAs encoded one of the proteins of the 140 kd glycoprotein complex.

The cDNA was then sequenced and found to encode a protein consisting of 803 amino acids. The protein included an amino terminal signal sequence and an apparent transmembrane α helix of 23 hydrophobic amino acids near the carboxy terminus. The sequence predicted a short cytosolic domain and a large extracellular domain containing multiple glycosylation sites consistent with the expected structure of a transmembrane glycoprotein.

The Impact

Hynes and his colleagues concluded that they had cloned a cDNA encoding a transmembrane protein that functioned to link fibronectin in the extracellular matrix to the cytoskeleton. They named the protein integrin "to denote its role as an integral membrane complex involved in the transmembrane association between the extracellular matrix and the cytoskeleton."

The initial cloning of integrin led to our current understanding of the molecular basis of stable cell junctions. At focal adhesions, integrins link the extracellular matrix to actin filaments. In addition, integrins mediate attachment of epithelial cells to the extracellular matrix at hemidesmosomes where they link the extracellular matrix to intermediate filaments. Thus, as suggested by Hynes and his colleagues, integrins play a general role in cell-matrix adhesion. Subsequent studies have shown that integrins also play an important role as signaling complexes in cells, relaying signals from outside of the cell to control multiple aspects of cell movement, proliferation, and survival (discussed in Chapter 15). The characterization of integrins thus opened the door to understanding not only the nature of cell attachment to the matrix but also to elucidating novel signaling mechanisms that regulate cell behavior.

Cell-Cell Interactions

Direct interactions between cells, as well as between cells and the extracellular matrix, are critical to the development and function of multicellular organisms. Some cell-cell interactions are transient, such as the interactions between cells of the immune system and the interactions that direct white blood cells to sites of tissue inflammation. In other cases, stable cell-cell junctions play a key role in the organization of cells in tissues. For example, several different types of stable cell-cell junctions are critical to the maintenance and function of epithelial cell sheets. In addition to mediating cell adhesion, specialized types of junctions provide mechanisms for rapid communication between cells. Plant cells also associate with their neighbors not only by interactions between their cell walls but also by specialized junctions between their plasma membranes.

Adhesion Junctions

Cell-cell adhesion is a selective process such that cells adhere only to other cells of specific types. This selectivity was first demonstrated in classical studies of embryo development, which showed that cells from one tissue (e.g., liver) specifically adhere to cells of the same tissue rather than to cells of a different tissue (e.g., brain). Such selective cell-cell adhesion is mediated by transmembrane proteins called **cell adhesion molecules**, which can be divided into four major groups: the **selectins**, the **integrins**, the **immunoglobulin (Ig) superfamily** (so named because they contain structural domains similar to immunoglobulins), and the **cadherins** (Table 14.2). Cell adhesion mediated by the selectins, integrins, and most cadherins requires Ca^{2+}, Mg^{2+} or Mn^{2+}, so many adhesive interactions between cells are divalent cation-dependent.

The selectins mediate transient interactions between leukocytes and endothelial cells or blood platelets. There are three members of the selectin family: L-selectin, which is expressed on leukocytes; E-selectin, which is expressed on endothelial cells; and P-selectin, which is expressed on platelets. As discussed in Chapter 13, the selectins recognize cell surface carbohydrates (see Figure 13.14). One of their critical roles is to initiate the interactions between leukocytes and endothelial cells during the migration of leukocytes from the circulation to sites of tissue inflammation (Figure 14.21). The selectins mediate the initial adhesion of leukocytes to endothelial cells. This is followed by the formation of more stable adhesions in which integrins on the surface of leukocytes bind to intercellular adhesion molecules (ICAMs), which are members of the Ig superfamily expressed on the surface of endothelial cells. The firmly attached leukocytes are then able to

TABLE 14.2 Cell Adhesion Molecules

Family	Ligands recognized	Stable cell junctions
Selectins	Carbohydrates	No
Integrins	Extracellular matrix	Focal adhesions and hemidesmosomes
	Members of Ig superfamily	No
Ig superfamily	Integrins	No
	Other Ig superfamily proteins	No
Cadherins	Other cadherins	Adherens junctions and desmosomes

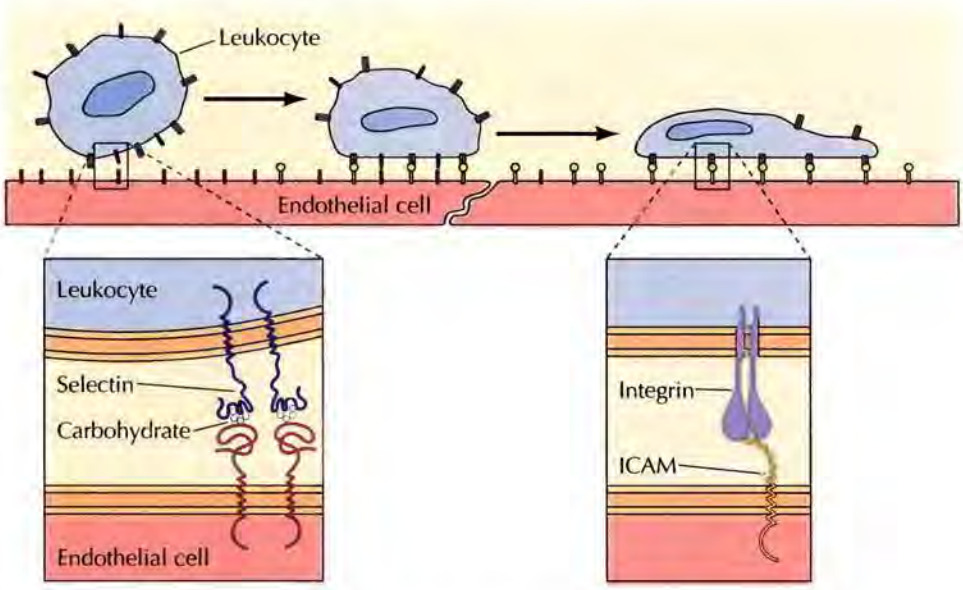


FIGURE 14.21 Adhesion between leukocytes and endothelial cells
Leukocytes leave the circulation at sites of tissue inflammation by interacting with the endothelial cells of capillary walls. The first step in this interaction is the binding of leukocyte selectins to carbohydrate ligands on the endothelial cell surface. This step is followed by more stable interactions between leukocyte integrins and members of the Ig superfamily (ICAMs) on endothelial cells.

penetrate the walls of capillaries and enter the underlying tissue by migrating between endothelial cells.

The binding of ICAMs to integrins is an example of a **heterophilic interaction** in which an adhesion molecule on the surface of one cell (e.g., an ICAM) recognizes a different molecule on the surface of another cell (e.g., an integrin). Other members of the Ig superfamily mediate **homophilic interactions** in which an adhesion molecule on the surface of one cell binds to the same molecule on the surface of another cell. Such homophilic binding can lead to selective adhesion between cells of the same type. For example, nerve cell adhesion molecules (N-CAMs) are members of the Ig superfamily expressed on nerve cells, and homophilic binding between N-CAMs contributes to the formation of selective associations between nerve cells during development. There are more than 100 members of the Ig superfamily, which mediate a variety of cell-cell interactions.

The fourth class of cell adhesion molecules is the cadherins. Cadherins are involved in selective adhesion between embryonic cells, formation of specific synapses in the nervous system, and are the proteins primarily responsible for the maintenance of stable junctions between cells in tissues. Cadherins are a large family of proteins (about 80 members) that share a highly conserved extracellular domain that mediates largely homophilic interactions. For example, E-cadherin is expressed on epithelial cells so homophilic interactions between E-cadherins lead to the selective adhesion of epithelial cells to one another. It is noteworthy that loss of E-cadherin can contribute to the development of cancers arising from epithelial tissues, illustrating the importance of cell-cell interactions in controlling cell behavior. Different members of the cadherin family, such as N-cadherin (neural

cadherin) and P-cadherin (placental cadherin), mediate selective adhesion of other cell types. There are several subfamilies of cadherins (classic cadherins, desmosomal cadherins, fat-like cadherins, and seven transmembrane domain cadherins) which differ primarily in their transmembrane and cytosolic domains.

The cell-cell interactions mediated by the selectins, integrins, and most members of the Ig superfamily are transient adhesions in which the cytoskeletons of adjacent cells are not linked to one another. Stable adhesion junctions involving the cytoskeletons of adjacent cells instead are based largely on cadherins. As discussed in Chapter 12, these cell-cell junctions are of two types: **adherens junctions** and **desmosomes.** At these junctions, classic and desmosomal cadherins are linked to actin bundles and intermediate filaments, respectively, by the interaction of their cytosolic tails with β-catenin or desmoplakin. The role of the cadherins in linking the cytoskeletons of adjacent cells is thus analogous to that of the integrins in forming stable junctions between cells and the extracellular matrix.

The basic structural unit of an adherens junction includes β-catenin and α-catenin in addition to the classic transmembrane cadherins (Figure 14.22). β-catenin binds directly to the cytosolic tail of the cadherins. α-catenin binds to β-catenin as well as to actin filaments and actin filament-binding proteins, such as vinculin. Thus α- and β-catenin directly link the actin cytoskeleton of one cell, through the transmembrane cadherins, to the actin cytoskeleton of an adjacent cell. Associated with the cadherins is the β-catenin related protein p120, which is not part of the structural link but is critical to maintain the stability of the adherens junction. β-catenin and p120

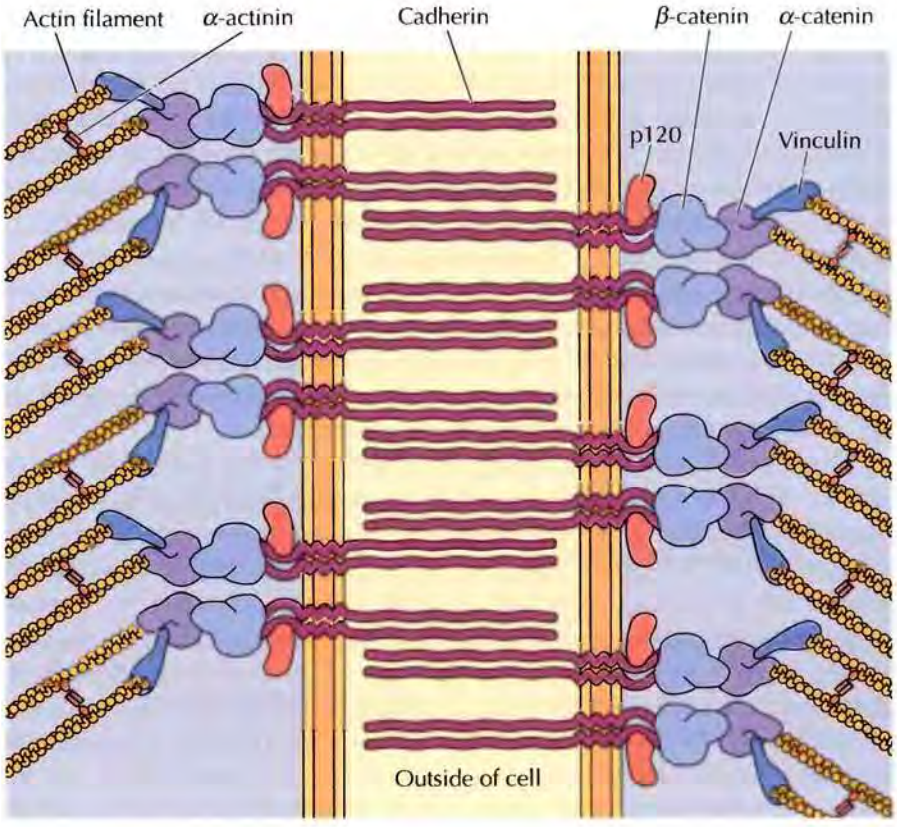


FIGURE 14.22 Adherens junctions In adherens junctions, cadherins link to actin filaments. β-catenin (an armadillo family protein) binds to the cytosolic tails of the cadherins. β-catenin also binds α-catenin, which binds both actin filaments and vinculin. This forms a direct link between the transmembrane cadherins and the actin cytoskeleton. A second armadillo family protein, p120, also binds the cytosolic tails of cadherins and regulates the stability of the junction.

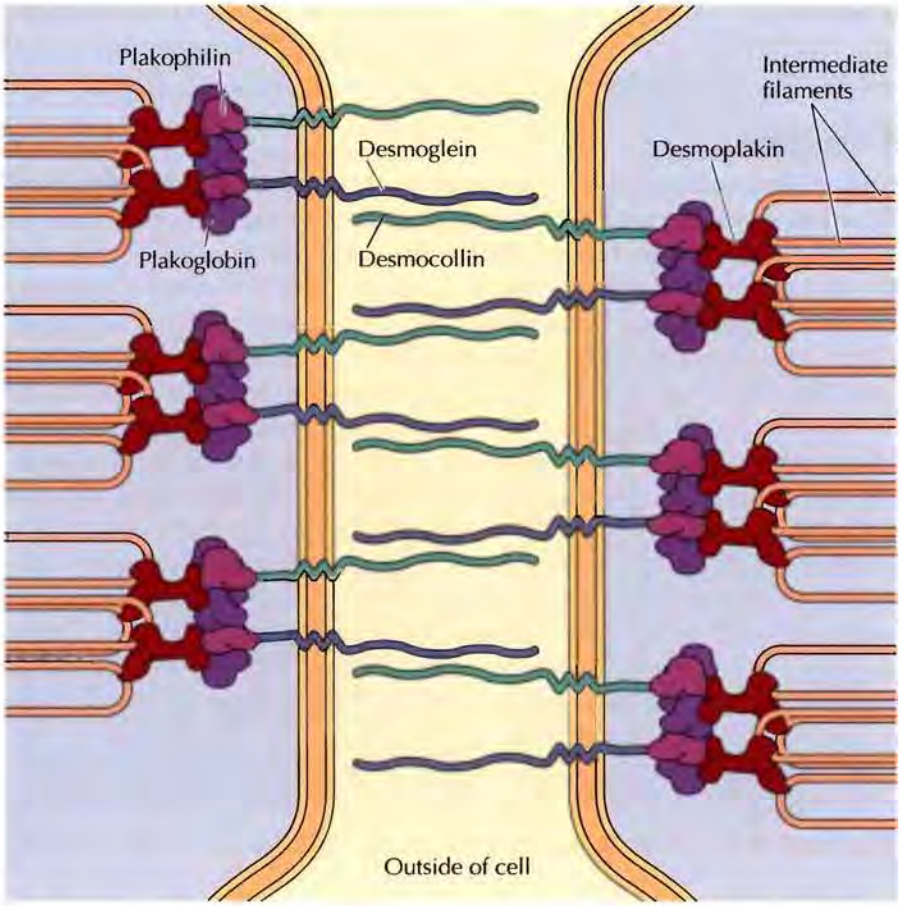


FIGURE 14.23 Desmosomes
In desmosomes, the desmosomal cadherins (desmoglein and desmocollin) link to intermediate filaments. The armadillo family proteins plakoglobin and plakophilin bind to the cytosolic tails of the cadherins. They also bind desmoplakin, which binds to intermediate filaments.

are members of the **armadillo protein family,** named after the *Drosophila* β-catenin protein. Despite its name, α-catenin is not part of the armadillo family but is more closely related to vinculin.

In addition to the cadherins, a second type of cell adhesion molecule, **nectin,** is also present at adherens junctions. There are four known nectins and a related family of nectin-like proteins, which are members of the Ig superfamily. Like cadherins, nectins interact in a homophilic manner, but nectins can also bind heterophilically to different nectins in the membrane of the apposing cell. Nectins are also involved in links to the actin cytoskeleton, but their major role appears to be in junction formation. In particular, it appears that cell adhesions are initially mediated by interactions between nectins, which then recruit cadherins to the adhesion sites. Cadherins then recruit actin-binding proteins, such as vinculin, to begin assembly of a mature adherens junction.

Desmosomes directly link the intermediate filament cytoskeletons of adjacent cells (Figure 14.23). The transmembrane cadherins, **desmoglein** and **desmocollin,** bind by heterophilic interactions across the junction. The armadillo family proteins plakoglobin and plakophilin bind to the cytosolic tails of the cadherins and provide a direct link to the intermediate filament-binding protein, desmoplakin. Desmoplakin is a member of the plakin family and is related to plectin, which functions analogously at hemidesmosomes (see Figure 14.19). The strength of desmosomal links between cells is

a property of both the intermediate filaments to which they are attached and the multiple interactions between plakoglobin, plakophilin, and desmoplakin that link the intermediate filaments to the cadherins.

Tight Junctions

Tight junctions are critically important to the function of epithelial cell sheets as barriers between fluid compartments. For example, the intestinal epithelium separates the lumen of the intestine from the underlying connective tissue, which contains blood capillaries. Tight junctions play two roles in allowing epithelia to fulfill such barrier functions. First, tight junctions form seals that prevent the free passage of molecules (including ions) between the cells of epithelial sheets. Second, tight junctions separate the apical and basolateral domains of the plasma membrane by preventing the free diffusion of lipids and membrane proteins between them. Consequently, specialized transport systems in the apical and basolateral domains are able to control the traffic of molecules between distinct extracellular compartments, such as the transport of glucose between the intestinal lumen and the blood supply (see Figure 13.32). While tight junctions are very effective seals of the extracellular space, they provide minimal adhesive strength between the apposing cells, so they are usually associated with adherens junctions and desmosomes in a **junctional complex** (Figure 14.24).

■ **The epithelial cell layers constitute significant barriers against invading microorganisms. Pathogenic bacteria have evolved strategies that disrupt junctional complexes, allowing them to penetrate between the cells of epithelial sheets.**

Tight junctions are the closest known contacts between adjacent cells. They were originally described as sites of apparent fusion between the outer leaflets of the plasma membranes, although it is now clear that the

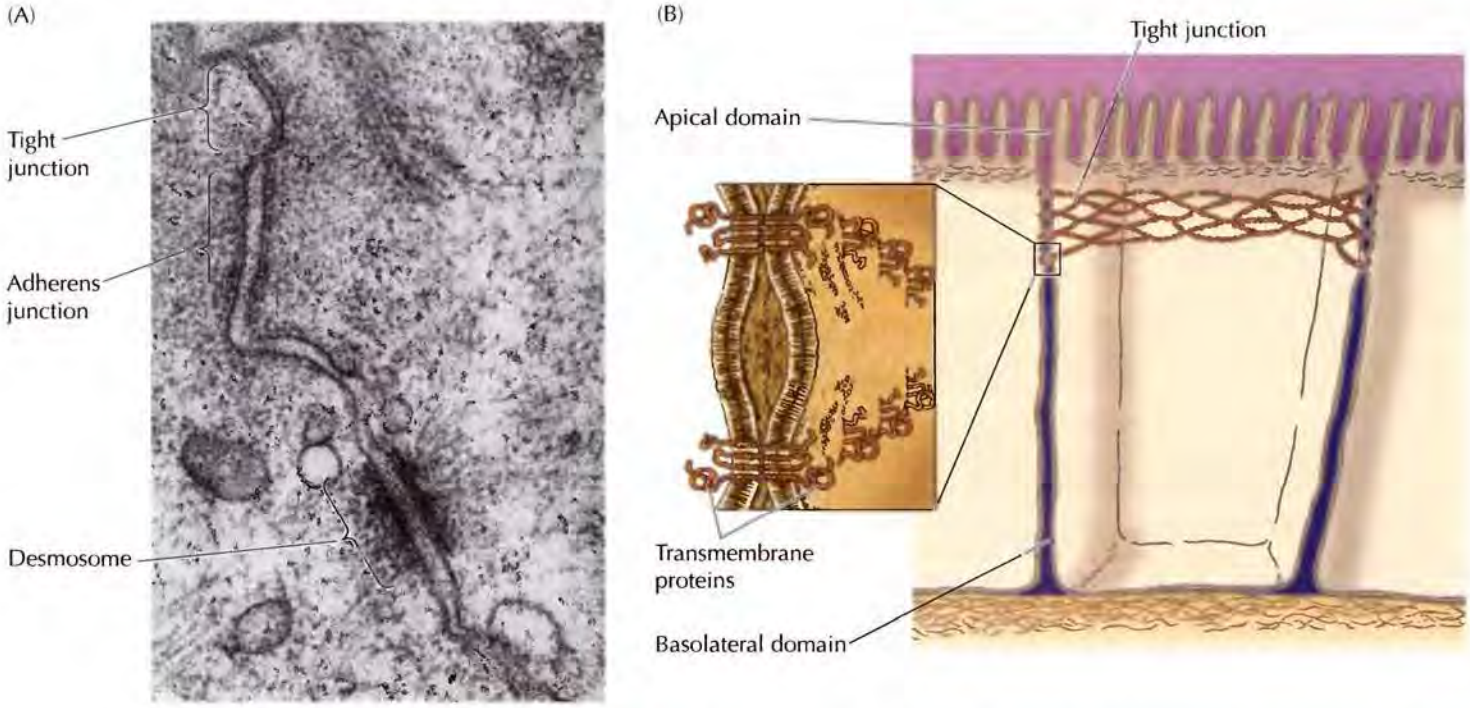


FIGURE 14.24 A junctional complex (A) Electron micrograph of epithelial cells joined by a junctional complex, including a tight junction, an adherens junction, and a desmosome. (B) Tight junctions are formed by interactions between strands of transmembrane proteins on adjacent cells. (A, Don Fawcett/Photo Researchers, Inc.)

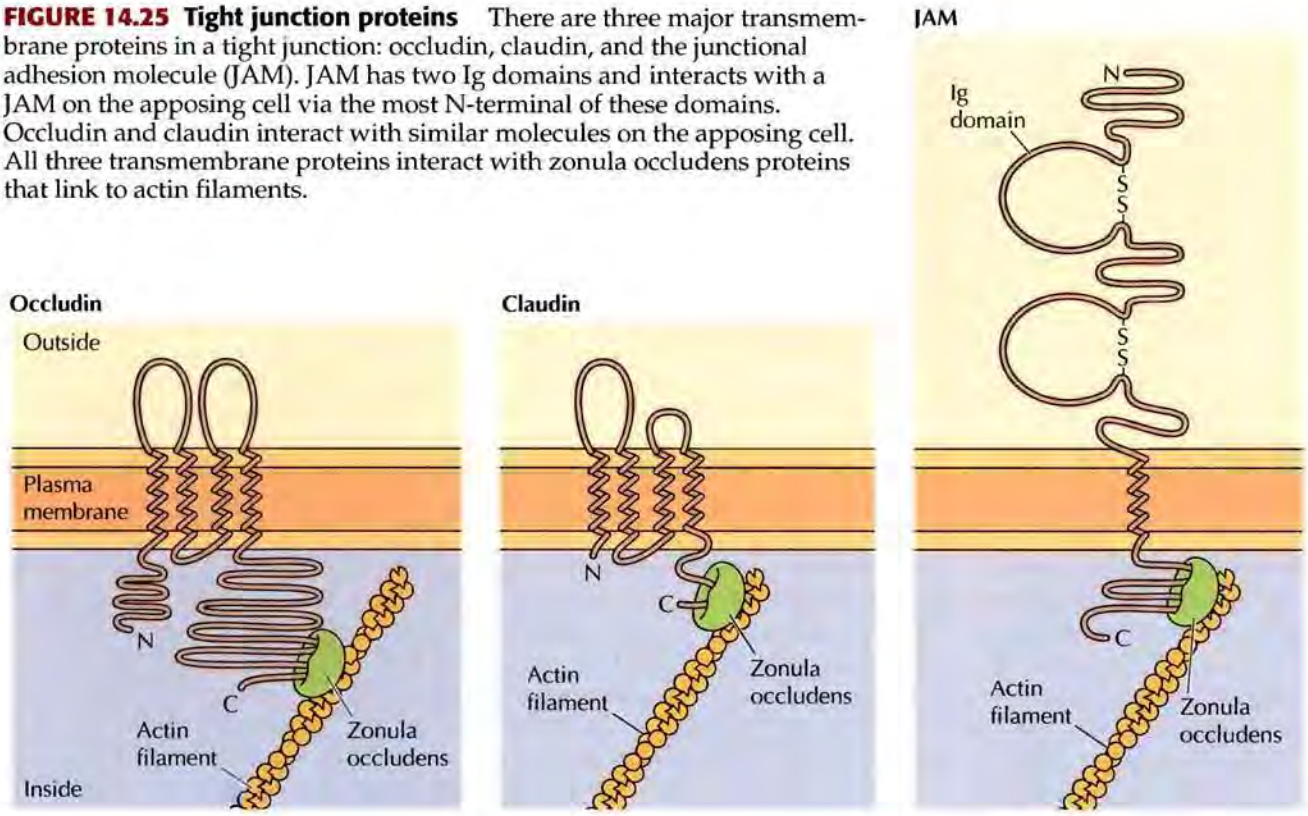


FIGURE 14.25 Tight junction proteins There are three major transmembrane proteins in a tight junction: occludin, claudin, and the junctional adhesion molecule (JAM). JAM has two Ig domains and interacts with a JAM on the apposing cell via the most N-terminal of these domains. Occludin and claudin interact with similar molecules on the apposing cell. All three transmembrane proteins interact with zonula occludens proteins that link to actin filaments.

membranes do not fuse. Instead, tight junctions are formed by a network of protein strands that continues around the entire circumference of the cell (see Figure 14.24B). Each strand in these networks is composed of transmembrane proteins of the claudin, occludin, and junctional adhesion molecule (JAM) families (Figure 14.25). All three of these proteins bind to similar proteins on adjacent cells, thereby sealing the space between their plasma membranes. The cytosolic tails of claudins, occludins, and JAMs are also associated with proteins of the zonula occludens family, which link the tight junction complex to the actin cytoskeleton and hold the tight junction in place on the plasma membrane. Nectins may also be present in tight junctions but their major role seems to be in recruiting claudins to initiate tight junction formation, analogous to their role in formation of adherens junctions.

Gap Junctions

The activities of individual cells in multicellular organisms need to be closely coordinated. This can be accomplished by signaling molecules that are released from one cell and act on another, as discussed in Chapter 15. However, within an individual tissue, such as the liver, cells are often linked by **gap junctions,** which provide direct connections between the cytoplasms of adjacent cells. Gap junctions are open channels through the plasma membrane, allowing ions and small molecules (less than approximately a thousand daltons) to diffuse freely between neighboring cells, but preventing the passage of proteins and nucleic acids. Consequently, gap junctions couple both the metabolic activities and the electric responses of the cells they connect. Most cells in animal tissues—including epithelial cells, endothelial cells, and the cells of cardiac and smooth muscle—

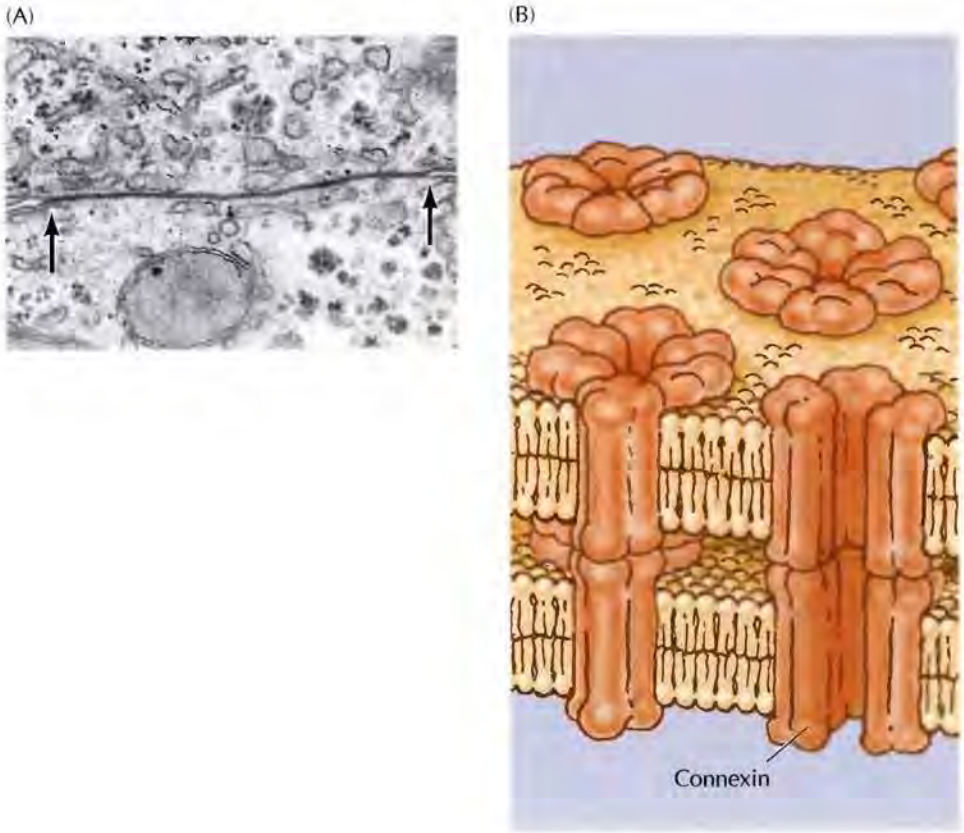


FIGURE 14.26 Gap junctions (A) Electron micrograph of a gap junction (arrows) between two liver cells. (B) Gap junctions consist of assemblies of six connexins, which form open channels through the plasma membranes of adjacent cells. (A, Don Fawcett and R. Wood/Photo Researchers, Inc.)

communicate by gap junctions. In electrically excitable cells, such as heart muscle cells, the direct passage of ions through gap junctions couples and synchronizes the contractions of neighboring cells. Gap junctions also allow the passage of some intracellular signaling molecules, such as cAMP and Ca^{2+}, between adjacent cells, potentially coordinating the responses of cells in tissues.

Gap junctions are constructed of transmembrane proteins of the **connexin** family, which consists of at least 21 different human proteins. Six connexins assemble to form a cylinder with an open aqueous pore in its center (Figure 14.26). Such an assembly of connexins, a **connexon**, in the plasma membrane of one cell then aligns with a connexon of an adjacent cell, forming an open channel between the two cytoplasms. The plasma membranes of the two cells are separated by a gap corresponding to the space occupied by the connexin extracellular domains—hence the term "gap junction," which was coined by electron microscopists. Many cells express more than one member of the connexin family, and combinations of different connexin proteins may give rise to gap junctions with varying properties.

Specialized assemblies of gap junctions occur on specific nerve cells in all eukaryotes and form an **electrical synapse.** The individual connexons within the electrical synapse can be opened or closed in response to several types of signals but, when open, allow the rapid passage of ions between the two nerve cells. The importance of gap junctions—especially in the nervous system—is illustrated by the number of human diseases associated with connexon mutations.

MOLECULAR MEDICINE

Gap Junction Diseases

The Diseases

Several unrelated human diseases have been found to result from mutations in genes that encode the connexin proteins of gap junctions. The first of these diseases to be described is the X-linked form of Charcot-Marie-Tooth disease (CMT), which was mapped to mutations in the gene encoding connexin 32 in 1993. CMT is an inherited disease leading to progressive degeneration of peripheral nerves, with slow loss of muscle control and eventual muscle degeneration. CMT can also be caused by mutations in several different genes encoding myelin proteins, defects which lead directly to degeneration of myelinated nerves. In addition, one form of CMT has been mapped to a mutation in the human gene for nuclear lamin A (see Chapter 9 Molecular Medicine, Nuclear Lamina Diseases).

The finding that CMT could be caused by mutations in a connexin implied that gap junctions played a critical role in myelinated nerves. Moreover, CMT was only the first of several inherited diseases that now have been traced back to mutations in genes encoding human connexins. The most common pathological consequences of connexin mutations are cataracts, skin disorders, and deafness.

Gap Junction Mutations in Human Disease

Disease	Connexin protein
Charcot-Marie-Tooth disease	Cx32/β1
Deafness	Cx26/β2, Cx30/β6, Cx31/β3, Cx32/β1, Cx43/α1
Skin diseases	
Erythrokeratoderma variabilis	Cx31/β3, Cx30.3/β4
Vohwinkel's syndrome	Cx26/β2
Cataracts	Cx46/α3, Cx50/α8

Cx proteins are named either by molecular weight or as a member of one subfamily.

Molecular and Cellular Basis

The human genome contains 21 genes encoding different connexins (Cx genes), which are divided into three subfamilies (α, β, and γ). Mutations in 8 of these genes have been identified as causes of human disease (see table). For example, mutations in *Cx32/β1* are associated with both Charcot-Marie-Tooth disease and deafness. The connexin proteins are widely expressed; Cx32/β1 is expressed in peripheral nerve, liver, and brain tissues, and Cx43α1 (which is also associated with deafness) is expressed in over 35 different human tissues. Thus it is somewhat surprising that there are fewer than ten different types of human diseases known to be caused by connexin mutations. Since most tissues express several different connexin proteins, the simple explanation for the lack of a defect in many tissues resulting from the loss of any single connexin is that other connexins compensate for the one that is missing. The question then is why any mutation in a single connexin gene causes a human disease.

The initial discovery of *Cx32/β1* mutations in CMT was based on eight different mutations in various families: six single base changes that resulted in nonconservative amino acid substitutions, one frameshift that yielded a shorter protein, and one mutation in the promoter region of the gene. These eight mutations in *Cx32/β1*, as well as 262 others that have been subsequently discovered, cause clinically identical CMT, suggesting that myelinated nerve might be a tissue particularly sensitive to gap junction defects. Three other tissues also appear sensitive to connexin mutations: the lens of the eye, the sensory epithelium of the inner ear, and the skin. The sensitivity of the lens is easiest to understand, because the lens fiber cells lose most of their organelles during development and fill up with crystalline protein to allow the passage of light. They must obtain nutrients and ions from the lens epithelial cells through gap junctions, and loss of this input leads to cataract formation. The sensitivity of the sensory epithelium of the inner ear is related to the need for the epithelial cells to rapidly exchange K^+ via gap junctional communication. Less clear is the basis for overgrowth of the outer layers of the skin, but it is thought that gap junctions play a role in the balance between proliferation and differentiation, and gap junctional mutations perturb the balance.

Each of these tissues expresses additional connexins, and their failure to compensate for loss of the mutant connexin appears to be based on two phenomena. Although gap junctions can be formed from combinations of different connexins, not all connexins are able to cooperate to form functional connexons. Thus the connexins in sensitive tissues may not be able to compensate effectively for one that is mutated. In addition, gap junction connexons are assembled not at the cell surface but in the Golgi apparatus or earlier in the secretory process. Thus a single mutant connexin that cannot be properly processed and exported might act as a dominant negative and interfere with the processing of the normal connexins expressed in that tissue. This has recently been reported for a mutant of Cx46/α3 that causes congenital cataracts: the mutant

(Continued on next page)

MOLECULAR MEDICINE

protein fails to exit the ERGIC or Golgi complex and traps normal connexins along with it. It thus appears that the relationship of particular disorders to connexin mutations may result from the gap junctional requirements of a given tissue, as well as the nature of interactions between the connexins that are expressed in that tissue.

Prevention and Treatment

The identification of several human diseases caused by mutations in connexin genes illustrates the importance of gap junctions to normal tissue function. Like other junctions critical to tissue structure or function, gap junctions would seem to be obvious locations for hereditary human diseases, yet such diseases are uncommon. Recent discoveries on the processing and assembly of gap junction connexons have started to explain the bases of these rare diseases. While it is thought that all the human connexin genes have been described, the tissue distribution of the proteins—especially during embryonic development—remains poorly understood. New knowledge about the distribution and interactions of the connexin proteins will provide a basis for understanding each of the pathologies resulting from mutations in connexin genes. Only then can possible compensatory therapies—for example, regulating cell proliferation in the skin or modulating the exchange of K^+ in the inner ear—be developed.

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Plasmodesmata

Adhesion between plant cells is mediated by their cell walls rather than by transmembrane proteins. In particular, a specialized pectin-rich region of the cell wall called the **middle lamella** acts as a glue to hold adjacent cells together. Because of the rigidity of plant cell walls, stable associations between plant cells do not require the formation of cytoskeletal links such as those provided by the desmosomes and adherens junctions of animal cells. However, adjacent plant cells communicate with each other through cytoplasmic connections called **plasmodesmata** (singular, plasmodesma) (Figure 14.27). Although distinct in structure, plasmodesmata function analogously to gap junctions as a means of direct communication between adjacent cells in tissues.

Plasmodesmata form from incomplete separation of daughter cells following plant cell mitosis. At each plasmodesma, the plasma membrane of one cell is continuous with that of its neighbor, creating a channel between the two cytosols (Figure 14.28). An extension of the smooth endoplasmic reticulum passes through the pore, leaving a ring of surrounding cytoplasm through which ions and small molecules are able to pass freely between the cells. Plasmodesmata are dynamic structures that can open or close in response to appropriate stimuli, permitting the regulated passage of macromolecules between adjacent cells. In addition, there is evidence that proteins and lipids can be targeted to plasmodesmata in response to specific signals. Plasmodesmata may thus play a key role in plant development by controlling the trafficking of regulatory molecules, such as transcription factors or RNAs, between cells.

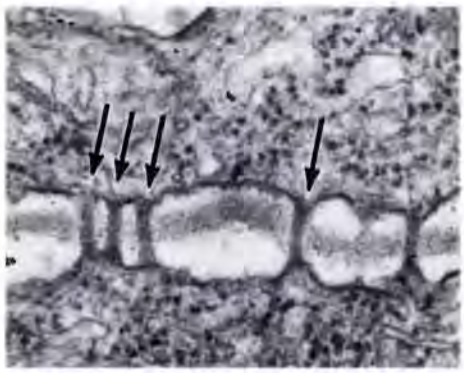

FIGURE 14.27 Plasmodesmata Electron micrograph of plasmodesmata (arrows). (E. H. Newcomb, University of Wisconsin/Biological Photo Service.)

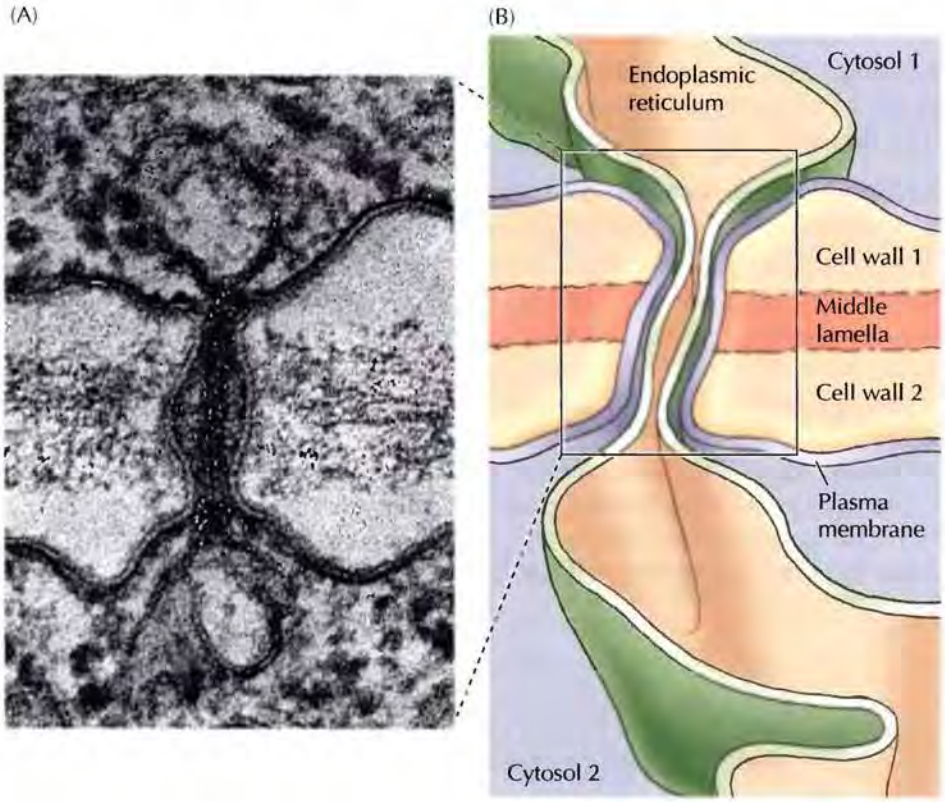

FIGURE 14.28 Structure of a plasmodesma In a plasmodesma, the plasma membranes of neighboring cells are continuous, forming cytoplasmic channels through the adjacent cell walls. An extension of the endoplasmic reticulum usually passes through the channel. (A, From Tilney, L., T. J. Cooke, P. S. Connelly and M. S. Tilney. 1991. *J. Cell Biol.* 112: 739–748.)

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

SUMMARY

CELL WALLS

Bacterial Cell Walls: The principal component of bacterial cell walls is a peptidoglycan consisting of polysaccharide chains cross-linked by short peptides.

Eukaryotic Cell Walls: The cell walls of fungi, algae, and higher plants are composed of fibrous polysaccharides (e.g., cellulose) embedded in a gel-like matrix of polysaccharides and proteins. Their rigid cell walls allow plant cells to expand rapidly by the uptake of water.

THE EXTRACELLULAR MATRIX AND CELL-MATRIX INTERACTIONS

Matrix Structural Proteins: The major structural proteins of the extracellular matrix are members of the large collagen protein family. Collagens form the fibrils that characterize the extracellular matrix of connective tissues, as well as forming networks in basal laminae.

Matrix Polysaccharides: Polysaccharides in the form of glycosaminoglycans and proteoglycans make up the bulk of the extracellular matrix. They bind to and modify the collagen fibrils and interact with all other matrix molecules.

KEY TERMS

peptidoglycan

chitin, cellulose, hemicellulose, pectin, primary cell wall, secondary cell wall, lignin, turgor pressure, auxin, cellulose synthase

extracellular matrix, basal lamina, collagen, collagen fibril, procollagen, elastic fiber, elastin

glycosaminoglycan (GAG), proteoglycan

KEY TERMS

fibronectin, laminin, entactin

integrin, focal adhesion, hemidesmosome, plakin, focal complex

cell adhesion molecule, selectin, integrin, immunoglobulin (Ig) superfamily, cadherin, heterophilic interaction, homophilic interaction, armadillo protein family, nectin, desmoglein, desmocollin

tight junction, junctional complex

gap junction, connexin, connexon, electrical synapse

middle lamella, plasmodesma

SUMMARY

Matrix Adhesion Proteins: Matrix adhesion proteins link the components of the extracellular matrix to one another and are the major binding sites for integrins, which mediate most cell-matrix adhesions.

Cell-Matrix Interactions: Integrins are the major cell surface receptors that attach cells to the extracellular matrix. At focal adhesions and hemidesmosomes, integrins provide stable links between the extracellular matrix and the actin and intermediate filament cytoskeletons, respectively.

CELL-CELL INTERACTIONS

Adhesion Junctions: Selective cell-cell interactions are mediated by four major groups of cell adhesion proteins: selectins, integrins, immunoglobulin (Ig) superfamily members, and cadherins. The cadherins link the cytoskeletons of adjacent cells at stable cell-cell junctions.

Tight Junctions: Tight junctions prevent the free passage of molecules between epithelial cells and separate the apical and basolateral domains of epithelial cell plasma membranes.

Gap Junctions: Gap junctions are open channels connecting the cytosols of adjacent animal cells. Electrical synapses are gap junctions that mediate signaling between cells of the nervous system.

Plasmodesmata: Adjacent plant cells are linked by cytoplasmic connections called plasmodesmata.

Questions

1. How do the cell walls and adjacent membranes differ between Gram-positive and Gram-negative bacteria?

2. An important function of the Na^+-K^+ pump in animal cells is the maintenance of osmotic equilibrium. Why is this unnecessary for plant cells?

3. How do hemicelluloses impart structural strength to plant cell walls?

4. What is the importance of selectively targeting different glucose transporters to the apical and basolateral domains of the plasma membrane of intestinal epithelial cells? What is the role of tight junctions in this process?

5. How would an inhibitor of the enzyme lysyl hydroxylase (the enzyme responsible for hydroxylating lysine residues in collagen) affect the stability of collagen synthesized by a cell?

6. Why do fibril-forming collagens not assemble into triple helices inside cells?

7. What property of GAGs allows them to form hydrated gels?

8. You are studying a transporter found only in the apical plasma membrane of epithelial cells. You treat the epithelial cells with a synthetic peptide that is similar to the extracellular domain of a claudin family protein. The transporter is now found in both the apical and basolateral domains of the plasma membrane. What is the most likely mechanism by which the peptide disturbs the localization of the transporter?

9. You have overexpressed a dominant negative E-cadherin lacking an extracellular domain in epithelial cells. How will this mutant affect cell-cell adhesion?

10. A mutation in an epithelial cell leads to the expression of $\alpha_6\beta_4$ integrin with a deleted cytoplasmic domain. How will this mutation affect adhesion of the epithelial cell to the basal lamina?

11. What is an electrical synapse? What function does it serve?

12. How are gap junctions and plasmodesmata similar? Are they likely to be analogous or homologous structures in animals and plants?

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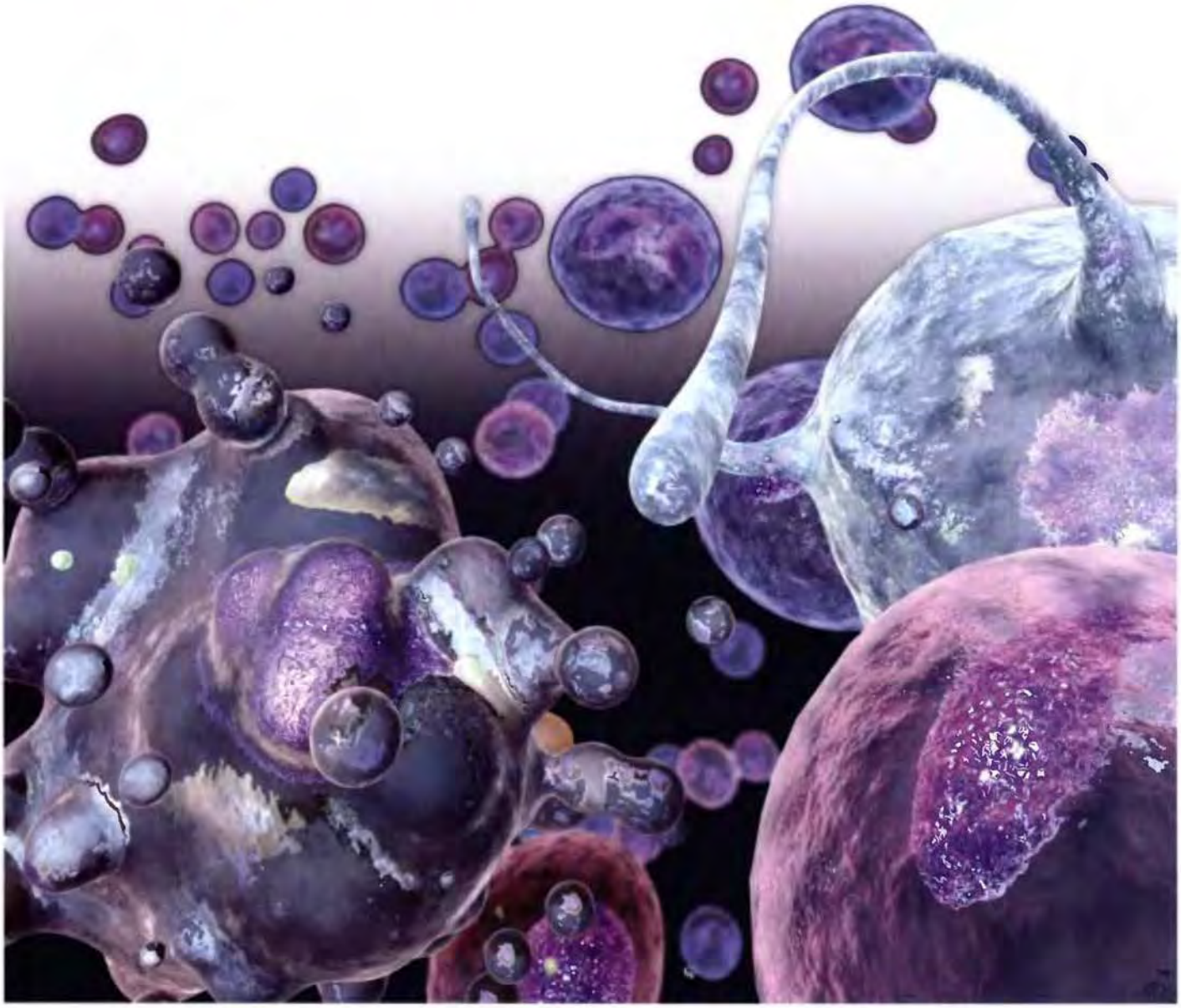
PART IV

Cell Regulation



CHAPTER 15 ■ Cell Signaling

CHAPTER 16 ■ The Cell Cycle

CHAPTER 17 ■ Cell Death and Cell Renewal

CHAPTER 18 ■ Cancer

CHAPTER 15

Cell Signaling

- ***Signaling Molecules and Their Receptors 600***
- ***Functions of Cell Surface Receptors 609***
- ***Pathways of Intracellular Signal Transduction 617***
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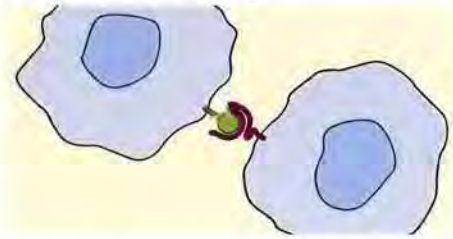
ALL CELLS RECEIVE AND RESPOND TO SIGNALS from their environment. Even the simplest bacteria sense and swim toward high concentrations of nutrients, such as glucose or amino acids. Many unicellular eukaryotes also respond to signaling molecules secreted by other cells, allowing cell-cell communication. Mating between yeast cells, for example, is signaled by peptides that are secreted by one cell and bind to receptors on the surface of another. It is in multicellular organisms, however, that cell-cell communication reaches its highest level of sophistication. Whereas the cells of prokaryotes and unicellular eukaryotes are largely autonomous, the behavior of each individual cell in multicellular plants and animals must be carefully regulated to meet the needs of the organism as a whole. This is accomplished by a variety of signaling molecules that are secreted or expressed on the surface of one cell and bind to receptors expressed by other cells, thereby integrating and coordinating the functions of the many individual cells that make up organisms as complex as human beings.

The binding of most signaling molecules to their receptors initiates a series of intracellular reactions that regulate virtually all aspects of cell behavior, including metabolism, movement, proliferation, survival, and differentiation. Understanding the molecular components of these pathways and how they are regulated has thus become a major area of research in contemporary cell biology. Interest in this area is further heightened by the fact that many cancers arise as a result of a breakdown in the signaling pathways that control normal cell proliferation and survival. In fact, many of our current insights into cell signaling mechanisms have come from the study of cancer cells—a striking example of the fruitful interplay between medicine and basic research in cell and molecular biology.

15.1 WEBSITE ANIMATION

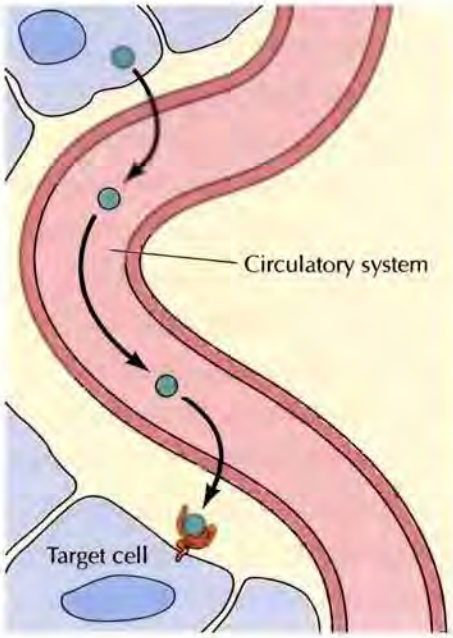
Signaling by Secreted Molecules In endocrine signaling, hormones act on distant target cells; in paracrine signaling, a molecule released from a cell acts on nearby targets; and in autocrine signaling, a cell produces a signaling molecule to which it also responds.

Direct Cell-Cell Signaling

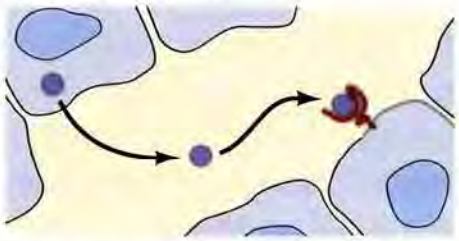


Signaling by Secreted Molecules

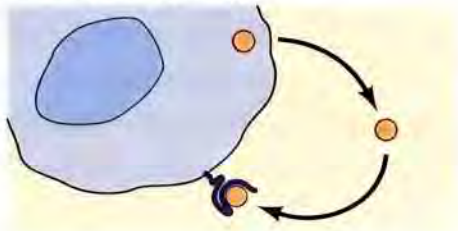
(A) Endocrine signaling



(B) Paracrine signaling



(C) Autocrine signaling



Signaling Molecules and Their Receptors

Many different kinds of molecules transmit information between the cells of multicellular organisms. Although all these molecules act as ligands that bind to receptors expressed by their target cells, there is considerable variation in the structure and function of the different types of molecules that serve as signal transmitters. Structurally, the signaling molecules used by plants and animals range in complexity from simple gases to proteins. Some of these molecules carry signals over long distances, whereas others act locally to convey information between neighboring cells. In addition, signaling molecules differ in their mode of action on their target cells. Some signaling molecules are able to cross the plasma membrane and bind to intracellular receptors in the cytoplasm or nucleus, whereas most bind to receptors expressed on the target cell surface. The sections that follow discuss the major types of signaling molecules and the receptors with which they interact. Subsequent discussion in this chapter focuses on the mechanisms by which cell surface receptors then function to regulate cell behavior.

Modes of Cell-Cell Signaling

Cell signaling can result either from the direct interaction of a cell with its neighbor or from the action of secreted signaling molecules (Figure 15.1). Signaling by direct cell-cell (or cell-matrix) interactions plays a critical role in regulating the behavior of cells in animal tissues. For example, the integrins and cadherins (which were discussed in the previous chapter) function not only as cell adhesion molecules but also as signaling molecules that regulate cell proliferation and survival in response to cell-cell and cell-matrix contacts. In addition, cells express a variety of cell surface receptors that interact with signaling molecules on the surface of neighboring cells. Signaling via such direct cell-cell interactions plays a critical role in regulating the many interactions between different types of cells that take place during embryonic development as well as in the maintenance of adult tissues.

The multiple varieties of signaling by secreted molecules are frequently divided into three general categories based on the distance over which signals are transmitted. In **endocrine signaling**, the signaling molecules (**hormones**) are secreted by specialized endocrine cells and carried through the circulation to act on target cells at distant body sites. A classic example is provided by the steroid hormone estrogen, which is produced by the ovary and stimulates development and maintenance of the female reproductive system and secondary sex characteristics. In animals, more than 50 different hormones are produced by endocrine glands, including the pituitary, thyroid, parathyroid, pancreas, adrenal glands, and gonads.

In contrast to hormones, some signaling molecules act locally to affect the behavior of nearby cells. In **paracrine signaling**, a molecule released by one cell acts on neighboring target cells. An example is provided by the action of neurotransmitters in carrying signals between nerve cells at a synapse. Finally, some cells respond to signaling molecules that they themselves produce. One important example of such **autocrine signaling** is the

FIGURE 15.1 Modes of cell-cell signaling Cell signaling can take place either through direct cell-cell contacts or through the action of secreted signaling molecules. (A) In endocrine signaling, hormones are carried through the circulatory system to act on distant target cells. (B) In paracrine signaling, a molecule released from one cell acts locally to affect nearby target cells. (C) In autocrine signaling, a cell produces a signaling molecule to which it also responds.

response of cells of the vertebrate immune system to foreign antigens. Certain types of T lymphocytes respond to antigenic stimulation by synthesizing a growth factor that drives their own proliferation, thereby increasing the number of responsive T lymphocytes and amplifying the immune response. It is also noteworthy that abnormal autocrine signaling frequently contributes to the uncontrolled growth of cancer cells (see Chapter 18). In this situation, a cancer cell produces a growth factor to which it also responds, thereby continuously driving its own unregulated proliferation.

Steroid Hormones and the Nuclear Receptor Superfamily

As already noted, all signaling molecules act by binding to receptors expressed by their target cells. In many cases these receptors are expressed on the target cell surface, but some receptors are intracellular proteins located in the cytosol or in the nucleus. These intracellular receptors respond to small hydrophobic signaling molecules that are able to diffuse across the plasma membrane. The **steroid hormones** are the classic examples of this group of signaling molecules, which also includes thyroid hormone, vitamin D_3, and retinoic acid (Figure 15.2).

The steroid hormones (including testosterone, estrogen, progesterone, the corticosteroids, and ecdysone) are all synthesized from cholesterol. **Testosterone, estrogen**, and **progesterone** are the sex steroids, which are produced by the gonads. The **corticosteroids** are produced by the adrenal gland. They include the **glucocorticoids**, which act on a variety of cells to stimulate production of glucose, and the **mineralocorticoids**, which act on

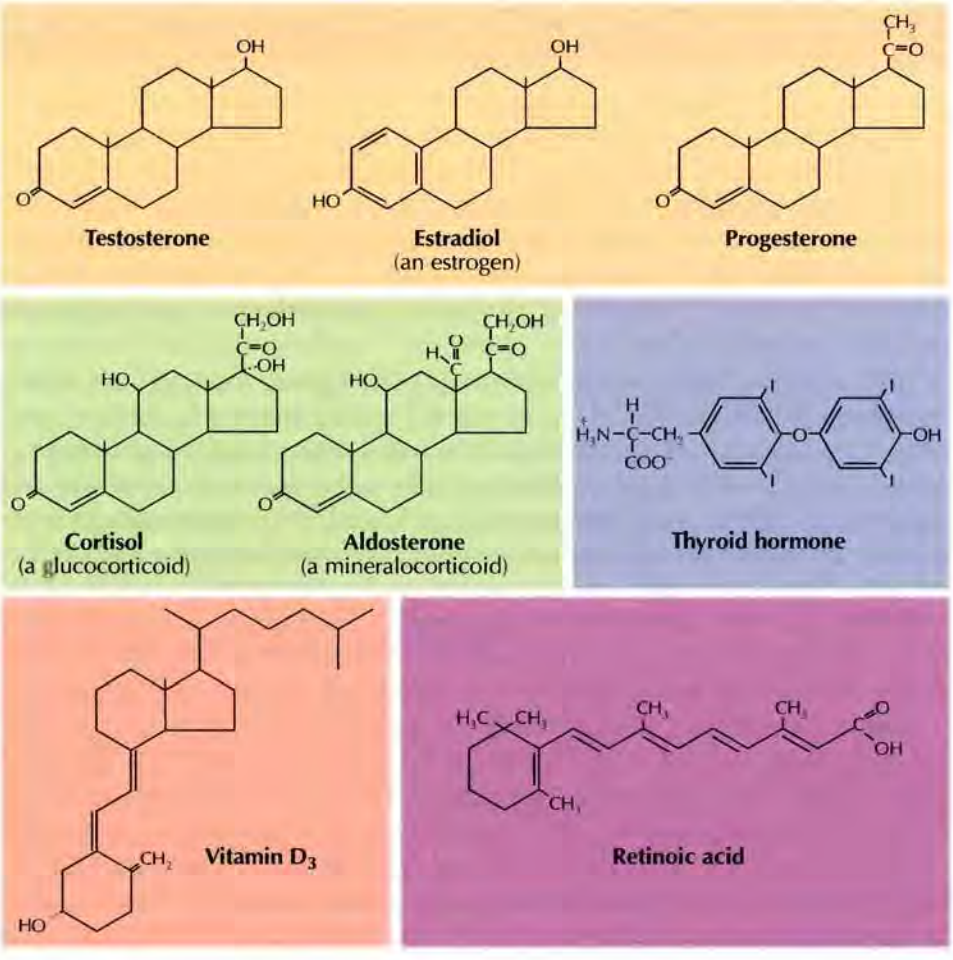


FIGURE 15.2 Structure of steroid hormones, thyroid hormone, vitamin D_3, and retinoic acid The steroids include the sex hormones (testosterone, estrogen, and progesterone), glucocorticoids, and mineralocorticoids.

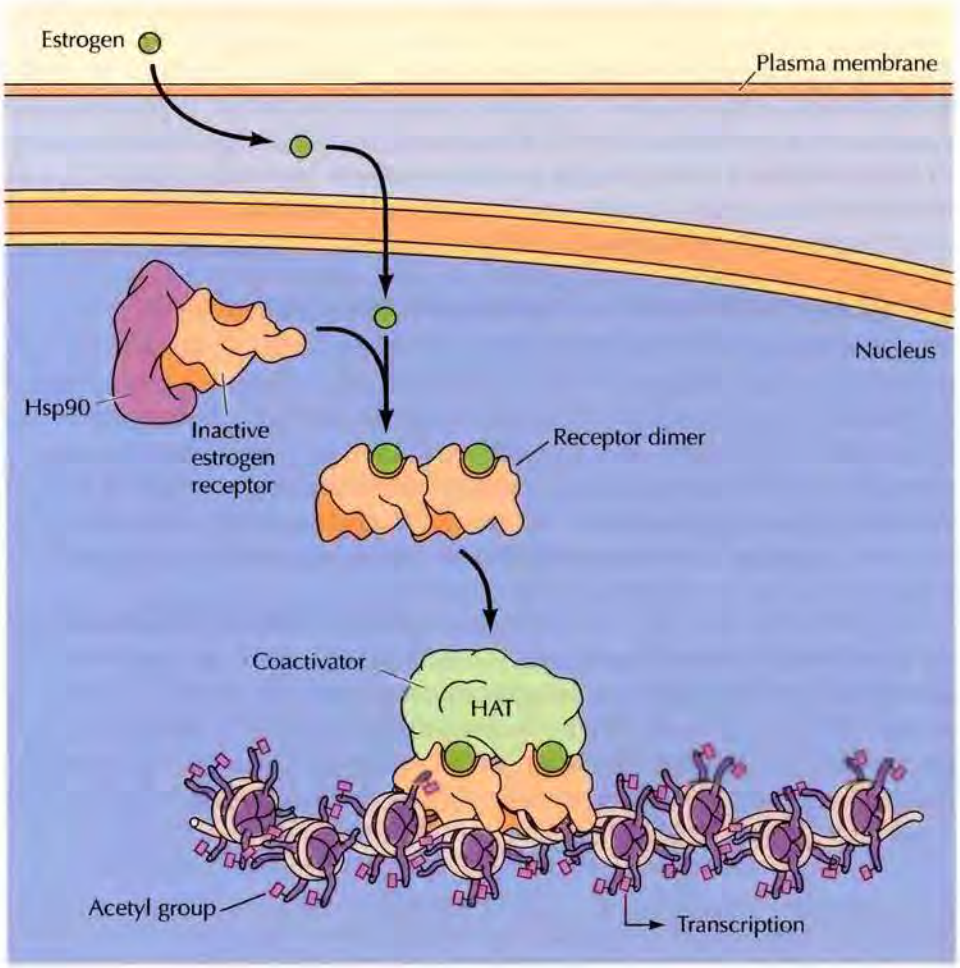


FIGURE 15.3 Estrogen action
Estrogen diffuses across the plasma membrane and binds to its receptor in the nucleus. In the absence of hormone, estrogen receptor is bound to Hsp90. Estrogen binding displaces the receptor from Hsp90 and allows the formation of receptor dimers, which bind DNA, associate with coactivators with histone acetyltransferase (HAT) activity, and stimulate transcription of their target genes.

the kidney to regulate salt and water balance. **Ecdysone** is an insect hormone that plays a key role in development by triggering the metamorphosis of larvae to adults. The **brassinosteroids** are plant-specific steroid hormones that control a number of developmental processes, including cell growth and differentiation.

Although thyroid hormone, vitamin D_3, and retinoic acid are structurally and functionally distinct from the steroids, they share a common mechanism of action in their target cells. **Thyroid hormone** is synthesized from tyrosine in the thyroid gland; it plays important roles in development and regulation of metabolism. **Vitamin D_3** regulates Ca^{2+} metabolism and bone growth. **Retinoic acid** and related compounds (**retinoids**) synthesized from vitamin A play important roles in vertebrate development.

Because of their hydrophobic character the steroid hormones, thyroid hormone, vitamin D_3, and retinoic acid are all able to enter cells by diffusing across the plasma membrane. Once inside the cell, they bind to intracellular receptors that are expressed by the hormonally responsive target cells. These receptors, which are members of a family of proteins known as the **nuclear receptor superfamily**, are transcription factors that contain related domains for ligand binding, DNA binding, and transcriptional activation. Ligand binding regulates their function as activators or repressors of their target genes, so the steroid hormones and related molecules directly regulate gene expression.

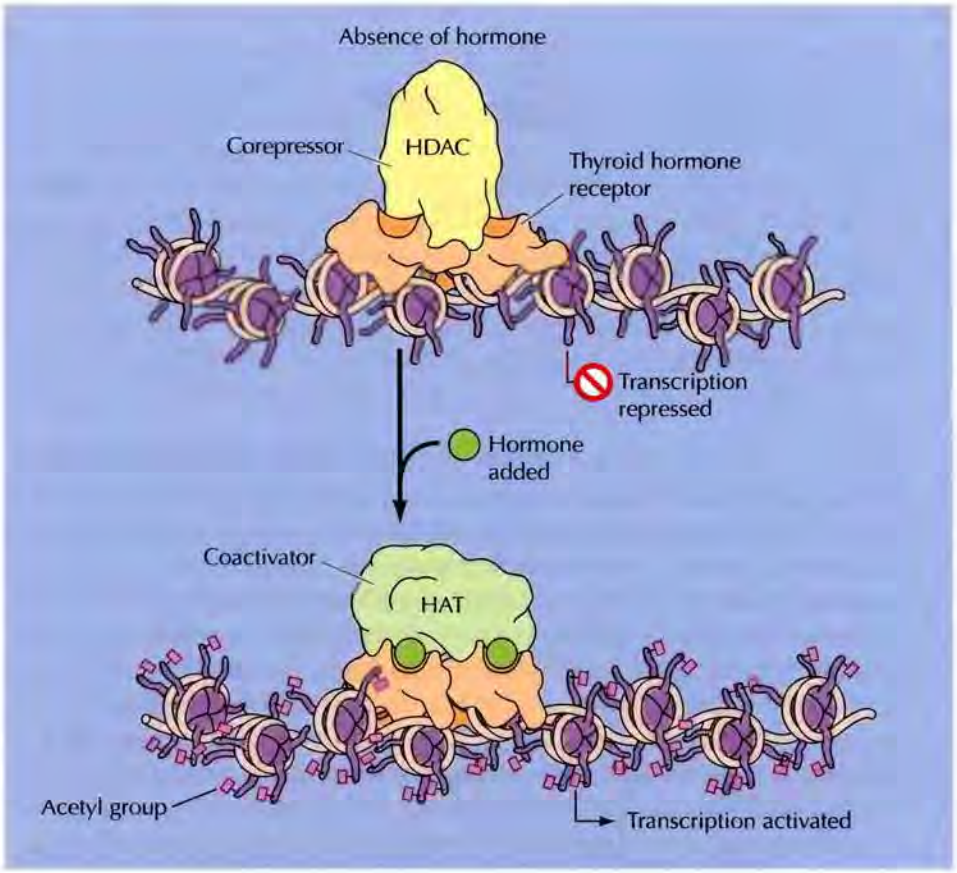


FIGURE 15.4 Gene regulation by the thyroid hormone receptor Thyroid hormone receptor binds DNA in either the presence or absence of hormone. However, hormone binding changes the function of the receptor from a repressor to an activator of target gene transcription. In the absence of hormone, the receptor associates with corepressors with histone deacetylase (HDAC) activity. In the presence of hormone, the receptor associates with coactivators with histone acetyltransferase (HAT) activity.

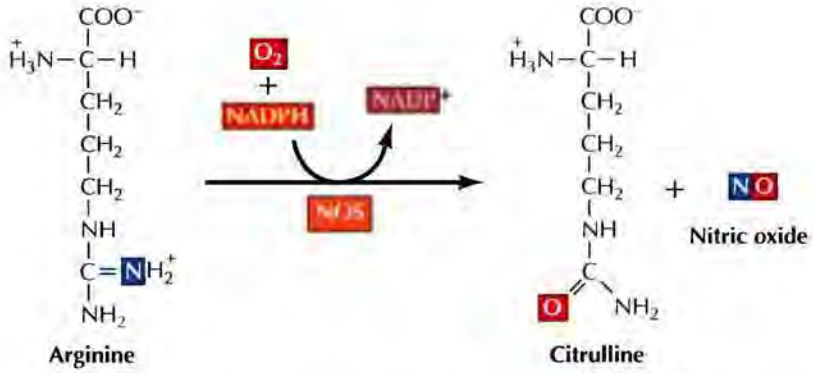
Ligand binding has distinct effects on different receptors. Some members of the steroid receptor superfamily are unable to bind to DNA in the absence of hormone. Estrogen receptor, for example, is bound to Hsp90 chaperones in the absence of hormone (Figure 15.3). The binding of estrogen induces a conformational change in the receptor, displacing Hsp90 and leading to the formation of receptor dimers that bind to regulatory DNA sequences and activate transcription of target genes. In other cases, the receptor binds DNA in either the presence or absence of hormone, but hormone binding alters the activity of the receptor as a transcriptional regulatory molecule. For example, in the absence of hormone, thyroid hormone receptor is associated with a corepressor complex and represses transcription of its target genes (Figure 15.4). Hormone binding induces a conformational change that results in the interaction of the receptor with coactivators rather than corepressors, leading to transcriptional activation of thyroid hormone-inducible genes.

Nitric Oxide and Carbon Monoxide

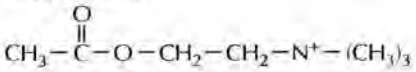
The simple gas nitric oxide (NO) is a major paracrine signaling molecule in the nervous, immune, and circulatory systems. Like the steroid hormones, NO is able to diffuse directly across the plasma membrane of its target cells. The molecular basis of NO action, however, is distinct from that of steroid action; rather than binding to a receptor that regulates transcription, NO alters the activity of intracellular target enzymes.

Nitric oxide is synthesized from the amino acid arginine by the enzyme nitric oxide synthase (Figure 15.5). Once synthesized, NO diffuses out of the

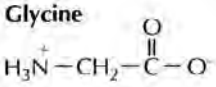
FIGURE 15.5 Synthesis of nitric oxide The enzyme nitric oxide synthase (NOS) catalyzes the formation of nitric oxide from arginine.



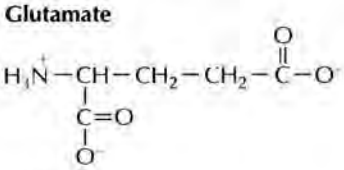
Acetylcholine



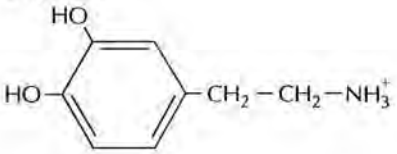
Glycine



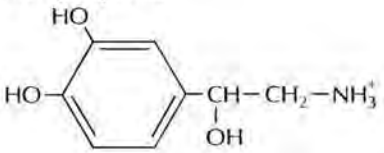
Glutamate



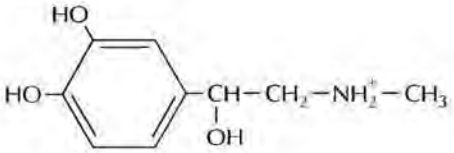
Dopamine



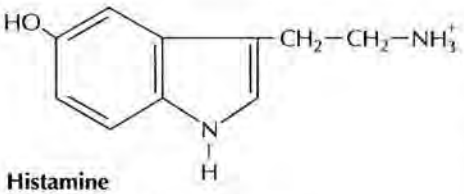
Norepinephrine



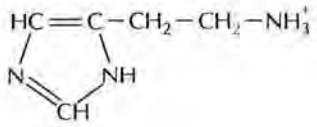
Epinephrine



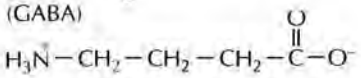
Serotonin



Histamine



γ-Aminobutyric acid (GABA)



cell and can act locally to affect nearby cells. Its action is restricted to such local effects because NO is extremely unstable, with a half-life of only a few seconds. The major intracellular target of NO is guanylyl cyclase. NO binds to a heme group at the active site of this enzyme, stimulating synthesis of the second messenger cyclic GMP (discussed later in this chapter). In addition, NO may directly modify some target proteins by nitrosylation of cysteine residues (see Figure 8.40). A well-characterized example of NO action is signaling the dilation of blood vessels. The first step in this process is the release of neurotransmitters, such as acetylcholine, from the termini of nerve cells in the blood vessel wall. These neurotransmitters act on endothelial cells to stimulate NO synthesis. NO then diffuses to neighboring smooth muscle cells where it activates guanylyl cyclase resulting in synthesis of cyclic GMP, which induces muscle cell relaxation and blood vessel dilation. For example, NO is responsible for signaling the dilation of blood vessels that leads to penile erection. It is also interesting to note that the medical use of nitroglycerin in treatment of heart disease is based on its conversion to NO, which dilates coronary blood vessels and increases blood flow to the heart.

Another simple gas, carbon monoxide (CO), also functions as a signaling molecule in the nervous system. CO is closely related to NO and appears to act similarly as a neurotransmitter and mediator of blood vessel dilation. The synthesis of CO in brain cells, like that of NO, is stimulated by neurotransmitters. In addition, CO can stimulate guanylate cyclase, which may also represent the major physiological target of CO signaling.

Neurotransmitters

The **neurotransmitters** carry signals between neurons or from neurons to other types of target cells (such as muscle cells). They are a diverse group of small hydrophilic molecules including acetylcholine, dopamine, epinephrine (adrenaline), serotonin, histamine, glutamate, glycine, and γ-aminobutyric acid (GABA) (Figure 15.6). The release of neurotransmitters is signaled by the arrival of an action potential at the terminus of a neuron (see Figure 13.22). The neurotransmitters then diffuse across the synaptic cleft and bind to receptors on the target cell surface. Note that some neurotransmitters can also act as hormones—for example, epinephrine functions both as a neurotransmitter and as a hormone produced by the adrenal gland to signal glycogen breakdown in muscle cells.

FIGURE 15.6 Structure of representative neurotransmitters The neurotransmitters are hydrophilic molecules that bind to cell surface receptors.

Because the neurotransmitters are hydrophilic molecules they are unable to cross the plasma membrane of their target cells. Therefore, in contrast to steroid hormones and NO or CO, the neurotransmitters act by binding to cell surface receptors. Many neurotransmitter receptors are ligand-gated ion channels, such as the acetylcholine receptor discussed in Chapter 13 (see Figure 13.23). Neurotransmitter binding to these receptors induces a conformational change that opens ion channels, directly resulting in changes in ion flux in the target cell. Other neurotransmitter receptors are coupled to G proteins—a major group of signaling molecules (discussed later in this chapter) that link cell surface receptors to a variety of intracellular responses. In the case of neurotransmitter receptors, the associated G proteins frequently act to indirectly regulate ion channel activity.

Peptide Hormones and Growth Factors

The widest variety of signaling molecules in animals are peptides, ranging in size from only a few to more than a hundred amino acids. This group of signaling molecules includes peptide hormones, neuropeptides, and a diverse array of polypeptide growth factors (Table 15.1). Well-known examples of **peptide hormones** include insulin, glucagon, and the hormones produced by the pituitary gland (growth hormone, follicle-stimulating hormone, prolactin, and others).

Neuropeptides are secreted by some neurons instead of the small-molecule neurotransmitters discussed in the previous section. Some of these

TABLE 15.1 Representative Peptide Hormones, Neuropeptides, and Growth Factors

Signaling molecule	Size[a]	Activities[b]
Peptide hormones		
Insulin	A = 21, B = 30	Regulation of glucose uptake; stimulation of cell proliferation
Glucagon	29	Stimulation of glucose synthesis
Growth hormone	191	General stimulation of growth
Follicle-stimulating hormone (FSH)	$\alpha = 92, \beta = 118$	Stimulation of the growth of oocytes and ovarian follicles
Prolactin	198	Stimulation of milk production
Neuropeptides and neurohormones		
Substance P	11	Sensory synaptic transmission
Oxytocin	9	Stimulation of smooth muscle contraction
Vasopressin	9	Stimulation of water reabsorption in the kidney
Enkephalin	5	Analgesic
β-Endorphin	31	Analgesic
Growth factors		
Nerve growth factor (NGF)	118	Differentiation and survival of neurons
Epidermal growth factor (EGF)	53	Proliferation of many types of cells
Platelet-derived growth factor (PDGF)	A = 125, B = 109	Proliferation of fibroblasts and other cell types
Interleukin-2	133	Proliferation of T lymphocytes
Erythropoietin	166	Development of red blood cells

[a] Size is indicated in number of amino acids. Some hormones and growth factors consist of two different polypeptide chains, which are designated either A and B or α and β.

[b] Most of these hormones and growth factors possess other activities in addition to those indicated.

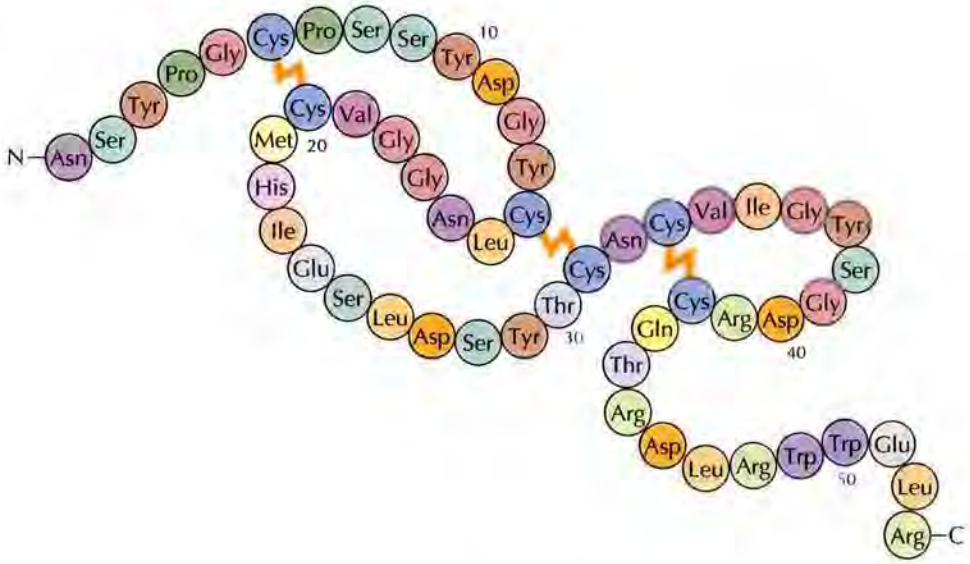


FIGURE 15.7 Structure of epidermal growth factor (EGF) EGF is a single polypeptide chain of 53 amino acids. Disulfide bonds between cysteine residues are indicated. (After G. Carpenter and S. Cohen, 1979. *Ann. Rev. Biochem.* 48: 193.)

peptides, such as the **enkephalins** and **endorphins**, function not only as neurotransmitters at synapses but also as **neurohormones** that act on distant cells. The enkephalins and endorphins have been widely studied because of their activity as natural analgesics that decrease pain responses in the central nervous system. Discovered during studies of drug addiction, they are naturally occurring compounds that bind to the same receptors on the surface of brain cells as morphine does.

The polypeptide **growth factors** include a wide variety of signaling molecules that control animal cell growth and differentiation. The first of these factors (**nerve growth factor**, or **NGF**) was discovered by Rita Levi-Montalcini in the 1950s. NGF is a member of a family of polypeptides (called **neurotrophins**) that regulate the development and survival of neurons. During the course of experiments on NGF, Stanley Cohen serendipitously discovered an unrelated factor (called **epidermal growth factor**, or **EGF**) that stimulates cell proliferation. EGF, a 53-amino-acid polypeptide (Figure 15.7), has served as the prototype of a large array of growth factors that play critical roles in controlling animal cell proliferation, both during embryonic development and in adult organisms.

A good example of growth factor action is provided by the activity of **platelet-derived growth factor** (**PDGF**) in wound healing. PDGF is stored in blood platelets and released during blood clotting at the site of a wound. It then stimulates the proliferation of fibroblasts in the vicinity of the clot, thereby contributing to regrowth of the damaged tissue. Members of another large group of polypeptide growth factors (called **cytokines**) regulate the development and differentiation of blood cells and control the activities of lymphocytes during the immune response. Other polypeptide growth factors (**membrane-anchored growth factors**) remain associated with the plasma membrane rather than being secreted into extracellular fluids, therefore functioning specifically as signaling molecules during direct cell-cell interactions.

Peptide hormones, neuropeptides, and growth factors are unable to cross the plasma membrane of their target cells, so they act by binding to cell surface receptors, as discussed later in this chapter. As might be expected from the critical roles of polypeptide growth factors in controlling cell proliferation, abnormalities in growth factor signaling are the basis for a variety of diseases, including many kinds of cancer. For example, abnormal expres-

sion of the EGF receptor is an important factor in the development of many human cancers, and inhibitors of the EGF receptor appear to be promising agents for cancer treatment (see Chapter 18).

Eicosanoids

Several types of lipids serve as signaling molecules that, in contrast to the steroid hormones, act by binding to cell surface receptors. The most important of these molecules are members of a class of lipids called the **eicosanoids**, which includes **prostaglandins**, **prostacyclin**, **thromboxanes**, and **leukotrienes** (Figure 15.8). The eicosanoids are rapidly broken down and therefore act locally in autocrine or paracrine signaling pathways. They stimulate a variety of responses in their target cells, including blood platelet aggregation, inflammation, and smooth-muscle contraction.

All eicosanoids are synthesized from arachidonic acid, which is formed from phospholipids. The first step in the pathway leading to synthesis of either prostaglandins or thromboxanes is the conversion of arachidonic acid to prostaglandin H_2. Interestingly, the enzyme that catalyzes this reaction (cyclooxygenase) is the target of aspirin and other nonsteroidal anti-inflammatory drugs (NSAIDs). By inhibiting synthesis of the prostaglandins, aspirin reduces inflammation and pain. By inhibiting synthesis of thromboxane, aspirin also reduces platelet aggregation and blood clotting.

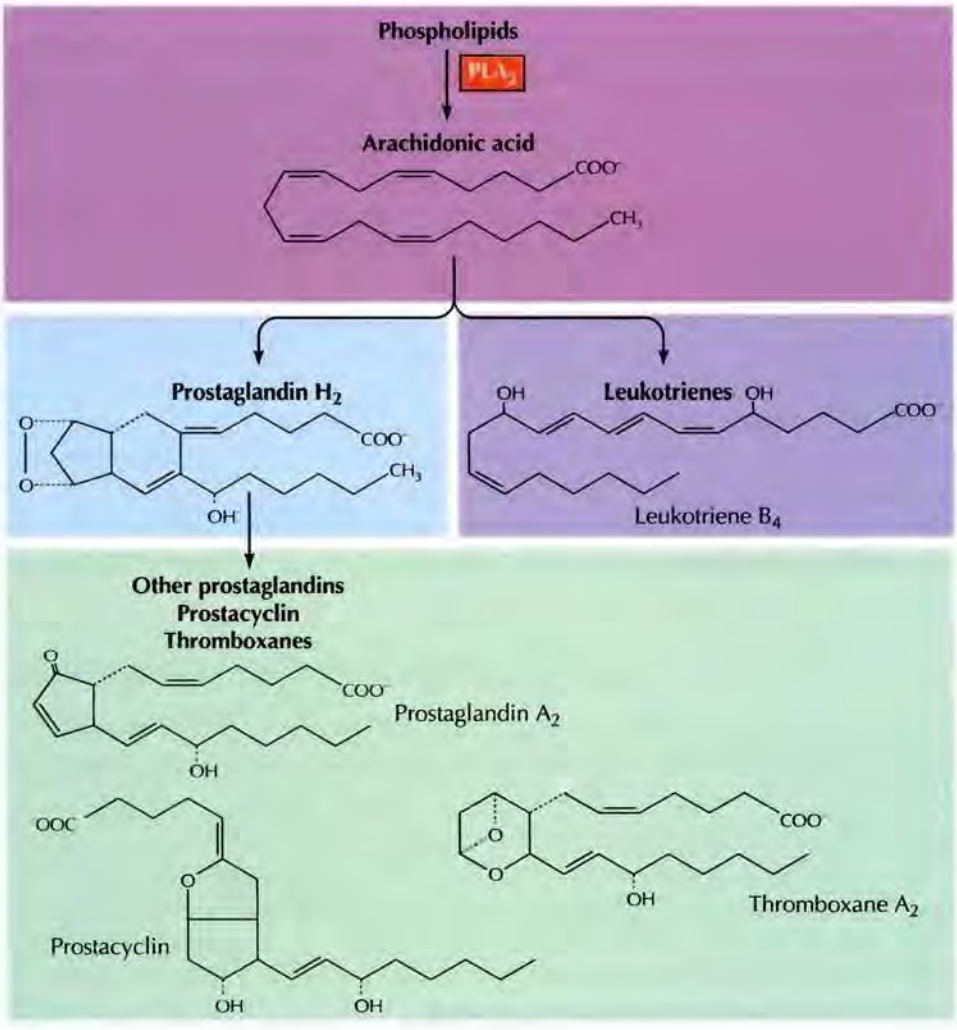


FIGURE 15.8 Synthesis and structure of eicosanoids The eicosanoids include the prostaglandins, prostacyclin, thromboxanes, and leukotrienes. They are synthesized from arachidonic acid, which is formed by the hydrolysis of phospholipids catalyzed by phospholipase A_2 (PLA_2). Arachidonic acid can then be metabolized via two alternative pathways—one pathway leads to synthesis of prostaglandins, prostacyclin, and thromboxanes, while the other pathway leads to synthesis of leukotrienes.

Because of this activity, small daily doses of aspirin are frequently prescribed for prevention of strokes. In addition, aspirin and NSAIDs have been found to reduce the frequency of colon cancer in both animal models and humans, apparently by inhibiting the synthesis of prostaglandins that act to stimulate cell proliferation and promote cancer development. It is noteworthy that there are two forms of cyclooxygenase: COX-1 and COX-2. COX-1 is thought to be principally responsible for the normal physiological production of prostaglandins and COX-2 for the increased prostaglandin production associated with inflammation and disease. Consequently, selective inhibitors of COX-2 have been developed based on the rationale that such drugs would be more effective and have fewer side effects than aspirin or conventional NSAIDs, which inhibit both COX-1 and COX-2. However, selective COX-2 inhibitors may also be associated with serious side effects, including an increased risk of cardiovascular disease.

Plant Hormones

Plant growth and development are regulated by a group of small molecules called **plant hormones**. The levels of these molecules within the plant are typically modified by environmental factors, such as light or infection, so they coordinate the responses of tissues in different parts of the plant to environmental signals.

The plant hormones are classically divided into five major groups: **auxins**, **gibberellins**, **cytokinins**, **abscisic acid**, and **ethylene** (Figure 15.9), although several additional plant hormones (such as the plant steroid hormones) have recently been discovered. The first plant hormone to be identified was auxin, with the early experiments leading to its discovery having been performed by Charles Darwin in the 1880s. One of the effects of auxins

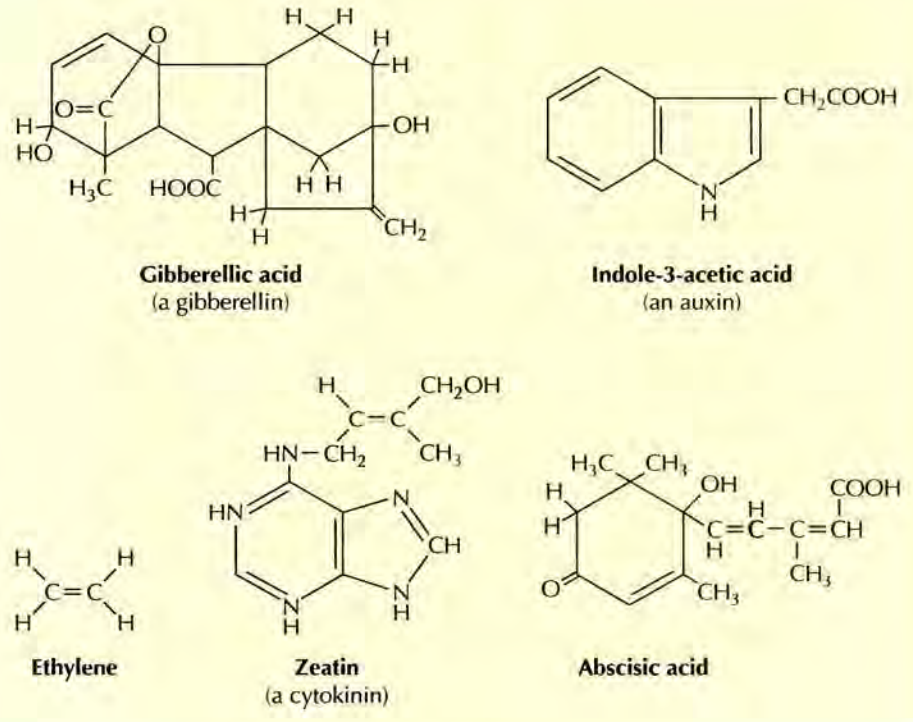


FIGURE 15.9 Structure of plant hormones

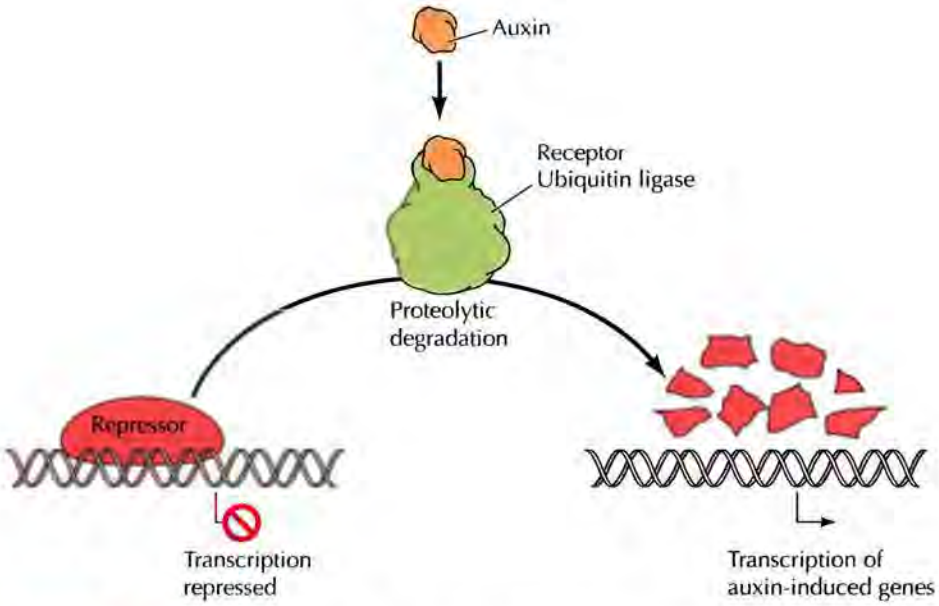


FIGURE 15.10 Auxin signaling Auxin binds to a receptor with ubiquitin ligase activity. This stimulates ubiquitination and proteolysis of a specific transcriptional repressor, leading to transcription of auxin-induced genes.

is to induce plant cell elongation by weakening the cell wall (see Figure 14.7). In addition, auxins regulate many other aspects of plant development, including cell division and differentiation. The other plant hormones likewise have multiple effects in their target tissues, including stem elongation (gibberellins), fruit ripening (ethylene), cell division (cytokinins), and the onset of dormancy (abscisic acid).

The signaling pathways triggered by these hormones in plants use a variety of mechanisms that are conserved in animal cells, as well as a number of elements that are unique to plants. For example, one well understood pathway, which has been elucidated by genetic analysis of *Arabidopsis thaliana*, signals the response of plant cells to ethylene. Elements of this pathway include an ethylene receptor on the plant cell surface, a protein kinase related to the Raf protein kinases of animal cells, and a novel transcription factor that regulates expression of ethylene-responsive genes. Recent studies have also elucidated the mechanism of auxin action (Figure 15.10). Auxin binds to a receptor associated with a ubiquitin ligase (see Figure 8.42), stimulating the ubiquitination and degradation of specific transcriptional repressors and resulting in auxin-mediated gene induction.

Functions of Cell Surface Receptors

As already reviewed, most ligands responsible for cell-cell signaling (including neurotransmitters, peptide hormones, and growth factors) bind to receptors on the surface of their target cells. Consequently, a major challenge in understanding cell-cell signaling is unraveling the mechanisms by which cell surface receptors transmit the signals initiated by ligand binding. As discussed in Chapter 13, some neurotransmitter receptors are ligand-gated ion channels that directly control ion flux across the plasma membrane. Other cell surface receptors, including the receptors for peptide hormones and growth factors, act instead by regulating the activity of

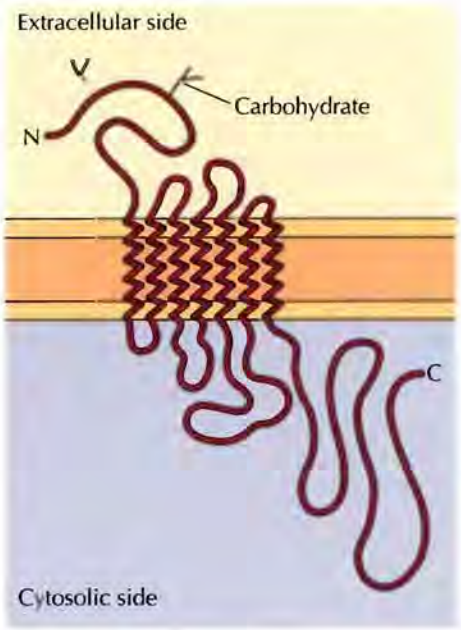


FIGURE 15.11 Structure of a G protein-coupled receptor The G protein-coupled receptor is characterized by seven transmembrane α helices.

■ **The G protein-coupled receptors responsible for our sense of smell (odorant receptors) are encoded by more than 500 genes in the human genome.**

intracellular proteins. These proteins then transmit signals from the receptor to a series of additional intracellular targets, frequently including transcription factors. Ligand binding to a receptor on the surface of the cell thus initiates a chain of intracellular reactions, ultimately reaching the target cell nucleus and resulting in programmed changes in gene expression. The functions of the major classes of cell surface receptors are discussed here, with the pathways of intracellular signaling downstream of these receptors being considered in the next section of this chapter.

G Protein-Coupled Receptors

The largest family of cell surface receptors transmits signals to intracellular targets via the intermediary action of guanine nucleotide-binding proteins called **G proteins**. More than a thousand such **G protein-coupled receptors** have been identified, including the receptors for eicosanoids, many neurotransmitters, neuropeptides, and peptide hormones. In addition, the G protein-coupled receptor family includes a large number of receptors that are responsible for smell, sight, and taste.

The G protein-coupled receptors are structurally and functionally related proteins characterized by seven membrane-spanning α helices (Figure 15.11). The binding of ligands to the extracellular domain of these receptors induces a conformational change that allows the cytosolic domain of the receptor to bind to a G protein associated with the inner face of the plasma membrane. This interaction activates the G protein, which then dissociates from the receptor and carries the signal to an intracellular target, which may be either an enzyme or an ion channel.

The discovery of G proteins came from studies of hormones (such as epinephrine) that regulate the synthesis of cyclic AMP (cAMP) in their target cells. As discussed later in the chapter, cAMP is an important second messenger that mediates cellular responses to a variety of hormones. In the 1970s Martin Rodbell and his colleagues made the key observation that GTP is required for hormonal stimulation of adenylyl cyclase (the enzyme responsible for cAMP formation). This finding led to the discovery that a guanine nucleotide-binding protein (called a G protein) is an intermediary in adenylyl cyclase activation (Figure 15.12). Since then an array of G proteins have been found to act as physiological switches that regulate the activities of a variety of intracellular targets in response to extracellular signals.

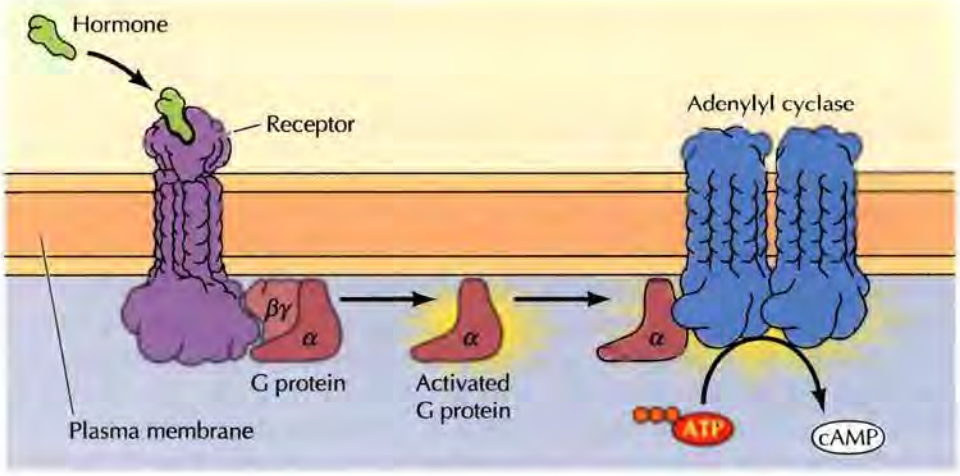
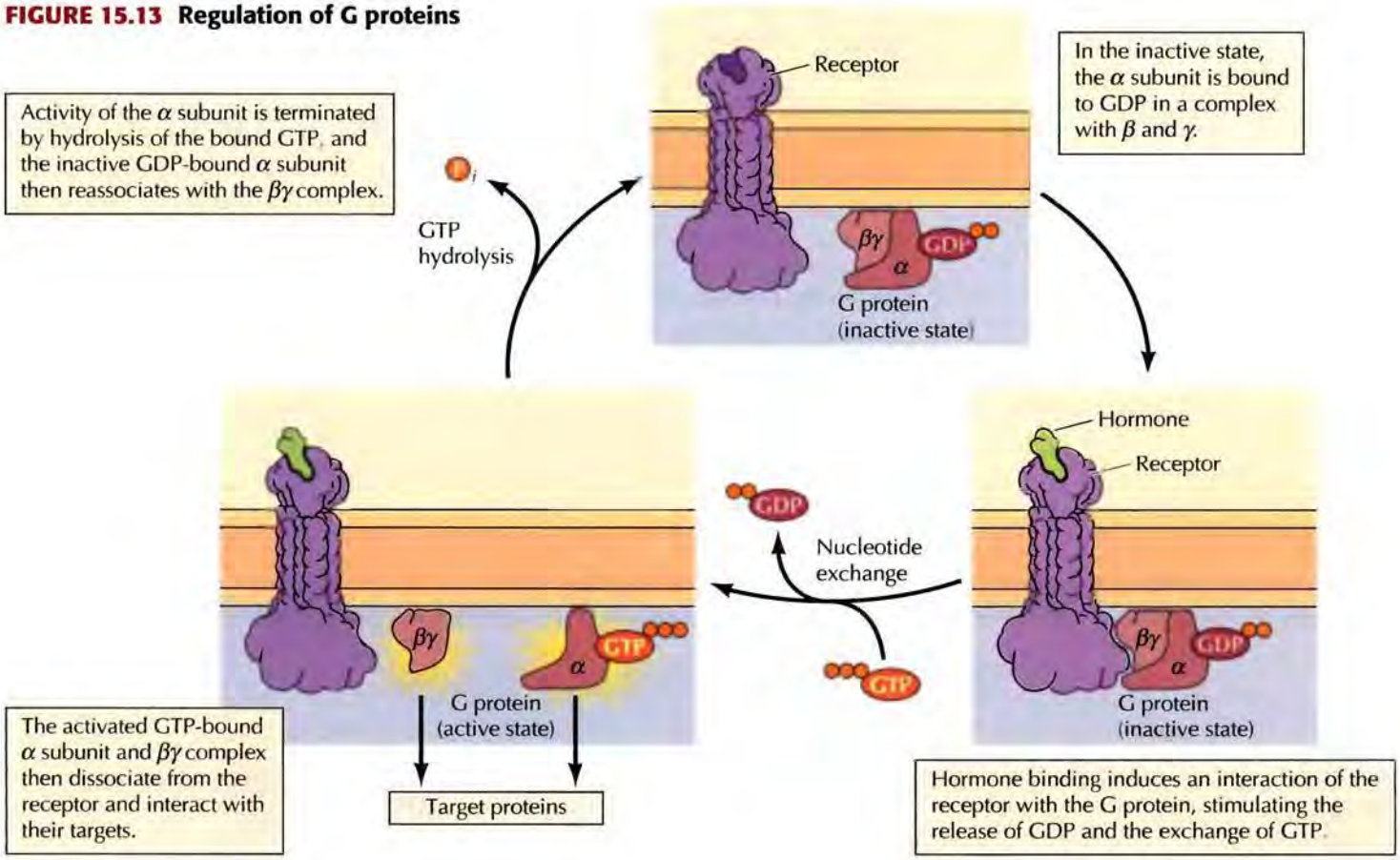


FIGURE 15.12 Hormonal activation of adenylyl cyclase Binding of hormone promotes the interaction of the receptor with a G protein. The activated G protein α subunit then dissociates from the receptor and stimulates adenylyl cyclase, which catalyzes the conversion of ATP to cAMP.

FIGURE 15.13 Regulation of G proteins



G proteins consist of three subunits designated α, β, and γ (Figure 15.13). They are frequently called **heterotrimeric G proteins** to distinguish them from other guanine nucleotide-binding proteins, such as the Ras proteins discussed later in the chapter. The α subunit binds guanine nucleotides, which regulate G protein activity. In the resting state, α is bound to GDP in a complex with β and γ. Hormone binding induces a conformational change in the receptor, such that the cytosolic domain of the receptor interacts with the G protein and stimulates the release of bound GDP and its exchange for GTP. The activated GTP-bound α subunit then dissociates from β and γ, which remain together and function as a $\beta\gamma$ complex. Both the active GTP-bound α subunit and the $\beta\gamma$ complex then interact with their targets to elicit an intracellular response. The activity of the α subunit is terminated by hydrolysis of the bound GTP, and the inactive α subunit (now with GDP bound) then reassociates with the $\beta\gamma$ complex, ready for the cycle to start anew.

Mammalian genomes encode 20 different α subunits, 5 β subunits, and 12 γ subunits. Different G proteins associate with different receptors, so this array of G proteins couples receptors to distinct intracellular targets. For example, the G protein associated with the epinephrine receptor is called G_s because its α subunit stimulates adenylyl cyclase (see Figure 15.12). Other G protein α and $\beta\gamma$ subunits act instead to inhibit adenylyl cyclase or to regulate the activities of other target enzymes.

In addition to regulating target enzymes, both the α and $\beta\gamma$ subunits of some G proteins directly regulate ion channels. A good example is provided

15.2 WEBSITE ANIMATION

Signal Transduction

The largest family of cell surface receptors transmits signals inside the cell by activating G proteins, which bind GTP and then activate effector proteins.

by the action of the neurotransmitter acetylcholine on heart muscle, which is distinct from its effects on nerve and skeletal muscle. The nicotinic acetylcholine receptor on nerve and skeletal muscle cells is a ligand-gated ion channel (see Figure 13.23). Heart muscle cells have a different acetylcholine receptor, which is G protein-coupled. This G protein is designated G_i because its α subunit *i*nhibits adenylyl cyclase. In addition, the G_i $\beta\gamma$ subunits act directly to open K^+ channels in the plasma membrane, which has the effect of slowing heart muscle contraction.

Receptor Protein-Tyrosine Kinases

In contrast to the G protein-coupled receptors, other cell surface receptors are directly linked to intracellular enzymes. The largest family of such enzyme-linked receptors is the **receptor protein-tyrosine kinases**, which phosphorylate their substrate proteins on tyrosine residues. This family includes the receptors for most polypeptide growth factors, so protein-tyrosine

KEY EXPERIMENT

The Src Protein-Tyrosine Kinase

Transforming Gene Product of Rous Sarcoma Virus Phosphorylates Tyrosine

Tony Hunter and Bartholomew M. Sefton

The Salk Institute, San Diego, CA

Proceedings of the National Academy of Science, USA, 1980, Volume 77, pages 1311–1315

Tony Hunter

Bartholomew Sefton

The Context

Following its isolation in 1911 Rous sarcoma virus (RSV) became the first virus that was generally accepted to cause tumors in animals (see the Molecular Medicine box in Chapter 1). Several features of RSV then made it an attractive model for studying the development of cancer. In particular, the small size of the RSV genome offered the hope of identifying specific viral genes responsible for inducing the abnormal proliferation that is characteristic of cancer cells. This goal was reached in the 1970s when it was established that a single RSV gene (called *src* for *sarc*oma) is required for tumor induction. Importantly, a closely related *src* gene was also found to be part of the normal genetic complement of a variety of vertebrates, including humans. Since the viral Src protein is responsible for driving the uncontrolled proliferation of cancer cells, it appeared that understanding Src function would yield crucial insights into the molecular bases of both cancer induction and the regulation of normal cell proliferation.

In 1977 Ray Erikson and his colleagues identified the Src protein by immunoprecipitation (see Figure 4.30) with antisera from animals bearing RSV-induced tumors. Shortly thereafter, it was found that incubation of Src immunoprecipitates with radioactive ATP resulted in phosphorylation of the immunoglobulin molecules. Src therefore appeared to be a protein kinase, clearly implicating protein phosphorylation in the control of cell proliferation.

All previously studied protein kinases phosphorylated serine or threonine residues, which were also the only phosphoamino acids to have been detected in animal cells. However, Walter Eckhardt and Tony Hunter had observed in 1979 that the oncogenic protein of another animal tumor virus (polyomavirus) was phosphorylated on a tyrosine residue. Hunter and Sefton therefore tested the possibility that Src might phosphorylate tyrosine, rather than serine/threonine, residues in its substrate proteins. Their experiments demonstrated that Src does indeed function as a protein-tyrosine kinase—an activity now recognized as playing a central role in cell signaling pathways.

phosphorylation has been particularly well studied as a signaling mechanism involved in the control of animal cell growth and differentiation. Indeed, the first protein-tyrosine kinase was discovered in 1980 during studies of the oncogenic proteins of animal tumor viruses—in particular, Rous sarcoma virus—by Tony Hunter and Bartholomew Sefton. The EGF receptor was then found to function as a protein-tyrosine kinase by Stanley Cohen and his colleagues, clearly establishing protein-tyrosine phosphorylation as a key signaling mechanism in the response of cells to growth factor stimulation.

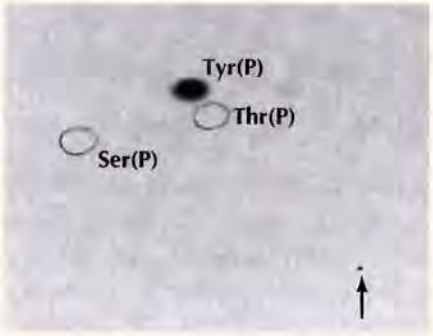
The human genome encodes 59 receptor protein-tyrosine kinases, including the receptors for EGF, NGF, PDGF, insulin, and many other growth factors. All these receptors share a common structural organization: an N-terminal extracellular ligand-binding domain, a single transmembrane α helix, and a cytosolic C-terminal domain with protein-tyrosine kinase activity (Figure 15.14). Most of the receptor protein-tyrosine kinases consist of single polypeptides, although the insulin receptor and some

KEY EXPERIMENT

The Experiments

Hunter and Sefton identified the amino acid phosphorylated by Src by incubating Src immunoprecipitates with ^{32}P-labeled ATP. The amino acid that was phosphorylated by Src in the substrate protein (in this case, immunoglobulin) therefore became radioactively labeled. The immunoglobulin was then isolated and hydrolyzed to yield individual amino acids, which were analyzed by electrophoresis and chromatography methods that separated phosphotyrosine, phosphoserine, and phosphothreonine (see figure). The radioactive amino acid detected in these experiments was phosphotyrosine, indicating that Src specifically phosphorylates tyrosine residues.

Further experiments showed that the normal cell Src protein, as well as viral Src, functions as a protein-tyrosine kinase in immunoprecipitation assays. In addition, Hunter and Sefton extended these *in vitro* experiments by demonstrating the presence of phosphotyrosine in proteins extracted from whole cells. In normal cells, phosphotyrosine accounted for only about 0.03% of total phosphoamino acids (the rest being phosphoserine and phosphothreonine), explaining why it had previously escaped detection. However, phosphotyrosine was about ten times more abundant in cells that were infected with RSV, suggesting that increased protein-tyrosine kinase activity of the viral Src protein was responsible for its ability to induce abnormal cell proliferation.



Identification of phosphotyrosine in immunoglobulin phosphorylated by Src. An immunoprecipitate containing RSV Src was incubated with [^{32}P]-ATP. The immunoglobulin was then isolated and hydrolyzed. Amino acids in the hydrolysate were separated by electrophoresis and chromatography on a cellulose thin-layer plate. The positions of ^{32}P-labeled amino acids were determined by exposing the plate to X-ray film. Broken lines indicate the positions of unlabeled phosphoamino acids that were included as markers. Note that the principal ^{32}P-labeled amino acid is phosphotyrosine.

The Impact

The discovery that Src was a protein-tyrosine kinase both identified a new protein kinase activity and established this activity as being related to the control of cell proliferation. The results of Hunter and Sefton were followed by demonstrations that many other tumor virus proteins also function as protein-tyrosine kinases, generalizing the link between protein-tyrosine phosphorylation and the abnormal proliferation of cancer cells. Stanley Cohen and his colleagues further found that the EGF receptor is a protein-tyrosine kinase, directly implicating protein-tyrosine phosphorylation in the control of normal cell proliferation. Continuing studies have identified numerous additional receptor and nonreceptor protein-tyrosine kinases that function in a variety of cell signaling pathways. Studies of the mechanism by which a virus causes cancer in chickens thus revealed a previously unknown enzymatic activity that plays a central role in the signaling pathways that regulate animal cell growth, survival, and differentiation. Moreover, as discussed in Chapter 18, protein-tyrosine kinases encoded by oncogenes have provided the most promising targets to date for development of specific drugs against cancer cells.

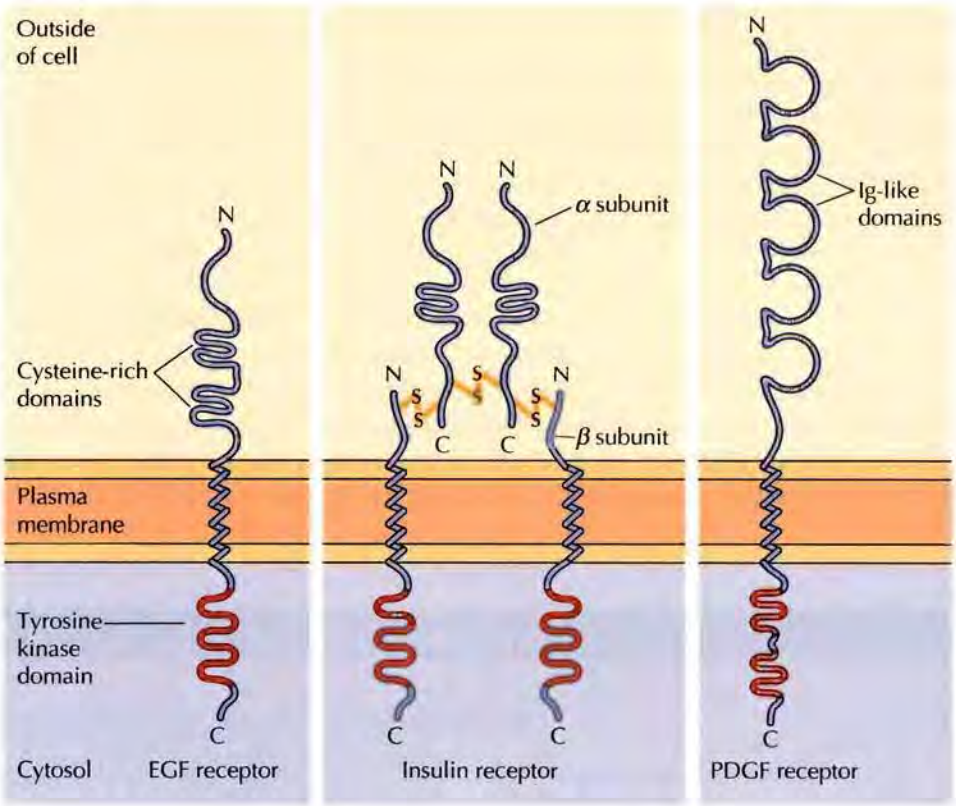


FIGURE 15.14 Organization of receptor protein-tyrosine kinases Each receptor consists of an N-terminal extracellular ligand-binding domain, a single transmembrane α helix, and a cytosolic C-terminal domain with protein-tyrosine kinase activity. The structures of three distinct subfamilies of receptor protein-tyrosine kinases are shown. The EGF receptor and insulin receptor both have cysteine-rich extracellular domains, whereas the PDGF receptor has immunoglobulin (Ig)-like domains. The PDGF receptor is also noteworthy in that its kinase domain is interrupted by an insert of approximately a hundred amino acids unrelated to those found in most other protein-tyrosine kinase catalytic domains. The insulin receptor is unusual in being a dimer of two pairs of polypeptide chains (designated α and β).

related receptors are dimers consisting of two polypeptide chains. The binding of ligands (e.g., growth factors) to the extracellular domains of these receptors activates their cytosolic kinase domains, resulting in phosphorylation of both the receptors themselves and intracellular target proteins that propagate the signal initiated by growth factor binding.

The first step in signaling from most receptor protein-tyrosine kinases is ligand-induced receptor dimerization (Figure 15.15). Some growth factors, such as PDGF and NGF, are themselves dimers consisting of two identical

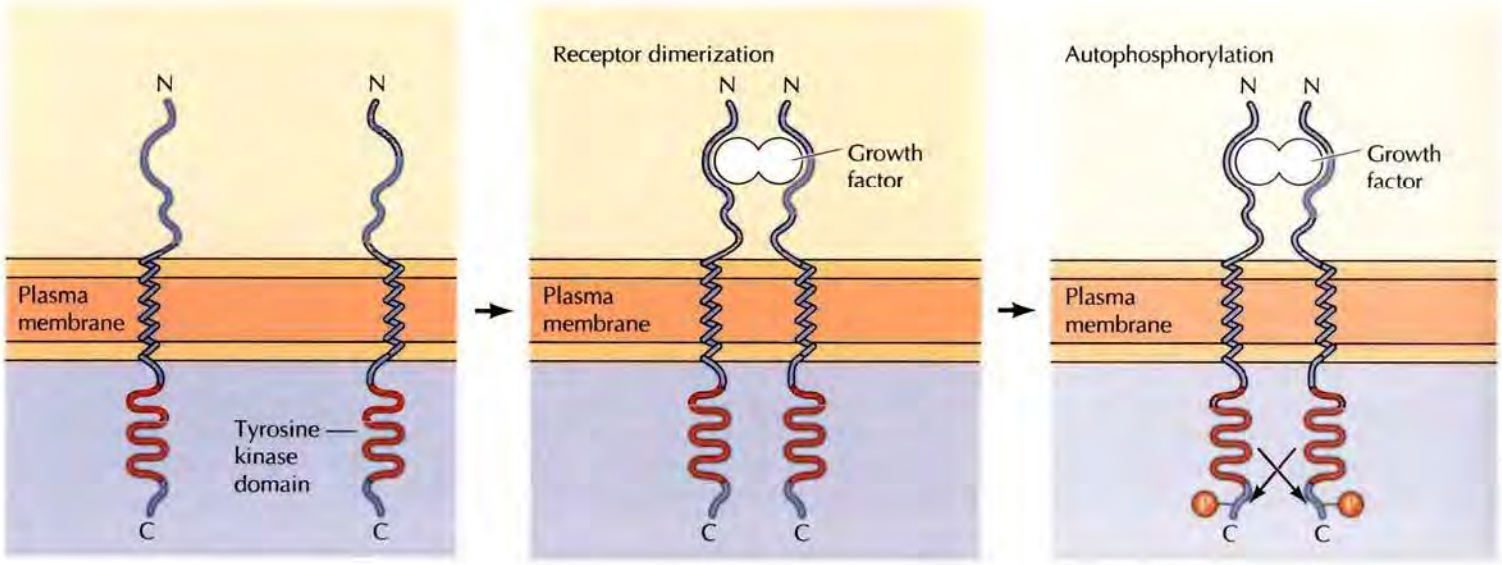


FIGURE 15.15 Dimerization and autophosphorylation of receptor protein-tyrosine kinases Growth factor binding induces receptor dimerization, which results in receptor autophosphorylation as the two polypeptide chains phosphorylate one another.

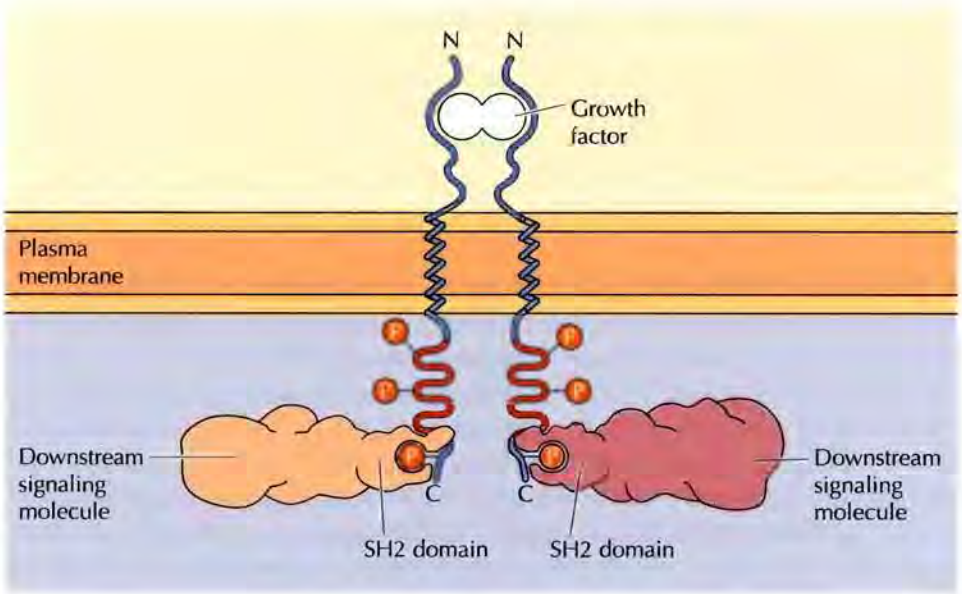


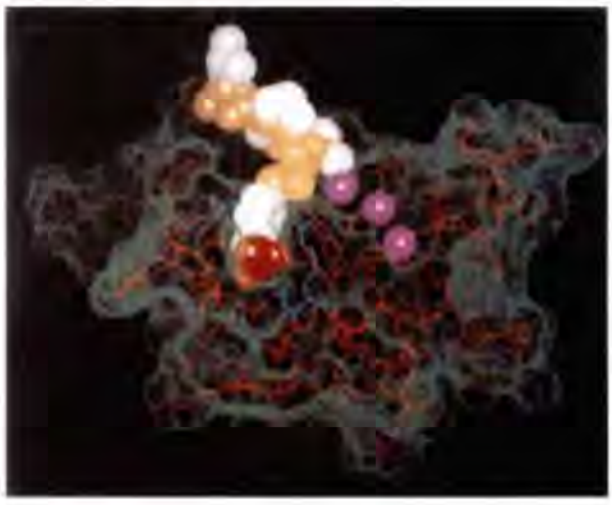
FIGURE 15.16 Association of downstream signaling molecules with receptor protein-tyrosine kinases SH2 domains bind to specific phosphotyrosine-containing peptides of the activated receptors.

polypeptide chains; these growth factors directly induce dimerization by simultaneously binding to two different receptor molecules. Other growth factors (such as EGF) are monomers but lead to receptor dimerization as a result of inducing conformational changes that promote protein-protein interactions between different receptor polypeptides.

Ligand-induced dimerization then leads to **autophosphorylation** of the receptor as the dimerized polypeptide chains cross-phosphorylate one another (see Figure 15.15). Such autophosphorylation plays two key roles in signaling from these receptors. First, phosphorylation of tyrosine residues within the catalytic domain increases protein kinase activity. Second, phosphorylation of tyrosine residues outside of the catalytic domain creates specific binding sites for additional proteins that transmit intracellular signals downstream of the activated receptors.

The association of these downstream signaling molecules with receptor protein-tyrosine kinases is mediated by protein domains that bind to specific phosphotyrosine-containing peptides (Figure 15.16). The first of these domains to be characterized are called **SH2 domains** (for *S*rc *h*omology *2*) because they were initially recognized in protein-tyrosine kinases related to Src, the oncogenic protein of Rous sarcoma virus. SH2 domains consist of approximately 100 amino acids and bind to specific short peptide sequences containing phosphotyrosine residues (Figure 15.17). Other proteins bind to phosphotyrosine-containing peptides via **PTB domains** (for *p*hospho*t*yrosine-*b*inding). The resulting association of SH2- or PTB-containing proteins with activated receptor protein-tyrosine kinases can have several effects: It localizes these proteins to the plasma membrane, leads to their association with other proteins, promotes their phosphorylation, and stimulates their enzymatic activities. The association of these proteins with autophosphorylated receptors thus represents the first step in the intracellular transmission of signals initiated by the binding of growth factors to the cell surface.

FIGURE 15.17 Complex between an SH2 domain and a phosphotyrosine peptide The polypeptide chain of the Src SH2 domain is shown in red with its surface indicated by green dots. Purple spheres indicate a groove on the surface. The three amino acid residues that interact with the phosphotyrosine are shown in blue. The phosphotyrosine-containing peptide is shown as a space-filling model. Yellow and white spheres indicate the backbone and side-chain atoms, respectively, and the phosphate group is shown in red. (From G. Waksman and 13 others, 1992. *Nature* 358: 646.)



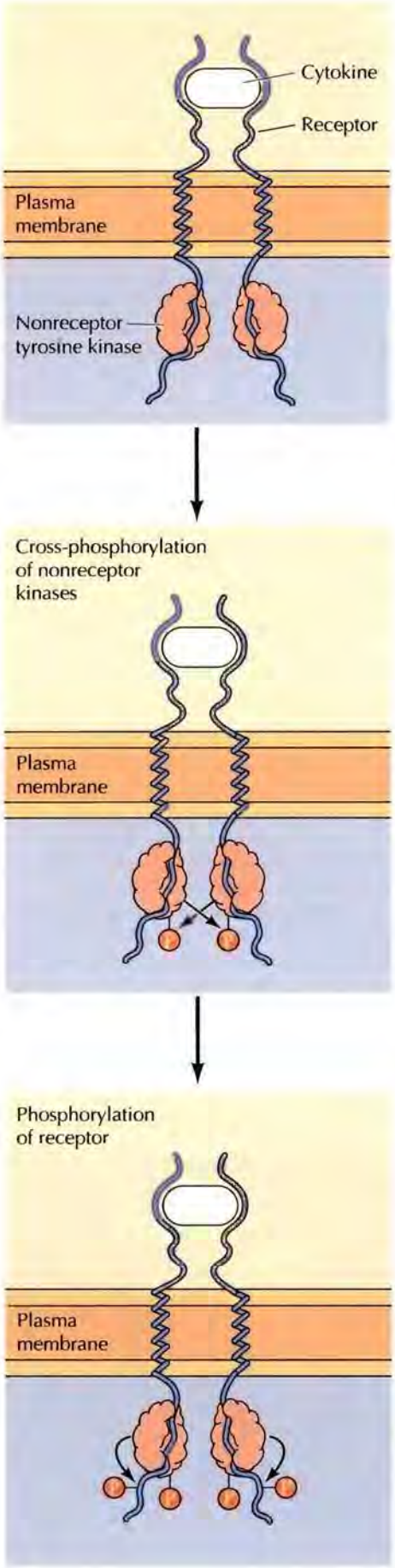


FIGURE 15.18 Signaling from cytokine receptors Ligand binding induces receptor dimerization and leads to the activation of associated nonreceptor protein-tyrosine kinases as a result of cross-phosphorylation. The activated kinases then phosphorylate tyrosine residues of the receptor, creating phosphotyrosine-binding sites for downstream signaling molecules.

Cytokine Receptors and Nonreceptor Protein-Tyrosine Kinases

Rather than possessing intrinsic enzymatic activity, many receptors act by stimulating intracellular protein-tyrosine kinases with which they are noncovalently associated. This family of receptors (called the **cytokine receptor superfamily**) includes the receptors for most cytokines (e.g., interleukin-2 and erythropoietin) and for some polypeptide hormones (e.g., growth hormone). Like receptor protein-tyrosine kinases, the cytokine receptors contain N-terminal extracellular ligand-binding domains, single transmembrane α helices, and C-terminal cytosolic domains. However, the cytosolic domains of the cytokine receptors are devoid of any known catalytic activity. Instead, the cytokine receptors function in association with **nonreceptor protein-tyrosine kinases**, which are activated as a result of ligand binding.

The first step in signaling from cytokine receptors is thought to be ligand-induced receptor dimerization and cross-phosphorylation of the associated nonreceptor protein-tyrosine kinases (Figure 15.18). These activated kinases then phosphorylate the receptor, providing phosphotyrosine-binding sites for the recruitment of downstream signaling molecules that contain SH2 domains. Combinations of cytokine receptors plus associated nonreceptor protein-tyrosine kinases thus function analogously to the receptor protein-tyrosine kinases discussed in the previous section.

The kinases associated with cytokine receptors belong to the **Janus kinase** (or **JAK**) family, which consists of four related nonreceptor protein-tyrosine kinases. Members of the JAK family appear to be universally required for signaling from cytokine receptors, indicating that JAK family kinases play a critical role in coupling these receptors to the tyrosine phosphorylation of intracellular targets.

Additional nonreceptor protein-tyrosine kinases belong to the **Src** family, which consists of Src and eight closely related proteins. As already noted, Src was initially identified as the oncogenic protein of Rous sarcoma virus and was the first protein shown to possess protein-tyrosine kinase activity, so it has played a pivotal role in experiments leading to our current understanding of cell signaling. Members of the Src family play key roles in signaling downstream of receptor protein-tyrosine kinases, from antigen receptors on B and T lymphocytes, and (as discussed later in this chapter) from integrins at sites of cell attachment to the extracellular matrix.

Receptors Linked to Other Enzymatic Activities

Although the vast majority of enzyme-linked receptors stimulate protein-tyrosine phosphorylation, some receptors are associated with other enzymatic activities. These receptors include protein-tyrosine phosphatases, protein-serine/threonine kinases, and guanylyl cyclases.

Protein-tyrosine phosphatases remove phosphate groups from phosphotyrosine residues, thus acting to counterbalance the effects of protein-tyrosine kinases. In many cases, protein-tyrosine phosphatases play negative regulatory roles in cell signaling pathways by terminating the signals initiated by protein-tyrosine phosphorylation. However, some protein-tyro-

sine phosphatases are cell surface receptors whose enzymatic activities play a positive role in cell signaling: 21 such receptor protein-tyrosine phosphatases are encoded in the human genome. A good example is provided by a receptor called CD45, which is expressed on the surface of T and B lymphocytes. Following antigen stimulation, CD45 dephosphorylates a specific phosphotyrosine that inhibits the enzymatic activity of Src family members. Thus the CD45 protein-tyrosine phosphatase acts (somewhat paradoxically) to stimulate nonreceptor protein-tyrosine kinases.

■ **Cytokine receptors are used by human immunodeficiency virus (HIV) as cell surface receptors for infection of immune cells.**

The receptors for **transforming growth factor β (TGF-β)** and related polypeptides are protein kinases that phosphorylate serine or threonine, rather than tyrosine, residues on their substrate proteins. TGF-β is the prototype of a family of polypeptide growth factors that control proliferation and differentiation of a variety of cell types. The cloning of the first receptor for a member of the TGF-β family in 1991 revealed that it is the prototype of a unique receptor family with a cytosolic **protein-serine/threonine kinase** domain. Since then, receptors for additional TGF-β family members have similarly been found to be protein-serine/threonine kinases. The binding of ligand to these receptors results in the association of two distinct types of polypeptide chains, which are encoded by different members of the TGF-β receptor family, to form heterodimers in which one of the receptor kinases phosphorylates the other. The activated TGF-β receptors then phosphorylate members of a family of transcription factors called Smads, which translocate to the nucleus and stimulate expression of target genes.

Some peptide ligands bind to receptors whose cytosolic domains are guanylyl cyclases, which catalyze formation of cyclic GMP. As discussed earlier, nitric oxide also acts by stimulating guanylyl cyclase, but the target of nitric oxide is an intracellular enzyme rather than a transmembrane receptor. The receptor **guanylyl cyclases** have an extracellular ligand-binding domain, a single transmembrane α helix, and a cytosolic domain with catalytic activity. Ligand binding stimulates cyclase activity, leading to the formation of cyclic GMP—a second messenger whose intracellular effects are discussed in the next section of this chapter.

Other receptors bind to cytoplasmic proteins with additional biochemical activities. For example, the cytokine tumor necrosis factor (TNF) induces cell death, perhaps (as discussed in Chapter 17) as a way of eliminating damaged or unwanted cells from tissues. The receptors for TNF and related death-signaling molecules are associated with specific proteases, which are activated in response to ligand binding. Activation of these receptor-associated proteases triggers the activation of additional downstream proteases, ultimately leading to degradation of a variety of intracellular proteins and death of the cell.

Pathways of Intracellular Signal Transduction

Many cell surface receptors stimulate intracellular target enzymes, which may be either directly linked or indirectly coupled to receptors by G proteins. These intracellular enzymes serve as downstream signaling elements that propagate and amplify the signal initiated by ligand binding. In most cases, a chain of reactions transmits signals from the cell surface to a variety of intracellular targets—a process called **intracellular signal transduction**. The targets of such signaling pathways frequently include transcription factors that function to regulate gene expression. Intracellular signaling pathways thus connect the cell surface to the nucleus, leading to changes in gene expression in response to extracellular stimuli.

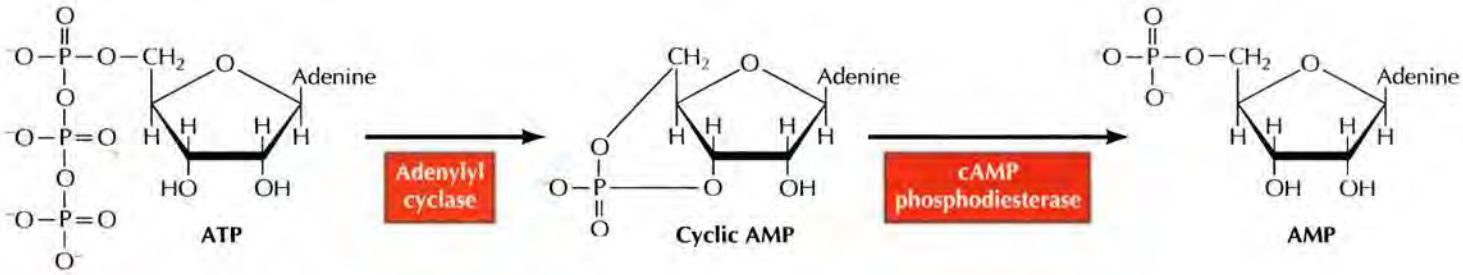


FIGURE 15.19 Synthesis and degradation of cAMP Cyclic AMP is synthesized from ATP by adenylyl cyclase and degraded to AMP by cAMP phosphodiesterase.

The cAMP Pathway: Second Messengers and Protein Phosphorylation

Intracellular signaling was first elucidated by studies of the action of hormones such as epinephrine, which signals the breakdown of glycogen to glucose in anticipation of muscular activity. In 1958 Earl Sutherland discovered that the action of epinephrine was mediated by an increase in the intracellular concentration of **cyclic AMP (cAMP)**, leading to the concept that cAMP is a **second messenger** in hormonal signaling (the first messenger being the hormone itself). Cyclic AMP is formed from ATP by the action of **adenylyl cyclase** and degraded to AMP by **cAMP phosphodiesterase** (Figure 15.19). As discussed earlier, the epinephrine receptor is coupled to adenylyl cyclase via a G protein that stimulates enzymatic activity, thereby increasing the intracellular concentration of cAMP (see Figure 15.12).

How does cAMP then signal the breakdown of glycogen? This and most other effects of cAMP in animal cells are mediated by the action of **cAMP-dependent protein kinase**, or **protein kinase A**, an enzyme discovered by Donal Walsh and Ed Krebs in 1968. The inactive form of protein kinase A is a tetramer consisting of two catalytic and two regulatory subunits (Figure 15.20). Cyclic AMP binds to the regulatory subunits, leading to their dissociation from the catalytic subunits. The free catalytic subunits are then enzymatically active and able to phosphorylate serine residues on their target proteins.

In the regulation of glycogen metabolism, protein kinase A phosphorylates two key target enzymes (Figure 15.21). The first is another protein kinase, phosphorylase kinase, which is phosphorylated and activated by protein kinase A. Phosphorylase kinase in turn phosphorylates and activates glycogen phosphorylase, which catalyzes the breakdown of glycogen to glucose-1-phosphate. In addition, protein kinase A phosphorylates the enzyme glycogen synthase, which catalyzes glycogen synthesis. In this case, however, phosphorylation inhibits enzymatic activity. Elevation of cAMP and activation of protein kinase A thus blocks further glycogen synthesis at the same time as it stimulates glycogen breakdown.

The chain of reactions leading from the epinephrine receptor to glycogen phosphorylase provides a good illustration of signal amplification during intracellular signal transduction. Each molecule of epinephrine activates only a single receptor. However, each receptor may activate up to a hundred molecules of G_s. Each molecule of G_s then stimulates the enzymatic

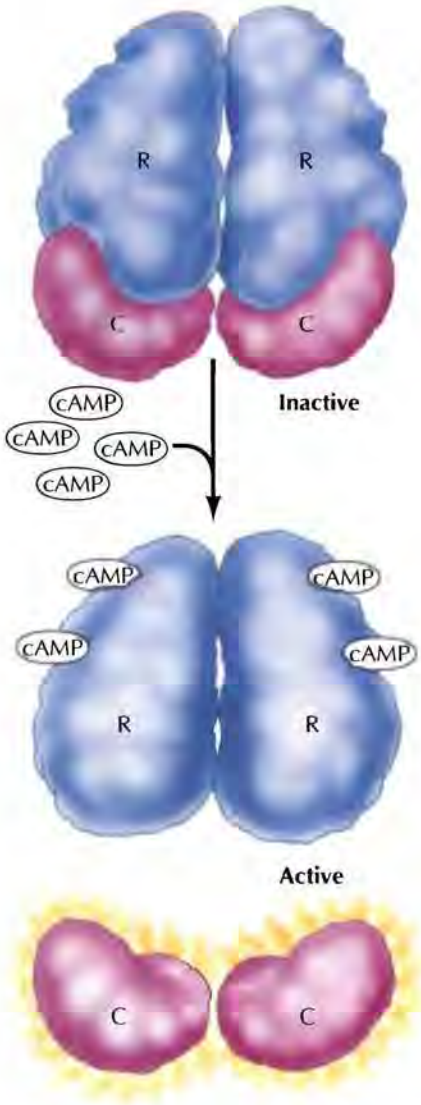


FIGURE 15.20 Regulation of protein kinase A The inactive form of protein kinase A consists of two regulatory (R) and two catalytic (C) subunits. Binding of cAMP to the regulatory subunits induces a conformational change that leads to dissociation of the catalytic subunits, which are then enzymatically active.

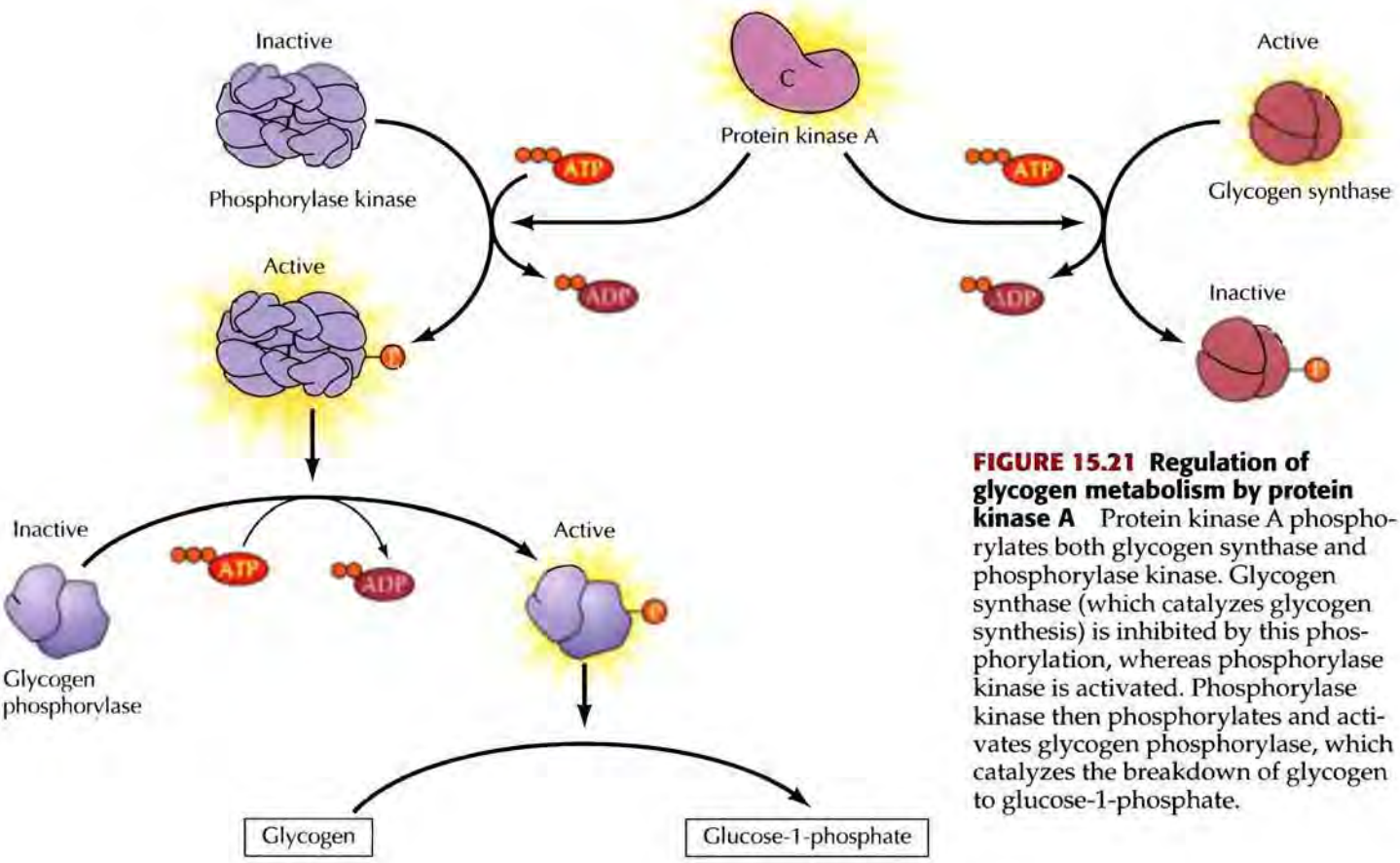


FIGURE 15.21 Regulation of glycogen metabolism by protein kinase A Protein kinase A phosphorylates both glycogen synthase and phosphorylase kinase. Glycogen synthase (which catalyzes glycogen synthesis) is inhibited by this phosphorylation, whereas phosphorylase kinase is activated. Phosphorylase kinase then phosphorylates and activates glycogen phosphorylase, which catalyzes the breakdown of glycogen to glucose-1-phosphate.

activity of adenylyl cyclase, which can catalyze the synthesis of many molecules of cAMP. Signal amplification continues as each molecule of protein kinase A phosphorylates many molecules of phosphorylase kinase, which in turn phosphorylates many molecules of glycogen phosphorylase. Hormone binding to a small number of receptors thus leads to activation of a much larger number of intracellular target enzymes.

In many animal cells, increases in cAMP activate the transcription of specific target genes that contain a regulatory sequence called the **cAMP response element**, or **CRE** (Figure 15.22). In this case, the signal is carried from the cytoplasm to the nucleus by the catalytic subunit of protein kinase A, which is able to enter the nucleus following its release from the regulatory subunit. Within the nucleus, protein kinase A phosphorylates a transcription factor called **CREB** (for CRE-*b*inding protein), leading to the recruitment of coactivators and transcription of cAMP-inducible genes. Such regulation of gene expression by cAMP plays important roles in controlling the proliferation, survival, and differentiation of a wide variety of animal cells, as well as being implicated in learning and memory.

It is important to recognize that protein kinases, such as protein kinase A, do not function in isolation within the cell. To the contrary, protein phosphorylation is rapidly reversed by the action of protein phosphatases. Some protein phosphatases are transmembrane receptors, as discussed in the preceding section. A number of others are cytosolic enzymes that remove phosphate groups from either phosphorylated tyrosine or serine/threonine residues in their substrate proteins. These protein phosphatases serve to terminate the responses initiated by receptor activation of protein kinases. For example, the serine residues of proteins that are phosphorylated by protein

15.3 WEBSITE ANIMATION

Signal Amplification

In a signal transduction cascade, each enzyme activated at a stage in the cascade may activate 100 molecules of the next enzyme, quickly amplifying the response to the receptor-bound ligand.

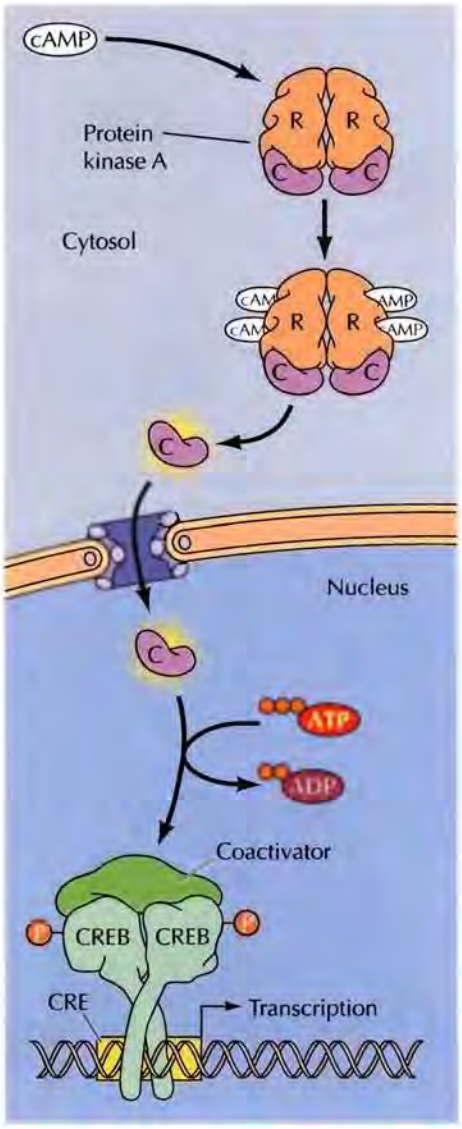


FIGURE 15.22 Cyclic AMP-inducible gene expression The free catalytic subunit of protein kinase A translocates to the nucleus and phosphorylates the transcription factor CREB (CRE-binding protein), leading to the recruitment of coactivators and expression of cAMP-inducible genes.

kinase A are usually dephosphorylated by the action of a phosphatase called protein phosphatase 1 (Figure 15.23). The levels of phosphorylation of protein kinase A substrates (such as phosphorylase kinase and CREB) are thus determined by a balance between the intracellular activities of protein kinase A and protein phosphatases.

Although most effects of cAMP are mediated by protein kinase A, cAMP can also directly regulate ion channels, independent of protein phosphorylation. Cyclic AMP functions in this way as a second messenger involved in sensing smells. Many of the odorant receptors in sensory neurons in the nose are G protein-coupled receptors that stimulate adenylyl cyclase, leading to an increase in intracellular cAMP. Rather than stimulating protein kinase A, cAMP in this system directly opens Na^+ channels in the plasma membrane, leading to membrane depolarization and initiation of a nerve impulse.

Cyclic GMP

Cyclic GMP (cGMP) is also an important second messenger in animal cells, although its roles are not as clearly understood as those of cAMP. Cyclic GMP is formed from GTP by guanylyl cyclases and degraded to GMP by a phosphodiesterase. As discussed earlier in this chapter, guanylyl cyclases are activated by nitric oxide and carbon monoxide as well as by peptide ligands. Stimulation of these guanylyl cyclases leads to elevated levels of cGMP, which then mediate biological responses, such as blood vessel dilation. The action of cGMP is frequently mediated by activation of cGMP-dependent protein kinases, although cGMP also regulates ion channels and phosphodiesterases.

One well-characterized role of cGMP is in the vertebrate eye, where it serves as the second messenger responsible for converting the visual signals received as light to nerve impulses. The photoreceptor in rod cells of the retina is a G protein-coupled receptor called **rhodopsin** (Figure 15.24). Rhodopsin is activated as a result of the absorption of light by the associated small molecule 11-*cis*-retinal, which then isomerizes to all-*trans*-retinal, inducing a conformational change in the rhodopsin protein. Rhodopsin

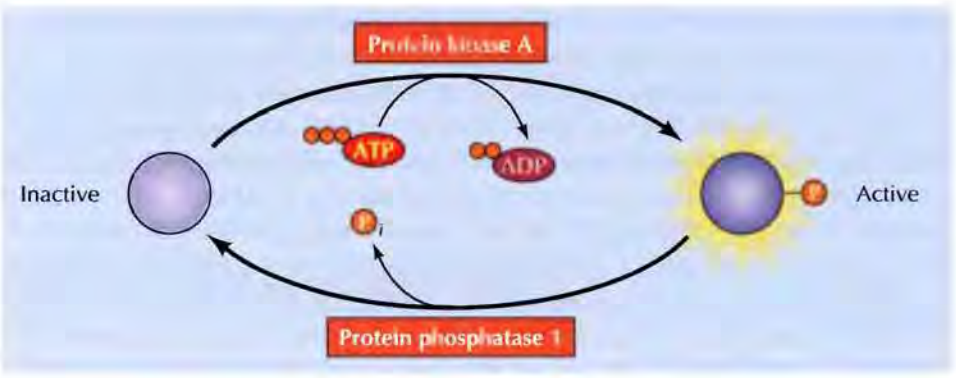


FIGURE 15.23 Regulation of protein phosphorylation by protein kinase A and protein phosphatase 1 The phosphorylation of target proteins by protein kinase A is reversed by the action of protein phosphatase 1.

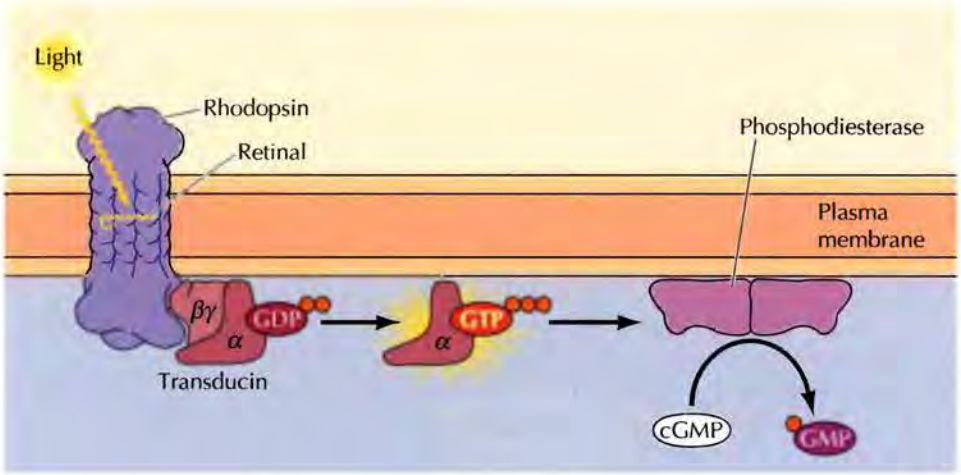


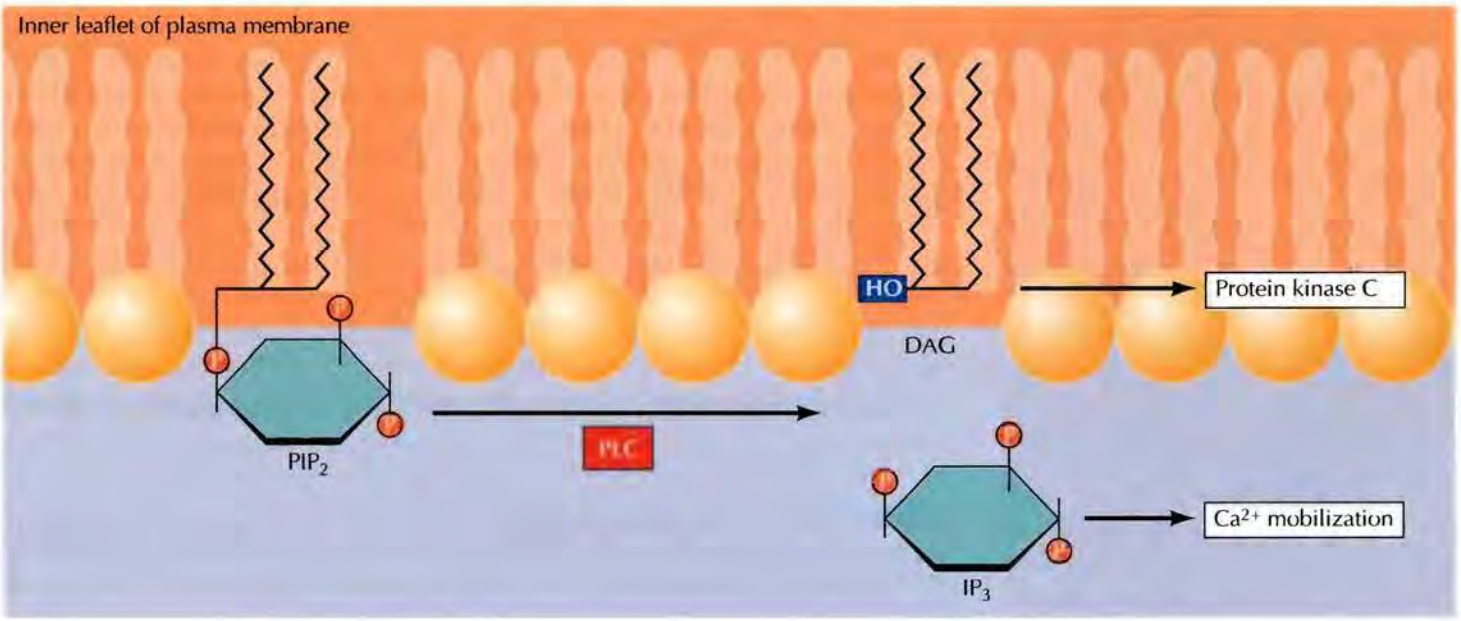
FIGURE 15.24 Role of cGMP in photoreception Absorption of light by retinal activates the G protein-coupled receptor rhodopsin. The α subunit of transducin then stimulates cGMP phosphodiesterase, leading to a decrease in intracellular levels of cGMP.

then activates the G protein **transducin**, and the α subunit of transducin stimulates the activity of **cGMP phosphodiesterase**, leading to a decrease in the intracellular level of cGMP. This change in cGMP level in retinal rod cells is translated to a nerve impulse by a direct effect of cGMP on ion channels in the plasma membrane, similar to the action of cAMP in sensing smells.

Phospholipids and Ca^{2+}

One of the most widespread pathways of intracellular signaling is based on the use of second messengers derived from the membrane phospholipid **phosphatidylinositol 4,5-bisphosphate (PIP_2)**. PIP_2 is a minor component of the plasma membrane, localized to the inner leaflet of the phospholipid bilayer (see Figure 13.2). A variety of hormones and growth factors stimulate the hydrolysis of PIP_2 by **phospholipase C**—a reaction that produces two distinct second messengers, **diacylglycerol** and **inositol 1,4,5-trisphosphate (IP_3)** (Figure 15.25). Diacylglycerol and IP_3 stimulate distinct down-

FIGURE 15.25 Hydrolysis of PIP_2 Phospholipase C (PLC) catalyzes the hydrolysis of phosphatidylinositol 4,5-bisphosphate (PIP_2) to yield diacylglycerol (DAG) and inositol trisphosphate (IP_3). Diacylglycerol activates members of the protein kinase C family, and IP_3 signals the release of Ca^{2+} from intracellular stores.



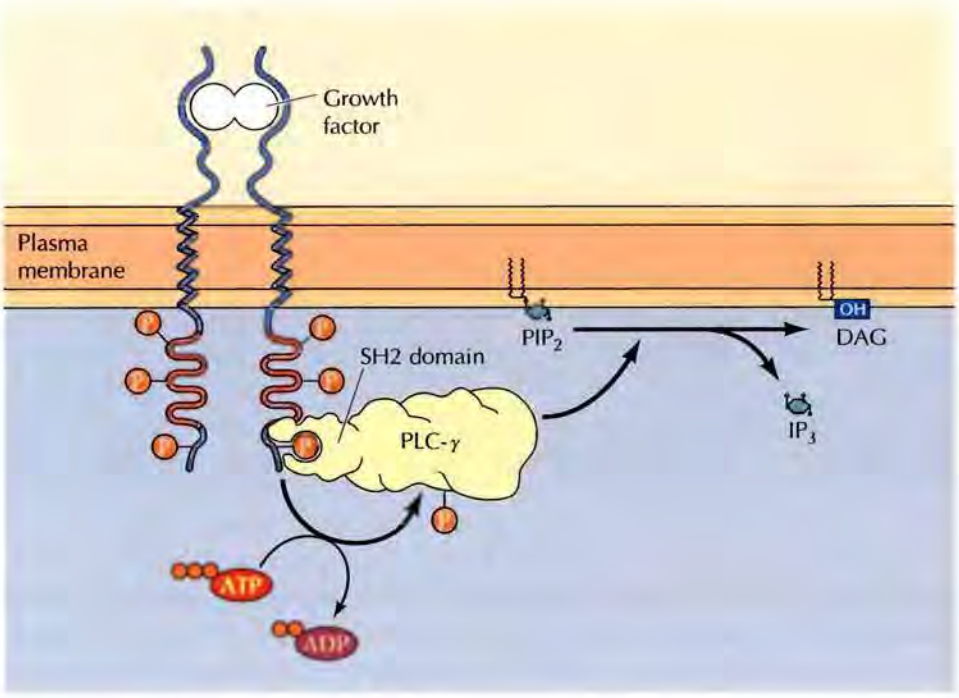
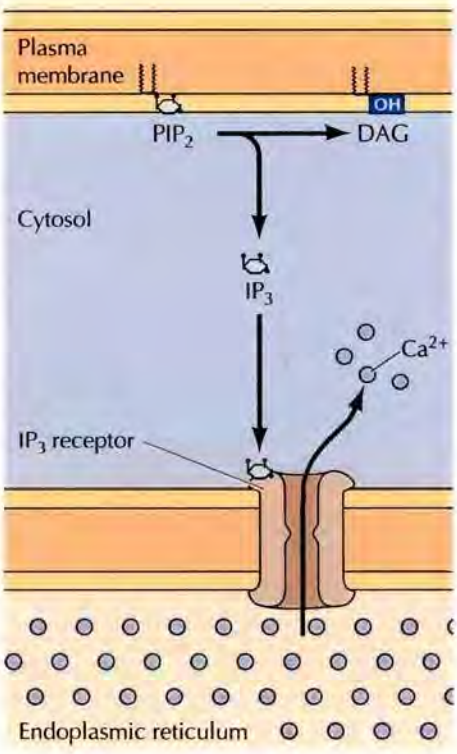


FIGURE 15.26 Activation of phospholipase C by protein-tyrosine kinases Phospholipase C-γ (PLC-γ) binds to activated receptor protein-tyrosine kinases via its SH2 domains. Tyrosine phosphorylation increases PLC-γ activity, stimulating the hydrolysis of PIP_2.

■ **Toxic snake venoms contain phospholipases. Hydrolysis of phospholipids by venom from rattlesnakes and cobras leads to the rupture of red blood cell membranes.**



stream signaling pathways (protein kinase C and Ca^{2+} mobilization, respectively), so PIP_2 hydrolysis triggers a two-armed cascade of intracellular signaling.

It is noteworthy that the hydrolysis of PIP_2 is activated downstream of both G protein-coupled receptors and protein-tyrosine kinases. This occurs because one form of phospholipase C (PLC-β) is stimulated by G proteins, whereas a second form of phospholipase C (PLC-γ) contains SH2 domains that mediate its association with activated receptor protein-tyrosine kinases (Figure 15.26). This interaction localizes PLC-γ to the plasma membrane as well as leading to its tyrosine phosphorylation, which increases its catalytic activity.

The diacylglycerol produced by hydrolysis of PIP_2 remains associated with the plasma membrane and activates protein-serine/threonine kinases belonging to the **protein kinase C** family, many of which play important roles in the control of cell growth and differentiation. The other second messenger produced by PIP_2 cleavage, IP_3, is a small polar molecule that is released into the cytosol, where it acts to signal the release of Ca^{2+} from intracellular stores (Figure 15.27). As noted in Chapter 13, the cytosolic concentration of Ca^{2+} is maintained at an extremely low level (about 0.1 μM) as a result of Ca^{2+} pumps that actively export Ca^{2+} from the cell. Ca^{2+} is pumped not only across the plasma membrane but also into the endoplasmic reticulum, which therefore serves as an intracellular Ca^{2+} store. IP_3 acts to release Ca^{2+} from the endoplasmic reticulum by binding to receptors that are ligand-gated Ca^{2+} channels. As a result, cytosolic Ca^{2+} levels increase to about 1 μM, which affects the activities of a variety of target proteins, including

FIGURE 15.27 Ca^{2+} mobilization by IP_3 Ca^{2+} is pumped from the cytosol into the endoplasmic reticulum, which serves as an intracellular Ca^{2+} store. IP_3 binds to receptors that are ligand-gated Ca^{2+} channels in the endoplasmic reticulum membrane, thereby allowing the efflux of Ca^{2+} into the cytosol.

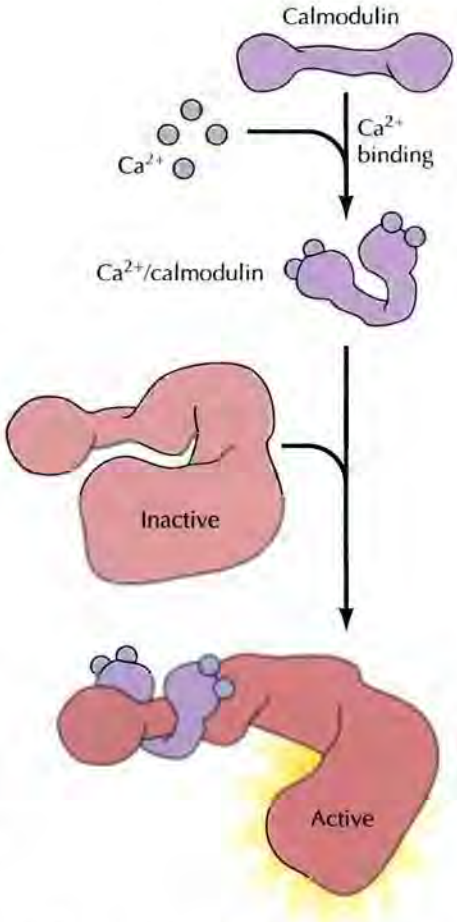


FIGURE 15.28 Function of calmodulin Calmodulin is a dumbbell-shaped protein with four Ca^{2+}-binding sites. The active Ca^{2+}/calmodulin complex binds to a variety of target proteins, including Ca^{2+}/ calmodulin-dependent protein kinases.

protein kinases and phosphatases. For example, some members of the protein kinase C family require Ca^{2+} as well as diacylglycerol for their activation, so these protein kinases are regulated jointly by both arms of the PIP_2 signaling pathway. In most cells, the transient increase in intracellular Ca^{2+} resulting from production of IP_3 triggers a more sustained increase caused by the entry of extracellular Ca^{2+} through channels in the plasma membrane. This entry of Ca^{2+} from outside the cell serves both to prolong the signal initiated by release of Ca^{2+} from the endoplasmic reticulum and to allow the stores of Ca^{2+} within the endoplasmic reticulum to be replenished.

Many of the effects of Ca^{2+} are mediated by the Ca^{2+}-binding protein **calmodulin**, which is activated when the concentration of cytosolic Ca^{2+} increases to about 0.5 μM (Figure 15.28). Ca^{2+}/calmodulin then binds to a variety of target proteins, including protein kinases. One example of such a Ca^{2+}/calmodulin-dependent protein kinase is myosin light-chain kinase, which signals actin-myosin contraction by phosphorylating one of the myosin light chains (see Figure 12.31). Other protein kinases that are activated by Ca^{2+}/calmodulin include members of the **CaM kinase** family, which phosphorylate a number of different proteins, including metabolic enzymes, ion channels, and transcription factors. One form of CaM kinase is particularly abundant in the nervous system where it regulates the synthesis and release of neurotransmitters. In addition, CaM kinases can regulate gene expression by phosphorylating transcription factors. Interestingly, one of the transcription factors phosphorylated by CaM kinase is CREB, which is phosphorylated at the same site by protein kinase A. This phosphorylation of CREB illustrates one of many intersections between the Ca^{2+} and cAMP signaling pathways. Other examples include the regulation of adenylyl cyclases and phosphodiesterases by Ca^{2+}/calmodulin, the regulation of Ca^{2+} channels by cAMP, and the phosphorylation of a number of target proteins by both protein kinase A and Ca^{2+}/calmodulin-dependent protein kinases. The cAMP and Ca^{2+} signaling pathways thus function coordinately to regulate many cellular responses.

The entry of extracellular Ca^{2+} is particularly important in the electrically excitable cells of nerve and muscle in which voltage-gated Ca^{2+} channels in the plasma membrane are opened by membrane depolarization (Figure 15.29). The resulting increases in intracellular Ca^{2+} then trigger the further release of Ca^{2+} from intracellular stores by activating distinct Ca^{2+} channels known as **ryanodine receptors**. One effect of increases in intracellular Ca^{2+} in neurons is to trigger the release of neurotransmitters, so Ca^{2+} plays a crit-

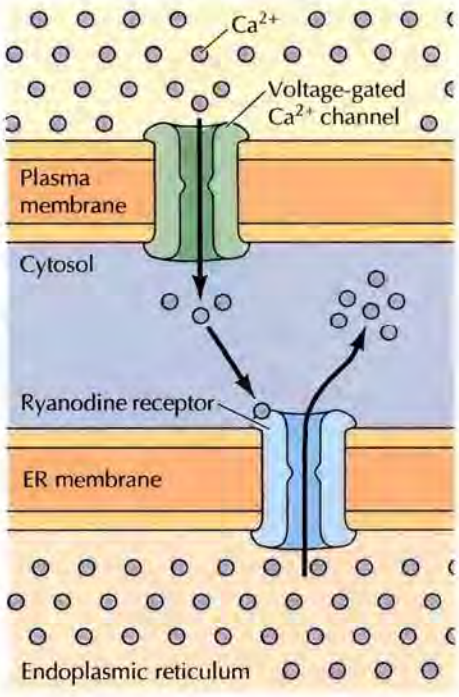


FIGURE 15.29 Regulation of intracellular Ca^{2+} in electrically excitable cells Membrane depolarization leads to the opening of voltage-gated Ca^{2+} channels in the plasma membrane causing the influx of Ca^{2+} from extracellular fluids. The resulting increase in intracellular Ca^{2+} then signals the further release of Ca^{2+} from intracellular stores by opening distinct Ca^{2+} channels (ryanodine receptors) in the endoplasmic reticulum membrane. In muscle cells, ryanodine receptors in the sarcoplasmic reticulum may also be opened directly in response to membrane depolarization.

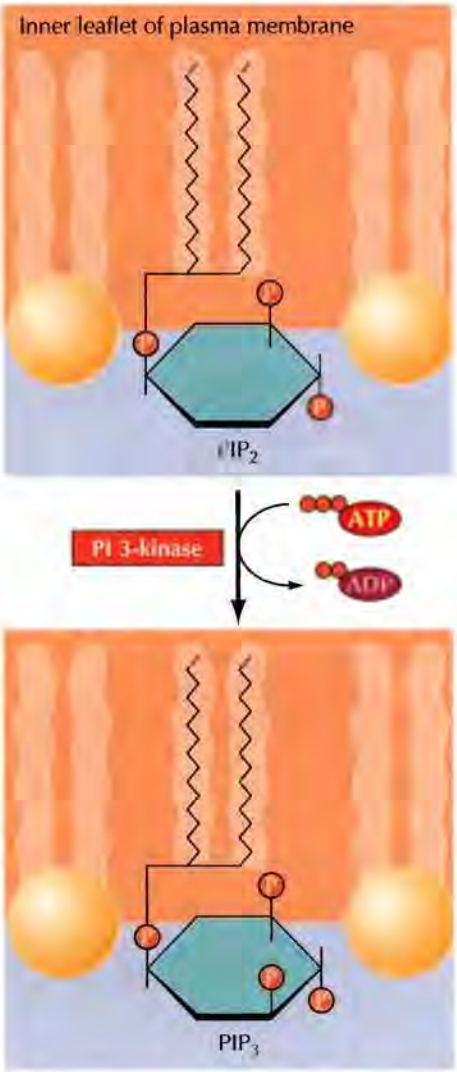


FIGURE 15.30 Activity of PI 3-kinase PI 3-kinase phosphorylates the 3 position of inositol, converting PIP_2 to PIP_3.

ical role in converting electric to chemical signals in the nervous system. In muscle cells Ca^{2+} is stored in the sarcoplasmic reticulum from which it is released by the opening of ryanodine receptors in response to changes in membrane potential. This release of stored Ca^{2+} leads to large increases in cytosolic Ca^{2+}, which trigger muscle contraction (see Chapter 12). Cells thus utilize a variety of mechanisms to regulate intracellular Ca^{2+} levels, making Ca^{2+} a versatile second messenger that controls a wide range of cellular processes.

The PI 3-Kinase/Akt and mTOR Pathways

PIP_2 not only serves as the source of diacylglycerol and IP_3 but is also the starting point of a distinct second messenger pathway that plays a key role in regulating cell growth and survival. In this pathway, PIP_2 is phosphorylated on the 3 position of inositol by the enzyme **phosphatidylinositide (PI) 3-kinase** (Figure 15.30). Like phospholipase C, one form of PI 3-kinase is activated by G proteins while a second form has SH2 domains and is activated by association with receptor protein-tyrosine kinases. Phosphorylation of PIP_2 yields the second messenger **phosphatidylinositol 3,4,5-trisphosphate (PIP_3)**.

A key target of PIP_3, which is critical for signaling cell proliferation and survival, is a protein-serine/threonine kinase called **Akt**. PIP_3 binds to a domain of Akt known as the pleckstrin homology domain (Figure 15.31). This interaction recruits Akt to the inner face of the plasma membrane where it is phosphorylated and activated by another protein kinase (called PDK1) that also contains a pleckstrin homology domain and binds PIP_3. The formation of PIP_3 thus results in the association of both Akt and PDK1 with the plasma membrane, leading to phosphorylation and activation of Akt. Activation of Akt also requires phosphorylation at a second site by a distinct protein kinase, which has recently been identified as a form of mTOR complexed with a protein called rictor. The mTOR/rictor complex is itself stimulated by growth factors, but its mechanism of activation remains to be understood.

Once activated, Akt phosphorylates a number of target proteins, including proteins that are direct regulators of cell proliferation and survival (discussed in Chapter 17), transcription factors, and other protein kinases. The critical transcription factors targeted by Akt include members of the Forkhead or FOXO family (Figure 15.32). Phosphorylation of FOXO by Akt creates a binding site for cytosolic chaperone proteins (14-3-3 proteins) that sequester FOXO in an inactive form in the cytoplasm. In the absence of growth factor signaling and Akt activity, FOXO is released from 14-3-3 and translocates to the nucleus, stimulating transcription of genes that inhibit cell proliferation or induce cell death. Another target of Akt is the protein kinase GSK-3β, which regulates metabolism as well as cell proliferation and survival. Like FOXO, GSK-3β is inhibited by Akt phosphorylation. The targets of GSK-3β include several transcription factors and the translation initiation factor eIF-2B. Phosphorylation of eIF-2B leads to a global downregulation of translation initiation (see Figure 8.20), so GSK-3β provides a link between growth factor signaling and control of cellular protein synthesis.

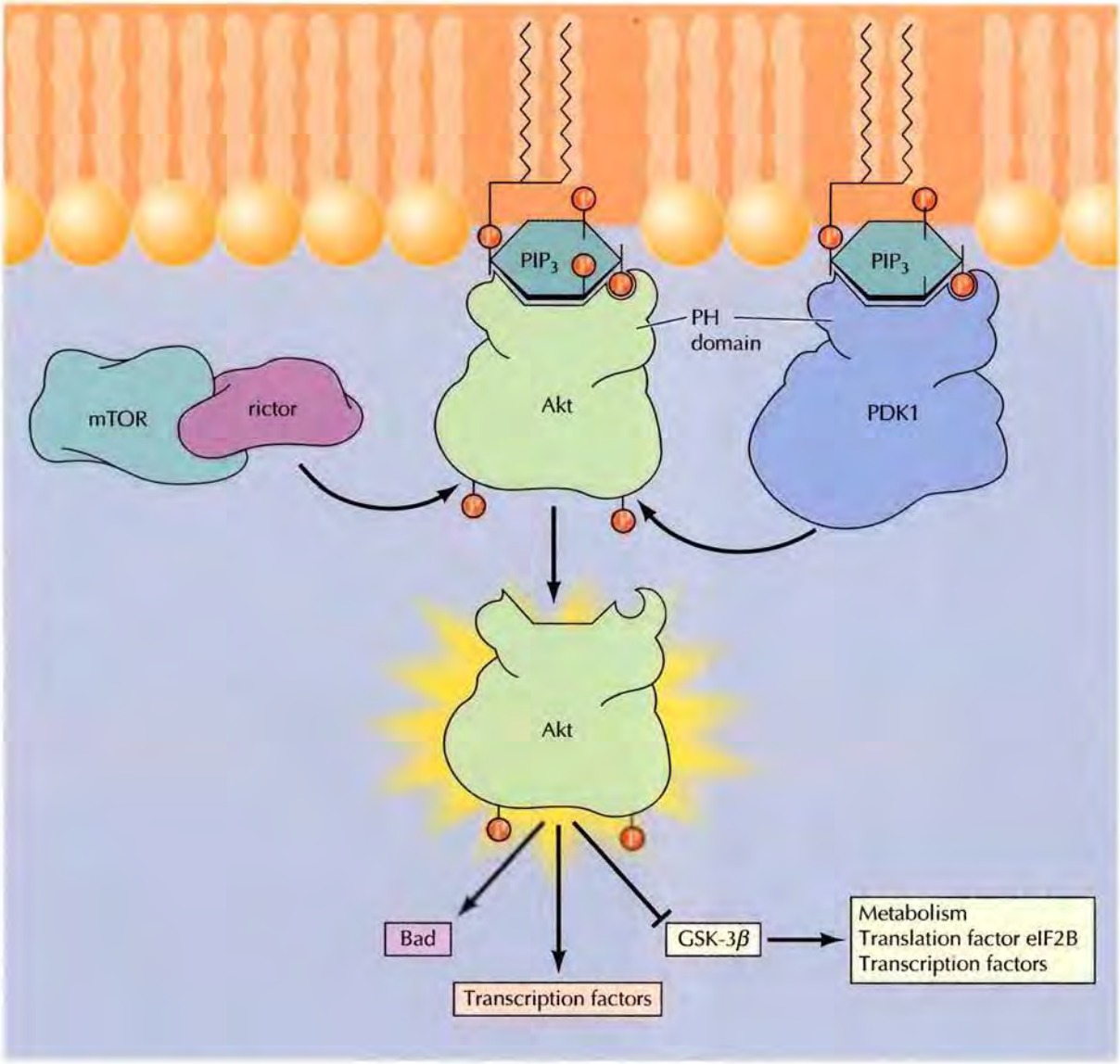


FIGURE 15.31 The PI 3-kinase/Akt pathway Akt is recruited to the plasma membrane by binding to PIP_3 via its pleckstrin homology (PH) domain. It is then activated as a result of phosphorylation by another protein kinase (PDK1) that also binds PIP_3, as well as by the mTOR/rictor complex. Akt then phosphorylates a number of target proteins, including direct regulators of cell survival (Bad, see Chapter 17), several transcription factors, and the protein kinase GSK-3β (which is inhibited by Akt phosphorylation). GSK-3β phosphorylates metabolic enzymes, transcription factors, and the translation initiation factor eIF-2B.

The **mTOR** pathway is a central regulator of cell growth that couples the control of protein synthesis to the availability of growth factors, nutrients, and energy (Figure 15.33). This is accomplished via the regulation of mTOR by multiple signals, including the PI 3-kinase/Akt pathway. The mTOR protein kinase exists in two distinct complexes in cells in which mTOR is associated with either rictor or raptor. As discussed above, the mTOR/rictor complex is one of the protein kinases that phosphorylates and activates Akt (see Figure 15.31). In contrast, the mTOR/raptor complex is activated downstream of Akt and functions to regulate cell size, at least in part by

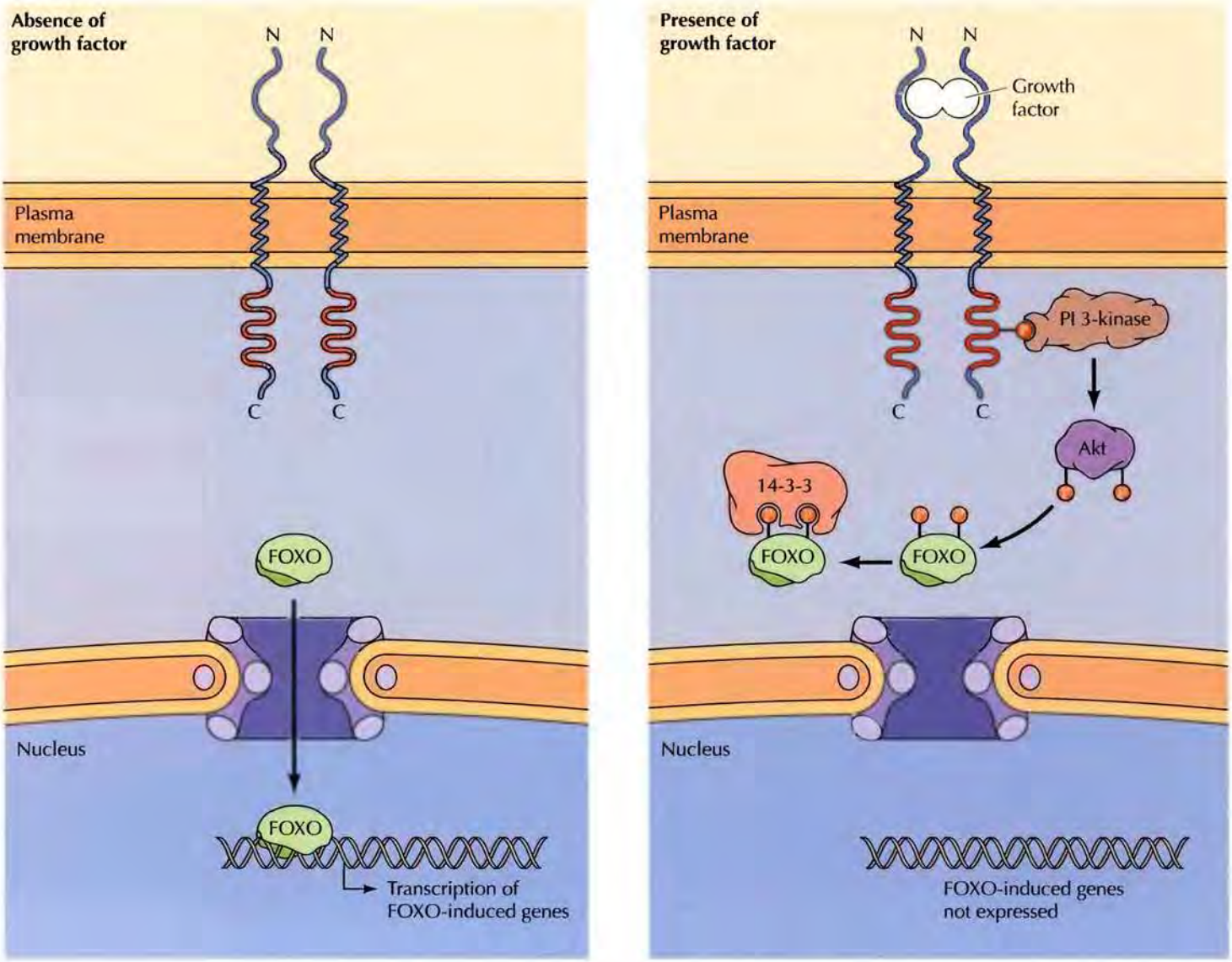


FIGURE 15.32 Regulation of FOXO In the absence of growth factor stimulation, the FOXO transcription factor translocates to the nucleus and induces target gene expression. Growth factor stimulation leads to activation of Akt, which phosphorylates FOXO. This creates binding sites for the cytosolic chaperone 14-3-3, which sequesters FOXO in an inactive form in the cytoplasm.

controlling protein synthesis. The mTOR/raptor complex is regulated by the Ras-related GTP-binding protein Rheb, which is in turn regulated by the GTPase-activating protein complex TSC1/2. Akt phosphorylates TSC2, leading to activation of mTOR/raptor in response to growth factor stimulation. In addition, TSC2 is regulated by another protein kinase called the AMP/activated kinase (AMPK). AMPK senses the energy state of the cell and is activated by a high ratio of AMP to ATP. Under these conditions, AMPK phosphorylates TSC2, leading to inhibition of mTOR/raptor when cellular energy stores are depleted. TSC2 is also regulated by the availability of amino acids, although the mechanism responsible remains to be established.

The mTOR/raptor complex phosphorylates at least two well-characterized targets that function to regulate protein synthesis: S6 kinase and eIF4E binding protein-1 (4E-BP1). S6 kinase controls translation by phosphorylating the ribosomal protein S6 as well as other proteins involved in translational regulation. The eIF4E binding protein controls translation by interacting with initiation factor eIF4E, which binds to the 5′ cap of mRNAs. In the absence of mTOR signaling, nonphosphorylated 4E-BPs bind to eIF4E and inhibit translation by interfering with the interaction of eIF4E with eIF4G (see Figure 8.11). Phosphorylation of 4E-BP1 by mTOR prevents its interaction with eIF4E, leading to increased rates of translation initiation.

■ **Rapamycin, an antibiotic produced by certain fungi, is a specific inhibitor of the mTOR/raptor complex and is used as an immunosuppressive drug in organ transplants.**

MAP Kinase Pathways

The MAP kinase pathway refers to a cascade of protein kinases that are highly conserved in evolution and play central roles in signal transduction in all eukaryotic cells ranging from yeasts to humans. The central elements in the pathway are a family of protein-serine/threonine kinases called the **MAP kinases** (for *m*itogen-*a*ctivated *p*rotein kinases) that are activated in response to a variety of growth factors and other signaling molecules. In yeasts, MAP kinase pathways control a variety of cellular responses, including mating, cell shape, and sporulation. In higher eukaryotes (including *C. elegans*, *Drosophila*, frogs, and mammals), MAP kinases are ubiquitous regulators of cell growth and differentiation.

The MAP kinases that were initially characterized in mammalian cells belong to the **ERK** (*e*xtracellular signal-*r*egulated *k*inase) family. ERK activation plays a central role in signaling cell proliferation induced by growth factors that act through either protein-tyrosine kinase or G protein-coupled receptors. Protein kinase C can also activate the ERK pathway, and both the Ca^{2+} and cAMP pathways intersect with ERK signaling, either stimulating or inhibiting the ERK pathway in different types of cells.

Activation of ERK is mediated by two upstream protein kinases, which are coupled to growth factor receptors by the **Ras** GTP-binding protein (Figure 15.34). Activation of Ras leads to activation of the **Raf** protein-serine/threonine kinase, which phosphorylates and activates a second protein kinase called **MEK** (for *M*AP kinase/*E*RK *k*inase). MEK is a dual-specificity protein kinase that activates members of the ERK family by phosphorylation of both threonine and tyrosine residues separated by one amino acid (e.g., threonine-183 and tyrosine-185 of ERK2). Once activated, ERK phosphorylates a variety of targets, including other protein kinases and transcription factors.

The central role of the ERK pathway in mammalian cells emerged from studies of the Ras proteins, which were first identified as the oncogenic proteins of tumor viruses that cause sarcomas in rats (hence the name Ras, from *ra*t *s*arcoma virus). Interest in Ras intensified considerably in 1982 when mutations in *ras* genes were first implicated in the development of

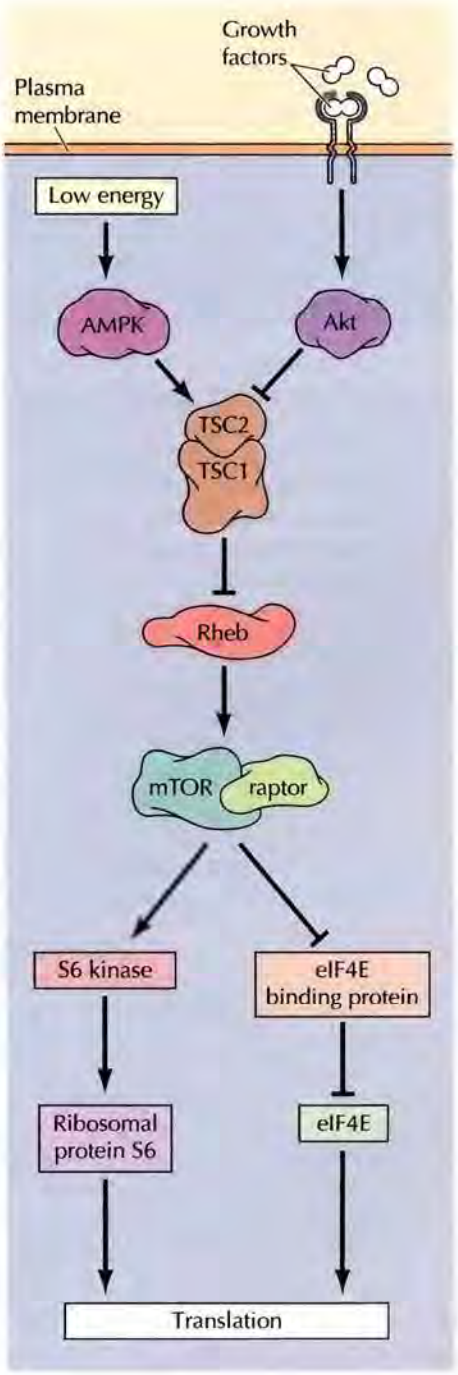


FIGURE 15.33 The mTOR pathway The mTOR/raptor protein kinase is activated by Rheb, which is inhibited by the TSC1/2 complex. Akt inhibits TSC1/2, leading to activation of Rheb and mTOR/raptor in response to growth factor stimulation. In contrast, AMPK activates TSC1/2, leading to inhibition of Rheb and mTOR/raptor if cellular energy stores are depleted. mTOR/raptor stimulates translation by phosphorylating S6 kinase (which phosphorylates ribosomal protein S6) and by phosphorylating eIF4E binding protein, relieving inhibition of translation initiation factor eIF4E.

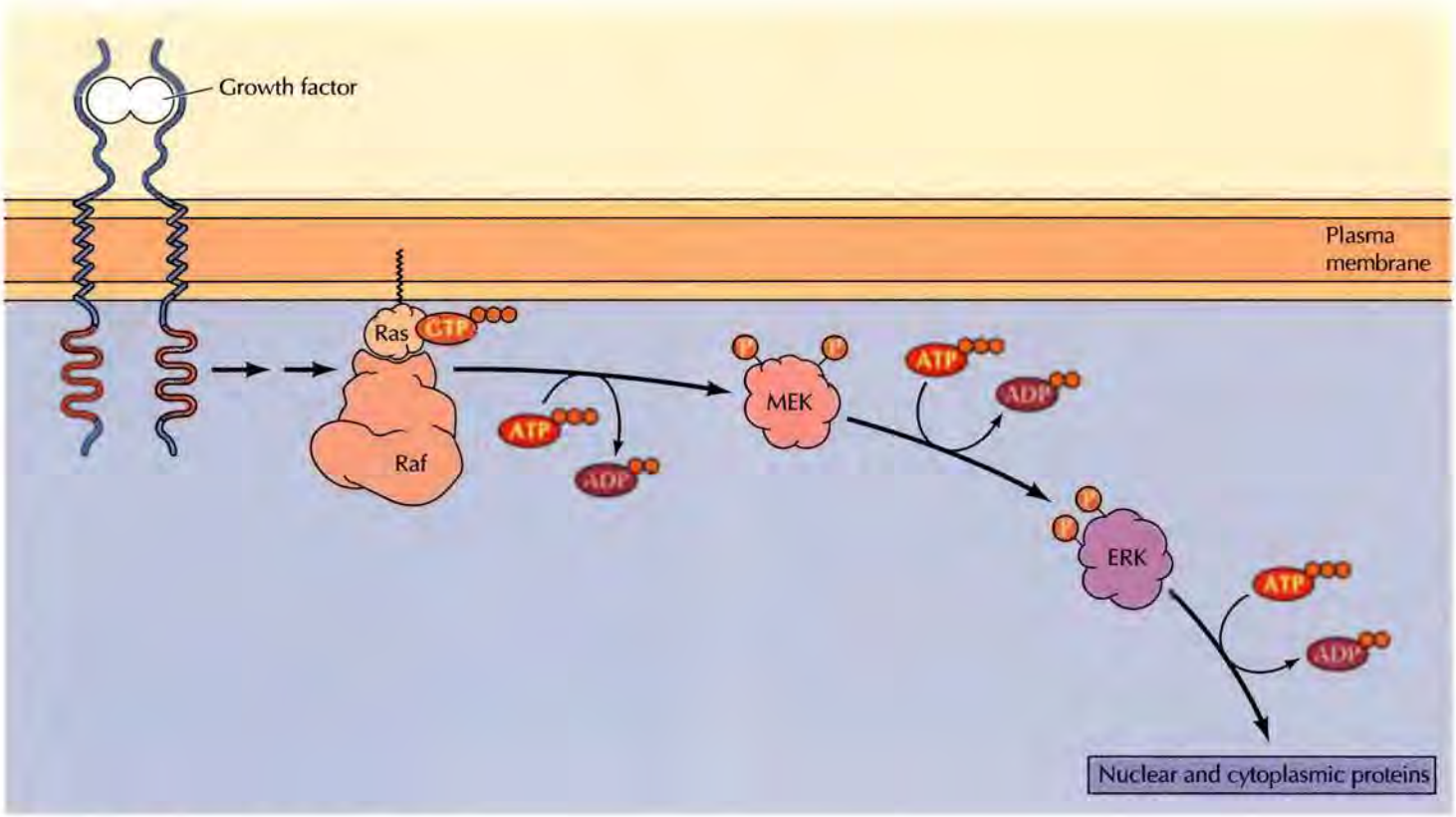


FIGURE 15.34 Activation of the ERK MAP kinases Stimulation of growth factor receptors leads to activation of the small GTP-binding protein Ras, which interacts with the Raf protein kinase. Raf phosphorylates and activates MEK, a dual-specificity protein kinase that activates ERK by phosphorylation on both threonine and tyrosine residues (Thr-183 and Tyr-185). ERK then phosphorylates a variety of nuclear and cytoplasmic target proteins.

human cancers (discussed in Chapter 18). The importance of Ras in intracellular signaling was then indicated by experiments showing that microinjection of active Ras protein directly induces proliferation of normal mammalian cells. Conversely, interference with Ras function by either microinjection of anti-Ras antibody or expression of a dominant negative Ras mutant blocks growth factor-induced cell proliferation. Thus Ras is not only capable of inducing the abnormal growth characteristic of cancer cells but also appears to be required for the response of normal cells to growth factor stimulation.

The Ras proteins are guanine nucleotide-binding proteins that function analogously to the α subunits of G proteins, alternating between inactive GDP-bound and active GTP-bound forms (Figure 15.35). In contrast to the G protein α subunits, however, Ras functions as a monomer rather than in association with $\beta\gamma$ subunits. Ras activation is mediated by **guanine nucleotide exchange factors** that stimulate the release of bound GDP and its exchange for GTP. Activity of the Ras-GTP complex is then terminated by GTP hydrolysis, which is stimulated by the interaction of Ras-GTP with **GTPase-activating proteins**. It is interesting to note that the mutations of *ras* genes in human cancers have the effect of inhibiting GTP hydrolysis by the Ras proteins. These mutated Ras proteins therefore remain continuously in the active GTP-bound form, driving the unregulated proliferation of cancer cells even in the absence of growth factor stimulation.

The Ras proteins are prototypes of a large family of approximately 50 related proteins frequently called small GTP-binding proteins because Ras and its relatives are about half the size of G protein α subunits. One member of this family is Rheb, which regulates mTOR signaling (see Figure 15.33). Other subfamilies of small GTP-binding proteins control a vast array

Cancer: Signal Transduction and the ras Oncogenes

The Disease

Cancer claims the lives of approximately one out of every four Americans, accounting for almost 600,000 deaths each year in the United States. There are more than a hundred different kinds of cancer, but some are more common than others. In this country the most common lethal cancers are those of the lung and colon/rectum, which together account for about 40% of all cancer deaths. Other major contributors to cancer mortality include cancers of the breast, pancreas, and prostate, which are responsible for approximately 7.2%, 5.6%, and 5.3% of U.S. cancer deaths, respectively.

The common feature of all cancers is the unrestrained proliferation of cancer cells, which eventually spread throughout the body, invading normal tissues and organs and leading to death of the patient. Surgery and radiotherapy are effective treatments for localized cancers but are unable to reach cancer cells that have spread to distant body sites. Treatment of these cancers therefore requires the use of chemotherapeutic drugs. Unfortunately, the commonly available chemotherapeutic agents are not specific for cancer cells. Most act by either damaging DNA or interfering with DNA synthesis, so they also kill rapidly dividing normal cells, such as the epithelial cells that line the digestive tract and the blood-forming cells of the bone marrow. The resulting toxicity of these drugs limits their effectiveness, and many cancers are not eliminated by doses of chemotherapy that can be tolerated by the patient. Consequently, although major progress has been made in cancer treatment, nearly half of all patients diagnosed with cancer ultimately die of their disease.

Molecular and Cellular Basis

The identification of viral genes that can convert normal cells to cancer cells, such as the *src* gene of RSV, provided the first demonstration that cancers can result from the action of specific genes (oncogenes). The subsequent discovery that viral oncogenes are related to genes of normal cells then engendered the hypothesis that non-virus-induced cancers (including most human cancers) might arise as a result of mutations in normal cell genes, giving rise to oncogenes of cellular rather than viral origin. Such cellular oncogenes were first identified in human cancers in 1981. The following year, human oncogenes of bladder, lung, and colon cancers were found to be related to the *ras* genes previously identified in rat sarcoma viruses.

Although many different genes are now known to play critical roles in cancer development, mutations of the *ras* genes remain one of the most common genetic abnormalities in human tumors. Mutated *ras* oncogenes are found in about 20% of all human cancers, including approximately 25% of lung cancers, 50% of colon cancers, and more than 90% of pancreatic cancers. Moreover, the action of *ras* oncogenes has clearly linked the development of human cancer to abnormalities in the signaling pathways that regulate cell proliferation. The mutations that convert normal *ras* genes to oncogenes substantially decrease GTP hydrolysis by the Ras proteins. Consequently, the mutated oncogenic Ras proteins remain locked in the active GTP-bound form, rather than alternating normally between inactive and active states in response to extracellular signals. The oncogenic Ras proteins thus continuously stimulate the ERK signaling pathway and drive cell proliferation, even in the absence of the growth factors that would be required to activate Ras and signal proliferation of normal cells.

Prevention and Treatment

The discovery of mutated oncogenes in human cancers raised the possibility of developing drugs specifically targeted against the oncogene proteins. In principle, such drugs might act selectively against cancer cells with less toxicity toward normal cells than that of conventional chemotherapeutic agents. Because *ras* is frequently mutated in human cancers,

A human colon polyp (an early stage of colon cancer). The *ras* oncogenes contribute to the development of about half of all colon cancers. (E. P. Ewing, Jr., Centers for Disease Control.)

the Ras proteins and other elements of Ras signaling pathways have attracted considerable interest as potential drug targets.

As discussed in Chapter 18, effective drugs that selectively target cancer cells have been recently developed against some protein-tyrosine kinase oncogenes, including the EGF receptor, that act upstream of Ras. A variety of additional drugs are under investigation as potential cancer treatments, including drugs targeted against Ras itself and against protein kinases activated downstream of Ras, such as Raf. The identification of oncogenes in human tumors has thus opened new strategies to rational development of drugs that act effectively and selectively against human cancer cells by targeting the signaling pathways that are responsible for cancer development.

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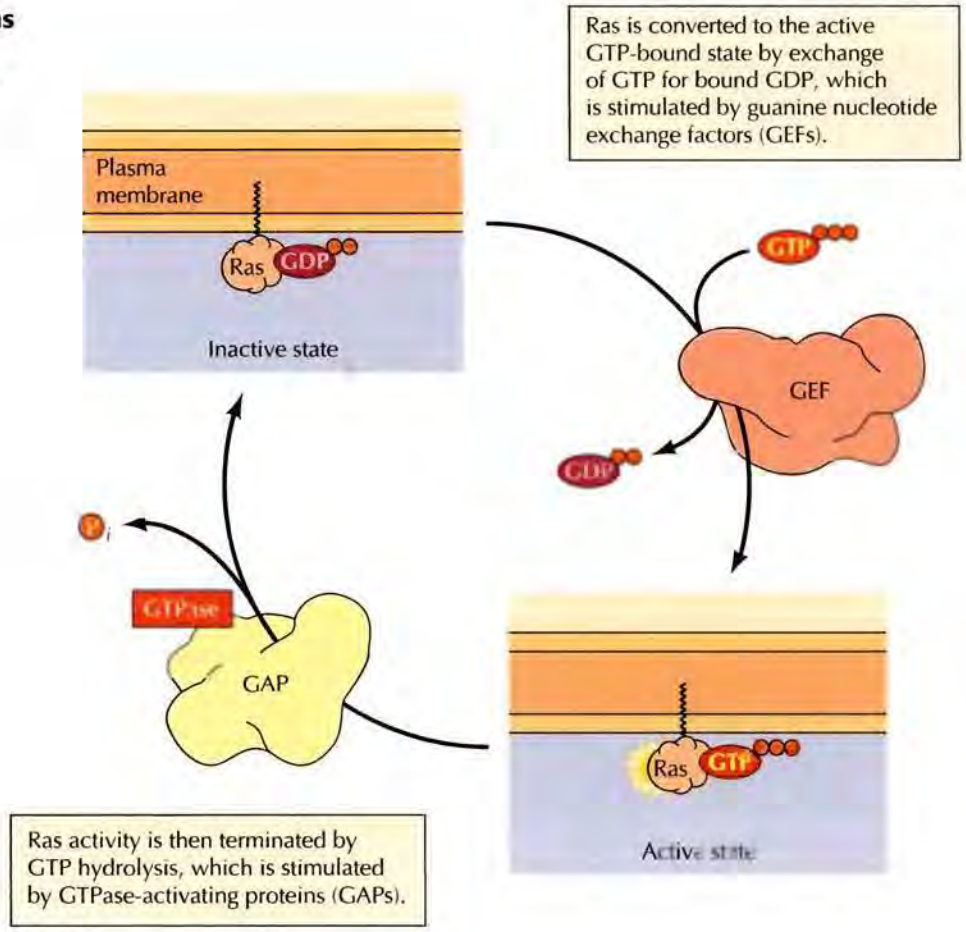
Sawyers, C. 2004. Targeted cancer therapy. *Nature* 432: 294–297.

FIGURE 15.35 Regulation of Ras proteins
Ras proteins alternate between inactive GDP-bound and active GTP-bound states.



of cellular activities. For example, the largest subfamily of small GTP-binding proteins (the Rab proteins) functions to regulate vesicle trafficking, as discussed in Chapter 10. Other small GTP-binding proteins are involved in the import and export of proteins from the nucleus (the Ran protein, discussed in Chapter 9) and organization of the cytoskeleton (the Rho subfamily, discussed later in this chapter).

The best understood mode of Ras activation is that mediated by receptor protein-tyrosine kinases (Figure 15.36). Autophosphorylation of these receptors results in their association with Ras guanine nucleotide exchange factors as a result of SH2-mediated protein interactions. One well-characterized example is provided by the guanine nucleotide exchange factor Sos, which is bound to the SH2-containing protein Grb2 in the cytosol of unstimulated cells. Tyrosine phosphorylation of receptors (or of other receptor-associated proteins) creates a binding site for the Grb2 SH2 domains. Association of Grb2 with activated receptors localizes Sos to the plasma membrane where it is able to interact with Ras proteins, which are anchored to the inner leaflet of the plasma membrane by lipids attached to the Ras C terminus (see Figure 13.9). Sos then stimulates guanine nucleotide exchange resulting in formation of the active Ras-GTP complex. In its active GTP-bound form Ras interacts with a number of effector proteins, including the Raf protein-serine/threonine kinase. This interaction with Ras recruits Raf from the cytosol to the plasma membrane and initiates Raf activation, which also involves phosphorylation of Raf by both protein-tyrosine and protein-serine/threonine kinases.

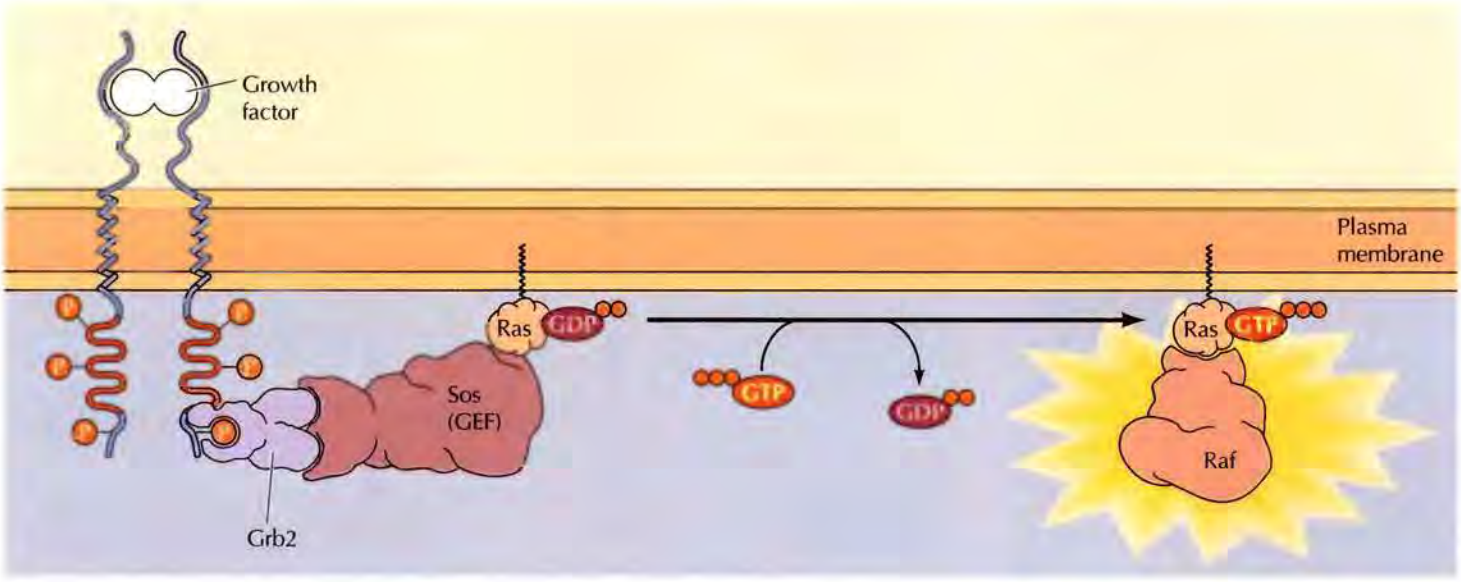


FIGURE 15.36 Ras activation downstream of receptor protein-tyrosine kinases A complex of Grb2 and the guanine nucleotide exchange factor Sos binds to a phosphotyrosine-containing sequence in the activated receptor via the Grb2 SH2 domain. This interaction recruits Sos to the plasma membrane where it can stimulate Ras GDP/GTP exchange. The activated Ras-GTP complex then binds to the Raf protein kinase.

As already noted, activation of Raf initiates a protein kinase cascade leading to ERK activation. ERK then phosphorylates a variety of target proteins, including other protein kinases. In addition, ERK regulates the mTOR pathway by phosphorylating TSC2 (see Figure 15.33). Importantly, a fraction of activated ERK translocates to the nucleus where it regulates transcription factors by phosphorylation (Figure 15.37). In this regard, it is notable that a primary response to growth factor stimulation is the rapid transcriptional induction of a family of approximately 100 genes called **immediate-early genes**. The induction of a number of immediate-early genes is mediated by a regulatory sequence called the **serum response element (SRE)**, which is recognized by a complex of transcription factors including the **serum response factor (SRF)** and **Elk-1**. ERK phosphorylates and activates Elk-1, providing a direct link between the ERK family of MAP kinases and immediate-early gene induction. Many immediate-early genes themselves encode transcription factors, so their induction in response to growth factor stimulation leads to altered expression of a battery of other downstream genes called **secondary response genes**. As discussed in Chapter 16, these alterations in gene expression directly link ERK signaling to the stimulation of cell proliferation induced by growth factors.

Both yeasts and mammalian cells have multiple MAP kinase pathways that control distinct cellular responses. Each cascade consists of three protein kinases: a terminal MAP kinase and two upstream kinases (analogous to Raf and MEK) that regulate its activity. In the yeast *S. cerevisiae* five different MAP kinase cascades regulate mating, sporulation, filamentation, cell wall remodeling, and response to high osmolarity. In mammalian cells, three major groups of MAP kinases have been identified. In addition to

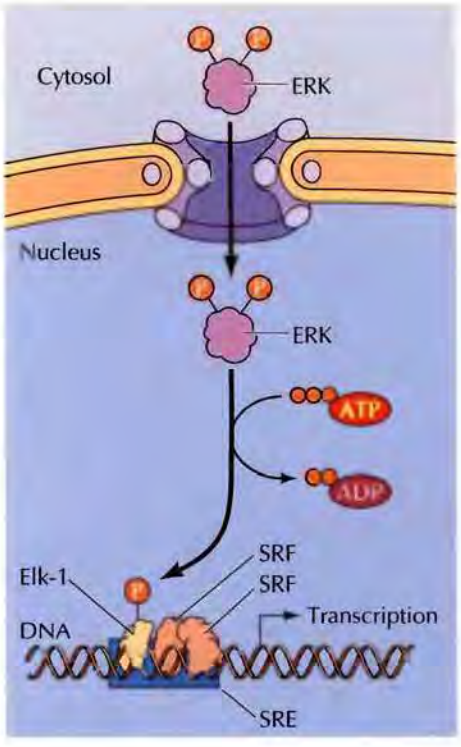


FIGURE 15.37 Induction of immediate-early genes by ERK Activated ERK translocates to the nucleus where it phosphorylates the transcription factor Elk-1. Elk-1 binds to the serum response element (SRE) in a complex with serum response factor (SRF). Phosphorylation stimulates the activity of Elk-1 as a transcriptional activator, leading to immediate-early gene induction.

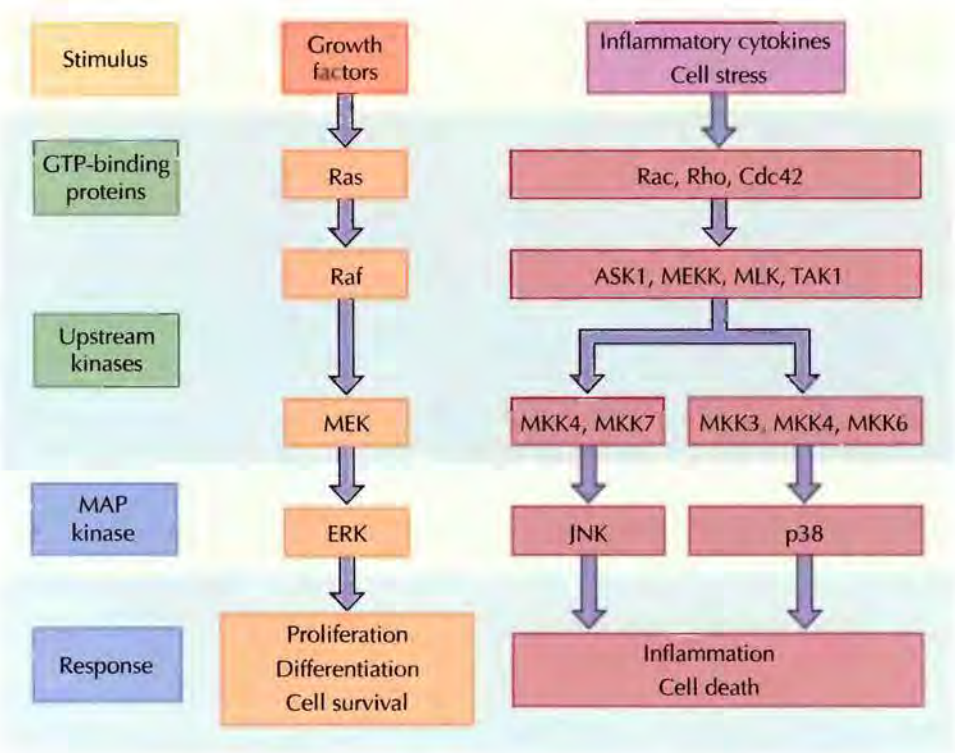


FIGURE 15.38 Pathways of MAP kinase activation in mammalian cells In addition to ERK, mammalian cells contain JNK and p38 MAP kinases. Activation of JNK and p38 is mediated by members of the Rho subfamily of small GTP-binding proteins (Rac, Rho, and Cdc42), which stimulate protein kinase cascades parallel to that responsible for ERK activation. The protein kinase cascades leading to JNK and p38 activation appear to be preferentially activated by inflammatory cytokines or cellular stress and generally lead to inflammation and cell death.

members of the ERK family these include the JNK and p38 MAP kinases, which are preferentially activated in response to inflammatory cytokines and cellular stress (e.g., ultraviolet irradiation) (Figure 15.38). The JNK and p38 MAP kinase cascades are activated by members of the Rho subfamily of small GTP-binding proteins (including Rac, Rho, and Cdc42) rather than by Ras. Whereas ERK signaling principally leads to cell proliferation, survival, and differentiation, the JNK and p38 MAP kinase pathways often lead to inflammation and cell death. Like ERK, the JNK and p38 MAP kinases can translocate to the nucleus and phosphorylate transcription factors that regulate gene expression. Multiple MAP kinase pathways thus function in all types of eukaryotic cells to control cellular responses to diverse environmental signals.

The specificity of MAP kinase signaling is maintained at least in part by the organization of the components of each MAP kinase cascade as complexes that are associated with **scaffold proteins**. For example, the JIP-1 scaffold protein organizes the JNK MAP kinase and its upstream activators MLK and MKK7 into a signaling cassette (Figure 15.39). As a result of the specific association of these protein kinases on the JIP-1 scaffold, activation of MLK by Rac leads to specific and efficient activation of MKK7, which in turn activates JNK. Distinct scaffold proteins are involved not only in the organization of other MAP kinase signaling cassettes but also in the association of other downstream signaling molecules with their receptors. The physical association of signaling pathway components as a result of their

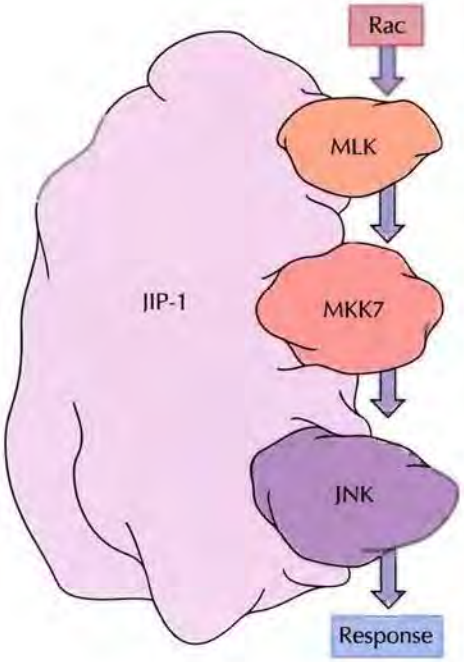


FIGURE 15.39 A scaffold protein for the JNK MAP kinase cascade The JIP-1 scaffold protein binds MLK, MKK7, and JNK, organizing these components of the JNK pathway into a signaling cassette.

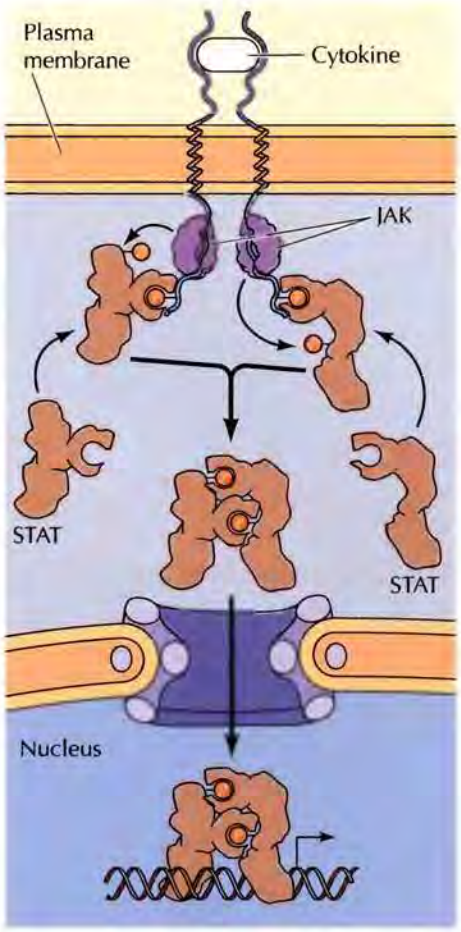


FIGURE 15.40 The JAK/STAT pathway The STAT proteins are transcription factors that contain SH2 domains that mediate their binding to phosphotyrosine-containing sequences. In unstimulated cells, STAT proteins are inactive in the cytosol. Stimulation of cytokine receptors leads to the binding of STAT proteins where they are phosphorylated by the receptor-associated JAK protein-tyrosine kinases. The phosphorylated STAT proteins then dimerize and translocate to the nucleus where they activate the transcription of target genes.

interaction with scaffold proteins is thought to play an important role in determining the specificity of signaling pathways within the cell.

The JAK/STAT and TGF-β/Smad Pathways

The PI 3-kinase and MAP kinase pathways are examples of indirect connections between the cell surface and the nucleus in which a cascade of protein kinases ultimately leads to transcription factor phosphorylation. The JAK/STAT and TGF-β/Smad pathways illustrate more direct connections between growth factor receptors and transcription factors in which the targeted transcription factors are phosphorylated directly by receptor-associated protein kinases.

The key elements in the **JAK/STAT pathway** are the **STAT proteins** (*s*ignal *t*ransducers and *a*ctivators of *t*ranscription), which were originally identified in studies of cytokine receptor signaling (Figure 15.40). The STAT proteins are a family of seven transcription factors that contain SH2 domains. They are inactive in unstimulated cells where they are localized to the cytoplasm. Stimulation of cytokine receptors leads to recruitment of STAT proteins, which bind via their SH2 domains to phosphotyrosine-containing sequences in the cytoplasmic domains of receptor polypeptides. Following their association with activated receptors the STAT proteins are phosphorylated by members of the JAK family of nonreceptor protein-tyrosine kinases, which are associated with cytokine receptors. Tyrosine phosphorylation promotes the dimerization of STAT proteins, which then translocate to the nucleus where they stimulate transcription of their target genes.

Further studies have shown that STAT proteins are also activated downstream of receptor protein-tyrosine kinases where their phosphorylation may be catalyzed either by the receptors themselves or by associated nonreceptor kinases. The STAT transcription factors thus serve as direct links between both cytokine and growth factor receptors on the cell surface and regulation of gene expression in the nucleus.

The receptors for members of the TGF-β family of growth factors are protein-serine/threonine kinases, which directly phosphorylate transcription factors of the **Smad** family (Figure 15.41). The receptors are composed of type I and type II polypeptides, which become associated following ligand binding. The type II receptor then phosphorylates the type I receptor, which in turn phosphorylates a Smad protein. The phosphorylated Smads then translocate to the nucleus and regulate gene expression. It should be noted that there are 42 different members of the TGF-β family in humans, which

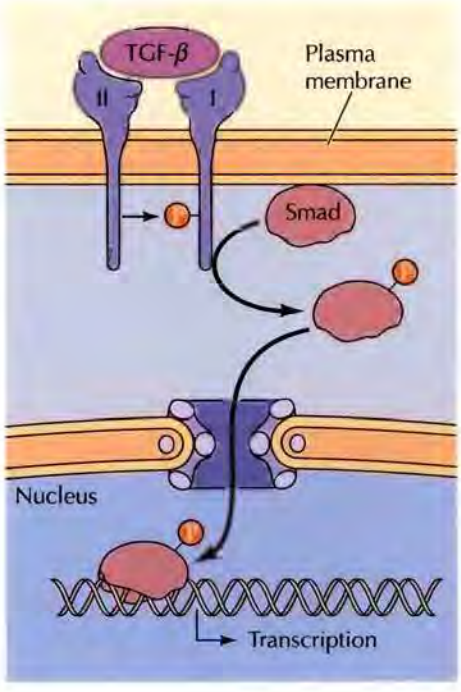


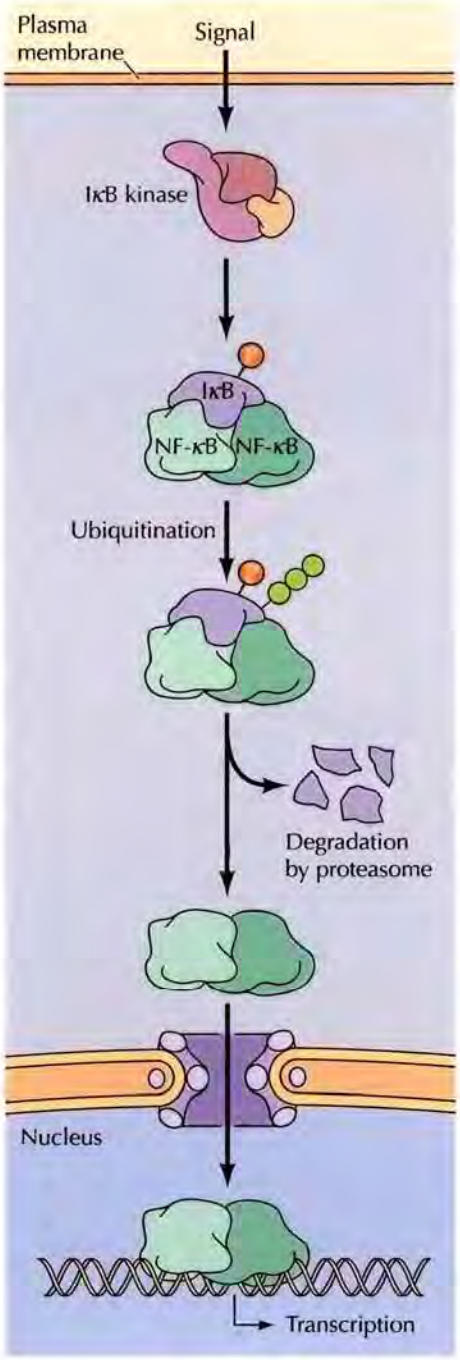
FIGURE 15.41 Signaling from TGF-β receptors TGF-β receptors are dimers of type I and II polypeptides. The type II receptor phosphorylates and activates type I, which then phosphorylates a Smad protein. Phosphorylated Smads translocate to the nucleus and activate transcription of target genes.

■ **Growth hormone functions by activating the JAK/STAT pathway.**

elicit different responses in their target cells. This is accomplished by combinatorial interactions of seven different type I receptors and five type II receptors, which lead to activation of different members of the Smad family (a total of 8 family members). Smad family members can also be phosphorylated by ERK, and this intersection between the TGFβ/Smad pathway and ERK plays a critical role in embryonic development.

NF-κB Signaling

NF-κB signaling is another example of a signaling pathway that directly targets a specific family of transcription factors. The **NF-κB** family consists of five transcription factors that play key roles in the immune system and in inflammation as well as in regulation of proliferation and survival of many types of animal cells. Members of this transcription factor family are activated in response to a variety of stimuli, including cytokines, growth factors, viral infection, and DNA damage. In unstimulated cells NF-κB proteins are bound to inhibitory **IκB** proteins that maintain NF-κB in an inactive state in the cytosol (Figure 15.42). Activation of NF-κB results from signals that activate the IκB kinase, which phosphorylates IκB. This phosphorylation targets IκB for ubiquitination and degradation by the proteasome, freeing NF-κB to translocate to the nucleus and induce expression of its target genes.



The Hedgehog, Wnt, and Notch Pathways

The **Hedgehog** and **Wnt** pathways are closely connected signaling systems that play key roles in determining cell fate during embryonic development. Both Hedgehog and Wnt pathways were first described in *Drosophila*, but members of the Hedgehog and Wnt families have been found to control a wide range of events that establish cell patterning during the development of both vertebrate and invertebrate embryos. Examples of the processes regulated by these signaling pathways include the determination of cell types and establishment of cell patterning during the development of limbs, the nervous system, the skeleton, lungs, hair, teeth, and gonads.

The *hedgehog* genes (one in *Drosophila* and three in mammals) encode secreted proteins that are modified by the addition of lipids. The functional receptor for Hedgehog consists of two transmembrane proteins, Patched and Smoothened (Figure 15.43). Hedgehog binds to Patched, which acts as a negative regulator of Smoothened. The binding of Hedgehog to Patched allows Smoothened to propagate an intracellular signal, leading to activation of a transcription factor called Cubitus interruptus (Ci) in *Drosophila* (Gli in mammals). In the absence of Hedgehog, Ci is maintained in a complex with a protein kinase called Fused and a kinesin-related protein called Coastal-2, which anchors the complex to microtubules. Within this complex, Ci is either completely degraded or cleaved to generate a transcriptional repressor (Ci75). Hedgehog signaling promotes the interaction of Smoothened with Coastal-2, leading to the release of full-length Ci (Ci155), which is then able to translocate to the nucleus and activate transcription of its target genes.

FIGURE 15.42 NF-κB signaling In the inactive state, homo- or heterodimers of NF-κB are bound to IκB in the cytoplasm. A variety of signals leads to activation of the IκB kinase, which phosphorylates IκB. This phosphorylation marks IκB for ubiquitination and degradation by the proteasome, allowing NF-κB to translocate to the nucleus and activate transcription of target genes.

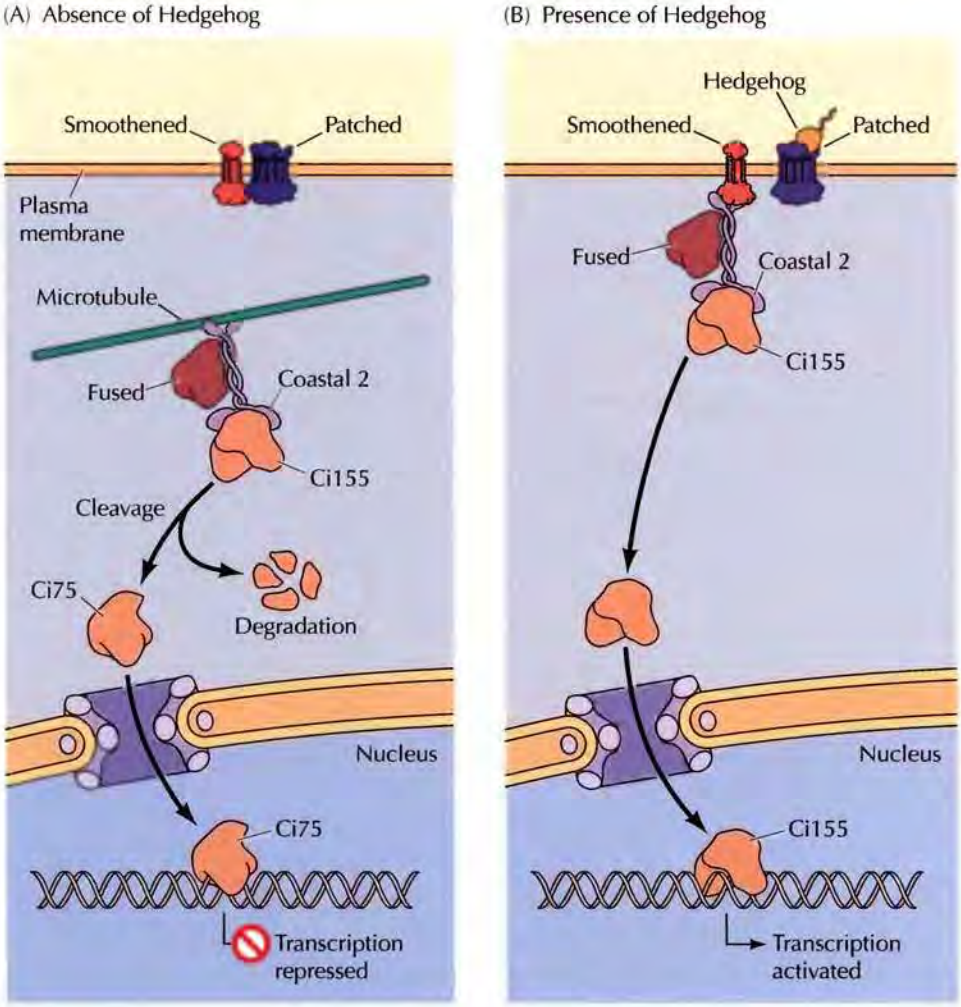


FIGURE 15.43 Hedgehog signaling In the absence of Hedgehog, the transcription factor Ci155 is anchored to microtubules by the kinesin-related protein Coastal-2 in association with the protein kinase Fused. Within this complex, Ci155 is degraded or cleaved to generate a transcriptional repressor (Ci75). The Hedgehog polypeptide binds to Patched on the surface of a target cell, relieving the inhibition of Smoothened by Patched and promoting the interaction of Smoothened with Coastal-2. This leads to the release of full-length Ci (Ci155), which translocates to the nucleus and activates transcription of its target genes.

The Wnt proteins are a family of secreted growth factors that bind to receptors of the LRP and Frizzled family, which are related to Smoothened (Figure 15.44). Signaling from LRP and Frizzled leads to activation of a cytoplasmic protein called Dishevelled, and inhibition of a complex of the proteins axin, APC, and the protein kinase GSK-3β. Within this complex, GSK-3β phosphorylates β-catenin, leading to its ubiquitination and degradation, so Wnt signaling results in increased β-catenin levels. β-catenin was discussed in Chapter 14 as a transmembrane protein that links cadherins to actin at adherens junctions (see Figure 14.22). Importantly, linking cadherins to actin is only one role of β-catenin. In Wnt signaling, β-catenin acts as a direct regulator of gene expression by forming a complex with members of the Tcf/LEF family of transcription factors. The association of β-catenin activates Tcf/LEF family members, leading to the expression of tar-

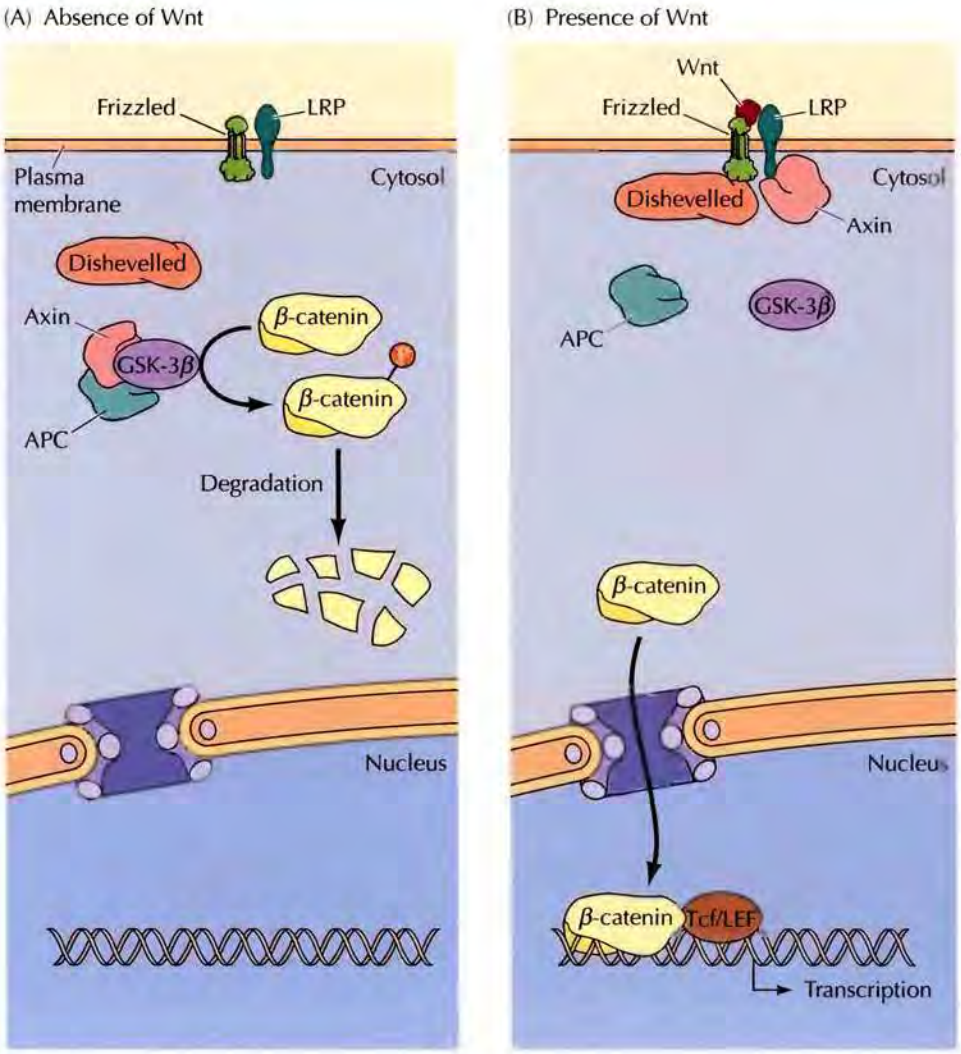


FIGURE 15.44 The Wnt pathway In the absence of Wnt, β-catenin is phosphorylated by GSK-3β in a complex with axin and APC, leading to its ubiquitination and degradation. Wnt polypeptides bind to Frizzled and LRP cell surface receptors, which activate Dishevelled and lead to inhibition of the axin/APC/GSK-3β complex. This results in stabilization of β-catenin, which translocates to the nucleus and forms a complex with Tcf/LEF transcription factors.

get genes encoding other cell signaling molecules and a variety of transcription factors that control cell fate.

The **Notch** pathway is another highly conserved signaling pathway that controls cell fate during animal development. Notch signaling is an example of direct cell-cell interactions during development. Notch is a large protein with a single transmembrane domain that serves as a receptor for signaling by transmembrane proteins (e.g., Delta) on the surface of adjacent cells (Figure 15.45). Stimulation of Notch initiates a novel and direct pathway of transcriptional activation. In particular, ligand binding leads to proteolytic cleavage of Notch, and the intracellular domain of Notch is then translocated into the nucleus. The Notch intracellular domain then interacts with a transcription factor (called Su[H] in *Drosophila*, or CSL in mammals) and converts it from a repressor to an activator of its target genes. As in the

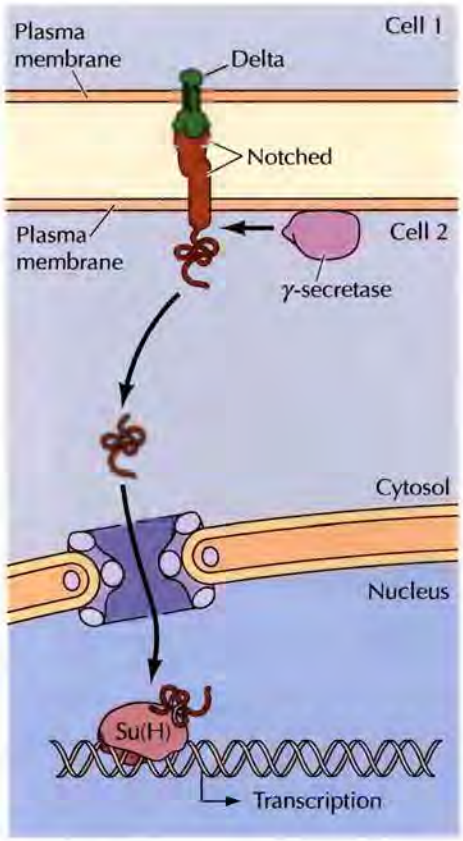


FIGURE 15.45 Notch signaling Notch serves as a receptor for direct cell-cell signaling by transmembrane proteins (e.g., Delta) on neighboring cells. The binding of Delta leads to proteolytic cleavage of Notch by γ-secretase. This releases the Notch intracellular domain, which translocates to the nucleus and interacts with a transcription factor (Su[H] or CSL) to induce gene expression.

Wnt signaling pathway, the Notch target genes include genes encoding other transcriptional regulatory proteins, which act to determine cell fate.

Signal Transduction and the Cytoskeleton

The preceding sections focused on signaling pathways that regulate changes in metabolism, gene expression, and cell behavior in response to hormones and growth factors. However, the functions of most cells are also directly affected by cell adhesion and the organization of the cytoskeleton. The receptors responsible for cell adhesion thus act to initiate intracellular signaling pathways that regulate other aspects of cell behavior, including gene expression. Conversely, growth factors frequently act to induce cytoskeletal alterations resulting in cell movement or changes in cell shape. Components of the cytoskeleton thus act as both receptors and targets in cell signaling pathways, integrating cell shape and movement with other cellular responses.

Integrins and Signal Transduction

As discussed in Chapters 12 and 14, the integrins are the major receptors responsible for the attachment of cells to the extracellular matrix. At two types of cell-matrix junctions (focal adhesions and hemidesmosomes), the integrins also interact with components of the cytoskeleton to provide a stable linkage between the extracellular matrix and adherent cells (see Figure 14.19). In addition to this structural role, the integrins serve as receptors that activate intracellular signaling pathways thereby controlling cell movement and other aspects of cell behavior (including cell proliferation and survival) in response to adhesive interactions.

Like members of the cytokine receptor superfamily, the integrins have short cytoplasmic tails that lack any intrinsic enzymatic activity. However, protein-tyrosine phosphorylation is an early response to the interaction of integrins with extracellular matrix components, suggesting that the integrins are linked to nonreceptor protein-tyrosine kinases. One mode of signaling downstream of integrins involves activation of a nonreceptor protein-tyrosine kinase called **FAK** (*f*ocal *a*dhesion *k*inase) (Figure 15.46). As its name implies, FAK is localized to focal adhesions and rapidly becomes tyrosine-phosphorylated following the binding of integrin to extracellular matrix components, such as fibronectin. Like other protein-tyrosine kinases, the activation of FAK involves autophosphorylation induced by the clustering of integrins bound to the extracellular matrix as well as by their interactions with actin. Autophosphorylation of FAK creates docking sites for signaling molecules containing SH2 domains, including members of the Src family of nonreceptor protein-tyrosine kinases that phosphorylate additional sites on FAK. As discussed earlier for growth factor receptors, tyrosine phosphorylation of FAK creates binding sites for the SH2 domains of other downstream signaling molecules, including phospholipase C-γ, PI 3-kinase, and the Grb2-

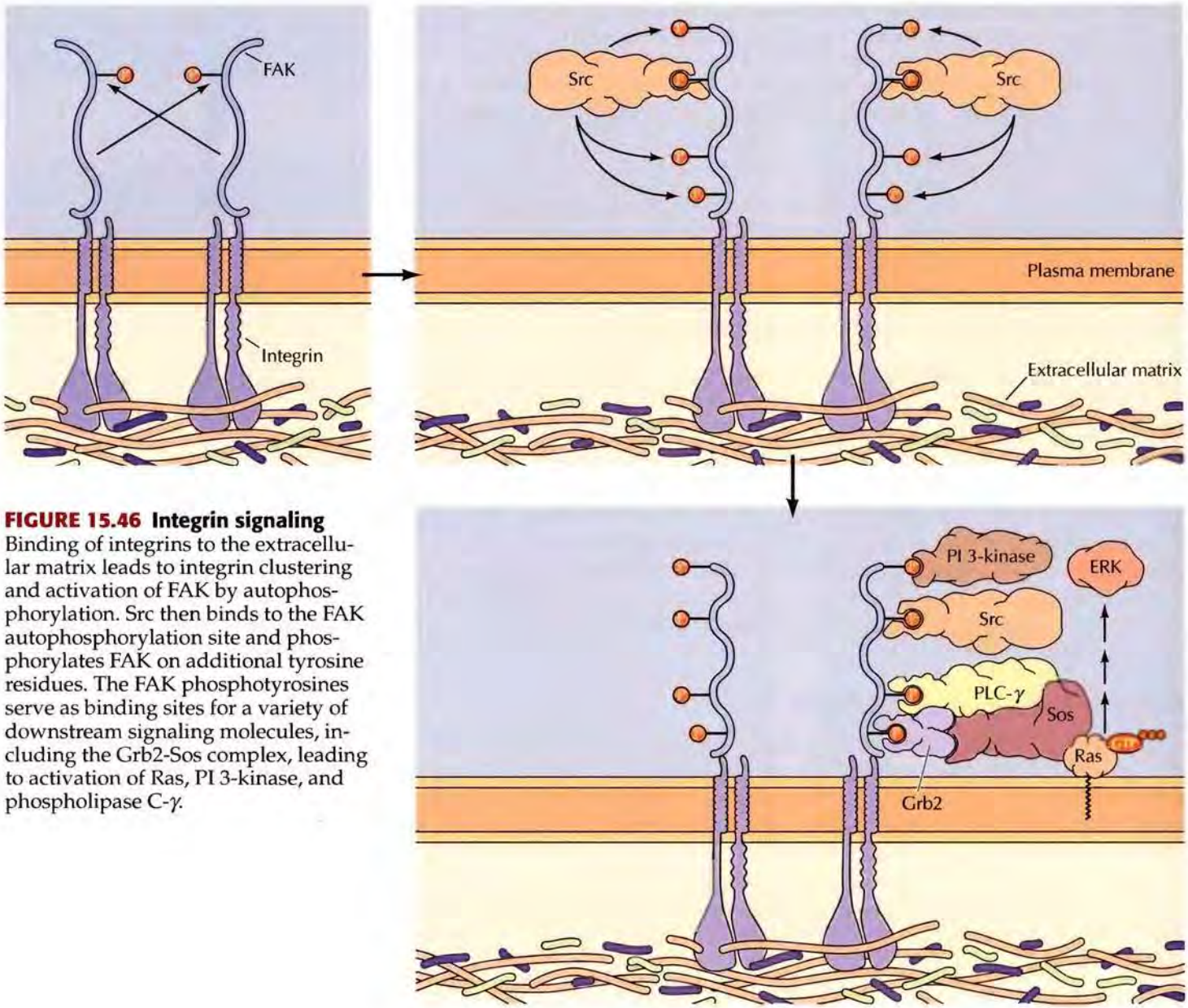


FIGURE 15.46 Integrin signaling Binding of integrins to the extracellular matrix leads to integrin clustering and activation of FAK by autophosphorylation. Src then binds to the FAK autophosphorylation site and phosphorylates FAK on additional tyrosine residues. The FAK phosphotyrosines serve as binding sites for a variety of downstream signaling molecules, including the Grb2-Sos complex, leading to activation of Ras, PI 3-kinase, and phospholipase C-γ.

Sos complex. Recruitment of the Sos guanine nucleotide exchange factor leads to activation of Ras, which in turn couples integrins to activation of the ERK pathway. Integrin activation of the FAK and Src nonreceptor protein-tyrosine kinases thus links cell adhesion to the same downstream signaling pathways that regulate gene expression, cell proliferation, and cell survival downstream of growth factor receptors. In addition, integrins can interact with and stimulate the activities of receptor protein-tyrosine kinases, such as the EGF receptor, leading to a parallel activation of the signaling pathways stimulated by growth factors and by cell adhesion.

Regulation of the Actin Cytoskeleton

Signaling from integrins as well as from growth factor receptors also plays a central role in the control of cell movement by regulating the dynamic behavior of the actin cytoskeleton. Cellular responses to growth factors as well as cell adhesion receptors frequently include changes in cell motility,

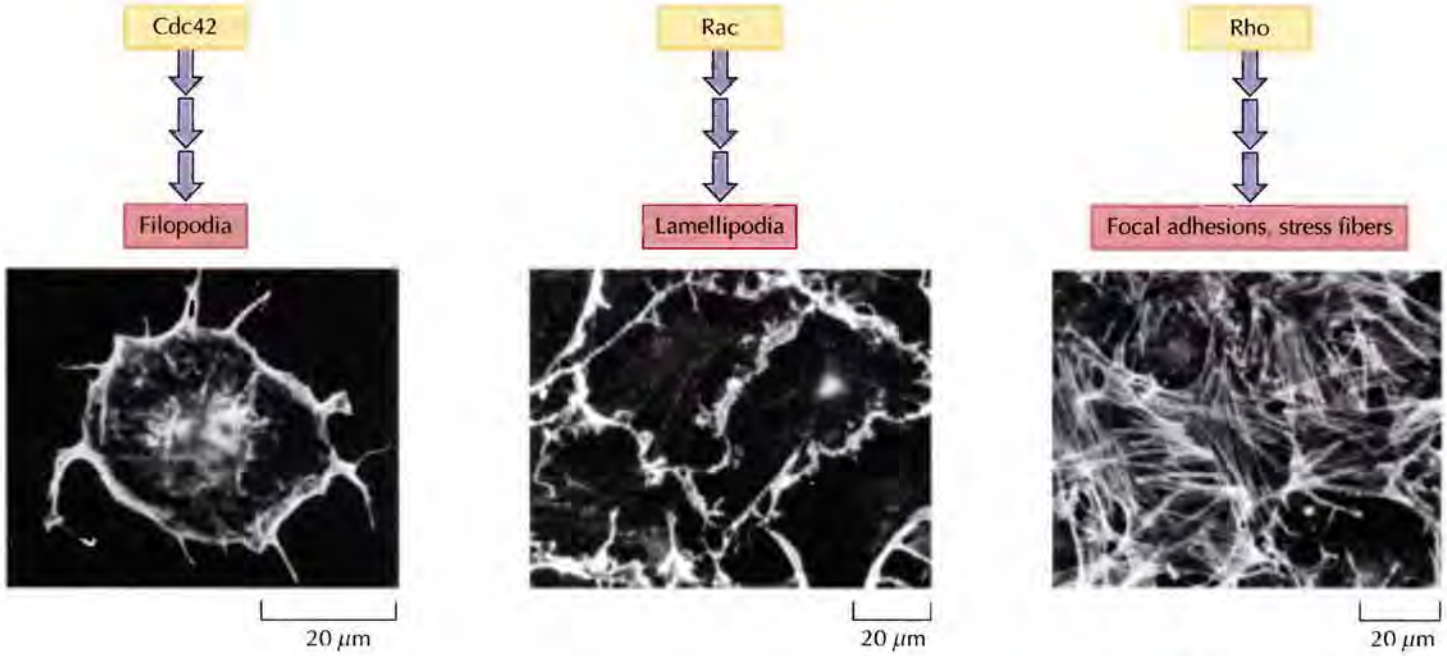


FIGURE 15.47 Regulation of actin remodeling by Rho family proteins Different members of the Rho family regulate the polymerization of actin to produce filopodia (Cdc42), lamellipodia (Rac), and stress fibers (Rho). Fluorescence micrographs illustrate the distribution of actin following microinjection of fibroblasts with Cdc42, Rac, and Rho. (From C. D. Nobes and A. Hall, 1995. *Cell* 81: 53.)

which play critical roles in processes such as wound healing and embryonic development. As discussed in Chapter 12, these aspects of cell behavior are governed principally by the dynamic assembly and disassembly of actin filaments underlying the plasma membrane. Remodeling of the actin cytoskeleton therefore represents a key element of the response of many cells to extracellular stimuli.

Members of the Rho subfamily of small GTP-binding proteins (including **Rho**, **Rac**, and **Cdc42**) play central roles in regulating the organization of the actin cytoskeleton and thus control a variety of cell processes, including cell motility and cell adhesion. The role of Rho family members in cytoskeletal remodeling was first demonstrated by experiments showing that microinjection of fibroblasts with Rho proteins induced cytoskeletal alterations, including the production of cell surface protrusions (filopodia and lamellipodia) and the formation of focal adhesions and stress fibers (Figure 15.47). Microinjection of cells with specific mutants of different Rho family members has shown that Cdc42 induces the formation of filopodia, Rac mediates the formation of lamellipodia, and Rho is responsible for the formation of stress fibers. Further studies have demonstrated that the activities of Rho family members are not restricted to fibroblasts: They also play similar roles in regulating the actin cytoskeleton in all types of eukaryotic cells.

Rho family members are activated by integrin signaling as well as by growth factor receptors. A large number of proteins then serve to mediate the cytoskeletal changes induced by Rho, Rac, and Cdc42. For example, a key target of Rho in regulating cytoskeletal alterations is a protein-serine/threonine kinase called Rho kinase (Figure 15.48). Activation of Rho kinase increases the phosphorylation of the light chain of myosin II by two

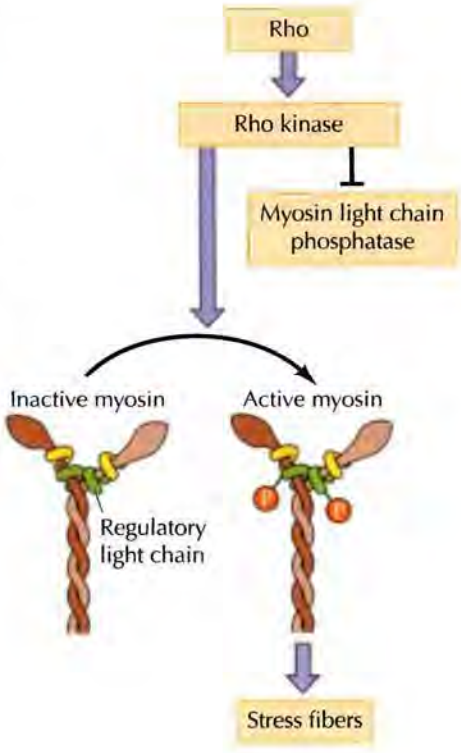


FIGURE 15.48 Regulation of myosin light chain phosphorylation by Rho Rho activates the Rho kinase, which phosphorylates the regulatory light chain of myosin II and inhibits myosin light chain phosphatase. The resulting increase in phosphorylation of the light chain activates myosin II, leading to assembly of actin-myosin filaments and the formation of stress fibers.

mechanisms: Rho kinase not only directly phosphorylates the myosin light chain but also phosphorylates and inhibits myosin light chain phosphatase. The resulting increase in myosin light chain phosphorylation activates myosin and leads to the assembly of actin-myosin filaments, resulting in the formation of stress fibers and focal adhesions. Additional targets of Rho may also contribute to stress fiber formation by promoting actin polymerization. Both Rac and Cdc42 lead to the formation of cell surface protrusions (filopodia and lamellipodia) by stimulating actin polymerization, acting through several targets that associate with the Arp2/3 complex to induce actin filament formation (see Figure 12.35).

Signaling Networks

We have so far discussed signaling in terms of linear pathways that transmit information from the environment to intracellular targets. However, signaling within the cell is far more complicated. First, the activities of individual pathways are regulated by feedback loops that control the extent and duration of signaling activity. In addition, signaling pathways do not operate in isolation; rather, there is frequent crosstalk between different pathways, so that intracellular signal transduction ultimately needs to be understood as an integrated network of connected pathways. Computational modeling of such signaling networks is currently a major challenge in cell biology, which will be necessary to understand the dynamic response of cells to their environment.

Feedback and Crosstalk

The activity of signaling pathways is controlled by **feedback loops**, which are similar in principal to feedback regulation of metabolic pathways (see Figure 8.36). A good example of a negative feedback loop is provided by the NF-κB pathway (Figure 15.49). NF-κB is activated by signals that lead to proteolysis of the inhibitor IκB, allowing NF-κB to translocate to the nucleus and induce expression of its target genes. One of the target genes induced by NF-κB encodes IκB, so NF-κB signaling leads to the synthesis of new IκB, which inhibits continued NF-κB activity. This regulation is critical because the extent and duration of NF-κB activity can determine the transcriptional response of the cell. For example, some target genes are induced by transient NF-κB activity, persisting for only 30–60 minutes, whereas the induction of other genes requires several hours of sustained NF-κB signaling.

Signaling by the ERK MAP kinase provides another example of the importance of the duration of signaling. In a well-studied model of cell differentiation in response to nerve growth factor (NGF), ERK signaling can lead either to cell proliferation or to neuronal differentiation depending on the duration of ERK activity. In particular, transient activation of ERK (for 30–60 minutes) stimulates cell proliferation, but sustained activation of ERK for 2–3 hours induces differentiation of the NGF-treated cells into neurons. Although the mechanism by which these differences in the duration of ERK activity lead to such distinct biological outcomes remains to be understood, it is clear that quantitative considerations of signaling activity are critical to cell response.

Crosstalk refers to the interaction of one signaling pathway with another. Several examples have already been noted in this chapter, including junctions between Ca^{2+} and cAMP signaling, between the cAMP and ERK pathways, between the ERK and TGFβ/Smad pathways, and between integrin signaling and receptor protein-tyrosine kinases. A novel example

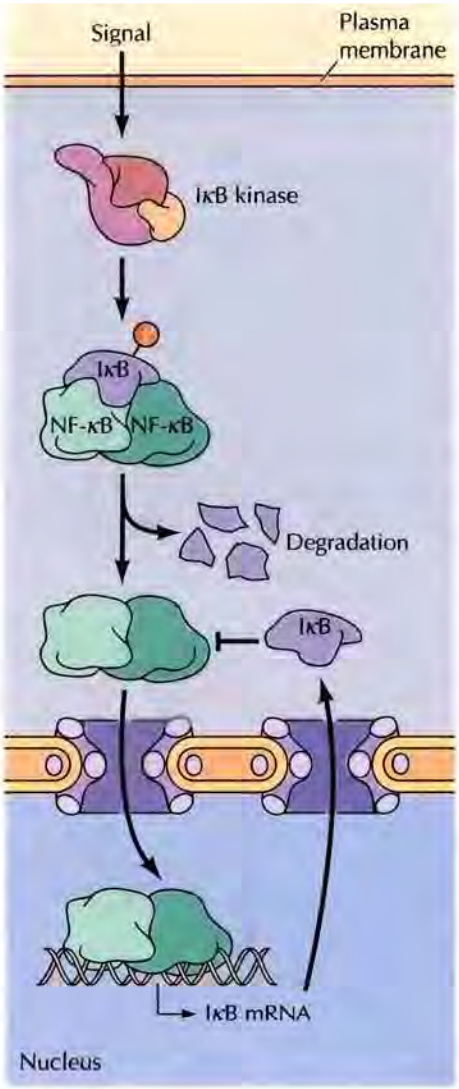


FIGURE 15.49 Feedback inhibition of NF-κB NF-κB is activated as a result of phosphorylation and degradation of IκB, allowing NF-κB to translocate to the nucleus and activate transcription of target genes. One of the genes activated by NF-κB encodes IκB, generating a feedback loop that inhibits NF-κB activity.

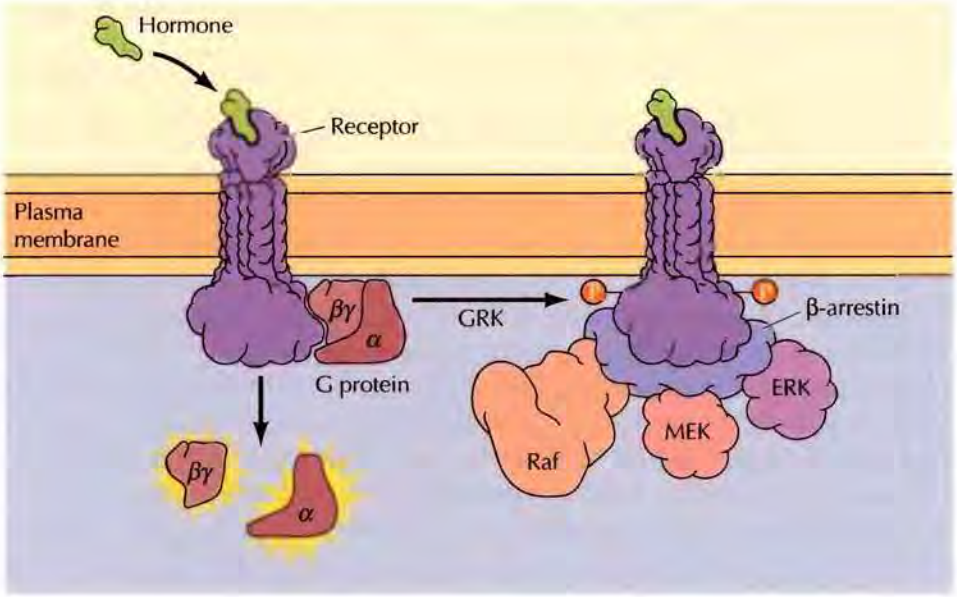


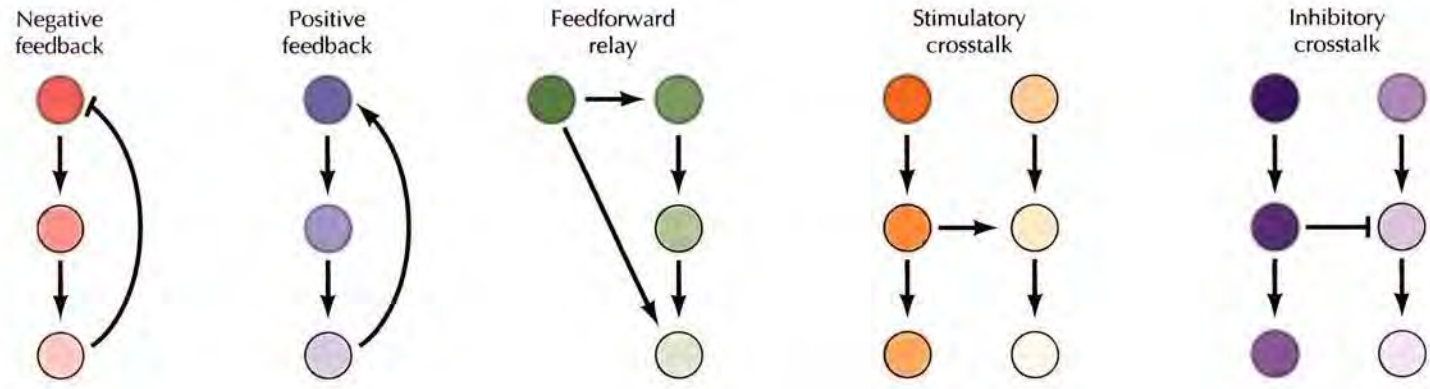
FIGURE 15.50 Crosstalk between G protein-coupled receptors and ERK signaling by β-arrestin Ligand binding stimulates G protein-coupled receptors, leading to activation of trimeric G proteins. The activity of the receptors is turned-off as a result of phosphorylation by GRKs and association of β-arrestin with the phosphorylated receptor. β-arrestin also acts as a scaffold protein for Raf, MEK, and ERK, linking G protein-coupled receptors to the ERK signaling pathway.

in which regulation of the duration of signaling is combined with crosstalk has come from recent studies of G protein-coupled receptors. These receptors are linked to MAP kinase signaling by **β-arrestins**, which were first identified as regulatory proteins that turn-off signaling from G protein-coupled receptors to G proteins (Figure 15.50). The signaling activity of these receptors is turned-off as a result of phosphorylation by a family of protein kinases called GRKs (*G* protein-coupled *r*eceptor *k*inases), followed by association of β-arrestin with the phosphorylated receptor. However, the β-arrestins do more than turn-off signaling to G proteins: They also act as signaling molecules themselves to stimulate additional downstream pathways, including nonreceptor protein-tyrosine kinases (e.g., Src family members) and MAP kinase pathways. In particular, β-arrestin-2 serves as a scaffold protein for Raf-MEK-ERK signaling, directly linking this MAP kinase pathway to G protein-coupled receptors.

Networks of Cellular Signal Transduction

The extensive crosstalk between individual signal transduction pathways means that multiple pathways interact with one another to form **signaling networks** within the cell. Some of the ways in which pathways can connect within a network are illustrated in Figure 15.51. The junctions between differ-

FIGURE 15.51 Elements of signaling networks In feedback loops, a downstream element of a pathway either inhibits (negative feedback) or stimulates (positive feedback) an upstream element. In feedforward relays, an upstream element of a pathway stimulates both its immediate target and another element further downstream. Crosstalk occurs when an element of one pathway either stimulates or inhibits an element of a second pathway.



ent pathways can be either positive (where one pathway stimulates the other) or negative (where one pathway inhibits the other). In addition to negative feedback loops, signaling networks can contain positive feedback loops and **feedforward relays** in which the activity of one component of a pathway stimulates a distant downstream component.

A full understanding of cell signaling will require the development of network models that predict the dynamic behavior of the interconnected signaling pathways that ultimately result in a biological response. In this view of cell signaling as an integrated system, mathematical models and computer simulations will clearly be needed to deal with the complexity of the problem. For example, many signaling pathways involve receptors that stimulate cascades of protein kinases that ultimately affect gene expression by regulating transcription factors. The human genome encodes approximately 1500 different receptors, almost 700 protein kinases and phosphatases, and nearly 2000 transcription factors, so the potential for cross-regulation between pathways formed from combinations of these elements is enormous. A recently developed network model for signaling in a mammalian neuron provides a graphical representation of this complexity (Figure 15.52). Although clearly a daunting task, understanding cell signaling in quantitative terms using mathematical and computational approaches that view the cell as an integrated biological system is a critically important problem at the cutting edge of research in cell biology.

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

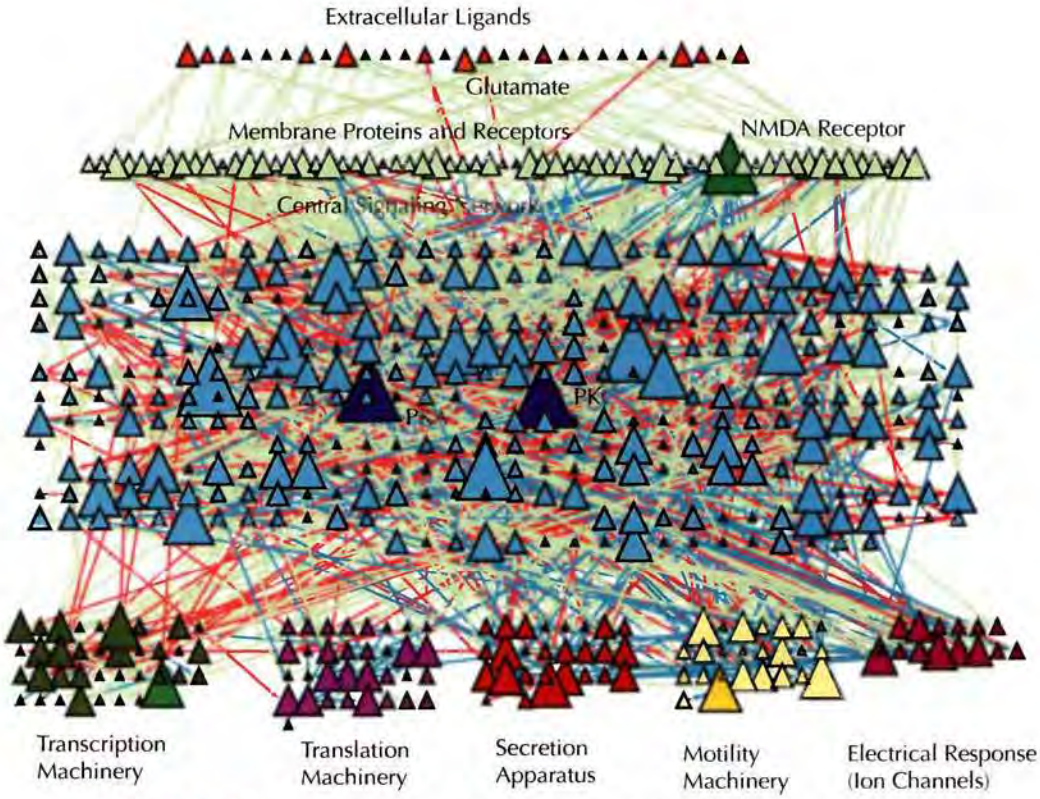


FIGURE 15.52 A mammalian signaling network The network depicts connections between 545 signaling elements in a mammalian neuron. Green arrows are stimulatory, red arrows are inhibitory, and blue arrows are neutral links between elements. (From A. Ma'ayan et al., 2005. *Science* 309: 1078.)

SUMMARY

SIGNALING MOLECULES AND THEIR RECEPTORS

Modes of Cell-Cell Signaling: Most signaling molecules are secreted by one cell and bind to receptors expressed by a target cell. Cell-cell signaling is divided into three general categories (endocrine, paracrine, and autocrine signaling) based on the distance over which signals are transmitted.

endocrine signaling, hormone, paracrine signaling, autocrine signaling

Steroid Hormones and the Nuclear Receptor Superfamily: The steroid hormones, thyroid hormone, vitamin D_3, and retinoic acid are small hydrophobic molecules that diffuse across the plasma membrane of their target cells and bind to intracellular receptors. Members of the nuclear receptor superfamily function as transcription factors to directly regulate gene expression in response to ligand binding.

steroid hormone, testosterone, estrogen, progesterone, corticosteroid, glucocorticoid, mineralocorticoid, ecdysone, brassinosteroid, thyroid hormone, vitamin D_3, retinoic acid, retinoid, nuclear receptor superfamily

Nitric Oxide and Carbon Monoxide: The simple gases nitric oxide and carbon monoxide are important paracrine signaling molecules in the nervous system and other cell types.

Neurotransmitters: Neurotransmitters are small hydrophilic molecules that carry signals between neurons or between neurons and other target cells at a synapse. Many neurotransmitters bind to ligand-gated ion channels.

neurotransmitter

Peptide Hormones and Growth Factors: The widest variety of signaling molecules in animals are peptides, ranging from only a few to more than a hundred amino acids. This group of molecules includes peptide hormones, neuropeptides, and growth factors.

peptide hormone, neuropeptide, enkephalin, endorphin, neurohormone, growth factor, nerve growth factor (NGF), neurotrophin, epidermal growth factor (EGF), platelet-derived growth factor (PDGF), cytokine, membrane-anchored growth factor

Eicosanoids: The eicosanoids are a class of lipids that function in paracrine and autocrine signaling.

eicosanoid, prostaglandin, prostacyclin, thromboxane, leukotriene

Plant Hormones: Small molecules known as plant hormones regulate plant growth and development.

plant hormone, auxin, gibberellin, cytokinin, abscisic acid, ethylene

FUNCTIONS OF CELL SURFACE RECEPTORS

G Protein-Coupled Receptors: The largest family of cell surface receptors, including the receptors for many hormones and neurotransmitters, transmit signals to intracellular targets via the intermediary action of G proteins.

G protein, G protein-coupled receptor, heterotrimeric G protein

Receptor Protein-Tyrosine Kinases: The receptors for most growth factors are protein-tyrosine kinases.

receptor protein-tyrosine kinase, autophosphorylation, SH2 domain, PTB domain

Cytokine Receptors and Nonreceptor Protein-Tyrosine Kinases: The receptors for many cytokines act in association with nonreceptor protein-tyrosine kinases.

cytokine receptor superfamily, nonreceptor protein-tyrosine kinase, Janus kinase (JAK)

KEY TERMS

KEY TERMS

protein-tyrosine phosphatase, transforming growth factor β (TGF-β), protein-serine/threonine kinase, guanylyl cyclase

intracellular signal transduction, cyclic AMP (cAMP), second messenger, adenylyl cyclase, cAMP phosphodiesterase, cAMP-dependent protein kinase, protein kinase A, cAMP response element (CRE), CREB

cyclic GMP (cGMP), rhodopsin, transducin, cGMP phosphodiesterase

phosphatidylinositol 4,5-bisphosphate (PIP_2), phospholipase C, diacylglycerol, inositol 1,4,5-trisphosphate (IP_3), protein kinase C, calmodulin, CaM kinase, ryanodine receptor

phosphatidylinositide (PI) 3-kinase, phosphatidylinositol 3,4,5-trisphosphate (PIP_3), Akt, mTOR

MAP kinase, ERK, Ras, Raf, MEK, guanine nucleotide exchange factor, GTPase-activating protein, immediate-early gene, serum response element (SRE), serum response factor (SRF), Elk-1, secondary response gene, scaffold protein

SUMMARY

Receptors Linked to Other Enzymatic Activities: Other kinds of cell surface receptors include protein-tyrosine phosphatases, protein-serine/threonine kinases, and guanylyl cyclases.

PATHWAYS OF INTRACELLULAR SIGNAL TRANSDUCTION

The cAMP Pathway: Second Messengers and Protein Phosphorylation: Cyclic AMP is an important second messenger in the response of animal cells to a variety of hormones and odorants. Most actions of cAMP are mediated by protein kinase A, which phosphorylates both metabolic enzymes and the transcription factor CREB.

Cyclic GMP: Cyclic GMP is also an important second messenger in animal cells. Its best-characterized role is in visual reception in the vertebrate eye.

Phospholipids and Ca^{2+}: Phospholipids and Ca^{2+} are common second messengers activated downstream of both G protein-coupled receptors and protein-tyrosine kinases. Hydrolysis of phosphatidylinositol 4,5-bisphosphate (PIP_2) yields diacylglycerol and inositol 1,4,5-trisphosphate (IP_3), which activate protein kinase C and mobilize Ca^{2+} from intracellular stores, respectively. Increased levels of cytosolic Ca^{2+} then activate a variety of target proteins, including Ca^{2+}/calmodulin-dependent protein kinases. In electrically excitable cells of nerve and muscle, levels of cytosolic Ca^{2+} are increased by the opening of voltage-gated Ca^{2+} channels in the plasma membrane and ryanodine receptors in the endoplasmic and sarcoplasmic reticula.

The PI 3-kinase/Akt and mTOR Pathways: In addition to being cleaved into diacylglycerol and IP_3, PIP_2 can be phosphorylated to the distinct second messenger PIP_3. This leads to activation of the protein-serine/threonine kinase Akt, which plays a key role in cell survival. One of the targets of Akt signaling is the protein kinase mTOR, which is a central regulator of cell growth and couples protein synthesis to the availability of growth factors, nutrients, and cellular energy.

MAP Kinase Pathways: The MAP kinase pathways are conserved chains of protein kinases activated downstream of a variety of extracellular signals. In animal cells, the best-characterized forms of MAP kinase are coupled to growth factor receptors by the small GTP-binding protein Ras, which initiates a protein kinase cascade leading to MAP kinase (ERK) activation. ERK then phosphorylates a variety of cytosolic and nuclear proteins, including transcription factors that mediate immediate-early gene induction. Other MAP kinase pathways mediate responses of mammalian cells to inflammation and stress. Components of MAP kinase pathways are organized by scaffold proteins, which play an important role in maintaining the specificity of MAP kinase signaling.

SUMMARY

KEY TERMS

The JAK/STAT and TGF-β/Smad Pathways: STAT proteins are transcription factors that contain SH2 domains and are activated directly by the JAK protein-tyrosine kinases associated with cytokine and growth factor receptors. Members of the TGF-β receptor family are protein-serine/threonine kinases that directly phosphorylate and activate Smad transcription factors.

JAK/STAT pathway, STAT protein, Smad

NF-κB Signaling: Members of the NF-κB family of transcription factors are activated in response to cytokines, growth factors, and a variety of other stimuli. Their activation is mediated by phosphorylation and degradation of inhibitory IκB subunits.

NF-κB, IκB

The Hedgehog, Wnt, and Notch Pathways: The Hedgehog, Wnt, and Notch pathways play key roles in determination of cell fate and patterning during animal development. The Hedgehog and Wnt signaling pathways both act by preventing degradation of transcription factors in complexes in the cytoplasm. Notch signaling is mediated by direct cell-cell interactions, which induce proteolytic cleavage of Notch, followed by translocation of the Notch intracellular domain to the nucleus where it interacts with a transcription factor to affect expression of target genes.

Hedgehog, Wnt, Notch

SIGNAL TRANSDUCTION AND THE CYTOSKELETON

Integrins and Signal Transduction: Binding of integrins to the extracellular matrix stimulates the FAK and Src nonreceptor protein-tyrosine kinases, leading to activation of phospholipase C, PI 3-kinase, and Ras/Raf/ERK signaling pathways.

FAK

Regulation of the Actin Cytoskeleton: Signaling from integrins as well as growth factor receptors induces alterations in cell movement and cell shape by remodeling the actin cytoskeleton. These cytoskeletal alterations are mediated by members of the Rho subfamily of small GTP-binding proteins.

Rho, Rac, Cdc42

SIGNALING NETWORKS

Feedback and Crosstalk: The activity of signaling pathways within the cell is regulated by feedback loops that control the extent and duration of signaling. Different signaling pathways also interact to regulate each others activity.

feedback loop, crosstalk, β-arrestin

Networks of Cellular Signal Transduction: The extensive crosstalk between individual pathways leads to the formation of complex signaling networks. A full understanding of signaling within the cell will require the development of quantitative network models.

signaling network, feedforward relay

Questions

1. What is the difference between paracrine and endocrine signaling?

2. How does signaling by hydrophobic molecules like steroid hormones differ from signaling by peptide hormones?

3. How does aspirin reduce inflammation and blood clotting?

4. Hormones that activate a receptor coupled to G_s stimulate the proliferation of thyroid cells. How would inhibitors of cAMP phosphodiesterase affect the proliferation of these cells?

5. The epinephrine receptor is coupled to G_s, whereas the acetylcholine receptor (on heart muscle cells) is coupled to G_i. Suppose you were to construct a recombinant molecule containing the extracellular sequences of the epinephrine receptor joined to the cytosolic sequences of the acetylcholine receptor. What effect would epinephrine have on cAMP levels in cells expressing such a recombinant receptor? How would acetylcholine affect these cells?

6. Platelet-derived growth factor (PDGF) is a dimer of two polypeptide chains. How would PDGF monomers affect signaling by the PDGF receptor?

7. You have generated a truncated version of the EGF receptor that lacks the tyrosine kinase domain. Expression of this truncated receptor inhibits the response of cells to EGF. Why?

8. How would overexpression of protein phosphatase 1 affect the induction of cAMP-inducible genes in response to hormone stimulation of target cells? Would protein phosphatase 1 affect the function of cAMP-gated ion channels?

9. How does the PI 3-kinase/Akt pathway regulate cellular protein synthesis in response to growth factor stimulation?

10. You are studying an immediate early gene that is induced via the Ras/Raf/MEK/ERK pathway in response to growth factor treatment of fibroblasts. How would expression of siRNA against Sos affect the induction of this gene?

11. A specific member of the STAT family induces certain liver genes in response to stimulation by a cytokine. How would the induction of these genes be affected if you overexpressed a dominant-negative mutant of JAK?

12. You are studying a gene that is induced by the Wnt signaling pathway. To determine the role played by β-catenin, you make different site-specific mutants of β-catenin. You find that changing a specific Lys residue to Arg leads to nuclear accumulation of β-catenin and constitutive expression of the gene (even in the absence of Wnt stimulation). What is the most likely mechanism by which this mutation alters β-catenin activity?

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CHAPTER 16

The Cell Cycle

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- ***The Events of M Phase 669***
- ***Meiosis and Fertilization 678***
- ***KEY EXPERIMENT:*** The Discovery of MPF 658
- ***KEY EXPERIMENT:*** The Identification of Cyclin 662

SELF-REPRODUCTION IS PERHAPS THE MOST FUNDAMENTAL characteristic of cells—as may be said for all living organisms. All cells reproduce by dividing in two, with each parental cell giving rise to two daughter cells on completion of each cycle of cell division. These newly formed daughter cells can themselves grow and divide, giving rise to a new cell population formed by the growth and division of a single parental cell and its progeny. In the simplest case, such cycles of growth and division allow a single bacterium to form a colony consisting of millions of progeny cells during overnight incubation on a plate of nutrient agar medium. In a more complex case, repeated cycles of cell growth and division result in the development of a single fertilized egg into the approximately 10^{14} cells that make up the human body.

The division of all cells must be carefully regulated and coordinated with both cell growth and DNA replication in order to ensure the formation of progeny cells containing intact genomes. In eukaryotic cells, progression through the cell cycle is controlled by a series of protein kinases that have been conserved from yeasts to mammals. In higher eukaryotes, this cell cycle machinery is itself regulated by the growth factors that control cell proliferation, allowing the division of individual cells to be coordinated with the needs of the organism as a whole. Not surprisingly, defects in cell cycle regulation are a common cause of the abnormal proliferation of cancer cells, so studies of the cell cycle and cancer have become closely interconnected, similar to the relationship between studies of cancer and the cell signaling pathways discussed in Chapter 15.

The Eukaryotic Cell Cycle

The division cycle of most cells consists of four coordinated processes: cell growth, DNA replication, distribution of the duplicated chromosomes to daughter cells, and cell division. In bacteria, cell growth and DNA replication take place throughout most of the cell cycle, and duplicated chromosomes are distributed to daughter cells in association with the plasma membrane. In eukaryotes, however, the cell cycle is more complex and consists of four discrete phases. Although cell growth is usually a continuous process, DNA is synthesized during only one phase of the cell cycle, and the replicated chromosomes are then distributed to daughter nuclei by a complex series of events preceding cell division. Progression between these stages of the cell cycle is controlled by a conserved regulatory apparatus, which not only coordinates the different events of the cell cycle but also links the cell cycle with extracellular signals that control cell proliferation.

16.1 WEBSITE ANIMATION

Phases of the Cell Cycle

The cell cycle of most eukaryotic cells is divided into four discrete phases: M, G_1, S, and G_2.

Phases of the Cell Cycle

A typical eukaryotic cell cycle is illustrated by human cells in culture, which divide approximately every 24 hours. As viewed in the microscope, the cell cycle is divided into two basic parts: **mitosis** and **interphase**. Mitosis (nuclear division) is the most dramatic stage of the cell cycle, corresponding to the separation of daughter chromosomes and usually ending with cell division (**cytokinesis**). However, mitosis and cytokinesis last only about an hour, so approximately 95% of the cell cycle is spent in interphase—the period between mitoses. During interphase, the chromosomes are decondensed and distributed throughout the nucleus, so the nucleus appears morphologically uniform. At the molecular level, however, interphase is the time during which both cell growth and DNA replication occur in an orderly manner in preparation for cell division.

The cell grows at a steady rate throughout interphase, with most dividing cells doubling in size between one mitosis and the next. In contrast, DNA is synthesized during only a portion of interphase. The timing of DNA synthesis thus divides the cycle of eukaryotic cells into four discrete phases (Figure 16.1). The **M phase** of the cycle corresponds to mitosis, which is usually followed by cytokinesis. This phase is followed by the **G_1 phase** (gap 1), which corresponds to the interval (gap) between mitosis and initiation of DNA replication. During G_1, the cell is metabolically active and continuously grows but does not replicate its DNA. G_1 is followed by **S phase** (synthesis) during which DNA replication takes place. The completion of DNA synthesis is followed by the **G_2 phase** (gap 2) during which cell growth continues and proteins are synthesized in preparation for mitosis.

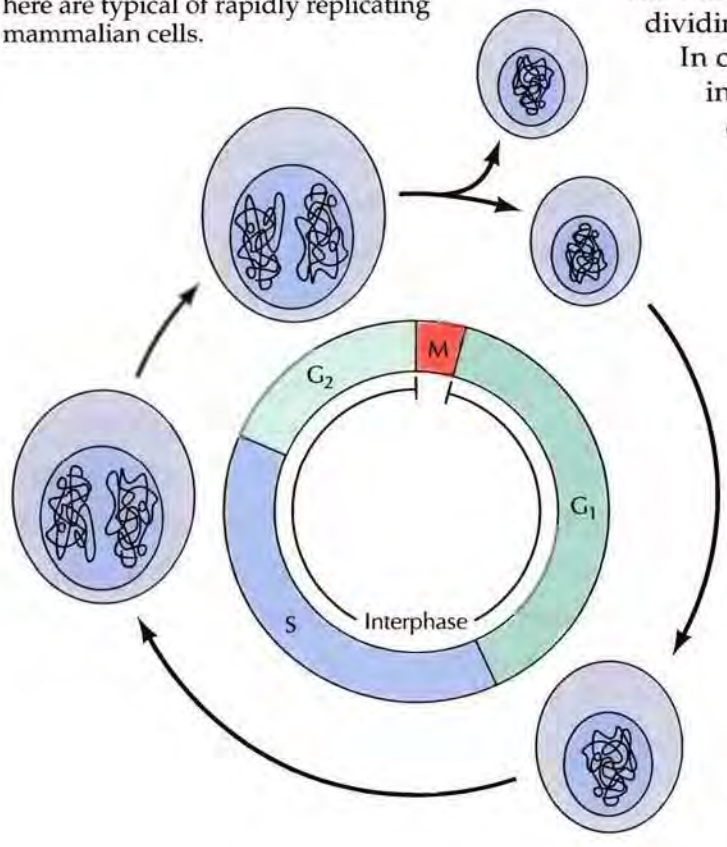


FIGURE 16.1 Phases of the cell cycle The division cycle of most eukaryotic cells is divided into four discrete phases: M, G_1, S, and G_2. M phase (mitosis) is usually followed by cytokinesis. S phase is the period during which DNA replication occurs. The cell grows throughout interphase, which includes G_1, S, and G_2. The relative lengths of the cell cycle phases shown here are typical of rapidly replicating mammalian cells.

The duration of these cell cycle phases varies considerably in different kinds of cells. For a typical rapidly proliferating human cell with a total cycle time of 24 hours, the G_1 phase might last about 11 hours, S phase about 8 hours, G_2 about 4 hours, and M about 1 hour. Other types of cells, however, can divide much more rapidly. Budding yeasts, for example, can progress through all four stages of the cell cycle in only about 90 minutes. Even shorter cell cycles (30 minutes or

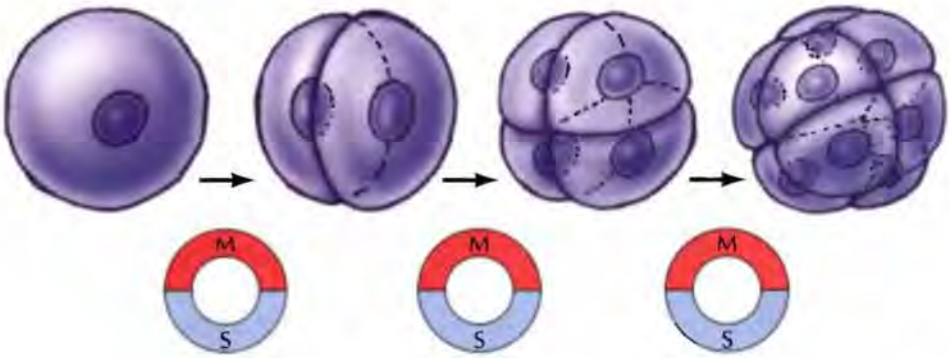


FIGURE 16.2 Embryonic cell cycles Early embryonic cell cycles rapidly divide the cytoplasm of the egg into smaller cells. The cells do not grow during these cycles, which lack G_1 and G_2 and consist simply of short S phases alternating with M phases.

16.2 WEBSITE ANIMATION

Embryonic Cell Cycles

During early embryonic cell cycles, the cells do not grow, and instead divide into progressively smaller cells.

less) occur in early embryo cells shortly after fertilization of the egg (Figure 16.2). In this case, however, cell growth does not take place. Instead, these early embryonic cell cycles rapidly divide the egg cytoplasm into smaller cells. There is no G_1 or G_2 phase, and DNA replication occurs very rapidly in these early embryonic cell cycles, which therefore consist of very short S phases alternating with M phases.

In contrast to the rapid proliferation of embryonic cells, some cells in adult animals cease division altogether (e.g., nerve cells) and many other cells divide only occasionally, as needed to replace cells that have been lost because of injury or cell death. Cells of the latter type include skin fibroblasts, as well as the cells of some internal organs, such as the liver. As discussed further in the next section, these cells exit G_1 to enter a quiescent stage of the cycle called $\mathbf{G_0}$, where they remain metabolically active but no longer proliferate unless called on to do so by appropriate extracellular signals.

Analysis of the cell cycle requires identification of cells at the different stages discussed above. Although mitotic cells can be distinguished microscopically, cells in other phases of the cycle (G_1, S, and G_2) must be identified by biochemical criteria. Cells in S phase can be readily identified because they incorporate radioactive thymidine, which is used exclusively for DNA synthesis (Figure 16.3). For example, if a population of rapidly pro-

FIGURE 16.3 Identification of S phase cells by incorporation of radioactive thymidine The cells were exposed to radioactive thymidine and analyzed by autoradiography. Labeled cells are indicated by arrows. (From D. W. Stacey et al., 1991. *Mol. Cell Biol.* 11: 4053.)

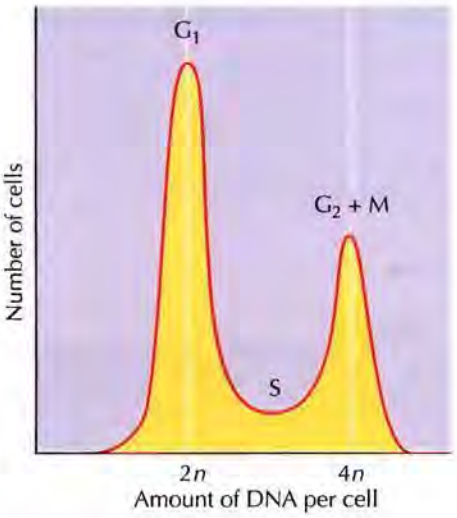


FIGURE 16.4 Determination of cellular DNA content A population of cells is labeled with a fluorescent dye that binds DNA. The cells are then passed through a flow cytometer, which measures the fluorescence intensity of individual cells. The data are plotted as cell number versus fluorescence intensity, which is proportional to DNA content. The distribution shows two peaks, corresponding to cells with DNA contents of $2n$ and $4n$; these cells are in the G_1 and G_2/M phases of the cycle, respectively. Cells in S phase have DNA contents between $2n$ and $4n$ and are distributed between these two peaks.

liferating human cells in culture is exposed to radioactive thymidine for a short period of time (e.g., 15 minutes) and then analyzed by autoradiography, about a third of the cells will be found to be radioactively labeled, corresponding to the fraction of cells in S phase.

Variations of such cell labeling experiments can also be used to determine the length of different stages of the cell cycle. For example, consider an experiment in which cells are exposed to radioactive thymidine for 15 minutes, after which the radioactive thymidine is removed and the cells are cultured for varying lengths of time prior to autoradiography. Radioactively labeled interphase cells that were in S phase during the time of exposure to radioactive thymidine will be observed for several hours as they progress through the remainder of S and G_2. In contrast, radioactively labeled mitotic cells will not be observed until 4 hours after labeling. This 4-hour lag time corresponds to the length of G_2—the minimum time required for a cell that incorporated radioactive thymidine at the end of S phase to enter mitosis.

Cells at different stages of the cell cycle can also be distinguished by their DNA content (Figure 16.4). For example, animal cells in G_1 are diploid (containing two copies of each chromosome), so their DNA content is referred to as $2n$ (n designates the haploid DNA content of the genome). During S phase, replication increases the DNA content of the cell from $2n$ to $4n$, so cells in S have DNA contents ranging from $2n$ to $4n$. DNA content then remains at $4n$ for cells in G_2 and M, decreasing to $2n$ after cytokinesis. Experimentally, cellular DNA content can be determined by incubation of cells with a fluorescent dye that binds to DNA, followed by analysis of the fluorescence intensity of individual cells in a **flow cytometer** or **fluorescence-activated cell sorter**, thereby distinguishing and allowing isolation of cells in the G_1, S, and G_2/M phases of the cell cycle.

Regulation of the Cell Cycle by Cell Growth and Extracellular Signals

The progression of cells through the division cycle is regulated by extracellular signals from the environment, as well as by internal signals that monitor and coordinate the various processes that take place during different cell cycle phases. An example of cell cycle regulation by extracellular signals is provided by the effect of growth factors on animal cell proliferation. In addition, different cellular processes, such as cell growth, DNA replication, and mitosis, all must be coordinated during cell cycle progression. This is accomplished by a series of control points that regulate progression through various phases of the cell cycle.

A major cell cycle regulatory point in many types of cells occurs late in G_1 and controls progression from G_1 to S. This regulatory point was first defined by studies of budding yeast (*Saccharomyces cerevisiae*), where it is known as **START** (Figure 16.5). Once cells have passed START, they are committed to entering S phase and undergoing one cell division cycle. However, passage through START is a highly regulated event in the yeast cell cycle where it is controlled by external signals, such as the availability of nutrients, as well as by cell size. For example, if yeasts are faced with a shortage of nutrients, they arrest their cell cycle at START and enter a resting state rather than proceeding to S phase. Thus START represents a decision point at which the cell determines whether sufficient nutrients are available to support progression through the rest of the division cycle. Polypeptide factors that signal yeast mating also arrest the cell cycle at

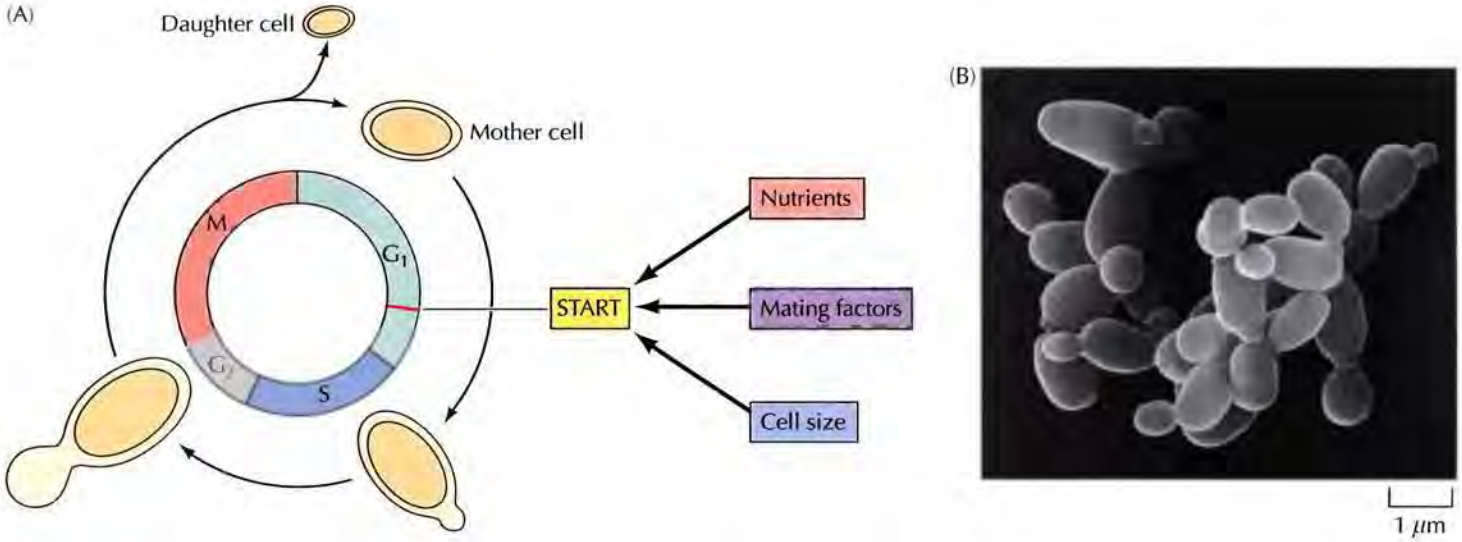


FIGURE 16.5 Regulation of the cell cycle of budding yeast (A) The cell cycle of *Saccharomyces cerevisiae* is regulated primarily at a point in late G_1 called START. Passage through START is controlled by the availability of nutrients, mating factors, and cell size. Note that these yeasts divide by budding. Buds form just after START and continue growing until they separate from the mother cell after mitosis. The daughter cell formed from the bud is smaller than the mother cell and therefore requires more time to grow during the G_1 phase of the next cell cycle. Although G_1 and S phases occur normally, the mitotic spindle begins to form during S phase, so the cell cycle of budding yeast lacks a distinct G_2 phase. (B) Scanning electron micrograph of *S. cerevisiae*. The size of the bud reflects the position of the cell in the cycle. (B, David M. Phillips/Visuals Unlimited.)

START, allowing haploid yeast cells to fuse with one another instead of progressing to S phase.

In addition to serving as a decision point for monitoring extracellular signals, START is the point at which cell growth is coordinated with DNA replication and cell division. The importance of this regulation is particularly evident in budding yeasts in which cell division produces progeny cells of very different sizes: a large mother cell and a small daughter cell. In order for yeast cells to maintain a constant size, the small daughter cell must grow more than the large mother cell does before they divide again. Thus cell size must be monitored in order to coordinate cell growth with other cell cycle events. This regulation is accomplished by a control mechanism that requires each cell to reach a minimum size before it can pass START. Consequently, the small daughter cell spends a longer time in G_1 and grows more than the mother cell.

The proliferation of most animal cells is similarly regulated in the G_1 phase of the cell cycle. In particular, a decision point in late G_1, called the **restriction point** in animal cells, functions analogously to START in yeasts (Figure 16.6). In contrast to yeasts, however, the passage of animal cells through the cell cycle is regulated primarily by the extracellular growth factors that signal cell proliferation, rather than by the availability of nutrients. In the presence of the appropriate growth factors, cells pass the restriction point and enter S phase. Once it has passed through the restriction point, the cell is committed to proceed through S phase and the rest of the cell

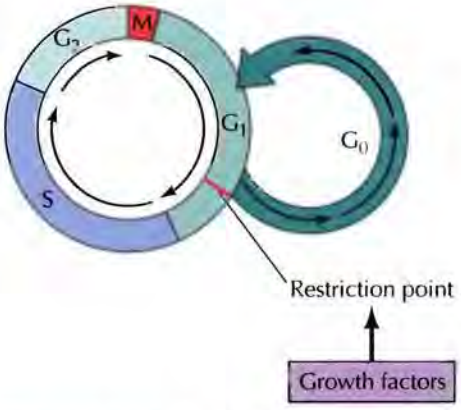


FIGURE 16.6 Regulation of animal cell cycles by growth factors The availability of growth factors controls the animal cell cycle at a point in late G_1 called the restriction point. If growth factors are not available during G_1, the cells enter a quiescent stage of the cycle called G_0.

FIGURE 16.7 Cell cycle of fission yeast (A) Fission yeasts grow by elongating at both ends and divide by forming a wall through the middle of the cell. In contrast to the cycle of budding yeasts, the cell cycle of fission yeasts has normal G_1, S, G_2, and M phases. Note that cytokinesis occurs in G_1. The length of the cell indicates its position in the cycle. (B) Light micrographs showing successive stages of mitosis and cytokinesis of *Schizosaccharomyces pombe*. (B, courtesy of C. F. Robinow, University of Western Ontario.)

cycle, even in the absence of further growth factor stimulation. On the other hand, if appropriate growth factors are not available in G_1, progression through the cell cycle stops at the restriction point. Such arrested cells then enter a quiescent stage of the cell cycle called G_0 in which they can remain for long periods of time without proliferating. G_0 cells are metabolically active, although they cease growth and have reduced rates of protein synthesis. As already noted, many cells in animals remain in G_0 unless called on to proliferate by appropriate growth factors or other extracellular signals. For example, skin fibroblasts are arrested in G_0 until they are stimulated to divide as required to repair damage resulting from a wound. The proliferation of these cells is triggered by platelet-derived growth factor, which is released from blood platelets during clotting and signals the proliferation of fibroblasts in the vicinity of the injured tissue.

Although the proliferation of most cells is regulated primarily in G_1, some cell cycles are instead controlled principally in G_2. One example is the cell cycle of the fission yeast *Schizosaccharomyces pombe* (Figure 16.7). In contrast to *Saccharomyces cerevisiae*, the cell cycle of *S. pombe* is regulated primarily by control of the transition from G_2 to M, which is the principal point at which cell size and nutrient availability are monitored. In animals, the primary example of cell cycle control in G_2 is provided by oocytes. Vertebrate oocytes can remain arrested in G_2 for long periods of time (several decades in humans) until their progression to M phase is triggered by hormonal stimulation. Extracellular signals can thus control cell proliferation by regulating progression from the G_2 to M as well as the G_1 to S phases of the cell cycle.

Cell Cycle Checkpoints

The controls discussed in the previous section regulate cell cycle progression in response to cell size and extracellular signals, such as nutrients and growth factors. In addition, the events that take place during different stages of the cell cycle must be coordinated with one another so that they

occur in the appropriate order. For example, it is critically important that the cell not begin mitosis until replication of the genome has been completed. The alternative would be a catastrophic cell division in which the daughter cells failed to inherit complete copies of the genetic material. In most cells, this coordination between different phases of the cell cycle is dependent on a series of **cell cycle checkpoints** that prevent entry into the next phase of the cell cycle until the events of the preceding phase have been completed.

Several cell cycle checkpoints function to ensure that incomplete or damaged chromosomes are not replicated and passed on to daughter cells (Figure 16.8). These checkpoints sense unreplicated or damaged DNA and coordinate further cell cycle progression with the completion of DNA replication or repair. For example, the checkpoint in G_2 prevents the initiation of mitosis until DNA replication is completed. This G_2 checkpoint senses unreplicated DNA, which generates a signal that leads to cell cycle arrest. Operation of the G_2 checkpoint therefore prevents the initiation of M phase before completion of S phase, so cells remain in G_2 until the genome has been completely replicated. Only then is the inhibition of G_2 progression relieved, allowing the cell to initiate mitosis and distribute the completely replicated chromosomes to daughter cells. In addition to sensing unreplicated DNA, the G_2 checkpoint senses DNA damage, such as that resulting from irradiation. If DNA damage is detected, arrest at the checkpoint allows time for the damage to be repaired, rather than being passed on to daughter cells.

DNA damage not only arrests the cell cycle in G_2 but also at checkpoints in G_1 and S phase. Arrest at the G_1 checkpoint allows repair of the damage to take place before the cell enters S phase, where the damaged DNA would be replicated. The S-phase checkpoint provides continual monitoring of the integrity of DNA to ensure that damaged DNA is repaired before it is replicated. In addition, the S-phase checkpoint provides a quality control monitor to promote the repair of any errors that occur during DNA replication, such as the incorporation of incorrect bases or incomplete replication of segments of DNA.

Cell cycle arrest at the G_1, S, and G_2 checkpoints is mediated by two related protein kinases, designated **ATM** and **ATR**, that recognize damaged or unreplicated DNA and are activated in response to DNA damage. ATM and ATR then activate a signaling pathway that leads not only to cell cycle arrest, but also to the activation of DNA repair and, in some cases, programmed cell death. The importance of checkpoint regulation is emphasized by the fact that these proteins were initially identified because mutations in the gene encoding ATM are responsible for the disease ataxia telangiectasia, which results in defects in the nervous and immune systems as well as a high frequency of cancer in affected individuals.

Another important cell cycle checkpoint that maintains the integrity of the genome occurs toward the end of mitosis (see Figure 16.8). This checkpoint, called the spindle assembly checkpoint, monitors the alignment of chromosomes on the mitotic spindle, thus ensuring that a complete set of chromosomes is distributed accurately to the daughter cells. For example, the failure of one or more chromosomes to align properly on the spindle causes mitosis to arrest at metaphase, prior to the segregation of the newly replicated chromosomes to daughter nuclei. As a result of the spindle assembly checkpoint, the chromosomes do not separate until a complete complement of chromosomes has been organized for distribution to each daughter cell.

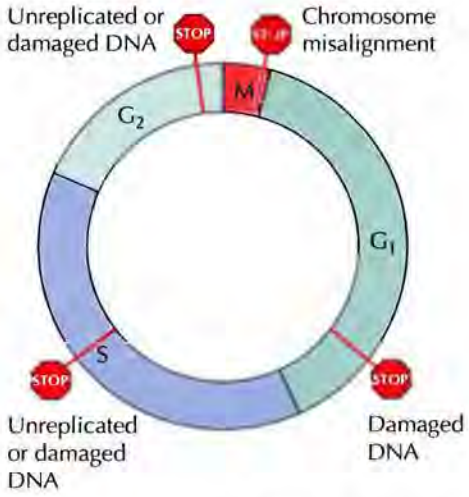


FIGURE 16.8 Cell cycle checkpoints Several checkpoints function to ensure that complete genomes are transmitted to daughter cells. DNA damage checkpoints in G_1, S, and G_2 lead to cell cycle arrest in response to damaged or unreplicated DNA. Another checkpoint, called the spindle assembly checkpoint, arrests mitosis if the chromosomes are not properly aligned on the mitotic spindle.

16.3 WEBSITE ANIMATION

Checkpoints in the Cell Cycle Several checkpoints function throughout the cell cycle to ensure that complete genomes are transmitted to daughter cells.

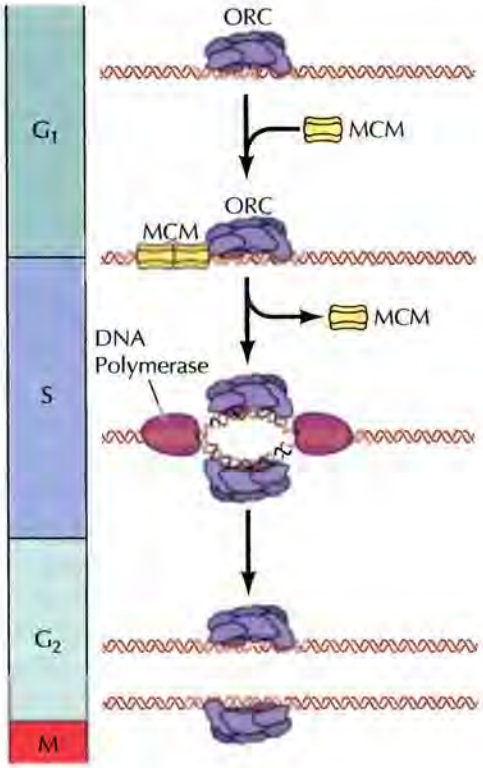


FIGURE 16.9 Restriction of DNA replication DNA replication is restricted to once per cell cycle by the MCM helicase proteins that bind to origins of replication together with ORC (origin recognition complex) proteins and are required for the initiation of DNA replication. MCM proteins are only able to bind to DNA in G_1, allowing DNA replication to initiate in S phase. Once initiation has occurred, the MCM proteins are displaced so that replication cannot initiate again until after mitosis.

Restricting DNA Replication to Once per Cell Cycle

The G_2 checkpoint prevents the initiation of mitosis prior to the completion of S phase, thereby ensuring that incompletely replicated DNA is not distributed to daughter cells. It is equally important to ensure that the genome is replicated only once per cell cycle. Thus, once a segment of DNA has been replicated in S phase, control mechanisms must exist to prevent re-initiation of DNA replication until the cell cycle has been completed and the cell has passed through mitosis. As discussed in Chapter 6, mammalian cells use thousands of origins to replicate their DNA, so the initiation of replication at each of these origins must be carefully controlled so that each segment of the genome is replicated only once during the S phase of each cell cycle.

The molecular mechanism that restricts DNA replication to once per cell cycle involves the action of the MCM helicase proteins that bind to replication origins together with the origin recognition complex (ORC) proteins (see Figure 6.15). The MCM proteins act as "licensing factors" that allow replication to initiate (Figure 16.9). Their binding to DNA is regulated during the cell cycle such that the MCM proteins are only able to bind to replication origins during G_1, allowing DNA replication to initiate when the cell enters S phase. Once initiation has occurred, the MCM proteins are displaced from the origin, so replication cannot initiate again until the cell passes through mitosis and enters G_1 phase of the next cell cycle. The association of MCM proteins with DNA during the S, G_2 and M phases of the cell cycle is blocked by activity of the protein kinases that regulate cell cycle progression, as discussed in the next section of this chapter.

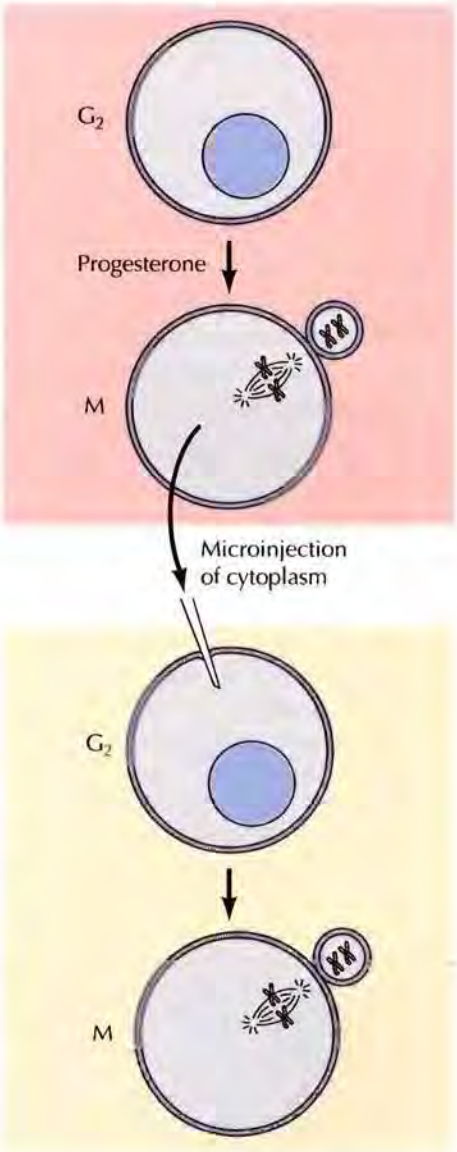


FIGURE 16.10 Identification of MPF Frog oocytes are arrested in the G_2 phase of the cell cycle, and entry into M phase of meiosis is triggered by the hormone progesterone. In the experiment diagrammed here, G_2-arrested oocytes were microinjected with cytoplasm extracted from oocytes that had undergone the transition from G_2 to M. Such cytoplasmic transfers induced the G_2 to M transition in the absence of hormonal stimulation, demonstrating that a cytoplasmic factor (MPF) is sufficient to induce entry into the M phase of meiosis.

Regulators of Cell Cycle Progression

One of the most exciting developments in contemporary cell biology has been the elucidation of the molecular mechanisms that control the progression of eukaryotic cells through the division cycle. Our current understanding of cell cycle regulation has emerged from a convergence of results obtained through experiments on organisms as diverse as yeasts, sea urchins, frogs, and mammals. These studies have revealed that the cell cycle of all eukaryotes is controlled by a conserved set of protein kinases, which are responsible for triggering the major cell cycle transitions.

Protein Kinases and Cell Cycle Regulation

Three initially distinct experimental approaches contributed to identification of the key molecules responsible for cell cycle regulation. The first of these avenues of investigation originated with studies of frog oocytes (Figure 16.10). These oocytes are arrested in the G_2 phase of the cell cycle until hormonal stimulation triggers their entry into the M phase of meiosis (discussed later in this chapter). In 1971, two independent teams of researchers (Yoshio Masui and Clement Markert, as well as Dennis Smith and Robert Ecker) found that oocytes arrested in G_2 could be induced to enter M phase by microinjection of cytoplasm from oocytes that had been hormonally stimulated. It thus appeared that a cytoplasmic factor present in hormone-treated oocytes was sufficient to trigger the transition from G_2 to M in oocytes that had not been exposed to hormone. Because the entry of oocytes into meiosis is frequently referred to as oocyte maturation, this cytoplasmic factor was called **maturation promoting factor (MPF)**. Further studies showed, however, that the activity of MPF is not restricted to the entry of oocytes into meiosis. To the contrary, MPF is also present in somatic cells, where it induces entry into M phase of the mitotic cycle. Rather than being specific to oocytes, MPF thus appeared to act as a general regulator of the transition from G_2 to M.

The second approach to understanding cell cycle regulation was the genetic analysis of yeasts, pioneered by Lee Hartwell and his colleagues in the early 1970s. Studying the budding yeast *Saccharomyces cerevisiae*, these investigators identified temperature-sensitive mutants that were defective in cell cycle progression. The key characteristic of these mutants (called *cdc* for *c*ell *d*ivision *c*ycle mutants) was that they underwent growth arrest at specific points in the cell cycle. For example, a particularly important mutant designated *cdc28* caused the cell cycle to arrest at START, indicating that the Cdc28 protein is required for passage through this critical regula-

KEY EXPERIMENT

The Discovery of MPF

Cytoplasmic Control of Nuclear Behavior during Meiotic Maturation of Frog Oocytes

Yoshio Masui and Clement L. Markert
Yale University, New Haven, CT
Journal of Experimental Zoology, 1971, Volume 177, Pages 129–146

Yoshio Masui

Clement Markert

The Context

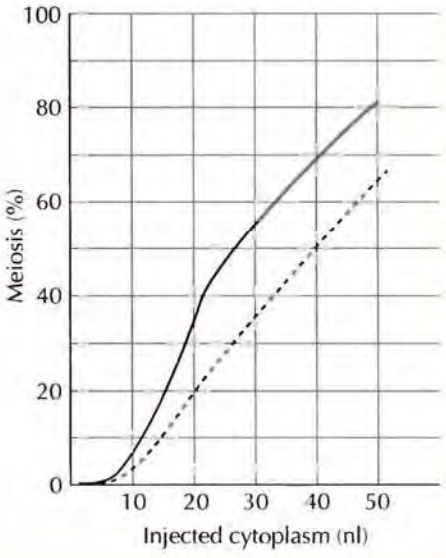
Nuclear transplantation and cell fusion experiments performed in the 1960s indicated that nuclei transferred into cells at different stages of the mitotic cell cycle generally adopt the behavior of the host cell. Thus it appeared that mitotic activity of the nucleus was regulated by the cytoplasm. However, the existence of postulated cytoplasmic factors that control the mitotic activity of nuclei remained to be demonstrated by a direct experimental approach. This demonstration was provided by the studies of Masui and Markert, who investigated the role of cytoplasmic factors in regulating nuclear behavior during meiosis of frog oocytes.

Several features of frog oocyte meiosis suggested that it is controlled by cytoplasmic factors. In particular, meiosis of frog oocytes is arrested at the end of the prophase of meiosis I. Treatment with the hormone progesterone triggers the resumption of meiosis, which is equivalent to the G_2 to M transition of somatic cells. The oocytes then undergo a second arrest at the metaphase of meiosis II, where they remain until fertilization. Masui and Markert hypothesized that the effects of both hormone treatment and fertilization on meiosis were due to changes in the cytoplasm that secondarily controlled behavior of the nucleus. They tested this hypothesis directly by transferring cytoplasm from hormone-stimulated oocytes into unstimulated oocytes. These experiments demonstrated that a cytoplasmic factor, which Masui and Markert named maturation promoting factor (MPF), is responsible for the induction of meiosis following hormone treatment.

The Experiments

Because of both their large size and their ability to survive injection with glass micropipettes, frog oocytes appeared to provide a particularly suitable experimental system for testing the activity of cytoplasmic factors. The basic design of Masui and Markert's experiments was to remove cytoplasm from donor oocytes that had been treated with progesterone to induce the resumption of meiosis. Varying amounts of this cytoplasm were then injected into untreated recipient oocytes. The key result was that cytoplasm removed six or more hours after hormone treatment of donor oocytes induced the resumption of meiosis in injected recipients (see figure). In contrast, injection of cytoplasm from control oocytes that had not been exposed to progesterone had no effect on the recipients. It thus appeared that hormone-treated oocytes contained a cytoplasmic factor that could induce the resumption of meiosis in recipients that had never been exposed to progesterone.

Control experiments ruled out the possibility that progesterone itself is the meiosis-inducing factor in donor cytoplasm. In particular, it was demonstrated that the injection of progesterone into recipient oocytes fails to induce meiosis. Only external application of the hormone is effective, indicating that progesterone acts on a cell surface receptor to activate a distinct cytoplasmic factor. Similar experiments performed independently by Dennis Smith and Robert Ecker (The interaction of steroids with *Rana pipiens* oocytes in the induction of maturation. *Dev. Biol.* 25: 232–247, 1971) led to the same conclusion.



Recipient oocytes were injected with the indicated amounts of cytoplasm from oocytes that had been treated with progesterone. Donor cytoplasm was either withdrawn from the central region of oocytes with a micropipette (solid line) or prepared by homogenization of whole oocytes (dashed line). Results are presented as the percentage of injected oocytes that were induced to resume meiosis.

KEY EXPERIMENT

Interestingly, the action of progesterone in this system is distinct from its action in most cells, where it diffuses across the plasma membrane and binds to an intracellular receptor (see Chapter 15). In oocytes, however, progesterone clearly acts on the cell surface to activate a distinct factor in the oocyte cytoplasm. Since the resumption of oocyte meiosis is commonly referred to as oocyte maturation, Masui and Markert coined the term "maturation promoting factor" for their newly discovered regulator of meiosis.

The Impact

Following its discovery in frog oocytes, MPF was also found to be present in somatic cells, where it induces the G_2 to M transition of mitosis. Thus MPF appeared to be a general regulator of entry into the M phase of both mitotic and meiotic cell cycles. The eventual purification of MPF from frog oocytes in 1988 then converged with yeast genetics and studies of sea urchin embryos to reveal the identity of this critical cell cycle regulator. Namely, MPF was found to be a dimer of cyclin B and the Cdk1 protein kinase. Further studies have established that both cyclin B and Cdk1 are members of large families of proteins, with different cyclins and Cdk1-related protein kinases functioning analogously to MPF in the regulation of other cell cycle transitions. The discovery of MPF in frog oocytes thus paved the way to understanding a cell cycle regulatory apparatus that is conserved throughout all eukaryotes.

tory point in G_1 (Figure 16.11). A similar collection of cell cycle mutants was isolated in the fission yeast *Schizosaccharomyces pombe* by Paul Nurse and his collaborators. These mutants included *cdc2*, which arrests the *S. pombe* cell cycle both in G_1 and at the G_2 to M transition (the major regulatory point in fission yeast). Comparative analysis showed that *S. cerevisiae cdc28* and *S. pombe cdc2* are functionally homologous genes, which are required for passage through START as well as for entry into mitosis in both species of yeasts. Further studies demonstrated that *cdc2* and *cdc28* encoded a protein kinase—the first indication of the prominent role of protein phosphorylation in regulating the cell cycle. In addition, related genes were identified in other eukaryotes, including humans. The protein kinase encoded by the

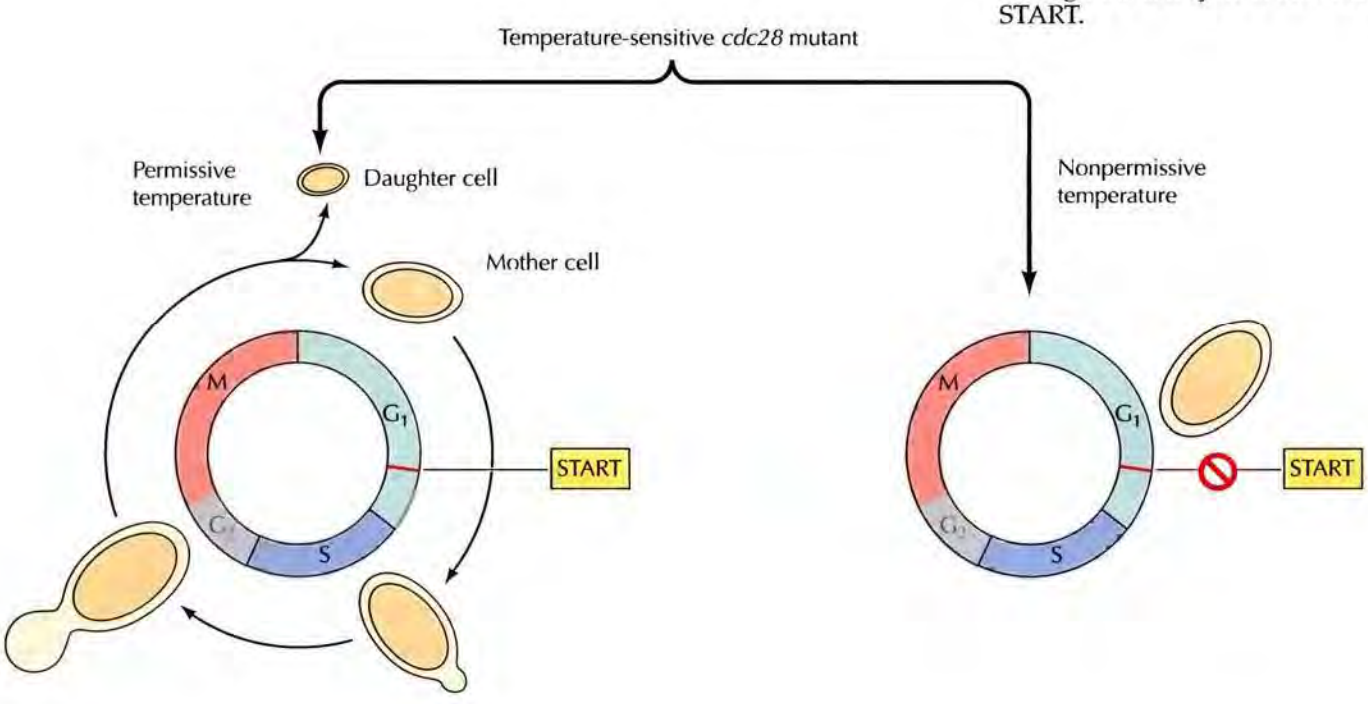


FIGURE 16.11 Properties of *S. cerevisiae cdc28* mutants The temperature-sensitive *cdc28* mutant replicates normally at the permissive temperature. At the nonpermissive temperature, however, progression through the cell cycle is blocked at START.

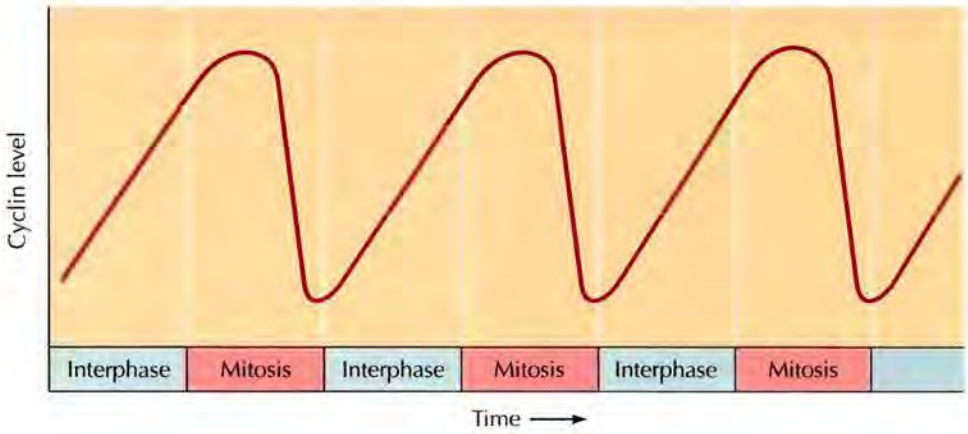


FIGURE 16.12 Accumulation and degradation of cyclins in sea urchin embryos The cyclins were identified as proteins that accumulate throughout interphase and are rapidly degraded toward the end of mitosis.

yeast *cdc2* and *cdc28* genes has since been shown to be a conserved cell cycle regulator in all eukaryotes, which is known as **Cdk1.**

The third line of investigation that eventually converged with the identification of MPF and yeast genetics emanated from studies of protein synthesis in early sea urchin embryos. Following fertilization, these embryos go through a series of rapid cell divisions. Intriguingly, studies with protein synthesis inhibitors had revealed that entry into M phase of these embryonic cell cycles requires new protein synthesis. In 1983, Tim Hunt and his colleagues identified two proteins that display a periodic pattern of accumulation and degradation in sea urchin and clam embryos. These proteins accumulate throughout interphase and are then rapidly degraded toward the end of each mitosis (Figure 16.12). Hunt called these proteins **cyclins** (the two proteins were designated cyclin A and cyclin B) and suggested that they might function to induce mitosis, with their periodic accumulation and destruction controlling entry and exit from M phase. Direct support for such a role of cyclins was provided in 1986, when Joan Ruderman and her colleagues showed that microinjection of cyclin A into frog oocytes is sufficient to trigger the G_2 to M transition.

These initially independent approaches converged dramatically in 1988 when MPF was purified from frog eggs in the laboratory of James Maller. Molecular characterization of MPF in several laboratories then showed that this conserved regulator of the cell cycle is composed of two key subunits: Cdk1 and cyclin B (Figure 16.13). Cyclin B is a regulatory subunit required for catalytic activity of the Cdk1 protein kinase, consistent with the notion that MPF activity is controlled by the periodic accumulation and degradation of cyclin B during cell cycle progression.

A variety of further studies have confirmed this role of cyclin B, as well as demonstrating the regulation of MPF by phosphorylation and dephosphorylation of Cdk1 (Figure 16.14). In mammalian cells, cyclin B is synthesized and forms complexes with Cdk1 during G_2. As these complexes form, Cdk1 is phosphorylated at two critical regulatory positions. One of these phosphorylations occurs on threonine-161 and is required for Cdk1 kinase activity. The second is a phosphorylation of tyrosine-15 and of the adjacent threonine-14 in vertebrates. Phosphorylation of tyrosine-15, catalyzed by a protein kinase called Wee1, inhibits Cdk1 activity and leads to the accumulation of inactive Cdk1/cyclin B complexes throughout G_2. The transition from G_2 to M is then brought about by activation of the Cdk1/cyclin B complex as a result of dephosphorylation of threonine-14 and tyrosine-15 by a protein phosphatase called Cdc25C.

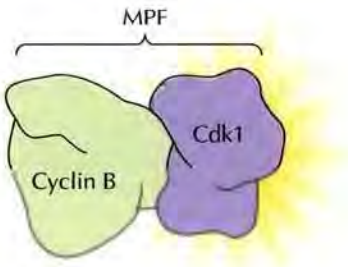


FIGURE 16.13 Structure of MPF MPF is a dimer consisting of cyclin B and the Cdk1 protein kinase.

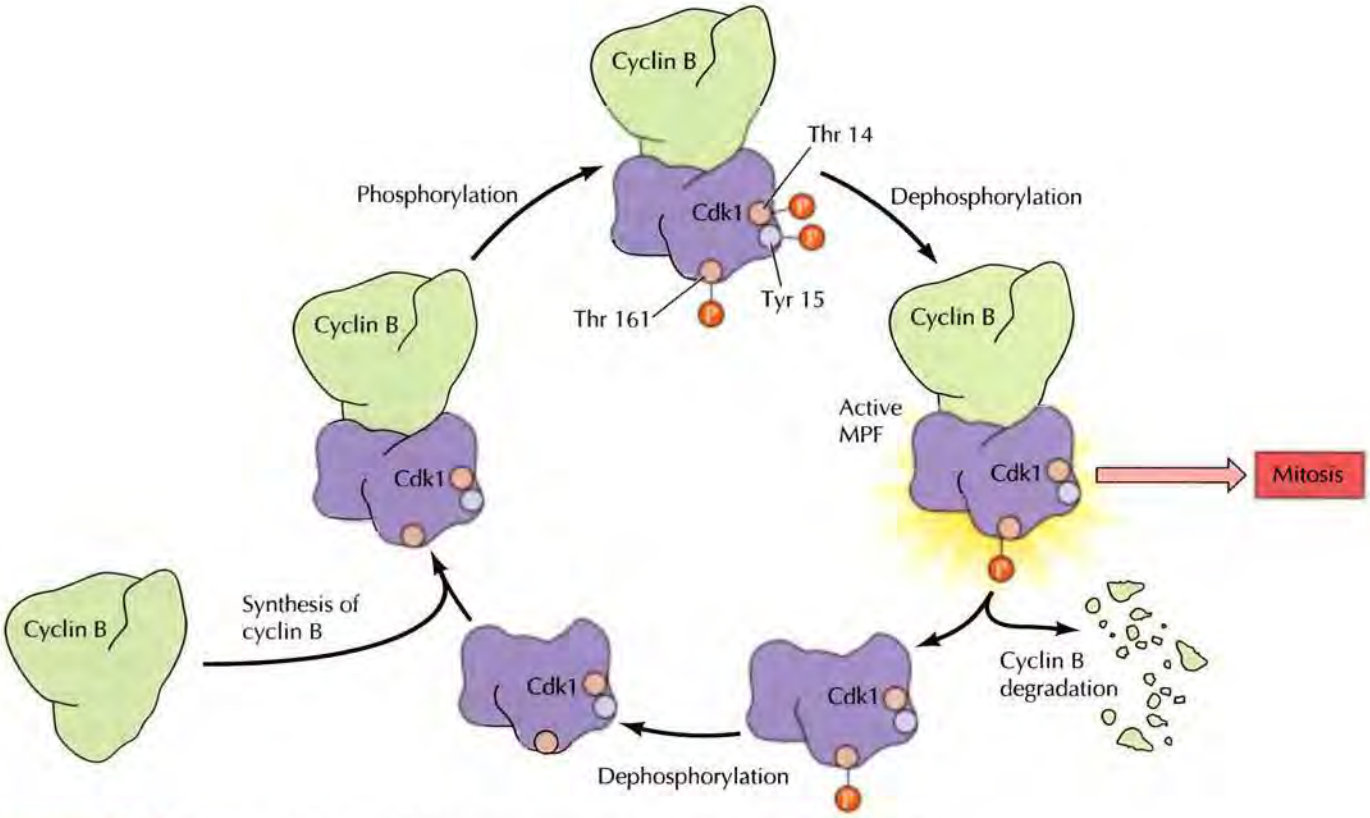


FIGURE 16.14 MPF regulation Cdk1 forms complexes with cyclin B during G_2. Cdk1 is then phosphorylated on threonine-161, which is required for Cdk1 activity, as well as on tyrosine-15 (and threonine-14 in vertebrate cells), which inhibits Cdk1 activity. Dephosphorylation of Thr14 and Tyr15 activates MPF at the G_2 to M transition. MPF activity is then terminated toward the end of mitosis by proteolytic degradation of cyclin B.

Once activated, the Cdk1 protein kinase phosphorylates a variety of target proteins that initiate the events of M phase, which are discussed later in this chapter. In addition, Cdk1 activity triggers the degradation of cyclin B, which occurs as a result of ubiquitin-mediated proteolysis. This proteolytic destruction of cyclin B then inactivates Cdk1, leading the cell to exit mitosis, undergo cytokinesis, and return to interphase.

16.4 WEBSITE ANIMATION

Cyclins, Cdks, and the Cell Cycle Maturation promoting factor (MPF)—consisting of cyclin B and Cdk1—regulates the transition from G_2 to M phase of the cell cycle.

Families of Cyclins and Cyclin-Dependent Kinases

The structure and function of MPF (Cdk1/cyclin B) provide not only a molecular basis for understanding entry and exit from M phase but also the foundation for elucidating the regulation of other cell cycle transitions. The insights provided by characterization of the Cdk1/cyclin B complex have thus had a sweeping impact on understanding cell cycle regulation. In particular, further research has established that both Cdk1 and cyclin B are members of families of related proteins, with different members of these families controlling progression through distinct phases of the cell cycle.

As discussed earlier, Cdk1 controls passage through START as well as entry into mitosis in yeasts. It does so, however, in association with distinct cyclins (Figure 16.15). In particular, the G_2 to M transition is driven by Cdk1 in association with the mitotic B-type cyclins (Clb1, Clb2, Clb3, and Clb4). Passage through START, however, is controlled by Cdk1 in association with a distinct class of cyclins called **G_1 cyclins** or **Cln's**. Cdk1 then associates

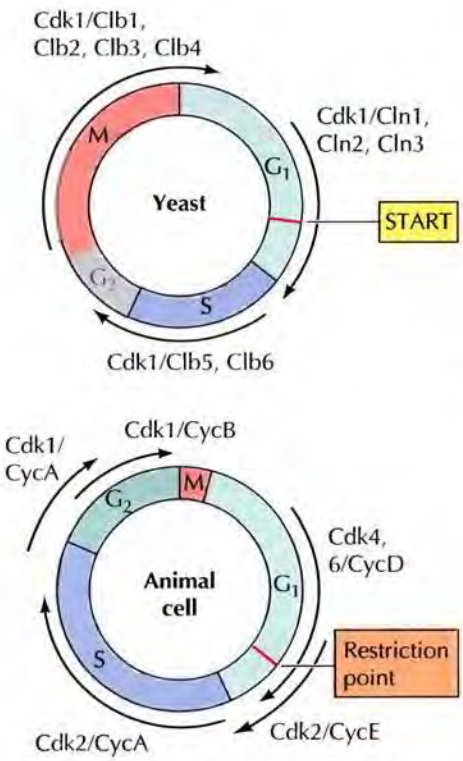


FIGURE 16.15 Complexes of cyclins and cyclin-dependent kinases In yeast, passage through START is controlled by Cdk1 in association with G_1 cyclins (Cln1, Cln2, and Cln3). Complexes of Cdk1 with distinct B-type cyclins (Clb's) then regulate progression through S phase and entry into mitosis. In animal cells, progression through the G_1 restriction point is controlled by complexes of Cdk4 and Cdk6 with D-type cyclins. Cdk2/cyclin E complexes function later in G_1 and are required for the G_1 to S transition. Cdk2/cyclin A complexes are then required for progression through S phase. Cdk1/cyclin A regulates progression to G_2, and Cdk1/cyclin B complexes drive the G_2 to M transition.

with different B-type cyclins (Clb5 and Clb6), which are required for progression through S phase. These associations of Cdk1 with distinct B-type and G_1 cyclins direct Cdk1 to phosphorylate different substrate proteins, as required for progression through specific phases of the cell cycle.

The cell cycles of higher eukaryotes are controlled not only by multiple cyclins but also by multiple Cdk1-related protein kinases, which are known as **Cdk's** for *c*yclin-*d*ependent *k*inases. These multiple members of the Cdk family associate with specific cyclins to drive progression through the different stages of the cell cycle (see Figure 16.15). For example, progression from G_1 to S is regulated principally by Cdk2, Cdk4, and Cdk6 in association with cyclins D and E. Complexes of Cdk4 and Cdk6 with the D-type cyclins (cyclins D1, D2, and D3) play a critical role in progression through

KEY EXPERIMENT

The Identification of Cyclin

Cyclin: A Protein Specified by Maternal mRNA in Sea Urchin Eggs That Is Destroyed at Each Cleavage Division

Tom Evans, Eric T. Rosenthal, Jim Youngblom, Dan Distel, and Tim Hunt

Marine Biological Laboratory, Woods Hole, MA

Cell, 1983, Volume 33, Pages 389–396

Tim Hunt

The Context

Tim Hunt and his colleagues undertook an analysis of protein synthesis in sea urchin eggs following fertilization, with the goal of identifying newly synthesized proteins that might be required for cell division during embryo development. A variety of earlier experiments had shown that translation of maternal mRNAs, stored in unfertilized eggs, was activated at fertilization, leading to striking changes in the profile of protein synthesis following fertilization. In addition, studies with inhibitors of translation had established that protein synthesis was required for cell division during early embryonic development. For example, inhibition of protein synthesis following fertilization of sea urchin eggs blocked nuclear envelope breakdown, chromosome condensation, and spindle formation. These results suggested that embryonic cell divisions required proteins that were synthesized from maternal mRNAs in fertilized eggs.

By studies of protein synthesis in fertilized eggs, Hunt and colleagues therefore hoped to identify newly synthesized proteins that were involved in cell division. Their experiments led to the identification of cyclins and elucidation of the fundamental mechanism that drives the division cycle of eukaryotic cells.

The Experiments

Unfertilized and fertilized sea urchin eggs were incubated in the presence of radioactive methionine, and cell extracts were analyzed by gel electrophoresis to identify newly synthesized proteins. Several prominent proteins were found to be synthesized in fertilized eggs, as had previously been seen in eggs of other organisms. Moreover, careful examination of the profile of radiolabeled proteins at different times after fertilization revealed a surprising

the restriction point in G_1. The E-type cyclins (E1 and E2) are expressed later in G_1, and Cdk2/cyclin E complexes are required for the G_1 to S transition and initiation of DNA synthesis. Complexes of Cdk2 with A-type cyclins (A1 and A2) function in the progression of cells through S phase. Cdk1 then regulates passage from S to G_2 and from G_2 to M in complexes with A- and B-type cyclins (B1, B2, and B3), respectively.

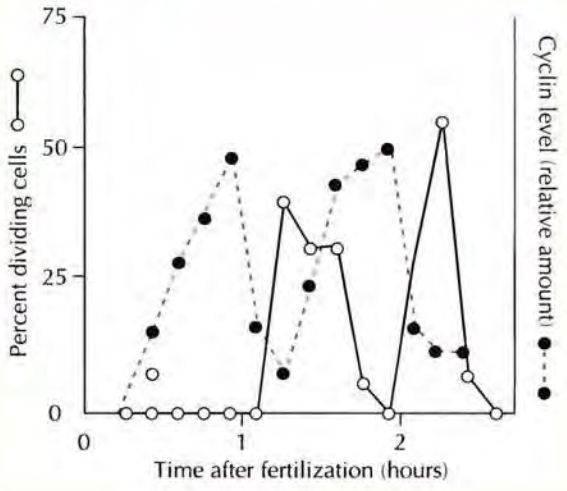
The roles of the Cdk/cyclin complexes summarized in Figure 16.15 are based on extensive research in many laboratories, principally using cultured cells as experimental models. However, recent studies of Cdk's and cyclins in genetically modified mice have yielded surprising results. In particular, embryonic cells from mice with mutated genes for the D-type and E-type cyclins as well as for Cdk2, Cdk4, and Cdk6 are still capable of proliferation. Moreover, mice lacking many of these genes are viable, although they may exhibit abnormalities in the development of specific cell types.

KEY EXPERIMENT

result. By analyzing the proteins present at 10 minute intervals after fertilization, Hunt and his colleagues identified a protein that was prominent in newly fertilized eggs, but then almost disappeared 85 minutes after fertilization. This protein again became abundant 95 minutes after fertilization, and declined again 30 minutes later. Because of these cyclic oscillations in its abundance, the protein was named "cyclin."

A further striking result was obtained when the oscillations in levels of cyclin were compared with the timing of cell divisions in the developing sea urchin embryos (see figure). The first division of the fertilized egg to a two-cell embryo took place 85 minutes after fertilization, coinciding with the decline of cyclin levels. The levels of cyclin then increased following this first cell division, and declined again 125 minutes after fertilization, coincident with the second cycle of embryonic cell division. Additional experiments demonstrated that cyclin was continuously synthesized after fertilization, but was rapidly degraded prior to each cell division.

Sea urchin eggs were fertilized and radioactive methionine was added 5 minutes after fertilization. Samples were then harvested at 10 minute intervals and analyzed by gel electrophoresis to determine the levels of cyclin. In parallel, samples were examined microscopically to quantitate the percentage of cells undergoing cell division.



The Impact

Based on its periodic degradation, coincident with the onset of embryonic cell division, Hunt and colleagues concluded that it was "difficult to believe that the behavior of the cyclins is not connected with the processes involved in cell division." However, in the absence of experiments demonstrating the biological activity of cyclin, they had no direct evidence for this conclusion. Nonetheless, the activity of MPF was known to oscillate similarly during the cell cycle, and additional experiments had demonstrated that the formation of active MPF required protein synthesis. These similarities between MPF and cyclin led Hunt and his colleagues to suggest a direct relationship between MPF, defined as a biological activity, and cyclin, defined as a protein whose levels oscillated during cell cycle progression.

Further experiments of Joan Ruderman and her colleagues demonstrated that injection of cyclin was sufficient to induce the entry of frog oocytes into meiosis, showing that cyclin had the biological activity of MPF. The biochemical relationship between cyclin and MPF was then established by the purification of MPF from frog eggs in the laboratory of James Maller and the subsequent characterization of MPF as a dimer of cyclin B and Cdk1. The periodic degradation of cyclin B, discovered by Tim Hunt and colleagues, controls the activity of Cdk1 during entry and exit from mitosis. Moreover, other cyclins play similar roles in regulating transitions between other phases of the cell cycle. The discovery of cyclin as an oscillating protein in sea urchin embryos was thus key to elucidating the molecular basis of cell cycle regulation.

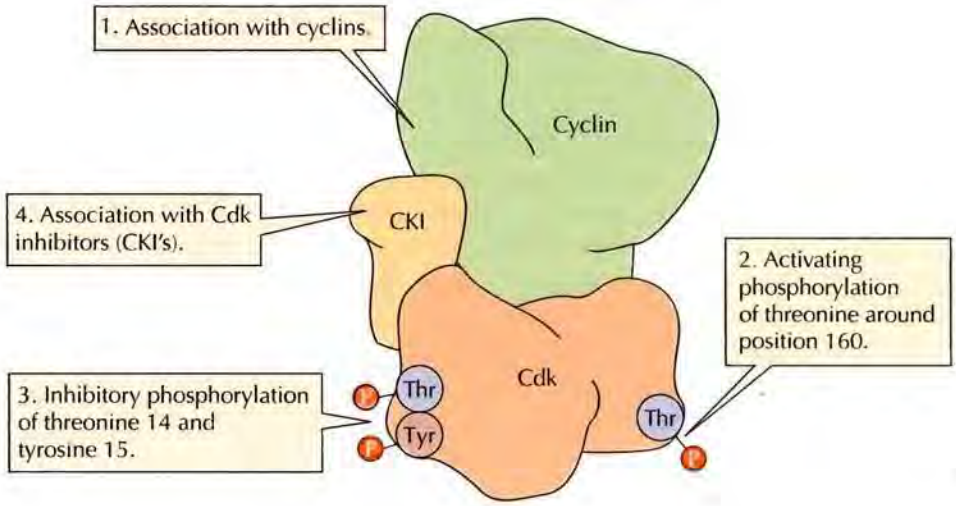


FIGURE 16.16 Mechanisms of Cdk regulation The activities of Cdk's are regulated by four molecular mechanisms.

These results suggest that there may be substantial redundancy between different cyclins and Cdk's, so that one can compensate for the absence of another. For example, recent work suggests that Cdk1 may be able to substitute for Cdk2 (in complexes with both cyclin A and cyclin E). However, further studies are clearly necessary to understand the extent to which these cell cycle regulators can substitute for one another.

The activity of Cdk's during cell cycle progression is regulated by four molecular mechanisms (Figure 16.16). As already discussed for Cdk1, the first level of regulation involves the association of Cdk's with their cyclin partners. Thus the formation of specific Cdk/cyclin complexes is controlled by cyclin synthesis and degradation. Second, activation of Cdk/cyclin complexes requires phosphorylation of a conserved Cdk threonine residue around position 160. This activating phosphorylation of the Cdk's is catalyzed by an enzyme called CAK (for Cdk-*a*ctivating *k*inase), which is itself composed of a Cdk (Cdk7) complexed with cyclin H. Complexes of Cdk7 and cyclin H are also associated with the transcription factor TFIIH, which is required for initiation of transcription by RNA polymerase II (see Chapter 6), so this member of the Cdk family participates in transcription as well as cell cycle regulation.

■ **Some of the cell cycle mutants of *S. pombe*, isolated by Paul Nurse and his colleagues at the University of Edinburgh, were identified because the cells divided when they were smaller in size than normal. These mutants were named *Wee*, the Scottish term for small.**

In contrast to the activating phosphorylation by CAK, the third mechanism of Cdk regulation involves inhibitory phosphorylation of tyrosine residues near the Cdk amino terminus, catalyzed by the Wee1 protein kinase. In particular, both Cdk1 and Cdk2 are inhibited by phosphorylation of tyrosine-15, and the adjacent threonine-14 in vertebrates. These Cdk's are then activated by dephosphorylation of these residues by members of the Cdc25 family of protein phosphatases.

In addition to regulation of the Cdk's by phosphorylation, their activities are also controlled by the binding of inhibitory proteins (called **Cdk inhibitors** or **CKIs**). In mammalian cells, two families of Cdk inhibitors are responsible for regulating different Cdk's (Table 16.1). Members of the Ink4 family specifically bind to and inhibit monomeric Cdk4 and Cdk6, so the Ink4 CKIs act to inhibit progression through G_1. In contrast, members of the Cip/Kip family bind to and inhibit the protein kinase activity of both Cdk1 (complexed with either cyclin A or cyclin B) and Cdk2 (complexed with either cyclin A or cyclin E), thereby inhibiting progression through all phases of the cell cycle. The Cip/Kip CKI's also bind to complexes of Cdk4 and Cdk6 with cyclin D, but their role in regulation of these complexes is

TABLE 16.1 Cdk Inhibitors

Inhibitor	Cdk or Cdk/cyclin complex	Cell cycle phase affected
Ink4 family (p15, p16, p18, p19)	Cdk4 and Cdk6	G_1
Cip/Kip family (p21, p27, p57)	Cdk1/cyclin A	G_2
	Cdk1/cyclin B	G_2/M
	Cdk2/cyclin A	S
	Cdk2/cyclin E	G_1

unclear. Control of Cdk inhibitors thus provides an additional mechanism for regulating Cdk activity. The combined effects of these multiple modes of Cdk regulation are responsible for controlling cell cycle progression in response both to checkpoint controls and to the variety of extracellular stimuli that regulate cell proliferation.

Growth Factors and the Regulation of G_1 Cdk's

As discussed earlier, the proliferation of animal cells is regulated largely by a variety of extracellular growth factors that control the progression of cells through the restriction point in late G_1. In the absence of growth factors, cells are unable to pass the restriction point and become quiescent, frequently entering the resting state known as G_0 from which they can reenter the cell cycle in response to growth factor stimulation. This control of cell cycle progression by extracellular growth factors implies that the intracellular signaling pathways stimulated downstream of growth factor receptors (discussed in the preceding chapter) ultimately act to regulate components of the cell cycle machinery.

One critical link between growth factor signaling and cell cycle progression is provided by the D-type cyclins (Figure 16.17). Cyclin D1 synthesis is induced in response to growth factor stimulation in part as a result of signaling through the Ras/Raf/MEK/ERK pathway, and cyclin D1 continues to be synthesized as long as growth factors are present. However, cyclin D1 is also rapidly degraded, so its intracellular concentration rapidly falls if growth factors are removed. Thus, as long as growth factors are present through G_1, complexes of Cdk4, 6/cyclin D1 drive cells through the restriction point. On the other hand, if growth factors are removed prior to this key regulatory point in the cell cycle, the levels of cyclin D1 rapidly fall and cells are unable to progress through G_1 to S, instead becoming quiescent and entering G_0. The inducibility and rapid turnover of cyclin D1 thus integrates growth factor signaling with the cell cycle machinery, allowing the availability of extracellular growth factors to control the progression of cells through G_1.

Since cyclin D1 is a critical target of growth factor signaling, it might be expected that defects in cyclin D1 regulation could contribute to the loss of growth regulation characteristic of cancer cells. Consistent with this expectation, many human cancers have been found to arise as a result of defects in cell cycle regulation, just as many others result from abnormalities in the intracellular signaling pathways activated by growth factor receptors (see Chapter 15). For example, mutations resulting in continual unregulated expression of cyclin D1 contribute to the development of a variety of human cancers, including lymphomas and breast cancers. Similarly, mutations that inactivate the Ink4 Cdk inhibitors that bind to Cdk4 and Cdk6 are commonly found in human cancer cells.

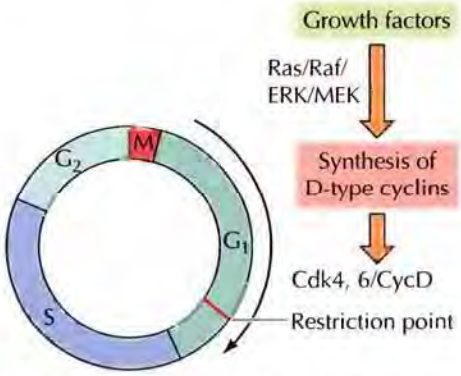


FIGURE 16.17 Induction of D-type cyclins Growth factors regulate cell cycle progression through the G_1 restriction point by inducing synthesis of D-type cyclins via the Ras/Raf/MEK/ERK signaling pathway.

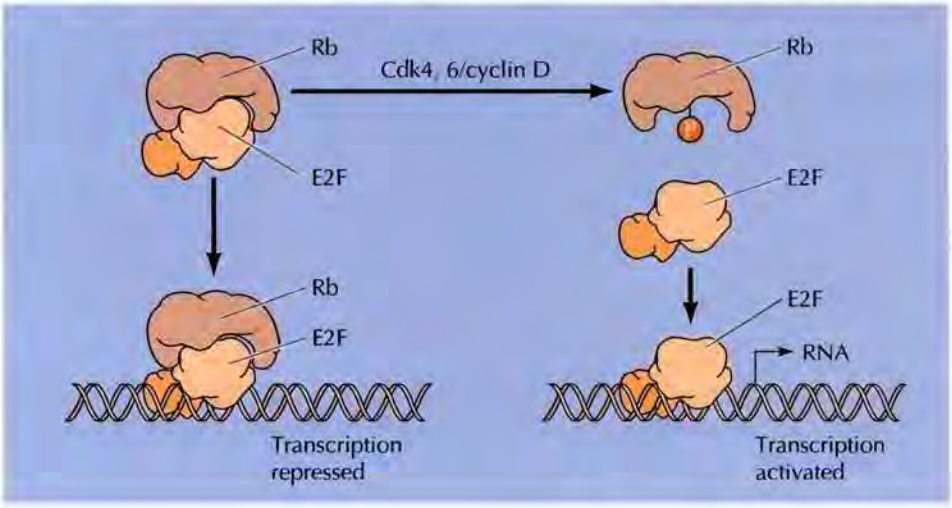


FIGURE 16.18 Cell cycle regulation of Rb and E2F In its underphosphorylated form, Rb binds to members of the E2F family, repressing transcription of E2F-regulated genes. Phosphorylation of Rb by Cdk4, 6/cyclin D complexes results in its dissociation from E2F in late G_1. E2F then stimulates expression of its target genes, which encode proteins required for cell cycle progression.

The connection between cyclin D, growth control, and cancer is further fortified by the fact that a key substrate protein of Cdk4, 6/cyclin D complexes is itself frequently mutated in a wide array of human tumors. This protein, designated **Rb**, was first identified as the product of a gene responsible for retinoblastoma, a rare inherited childhood eye tumor (see Chapter 18). Further studies then showed that mutations resulting in the absence of functional Rb protein are not restricted to retinoblastoma but also contribute to a variety of common human cancers. Rb is the prototype of a **tumor suppressor gene**—a gene whose inactivation leads to tumor development. Whereas oncogene proteins such as Ras (see Chapter 15) and cyclin D drive cell proliferation, the proteins encoded by many tumor suppressor genes (including Rb and the Ink4 Cdk inhibitors) act as brakes that slow down cell cycle progression.

Further studies have revealed that Rb and related members of the Rb family play a key role in coupling the cell cycle machinery to the expression of genes required for cell cycle progression and DNA synthesis (Figure 16.18). The activity of Rb proteins is regulated by changes in phosphorylation as cells progress through the cycle. In particular, Rb becomes phosphorylated by Cdk4, 6/cyclin D complexes as cells pass through the restriction point in G_1. In its underphosphorylated form (present in G_0 or early G_1), Rb binds to members of the **E2F** family of transcription factors, which regulate expression of several genes involved in cell cycle progression, including the gene encoding cyclin E. E2F binds to its target sequences in either the presence or absence of Rb. However, Rb acts as a repressor, so the Rb/E2F complex suppresses transcription of E2F-regulated genes. Phosphorylation of Rb by Cdk4, 6/cyclin D complexes results in its dissociation from E2F, which then activates transcription of its target genes. Rb thus acts as a molecular switch that converts E2F from a repressor to an activator of genes required for cell cycle progression. The control of Rb by Cdk4, 6/cyclin D phosphorylation in turn couples this critical regulation of gene expression to the availability of growth factors in G_1.

Progression through the restriction point and entry into S phase is mediated by the activation of Cdk2/cyclin E complexes (Figure 16.19). This results in part from the synthesis of cyclin E, which is stimulated by E2F following phosphorylation of Rb. In addition, the activity of Cdk2/cyclin E is inhibited in G_0 or early G_1 by the Cdk inhibitor p27, which belongs to the Cip/Kip family (see Table 16.1). This inhibition of Cdk2 by p27 is relieved by multiple mechanisms as cells progress through G_1. First, growth factor

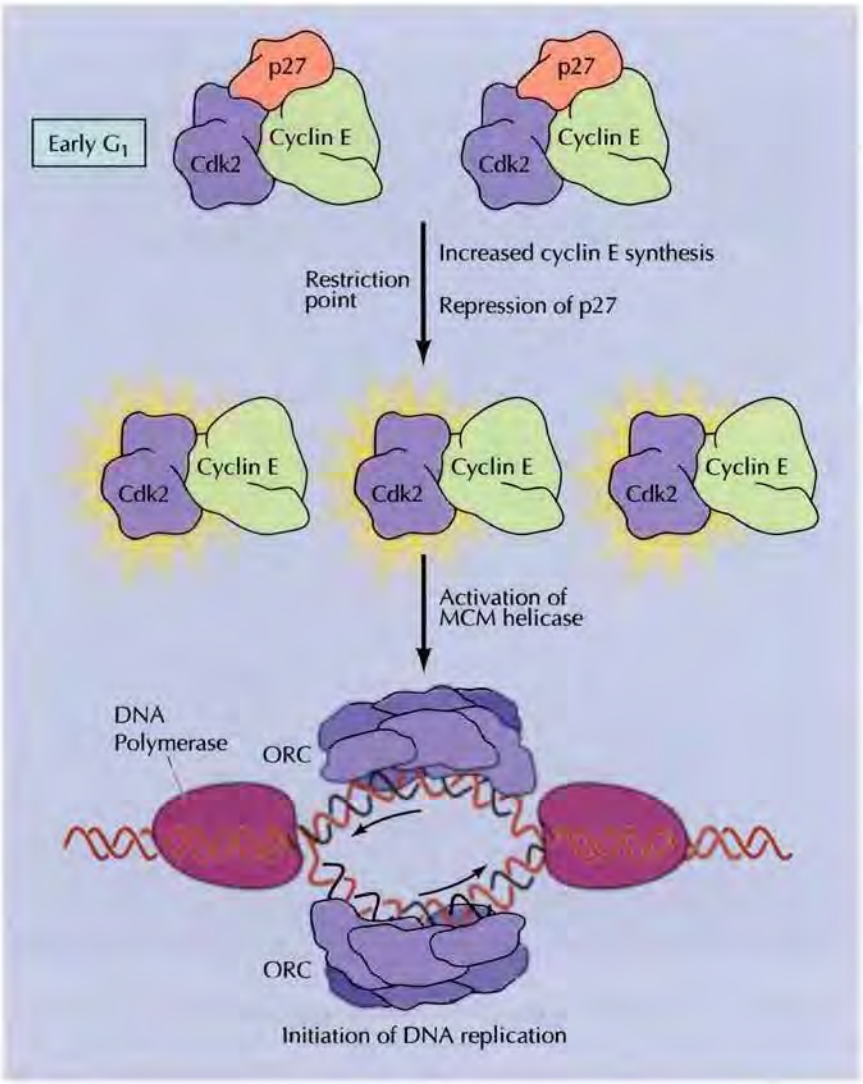


FIGURE 16.19 Cdk2/cyclin E and entry into S phase In early G_1, Cdk2/cyclin E complexes are inhibited by the Cdk inhibitor p27. Passage through the restriction point induces the synthesis of cyclin E via activation of E2F. In addition, growth factor signaling reduces the levels of p27 by inhibiting its transcription and translation. The resulting activation of Cdk2/cyclin E leads to activation of the MCM helicase and initiation of DNA replication.

signaling via both the Ras/Raf/MEK/ERK and PI 3-kinase/Akt pathways reduces the transcription and translation of p27, lowering the levels of p27 within the cell. In addition, increased synthesis of cyclin D leads to the binding of p27 to Cdk4, 6/cyclin D complexes, sequestering it from binding to Cdk2/cyclin E. Once Cdk2 becomes activated, it brings about the complete degradation of p27 by phosphorylating it and targeting it for ubiquitination. This positive autoregulation then results in full activation of Cdk2/cyclin E complexes. Cdk2 also phosphorylates Rb, completing its inactivation. Cdk2/cyclin E complexes then initiate S phase by activating the MCM helicase proteins at replication origins (see Figure 16.9), leading to the initiation of DNA synthesis.

DNA Damage Checkpoints

Cell proliferation is regulated not only by growth factors but also by a variety of signals that act to inhibit cell cycle progression. DNA damage checkpoints play a critical role in maintaining the integrity of the genome by arresting cell cycle progression in response to damaged or incompletely replicated DNA. These checkpoints, which are operative in G_1, S, and G_2 phases of the cell cycle, serve to halt cell cycle progression and allow time for the damage to be repaired before DNA replication or cell division proceeds (see Figure 16.8).

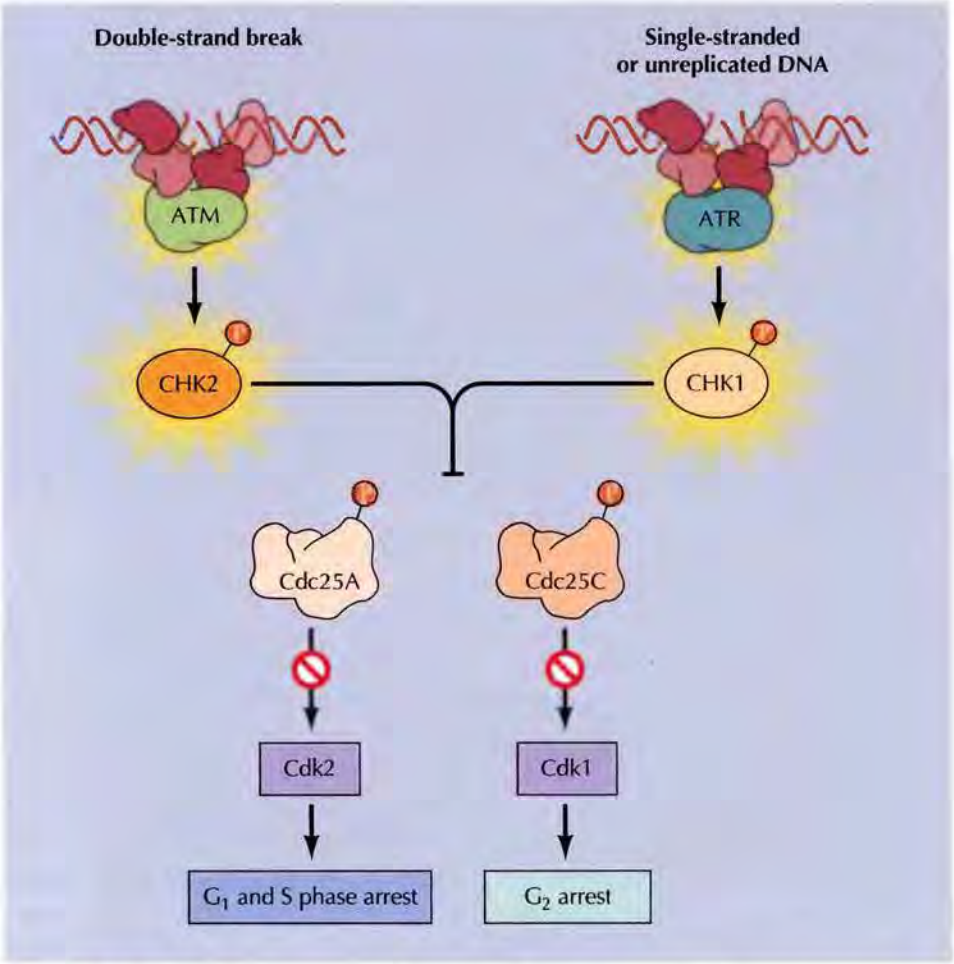


FIGURE 16.20 Cell cycle arrest at the DNA damage checkpoints The ATM and ATR protein kinases are activated in complexes of proteins that recognize unreplicated or damaged DNA. ATM is activated principally by double-strand breaks and ATR by single-stranded or unreplicated DNA. ATM and ATR then phosphorylate and activate the CHK2 and CHK1 protein kinases, respectively. CHK1 and CHK2 phosphorylate and inhibit the Cdc25A and Cdc25C protein phosphatases. Cdc25A and Cdc25C are required to activate Cdk2 and Cdk1, respectively, so their inhibition leads to arrest at the DNA damage checkpoints in G_1, S, and G_2.

Cell cycle arrest at the DNA damage checkpoints is initiated by the ATM or ATR protein kinases, which are components of protein complexes that recognize damaged or unreplicated DNA (Figure 16.20). ATM is activated principally by double-strand breaks, while ATR is activated by single-stranded or unreplicated DNA. Once activated by DNA damage, ATM and ATR phosphorylate and activate the **checkpoint kinases** CHK2 and CHK1, respectively, bringing about cell cycle arrest.

Both CHK1 and CHK2 phosphorylate and inhibit Cdc25 phosphatases, which are required to activate Cdk/cyclin complexes by removing inhibitory phosphorylations (see Figure 16.16). At the G_1 and S phase checkpoints, CHK1 and CHK2 phosphorylate Cdc25A, which is required to activate complexes of Cdk2 and cyclins A or E. Phosphorylation leads to the rapid degradation of Cdc25A, resulting in inhibition of Cdk2. At the G_2 checkpoint, CHK1 and CHK2 phosphorylate and inhibit Cdc25C, which is responsible for activating Cdk1/cyclin B complexes. In the absence of Cdk1 activation, progression to mitosis is blocked and the cell remains arrested in G_2.

In mammalian cells, arrest at the G_1 checkpoint is also mediated by the action of an additional protein known as **p53**, which is phosphorylated by both ATM and CHK2 (Figure 16.21). Phosphorylation stabilizes p53, which is otherwise rapidly degraded, resulting in a rapid increase in p53 levels in response to damaged DNA. The p53 protein is a transcription factor, and its increased expression leads to the induction of the Cip/Kip family Cdk inhibitor p21. The p21 protein inhibits Cdk2/cyclin E complexes, leading to cell cycle arrest in G_1.

Interestingly, the gene encoding p53 is frequently mutated in human cancers. Loss of p53 function as a result of these mutations prevents G_1 arrest in response to DNA damage, so the damaged DNA is replicated and passed on to daughter cells instead of being repaired. This inheritance of damaged DNA results in an increased frequency of mutations and general instability of the cellular genome, which contributes to cancer development. Mutations in the *p53* gene are among the most common genetic alterations in human cancers (see Chapter 18), illustrating the critical importance of cell cycle regulation in the life of multicellular organisms.

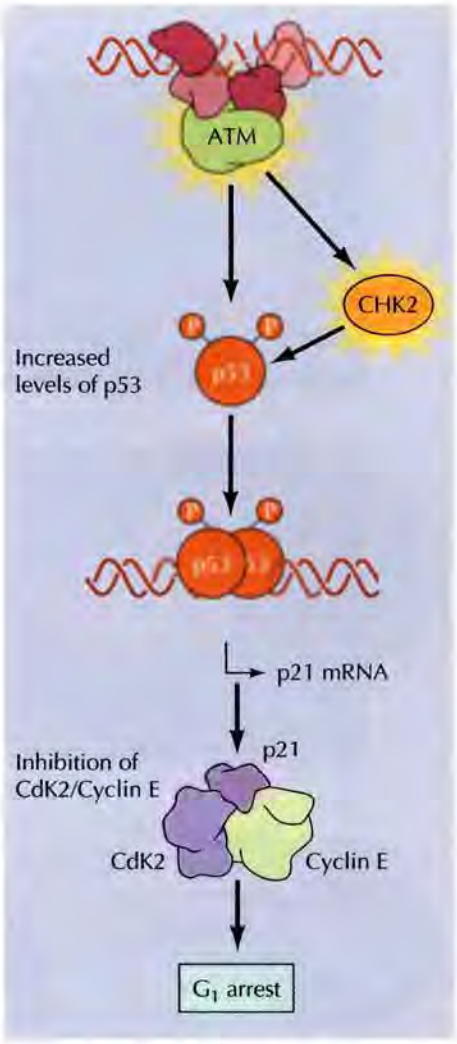


FIGURE 16.21 Role of p53 in G_1 arrest The protein p53 plays a key role in cell cycle arrest at the G_1 checkpoint in mammalian cells. Phosphorylation by ATM and CHK2 stabilize p53, resulting in rapid increases in p53 levels in response to DNA damage. The protein p53 then activates transcription of the gene encoding the Cdk inhibitor p21, leading to inhibition of Cdk2/cyclin E complexes and cell cycle arrest.

The Events of M Phase

M phase is the most dramatic period of the cell cycle, involving a major reorganization of virtually all cell components. During mitosis (nuclear division), the chromosomes condense, the nuclear envelope of most cells breaks down, the cytoskeleton reorganizes to form the mitotic spindle, and the chromosomes move to opposite poles. Chromosome segregation is then usually followed by cell division (cytokinesis). Although some of these events have been discussed in previous chapters, they are reviewed here in the context of a coordinated view of M phase and the action of MPF (Cdk1/cyclin B).

Stages of Mitosis

Although many of the details of mitosis vary among different organisms, the fundamental processes that ensure the faithful segregation of sister chromatids are conserved in all eukaryotes. These basic events of mitosis include chromosome condensation, formation of the mitotic spindle, and attachment of chromosomes to the spindle microtubules. Sister chromatids then separate from each other and move to opposite poles of the spindle, followed by the formation of daughter nuclei.

Mitosis is conventionally divided into four stages—**prophase**, **metaphase**, **anaphase**, and **telophase**—which are illustrated for an animal cell in Figures 16.22 and 16.23. The beginning of prophase is marked by the appearance of condensed chromosomes, each of which consists of two sister chromatids (the daughter DNA molecules produced in S phase). These newly replicated DNA molecules remain intertwined throughout S and G_2, becoming untangled during the process of chromatin condensation. The condensed sister chromatids are then held together at the **centromere**, which (as discussed in Chapter 5) is a DNA sequence to which proteins bind to form the **kinetochore**—the site of eventual attachment of the spindle microtubules. In addition to chromosome condensation, cytoplasmic changes leading to the development of the mitotic spindle initiate during prophase. The **centrosomes** (which had duplicated during interphase) separate and move to opposite sides of the nucleus. There they serve as the two poles of the **mitotic spindle**, which begins to form during late prophase.

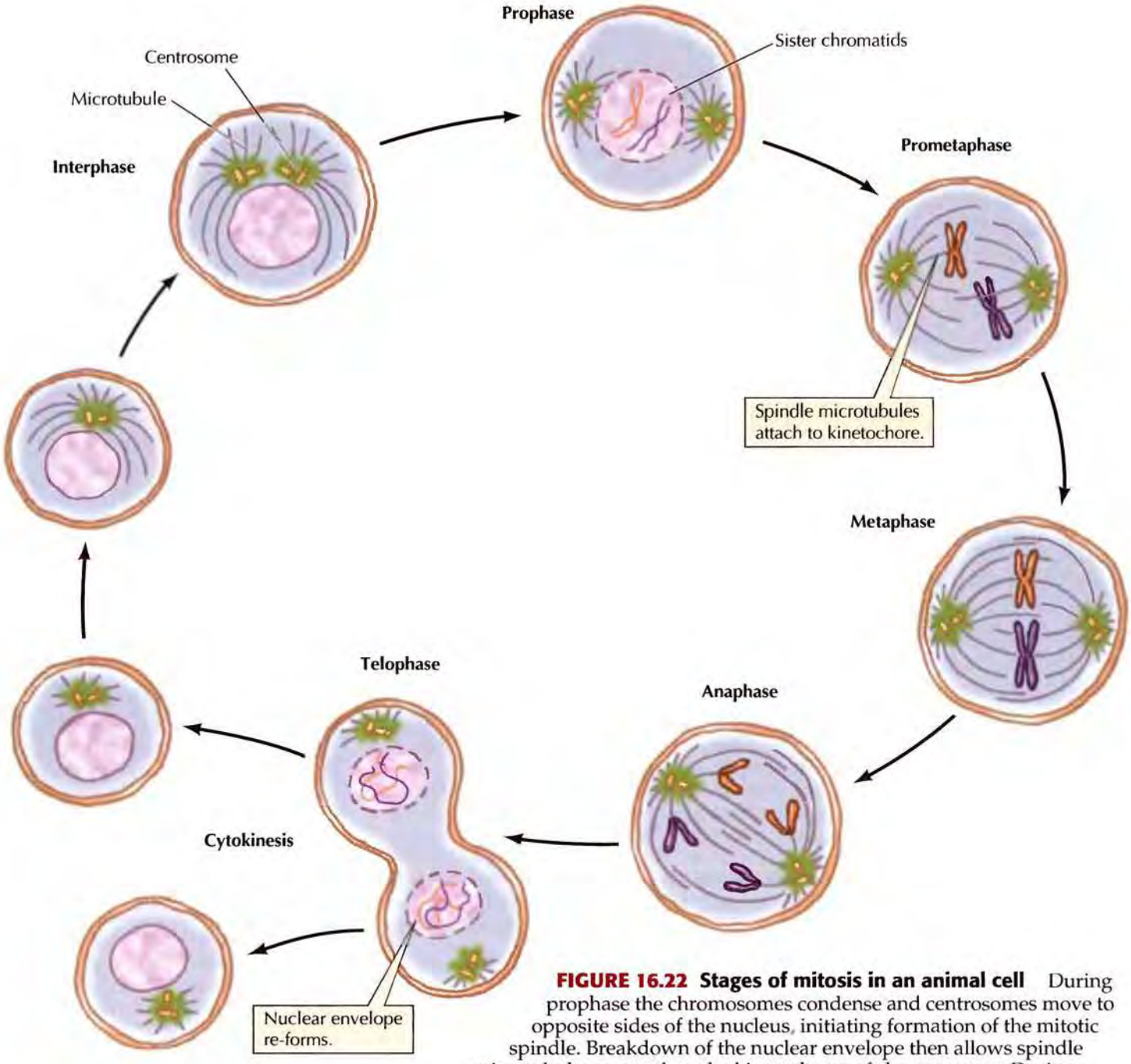


FIGURE 16.22 Stages of mitosis in an animal cell During prophase the chromosomes condense and centrosomes move to opposite sides of the nucleus, initiating formation of the mitotic spindle. Breakdown of the nuclear envelope then allows spindle microtubules to attach to the kinetochores of chromosomes. During prometaphase the chromosomes shuffle back and forth between the centrosomes and the center of the cell, eventually aligning in the center of the spindle (metaphase). At anaphase, the sister chromatids separate and move to opposite poles of the spindle. Mitosis then ends with re-formation of nuclear envelopes and chromosome decondensation during telophase, and cytokinesis yields two interphase daughter cells. Note that each daughter cell receives one centrosome, which duplicates prior to the next mitosis.

16.5 WEBSITE ANIMATION

Mitosis in an Animal Cell Mitosis, which is the division of the nucleus, consists of four phases—prophase, metaphase, anaphase, and telophase—and is followed by the division of the cytoplasm, called cytokinesis.

In higher eukaryotes the end of prophase corresponds to the breakdown of the nuclear envelope. However, this disassembly of the nucleus is not a universal feature of mitosis and does not occur in all cells. Some unicellular eukaryotes (e.g., yeasts) undergo so-called closed mitosis in which the nuclear envelope remains intact (Figure 16.24). In closed mitosis the daughter chromosomes migrate to opposite poles of the nucleus, which then divides

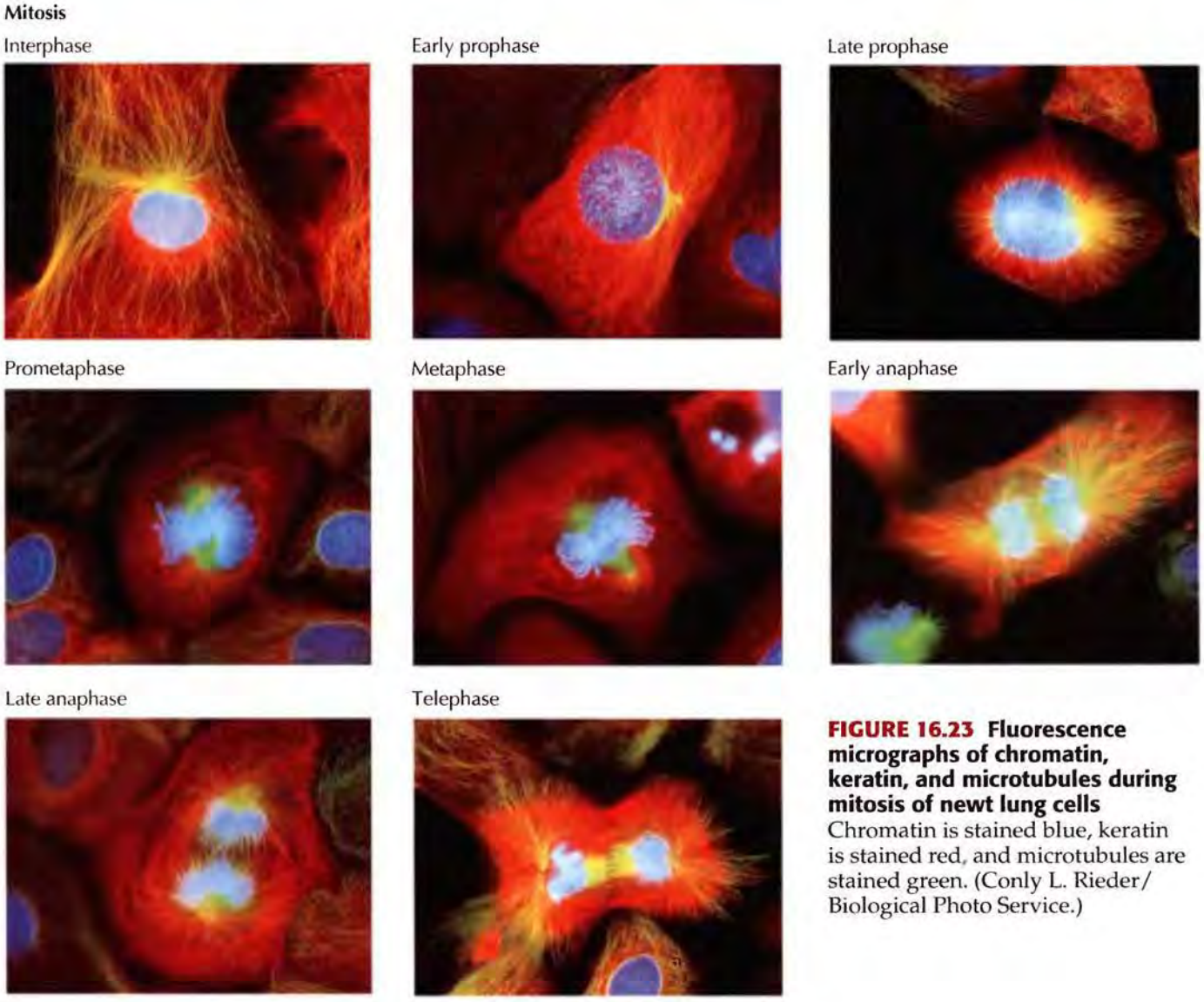


FIGURE 16.23 Fluorescence micrographs of chromatin, keratin, and microtubules during mitosis of newt lung cells
Chromatin is stained blue, keratin is stained red, and microtubules are stained green. (Conly L. Rieder/ Biological Photo Service.)

in two. In these cells the spindle pole bodies are embedded within the nuclear envelope, and the nucleus divides in two following migration of daughter chromosomes to opposite poles of the spindle.

Following completion of prophase, the cell enters **prometaphase**—a transition period between prophase and metaphase. During prometaphase the microtubules of the mitotic spindle attach to the kinetochores of condensed chromosomes. The kinetochores of sister chromatids are oriented on opposite sides of the chromosome, so they attach to microtubules emanating from opposite poles of the spindle. The chromosomes shuffle back and forth until they eventually align on the metaphase plate in the center of the spindle. At this stage, the cell has reached metaphase.

Most cells remain only briefly at metaphase before proceeding to anaphase. The transition from metaphase to anaphase is triggered by breakage of the link between sister chromatids, which then separate and move to opposite poles of the spindle. Mitosis ends with telophase, during which nuclei re-form and the chromosomes decondense. Cytokinesis usually begins during late anaphase and is almost complete by the end of telophase, resulting in the formation of two interphase daughter cells.

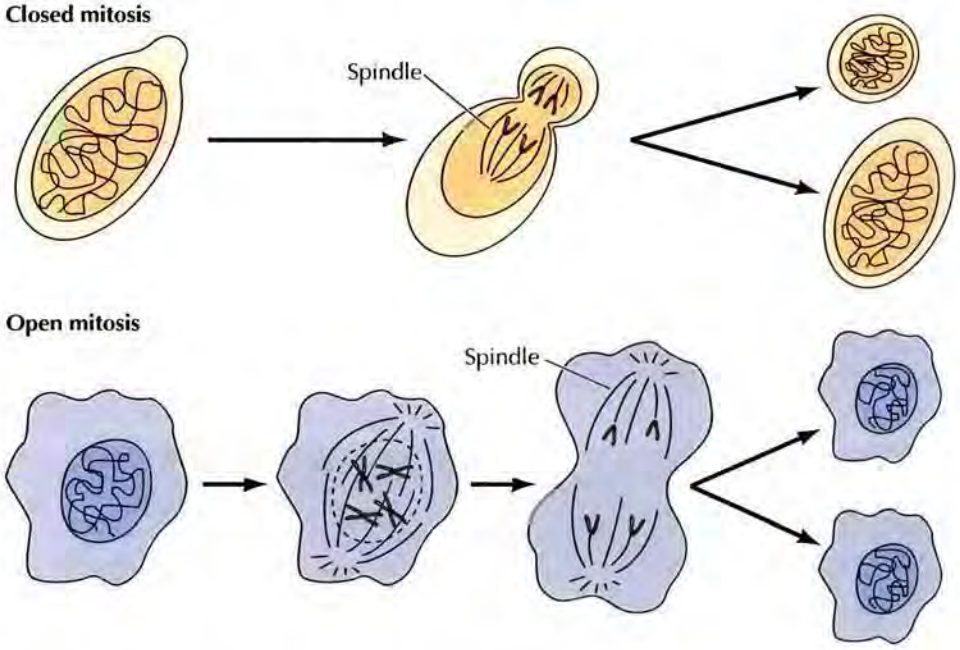


FIGURE 16.24 Closed and open mitosis In closed mitosis, the nuclear envelope remains intact and chromosomes migrate to opposite poles of a spindle within the nucleus. In open mitosis, the nuclear envelope breaks down and then re-forms around the two sets of separated chromosomes.

Cdk1/Cyclin B and Progression to Metaphase

Mitosis involves dramatic changes in multiple cellular components, leading to a major reorganization of the entire structure of the cell. As discussed earlier in this chapter, these events are initiated by activation of the Cdk1/cyclin B protein kinase (MPF). Cdk1/cyclin B not only acts as a master regulator of the M phase transition, phosphorylating and activating other downstream protein kinases but also acts directly by phosphorylating some of the structural proteins involved in this cellular reorganization (Figure 16.25).

The condensation of interphase chromatin to form the compact chromosomes of mitotic cells is a key event in mitosis, critical in enabling the chro-

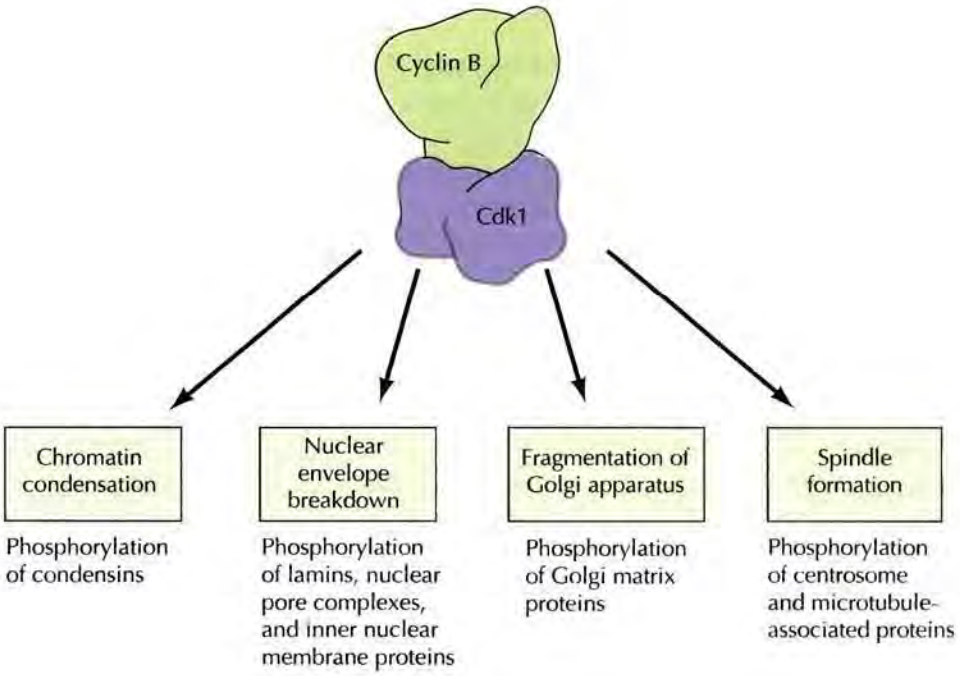


FIGURE 16.25 Targets of Cdk1/cyclin B The Cdk1/cyclin B complex induces multiple nuclear and cytoplasmic changes at the onset of M phase both by activating other protein kinases and by phosphorylating proteins such as condensins, components of the nuclear envelope, Golgi matrix proteins, and proteins associated with centrosomes and microtubules.

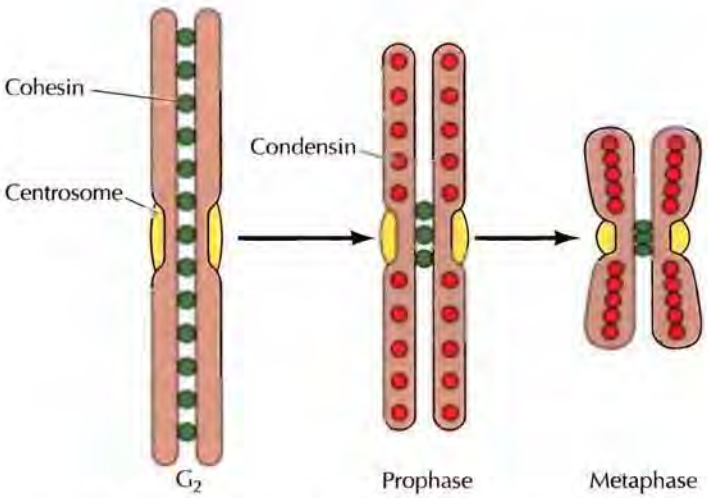


FIGURE 16.26 The action of cohesins and condensins Cohesins bind to DNA during S phase and maintain the linkage between sister chromatids following DNA replication in S and G_2. As the cell enters M phase, the cohesins are replaced by condensins along most of the chromosome, remaining only at the centromere. Phosphorylation by Cdk1 activates the condensins, which drive chromatin condensation.

mosomes to move along the mitotic spindle without becoming broken or tangled with one another. As discussed in Chapter 5, the chromatin in interphase nuclei condenses nearly a thousandfold during the formation of metaphase chromosomes. Such highly condensed chromatin cannot be transcribed, so transcription ceases as chromatin condensation takes place. Despite the fundamental importance of this event, we do not fully understand either the structure of metaphase chromosomes or the molecular mechanism of chromatin condensation. However, it has been established that chromatin condensation is driven by protein complexes called **condensins**, which are members of a class of "structural maintenance of chromatin" (SMC) proteins that play key roles in the organization of eukaryotic chromosomes.

Both condensins and another family of SMC proteins, called **cohesins**, contribute to chromosome segregation during mitosis (Figure 16.26). Cohesins bind to DNA in S phase and maintain the linkage between sister chromatids following DNA replication. As the cell enters M phase, the condensins are activated by Cdk1/cyclin B phosphorylation. The condensins then replace the cohesins along most of the length of the chromosome, so that the sister chromatids remain linked only at the centromere. The condensins also induce chromatin condensation, leading to the formation of metaphase chromosomes.

Breakdown of the nuclear envelope, which is one of the most dramatic events of mitosis, involves changes in all of its components: the nuclear membranes fragment, the nuclear pore complexes dissociate, and the nuclear lamina depolymerizes. Depolymerization of the nuclear lamina (the meshwork of filaments underlying the nuclear membrane) results from phosphorylation of the lamins by Cdk1 (Figure 16.27). Phosphorylation causes the lamin filaments to break down into individual lamin dimers, leading directly to depolymerization of the nuclear lamina. Cdk1 also phosphorylates several proteins in the inner nuclear membrane and the nuclear pore complex, leading to disassembly of nuclear pore complexes and

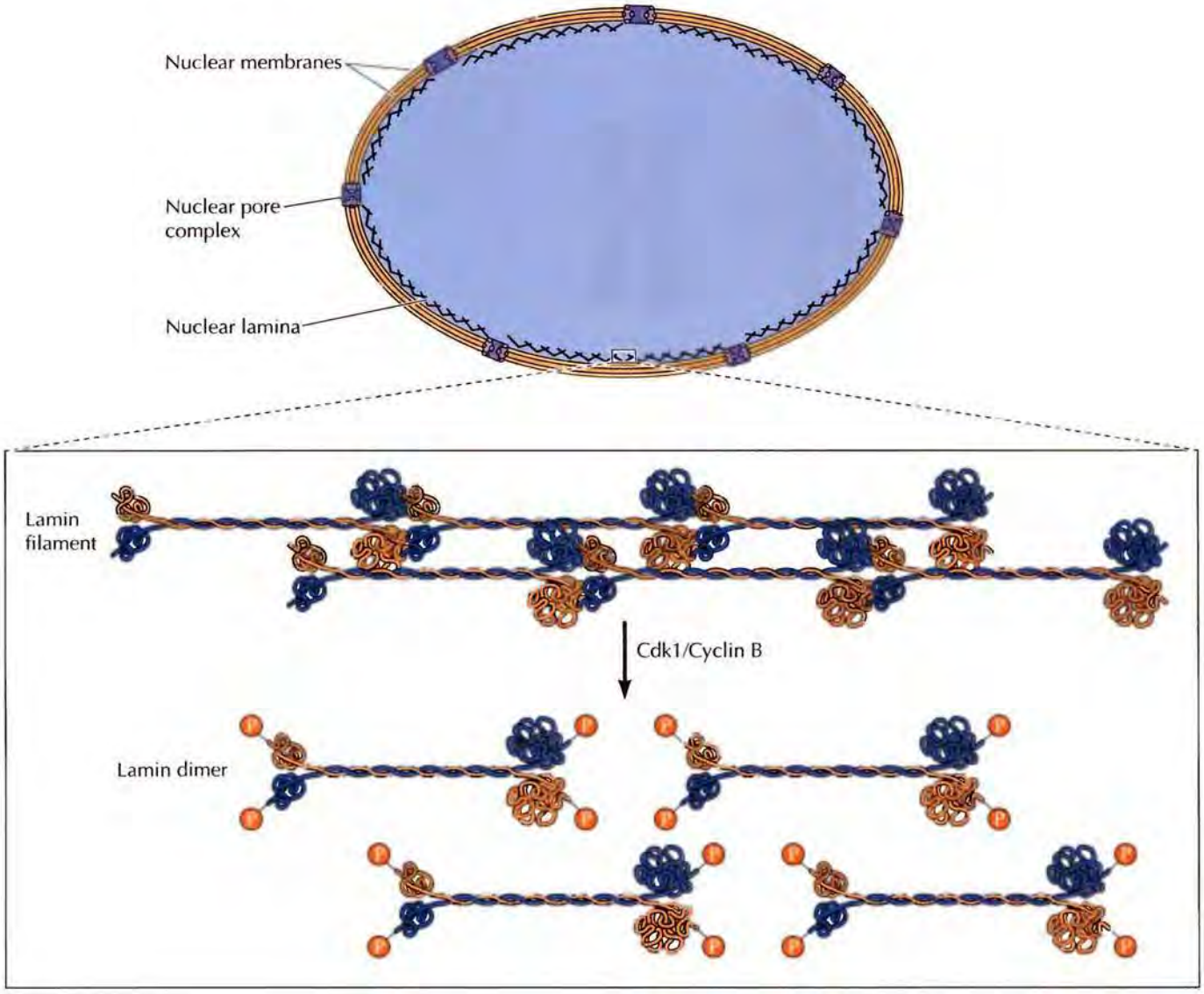


FIGURE 16.27 Breakdown of the nuclear envelope Cdk1/cyclin B phosphorylates the nuclear lamins as well as proteins of the nuclear pore complex and inner nuclear membrane. Phosphorylation of the lamins causes the filaments that form the nuclear lamina to dissociate into free lamin dimers.

detachment of the inner nuclear membrane from lamins and chromatin. Current evidence suggests that the nuclear membrane is then absorbed into the endoplasmic reticulum, which remains as an intact network and is distributed to daughter cells at mitosis.

The Golgi apparatus fragments into small vesicles at mitosis, which may either be absorbed into the endoplasmic reticulum or distributed directly to daughter cells at cytokinesis. The breakdown of these membranes is in part thought to be mediated by Cdk1 phosphorylation of Golgi matrix proteins (such as GM130 and GRASP-65), which are required for the docking of COPI-coated vesicles to the Golgi membrane. Phosphorylation of these proteins by Cdk1 inhibits vesicle docking and fusion, leading to fragmentation of the Golgi apparatus.

The reorganization of the cytoskeleton that culminates in formation of the mitotic spindle results from the dynamic instability of microtubules (see Chapter 12). At the beginning of prophase, activation of Cdk1 leads to separation of the centrosomes, which were duplicated during S phase. The centrosomes then move to opposite sides of the nucleus and undergo a process

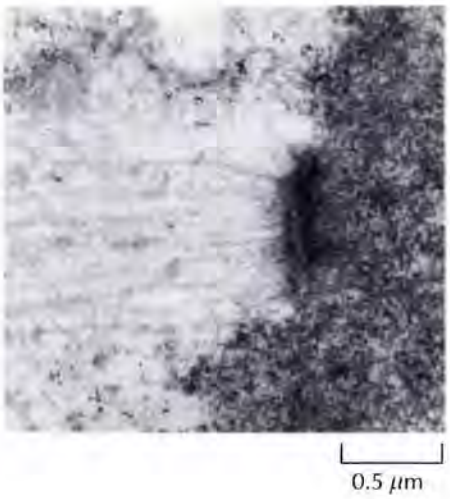


FIGURE 16.28 Electron micrograph of microtubules attached to the kinetochore of a chromosome (Conly L. Rieder/Biological Photo Service.)

of maturation during which they enlarge and recruit γ-tubulin and other proteins needed for spindle assembly. Centrosome maturation and spindle assembly involves the activity of protein kinases of the **Aurora** and **Polo-like kinase** families, which are located at the centrosome. Like Cdk1, Aurora and Polo-like kinases are activated in mitotic cells, and they play important roles in spindle formation and kinetochore function, as well as in cytokinesis. The rate of microtubule turnover increases five- to tenfold during mitosis, resulting in depolymerization and shrinkage of the interphase microtubules. This increased turnover is thought to result from phosphorylation of microtubule-associated proteins, either by Cdk1 or other mitotic protein kinases, such as the Aurora or Polo-like kinases. The number of microtubules emanating from the centrosomes also increases, so the interphase microtubules are replaced by large numbers of short microtubules radiating from the centrosomes.

The breakdown of the nuclear envelope then allows some of the spindle microtubules to attach to chromosomes at their kinetochores (Figure 16.28), initiating the process of chromosome movement that characterizes prometaphase. The proteins assembled at the kinetochore include microtubule motors that direct the movement of chromosomes toward the minus ends of the spindle microtubules, which are anchored in the centrosome. The action of these proteins, which draw chromosomes toward the centrosome, is opposed by plus-end directed motor proteins and by the growth of the spindle microtubules, which pushes the chromosomes away from the spindle poles. Consequently, the chromosomes in prometaphase shuffle back and forth between the centrosomes and the center of the spindle.

Microtubules from opposite poles of the spindle eventually attach to the two kinetochores of sister chromatids (which are located on opposite sides of the chromosome), and the balance of forces acting on the chromosomes leads to their alignment on the metaphase plate in the center of the spindle (Figure 16.29). As discussed in Chapter 12, the spindle consists of kinetochore and chromosomal microtubules, which are attached to the chromosomes, as well as polar microtubules, which overlap with one another in the center of the cell. In addition, short astral microtubules radiate outward from the centrosomes toward the cell periphery.

The Spindle Assembly Checkpoint and Progression to Anaphase

As discussed earlier in this chapter, the **spindle assembly checkpoint** monitors the alignment of chromosomes on the metaphase spindle. Once this has been accomplished, the cell proceeds to initiate anaphase and complete mitosis. The progression from metaphase to anaphase results from ubiquitin-mediated proteolysis of key regulatory proteins, triggered by acti-

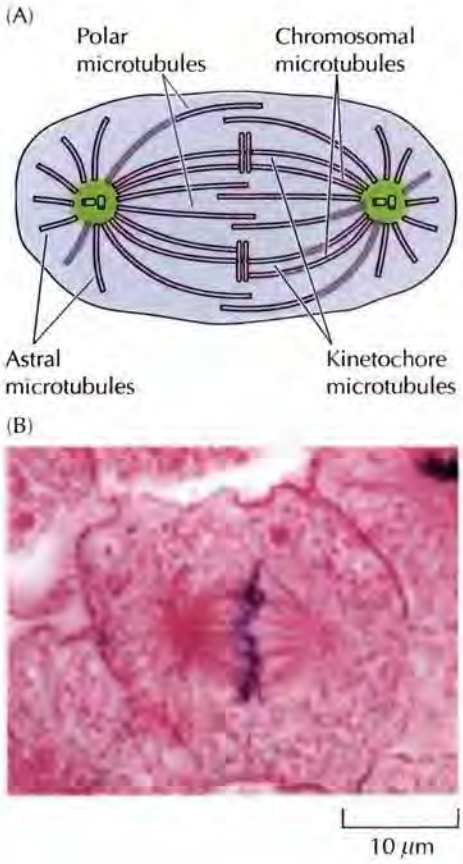


FIGURE 16.29 The metaphase spindle (A) The spindle consists of four kinds of microtubules. Kinetochore and chromosomal microtubules are attached to chromosomes; polar microtubules overlap in the center of the cell; and astral microtubules radiate from the centrosome to the cell periphery. (B) A whitefish cell at metaphase. (B, Michael Abbey/Photo Researchers, Inc.)

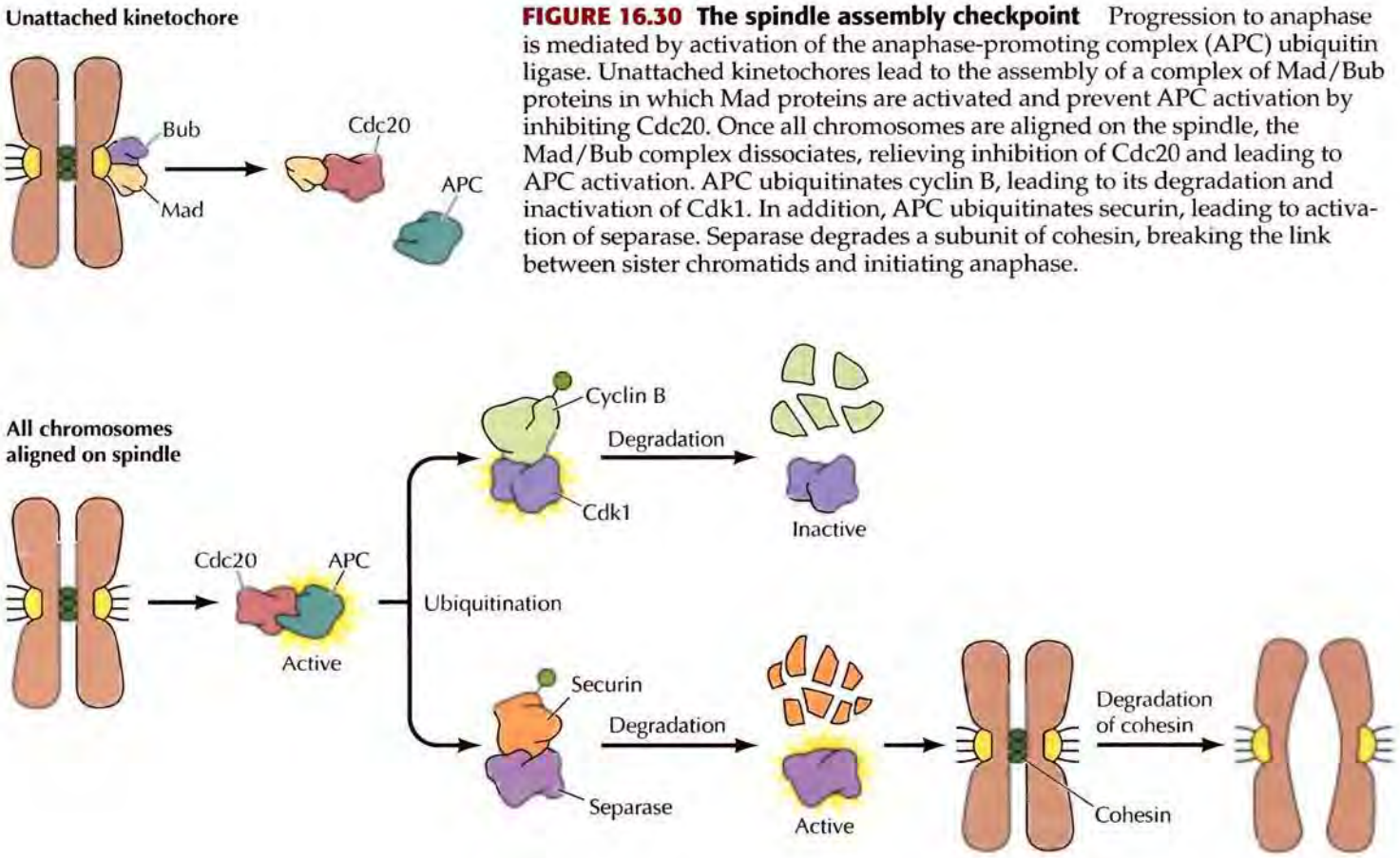


FIGURE 16.30 The spindle assembly checkpoint Progression to anaphase is mediated by activation of the anaphase-promoting complex (APC) ubiquitin ligase. Unattached kinetochores lead to the assembly of a complex of Mad/Bub proteins in which Mad proteins are activated and prevent APC activation by inhibiting Cdc20. Once all chromosomes are aligned on the spindle, the Mad/Bub complex dissociates, relieving inhibition of Cdc20 and leading to APC activation. APC ubiquitinates cyclin B, leading to its degradation and inactivation of Cdk1. In addition, APC ubiquitinates securin, leading to activation of separase. Separase degrades a subunit of cohesin, breaking the link between sister chromatids and initiating anaphase.

vation of an E3 ubiquitin ligase (see Figure 8.42) called the **anaphase-promoting complex**. Activation of the anaphase-promoting complex is induced at the beginning of mitosis, so the activation of Cdk1/cyclin B ultimately triggers its own destruction. The anaphase-promoting complex remains inhibited, however, until the cell passes the spindle assembly checkpoint, after which activation of the ubiquitin degradation system brings about the transition from metaphase to anaphase and progression through the rest of mitosis.

The spindle assembly checkpoint is remarkable in that the presence of even a single unaligned chromosome is sufficient to inhibit activation of the anaphase-promoting complex. The checkpoint is mediated by a complex of proteins, called the Mad/Bub proteins, that bind to Cdc20—a required component of the anaphase-promoting complex (Figure 16.30). The Mad/Bub proteins are assembled in a complex at unattached kinetochores. It appears that the Mad proteins are activated in this complex, and then released in an active form that inhibits Cdc20, maintaining the anaphase-promoting complex in an inactive state. Once microtubules have attached to the kinetochores, the Mad/Bub complex disassembles and inhibition of Cdc20 is relieved, leading to anaphase-promoting complex activation.

Activation of the anaphase-promoting complex results in ubiquitination and degradation of two key target proteins. The onset of anaphase results from proteolytic degradation of a component of the cohesins, which maintain the connection between sister chromatids while they are aligned on the metaphase plate (see Figure 16.26). Cohesin degradation is not catalyzed directly by the anaphase-promoting complex, which instead degrades a

■ **Abnormalities in chromosome segregation resulting from failures of the spindle-assembly checkpoint are common in cancer cells and are thought to play an important role in the development of many tumors.**

protein called securin that is a regulatory subunit of a protease called separase. Degradation of securin results in the activation of separase, which in turn degrades cohesin. Cleavage of cohesin breaks the linkage between sister chromatids, allowing them to segregate by moving to opposite poles of the spindle (Figure 16.31). The separation of chromosomes during anaphase then proceeds as a result of the action of several types of motor proteins associated with the spindle microtubules (see Figures 12.57 and 12.58).

The other key regulatory protein targeted for ubiquitination and degradation by the anaphase-promoting complex is cyclin B. Degradation of cyclin B leads to inactivation of Cdk1, which is required for the cell to exit mitosis and return to interphase. Many of the cellular changes involved in these transitions are simply the reversal of the events induced by Cdk1 during entry into mitosis. For example, reassembly of the nuclear envelope, chromatin decondensation, and the return of microtubules to an interphase state probably result directly from loss of Cdk1 activity and dephosphorylation of proteins that had been phosphorylated by Cdk1 at the beginning of mitosis. As discussed next, inactivation of Cdk1 also triggers cytokinesis.

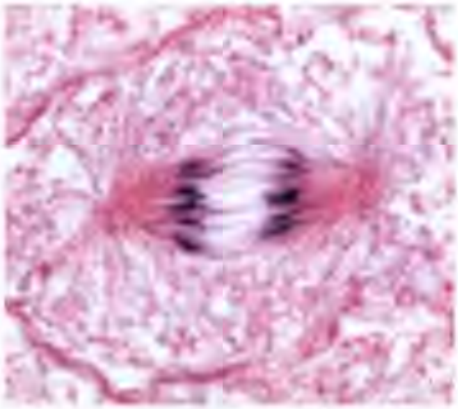


FIGURE 16.31 A whitefish cell at anaphase (Michael Abbey/Photo Researchers, Inc.)

Cytokinesis

The completion of mitosis is usually accompanied by cytokinesis, giving rise to two daughter cells. Cytokinesis usually initiates shortly after the onset of anaphase and is triggered by the inactivation of Cdk1, thereby coordinating nuclear and cytoplasmic division of the cell. As discussed in Chapter 12, cytokinesis of yeast and animal cells is mediated by a **contractile ring** of actin and myosin II filaments that forms beneath the plasma membrane (Figure 16.32). The location of this ring is determined by the position of the mitotic spindle, so the cell is eventually cleaved in a plane that passes through the metaphase plate perpendicular to the spindle. Cleavage proceeds as contraction of the actin-myosin filaments pulls the plasma membrane inward, eventually pinching the cell in half. The bridge between the two daughter cells is then broken, and the plasma membrane is resealed.

The mechanism of cytokinesis is different for higher plant cells. Rather than being pinched in half by a contractile ring, these cells divide by forming new cell walls and plasma membranes inside the cell (Figure 16.33). In

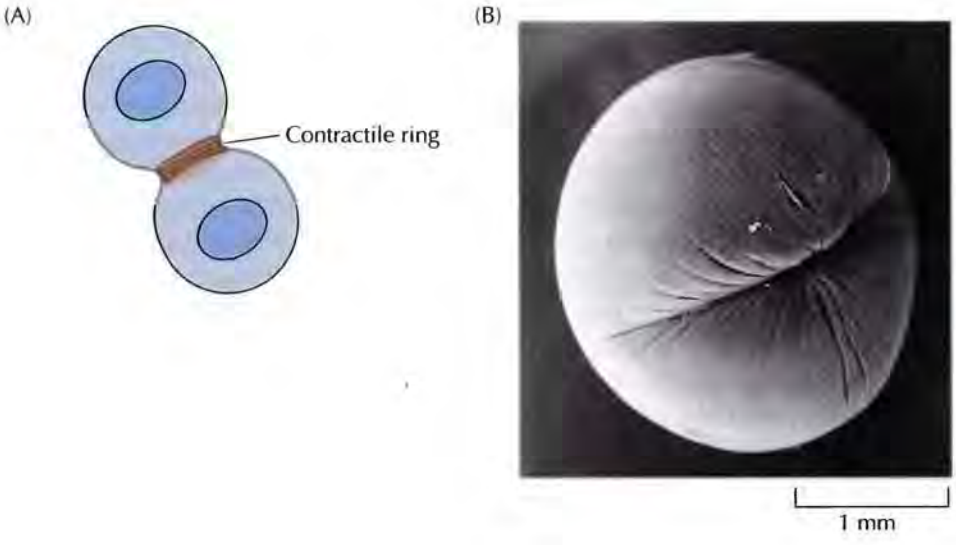
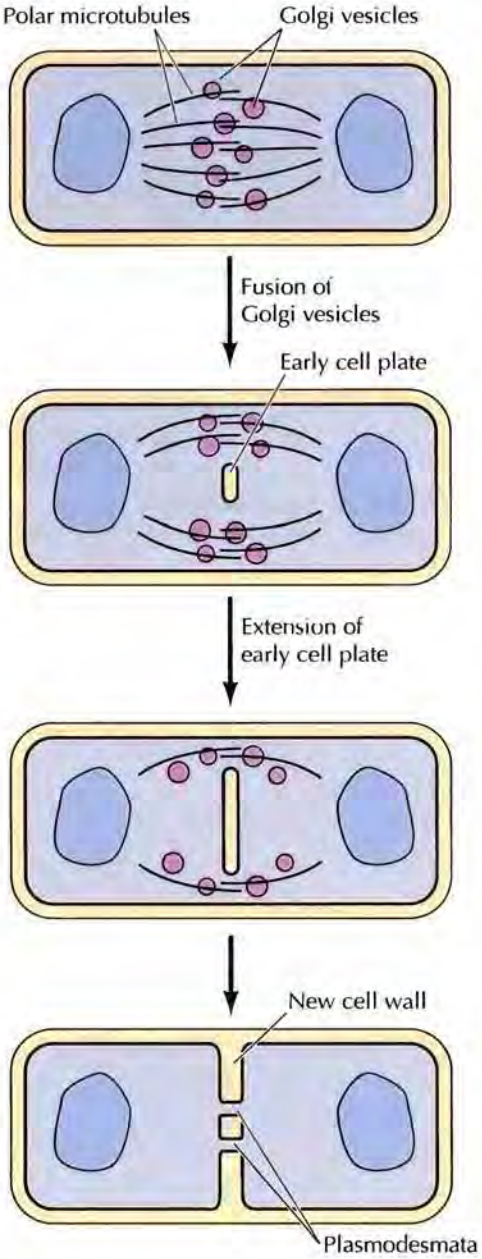


FIGURE 16.32 Cytokinesis of animal cells (A) Cytokinesis results from contraction of a ring of actin and myosin filaments, which pinches the cell in two. (B) Scanning electron micrograph of a frog egg undergoing cytokinesis. (B, David M. Phillips/Visuals Unlimited.)



16.6 WEBSITE ANIMATION

Cytokinesis in Higher Plants During cell division, a plant cell divides its cytoplasm by depositing Golgi vesicles containing cell-wall precursors at the former site of the metaphase plate, building a larger and larger disc-like structure that grows toward and fuses with the plasma membrane.

FIGURE 16.33 Cytokinesis in higher plants Golgi vesicles carrying cell wall precursors associate with polar microtubules at the former site of the metaphase plate. Fusion of these vesicles yields a membrane-enclosed, disklike structure (the early cell plate) that expands outward and fuses with the parental plasma membrane. The daughter cells remain connected at plasmodesmata.

early telophase, vesicles carrying cell wall precursors from the Golgi apparatus associate with remnants of the spindle microtubules and accumulate at the former site of the metaphase plate. These vesicles then fuse to form a large, membrane-enclosed, disklike structure, and their polysaccharide contents assemble to form the matrix of a new cell wall (called a cell plate). The cell plate expands outward, perpendicular to the spindle, until it reaches the plasma membrane. The membrane surrounding the cell plate then fuses with the parental plasma membrane, dividing the cell in two. Connections between the daughter cells (plasmodesmata, see Figure 14.28) are formed as a result of incomplete vesicle fusion during cytokinesis.

Meiosis and Fertilization

The somatic cell cycles discussed so far in this chapter result in diploid daughter cells with identical genetic complements. Meiosis, in contrast, is a specialized kind of cell cycle that reduces the chromosome number by half, resulting in the production of haploid daughter cells. Unicellular eukaryotes, such as yeasts, can undergo meiosis as well as reproducing by mitosis. Diploid *Saccharomyces cerevisiae*, for example, undergo meiosis and produce spores when faced with unfavorable environmental conditions. In multicellular plants and animals, however, meiosis is restricted to the germ cells, where it is key to sexual reproduction. Whereas somatic cells undergo mitosis to proliferate, the germ cells undergo meiosis to produce haploid gametes (the sperm and the egg). The development of a new progeny organism is then initiated by the fusion of these gametes at fertilization.

The Process of Meiosis

In contrast to mitosis, **meiosis** results in the division of a diploid parental cell into haploid progeny, each containing only one member of the pair of homologous chromosomes that were present in the diploid parental cell (Figure 16.34). This reduction in chromosome number is accomplished by two sequential rounds of nuclear and cell division (called meiosis I and meiosis II), which follow a single round of DNA replication. Like mitosis, meiosis I initiates after S phase has been completed and the parental chromosomes have replicated to produce identical sister chromatids. The pattern of chromosome segregation in meiosis I, however, is dramatically different from that of mitosis. During meiosis I, homologous chromosomes first pair with one another and then segregate to different daughter cells. Sister chromatids remain together, so completion of meiosis I results in the formation of daughter cells containing a single member of each chromosome pair (consisting of two sister chromatids). Meiosis I is followed by meiosis II, which resembles mitosis in that the sister chromatids separate and segregate to different daughter cells. Completion of meiosis II thus results in the production of four haploid daughter cells, each of which contains only one copy of each chromosome.

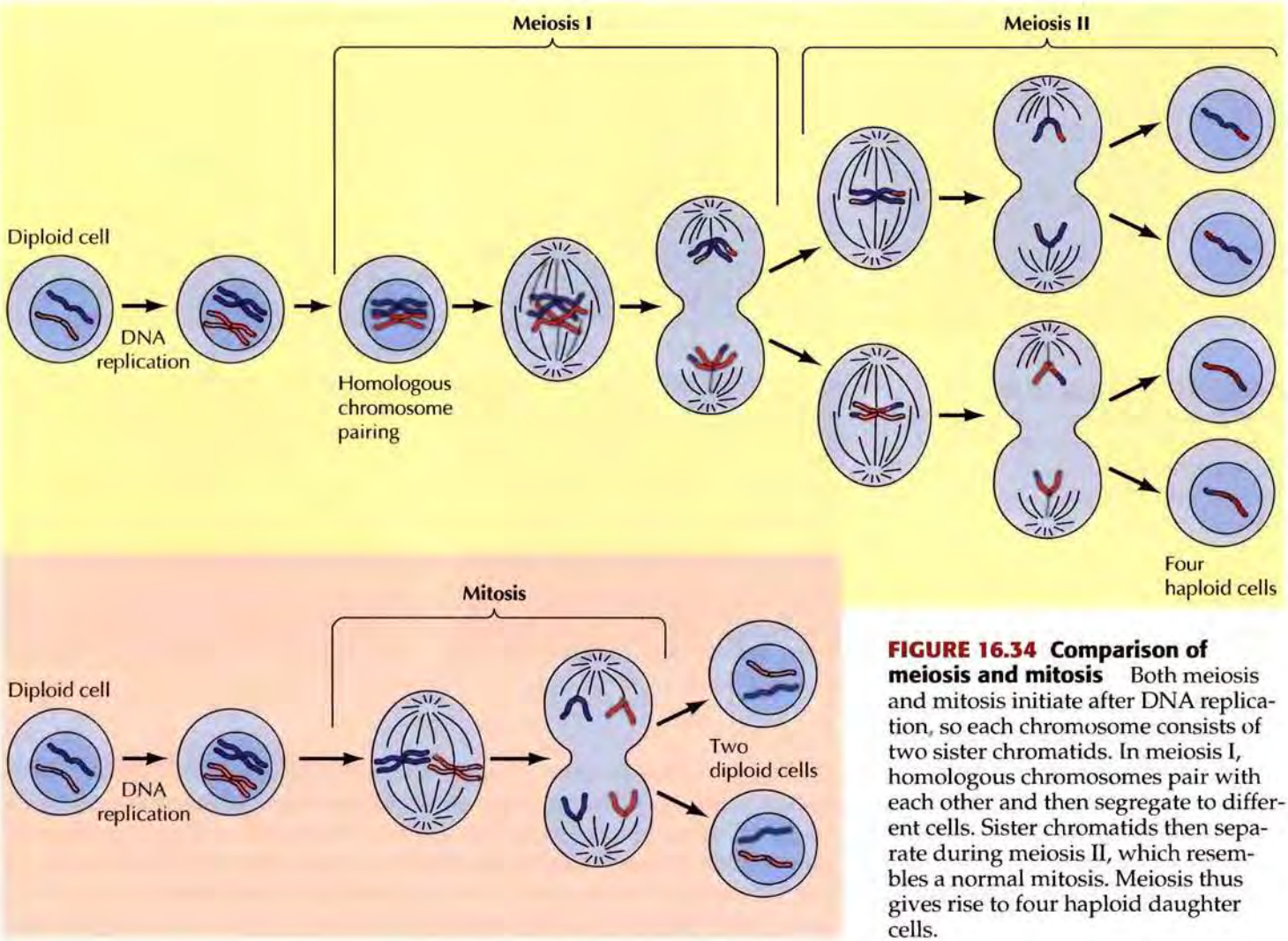


FIGURE 16.34 Comparison of meiosis and mitosis Both meiosis and mitosis initiate after DNA replication, so each chromosome consists of two sister chromatids. In meiosis I, homologous chromosomes pair with each other and then segregate to different cells. Sister chromatids then separate during meiosis II, which resembles a normal mitosis. Meiosis thus gives rise to four haploid daughter cells.

The pairing of homologous chromosomes after DNA replication results from recombination between chromosomes of paternal and maternal origin, so genetic recombination is critically linked to chromosome segregation during meiosis. Recombination between homologous chromosomes takes place during an extended prophase of meiosis I, which is divided into five stages (**leptotene**, **zygotene**, **pachytene**, **diplotene**, and **diakinesis**) on the basis of chromosome morphology (Figure 16.35). Recombination occurs at a high frequency during meiosis, and is initiated by double strand breaks that are induced early in meiotic prophase (leptotene) by a highly conserved endonuclease called Spo11. As discussed in Chapter 6, the formation of double strand breaks leads to the formation of single strand regions that invade a homologous chromosome by complementary base pairing (see Figure 6.32). The close association of homologous chromosomes (**synapsis**) begins during the zygotene stage. During this stage, a zipperlike protein structure, called the **synaptomenal complex**, forms along the length of the paired chromosomes (Figure 16.36). This complex keeps the homologous chromosomes closely associated and aligned with one another through the pachytene stage, which can persist for several days. Recombination between homologous chromosomes is completed by the end of pachytene,

16.7 WEBSITE ANIMATION

Meiosis In meiosis, a cell divides to produce daughter cells with half the number of chromosomes as the parent cell.

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Meiosis I and Mitosis Compared One difference between mitosis and meiosis can be seen at metaphase—in mitosis, homologous chromosomes line up separately on the metaphase plate, whereas in metaphase of meiosis I, homologous chromosomes line up in pairs at the metaphase plate.

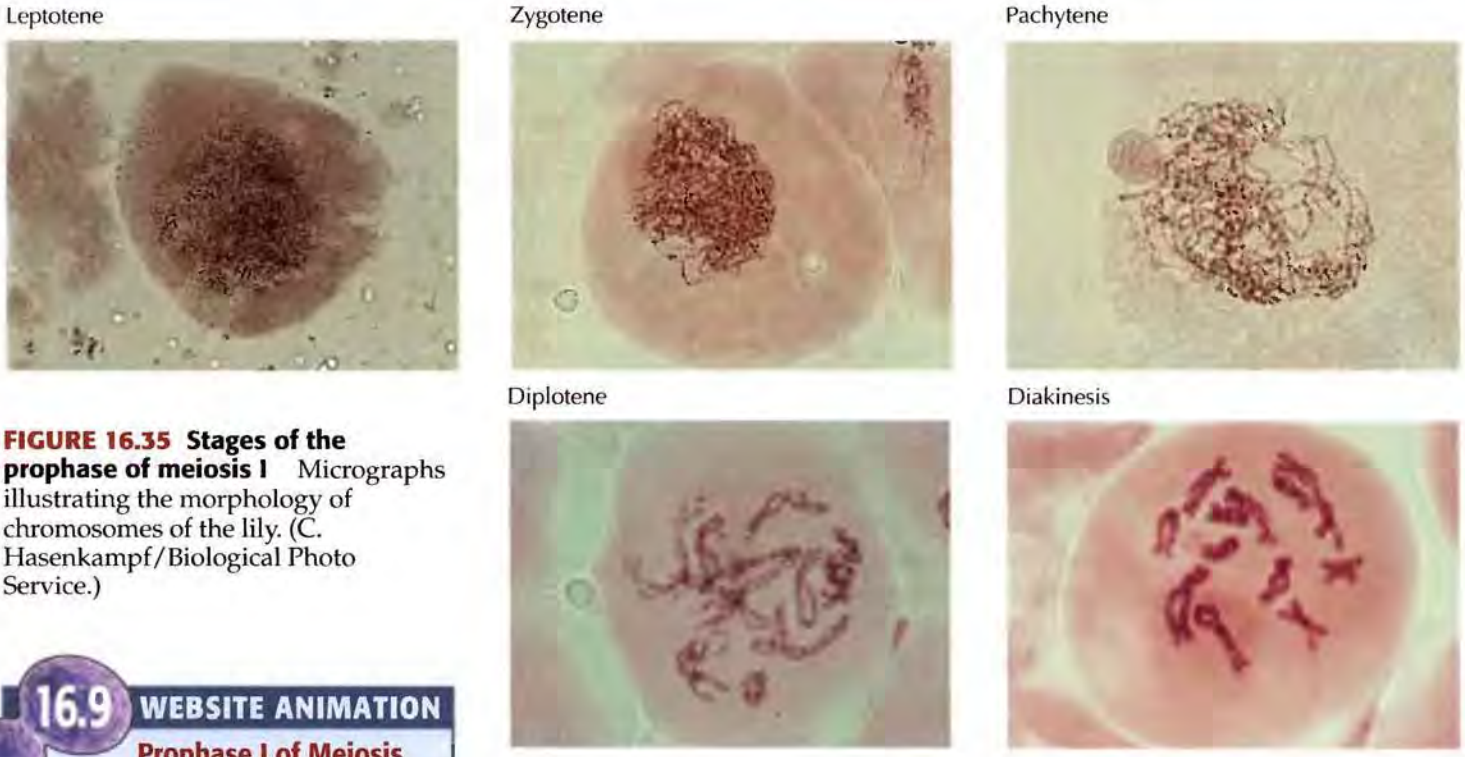

FIGURE 16.35 Stages of the prophase of meiosis I Micrographs illustrating the morphology of chromosomes of the lily. (C. Hasenkampf/Biological Photo Service.)

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Prophase I of Meiosis

Prophase I of meiosis consists of five stages, during which chromosomes condense and homologous chromosomes pair with each other and recombine.

leaving the chromosomes linked at the sites of crossing over (**chiasmata**; singular, chiasma). The synaptonemal complex disappears at the diplotene stage and the homologous chromosomes separate along their length. Importantly, however, they remain associated at the chiasmata, which is critical for their correct alignment at metaphase. At this stage, each chromosome pair (called a bivalent) consists of four chromatids with clearly evident chiasmata (Figure 16.37). Diakinesis, the final stage of prophase I, represents the transition to metaphase during which the chromosomes become fully condensed.

At metaphase I, the bivalent chromosomes align on the spindle. In contrast to mitosis (see Figure 16.29), the kinetochores of sister chromatids are adjacent to each other and oriented in the same direction, while the kinetochores of homologous chromosomes are pointed toward opposite spindle

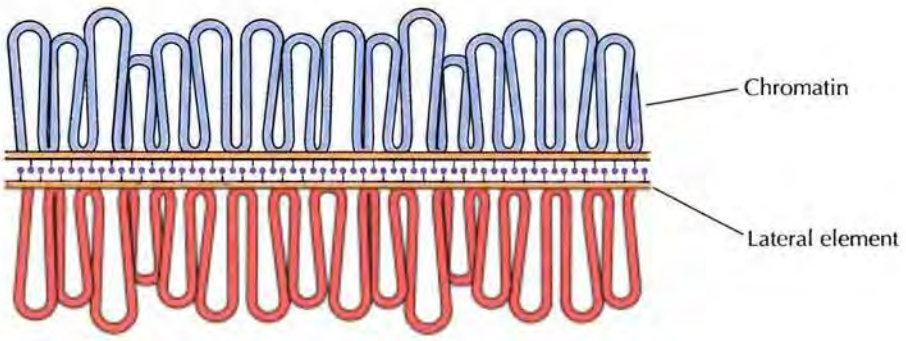


FIGURE 16.36 The synaptonemal complex Chromatin loops are attached to lateral elements, which are joined to each other by a zipperlike structure.

FIGURE 16.37 A bivalent chromosome at the diplotene stage The bivalent chromosome consists of paired homologous chromosomes. Sister chromatids of each chromosome are joined at the centromere. Chromatids of homologous chromosomes are joined at chiasmata, which are the sites at which genetic recombination has occurred. (B. John/Visuals Unlimited.)

poles (Figure 16.38). Consequently, microtubules from the same pole of the spindle attach to sister chromatids, while microtubules from opposite poles attach to homologous chromosomes. Anaphase I is initiated by disruption of the chiasmata at which homologous chromosomes are joined. The homologous chromosomes then separate, while sister chromatids remain associated at their centromeres. At completion of meiosis I each daughter cell has therefore acquired one member of each homologous pair, consisting of two sister chromatids.

Meiosis II initiates immediately after cytokinesis, usually before the chromosomes have fully decondensed. In contrast to meiosis I, meiosis II resembles a normal mitosis. At metaphase II, the chromosomes align on the spindle with microtubules from opposite poles of the spindle attached to the kinetochores of sister chromatids. The link between the centromeres of sister chromatids is broken at anaphase II, and sister chromatids segregate to opposite poles. Cytokinesis then follows, giving rise to haploid daughter cells.

Regulation of Oocyte Meiosis

Vertebrate oocytes (developing eggs) have been particularly useful models for research on the cell cycle in part because of their large size and ease of manipulation in the laboratory. A notable example, discussed earlier in this chapter, is provided by the discovery and subsequent purification of MPF (Cdk1/cyclin B) from frog oocytes. Meiosis of these oocytes, like those of other species, is regulated at two unique points in the cell cycle, and studies of oocyte meiosis have illuminated novel mechanisms of cell cycle control.

The first regulatory point in oocyte meiosis is in the diplotene stage of the first meiotic division (Figure 16.39). Oocytes can remain arrested at this stage for long periods of time—up to 50 years in humans. During this diplotene arrest, the oocyte chromosomes decondense and are actively transcribed. This transcriptional activity is reflected in the tremendous growth of oocytes during this period. Human oocytes, for example, are about 100 μm in diameter (more than a hundred times the volume of a typical somatic cell). Frog oocytes are even larger, with diameters of approximately 1 mm. During this period of cell growth, the oocytes accumulate stockpiles of materials, including RNAs and proteins that are needed to support early development of the embryo. As noted earlier in this chapter, early embryonic cell cycles then occur in the absence of cell growth, rapidly dividing the fertilized egg into smaller cells (see Figure 16.2).

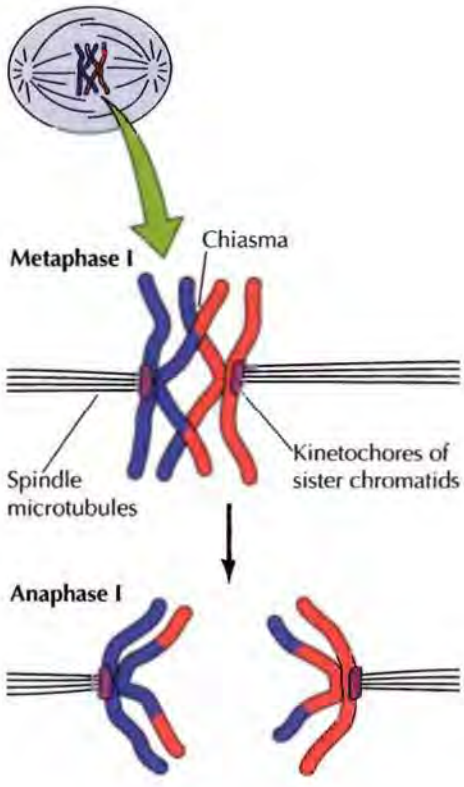


FIGURE 16.38 Chromosome segregation in meiosis I At metaphase I, the kinetochores of sister chromatids are either fused or adjacent to one another. Microtubules from the same pole of the spindle therefore attach to the kinetochores of sister chromatids, while microtubules from opposite poles attach to the kinetochores of homologous chromosomes. Chiasmata are disrupted at anaphase I, and homologous chromosomes move to opposite poles of the spindle.

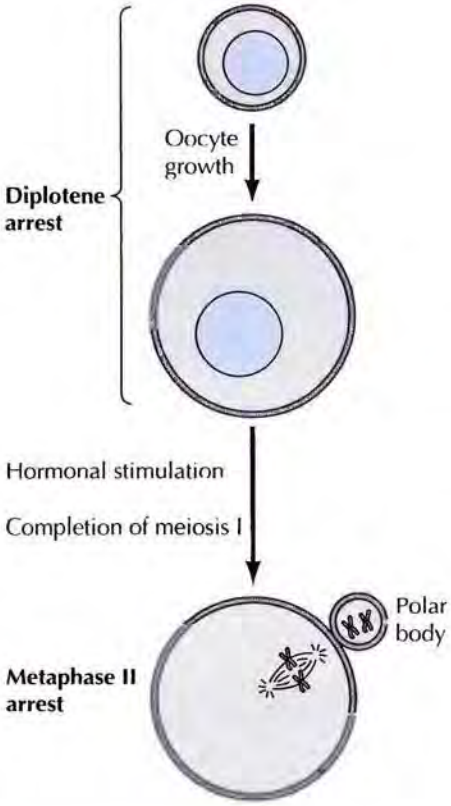


FIGURE 16.39 Meiosis of vertebrate oocytes Meiosis is arrested at the diplotene stage, during which oocytes grow to a large size. Oocytes then resume meiosis in response to hormonal stimulation and complete the first meiotic division, with asymmetric cytokinesis giving rise to a small polar body. Most vertebrate oocytes are then arrested again at metaphase II.

Oocytes of different species vary as to when meiosis resumes and fertilization takes place. In some animals, oocytes remain arrested at the diplotene stage until they are fertilized, only then proceeding to complete meiosis. However, the oocytes of most vertebrates (including frogs, mice, and humans) resume meiosis in response to hormonal stimulation and proceed through meiosis I prior to fertilization. Cell division following meiosis I is asymmetric, resulting in the production of a small **polar body** and an oocyte that retains its large size. The oocyte then proceeds to enter meiosis II without having re-formed a nucleus or decondensed its chromosomes. Most vertebrate oocytes are then arrested again at metaphase II, where they remain until fertilization.

Like the M phase of somatic cells, the meiosis of oocytes is controlled by the activity of Cdk1/cyclin B complexes. The regulation of Cdk1 during oocyte meiosis, however, displays unique features that are responsible for progression from meiosis I to meiosis II and for metaphase II arrest (Figure 16.40). Hormonal stimulation of diplotene-arrested oocytes initially triggers the resumption of meiosis by activating Cdk1, as at the G_2 to M transition of somatic cells. As in mitosis, Cdk1 then induces chromosome condensation, nuclear envelope breakdown, and formation of the spindle. Activation of

■ **Animals can be cloned by the procedure of somatic cell nuclear transfer in which the nucleus of a somatic cell is transferred to a metaphase II oocyte from which the normal chromosomes have been removed. The oocyte is then stimulated to divide and, upon implantation into a surrogate mother, can give rise to an animal genetically identical to the donor of the somatic cell nucleus. Since the cloning of Dolly the sheep in 1997, this technology has been used to create cloned offspring of several mammalian species. It also offers the potential of therapeutic cloning for the treatment of a variety of human diseases.**

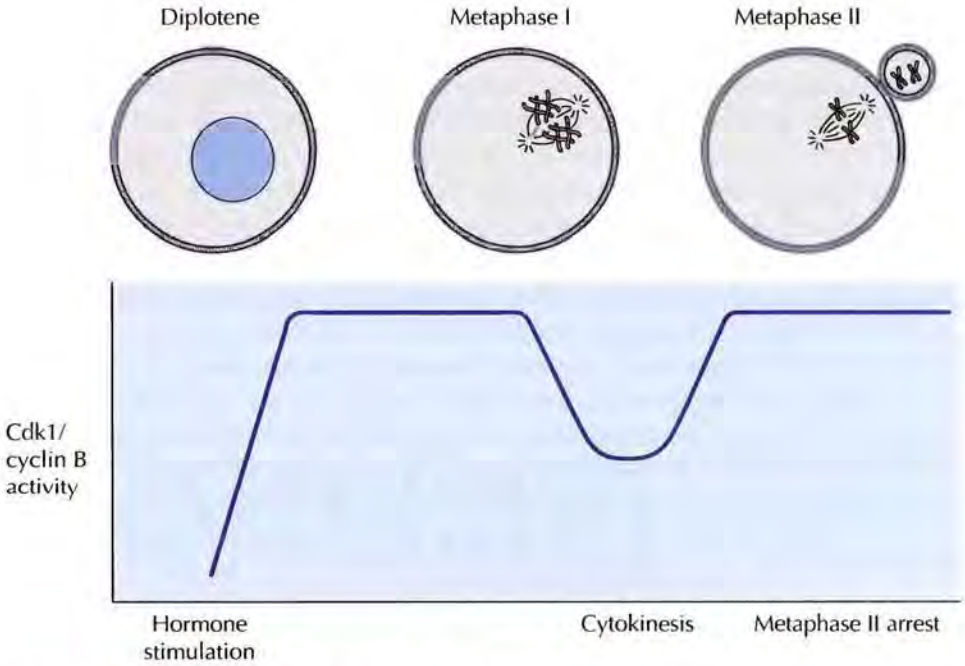


FIGURE 16.40 Activity of Cdk1/cyclin B during oocyte meiosis Hormonal stimulation of diplotene oocytes activates Cdk1/cyclin B, resulting in progression to metaphase I. Cdk1/cyclin B activity only partially falls at the transition from metaphase I to anaphase I, and the oocyte remains in M phase. Following completion of meiosis I, Cdk1/cyclin B activity again rises and remains high during metaphase II arrest.

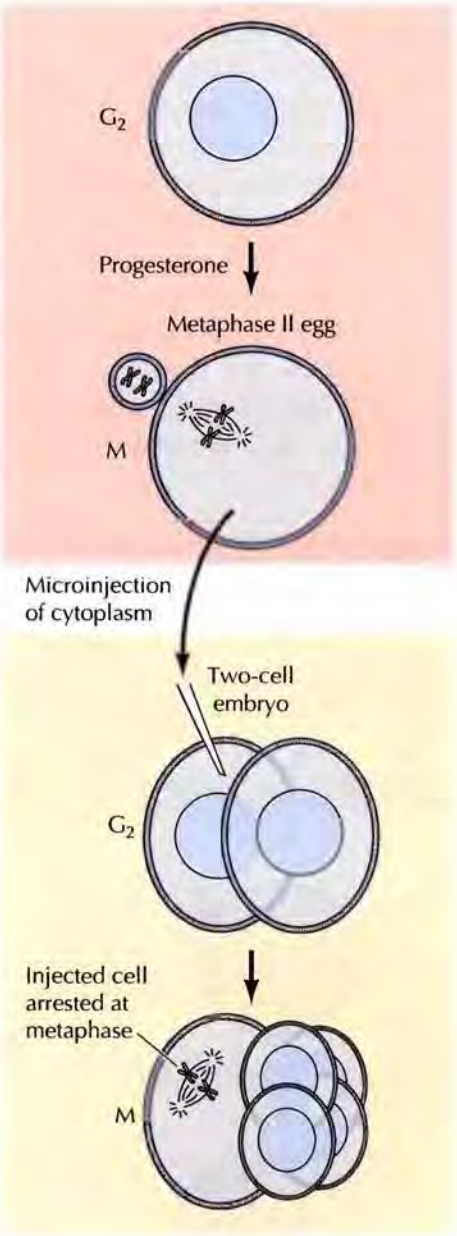


FIGURE 16.41 Identification of cytostatic factor Cytoplasm from a metaphase II egg is microinjected into one cell of a two-cell embryo. The injected embryo cell arrests at metaphase, while the uninjected cell continues to divide. A factor in metaphase II egg cytoplasm (cytostatic factor) therefore has induced metaphase arrest of the injected embryo cell.

the anaphase-promoting complex then leads to the metaphase to anaphase transition of meiosis I, accompanied by a decrease in the activity of Cdk1. However, in contrast to mitosis, Cdk1 activity is only partially decreased, so the oocyte remains in M phase, chromatin remains condensed, and nuclear envelopes do not re-form. Following cytokinesis, Cdk1 activity again rises and remains high while the egg is arrested at metaphase II. A regulatory mechanism unique to oocytes thus acts to maintain Cdk1 activity during the metaphase to anaphase transition of meiosis I and subsequent metaphase II arrest, preventing the inactivation of Cdk1 that would result from cyclin B proteolysis during a normal M phase.

The factor responsible for metaphase II arrest was first identified by Yoshio Masui and Clement Markert in 1971, in the same series of experiments that led to the discovery of MPF. In this case, however, cytoplasm from an egg arrested at metaphase II was injected into an early embryo cell that was undergoing mitotic cell cycles (Figure 16.41). This injection of egg cytoplasm caused the embryonic cell to arrest at metaphase, indicating that metaphase arrest was induced by a cytoplasmic factor present in the egg. Because this factor acted to arrest mitosis, it was called **cytostatic factor (CSF)**.

More recent experiments have identified a protein-serine/threonine kinase known as **Mos** as an essential component of CSF. Mos is specifically synthesized in oocytes around the time of completion of meiosis I and is required for the maintenance of Cdk1/cyclin B activity during the metaphase to anaphase transition of meiosis I as well as during metaphase II arrest. The action of Mos results from activation of the ERK MAP kinase, which plays a central role in the cell signaling pathways discussed in the previous chapter. In oocytes, however, ERK plays a different role: It activates another protein kinase called Rsk, which maintains the activity of MPF both by stimulating cyclin B synthesis and by inhibiting cyclin B degradation (Figure 16.42). Inhibition of cyclin B degradation is mediated

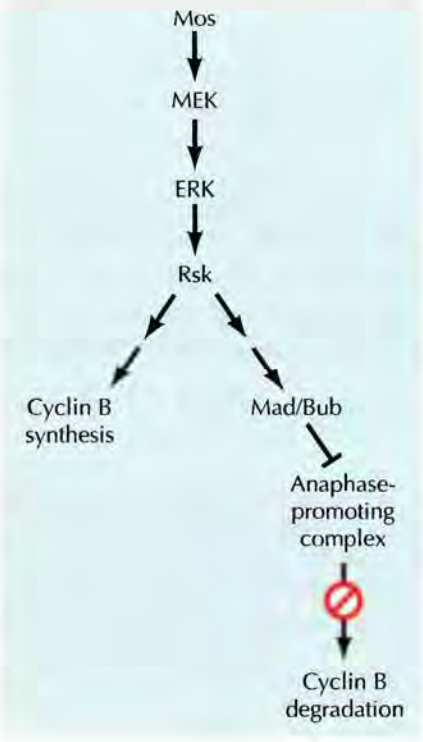


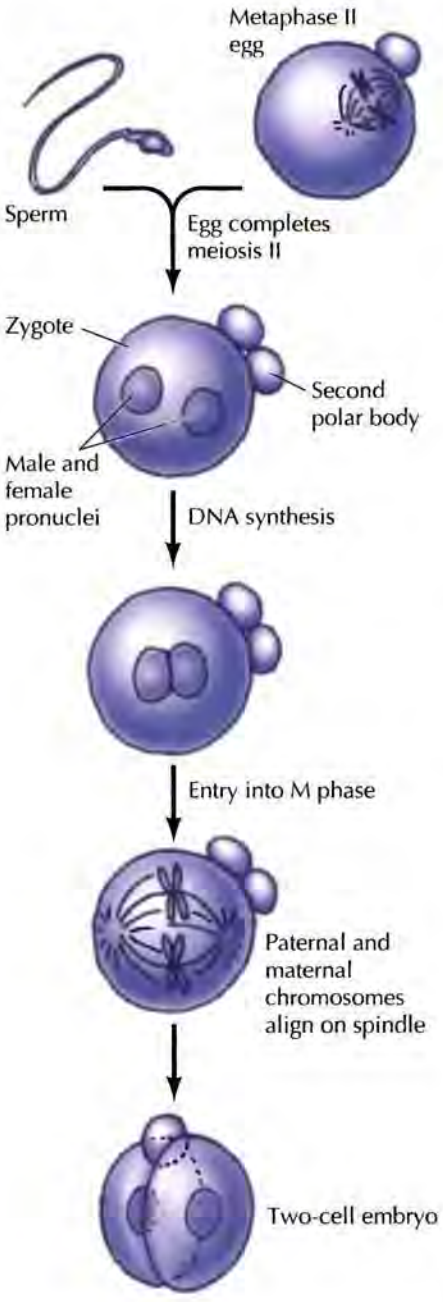
FIGURE 16.42 Maintenance of Cdk1/cyclin B activity by the Mos protein kinase The Mos protein kinase maintains Cdk1/cyclin B activity both by stimulating the synthesis of cyclin B and by inhibiting the degradation of cyclin B by the anaphase-promoting complex. The action of Mos is mediated by MEK, ERK, and Rsk protein kinases, and inhibition of the anaphase-promoting complex is mediated by Mad/Bub proteins.

16.10 WEBSITE ANIMATION

Polar Body Formation

During meiosis in female vertebrates, the meiotic divisions are often unequal, resulting in a single large egg and much smaller polar bodies.

FIGURE 16.43 Fertilization
Scanning electron micrograph of a human sperm fertilizing an egg. (David M. Philips/Visuals Unlimited.)



by inhibition of the anaphase-promoting complex by the Mad/Bub proteins, similar to arrest at the spindle assembly checkpoint (see Figure 16.30). Mos thus maintains Cdk1/cyclin B activity during oocyte meiosis, leading to the arrest of oocytes at metaphase II. Oocytes can remain arrested at this point in the meiotic cell cycle for several days, awaiting fertilization.

Fertilization

At **fertilization**, the sperm binds to a receptor on the surface of the egg and fuses with the egg plasma membrane, initiating the development of a new diploid organism containing genetic information derived from both parents (Figure 16.43). Not only does fertilization lead to the mixing of paternal and maternal chromosomes, but it also induces a number of changes in the egg cytoplasm that are critical for further development. These alterations activate the egg, leading to the completion of oocyte meiosis and initiation of the mitotic cell cycles of the early embryo.

A key signal resulting from the binding of a sperm to its receptor on the plasma membrane of the egg is an increase in the level of Ca^{2+} in the egg cytoplasm, probably as a consequence of stimulation of the hydrolysis of phosphatidylinositol 4,5-bisphosphate (PIP_2) (see Figure 15.27). One effect of this elevation in intracellular Ca^{2+} is the induction of surface alterations that prevent additional sperm from entering the egg. Because eggs are usually exposed to large numbers of sperm at one time, this is a critical event in ensuring the formation of a normal diploid embryo. These surface alterations result from the Ca^{2+}-induced exocytosis of secretory vesicles that are present in large numbers beneath the egg plasma membrane. Release of the

FIGURE 16.44 Fertilization and completion of meiosis Fertilization induces the transition from metaphase II to anaphase II, leading to completion of oocyte meiosis and emission of a second polar body (which usually degenerates). The sperm nucleus decondenses, so the fertilized egg (zygote) contains two haploid nuclei (male and female pronuclei). In mammals, the pronuclei replicate DNA as they migrate toward each other. They then initiate mitosis, with male and female chromosomes aligning on a common spindle. Completion of mitosis and cytokinesis thus gives rise to a two-cell embryo, with each cell containing a diploid genome.

contents of these vesicles alters the extracellular coat of the egg so as to block the entry of additional sperm.

The increase in cytosolic Ca^{2+} following fertilization also signals the completion of meiosis (Figure 16.44). In eggs arrested at metaphase II, the metaphase to anaphase transition is triggered by activation of the anaphase-promoting complex, resulting from Ca^{2+}-dependent phosphorylation and degradation of an inhibitory protein. The resultant degradation of cyclin B and condensin leads to completion of the second meiotic division, with asymmetric cytokinesis (as in meiosis I) giving rise to a second small polar body.

Following completion of oocyte meiosis, the fertilized egg (now called a **zygote**) contains two haploid nuclei (called **pronuclei**), one derived from each parent. In mammals, the two pronuclei then enter S phase and replicate their DNA as they migrate toward each other. As they meet, the zygote enters M phase of its first mitotic division. The two nuclear envelopes break down, and the condensed chromosomes of both paternal and maternal origin align on a common spindle. Completion of mitosis then gives rise to two embryonic cells, each containing a new diploid genome. These cells then commence the series of embryonic cell divisions that eventually lead to the development of a new organism.

■ ***In vitro*** **fertilization (IVF) is widely used to help infertile couples. A variety of reproductive disorders can be dealt with by this procedure in which metaphase II eggs are recovered from the ovary, fertilized *in vitro* and then returned to the Fallopian tube or uterus of the mother. The first baby resulting from *in vitro* fertilization was born in 1978, and tens of thousands of infertile couples have since been treated by IVF.**

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

SUMMARY

THE EUKARYOTIC CELL CYCLE

Phases of the Cell Cycle: Eukaryotic cell cycles are divided into four discrete phases: M, G_1, S, and G_2. M phase consists of mitosis, which is usually followed by cytokinesis. S phase is the period of DNA replication.

mitosis, interphase, cytokinesis, M phase, G_1 phase, S phase, G_2 phase, flow cytometer, fluorescence-activated cell sorter

Regulation of the Cell Cycle by Cell Growth and Extracellular Signals: Extracellular signals and cell size regulate progression through specific control points in the cell cycle.

START, restriction point

Cell Cycle Checkpoints: Checkpoints and feedback controls coordinate the events that take place during different phases of the cell cycle and arrest cell cycle progression if DNA is damaged.

cell cycle checkpoint, ATM, ATR

Restricting DNA Replication to Once per Cell Cycle: Once DNA replication has taken place, initiation of a new S phase is prevented until the cell has passed through mitosis.

REGULATORS OF CELL CYCLE PROGRESSION

Protein Kinases and Cell Cycle Regulation: MPF is the key molecule responsible for regulating the G_2 to M transition in all eukaryotes. MPF is a dimer of cyclin B and the Cdk1 protein kinase.

maturation promoting factor (MPF), Cdk1, cyclin

Families of Cyclins and Cyclin-Dependent Kinases: Distinct pairs of cyclins and Cdk1-related protein kinases regulate progression through different stages of the cell cycle. The activity of Cdk's is regulated by association with cyclins, activating and inhibitory phosphorylations, and the binding of Cdk inhibitors.

G_1 cyclin, Cln, Cdk, Cdk inhibitor (CKI)

KEY TERMS

KEY TERMS

Rb, tumor suppressor gene, E2F

checkpoint kinase, p53

prophase, metaphase, anaphase, telophase, centromere, kinetochore, centrosome, mitotic spindle, prometaphase

condensin, cohesin, Aurora kinase, Polo-like kinase

spindle assembly checkpoint, anaphase-promoting complex

contractile ring

SUMMARY

Growth Factors and the Regulation of G_1 Cdk's: Growth factors stimulate animal cell proliferation by inducing synthesis of the D-type cyclins. Cdk4, 6/cyclin D complexes then act to drive cells through the restriction point in G_1. A key substrate of Cdk4, 6/cyclin D complexes is the tumor suppressor protein Rb, which regulates transcription of genes required for cell cycle progression, including cyclin E. Activation of Cdk2/cyclin E complexes is then responsible for entry into S phase.

DNA Damage Checkpoints: DNA damage or incompletely replicated DNA arrest cell cycle progression in G_1, S, and G_2. Cell cycle arrest is mediated by protein kinases that are activated by DNA damage and inhibit Cdc25 phosphatases, which are required for Cdk activation. In mammalian cells, arrest at the G_1 checkpoint is also mediated by p53, which induces synthesis of the Cdk inhibitor p21.

THE EVENTS OF M PHASE

Stages of Mitosis: Mitosis is conventionally divided into four stages: prophase, metaphase, anaphase, and telophase. The basic events of mitosis include chromosome condensation, formation of the mitotic spindle, nuclear envelope breakdown, and attachment of spindle microtubules to chromosomes at the kinetochore. Sister chromatids then separate and move to opposite poles of the spindle. Finally, nuclei re-form, the chromosomes decondense, and cytokinesis divides the cell in half.

Cdk1/Cyclin B and Progression to Metaphase: M phase is initiated by activation of Cdk1/cyclin B, which phosphorylates other protein kinases, as well as the nuclear lamins and other proteins of the nuclear envelope, condensins, and Golgi matrix proteins. Activation of Cdk1/cyclin B is responsible for chromatin condensation, nuclear envelope breakdown, fragmentation of the Golgi apparatus, and reorganization of microtubules to form the mitotic spindle. The attachment of spindle microtubules to the kinetochores of sister chromatids then leads to their alignment on the metaphase plate.

The Spindle Assembly Checkpoint and Progression to Anaphase: Activation of a ubiquitin ligase called the anaphase-promoting complex leads to degradation of key regulatory proteins at the metaphase to anaphase transition. The activity of the anaphase-promoting complex is inhibited until the cell passes the spindle assembly checkpoint and all chromosomes are properly aligned on the spindle. Ubiquitin-mediated proteolysis initiated by the anaphase-promoting complex then leads to the degradation of cohesin, breaking the link between sister chromatids at the onset of anaphase. The anaphase-promoting complex also ubiquitinates cyclin B, leading to inactivation of Cdk1 and exit from mitosis.

Cytokinesis: Inactivation of Cdk1/cyclin B also triggers cytokinesis. In yeast and animal cells, cytokinesis results from contraction of a ring of actin and myosin filaments. In higher plant cells, cytokinesis results from the formation of a new cell wall and plasma membrane inside the cell.

SUMMARY

MEIOSIS AND FERTILIZATION

The Process of Meiosis: Meiosis is a specialized cell cycle that gives rise to haploid daughter cells. A single round of DNA synthesis is followed by two sequential cell divisions. During meiosis I, homologous chromosomes first form pairs and then segregate to different daughter cells. Meiosis II then resembles a normal mitosis in which sister chromatids separate.

Regulation of Oocyte Meiosis: Meiosis of vertebrate oocytes is regulated at two unique points in the cell cycle: the diplotene stage of meiosis I and metaphase of meiosis II. Metaphase II arrest results from inhibition of the anaphase-promoting complex by a protein kinase expressed in oocytes.

Fertilization: Fertilization triggers the resumption of oocyte meiosis by Ca^{2+}-dependent activation of the anaphase-promoting complex. The fertilized egg then contains two haploid nuclei, which form a new diploid genome and initiate embryonic cell divisions.

KEY TERMS

meiosis, leptotene, zygotene, pachytene, diplotene, diakinesis, synapsis, synaptomenal complex, chiasma

polar body, cytostatic factor (CSF), Mos

fertilization, zygote, pronucleus

Questions

1. In what ways are cells in G_0 and G_1 similar? How do they differ?

2. Consider a mammalian cell line that divides every 30 hours. Microscopic observation indicates that 3.3% of the cells are in mitosis at any given time. Analysis in the flow cytometer establishes that 53.3% of the cells have DNA contents of $2n$, 16.7 % have DNA contents of $4n$, and 30% have DNA contents ranging between $2n$ and $4n$. What are the lengths of the G_1, S, G_2, and M phases of the cycle of these cells?

3. Radiation damages DNA and arrests cell cycle progression at checkpoints in G_1, S, and G_2. Why is this advantageous for the cell?

4. What are the mechanisms that regulate the activity of cyclin-dependent kinases?

5. The spindle assembly checkpoint delays the onset of anaphase until all chromosomes are properly aligned on the spindle. What would be the result if a failure of this checkpoint allowed anaphase to initiate while one chromosome was attached to microtubules from only a single centrosome?

6. What cellular processes would be affected by expression of siRNA targeted against Cdk7?

7. The Cdk inhibitor p16 binds specifically to Cdk4, 6/cyclin D complexes. What would be the predicted effect of overexpression of p16 on cell cycle progression? Would overexpression of p16 affect a tumor cell lacking functional Rb protein?

8. *In vitro* mutagenesis of cloned lamin cDNAs has been used to generate mutants that cannot be phosphorylated by Cdk1. How would expression of these mutant lamins affect nuclear envelope breakdown at the end of prophase?

9. What substrates are phosphorylated by Cdk1/cyclin B to initiate mitosis?

10. A mutant of cyclin B resistant to degradation by the anaphase-promoting complex has been generated. How would expression of this mutant cyclin B affect the events at the metaphase to anaphase transition?

11. How does the activity of anaphase-promoting complex lead to the separation of sister chromatids?

12. Homologous recombination has been used to inactivate the *mos* gene in mice. What effect would you expect this to have on oocyte meiosis?

13. How does the fertilized egg provide a long-lasting block to additional sperm entering the egg?

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CHAPTER 17

Cell Death and Cell Renewal

- ***Programmed Cell Death 690***
- ***Stem Cells and the Maintenance of Adult Tissues 700***
- ***Embryonic Stem Cells and Therapeutic Cloning 709***
- ***KEY EXPERIMENT:*** Identification of Genes Required for Programmed Cell Death 694
- ***KEY EXPERIMENT:*** Culture of Embryonic Stem Cells 710

CELL DEATH AND CELL PROLIFERATION are balanced throughout the life of multicellular organisms. Animal development begins with the rapid proliferation of embryonic cells, which then differentiate to produce the many specialized types of cells that make up adult tissues and organs. Whereas the nematode *C. elegans* consists of only 959 somatic cells, humans possess a total of approximately 10^{14} cells, consisting of more than 200 differentiated cell types. Starting from only a single cell—the fertilized egg—all the diverse cell types of the body are produced and organized into tissues and organs. This complex process of development involves not only cell proliferation and differentiation but also cell death. Although cells can die as a result of unpredictable traumatic events, such as exposure to toxic chemicals, most cell deaths in multicellular organisms occur by a normal physiological process of programmed cell death, which plays a key role both in embryonic development and in adult tissues.

In adult organisms, cell death must be balanced by cell renewal, and most tissues contain stem cells that are able to replace cells that have been lost. Abnormalities of cell death are associated with a wide variety of illnesses, including cancer, autoimmune disease, and neurodegenerative disorders, such as Parkinson's and Alzheimer's disease. Conversely, the ability of stem cells to proliferate and differentiate into a wide variety of cell types has generated enormous interest in the possible use of these cells, particularly embryonic stem cells, to replace damaged tissues. The mechanisms and regulation of cell death and cell renewal have therefore become areas of research at the forefront of biology and medicine.

Programmed Cell Death

Programmed cell death is carefully regulated so that the fate of individual cells meets the needs of the organism as a whole. In adults, programmed cell death is responsible for balancing cell proliferation and maintaining constant cell numbers in tissues undergoing cell turnover. For example, about 5×10^{11} blood cells are eliminated daily in humans by programmed cell death, balancing their continual production in the bone marrow. In addition, programmed cell death provides a defense mechanism by which damaged and potentially dangerous cells can be eliminated for the good of the organism as a whole. Virus-infected cells frequently undergo programmed cell death, thereby preventing the production of new virus particles and limiting spread of the virus through the host organism. Other types of cellular insults, such as DNA damage, also induce programmed cell death. In the case of DNA damage, programmed cell death may eliminate cells carrying potentially harmful mutations, including cells with mutations that might lead to the development of cancer.

During development, programmed cell death plays a key role by eliminating unwanted cells from a variety of tissues. For example, programmed cell death is responsible for the elimination of larval tissues during amphibian and insect metamorphosis, as well as for the elimination of tissue between the digits during the formation of fingers and toes. Another well-characterized example of programmed cell death is provided by development of the mammalian nervous system. Neurons are produced in excess, and up to 50% of developing neurons are eliminated by programmed cell death. Those that survive are selected for having made the correct connections with their target cells, which secrete growth factors that signal cell survival by blocking the neuronal cell death program. The survival of many other types of cells in animals is similarly dependent on growth factors or contacts with neighboring cells or the extracellular matrix, so programmed cell death is thought to play an important role in regulating the associations between cells in tissues.

The Events of Apoptosis

In contrast to the accidental death of cells that results from an acute injury, programmed cell death is an active process, which usually proceeds by a distinct series of cellular changes known as **apoptosis**, first described in 1972. During apoptosis, chromosomal DNA is usually fragmented as a result of cleavage between nucleosomes (Figure 17.1). The chromatin condenses and the nucleus then breaks up into small pieces. Finally, the cell itself shrinks and breaks up into membrane-enclosed fragments called apoptotic bodies.

Apoptotic cells and cell fragments are efficiently recognized and phagocytosed by both macrophages and neighboring cells, so cells that die by apoptosis are rapidly removed from tissues. In contrast, cells that die as a result of acute injury swell and lyse, releasing their contents into the extracellular space and causing inflammation. The removal of apoptotic cells is mediated by the expression of so-called "eat me" signals on the cell surface. These signals include phosphatidylserine, which is normally restricted to the inner leaflet of the plasma membrane (see Figure 13.2). During apoptosis, phosphatidylserine becomes expressed on the cell surface where it is recognized by receptors expressed by phagocytic cells (Figure 17.2).

Pioneering studies of programmed cell death during the development of *C. elegans* provided the critical initial insights that led to understanding the

■ **The term apoptosis is derived from the Greek word describing the falling of leaves from a tree or petals from a flower. It was coined to differentiate this form of programmed cell death from the accidental cell deaths caused by inflammation or injury.**

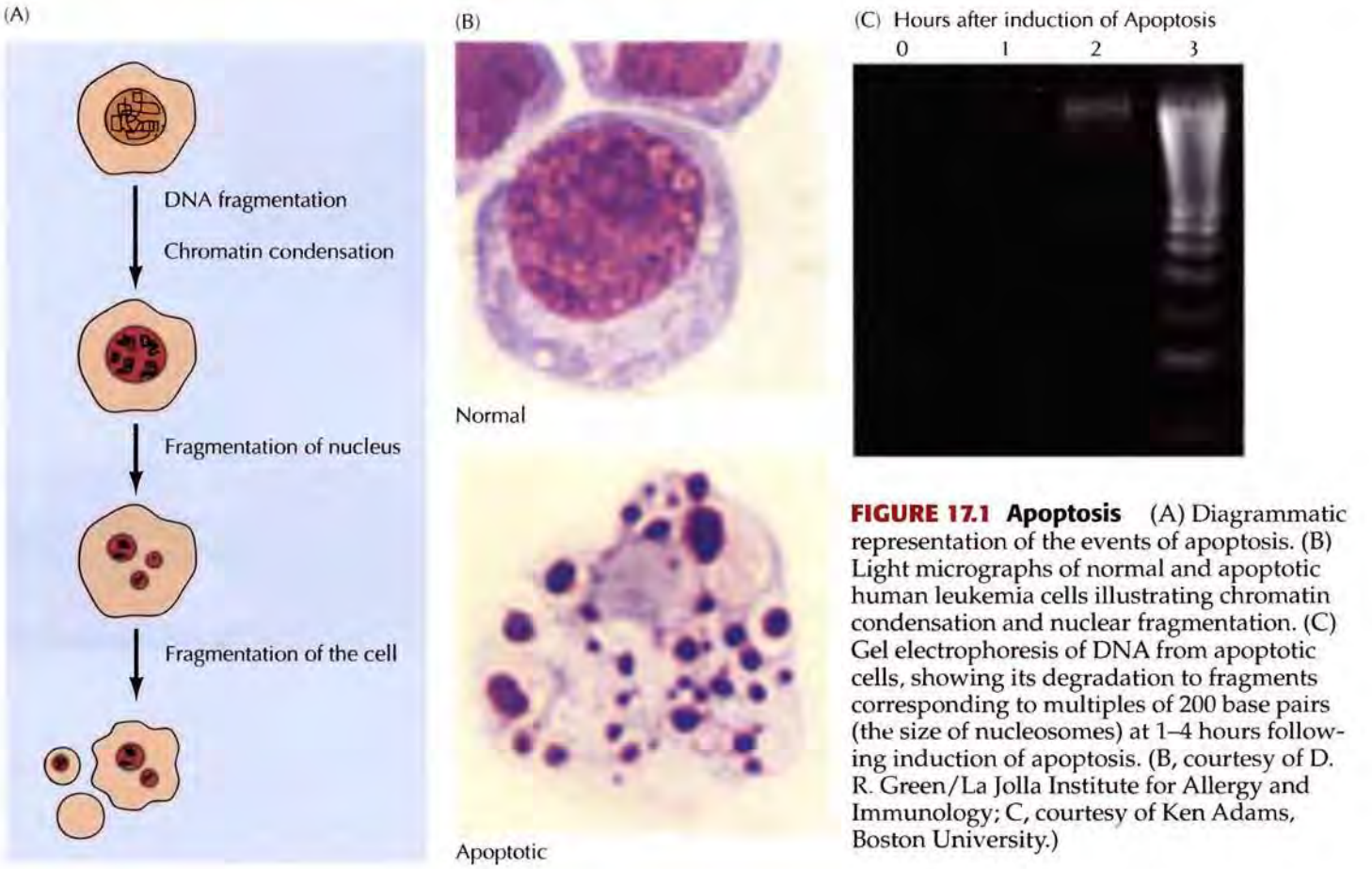


FIGURE 17.1 Apoptosis (A) Diagrammatic representation of the events of apoptosis. (B) Light micrographs of normal and apoptotic human leukemia cells illustrating chromatin condensation and nuclear fragmentation. (C) Gel electrophoresis of DNA from apoptotic cells, showing its degradation to fragments corresponding to multiples of 200 base pairs (the size of nucleosomes) at 1–4 hours following induction of apoptosis. (B, courtesy of D. R. Green/La Jolla Institute for Allergy and Immunology; C, courtesy of Ken Adams, Boston University.)

molecular mechanism of apoptosis. These studies in the laboratory of Robert Horvitz initially identified three genes that play key roles in regulating and executing apoptosis. During normal nematode development, 131 somatic cells out of a total of 1090 are eliminated by programmed cell death, yielding the 959 somatic cells in the adult worm. The death of these cells is highly specific, such that the same cells always die in developing embryos. Based on this developmental specificity, Robert Horvitz undertook a genetic analysis of cell death in *C. elegans* with the goal of identifying the genes responsible for these developmental cell deaths. In 1986 mutagenesis of *C. elegans* identified two genes, *ced-3* and *ced-4*, that were required for developmental cell death. If either *ced-3* or *ced-4* was inactivated by mutation, the normal programmed cell deaths did not take place. A third gene, *ced-9*, functioned as a negative regulator of apoptosis. If *ced-9* was inactivated by mutation, the cells that would normally survive failed to do so. Instead, they also underwent apoptosis, leading to death of the developing animal. Conversely, if *ced-9* was expressed at an abnormally high level, the normal programmed cell deaths failed to occur. Further studies indicated that the proteins encoded by these genes acted in a pathway with Ced-4 acting to stimulate Ced-3, and Ced-9 inhibiting Ced-4 (Figure 17.3). Genes related to *ced-3*, *ced-4*, and *ced-9* have also been identified in *Drosophila* and mammals and found to encode proteins that represent conserved effectors and regulators of apoptosis induced by a variety of stimuli.

17.1 WEBSITE ANIMATION

Apoptosis During apoptosis, chromosomal DNA is usually fragmented, the chromatin condenses, the nucleus breaks up, and the cell shrinks and breaks into apoptotic bodies.

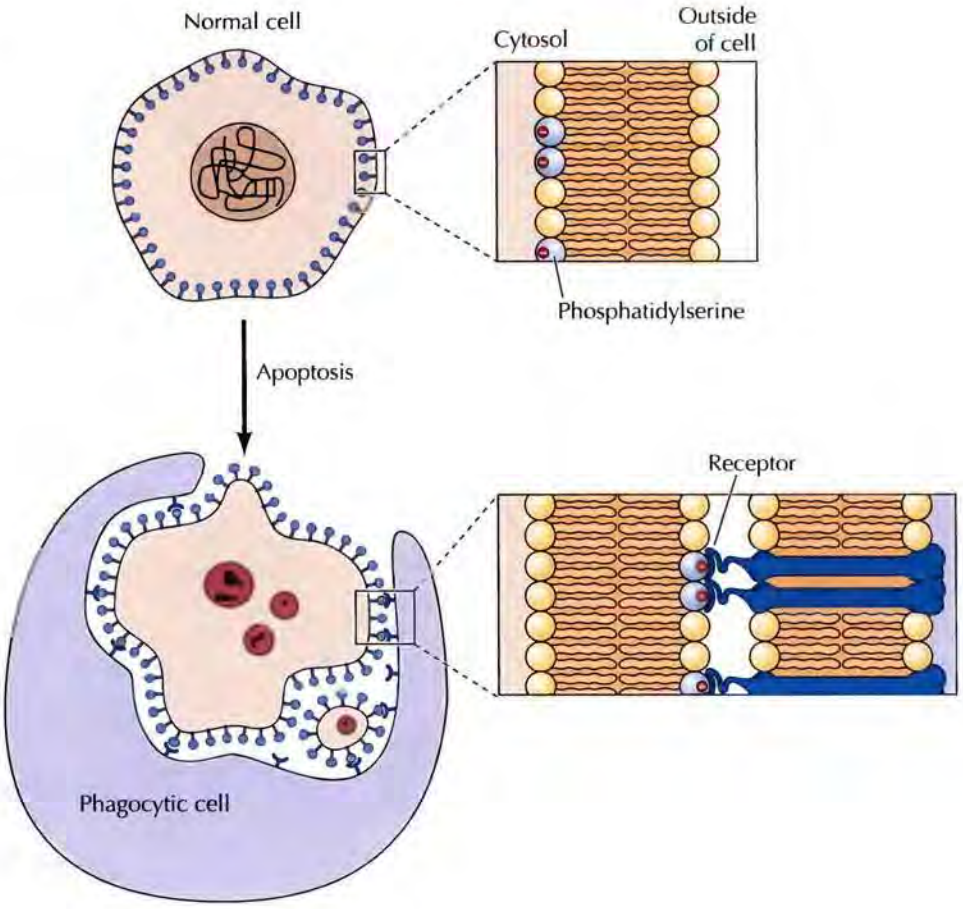


FIGURE 17.2 Phagocytosis of apoptotic cells Apoptotic cells and cell fragments are recognized and engulfed by phagocytic cells. One of the signals recognized by phagocytes is phosphatidylserine on the cell surface. In normal cells, phosphatidylserine is restricted to the inner leaflet of the plasma membrane, but it becomes expressed on the cell surface during apoptosis.

Caspases: The Executioners of Apoptosis

The molecular cloning and nucleotide sequencing of the *ced-3* gene indicated that it encoded a protease, providing the first insight into the molecular mechanism of apoptosis. We now know that Ced-3 is the prototype of a family of more than a dozen proteases, known as **caspases** because they have cysteine (C) residues at their active sites and cleave after aspartic acid (Asp) residues in their substrate proteins. The caspases are the ultimate effectors or executioners of programmed cell death, bringing about the events of apoptosis by cleaving nearly 100 different cell target proteins (Figure 17.4). Key targets of the caspases include an inhibitor of a DNase, which when activated is responsible for fragmentation of nuclear DNA. In addition, caspases cleave both nuclear lamins, leading to fragmentation of the nucleus, and cytoskeletal proteins, leading to disruption of the cytoskeleton, membrane blebbing, and cell fragmentation.

Ced-3 is the only caspase in *C. elegans*. However, *Drosophila* and mammals contain families of at least seven caspases that function in different

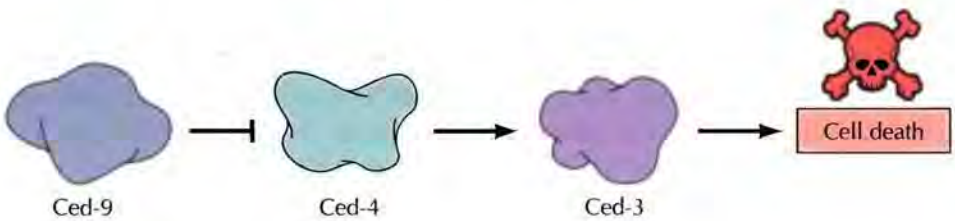


FIGURE 17.3 Programmed cell death in *C. elegans* Genetic analysis identified three genes that play key roles in programmed cell death during development of *C. elegans*. Two genes, *ced-3* and *ced-4*, are required for cell death, whereas *ced-9* inhibits cell death. The Ced-9 protein acts upstream of Ced-4, which activates Ced-3.

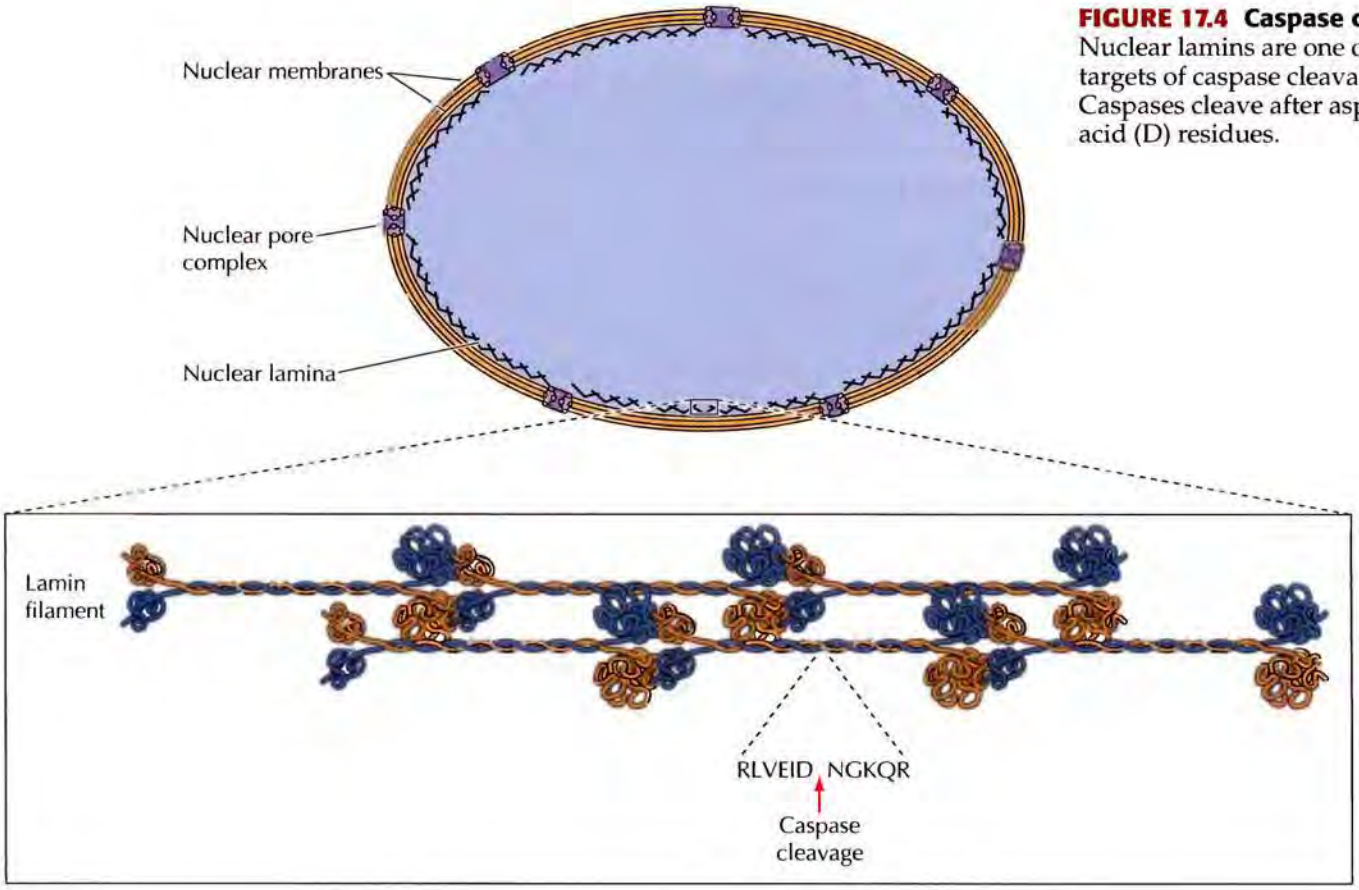


FIGURE 17.4 Caspase cleavage Nuclear lamins are one of the targets of caspase cleavage. Caspases cleave after aspartic acid (D) residues.

aspects of apoptosis. These caspases are generally classified as either *initiator* or *effector* caspases, and they function in a cascade that ultimately brings about death of the cell. All caspases are synthesized as inactive precursors that can be converted to the active form by proteolytic cleavage, catalyzed by other caspases. The activation of an initiator caspase therefore starts off a chain reaction leading to activation of downstream effector caspases and death of the cell. Regulation of caspases is thus central to determining cell survival.

Genetic analysis in *C. elegans* initially suggested that Ced-4 functioned as an activator of the caspase Ced-3. Subsequent studies have shown that Ced-4 and its mammalian homolog (Apaf-1) bind to caspases and promote their activation. In mammalian cells, the key initiator caspase (caspase-9) is activated by binding to Apaf-1 in a multisubunit complex called the **apoptosome** (Figure 17.5). Formation of this complex in mammals also requires cytochrome *c*, which is released from mitochondria by stimuli that trigger apoptosis (discussed in the following section). Once activated in the apoptosome, cas-

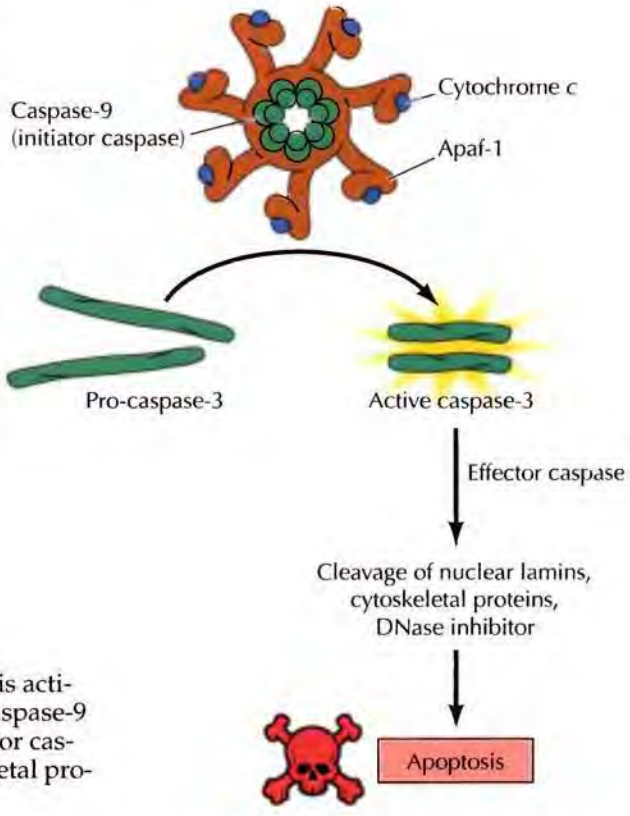


FIGURE 17.5 Caspase activation The mammalian initiator caspase-9 is activated as a complex with Apaf-1 and cytochrome *c* in the apoptosome. Caspase-9 then cleaves and activates effector caspases, such as caspase-3. The effector caspases cleave a variety of cell proteins, including nuclear lamins, cytoskeletal proteins, and an inhibitor of DNase, leading to death of the cell.

KEY EXPERIMENT

Identification of Genes Required for Programmed Cell Death

Genetic Control of Programmed Cell Death in the Nematode *C. elegans*

Hilary M. Ellis and H. Robert Horvitz
Massachusetts Institute of Technology, Cambridge, MA
Cell, 1986, Volume 44, pages 817–829

H. Robert Horvitz

The Context

By the 1960s cell death was recognized as a normal event during animal development, implying that it was a carefully regulated process with specific cells destined to die. The simple nematode *C. elegans*, which has been a critically important model system in developmental biology, proved to be key to understanding both the regulation and the mechanism of such programmed cell deaths. Microscopic analysis in the 1970s established a complete map of *C. elegans* development so that the embryonic origin and fate of each cell was known. Importantly, *C. elegans* development included a very specific pattern of programmed cell deaths. In particular, John Sulston and H. Robert Horvitz reported in 1977 that the development of adult worms (consisting of 959 somatic cells) involved the programmed death of 131 cells out of 1090 that were initially produced. The same cells died in all embryos, indicating that the death of these cells was a normal event during development, with cell death being a specific developmental fate. It was also notable that all of these dying cells underwent a similar series of morphological changes, suggesting that these programmed cell deaths occurred by a common mechanism.

Based on these considerations, Horvitz undertook a genetic analysis with the goal of characterizing the mechanism and regulation of programmed cell death during *C. elegans* development. In the experiments reported in this 1986 paper, Hilary Ellis and Horvitz identified two genes that were required for all of the programmed cell deaths that took place during development of the nematode. The identification and characterization of these genes was a critical first step leading to our current understanding of the molecular biology of apoptosis.

The Experiments

Cells undergoing programmed cell death in *C. elegans* can readily be identified as highly refractile cells by microscopic examination, so Ellis and Horvitz were able to use this as an assay to screen for mutant animals in which the normal cell deaths did not occur. To isolate mutants that displayed abnormalities in cell death, they treated nematodes with the chemical mutagen ethyl methanesulfonate, which reacts with DNA. The progeny of approximately 4000 worms were examined to identify dying cells, and two mutant strains were found in which the expected cell deaths did not take place (see figure). Both of these mutant strains harbored recessive mutations of the same gene, which was called *ced-3*. Further studies indicated that the mutations in *ced-3* blocked all of the 131 programmed cell deaths that would normally occur during development.

Continuing studies identified an additional mutation that was similar to *ced-3* in preventing programmed cell death. However, this mutation was in a different gene, which was located on a different chromosome than *ced-3*. This second gene was called *ced-4*. Similar to mutations in *ced-3*, recessive mutations in *ced-4* were found to block all programmed cell deaths in the worm.

The Impact

The isolation of *C. elegans* mutants by Ellis and Horvitz provided the first identification of genes that were involved in the process of programmed cell death. The proteins encoded by the *ced-3* and *ced-4* genes, as well as by the *ced-9* gene (which was subsequently identified by Horvitz and colleagues), proved to be prototypes of the central regulators

(A)

(B)

Photomicrographs of a normal worm (A) and a *ced-3* mutant (B). Dying cells are highly refractile and are indicated by arrows in panel A. These cells are not present in the mutant animal.

KEY EXPERIMENT

and effectors of apoptosis that are highly conserved in evolution. The cloning and sequencing of *ced-3* revealed that it was related to a protease that had been previously identified in mammalian cells, and which became the first member of the caspase family. The *C. elegans ced-9* gene was related to the *bcl-2* oncogene, first isolated from a human B cell lymphoma, which had the unusual property of inhibiting apoptosis rather than stimulating cell proliferation. And *ced-4* was found to encode an adaptor protein related to mammalian Apaf-1, which is required for caspase activation. The identification of these genes in *C. elegans* thus led the way to understanding the molecular basis of apoptosis, with broad implications both for development and for the maintenance of normal adult tissues. Since abnormalities of apoptosis contribute to a wide variety of diseases, including cancer, autoimmune disease, and neurodegenerative disorders, the seminal findings of Ellis and Horvitz have impacted a wide range of areas in biology and medicine.

pase-9 cleaves and activates downstream effector caspases, such as caspase-3 and caspase-7, eventually resulting in cell death.

Central Regulators of Apoptosis: The Bcl-2 Family

The third gene identified as a key regulator of programmed cell death in *C. elegans*, *ced-9*, was found to be closely related to a mammalian gene called ***bcl-2***, which was first identified in 1985 as an oncogene that contributed to the development of human B cell lymphomas (cancers of B lymphocytes). In contrast to other oncogenes proteins, such as Ras, that stimulate cell proliferation (see Molecular Medicine Chapter 15), Bcl-2 was found to inhibit apoptosis. Ced-9 and Bcl-2 were thus similar in function, and the role of Bcl-2 as a regulator of apoptosis first focused attention on the importance of cell survival in cancer development. As discussed further in the next chapter, we now recognize that cancer cells are generally defective in the normal process of programmed cell death and that their inability to undergo apoptosis is as important as their uncontrolled proliferation in the development of malignant tumors.

Mammals encode a family of approximately 20 proteins related to Bcl-2, which are divided into three functional groups (Figure 17.6). Some members of the Bcl-2 family (antiapoptotic family members)—like Bcl-2 itself—function as inhibitors of apoptosis and programmed cell death. Other members of the Bcl-2 family, however, are proapoptotic proteins that act to induce caspase activation and promote cell death. There are two groups of these proapoptotic proteins, which differ in function as well as in their extent of homology to Bcl-2. Bcl-2 and the other antiapoptotic family members share

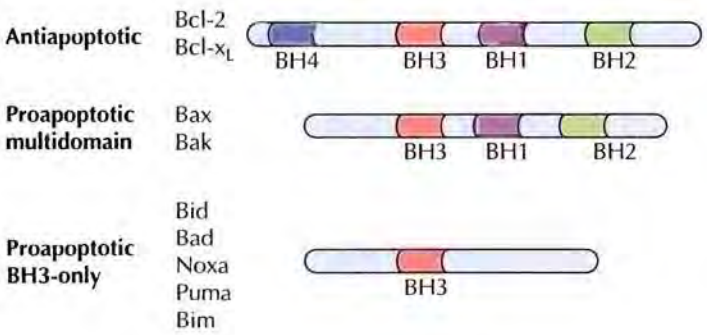


FIGURE 17.6 The Bcl-2 family The Bcl-2 family of proteins is divided into three functional groups. Antiapoptotic proteins (e.g., Bcl-2 and Bcl-x_L) have four Bcl-2 homology domains (BH1–BH4). The multidomain proapoptotic proteins (e.g., Bax and Bak) have three homology domains (BH1–BH3), whereas the BH3-only proapoptotic proteins (e.g., Bid, Bad, Noxa, Puma, and Bim) have only one homology domain (BH3).

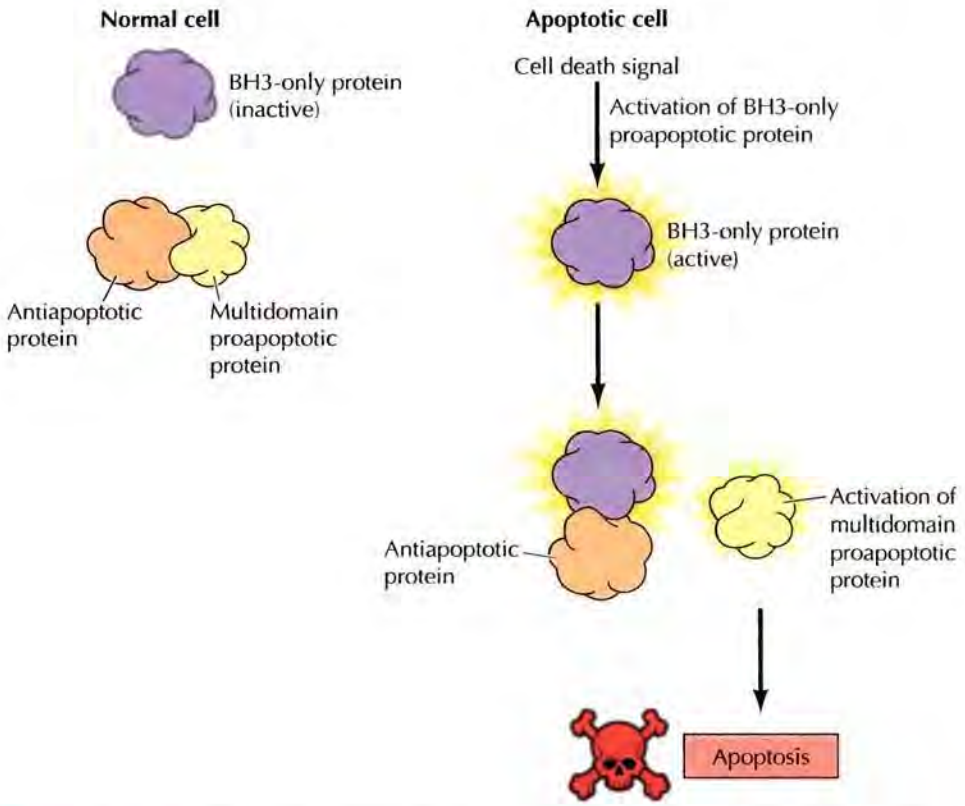


FIGURE 17.7 Regulatory interactions between Bcl-2 family members In normal cells, the BH3-only proapoptotic proteins are inactive, and the multidomain proapoptotic proteins are inhibited by interaction with antiapoptotic proteins. Cell death signals activate the BH3-only proteins, which then interact with the antiapoptotic proteins, leading to activation of the multidomain proapoptotic proteins and cell death.

four conserved regions called Bcl-2 homology (BH) domains. One group of proapoptotic family members called the "multidomain" proapoptotic proteins have 3 BH domains (BH1, BH2, and BH3), whereas the second group, the "BH3-only" proteins, have only the BH3 domain.

The fate of the cell—life or death—is determined by the balance of activity of proapoptotic and antiapoptotic Bcl-2 family members, which act to regulate one another (Figure 17.7). The multidomain proapoptotic family members, such as Bax and Bak, are the downstream effectors that directly induce apoptosis. They are inhibited by interactions with the antiapoptotic family members, such as Bcl-2. The BH3-only family members are upstream members of the cascade, regulated by the signals that induce cell death (e.g., DNA damage) or cell survival (e.g., growth factors). When activated, the BH3-only family members antagonize the antiapoptotic Bcl-2 family members, activating the multidomain proapoptotic proteins and tipping the balance in favor of caspase activation and cell death.

In mammalian cells, members of the Bcl-2 family act at mitochondria, which play a central role in controlling programmed cell death (Figure 17.8). When activated, Bax and Bak form oligomers in the mitochondrial outer membrane. Formation of these Bax or Bak oligomers leads to the release of cytochrome *c* from the mitochondrial intermembrane space, either by forming pores or by interacting with other mitochondrial outer membrane pro-

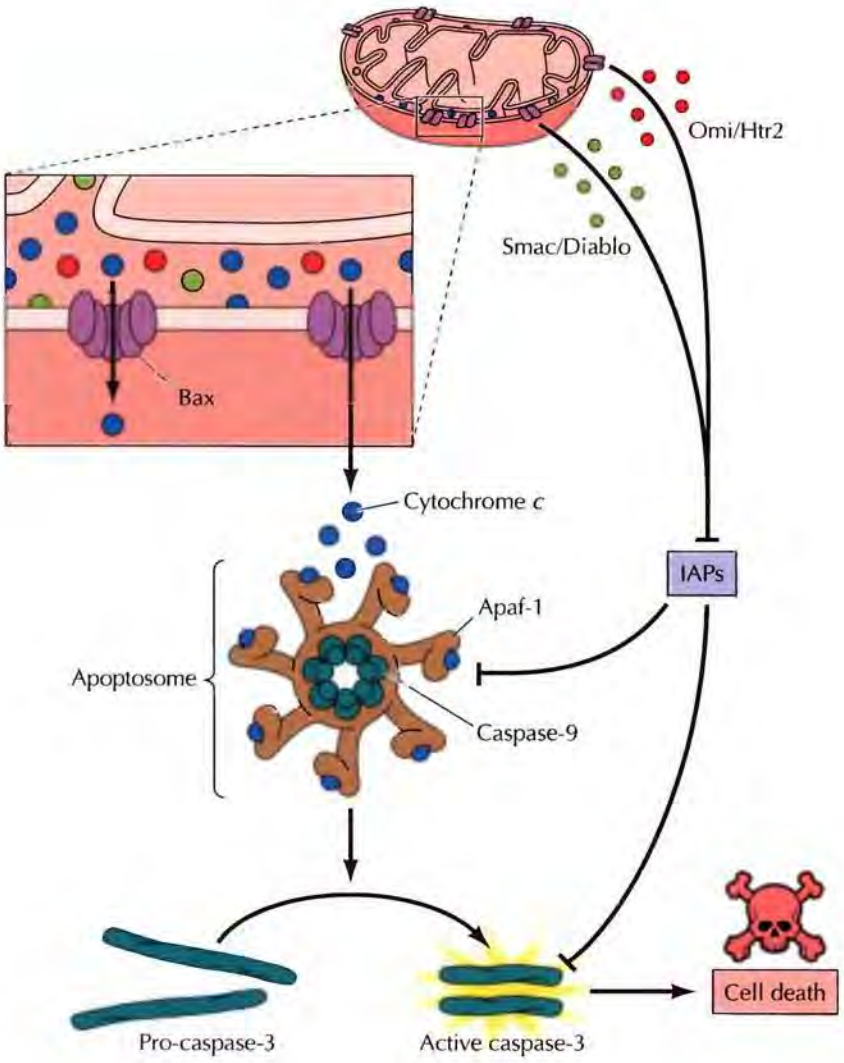


FIGURE 17.8 The mitochondrial pathway of apoptosis In mammalian cells, many cell death signals induce apoptosis as a result of damage to mitochondria. When active, the proapoptotic multidomain Bcl-2 family proteins (e.g., Bax) form oligomers in the outer membrane of mitochondria, resulting in the release of cytochrome *c* and other proapoptotic molecules from the intermembrane space. Release of cytochrome *c* leads to the formation of apoptosomes containing Apaf-1 and caspase-9 in which caspase-9 is activated. Caspase-9 then activates downstream caspases, such as caspase-3, by proteolytic cleavage. Smac/Diablo and Omi/Htr2 promote cell death by interfering with the action of IAPs, which are inhibitors of the caspases.

teins. The release of cytochrome *c* from mitochondria then triggers caspase activation. In particular, the key initiator caspase in mammalian cells (caspase-9) is activated by forming a complex with Apaf-1 in the apoptosome. In mammals, formation of this complex also requires cytochrome *c*. Under normal conditions of cell survival, cytochrome *c* is localized to the mitochondrial intermembrane space (see Figure 11.9) while Apaf-1 and caspase-9 are found in the cytosol, so caspase-9 remains inactive. Activation of Bax or Bak results in the release of cytochrome *c* to the cytosol, where it binds to Apaf-1 and triggers apoptosome formation and caspase-9 activation.

Caspases are also regulated by a family of proteins called the **IAP**, for *i*nhibitor of *ap*optosis, family. Members of the IAP family directly interact with caspases and suppress apoptosis by either inhibiting caspase activity or by targeting caspases for ubiquitination and degradation in the proteasome. IAPs are present in both *Drosophila* and mammals (but not *C. elegans*), and regulation of their activity or expression provides another mechanism for regulating apoptosis. In mammalian cells, the permeabilization of mitochondria by Bax or Bak results not only in the release of cytochrome *c* but also of other proteins that promote apoptosis. These include two proteins (called Smac/Diablo and Omi/Htr2) that stimulate caspase activity by interfering with the action of IAPs (see Figure 17.8). As discussed below,

■ **IAPs were first discovered in virus-infected insect cells as viral proteins that inhibited apoptosis of the host cell in response to viral infection.**

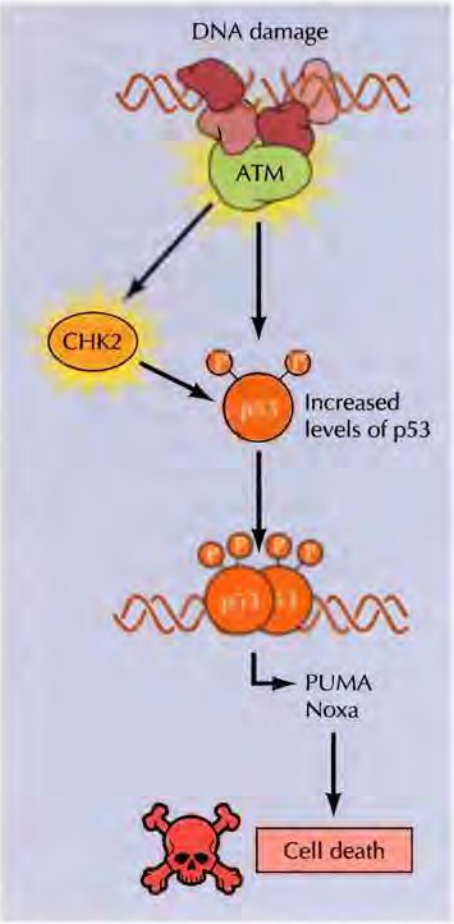


FIGURE 17.9 Role of p53 in DNA damage-induced apoptosis DNA damage leads to activation of the ATM and CHK2 protein kinases, which phosphorylate and stabilize p53 resulting in rapid increases in p53 levels. The protein p53 then activates transcription of genes encoding the proapoptotic BH3-only proteins PUMA and Noxa, leading to cell death.

IAPs as well as members of the Bcl-2 family are targets of the signaling pathways that control survival of mammalian cells.

Signaling Pathways that Regulate Apoptosis

Regulation of programmed cell death is mediated by the integrated activity of a variety of signaling pathways, some acting to induce cell death and others acting to promote cell survival. These signals control the fate of individual cells, so that cell survival or elimination is determined by the needs of the organism as a whole. As illustrated by the following examples, the multiple signaling pathways that regulate apoptosis converge on regulation of the BH3-only members of the Bcl-2 family.

One important role of apoptosis is the elimination of damaged cells. Cells with damaged genomes are particularly dangerous because of the possibility that they will have suffered mutations that may lead to the development of cancer. DNA damage is thus one of the principal triggers of programmed cell death, leading to the elimination of cells carrying potentially harmful mutations. As discussed in Chapter 16, several cell cycle checkpoints halt cell cycle progression in response to damaged DNA, allowing time for the damage to be repaired. In mammalian cells, a major pathway leading to cell cycle arrest in response to DNA damage is mediated by the transcription factor **p53**. The ATM and CHK protein kinases, which are activated by DNA damage, phosphorylate and stabilize p53. The resulting increase in p53 leads to transcriptional activation of p53 target genes. These include the Cdk inhibitor p21, which inhibits Cdk1/cyclin and Cdk2/cyclin complexes, halting cell cycle progression. However, activation of p53 by DNA damage can also lead to apoptosis (Figure 17.9). The induction of apoptosis by p53 results, at least in part, from transcriptional activation of genes encoding BH3-only propapoptotic Bcl-2 family members, such as PUMA and Noxa. Thus p53 mediates both cell cycle arrest and apoptosis in response to DNA damage. Whether DNA damage in a given cell leads to apoptosis or reversible cell cycle arrest may depend on the extent of damage and the resulting level of p53 induction, as well as the influence of other life/death signals being received by the cell.

Rather than inducing programmed cell death, many signaling pathways act to promote cell survival by inhibiting apoptosis. These signaling pathways control the fate of a wide variety of cells whose survival is dependent on extracellular growth factors or cell-cell interactions. As already noted, a well-characterized example of programmed cell death in development is provided by the vertebrate nervous system. About 50% of neurons die by apoptosis, with the survivors having received sufficient amounts of survival signals from their target cells. These survival signals are polypeptide growth factors related to nerve growth factor (NGF), which induces both neuronal survival and differentiation by activating a receptor protein-tyrosine kinase. Other types of cells are similarly dependent upon growth factors or cell contacts that activate nonreceptor protein-tyrosine kinases associated with integrins. Indeed, most cells in higher animals are programmed

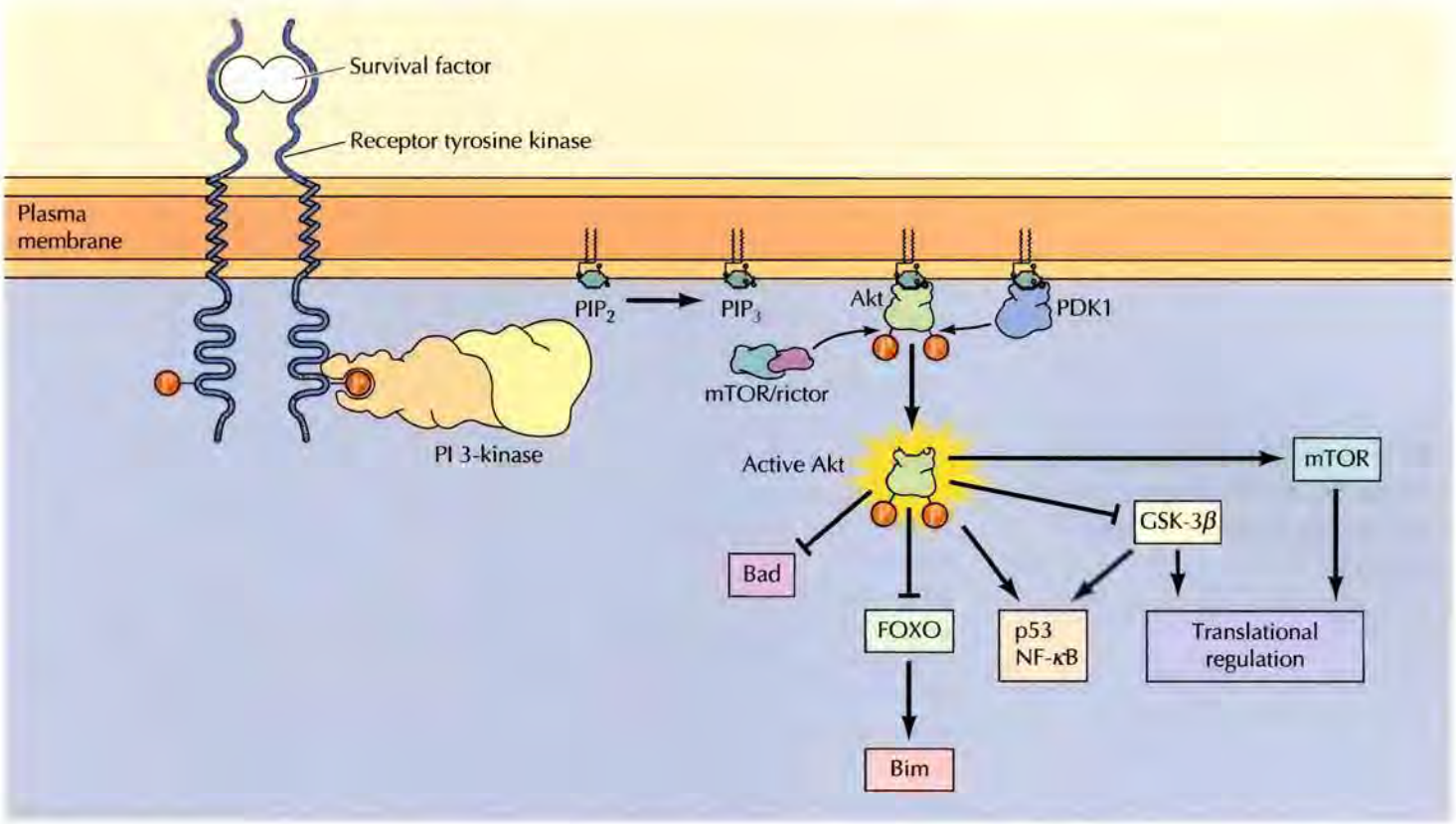


FIGURE 17.10 The PI 3-kinase pathway and cell survival Many growth factors that signal cell survival activate receptor protein-tyrosine kinases, leading to activation of PI 3-kinase, formation of PIP_3, and activation of the protein kinase, Akt. Akt then phosphorylates a number of proteins that contribute to cell survival. Phosphorylation of the BH3-only protein Bad maintains it in an inactive state, as does phosphorylation of the FOXO transcription factor. In the absence of Akt signaling, activation of Bad promotes apoptosis and activation of FOXO stimulates transcription of another BH3-only protein, Bim. Additional targets of Akt that have been implicated in regulation of apoptosis include the protein kinase GSK-3β and additional transcription factors, such as p53 and NF-κB, both of which are regulated by Akt and GSK-3β phosphorylation. Translational regulation by GSK-3β and by the mTOR pathway (see Figure 15.33) may also affect cell survival.

to undergo apoptosis unless cell death is actively suppressed by survival signals from other cells.

One of the major intracellular signaling pathways responsible for promoting cell survival is initiated by the enzyme **PI 3-kinase**, which is activated by either protein-tyrosine kinases or G protein-coupled receptors. PI 3-kinase phosphorylates the membrane phospholipid PIP_2 to form PIP_3, which activates the protein-serine/threonine kinase **Akt** (see Figures 15.30 and 15.31). Akt then phosphorylates a number of proteins that regulate apoptosis (Figure 17.10). One key substrate for Akt is the proapoptotic BH3-only Bcl-2 family member called Bad. Phosphorylation of Bad by Akt creates a binding site for 14-3-3 chaperone proteins that sequester Bad in an inactive form, so phosphorylation of Bad by Akt inhibits apoptosis and promotes cell survival. Bad is similarly phosphorylated by protein kinases of other growth factor-induced signaling pathways, including the Ras/Raf/MEK/ERK pathway, so it serves as a convergent regulator of growth factor signaling in mediating cell survival.

Other targets of Akt, including the FOXO transcription factors, also play key roles in cell survival. Phosphorylation of FOXO by Akt creates a binding site for 14-3-3 proteins, which sequester FOXO in an inactive form in the cytoplasm (see Figure 15.32). In the absence of growth factor signaling and Akt activity, FOXO is released from 14-3-3 and translocates to the nucleus, stimulating transcription of proapoptotic genes, including the gene encoding the BH3-only protein, Bim. Akt and its downstream target GSK-3β also regulate other transcription factors with roles in cell survival, including p53 and NF-κB, which may control the expression of IAPs as well as Bcl-2 family members. Translational regulation by both GSK-3β and the mTOR pathway (see Figure 15.33) may also affect cell survival.

In contrast to the growth factor signaling pathways that promote cell survival, some secreted polypeptides signal programmed cell death by activating receptors that directly induce apoptosis of the target cell. These cell death signals are polypeptides belonging to the **tumor necrosis factor (TNF)** family. They bind to members of the TNF receptor family, which can signal apoptosis in a variety of cell types. One of the best characterized members of this family is the cell surface receptor called Fas, which plays important roles in controlling cell death in the immune system. For example, apoptosis induced by activation of Fas is responsible for killing target cells of the immune system, such as cancer cells or virus-infected cells, as well as for eliminating excess lymphocytes at the end of an immune response.

■ **Therapies based on a member of the TNF family are under clinical trial for the treatment of certain cancers.**

The cell death receptors signal apoptosis by directly activating an initiator caspase (Figure 17.11). TNF and related family members consist of three identical polypeptide chains, and their binding induces receptor trimerization. The cytoplasmic portions of the receptors bind adaptor molecules that in turn bind an initiator caspase called caspase-8. This leads to activation of caspase-8, which can then cleave and activate downstream effector caspases. In some cells, caspase-8 activation and subsequent activation of caspases-3 and -7 is sufficient to induce apoptosis directly. In other cells, however, amplification of the signal is needed. This results from caspase-8 cleavage of the proapoptotic BH3-only protein Bid, leading to Bid activation, permeabilization of mitochondria, and activation of caspase-9, thus amplifying the caspase cascade initiated by direct activation of caspase-8 at cell death receptors.

Stem Cells and the Maintenance of Adult Tissues

Early development is characterized by the rapid proliferation of embryonic cells, which then differentiate to form the specialized cells of adult tissues and organs. As cells differentiate, their rate of proliferation usually decreases, and most cells in adult animals are arrested in the G_0 stage of the cell cycle. However, cells are lost either due to injury or programmed cell death throughout life. In order to maintain a constant number of cells in adult tissues and organs, cell death must be balanced by cell proliferation. In order to maintain this balance, most tissues contain cells that are able to proliferate as required to replace cells that have died. Moreover, in some tissues a subpopulation of cells divide continuously throughout life to replace cells that have a high rate of turnover in adult animals. Cell death and cell renewal are thus carefully balanced to maintain properly sized and functioning adult tissues and organs.

Proliferation of Differentiated Cells

Most types of differentiated cells in adult animals are no longer capable of proliferation. If these cells are lost, they are replaced by the proliferation of less differentiated cells derived from self-renewing stem cells, as discussed in the following section. Other types of differentiated cells, however, retain the ability to proliferate as needed to repair damaged tissue throughout the life of the organism. These cells enter the G_0 stage of the cell cycle but resume proliferation as needed to replace cells that have been injured or have died.

Cells of this type include fibroblasts, which are dispersed in connective tissues where they secrete collagen (Figure 17.12). Skin fibroblasts are normally arrested in G_0 but rapidly proliferate if needed to repair damage

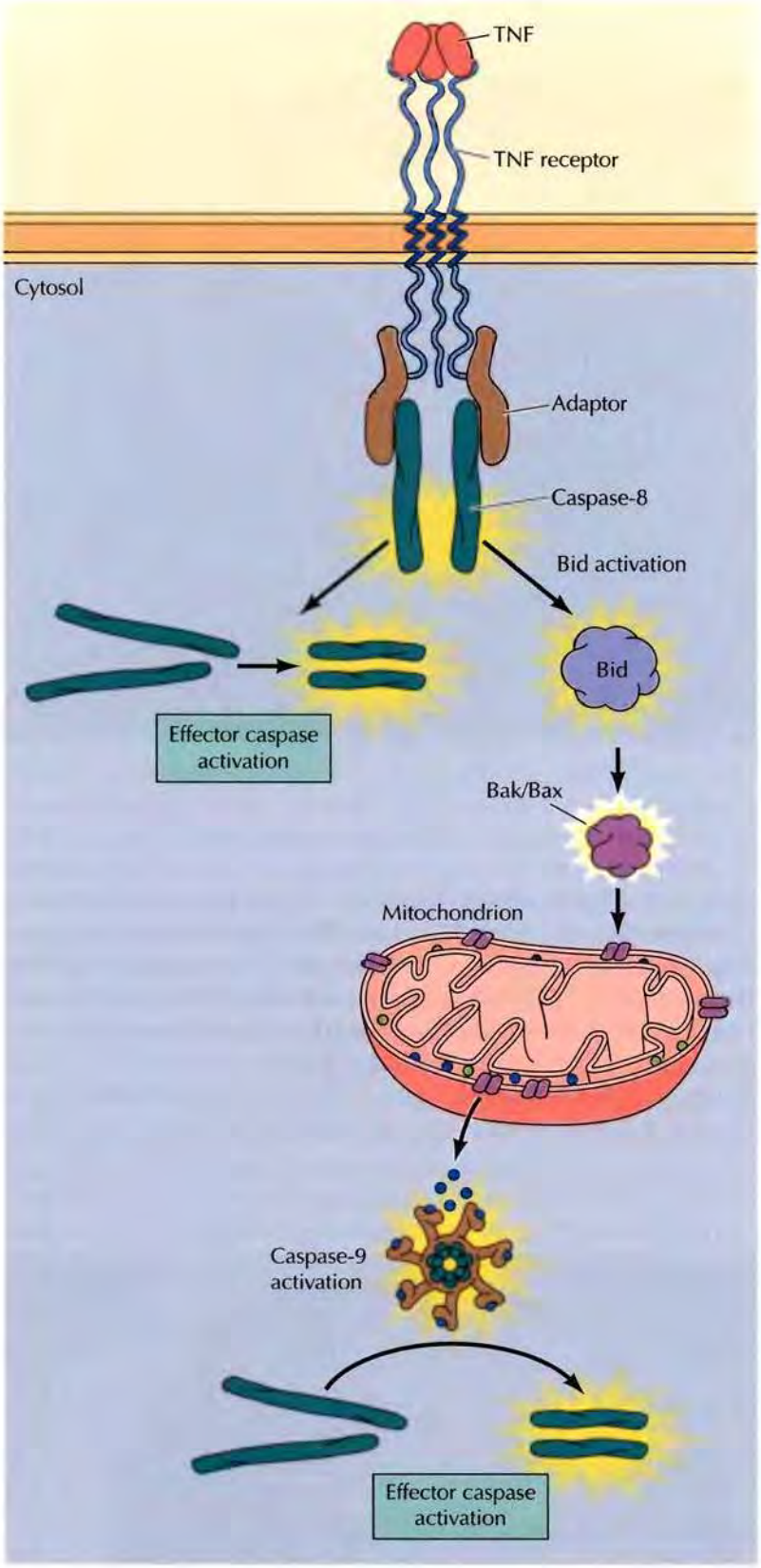


FIGURE 17.11 Cell death receptors TNF and other cell death receptor ligands consist of three polypeptide chains, so their binding to cell death receptors induces receptor trimerization. Caspase-8 is recruited to the receptor and activated via interaction with adaptor molecules. Once activated, caspase-8 can directly cleave and activate effector caspases. In addition, caspase-8 cleaves the BH3-only protein Bid, which activates the mitochondrial pathway of apoptosis, leading to caspase-9 activation.

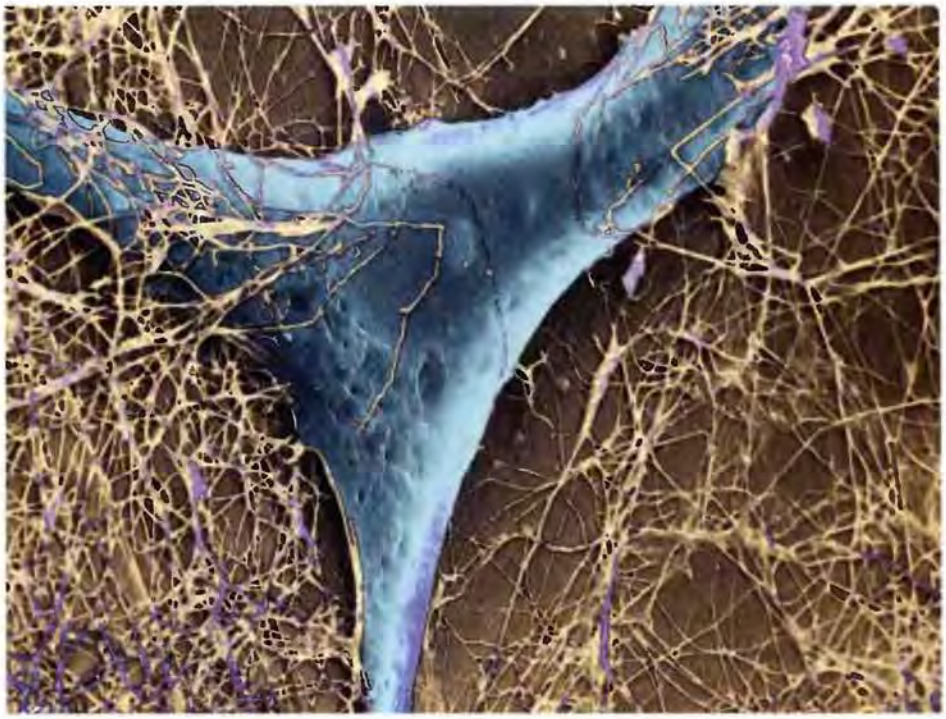


FIGURE 17.12 Skin fibroblasts
Scanning electron micrograph of a fibroblast surrounded by collagen fibrils. (© CMEABG-UCBL/Photo Researchers Inc.)

resulting from a cut or wound. Blood clotting at the site of a wound leads to the release of platelet-derived growth factor (PDGF) from blood platelets. As discussed in Chapter 15, PDGF activates a receptor protein-tyrosine kinase, stimulating both the proliferation of fibroblasts and their migration into the wound where their proliferation and secretion of collagen contributes to repair and regrowth of the damaged tissue.

The endothelial cells that line blood vessels (Figure 17.13) are another type of fully differentiated cell that remains capable of proliferation. Prolifera-

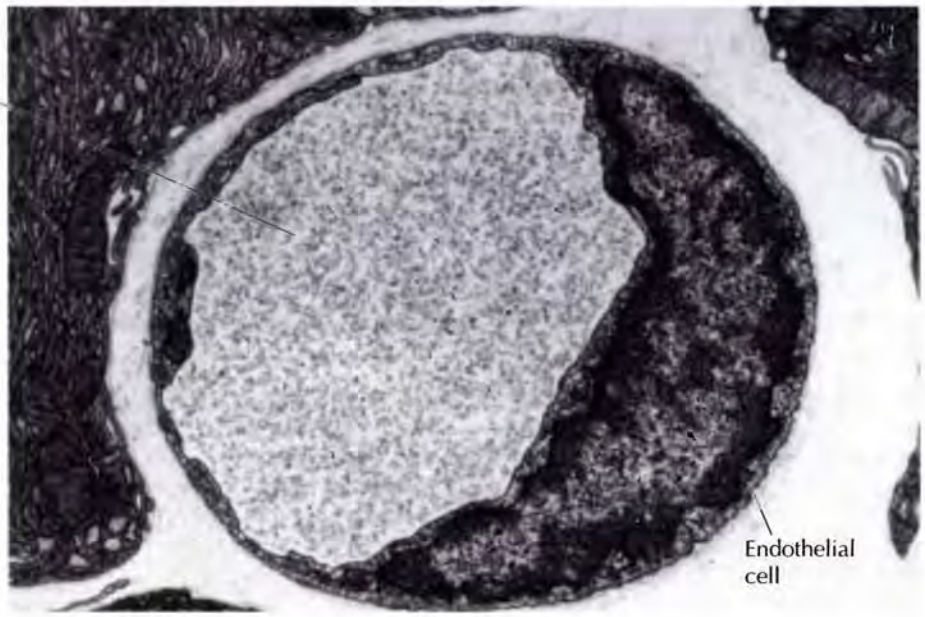


FIGURE 17.13 Endothelial cells
Electron micrograph of a capillary. The capillary is lined by a single endothelial cell surrounded by a thin basal lamina. (© Dr. Don W. Fawcett/Visuals Unlimited.)

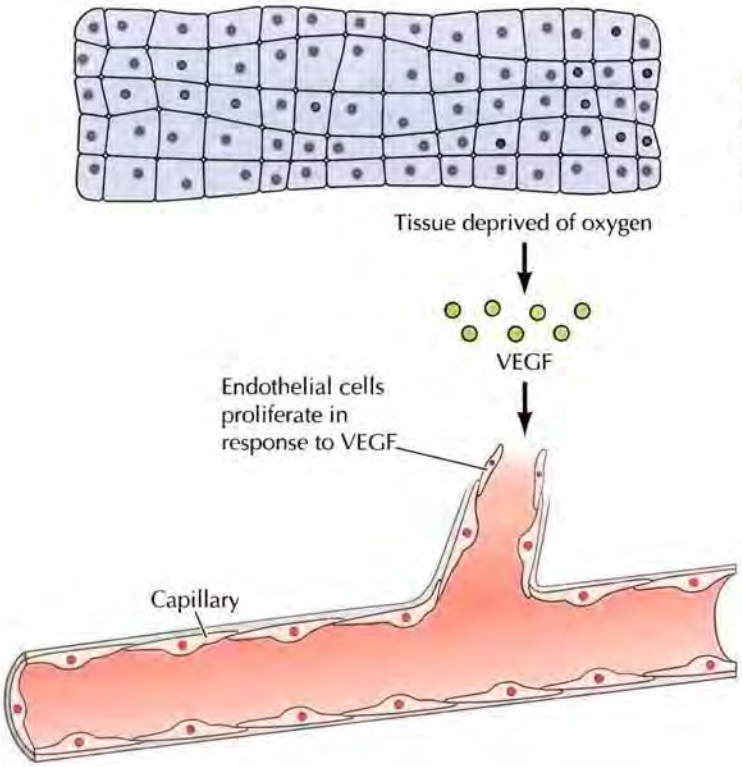


FIGURE 17.14 Proliferation of endothelial cells Endothelial cells are stimulated to proliferate by vascular endothelial growth factor (VEGF). VEGF is secreted by cells deprived of oxygen, leading to the outgrowth of new capillaries into tissues lacking an adequate blood supply.

tion of endothelial cells allows them to form new blood vessels as needed for repair and regrowth of damaged tissue. Endothelial cell proliferation and the resulting formation of blood capillaries is triggered by a growth factor (vascular endothelial growth factor or VEGF) produced by cells of the tissue, which the new capillaries will invade. The production of VEGF is in turned triggered by a lack of oxygen, so the result is a regulatory system in which tissues that have a low oxygen supply resulting from insufficient circulation stimulate endothelial cell proliferation and recruit new capillaries (Figure 17.14). Smooth muscle cells, which form the walls of larger blood vessels (e.g., arteries) as well as the contractile portions of the digestive and respiratory tracts and other internal organs, are also capable of resuming proliferation in response to growth factor stimulation. In contrast, differentiated skeletal and cardiac muscle cells are no longer able to divide.

The epithelial cells of some internal organs, such as the liver and pancreas, are also able to proliferate to replace damaged tissue. A striking example is provided by liver cells, which are normally arrested in the G_0 phase of the cell cycle. However, if large numbers of liver cells are lost (e.g., by surgical removal of part of the liver), the remaining cells are stimulated to proliferate to replace the missing tissue (Figure 17.15). For example, surgical removal of two-thirds of the liver of a rat is followed by rapid proliferation of the remaining cells, leading to regeneration of the entire liver within a few days.

Stem Cells

Most fully differentiated cells in adult animals, however, are no longer capable of cell division. Nonetheless, they can be replaced by the proliferation of a subpopulation of less differentiated self-renewing cells called **stem cells** that are present

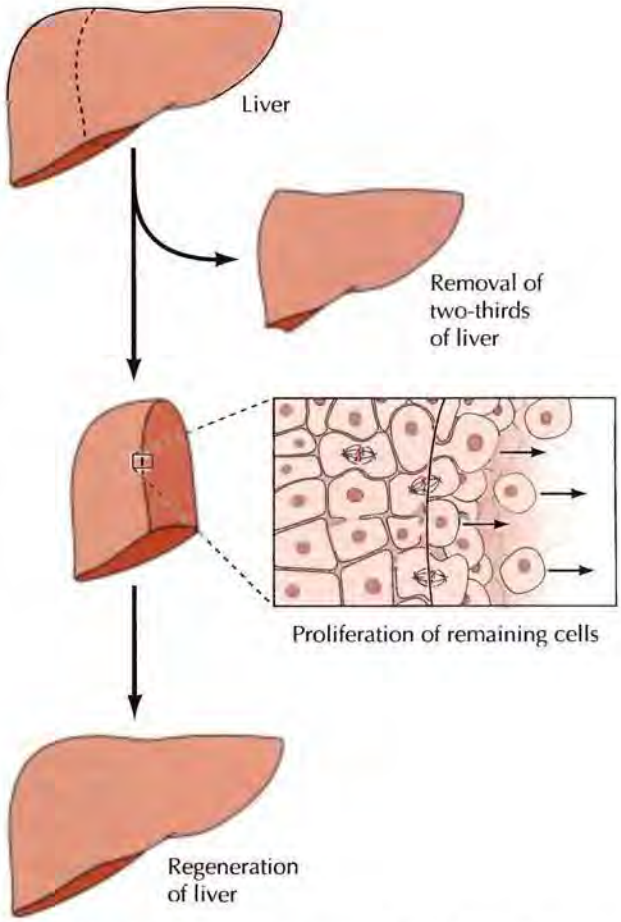


FIGURE 17.15 Liver regeneration Liver cells are normally arrested in G_0 but resume proliferation to replace damaged tissue. If two-thirds of the liver of a rat is surgically removed, the remaining cells proliferate to regenerate the entire liver in a few days.

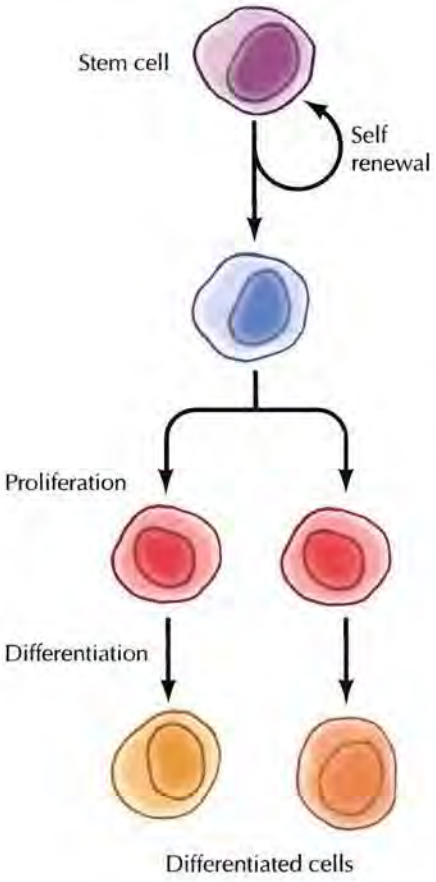


FIGURE 17.16 Stem cell proliferation Stem cells divide to form one daughter cell that remains a stem cell and a second that proliferates and then differentiates.

in most adult tissues. Because they retain the capacity to proliferate and replace differentiated cells throughout the lifetime of an animal, stem cells play a critical role in the maintenance of most tissues and organs.

The key property of stem cells is that they divide to produce one daughter cell that remains a stem cell and one that divides and differentiates (Figure 17.16). Because the division of stem cells produces new stem cells as well as differentiated daughter cells, stem cells are self-renewing populations that can serve as a source for the production of differentiated cells throughout life. The role of stem cells is particularly evident in the case of several types of differentiated cells, including blood cells, epithelial cells of the skin, and epithelial cells lining the digestive tract—all of which have short life spans and must be replaced by continual cell proliferation in adult animals. In all of these cases, the fully differentiated cells do not themselves proliferate; instead, they are continually renewed by the proliferation of stem cells that then differentiate to maintain a stable number of differentiated cells. Stem cells have also been identified in a variety of other adult tissues, including skeletal muscle and the nervous system, where they may function to replace damaged tissue.

Stem cells were first identified in the hematopoietic (blood-forming) system by Ernest McCulloch and James Till in 1961 in experiments showing that single cells derived from mouse bone marrow could proliferate and give rise to multiple differentiated types of blood cells. Hematopoietic stem cells are well-characterized and the production of blood cells provides a good example of the role of stem cells in maintaining differentiated cell populations. There are several distinct types of blood cells with specialized functions: erythrocytes (red blood cells) that transport O_2 and CO_2; granulocytes and macrophages, which are phagocytic cells; platelets (which are fragments of megakaryocytes) that function in blood coagulation; and lymphocytes that are responsible for the immune response. All these cells have limited life spans ranging from less than a day to a few months, and all are derived from the same population of hematopoietic stem cells. More than 100 billion blood cells are lost every day in humans, and must be continually produced from hematopoietic stem cells in the bone marrow (Figure 17.17). Descendants of the hematopoietic stem cell continue to proliferate and undergo several rounds of division as they become committed to specific differentiation pathways that are determined by growth factors that channel precursor cells along specific pathways of blood cell differentiation. Once they become fully differentiated, blood cells cease proliferation, so the maintenance of differentiated blood cell populations is dependent on continual division of the self-renewing hematopoietic stem cell.

The intestine provides an excellent example of stem cells in the self-renewal of an epithelial tissue. The intestine is lined by a single layer of epithelial cells that are responsible for the digestion of food and absorption of nutrients. These intestinal epithelial cells are exposed to an extraordinarily harsh environment and have a lifetime of only a few days before they die by apoptosis and are shed into the digestive tract. Renewal of the intestinal epithelium is therefore a continual process throughout life. New cells are derived from the continuous but slow division of stem cells at the bot-

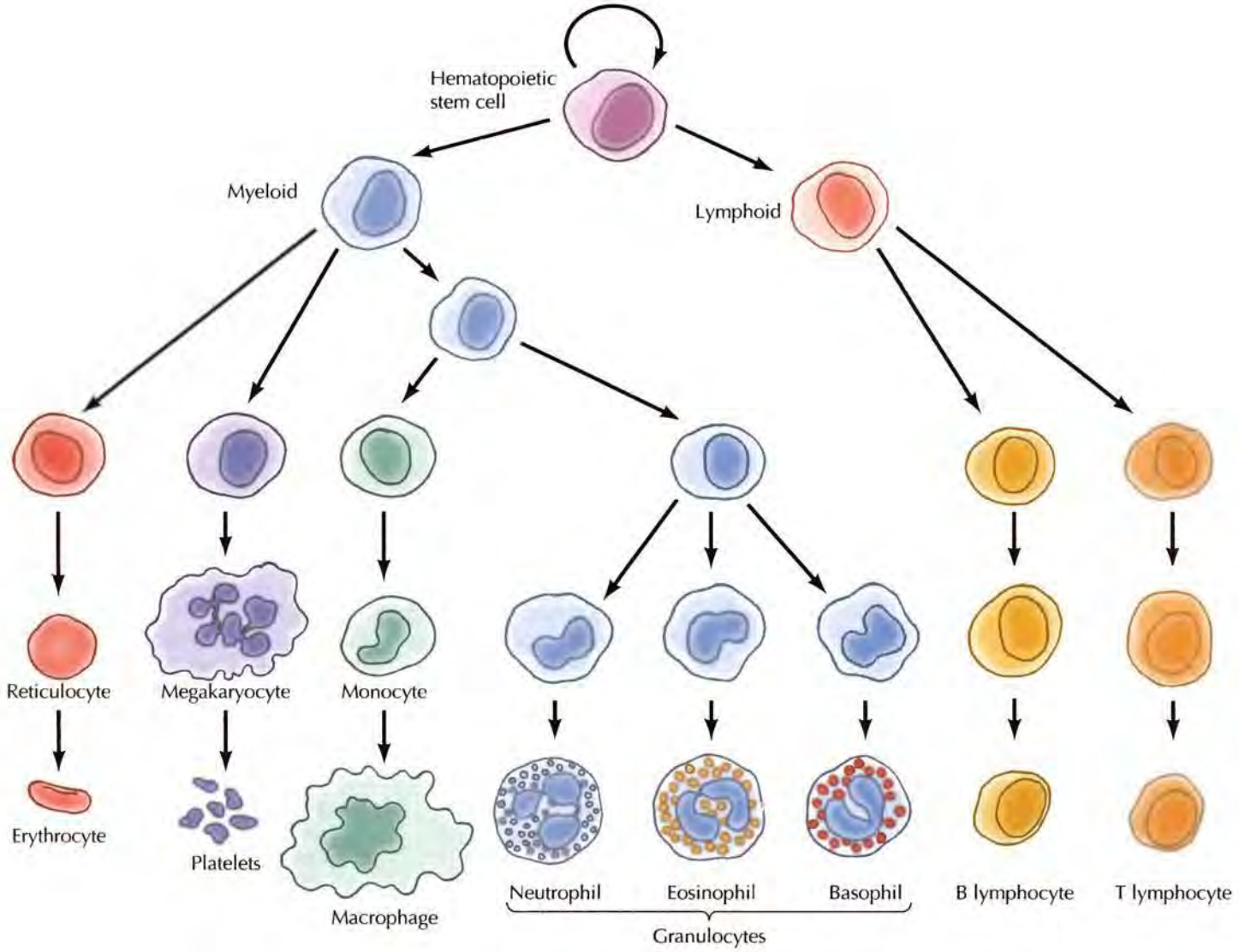


FIGURE 17.17 Formation of blood cells All of the different types of blood cells develop from a hematopoietic stem cell in the bone marrow. The precursors of differentiated cells undergo several rounds of cell division before they differentiate.

tom of intestinal crypts (Figure 17.18). The stem cells give rise to a population of transit-amplifying cells, which divide rapidly and occupy about two-thirds of the crypt. The transit-amplifying cells proliferate for 3–4 cell divisions and then differentiate into the three cell types of the colon surface epithelium: absorptive epithelial cells and two types of secretory cells, called goblet cells and enteroendocrine cells. The small intestine also contains a fourth cell type, Paneth cells, which secrete antibacterial agents. It is noteworthy that all of the cells in an adult crypt are derived from the continuous division of a single self-renewing stem cell.

Stem cells that are responsible for continuous renewal of the skin and hair are also well-characterized. Like the lining of the intestine, the skin and hair are exposed to a harsh external environment—including ultraviolet radiation from sunlight—and are continuously renewed throughout life. In humans, the epidermis turns over every two weeks, with cells being

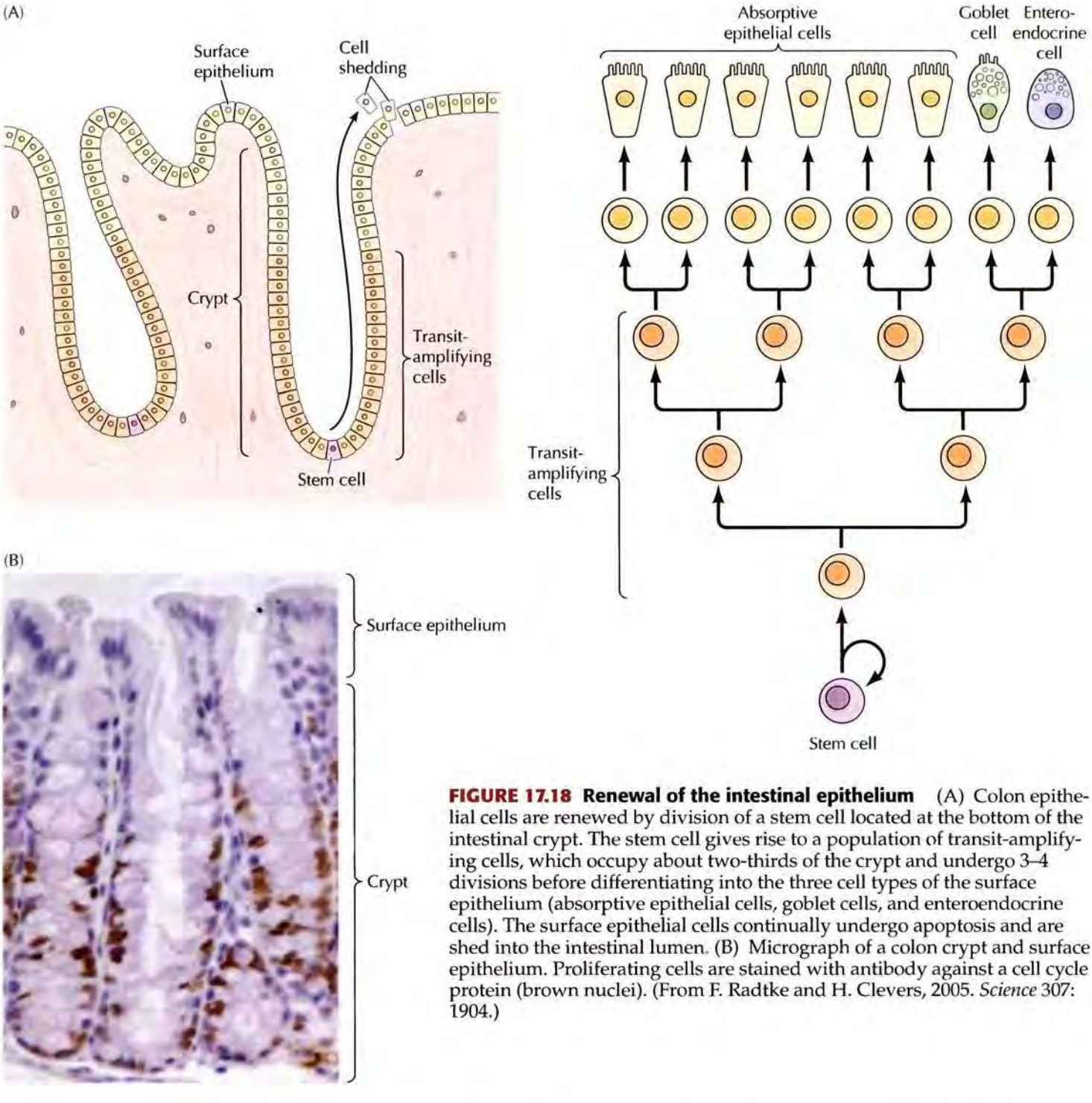


FIGURE 17.18 Renewal of the intestinal epithelium (A) Colon epithelial cells are renewed by division of a stem cell located at the bottom of the intestinal crypt. The stem cell gives rise to a population of transit-amplifying cells, which occupy about two-thirds of the crypt and undergo 3–4 divisions before differentiating into the three cell types of the surface epithelium (absorptive epithelial cells, goblet cells, and enteroendocrine cells). The surface epithelial cells continually undergo apoptosis and are shed into the intestinal lumen. (B) Micrograph of a colon crypt and surface epithelium. Proliferating cells are stained with antibody against a cell cycle protein (brown nuclei). (From F. Radtke and H. Clevers, 2005. *Science* 307: 1904.)

sloughed from the surface. The epidermis is a multi-layered epithelium, which is maintained by epidermal stem cells residing in a single basal layer (Figure 17.19). The epidermal stem cells give rise to transit-amplifying cells, which undergo 3–6 divisions before differentiating and moving outward to the surface of the skin. The stem cells responsible for producing hair reside in a region of the hair follicle called the bulge. The bulge stem cells give rise to transit-amplifying matrix cells, which proliferate and differentiate to form the hair shaft. If the skin is injured, stem cells of the bulge can also give rise to epidermis, demonstrating their activity as multipotent stem cells from which both skin and hair can be derived.

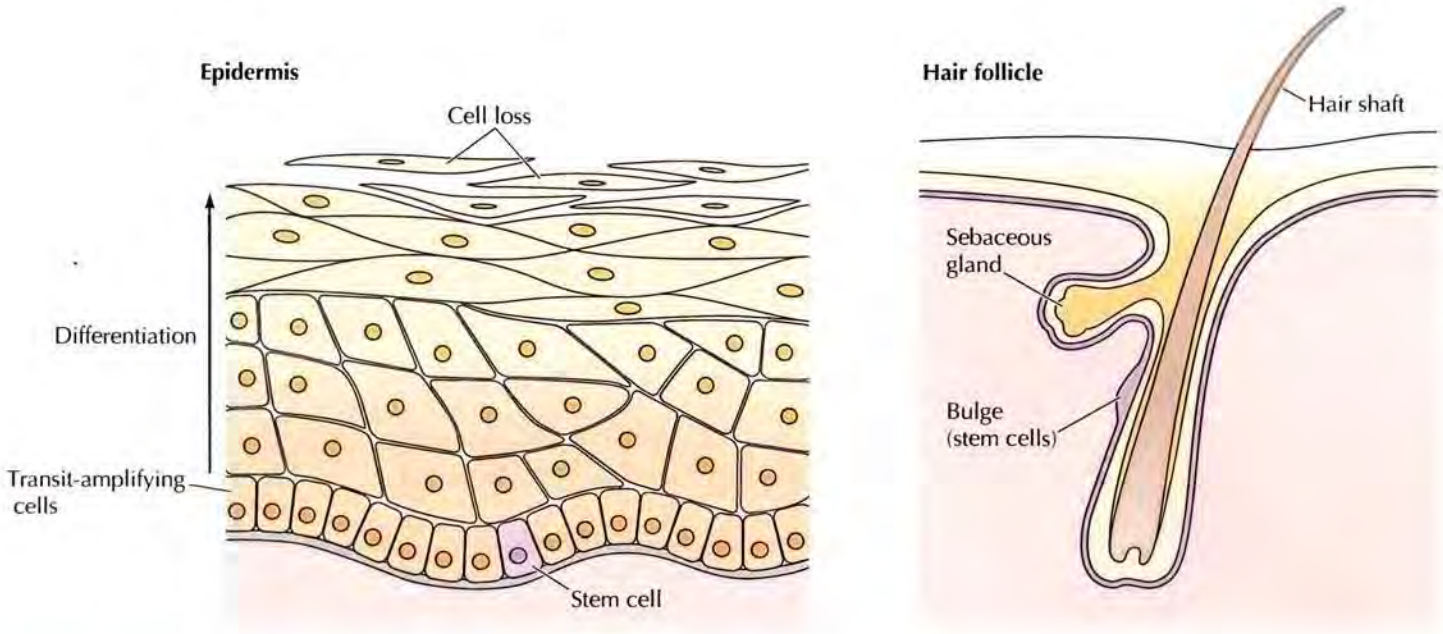


FIGURE 17.19 Stem cells of the skin The epidermis consists of multiple layers of epithelial cells. Cells from the surface are continually lost and replaced by epidermal stem cells in the basal layer. The stem cell gives rise to transit-amplifying cells, which undergo several divisions in the basal layer before differentiating and moving to the surface of the skin. Stem cells of hair follicles reside in a region beneath the sebaceous gland called the bulge. The bulge stem cells not only give rise to cells that form the hair shaft but can also give rise to epidermis if the skin is injured.

Skeletal muscle provides an example of the role of stem cells in the repair of damaged tissue, in contrast to the continual cell renewal just described in the hematopoietic system, intestinal epithelium, and skin. Skeletal muscle is composed of large multinucleated cells (muscle fibers) formed by cell fusion during development (see Figure 12.21). Although skeletal muscle is normally a stable tissue with little cell turnover, it is able to regenerate rapidly in response to injury or exercise. This regeneration is mediated by proliferation of satellite cells, which are the stem cells of adult muscle. Satellite cells are located beneath the basal lamina of muscle fibers (Figure 17.20). They are normally quiescent, arrested in the G_0 phase of the cell cycle, but are activated to proliferate in response to injury or exercise. Once activated, the satellite cells give rise to progeny that undergo several divisions and then differentiate and fuse to form new muscle fibers. The continuing capacity of skeletal muscle to regenerate throughout life is due to self-renewal of the satellite stem cell population.

Stem cells have also been identified in many other adult tissues, including the brain, retina, heart, lung, kidney, liver, and pancreas, and it is possible that most—if not all—tissues contain stem cells with the potential of replacing cells that are lost during the lifetime of the organism. It is also possible that stem cells from one tissue might be able to differentiate into cells giving rise to other tissues—a phenomenon that has been called "developmental plasticity." For example, hematopoietic stem cells derived from bone marrow have been reported to give rise to nonhematopoietic tissues, such as skin or lung. However, the extent to which stem cells from

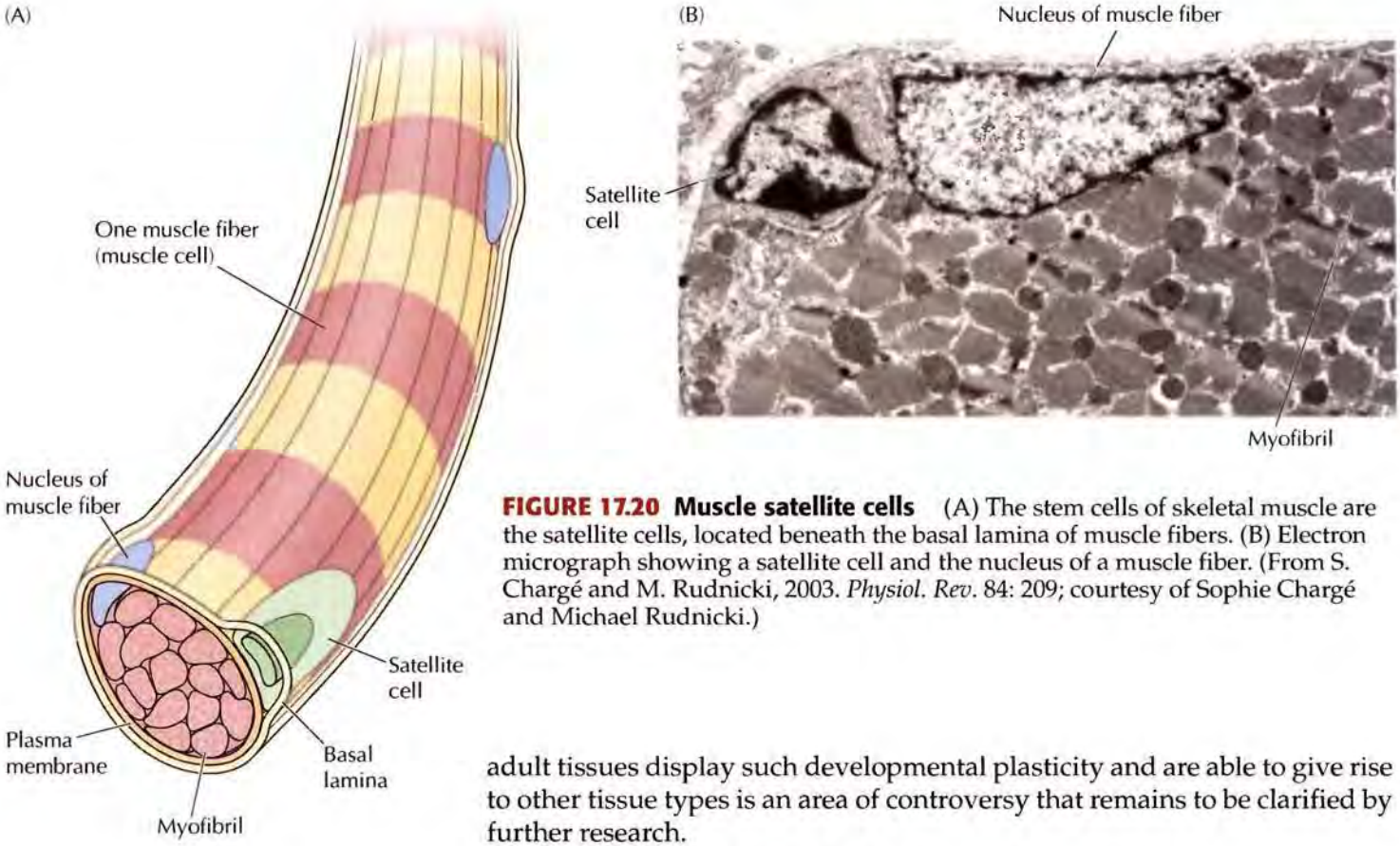


FIGURE 17.20 Muscle satellite cells (A) The stem cells of skeletal muscle are the satellite cells, located beneath the basal lamina of muscle fibers. (B) Electron micrograph showing a satellite cell and the nucleus of a muscle fiber. (From S. Chargé and M. Rudnicki, 2003. *Physiol. Rev.* 84: 209; courtesy of Sophie Chargé and Michael Rudnicki.)

adult tissues display such developmental plasticity and are able to give rise to other tissue types is an area of controversy that remains to be clarified by further research.

Medical Applications of Adult Stem Cells

The ability of adult stem cells to repair damaged tissue clearly suggests their potential utility in clinical medicine. If these stem cells could be isolated and propagated in culture, they could in principal be used to replace damaged tissue and treat a variety of disorders, such as diabetes or neurodegenerative diseases like Parkinson's or Alzheimer's disease. In some cases, the use of stem cells derived from adult tissues may be the optimal approach for such stem cell therapies, although the use of embryonic stem cells (discussed in the next section of this chapter) is likely to provide a more versatile approach to treatment of a wider variety of disorders.

A well-established clinical application of adult stem cells is **bone marrow transplantation**, which plays an important role in the treatment of a variety of cancers. As discussed in Chapter 18, most cancers are treated by chemotherapy with drugs that kill rapidly dividing cells by damaging DNA or inhibiting DNA replication. These drugs do not act selectively against cancer cells but are also toxic to those normal tissues that are dependent on continual renewal by stem cells, such as blood, skin, hair, and the intestinal epithelium. The hematopoietic stem cells are among the most rapidly dividing cells of the body, so the toxic effects of anticancer drugs on these cells frequently limit the effectiveness of chemotherapy in cancer treatment. Bone marrow transplantation provides an approach to bypassing this toxicity, thereby allowing the use of higher drug doses to treat the patient's cancer more effectively. In this procedure, the patient is treated with high doses of chemotherapy that would normally not be tolerated because of toxic effects on the hematopoietic system (Figure 17.21). The potentially lethal damage is repaired, however, by transferring new

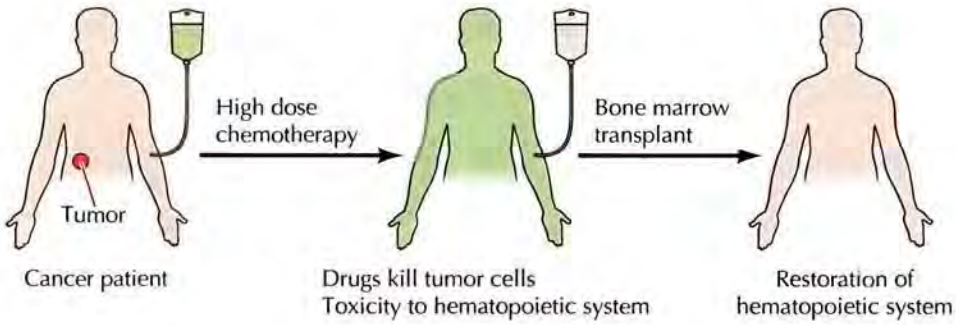


FIGURE 17.21 Bone marrow transplantation A cancer patient is treated with high doses of chemotherapy, which effectively kill tumor cells but normally would not be tolerated because of potentially lethal damage to the hematopoietic system. This damage is then repaired by transplantation of new hematopoietic stem cells (e.g., from bone marrow).

hematopoietic stem cells (obtained either from bone marrow or peripheral blood) to the patient following completion of chemotherapy, so that a normal hematopoietic system is restored. In some cases, the stem cells are obtained from the patient prior to chemotherapy, stored, and then returned to the patient once chemotherapy is completed. However, it is important to ensure that these cells are not contaminated with cancer cells. Alternatively, the stem cells to be transplanted can be obtained from a healthy donor (usually a close relative) whose tissue type closely matches the patient. In addition to their use in cancer treatment, hematopoietic stem cell transfers are used to treat patients with diseases of the hematopoietic system, such as aplastic anemia, hemoglobin disorders, and immune deficiencies.

■ **Stored umbilical cord blood from an unrelated donor can also be used as a source of hematopoietic stem cells for transplantation.**

Epithelial stem cells have also found clinical application in the form of skin grafts that are used to treat patients with burns, wounds, and ulcers. One approach to these procedures is to culture epidermal skin cells to form an epithelial sheet, which can then be transferred to the patient. Because the patient's own skin can be used for this procedure, it eliminates the potential complication of graft rejection by the immune system. The possibilities of using adult stem cells for similar replacement therapies of other diseases, including diabetes and Parkinson's disease, are being actively pursued. However, these clinical applications of adult stem cells are limited by the difficulties in isolating and culturing the appropriate stem cell populations, as well as by the fact that stem cells have not yet been identified in many adult tissues.

Embryonic Stem Cells and Therapeutic Cloning

While adult stem cells are difficult to isolate and culture, it is relatively straightforward to isolate and propagate the stem cells of early embryos (**embryonic stem cells**). These cells can be grown indefinitely as pure stem cell populations while maintaining the ability to give rise to all of the differentiated cell types of adult organisms. Consequently, there has been an enormous interest in embryonic stem cells from the standpoints of both basic science and clinical applications.

KEY EXPERIMENT

Culture of Embryonic Stem Cells

Isolation of a Pluripotent Cell Line from Early Mouse Embryos Cultured in Medium Conditioned by Teratocarcinoma Stem Cells

Gail R. Martin

University of California, San Francisco, CA

Proceedings of the National Academy of Science, USA, 1981, Volume 78, pages 7634–7638

Gail R. Martin

The Context

The cells of early embryos are unique in their ability to proliferate and differentiate into all of the types of cells that make up the tissues and organs of adult animals. In 1970 it was found that early mouse embryos frequently developed into tumors if they were removed from the uterus and transplanted to an abnormal site. These tumors, called teratocarcinomas, contained cells that were capable of forming an array of different tissues as they grew within the animal. In addition, cells from teratocarcinomas (called embryonal carcinoma cells) could be isolated and grown in tissue culture. These cells resembled normal embryo cells and could be induced to differentiate into a variety of cell types in culture. Some embryonal carcinoma cells could also participate in normal development of a mouse if they were injected into early mouse embryos (blastocysts) that were then implanted into a foster mother.

The ability of embryonal carcinoma cells to differentiate into a variety of cell types and to participate in normal mouse development suggested that these tumor-derived cells might be closely related to normal embryonic stem cells. However, the events that occurred during the establishment of teratocarcinomas in mice were unknown. Gail Martin hypothesized that the embryonal carcinoma cells found in teratocarcinomas were essentially normal embryo cells that proliferated abnormally simply because, when they were removed from the uterus and transplanted to an abnormal site, they did not receive the appropriate signals to induce normal differentiation. Based on this hypothesis, she attempted to culture cells from mouse embryos with the goal of isolating normal embryonic stem cell lines. Her experiments, together with similar work by Martin Evans and Matthew Kaufman (Establishment in culture of pluripotential cells from mouse embryos, *Nature*, 1981, 292: 154–156), demonstrated that stem cells could be cultured directly from normal mouse embryos. The isolation of these embryonic stem cell lines paved the way to genetic manipulation and analysis of mouse development, as well as to the possible use of human embryonic stem cells in transplantation therapy.

The Experiments

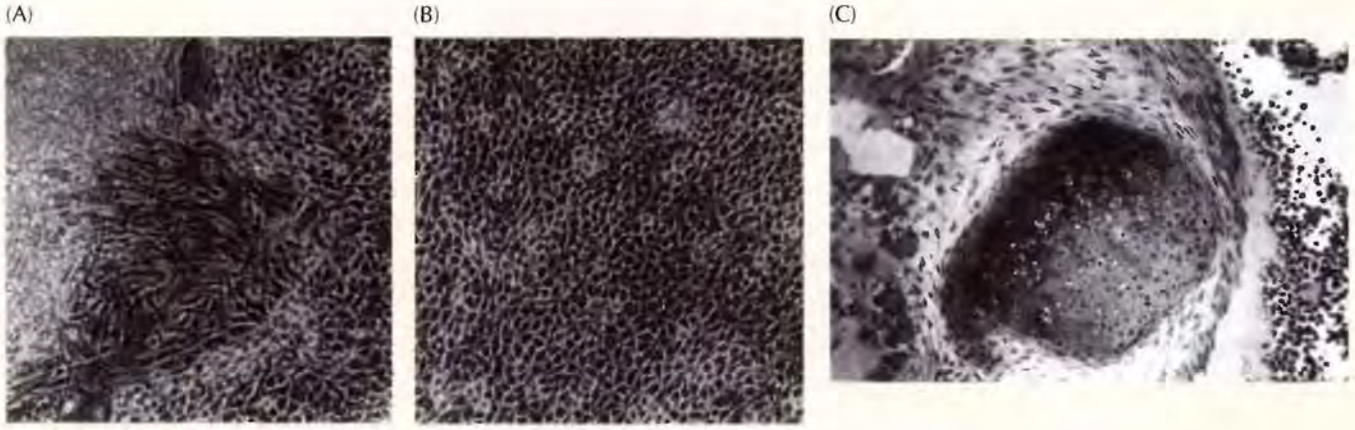
Based on the premise that embryonal carcinoma cells were derived from normal embryonic stem cells, Martin attempted to culture cells from normal

Embryonic Stem Cells

Embryonic stem cells were first cultured from mouse embryos in 1981 (Figure 17.22). They can be propagated indefinitely in culture and, if reintroduced into early embryos, are able to give rise to cells in all tissues of the mouse. Thus they retain the capacity to develop into all of the different types of cells in adult tissues and organs. In addition, they can be induced to differentiate into a variety of different types of cells in culture.

As discussed in Chapter 4, mouse embryonic stem cells have been an important experimental tool in cell biology because they can be used to introduce altered genes into mice (see Figure 4.36). Moreover, they provide an outstanding model system for studying the molecular and cellular events associated with embryonic cell differentiation, so embryonic stem cells have long been of considerable interest to cell and developmental biol-

KEY EXPERIMENT



Embryonic stem cells differentiate in culture to a variety of cell types, including neuron-like cells (A), endodermal cells (B), and cartilage (C).

mouse blastocysts. Starting with cells from approximately 30 embryos, she initially isolated four colonies of growing cells after a week of culture. These cells could be repeatedly passaged into mass cultures, and new cell lines could be reproducibly derived when the experiment was repeated with additional mouse embryos.

The cell lines derived from normal embryos (embryonic stem cells) closely resembled the embryonal carcinoma cells derived from tumors. Most importantly, the embryonic stem cells could be induced to differentiate in culture into a variety of cell types, including endodermal cells, cartilage, and neuron-like cells (see figure). Moreover, if the embryonic stem cells were injected into a mouse, they formed tumors containing multiple differentiated cell types. It thus appeared that embryonic stem cell lines, which retained the ability to differentiate into a wide array of cell types, could be established in culture from normal mouse embryos.

The Impact

The establishment of embryonic stem cell lines has had a major impact on studies of mouse genetics and development as well as opening new possibilities for the treatment of a variety of human diseases. Subsequent experiments demonstrated that embryonic stem cells could participate in normal mouse development following their injection into mouse embryos. Since gene transfer techniques could be used to introduce or mutate genes in cultured embryonic stem cells, these cells have been used to investigate the role of a variety of genes in mouse development. As discussed in Chapter 4, any gene of interest can be inactivated in embryonic stem cells by homologous recombination with a cloned DNA, and the role of that gene in mouse development can then be determined by introducing the altered embryonic stem cells into mouse embryos.

In 1998 two groups of researchers developed the first lines of human embryonic stem cells. Because of the proliferative and differentiative capacity of these cells, they offer the hope of providing new therapies for the treatment of a variety of diseases. Although a number of technical problems and ethical concerns need to be addressed, transplantation therapies based on the use of embryonic stem cells may provide the best hope for eventual treatment of diseases such as Parkinson's, Alzheimer's, diabetes, and spinal cord injuries.

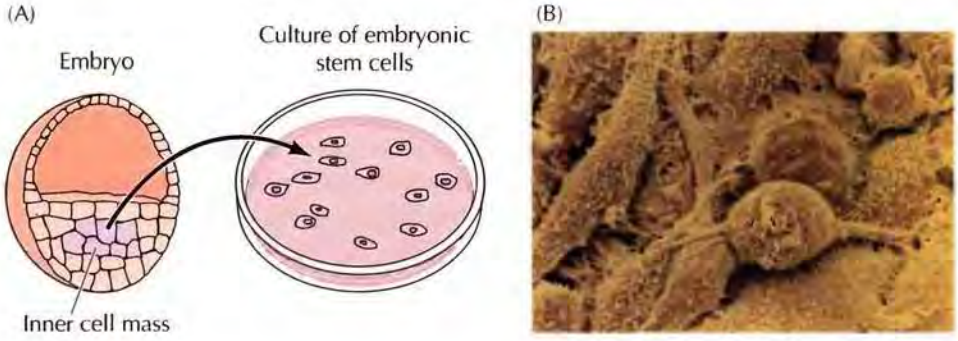


FIGURE 17.22 Culture of mammalian embryonic stem cells
(A) Embryonic stem cells are cultured from the inner cell mass of an early embryo (blastocyst). (B) Scanning electron micrograph of cultured embryonic stem cells. (Yorgos Nikas/Photo Researchers Inc.)

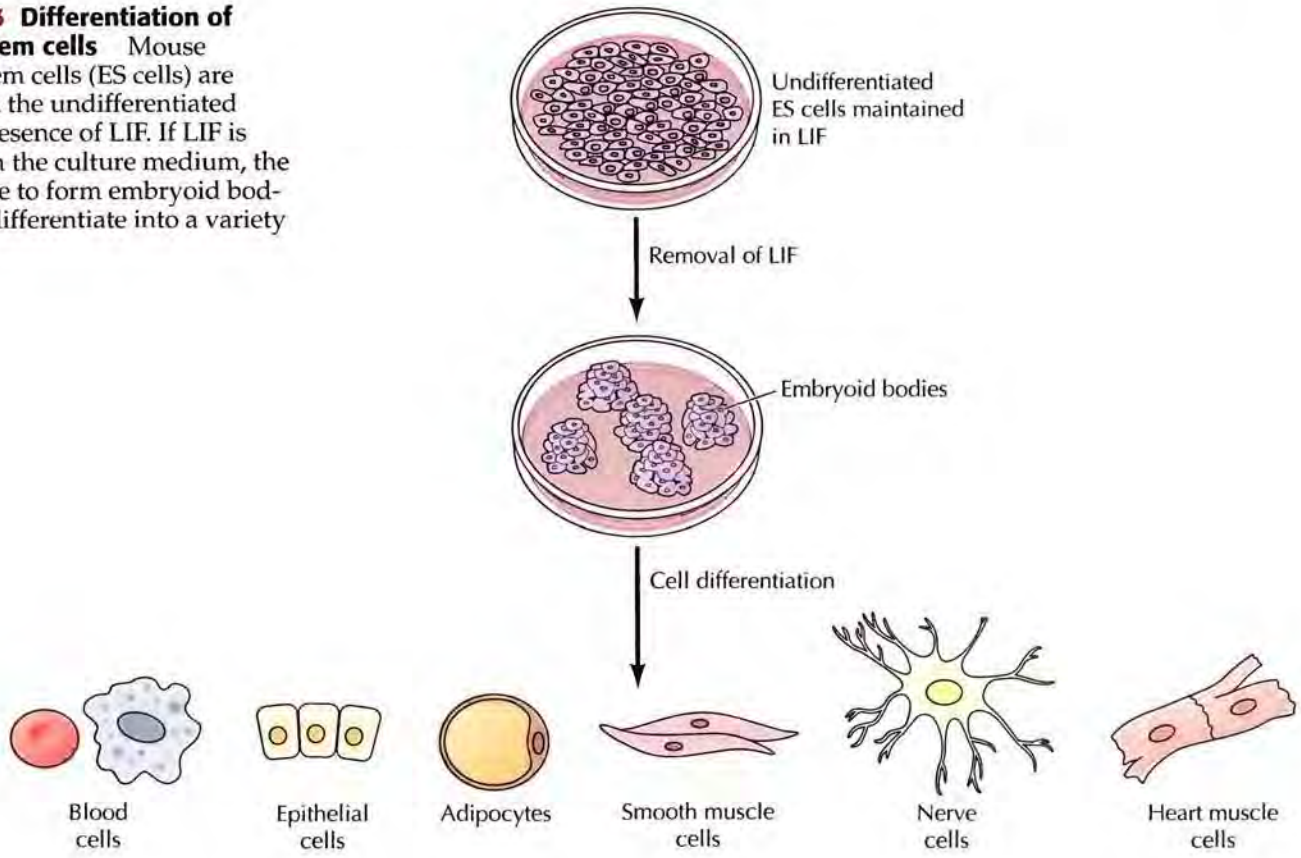


FIGURE 17.23 Differentiation of embryonic stem cells Mouse embryonic stem cells (ES cells) are maintained in the undifferentiated state in the presence of LIF. If LIF is removed from the culture medium, the cells aggregate to form embryoid bodies and then differentiate into a variety of cell types.

ogists. Interest in these cells reached a new peak of intensity, however, in 1998 when two groups of researchers reported the isolation of stem cells from human embryos, raising the possibility of using embryonic stem cells in clinical transplantation therapies.

Mouse embryonic stem cells are grown in the presence of a growth factor called LIF (for *l*eukemia *i*nhibitory *f*actor), which signals through the JAK/STAT pathway (see Figure 15.40) and is required to maintain these cells in their undifferentiated state (Figure 17.23). If LIF is removed from the medium, the cells aggregate into structures that resemble embryos (embryoid bodies) and then differentiate into a wide range of cell types, including neurons, adipocytes, blood cells, epithelial cells, vascular smooth muscle cells, and even beating heart muscle cells. Human embryonic stem cells do not require LIF but are similarly maintained in the undifferentiated state by other growth factors, which are not yet fully characterized.

Importantly, embryonic stem cells can be directed to differentiate along specific pathways by the addition of appropriate growth factors to the culture medium. It may thus be possible to derive populations of specific types of cells, such as heart cells or nerve cells, for transplantation therapy. For example, methods have been developed to direct the differentiation of mouse embryonic stem cells into neurons that have then been used for transplantation therapy in rodent models of myelin disease and Parkinson's disease. A great deal of current research is therefore focused on the development of culture conditions to promote the differentiation of embryonic stem cells along specific pathways, thereby producing populations of differentiated cells that can be used for transplantation therapy of a variety of diseases.

Somatic Cell Nuclear Transfer

The isolation of human embryonic stem cells in 1998 followed the first demonstration that the nucleus of an adult mammalian cell could give rise to a viable cloned animal. In 1997 Ian Wilmut and his colleagues initiated a new era of regenerative medicine with the cloning of Dolly the sheep (Figure 17.24). Dolly arose from the nucleus of a mammary epithelial cell that was transplanted into an unfertilized egg in place of the normal egg nucleus—a process called **somatic cell nuclear transfer**. It is interesting to note that this type of experiment was first carried out in frogs in the 1950s. The fact that it took over forty years before it was successfully performed in mammals attests to the technical difficulty of the procedure. Since the initial success of Wilmut and his colleagues, transfer of nuclei from adult somatic cells into enucleated eggs has been used to create cloned offspring of a variety of

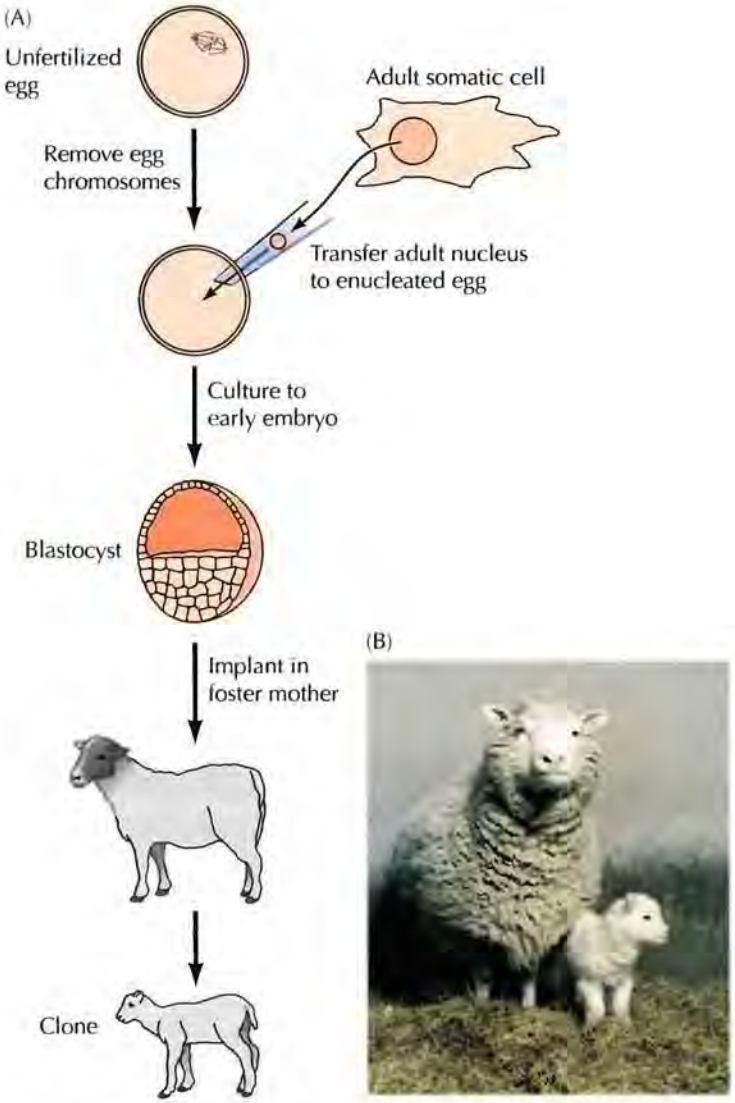


FIGURE 17.24 Cloning by somatic cell nuclear transfer (A) The nucleus of an adult somatic cell is transferred to an unfertilized egg from which the normal egg chromosomes have been removed (an enucleated egg). The egg is then cultured to an early embryo and transferred to a foster mother, who then gives birth to a clone of the donor of the adult nucleus. (B) Dolly (the adult sheep, left) was the first cloned mammal. She is shown with her lamb, Bonnie, who was produced by normal reproduction. (Photograph by Roddy Field; courtesy of T. Wakayama and R. Yanagimachi.)

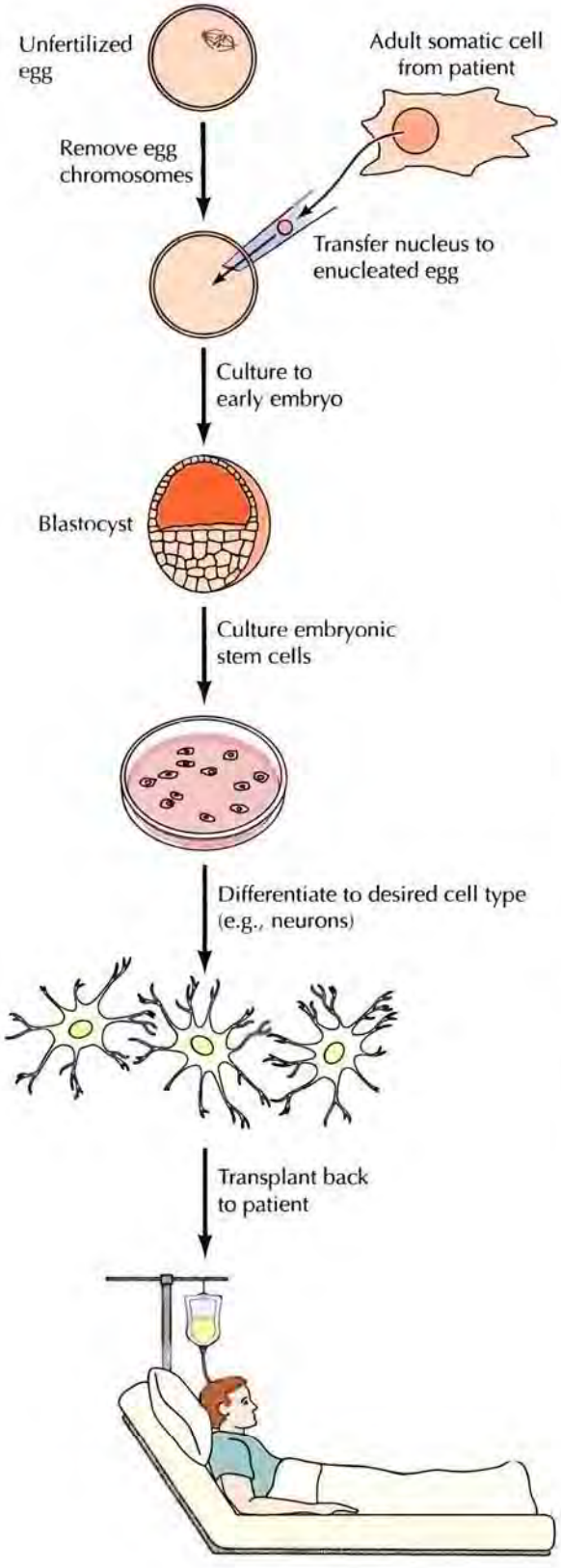


FIGURE 17.25 Therapeutic cloning In therapeutic cloning, the nucleus of a patient's cell would be transferred to an enucleated egg, which would be cultured to an early embryo. Embryonic stem cells would then be derived, differentiated into the desired cell type and transplanted back into the patient. The transplanted cells would be genetically identical to the recipient (who was the donor of the adult nucleus), so complications of immune rejection would be avoided.

mammalian species, including sheep, mice, pigs, cattle, goats, rabbits, and cats. However, cloning by somatic cell nuclear transfer in mammals remains an extremely inefficient procedure, such that only 1–2% of embryos generally give rise to live offspring.

Animal cloning by somatic cell nuclear transfer, together with the properties of embryonic stem cells, opens the possibility of **therapeutic cloning** (Figure 17.25). In therapeutic cloning, a nucleus from an adult human cell would be transferred to an enucleated egg, which would then be used to produce an early embryo in culture. Embryonic stem cells could then be cultured from the cloned embryo and used to generate appropriate types of differentiated cells for transplantation therapy. The major advantage provided by therapeutic cloning is that the embryonic stem cells derived by this procedure would be genetically identical to the recipient of the transplant, who was the donor of the adult somatic cell nucleus. This bypasses the barrier of the immune system in rejecting the transplanted tissue.

The application of therapeutic cloning combined with gene transfer has been demonstrated in mice by correction of an inherited immunodeficiency (Figure 17.26). The genetic defect treated in these experiments was the absence of a functional gene encoding the RAG2 protein, which is required for the gene rearrangements that produce functional antibodies and T cell receptors during the development of lymphocytes (see Figure 6.40). Nuclei from

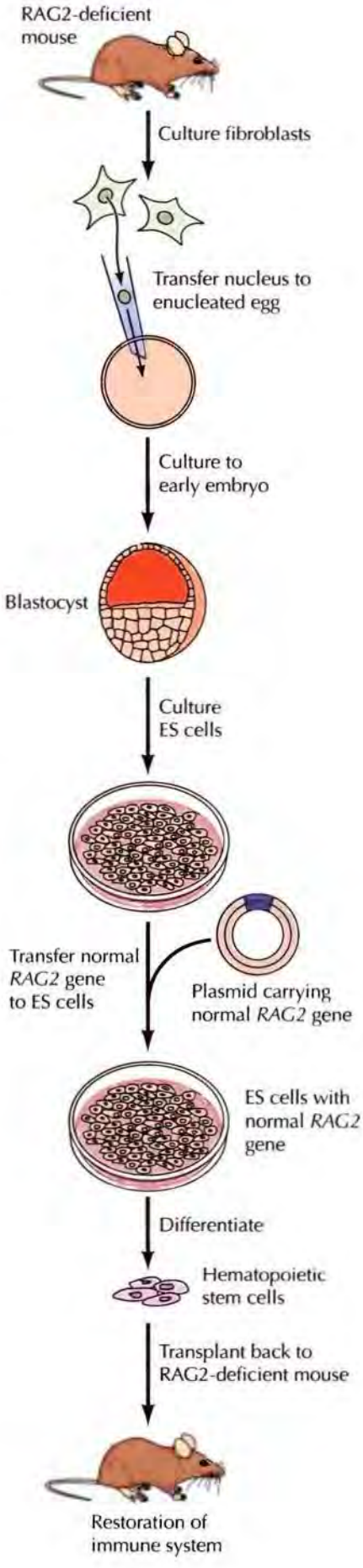


FIGURE 17.26 Correction of an inherited immunodeficiency in mice by therapeutic cloning Fibroblasts from immunodeficient mice lacking RAG2 were used as donors for nuclear transfer and derivation of embryonic stem (ES) cells. The defect was then repaired by introduction of a cloned functional *RAG2* gene into the ES cells. The ES cells were then differentiated to hematopoietic stem cells and transferred back to the RAG2-deficient mice, at least partially correcting the immunodeficiency. (After W. M. Rideout et al., 2002. *Cell* 109: 17.)

fibroblasts of RAG2-deficient mice were transferred to an enucleated egg, which was used as a source of embryonic stem cells. A functional *RAG2* gene was then introduced into the embryonic stem cells in culture, correcting the genetic defect. These corrected embryonic stem cells were then differentiated into hematopoietic stem cells, and transplantation of these cells into RAG2-deficient mice was found to at least partially rescue the immunodeficiency.

The possibility of therapeutic cloning provides the most general approach to treatment of the wide variety of devastating disorders for which stem cell transplantation therapy could be applied, including Parkinson's disease, Alzheimer's disease, diabetes, and spinal cord injuries. Although some success has been achieved in animal models, there remain major obstacles that need to be overcome before therapeutic cloning could be applied to humans. Substantial improvements are needed in the methods used to generate embryos by nuclear transfer, and it will be necessary to develop procedures that reliably differentiate embryonic stem cells into the appropriate cell types prior to transplantation. The development of therapeutic cloning also raises ethical concerns, not only with respect to the possibility of cloning human beings (**reproductive cloning**), but also with respect to the destruction of embryos that serve as the source of embryonic stem cells. While many challenges remain, continuing research in this area holds great promise of opening new approaches to the treatment of a broad array of human diseases.

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

KEY TERMS

programmed cell death, apoptosis

caspase, apoptosome

Bcl-2, IAP

p53, PI 3-kinase, Akt, tumor necrosis factor (TNF)

stem cell

bone marrow transplantation

SUMMARY

PROGRAMMED CELL DEATH

The Events of Apoptosis: Programmed cell death plays a key role in both the maintenance of adult tissues and embryonic development. In contrast to the accidental death of cells from an acute injury, programmed cell death takes place by the active process of apoptosis. Apoptotic cells and cell fragments are then efficiently removed by phagocytosis. Genes responsible for the regulation and execution of apoptosis were initially identified by genetic analysis of *C. elegans*.

Caspases: The Executioners of Apoptosis: The caspases are a family of proteases that are the effectors of apoptosis. Caspases are classified as either initiator or effector caspases, and both function in a cascade leading to cell death. In mammalian cells, the major initiator caspase is activated in a complex called the apoptosome, which also requires cytochrome *c* released from mitochondria.

Central Regulators of Apoptosis: the Bcl-2 Family: Members of the Bcl-2 family are central regulators of caspase activation and apoptosis. Some members of the Bcl-2 family function to inhibit apoptosis (antiapoptotic) whereas others act to promote apoptosis (proapoptotic). Signals that control programmed cell death alter the balance between proapoptotic and antiapoptotic Bcl-2 family members, which regulate one another. In mammalian cells, proapoptotic Bcl-2 family members act at mitochondria, where they promote the release of cytochrome *c*, leading to caspase activation. Caspases are also regulated directly by inhibitory IAP proteins.

Signaling Pathways that Regulate Apoptosis: A variety of signaling pathways regulate apoptosis by controlling the expression or activity of proapoptotic members of the Bcl-2 family. These pathways include DNA damage-induced activation of the tumor suppressor p53, growth factor-stimulated activation of PI 3-kinase/Akt signaling, and activation of death receptors by polypeptides that induce programmed cell death.

STEM CELLS AND THE MAINTENANCE OF ADULT TISSUES

Proliferation of Differentiated Cells: Most cells in adult animals are arrested in the G_0 stage of the cell cycle. A few types of differentiated cells, including skin fibroblasts, endothelial cells, smooth muscle cells, and liver cells are able to resume proliferation as required to replace cells that have been lost because of injury or cell death.

Stem Cells: Most differentiated cells do not themselves proliferate but can be replaced via the proliferation of stem cells. Stem cells divide to produce one daughter cell that remains a stem cell and another that divides and differentiates. Stem cells have been identified in a wide variety of adult tissues, including the hematopoietic system, skin, intestine, skeletal muscle, brain, and heart.

Medical Applications of Adult Stem Cells: The ability of stem cells to repair damaged tissue suggests their potential use in clinical medicine. Adult stem cells are used to repair damage to the hematopoietic system in bone marrow transplantation, and epidermal stem cells can be used for skin grafts. However, clinical applications of adult stem cells are limited by difficulties in isolating and culturing these cells.

SUMMARY

EMBRYONIC STEM CELLS AND THERAPEUTIC CLONING

Embryonic Stem Cells: Embryonic stem cells are cultured from early embryos. They can be readily grown in the undifferentiated state in culture while retaining the ability to differentiate into a wide variety of cell types, so they may offer considerable advantages over adult stem cells for many clinical applications.

Somatic Cell Nuclear Transfer: Mammals have been cloned by somatic cell nuclear transfer in which the nucleus of an adult somatic cell is transplanted into an enucleated egg. This opens the possibility of therapeutic cloning in which embryonic stem cells would be derived from a cloned embryo and used for transplantation therapy of the donor of the adult nucleus. Although many obstacles need to be overcome, the possibility of therapeutic cloning holds great promise for the development of new treatments for a variety of devastating diseases.

KEY TERMS

embryonic stem cell

somatic cell nuclear transfer, therapeutic cloning, reproductive cloning

Questions

1. Why is cell death via apoptosis more advantageous to multicellular organisms than cell death via acute injury?

2. What molecular mechanisms regulate caspase activity?

3. You have expressed mutants of nuclear lamins in human fibroblasts. The Asp residue in the caspase cleavage site has been mutated to Glu in these lamins. How would these mutant lamins affect the progression of apoptosis?

4. How do Bcl-2 family proteins regulate apoptosis in mammalian cells?

5. How does p53 activation in response to DNA damage affect cell cycle progression and cell survival?

6. You have constructed a Bad mutant in which the Akt phosphorylation site has been mutated such that Akt no longer phosphorylates it. How would expression of this mutant affect cell survival?

7. How would expression of siRNA targeted against 14-3-3 proteins affect apoptosis?

8. You are considering treatment of a leukemic patient with TNF. Upon further analysis you determine that the leukemic cells have an inactivating mutation of caspase-8. Will treatment with TNF be an effective therapy for this patient?

9. How would siRNA against Ced-3 affect the development of *C. elegans*?

10. You have isolated a polypeptide from a toxic plant, which localizes to mitochondria after endocytosis by mammalian cells. The polypeptide aggregates and forms large channels in the mitochondrial outer membrane, releasing proteins from the intermembrane space into the cytoplasm. How will treatment with this polypeptide affect mammalian cells in culture?

11. Many adult tissues contain terminally differentiated cells that are incapable of proliferation. However, these tissues can regenerate following damage. What gives these tissues their regenerative capabilities?

12. What is the critical property of stem cells?

13. What are the potential advantages of embryonic stem cells as compared to adult stem cells for therapeutic applications?

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CHAPTER 18

Cancer

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CANCER IS A PARTICULARLY APPROPRIATE TOPIC for the concluding chapter of this book because it results from a breakdown of the regulatory mechanisms that govern normal cell behavior. As discussed in preceding chapters, the proliferation, differentiation, and survival of individual cells in multicellular organisms are carefully regulated to meet the needs of the organism as a whole. This regulation is lost in cancer cells, which grow and divide in an uncontrolled manner, ultimately spreading throughout the body and interfering with the function of normal tissues and organs.

Because cancer results from defects in fundamental cell regulatory mechanisms, it is a disease that ultimately has to be understood at the molecular and cellular levels. Indeed, understanding cancer has been an objective of molecular and cellular biologists for many years. Importantly, studies of cancer cells have also illuminated the mechanisms that regulate normal cell behavior. In fact, many of the proteins that play key roles in cell signaling, regulation of the cell cycle, and control of programmed cell death were first identified because abnormalities in their activities led to the uncontrolled proliferation of cancer cells. The study of cancer has thus contributed significantly to our understanding of normal cell regulation, as well as vice versa.

The Development and Causes of Cancer

The fundamental abnormality resulting in the development of cancer is the continual unregulated proliferation of cancer cells. Rather than responding appropriately to the signals that control normal cell behavior, cancer cells grow and divide in an uncontrolled manner, invading normal tissues and organs and eventually spreading throughout the body. The generalized loss of growth control exhibited by cancer cells is the net result of accumulated abnormalities in multiple cell regulatory systems and is reflected in several aspects of cell behavior that distinguish cancer cells from their normal counterparts.

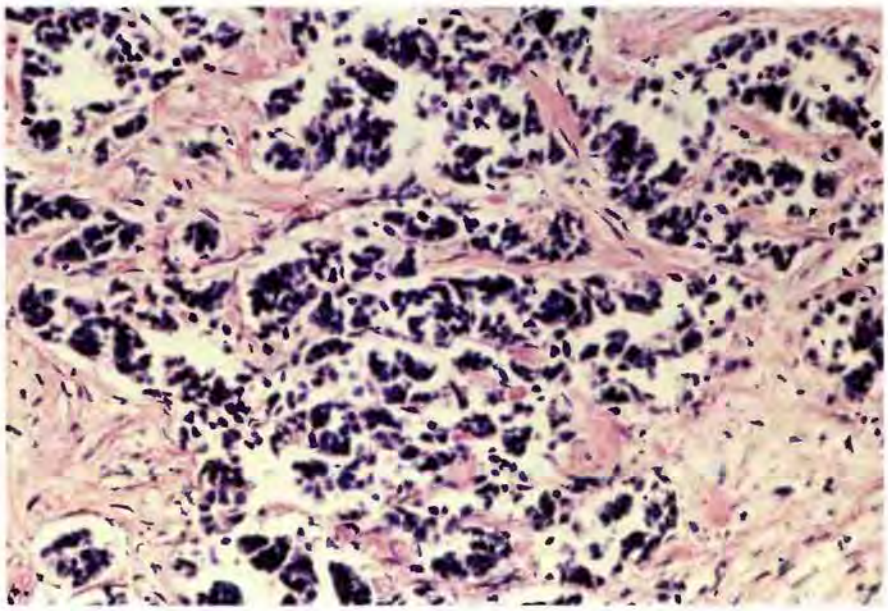


FIGURE 18.1 A cancer of the pancreas Light micrograph of a section through a pancreas showing pancreatic cancer. The cancer cells have dark purple nuclei and are invading the normal tissue (pink). (Astrid and Hanns-Frieder Michler/SPL/Photo Researchers, Inc.)

■ **The earliest recorded description of cancer is found in an Egyptian papyrus dating back to around 3000 BC. Hippocrates is believed to be the originator of the word cancer. It is thought that the shape of tumors reminded him of crabs and he used the words *carcinos*, *carcinoma* and *cancer*, which refer to crabs in Greek, to describe the tumors.**

Types of Cancer

Cancer can result from abnormal proliferation of any of the different kinds of cells in the body, so there are more than a hundred distinct types of cancer, which can vary substantially in their behavior and response to treatment. The most important issue in cancer pathology is the distinction between benign and malignant tumors (Figure 18.1). A **tumor** is any abnormal proliferation of cells, which may be either benign or malignant. A **benign tumor**, such as a common skin wart, remains confined to its original location, neither invading surrounding normal tissue nor spreading to distant body sites. A **malignant tumor**, however, is capable of both invading surrounding normal tissue and spreading throughout the body via the circulatory or lymphatic systems (**metastasis**). Only malignant tumors are properly referred to as cancers, and it is their ability to invade and metasta-

TABLE 18.1 Most Frequent Cancers in the United States

Cancer site	Cases per year	Deaths per year
Prostate	232,100 (16.9%)	30,400 (5.3%)
Breast	212,900 (15.5%)	40,900 (7.2%)
Lung	172,600 (12.6%)	163,500 (28.7%)
Colon/rectum	145,300 (10.6%)	56,300 (9.9%)
Lymphomas	63,700 (4.6%)	20,600 (3.6%)
Bladder	63,200 (4.6%)	13,200 (2.3%)
Skin (melanoma)	59,600 (4.3%)	7800 (1.4%)
Uterus	51,200 (3.7%)	11,000 (1.9%)
Kidney	36,200 (2.6%)	12,700 (2.2%)
Leukemias	34,800 (2.5%)	22,600 (4.0%)
Pancreas	32,200 (2.3%)	31,800 (5.6%)
Subtotal	1,103,800 (80.4%)	410,800 (72.1%)
All sites	1,373,000 (100%)	570,000 (100%)

Source: American Cancer Society, *Cancer Facts and Figures—2005.*

size that makes cancer so dangerous. Whereas benign tumors can usually be removed surgically, the spread of malignant tumors to distant body sites frequently makes them resistant to such localized treatment.

Both benign and malignant tumors are classified according to the type of cell from which they arise. Most cancers fall into one of three main groups: carcinomas, sarcomas, and leukemias or lymphomas. **Carcinomas**, which include approximately 90% of human cancers, are malignancies of epithelial cells. **Sarcomas**, which are rare in humans, are solid tumors of connective tissues, such as muscle, bone, cartilage, and fibrous tissue. **Leukemias** and **lymphomas**, which account for approximately 7% of human malignancies, arise from the blood-forming cells and from cells of the immune system, respectively. Tumors are further classified according to tissue of origin (e.g., lung or breast carcinomas) and the type of cell involved. For example, fibrosarcomas arise from fibroblasts, and erythroid leukemias from precursors of erythrocytes (red blood cells).

Although there are many kinds of cancer, only a few occur frequently (Table 18.1). More than a million cases of cancer are diagnosed annually in the United States, and more than 500,000 Americans die of cancer each year. Cancers of 11 different body sites account for more than 80% of this total cancer incidence. The four most common cancers accounting for more than half of all cancer cases are those of the prostate, breast, lung, and colon/rectum. Lung cancer, by far the most lethal, is responsible for nearly 30% of all cancer deaths.

Although cancer has been with us throughout the history of mankind, it has become a leading cause of death only in the last century. Prior to 1900, most deaths were due to infectious diseases, such as pneumonia and tuberculosis, and life expectancy was less than 50 years. Cancer was a rare disease that accounted for only a small percentage of deaths.

The Development of Cancer

One of the fundamental features of cancer is tumor clonality—the development of tumors from single cells that begin to proliferate abnormally. The single-cell origin of many tumors has been demonstrated by analysis of X chromosome inactivation (Figure 18.2). As discussed in Chapter 7, one member of the X chromosome pair is inactivated by being converted to heterochromatin in female cells. X inactivation occurs randomly during embryonic development, so one X chromosome is inactivated in some cells, while the other X chromosome is inactivated in other cells. Thus, if a female is heterozygous for an X chromosome gene, different alleles will be expressed in different cells. Normal tissues are composed of mixtures of cells with different inactive X chromosomes, so expression of both alleles is detected in normal tissues of heterozygous females. In contrast, tumor tissues generally express only one allele of a heterozygous X chromosome gene. The implication is that all of the cells constituting such a tumor were derived from a single cell of origin in which the pattern of X inactivation was fixed before the tumor began to develop.

The clonal origin of tumors does not, however, imply that the original progenitor cell that gives rise to a tumor has initially acquired all of the characteristics of a cancer cell. On the contrary, the development of cancer is a multistep process in which cells gradually become malignant through a progressive series of alterations. One indication of the multistep development of cancer is that most cancers develop late in life. The incidence of

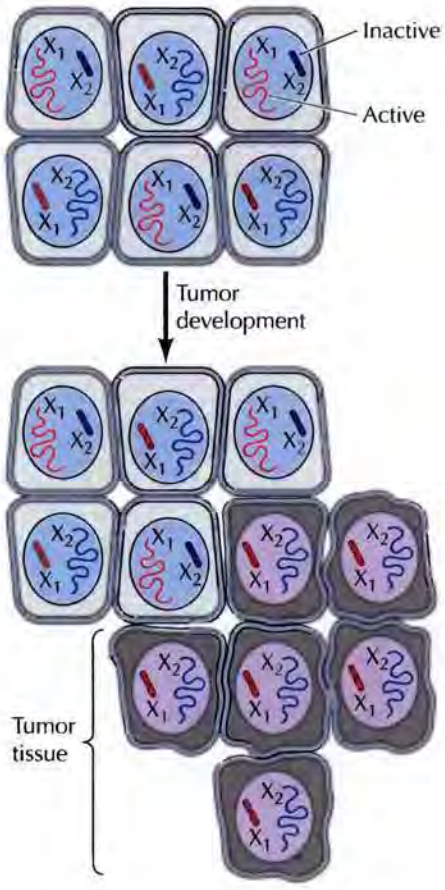


FIGURE 18.2 Tumor clonality Normal tissue is a mosaic of cells in which different X chromosomes (X_1 and X_2) have been inactivated. Tumors develop from a single initially altered cell, so each tumor cell displays the same pattern of X inactivation (X_1 inactive, X_2 active).

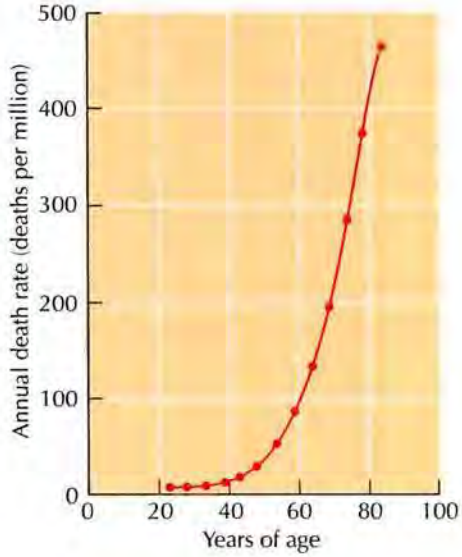
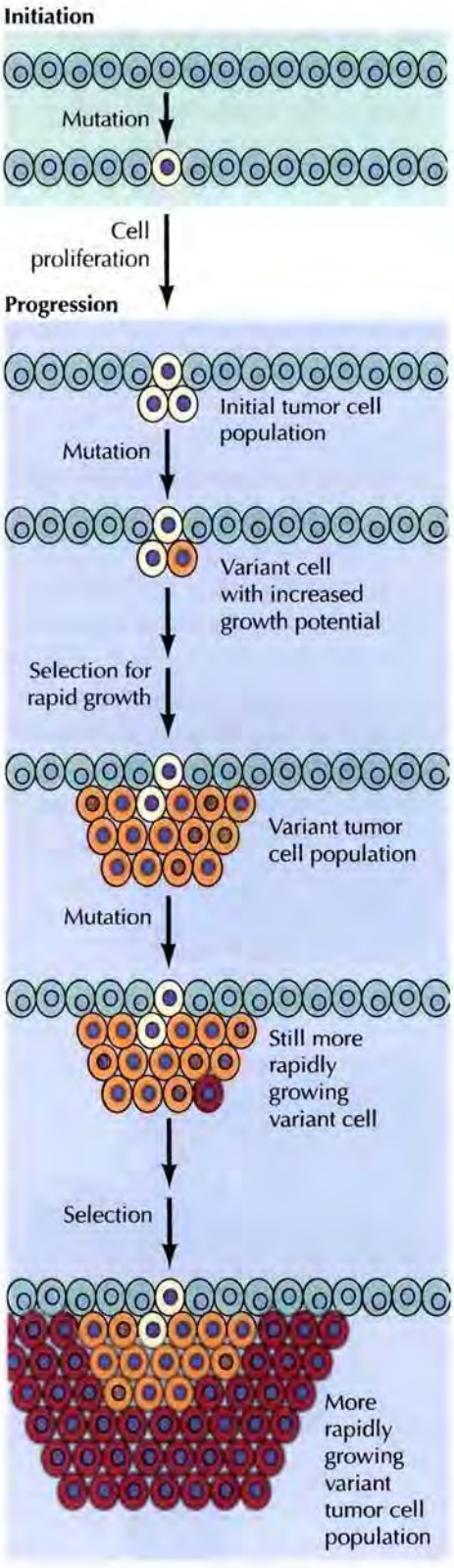


FIGURE 18.3 Increased rate of colon cancer with age Annual death rates from colon cancer in the United States. (Data from J. Cairns, 1978. *Cancer: Science and Society*, New York: W. H. Freeman.)



colon cancer, for example, increases more than tenfold between the ages of 30 and 50, and another tenfold between the ages of 50 and 70 (Figure 18.3). Such a dramatic increase of cancer incidence with age suggests that most cancers develop as a consequence of multiple abnormalities, which accumulate over periods of many years.

At the cellular level, the development of cancer is viewed as a multistep process involving mutation and selection for cells with progressively increasing capacity for proliferation, survival, invasion, and metastasis (Figure 18.4). The first step in the process, **tumor initiation**, is thought to be the result of a genetic alteration leading to abnormal proliferation of a single cell. Cell proliferation then leads to the outgrowth of a population of clonally derived tumor cells. **Tumor progression** continues as additional mutations occur within cells of the tumor population. Some of these mutations confer a selective advantage to the cell, such as more rapid growth, and the descendants of a cell bearing such a mutation will consequently become dominant within the tumor population. The process is called clonal selection, since a new clone of tumor cells has evolved on the basis of its increased growth rate or other properties (such as survival, invasion, or metastasis) that confer a selective advantage. Clonal selection continues throughout tumor development, so tumors continuously become more rapid-growing and increasingly malignant.

Studies of colon carcinomas have provided a clear example of tumor progression during the development of a common human malignancy (Figure 18.5). The earliest stage in tumor development is increased proliferation of colon epithelial cells. One of the cells within this proliferative cell population is then thought to give rise to a small benign neoplasm (an **adenoma** or **polyp**). Further rounds of clonal selection lead to the growth of adenomas of increasing size and proliferative potential. Malignant carcinomas then

FIGURE 18.4 Stages of tumor development The development of cancer initiates when a single mutated cell begins to proliferate abnormally. Additional mutations followed by selection for more rapidly growing cells within the population then result in progression of the tumor to increasingly rapid growth and malignancy.

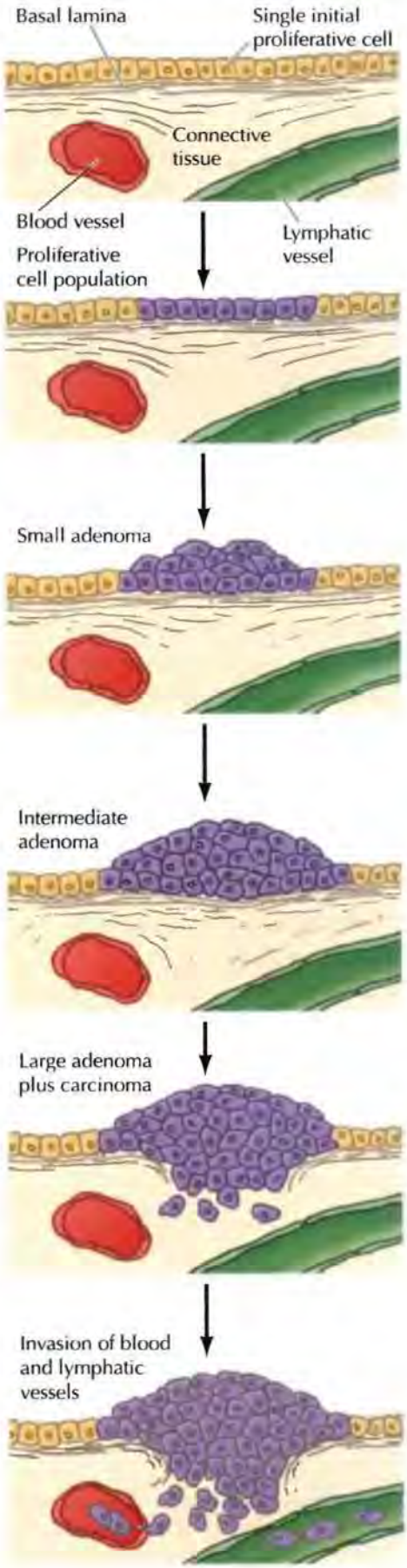


FIGURE 18.5 Development of colon carcinomas A single initially altered cell gives rise to a proliferative cell population, which progresses first to benign adenomas of increasing size and then to malignant carcinoma. The cancer cells invade the underlying connective tissue and penetrate blood and lymphatic vessels, thereby spreading throughout the body.

arise from the benign adenomas, indicated by invasion of the tumor cells through the basal lamina into underlying connective tissue. The cancer cells then continue to proliferate and spread through the connective tissues of the colon wall. Eventually the cancer cells penetrate the wall of the colon and invade other abdominal organs, such as the bladder or small intestine. In addition, the cancer cells invade blood and lymphatic vessels, allowing them to metastasize throughout the body.

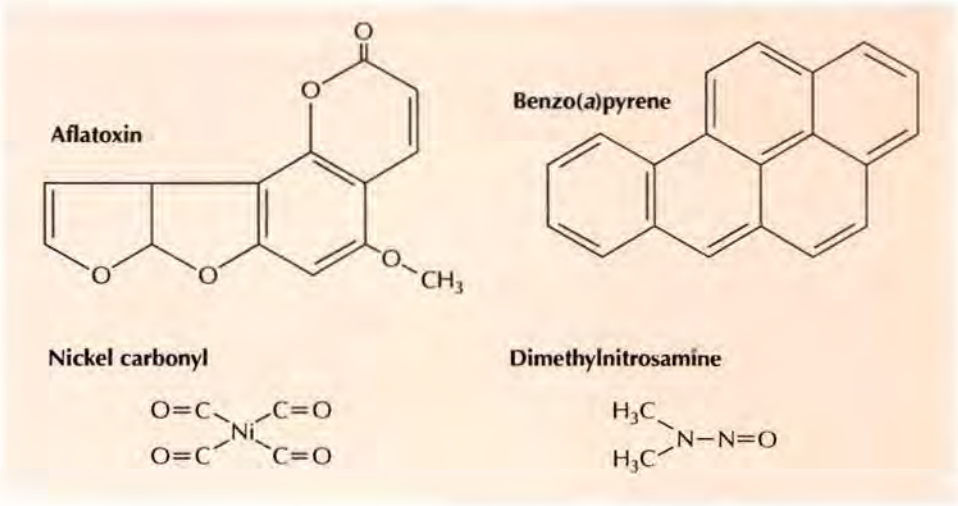
Causes of Cancer

Substances that cause cancer, called **carcinogens**, have been identified both by studies in experimental animals and by epidemiological analysis of cancer frequencies in human populations (e.g., the high incidence of lung cancer among cigarette smokers). Since the development of malignancy is a complex multistep process, many factors may affect the likelihood that cancer will develop, and it is overly simplistic to speak of single causes of most cancers. Nonetheless, many agents, including radiation, chemicals, and viruses, have been found to induce cancer in both experimental animals and humans.

Radiation and many chemical carcinogens (Figure 18.6) act by damaging DNA and inducing mutations. Some of the carcinogens that contribute to human cancers include solar ultraviolet radiation (the major cause of skin cancer), carcinogenic chemicals in tobacco smoke, and aflatoxin (a potent liver carcinogen produced by some molds that contaminate improperly stored supplies of peanuts and other grains). The carcinogens in tobacco smoke (including benzo(*a*)pyrene, dimethylnitrosamine, and nickel compounds) are the major identified causes of human cancer. Smoking is the undisputed cause of 80 to 90% of lung cancers, as well as being implicated in cancers of the oral cavity, pharynx, larynx, esophagus, and other sites. In total, it is estimated that smoking is responsible for nearly one-third of all cancer deaths—an impressive toll for a single carcinogenic agent.

Other carcinogens contribute to cancer development by stimulating cell proliferation, rather than by inducing mutations. Such compounds are referred to as **tumor promoters**, since the increased cell division they induce facilitates the outgrowth of a proliferative cell population during early stages of tumor development. Hormones, particularly estrogens, are important as tumor promoters in the development of some human cancers. The proliferation of cells of the uterine endometrium, for example, is stimulated by estrogen, and exposure to excess estrogen significantly increases the likelihood that a woman will develop endometrial cancer. The risk of endometrial cancer is therefore substantially increased by long-term postmenopausal estrogen replacement therapy with high doses of estrogen alone. Fortunately, this risk is minimized by administration of progesterone to counteract the stimulatory effect of estrogen on endometrial cell proliferation. However, oral contraceptives and hormone replacement therapy with combinations of estrogen and progesterone still lead to an increased risk of breast cancer.

FIGURE 18.6 Structure of representative chemical carcinogens



18.1 WEBSITE ANIMATION

Metastasis of a Cancer

A cancer begins when one altered cell gives rise to a proliferative cell population, forming a primary tumor from which cells can break away and form secondary tumors elsewhere in the body.

■ **Barry Marshall and Robin Warren were the first to establish a link between *Heliobacter pylori* and stomach ulcers. To prove that *H. pylori* is the causative agent for stomach ulcers, Marshall deliberately infected himself with a pure culture of the microbe and studied the course of the infection. Thankfully, the illness resolved itself spontaneously.**

Some viruses also cause cancer both in experimental animals and in humans, and infection with the bacterium *Heliobacter pylori* causes stomach cancer. The common human cancers caused by viruses include liver cancer and cervical carcinoma, which together account for 10 to 20% of worldwide cancer incidence. These viruses are important not only as causes of human cancer but, as discussed later in this chapter, studies of tumor viruses have played a key role in elucidating the molecular events responsible for the development of cancers induced by both viral and nonviral carcinogens.

Properties of Cancer Cells

The uncontrolled growth of cancer cells results from accumulated abnormalities affecting many of the cell regulatory mechanisms that have been discussed in preceding chapters. This relationship is reflected in several aspects of cell behavior that distinguish cancer cells from their normal counterparts. Cancer cells typically display abnormalities in the mechanisms that regulate normal cell proliferation, differentiation, and survival. Taken together, these characteristic properties of cancer cells provide a description of malignancy at the cellular level.

The uncontrolled proliferation of cancer cells *in vivo* is mimicked by their behavior in cell culture. A primary distinction between cancer cells and normal cells in culture is that normal cells display **density-dependent inhibition** of cell proliferation (Figure 18.7). Normal cells proliferate until they reach a finite cell density, which is determined in part by the availability of growth factors added to the culture medium (usually in the form of serum). They then cease proliferating and become quiescent, arrested in the G_0 stage of the cell cycle (see Figure 16.6). The proliferation of most cancer cells, however, is not sensitive to density-dependent inhibition. Rather than responding to the signals that cause normal cells to cease proliferation and enter G_0, tumor cells generally continue growing to high cell densities in culture, mimicking their uncontrolled proliferation *in vivo*.

A related difference between normal cells and cancer cells is that many cancer cells have reduced requirements for extracellular growth factors. As discussed in Chapter 15, the proliferation of most cells is controlled, at least in part, by polypeptide growth factors. For some cell types, particularly fibroblasts, the availability of serum growth factors is the principal determinant of their proliferative capacity in culture. The growth factor require-

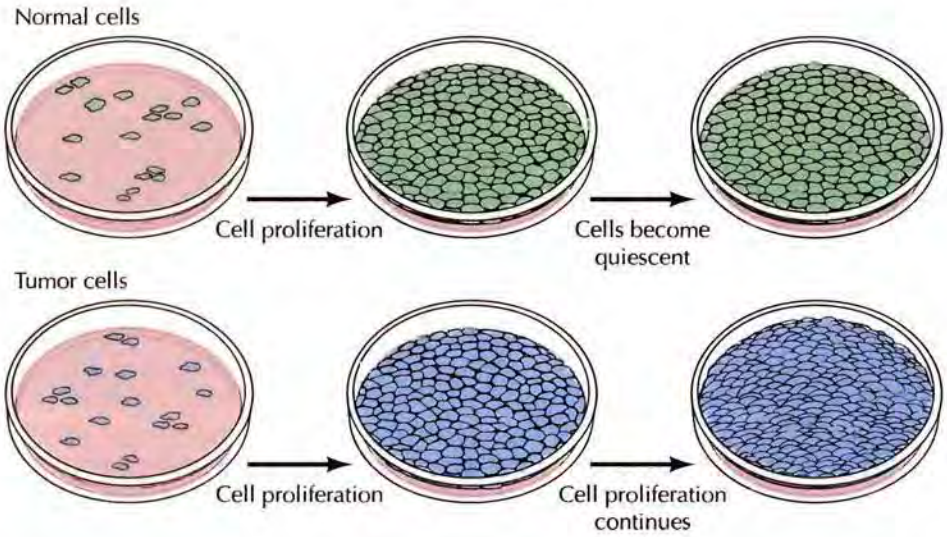


FIGURE 18.7 Density-dependent inhibition Normal cells proliferate in culture until they reach a finite cell density at which point they become quiescent. Tumor cells, however, continue to proliferate independent of cell density.

18.2 WEBSITE ANIMATION

Density-Dependent Inhibition In culture, normal cells proliferate until they reach a certain cell density, but tumor cells continue to proliferate independent of cell density.

ments of these cells are closely related to the phenomenon of density-dependent inhibition, since the density at which normal fibroblasts become quiescent is proportional to the concentration of serum growth factors in the culture medium.

The growth factor requirements of many tumor cells are reduced compared to their normal counterparts, contributing to the unregulated proliferation of tumor cells both *in vitro* and *in vivo*. In some cases, cancer cells produce growth factors that stimulate their own proliferation (Figure 18.8). Such abnormal production of a growth factor by a responsive cell leads to continuous autostimulation of cell division (**autocrine growth stimulation**), and the cancer cells are therefore less dependent on growth factors from other, physiologically normal sources. In other cases, the reduced growth factor dependence of cancer cells results from abnormalities in intracellular signaling systems, such as unregulated activity of growth factor receptors or other proteins (e.g., Ras proteins or protein kinases) that were discussed in Chapter 15 as elements of signal transduction pathways leading to cell proliferation.

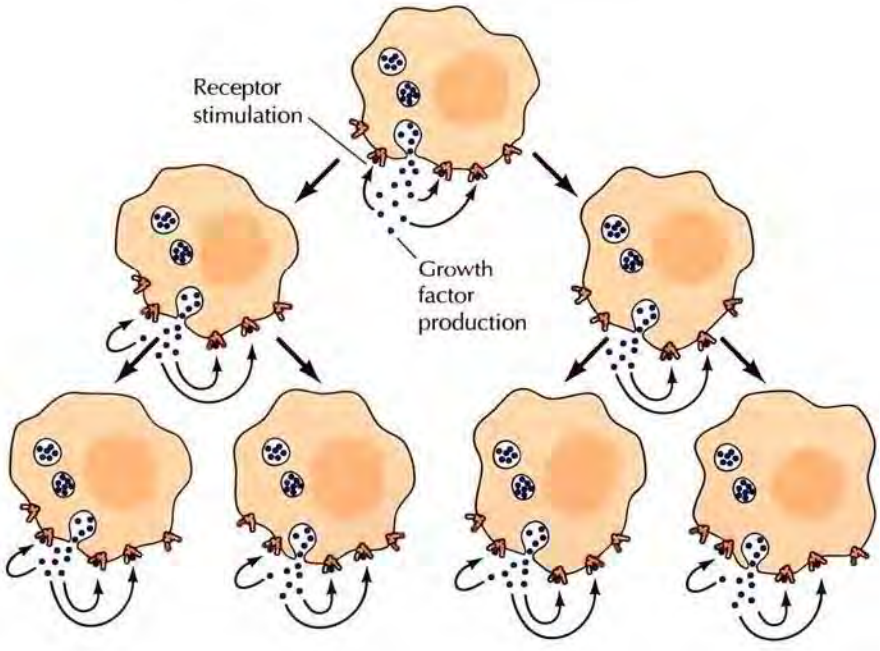


FIGURE 18.8 Autocrine growth stimulation A cell produces a growth factor to which it also responds, resulting in continuous stimulation of cell proliferation.

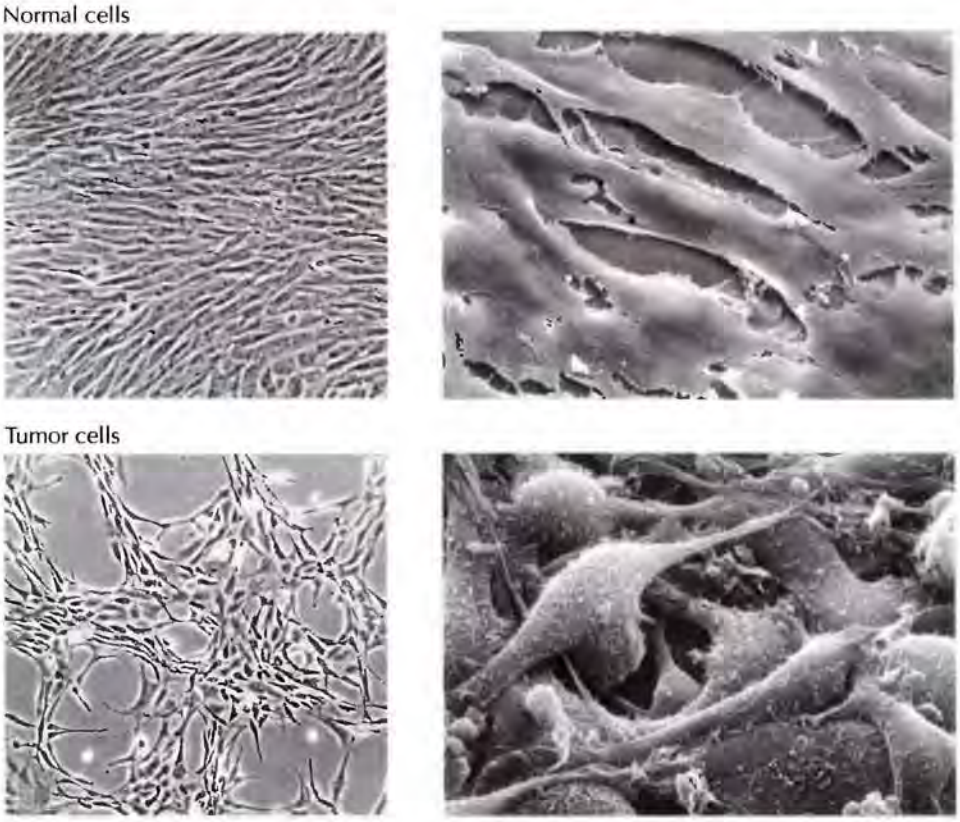


FIGURE 18.9 Contact inhibition Light micrographs (left) and scanning electron micrographs (right) of normal fibroblasts and tumor cells. The migration of normal fibroblasts is inhibited by cell contact, so they form an orderly side-by-side array on the surface of a culture dish. Tumor cells, however, are not inhibited by cell contact, so they migrate over one another and grow in a disordered, multilayered pattern. (Courtesy of Lan Bo Chen, Dana-Farber Cancer Institute.)

Cancer cells are also less stringently regulated than normal cells by cell-cell and cell-matrix interactions. Most cancer cells are less adhesive than normal cells, often as a result of reduced expression of cell surface adhesion molecules. For example, loss of E-cadherin, the principal adhesion molecule of epithelial cells, is important in the development of carcinomas (epithelial cancers). As a result of reduced expression of cell adhesion molecules, cancer cells are comparatively unrestrained by interactions with other cells and tissue components, contributing to the ability of malignant cells to invade and metastasize. The reduced adhesiveness of cancer cells also results in morphological and cytoskeletal alterations: Many tumor cells are rounder than normal, in part because they are less firmly attached to either the extracellular matrix or neighboring cells.

A striking difference in the cell-cell interactions displayed by normal cells and those of cancer cells is illustrated by the phenomenon of **contact inhibition** (Figure 18.9). Normal fibroblasts migrate across the surface of a culture dish until they make contact with a neighboring cell. Further cell migration is then inhibited, and normal cells adhere to each other, forming an orderly array of cells on the culture dish surface. Tumor cells, in contrast, continue moving after contact with their neighbors, migrating over adjacent cells, and growing in disordered, multilayered patterns. Not only the movement but also the proliferation of many normal cells is inhibited by cell-cell contact, and cancer cells are characteristically insensitive to such contact inhibition of growth.

Two additional properties of cancer cells affect their interactions with other tissue components, thereby playing important roles in invasion and metastasis. First, malignant cells generally secrete proteases that digest extracellular matrix components, allowing the cancer cells to invade adja-

cent normal tissues. Secretion of collagenase, for example, appears to be an important determinant of the ability of carcinomas to digest and penetrate through basal laminae to invade underlying connective tissue (see Figure 18.5). Second, cancer cells secrete growth factors that promote the formation of new blood vessels (**angiogenesis**). Angiogenesis is needed to support the growth of a tumor beyond the size of about a million cells, at which point new blood vessels are required to supply oxygen and nutrients to the proliferating tumor cells. Such blood vessels are formed in response to growth factors, secreted by the tumor cells, that stimulate proliferation of endothelial cells in the walls of capillaries in surrounding tissue, resulting in the outgrowth of new capillaries into the tumor (see Figure 17.14). The formation of such new blood vessels is important not only in supporting tumor growth, but also in metastasis. The actively growing new capillaries formed in response to angiogenic stimulation are easily penetrated by the tumor cells, providing a ready opportunity for cancer cells to enter the circulatory system and begin the metastatic process.

Another general characteristic of most cancer cells is that they fail to differentiate normally. Such defective differentiation is closely coupled to abnormal proliferation since, as discussed in Chapter 17, most fully differentiated cells cease cell division. Rather than carrying out their normal differentiation program, cancer cells are usually blocked at an early stage of differentiation, consistent with their continued active proliferation.

The leukemias provide a particularly good example of the relationship between defective differentiation and malignancy. All of the different types of blood cells are derived from a common hematopoietic stem cell in the bone marrow (see Figure 17.17). Descendants of these cells then become committed to specific differentiation pathways. Some cells, for example, differentiate to form erythrocytes whereas others differentiate to form lymphocytes, granulocytes, or macrophages. Progenitor cells of each of these types undergo several rounds of division as they differentiate, but once they become fully differentiated, cell division ceases. Leukemic cells, in contrast, fail to undergo terminal differentiation (Figure 18.10). Instead, they become arrested at early stages

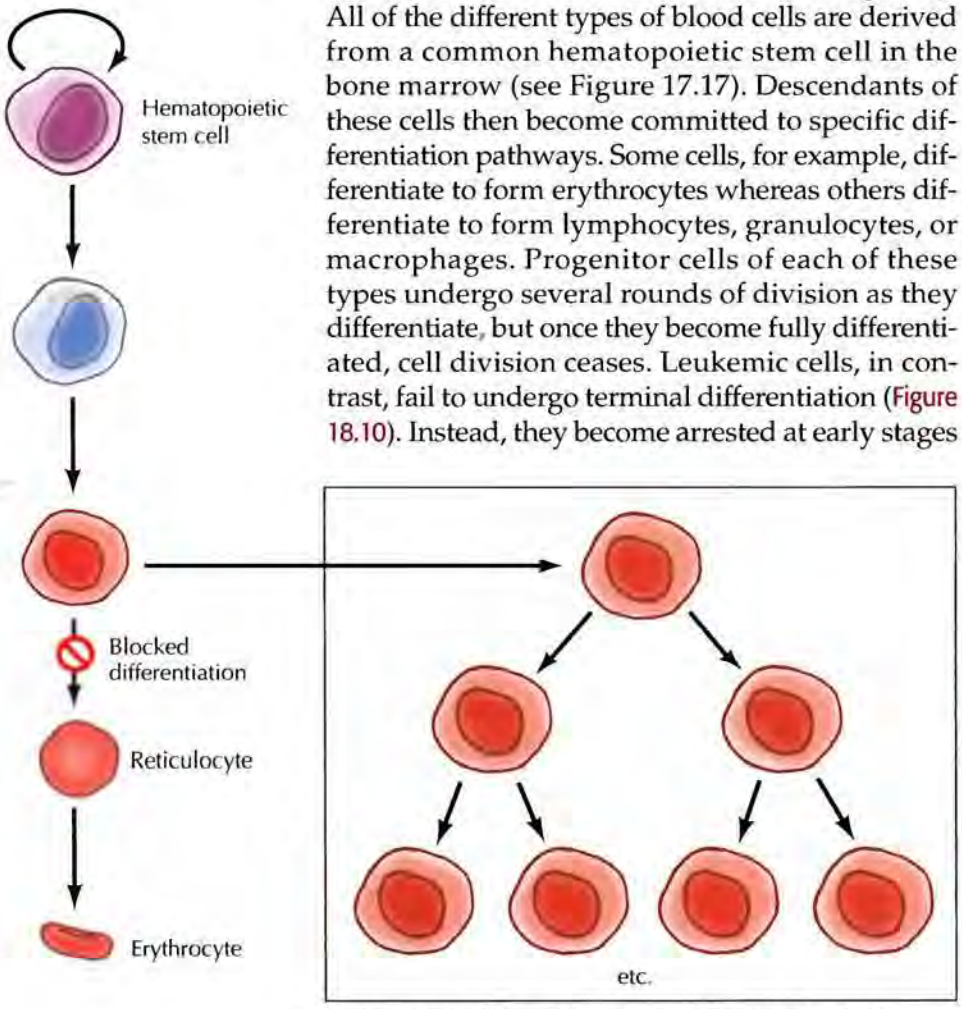


FIGURE 18.10 Defective differentiation and leukemia Different types of blood cells develop from the hematopoietic stem cell in the bone marrow. The precursors of differentiated cells undergo several rounds of cell division as they mature, but cell division ceases at the terminal stages of differentiation. The differentiation of leukemic cells is blocked at early stages of maturation, consistent with their continued proliferation.

of maturation at which they retain their capacity for proliferation and continue to reproduce.

As discussed in Chapter 17, **programmed cell death** or **apoptosis** is an integral part of the differentiation program of many cell types, including blood cells. Many cancer cells fail to undergo apoptosis and therefore exhibit increased life spans compared to their normal counterparts. This failure of cancer cells to undergo programmed cell death contributes substantially to tumor development. For example, the survival of many normal cells is dependent on signals from either growth factors or from the extracellular matrix that prevent apoptosis. In contrast, tumor cells are often able to survive in the absence of growth factors required by their normal counterparts. Such a failure of tumor cells to undergo apoptosis when deprived of normal environmental signals may be important not only in primary tumor development but also in the survival and growth of metastatic cells in abnormal tissue sites. Normal cells also undergo apoptosis following DNA damage, while many cancer cells fail to do so. In this case, the failure to undergo apoptosis contributes to the resistance of cancer cells to irradiation and many chemotherapeutic drugs, which act by damaging DNA. In addition to evading apoptosis, cancer cells generally acquire the capacity for unlimited replication as a result of expression of telomerase, which is required to maintain the ends of eukaryotic chromosomes (see Figure 6.16). Abnormal cell survival, as well as cell proliferation, thus plays a major role in the unrelenting growth of cancer cells in an animal.

Transformation of Cells in Culture

The study of tumor induction by radiation, chemicals, or viruses requires experimental systems in which the effects of a carcinogenic agent can be reproducibly observed and quantitated. Although the activity of carcinogens can be assayed in intact animals, such experiments are difficult to quantitate and control. The development of *in vitro* assays to detect the conversion of normal cells to tumor cells in culture, a process called **cell transformation**, therefore represented a major advance in cancer research. Such assays are designed to detect transformed cells, which display the *in vitro* growth properties of tumor cells, following exposure of a culture of normal cells to a carcinogenic agent. Their application has allowed experimental analysis of cell transformation to reach a level of sophistication that could not have been attained by studies in whole animals alone.

The first and most widely used assay of cell transformation is the focus assay, which was developed by Howard Temin and Harry Rubin in 1958. The focus assay is based on the ability to recognize a group of transformed cells as a morphologically distinct "focus" against a background of normal cells on the surface of a culture dish (Figure 18.11). The focus assay takes advantage of three properties of transformed cells: altered morphology, loss of contact inhibition, and loss of density-dependent inhibition of growth. The result is the formation of a colony of morphologically altered transformed cells that overgrow the background of normal cells in the culture. Such foci of transformed cells can usually be detected and quantified within a week or two after exposure to a carcinogenic agent. In general, cells transformed *in vitro* are able to form tumors following inoculation into suscepti-

FIGURE 18.11 The focus assay A focus of chicken embryo fibroblasts induced by Rous sarcoma virus. (From H. M. Temin and H. Rubin, 1958. *Virology* 6: 669.)

ble animals, supporting *in vitro* transformation as a valid indicator of the formation of cancer cells.

Tumor Viruses

Members of several families of animal viruses, called **tumor viruses**, are capable of directly causing cancer in either experimental animals or humans (Table 18.2). The viruses that cause human cancer include hepatitis B and C viruses (liver cancer), papillomaviruses (cervical and other anogenital cancers), Epstein-Barr virus (Burkitt's lymphoma and nasopharyngeal carcinoma), Kaposi's sarcoma-associated herpesvirus (Kaposi's sarcoma), and human T-cell lymphotropic virus (adult T-cell leukemia). In addition, HIV is indirectly responsible for the cancers that develop in AIDS patients as a result of immunodeficiency.

As already noted, tumor viruses not only are important as causes of human disease but have also played a critical role in cancer research by serving as models for cellular and molecular studies of cell transformation. The small size of their genomes has made tumor viruses readily amenable to molecular analysis, leading to the identification of viral genes responsible for cancer induction and paving the way to our current understanding of cancer at the molecular level.

Hepatitis B and C Viruses

The **hepatitis B** and **C viruses** are the principal causes of liver cancer, which is the third leading cause of cancer deaths worldwide. Both hepatitis B and C viruses specifically infect liver cells and can lead to long term chronic infections of the liver. Such chronic infections are associated with a high risk of liver cancer, which eventually develops in 10–20% of people chronically infected with hepatitis B and in about 5% of people chronically infected with hepatitis C virus.

The molecular mechanisms by which the hepatitis viruses cause liver cancer are not yet understood. Hepatitis B virus is a DNA virus with a genome of only 3 kb. Cell transformation by hepatitis B virus may be mediated by a viral gene (called the *X* gene) that affects expression of several cellular genes that drive cell proliferation. In addition, the development of cancers induced by hepatitis B virus is driven by the continual proliferation of liver cells that results from chronic tissue damage and inflammation.

TABLE 18.2 Tumor Viruses

Virus family	Human tumors	Genome size (kb)
DNA Genomes		
Hepatitis B viruses	Liver cancer	3
SV40 and polyomavirus	None	5
Papillomaviruses	Cervical carcinoma	8
Adenoviruses	None	35
Herpesviruses	Burkitt's lymphoma, nasopharyngeal carcinoma, Kaposi's sarcoma	100–200
RNA Genomes		
Hepatitis C virus	Liver cancer	10
Retroviruses	Adult T-cell leukemia	9–10

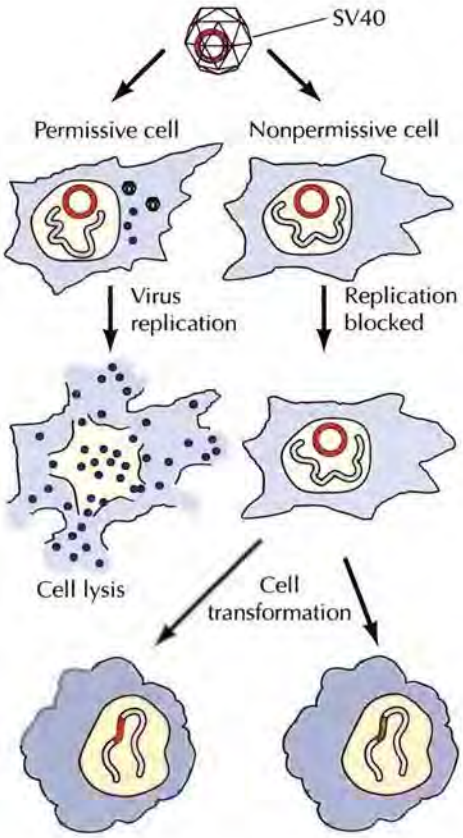


FIGURE 18.12 SV40 replication and transformation Infection of a permissive cell results in virus replication, cell lysis, and release of progeny virus particles. In a nonpermissive cell, virus replication is blocked, allowing some cells to become permanently transformed.

Hepatitis C virus is an RNA virus with a genome of approximately 10 kb. Cell proliferation in response to chronic inflammation is a major contributor to cancer development in people chronically infected with hepatitis C virus, although it is also possible that some hepatitis C virus proteins directly stimulate the proliferation of infected liver cells.

SV40 and Polyomavirus

Although neither **simian virus 40 (SV40)** nor **polyomavirus** are associated with human cancer, they have been critically important as models for understanding the molecular basis of cell transformation. The utility of these viruses in cancer research has stemmed from the availability of good cell culture assays for both virus replication and transformation, as well as from the small size of their genomes (approximately 5 kb).

SV40 and polyomavirus do not induce tumors or transform the cells of their natural host species—monkeys and mice, respectively. In cells of their natural hosts (permissive cells), infection leads to virus replication, cell lysis, and release of progeny virus particles (Figure 18.12). Since a permissive cell is killed as a consequence of virus replication, it cannot become transformed. The transforming potential of these viruses is revealed, however, by infection of nonpermissive cells in which virus replication is blocked. In this case, the viral genome sometimes integrates into cellular DNA, and expression of specific viral genes results in transformation of the infected cell.

The SV40 and polyomavirus genes that lead to cell transformation are the same viral genes that function in early stages of lytic infection. The genomes of SV40 and polyomavirus are divided into early and late regions. The early region is expressed immediately after infection and is required for synthesis of viral DNA. The late region is not expressed until after viral DNA replication has begun and includes genes encoding structural components of the virus particle. The early region of SV40 encodes two proteins (called small and large T antigens) of about 17 kd and 94 kd, respectively (Figure 18.13). Their mRNAs are generated by alternative splicing of a single early-region primary transcript. Polyomavirus likewise encodes small and large T antigens, as well as a third early-region protein of about 55 kd designated middle T. Transfection of cells with cDNAs for individual early-region proteins has shown that SV40 large T is sufficient to induce transformation, whereas middle T is primarily responsible for transformation by polyomavirus.

During lytic infection these early-region proteins fulfill multiple functions required for virus replication. SV40 T antigen, for example, binds to the SV40 origin and initiates viral DNA replication (see Chapter 6). In addition, the early-region proteins of SV40 and polyomavirus stimulate host cell gene expression and DNA synthesis. Since virus replication is dependent on host cell enzymes (e.g., DNA polymerase), such stimulation of the host cell is a critical event in the viral life cycle. Most cells in an animal are nonproliferating and therefore must be stimulated to divide in order to induce the enzymes needed for viral DNA replication. This stimulation of cell pro-

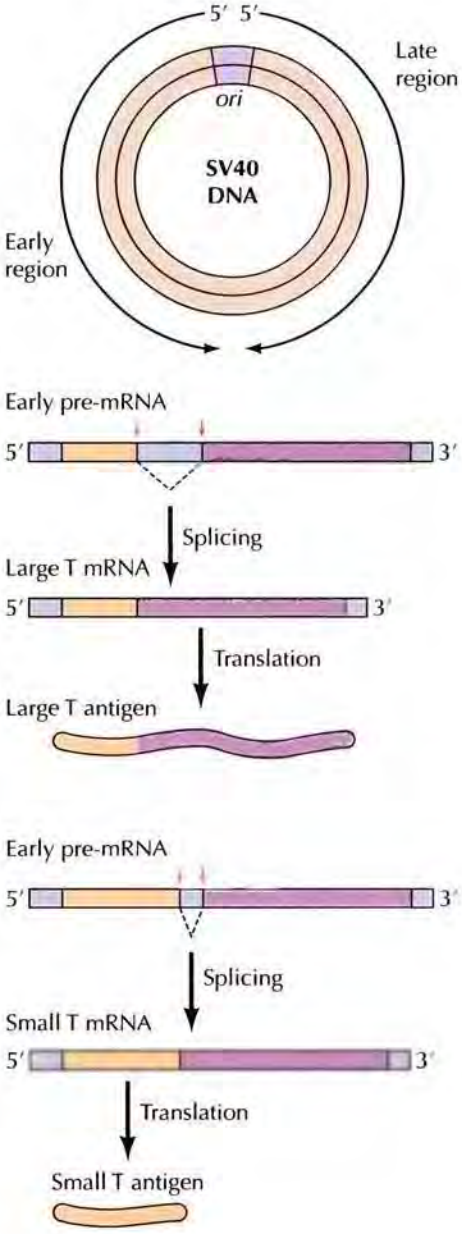


FIGURE 18.13 The SV40 genome The genome is divided into early and late regions. Large and small T antigens are produced by alternative splicing of early-region pre-mRNA.

liferation by the early gene products can lead to transformation if the viral DNA becomes stably integrated and expressed in a nonpermissive cell.

As discussed later in this chapter, both SV40 and polyomavirus early-region proteins induce transformation by interacting with host proteins that regulate cell proliferation. For example, SV40 T antigen binds to and inactivates the host cell tumor suppressor proteins Rb and p53, which are key regulators of cell proliferation and survival.

Papillomaviruses

The **papillomaviruses** are small DNA viruses (genomes of approximately 8 kb) that induce both benign and malignant tumors in humans and a variety of other animal species. Approximately 60 different types of human papillomaviruses, which infect epithelial cells of several tissues, have been identified. Some of these viruses cause only benign tumors (such as warts), whereas others are causative agents of malignant carcinomas, particularly cervical and other anogenital cancers. The mortality from cervical cancer is relatively low in the United States, in large part as a result of early detection and curative treatment made possible by the Pap smear. In other parts of the world, however, cervical cancer remains common; it is responsible for 5 to 10% of worldwide cancer incidence.

Cell transformation by human papillomaviruses results from expression of two early-region genes, *E6* and *E7* (Figure 18.14). Like SV40 T antigen, the E6 and E7 proteins induce transformation by interacting with host cell proteins that control cell proliferation and survival, including Rb and p53. In particular, E7 binds to Rb, and E6 stimulates the degradation of p53 by ubiquitin-mediated proteolysis.

Adenoviruses

The **adenoviruses** are a large family of DNA viruses with genomes of about 35 kb. In contrast to the papillomaviruses, the adenoviruses are not associated with naturally occurring cancers in either humans or other animals. However, they are widely studied and important models in experimental cancer biology.

■ **The Pap smear was developed by George Papanicolaou in the 1930s. Uterine cells are collected and smeared on a microscope slide. Since cancer cells have altered morphology they can be readily detected upon microscopic examination.**

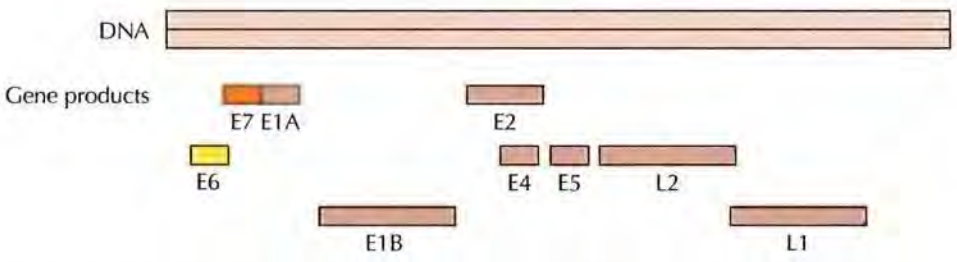


FIGURE 18.14 The genome of a human papillomavirus Gene products are designated E (early) or L (late). Transformation results from the action of E6 and E7.

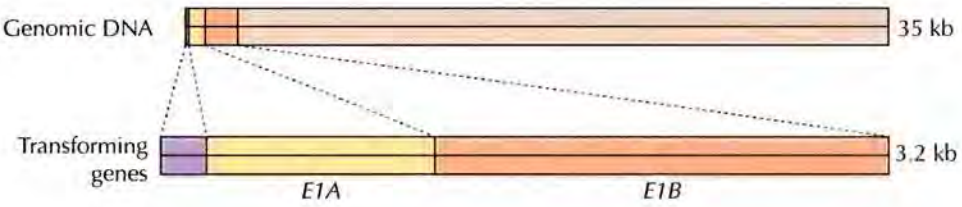


FIGURE 18.15 The adenovirus genome Two early-region genes, *E1A* and *E1B*, are responsible for induction of transformation.

Like SV40 and polyomaviruses, the adenoviruses are lytic in cells of their natural host species but can induce transformation in nonpermissive hosts. Transformation by the adenoviruses results from expression of two early-region genes, *E1A* and *E1B*, which are required for virus replication in permissive cells (Figure 18.15). These transforming proteins inactivate the Rb and p53 tumor suppressor proteins, with E1A binding to Rb and E1B binding to p53. It thus appears that SV40, papillomaviruses, and adenoviruses all induce transformation by related pathways in which interfering with the activities of Rb and p53 plays a central role.

Herpesviruses

The **herpesviruses** are among the most complex animal viruses, with genomes of 100 to 200 kb. Several herpesviruses induce tumors in animal species, including frogs, chickens, and monkeys. In addition, two members of the herpesvirus family, **Kaposi's sarcoma-associated herpesvirus** and **Epstein-Barr virus**, cause human cancers. Kaposi's sarcoma-associated herpesvirus plays a critical role in the development of Kaposi's sarcomas, and Epstein-Barr virus has been implicated in several human malignancies, including Burkitt's lymphoma in some regions of Africa, B-cell lymphomas in AIDS patients and other immunosuppressed individuals, and nasopharyngeal carcinoma in China.

■ **Denis Burkitt was a surgeon in Uganda when he came across several cases of patients with tumors of the head and neck and identified the disease as a previously uncharacterized lymphoma. Michael Epstein and Yvonne Barr subsequently isolated a virus from Burkitt's lymphoma tissue, providing one of the first links between viruses and human cancer.**

In addition to its association with these human malignancies, Epstein-Barr virus is able to transform human B lymphocytes in culture. Partly because of the complexity of the genome, however, the molecular biology of Epstein-Barr virus replication and transformation remains to be fully understood. The main Epstein-Barr virus transforming protein (LMP1) mimics a cell surface receptor on B lymphocytes and functions by activating signaling pathways that stimulate cell proliferation and inhibit apoptosis. Several other viral genes may also contribute to transformation of lymphocytes, but their functions in the transformation process have not been established.

Kaposi's sarcoma-associated herpesvirus DNA is regularly found in Kaposi's sarcoma cells, and several viral genes that affect cell proliferation and survival are expressed in these tumors. A noteworthy feature of Kaposi's sarcoma cells is that they secrete a variety of growth factors that drive their own proliferation. Interestingly, the transforming proteins of Kaposi's sarcoma-associated herpesvirus appear to act at least in part by stimulating growth factor secretion.

Retroviruses

Retroviruses cause cancer in a variety of animal species, including humans. One human retrovirus, human T-cell lymphotropic virus type I (HTLV-I), is the causative agent of adult T-cell leukemia, which is common in parts of Japan, the Caribbean, and Africa. Transformation of T lymphocytes by HTLV-I results from expression of the viral gene *tax*, which encodes a regu-

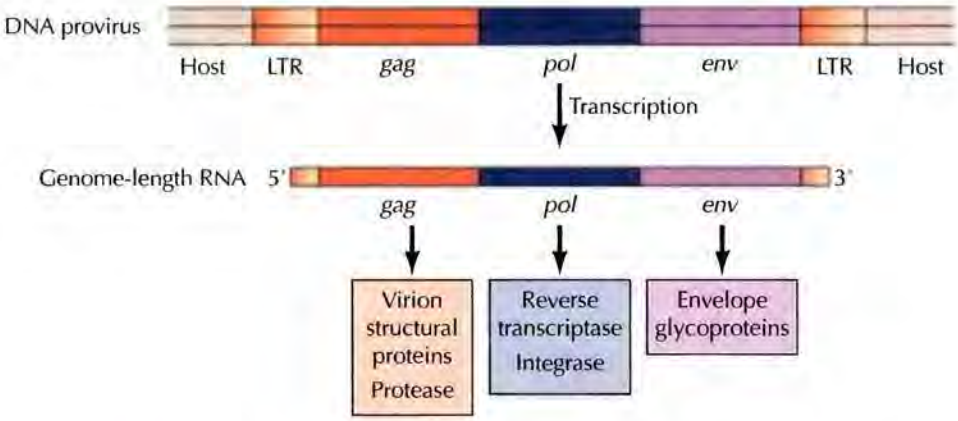


FIGURE 18.16 A typical retrovirus genome The DNA provirus, integrated into cellular DNA, is transcribed to yield genome-length RNA. This primary transcript serves as the genomic RNA for progeny virus particles and as mRNA for the *gag* and *pol* genes. In addition, the full-length RNA is spliced to yield mRNA for *env*. The *gag* gene encodes the viral protease and structural proteins of the virus particle, *pol* encodes reverse transcriptase and integrase, and *env* encodes envelope glycoproteins.

latory protein affecting expression of several cellular growth control genes. AIDS is caused by another retrovirus, HIV. In contrast to HTLV-I, HIV does not cause cancer by directly converting a normal cell into a tumor cell. However, AIDS patients suffer a high incidence of some malignancies, particularly lymphomas and Kaposi's sarcoma. These cancers, which are also common among other immunosuppressed individuals, are associated with infection by other viruses (e.g., Epstein-Barr virus and Kaposi's sarcoma-associated herpesvirus) and apparently develop as a secondary consequence of immunosuppression in AIDS patients.

Different retroviruses differ substantially in their oncogenic potential. Most retroviruses contain only three genes (*gag*, *pol*, and *env*) that are required for virus replication but play no role in cell transformation (Figure 18.16). Retroviruses of this type induce tumors only rarely, if at all, as a consequence of mutations resulting from the integration of proviral DNA within or adjacent to cellular genes.

Other retroviruses, however, contain specific genes responsible for induction of cell transformation and are potent carcinogens. The prototype of these highly oncogenic retroviruses is **Rous sarcoma virus (RSV)**, first isolated from a chicken sarcoma by Peyton Rous in 1911. More than 50 years later, studies of RSV led to identification of the first viral oncogene, which has provided a model for understanding many aspects of tumor development at the molecular level.

Oncogenes

Cancer results from alterations in critical regulatory genes that control cell proliferation, differentiation, and survival. Studies of tumor viruses revealed that specific genes (called **oncogenes**) are capable of inducing cell transformation, thereby providing the first insights into the molecular basis of cancer. However, the majority (approximately 80%) of human cancers are not induced by viruses and apparently arise from other causes, such as

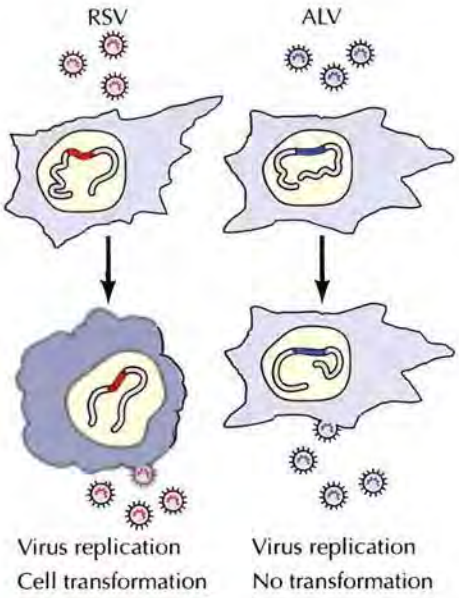


FIGURE 18.17 Cell transformation by RSV and ALV Both RSV and ALV infect and replicate in chicken embryo fibroblasts, but only RSV induces cell transformation.

radiation and chemical carcinogens. Therefore, in terms of our overall understanding of cancer, it has been critically important that studies of viral oncogenes also led to the identification of cellular oncogenes, which are involved in the development of non-virus-induced cancers. The key link between viral and cellular oncogenes was provided by studies of the highly oncogenic retroviruses.

Retroviral Oncogenes

Viral oncogenes were first defined in RSV, which transforms chicken embryo fibroblasts in culture and induces large sarcomas within 1 to 2 weeks after inoculation into chickens (Figure 18.17). In contrast, the closely related avian leukosis virus (ALV) replicates in the same cells as RSV without inducing transformation. This difference in transforming potential suggested the possibility that RSV contains specific genetic information responsible for transformation of infected cells. A direct comparison of the genomes of RSV and ALV was consistent with this hypothesis: The genomic RNA of RSV is about 10 kb, whereas that of ALV is smaller, about 8.5 kb.

In the early 1970s, Peter Vogt and Steven Martin isolated both deletion mutants and temperature-sensitive mutants of RSV that were unable to induce transformation. Importantly, these mutants still replicated normally in infected cells, indicating that RSV contains genetic information that is required for transformation but not for virus replication. Further analysis demonstrated that both the deletion and the temperature-sensitive RSV mutants define a single gene responsible for the ability of RSV to induce tumors in birds and transform fibroblasts in culture. Because RSV causes *sarc*omas, its oncogene was called ***src***. The *src* gene is an addition to the genome of RSV; it is not present in ALV (Figure 18.18). It encodes a 60-kd protein that was the first protein-tyrosine kinase to be identified (see the Key Experiment in Chapter 15).

More than 40 different highly oncogenic retroviruses have been isolated from a variety of animals, including chickens, turkeys, mice, rats, cats, and monkeys. All of these viruses, like RSV, contain at least one oncogene (in some cases two) that is not required for virus replication but is responsible for cell transformation. In some cases, different viruses contain the same oncogenes, but more than two dozen distinct oncogenes have been identi-

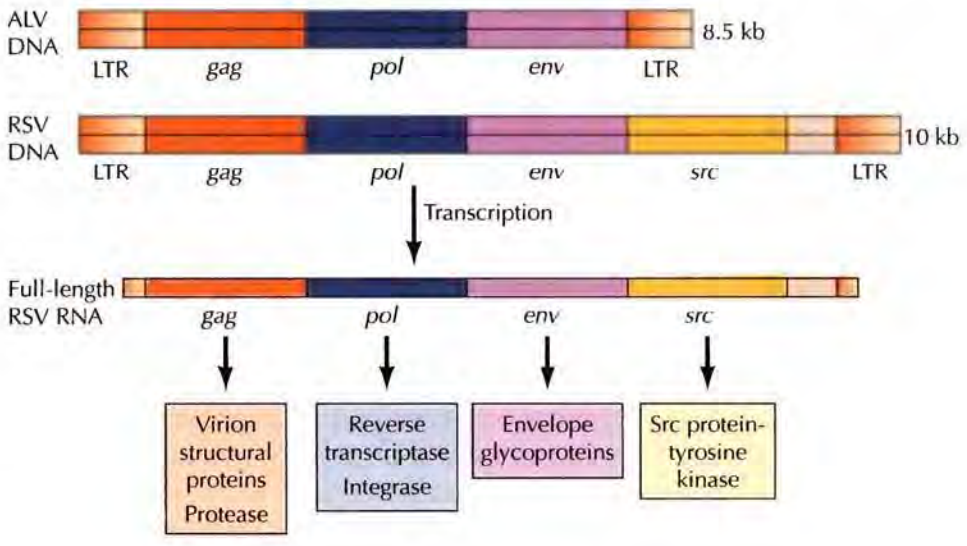


FIGURE 18.18 The RSV genome RSV contains an additional gene, *src*, that is not present in ALV and encodes the Src protein-tyrosine kinase.

TABLE 18.3 Retroviral Oncogenes

Oncogene	Virus	Species
abl	Abelson leukemia	Mouse
akt	AKT8	Mouse
cbl	Cas NS-1	Mouse
crk	CT10 sarcoma	Chicken
*erb*A	Avian erythroblastosis-ES4	Chicken
*erb*B	Avian erythroblastosis-ES4	Chicken
ets	Avian erythroblastosis-E26	Chicken
fes	Gardner-Arnstein feline sarcoma	Cat
fgr	Gardner-Rasheed feline sarcoma	Cat
fms	McDonough feline sarcoma	Cat
fos	FBJ murine osteogenic sarcoma	Mouse
fps	Fujinami sarcoma	Chicken
jun	Avian sarcoma-17	Chicken
kit	Hardy-Zuckerman feline sarcoma	Cat
maf	Avian sarcoma AS42	Chicken
mos	Moloney sarcoma	Mouse
mpl	Myeloproliferative leukemia	Mouse
myb	Avian myeloblastosis	Chicken
myc	Avian myelocytomatosis	Chicken
p3k	Avian sarcoma-16	Chicken
qin	Avian sarcoma-31	Chicken
raf	3611 murine sarcoma	Mouse
*ras*H	Harvey sarcoma	Rat
*ras*K	Kirsten sarcoma	Rat
rel	Reticuloendotheliosis	Turkey
ros	UR2 sarcoma	Chicken
sea	Avian erythroblastosis-S13	Chicken
sis	Simian sarcoma	Monkey
ski	Avian SK	Chicken
src	Rous sarcoma	Chicken
yes	Y73 sarcoma	Chicken

fied among this group of viruses (Table 18.3). Like *src*, many of these genes (such as ***ras*** and ***raf***) encode proteins that are now recognized as key components of signaling pathways that stimulate cell proliferation (see Figure 15.34).

Proto-Oncogenes

An unexpected feature of retroviral oncogenes is their lack of involvement in virus replication. Since most viruses are streamlined to replicate as efficiently as possible, the existence of viral oncogenes that are not an integral part of the virus life cycle seems paradoxical. Scientists were thus led to question where the retroviral oncogenes had originated and how they had become incorporated into viral genomes—a line of investigation that ultimately led to the identification of cellular oncogenes in human cancers.

The first clue to the origin of oncogenes came from the way in which the highly oncogenic retroviruses were isolated. The isolation of Abelson leukemia virus is a typical example (Figure 18.19). More than 150 mice were inoculated with a nontransforming virus containing only the *gag*, *pol*, and

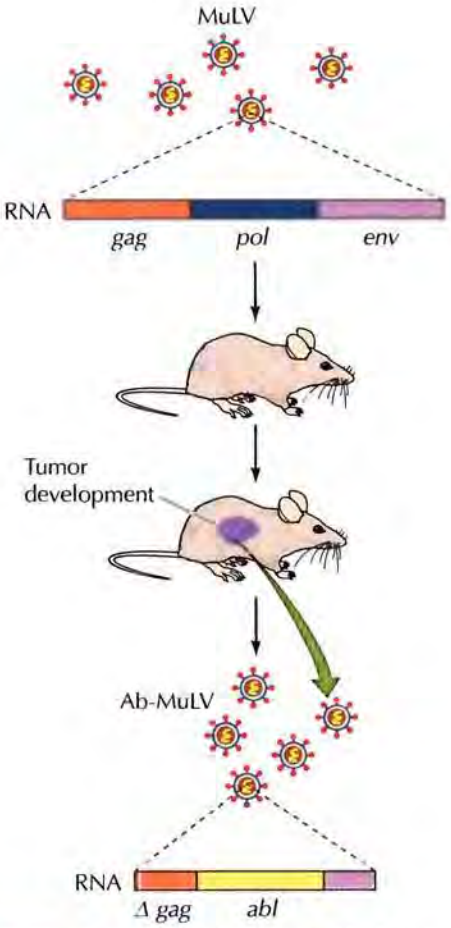


FIGURE 18.19 Isolation of Abelson leukemia virus The highly oncogenic virus Ab-MuLV was isolated from a rare tumor that developed in a mouse that had been inoculated with a nontransforming virus (Moloney murine leukemia virus, or MuLV). MuLV contains only the *gag*, *pol*, and *env* genes required for virus replication. In contrast, Ab-MuLV has acquired a new oncogene (*abl*), which is responsible for its transforming activity. The *abl* oncogene replaced some of the viral replicative genes and is fused with a partially deleted *gag* gene (designated *Δ gag*) in the Ab-MuLV genome.

env genes required for virus replication. One of these mice developed a lymphoma from which a new, highly oncogenic virus (Abelson leukemia virus), which now contained an oncogene (*abl*), was isolated. The scenario suggested the hypothesis that the retroviral oncogenes are derived from genes of the host cell, and that occasionally such a host cell gene becomes incorporated into a viral genome, yielding a new, highly oncogenic virus as the product of a virus-host recombination event.

The critical prediction of this hypothesis was that normal cells contain genes that are closely related to the retroviral oncogenes. This was definitively demonstrated in 1976 by Harold Varmus, J. Michael Bishop, and their colleagues, who showed that a cDNA probe for the *src* oncogene of RSV hybridized to closely related sequences in the DNA of normal chicken cells. Moreover, *src*-related sequences were also found in normal DNAs of a wide range of other vertebrates (including humans) and thus appeared to be highly conserved in evolution. Similar experiments with probes for the oncogenes of other highly oncogenic retroviruses have yielded comparable results, and it is now firmly established that the retroviral oncogenes were derived from closely related genes of normal cells.

The normal-cell genes from which the retroviral oncogenes originated are called **proto-oncogenes**. They are important cell regulatory genes, in many cases encoding proteins that function in the signal transduction pathways controlling normal cell proliferation (e.g., *src*, *ras*, and *raf*). The oncogenes are abnormally expressed or mutated forms of the corresponding proto-oncogenes. As a consequence of such alterations, the oncogenes induce abnormal cell proliferation and tumor development.

An oncogene incorporated into a retroviral genome differs in several respects from the corresponding proto-oncogene. First, the viral oncogene is transcribed under the control of viral promoter and enhancer sequences, rather than being controlled by the normal transcriptional regulatory sequences of the proto-oncogene. Consequently, oncogenes are usually expressed at much higher levels than the proto-oncogenes and are sometimes transcribed in inappropriate cell types. In some cases, such abnormalities of gene expression are sufficient to convert a normally functioning proto-oncogene into an oncogene that drives cell transformation.

In addition to such alterations in gene expression, oncogenes frequently encode proteins that differ in structure and function from those encoded by their normal homologs. Many oncogenes, such as *raf*, are expressed as fusion proteins with viral sequences at the amino terminus (Figure 18.20). Recombination events leading to the generation of such fusion proteins often occur during the capture of proto-oncogenes by retroviruses, and sequences from both the amino and carboxy termini of proto-oncogenes are frequently deleted during the process. Such deletions may result in the loss of regulatory domains that control the activity of the proto-oncogene proteins thereby generating oncogene proteins that function in an unregulated manner. For example, the viral *raf* oncogene encodes a fusion protein in which amino-terminal sequences of the normal Raf protein have been deleted. These amino-terminal sequences are critical to the normal regulation of Raf protein kinase activity, and their deletion results in unregulated constitutive activity of the oncogene-encoded Raf protein. This unregulated Raf activity drives cell proliferation, resulting in transformation.

Many other oncogenes differ from the corresponding proto-oncogenes by point mutations, resulting in single amino acid substitutions in the oncogene products. In some cases, such amino acid substitutions (like the deletions already discussed) lead to unregulated activity of the oncogene proteins. An important example of such point mutations is provided by the *ras*

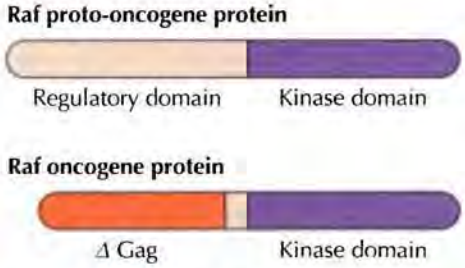


FIGURE 18.20 The Raf oncogene protein The Raf proto-oncogene protein consists of an amino-terminal regulatory domain and a carboxy-terminal protein kinase domain. In the viral Raf oncogene protein, the regulatory domain has been deleted and replaced by partially deleted viral Gag sequences (Δ Gag). As a result, the Raf kinase domain is constitutively active, causing cell transformation.

The Discovery of Proto-Oncogenes

DNA Related to the Transforming Gene(s) of Avian Sarcoma Viruses Is Present in Normal Avian DNA

Dominique Stehelin, Harold E. Varmus, J. Michael Bishop and Peter K. Vogt

Department of Microbiology, University of California, San Francisco (DS, HEV, and JMB) and Department of Microbiology, University of California, Los Angeles (PKV)

Nature, Volume 260, 1976, pages 170–173

J. Michael Bishop

Harold Varmus

The Context

Genetic analysis of RSV defined the first viral oncogene (*src*) as a gene that was specifically responsible for cell transformation but was not required for virus replication. The origin of highly oncogenic retroviruses from tumors of infected animals then suggested the hypothesis that retroviral oncogenes are derived from related genes of host cells. Consistent with this suggestion, normal cells of several species were found to contain retrovirus-related DNA sequences that could be detected by nucleic acid hybridization. However, it was unclear whether these sequences were related to the retroviral oncogenes or to the genes required for virus replication.

Harold Varmus, J. Michael Bishop, and their colleagues resolved this critical issue by exploiting the genetic characterization of the *src* oncogene. In particular, Peter Vogt had previously isolated transformation-defective mutants of RSV that had sustained deletions of approximately 1.5 kb, corresponding to most or all of the *src* gene. Stehelin and collaborators used these mutants to prepare a cDNA probe that specifically represented *src* sequences. The use of this defined probe in nucleic acid hybridization experiments allowed them to definitively demonstrate that normal cells contain *src*-related DNA sequences.

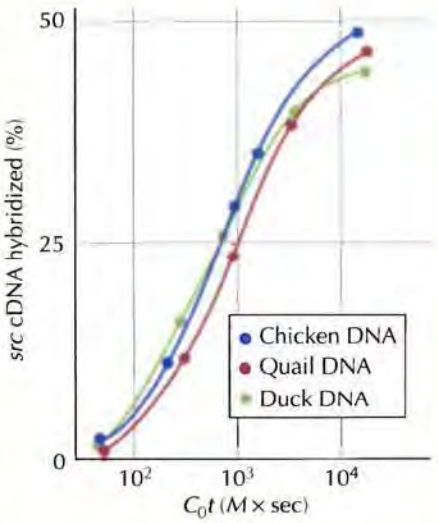
The Experiments

First reverse transcriptase was used to synthesize a radioactive cDNA probe composed of short single-stranded DNA fragments complementary to the entire genomic RNA of RSV. This probe was then hybridized to an excess of RNA isolated from a transformation-defective deletion mutant. Fragments of cDNA that were complementary to the viral replication genes hybridized to the transformation-defective RSV RNA, forming RNA-DNA duplexes. In contrast, cDNA fragments that were complementary to *src* were unable to hybridize and remained single-stranded. This single-stranded DNA was then isolated to provide a specific probe for *src* oncogene sequences. As predicted from the size of deletions in the transformation-defective RSV mutants, the *src*-specific probe was homologous to about 1.5 kb of RSV RNA.

The radioactive *src* cDNA was then used as a hybridization probe to attempt to detect related DNA sequences in normal avian cells. Strikingly, the *src* cDNA hybridized extensively to normal chicken DNA, as well as to DNA of other avian species (see figure). These experiments thus demonstrated that normal cells contain DNA sequences that are closely related to the *src* oncogene, supporting the hypothesis that retroviral oncogenes originated from cellular genes that became incorporated into viral genomes.

The Impact

Stehelin and colleagues concluded their 1976 paper by raising the possibility that the cellular *src* sequences are "involved in the normal regulation of cell growth and development or the transformation of cell behavior by physical, chemical or viral agents." This prediction has been strikingly substantiated, and the discovery of cellular *src* sequences has opened new doors to understanding both the regulation of normal cell proliferation and the molecular basis of human cancer. Studies of oncogene and proto-oncogene proteins, including Src itself, have proven critical in unraveling the signaling pathways that control the proliferation and differentiation of normal cells. The discovery of the *src* proto-oncogene further suggested that non-virus-induced tumors could arise as a result of mutations in related cellular genes, leading directly to the discovery of oncogenes in human tumors. By unifying studies of tumor viruses, normal cells, and non-virus-induced tumors, the results of Varmus, Bishop, and their colleagues have had an impact on virtually all aspects of studies of cell regulation and cancer research.



Hybridization of *src*-specific cDNA to normal chicken, quail, and duck DNA.

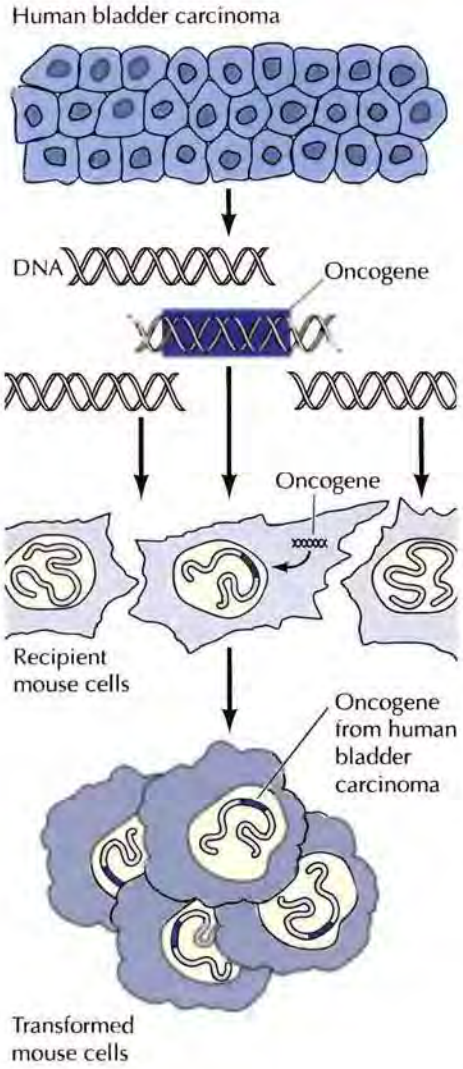


FIGURE 18.21 Detection of a human tumor oncogene by gene transfer DNA extracted from a human bladder carcinoma induced transformation of recipient mouse cells in culture. Transformation resulted from integration and expression of an oncogene derived from the human tumor.

oncogenes, which are discussed in the next section in terms of their role in human cancers.

Oncogenes in Human Cancer

Understanding the origin of retroviral oncogenes raised the question as to whether non-virus-induced tumors contain cellular oncogenes that were generated from proto-oncogenes by mutations or by DNA rearrangements during tumor development. Direct evidence for the involvement of cellular oncogenes in human tumors was first obtained by gene transfer experiments in the laboratories of Robert Weinberg and of one of the authors (GMC) in 1981. DNA of a human bladder carcinoma was found to efficiently induce transformation of recipient mouse cells in culture, indicating that the human tumor contained a biologically active cellular oncogene (Figure 18.21). Both gene transfer assays and alternative experimental approaches have since led to the detection of active cellular oncogenes in human tumors of many different types (Table 18.4).

Some of the oncogenes identified in human tumors are cellular homologs of oncogenes that were previously characterized in retroviruses, whereas others are new oncogenes first discovered in human cancers. The first human oncogene identified in gene transfer assays was subsequently identified as the human homolog of the *ras*H oncogene of Harvey sarcoma virus (see Table 18.3). Three closely related members of the *ras* gene family (*ras*H, *ras*K, and *ras*N) are the oncogenes most frequently encountered in human tumors. These genes are involved in approximately 20% of all human malignancies, including about 50% of colon and 25% of lung carcinomas.

The *ras* oncogenes are not present in normal cells; rather, they are generated in tumor cells as a consequence of mutations that occur during tumor development. The *ras* oncogenes differ from their proto-oncogenes by point mutations resulting in single amino acid substitutions at critical positions. The first such mutation discovered was the substitution of valine for glycine at position 12 (Figure 18.22). Other amino acid substitutions at position 12 (as well as at positions 13 and 61) are also frequently encountered in *ras* oncogenes in human tumors. In animal models, it has been shown that mutations that convert *ras* proto-oncogenes to oncogenes are caused by chemical carcinogens, providing a direct link between the mutagenic action of carcinogens and cell transformation.

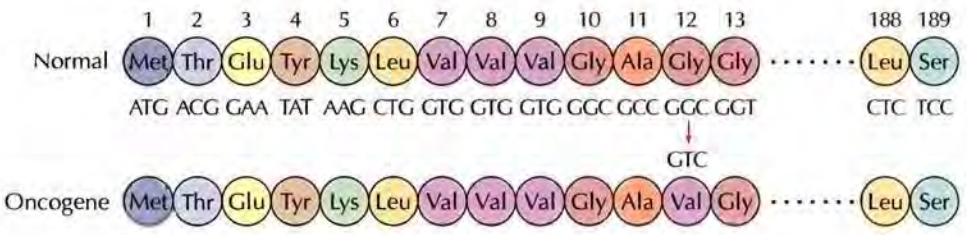


FIGURE 18.22 Point mutations in *ras* oncogenes A single nucleotide change, which alters codon 12 from GGC (Gly) to GTC (Val), is responsible for the transforming activity of the *ras*H oncogene detected in bladder carcinoma DNA.

TABLE 18.4 Representative Oncogenes of Human Tumors

Oncogene	Type of cancer	Activation mechanism
abl	Chronic myeloid leukemia, acute lymphocytic leukemia	Translocation
akt	Breast, ovarian and pancreatic carcinomas	Amplification
bcl-2	Follicular B-cell lymphoma	Translocation
*CCND*1	Parathyroid adenoma, B-cell lymphoma	Translocation
*CCND*1	Squamous cell, bladder, breast, esophageal, liver, and lung carcinomas	Amplification
cdk4	Melanomas	Point mutation
CTNNB1 (β-catenin)	Colon carcinoma	Point mutation
*erb*B	Gliomas, many carcinomas	Amplification
*erb*B	Lung carcinomas	Point mutation
*erb*B-2	Breast and ovarian carcinomas	Amplification
gli	Glioblastoma	Amplification
kit	Gastrointestinal stromal tumors	Point mutation
c-*myc*	Burkitt's lymphoma	Translocation
c-*myc*	Breast and lung carcinomas	Amplification
L-*myc*	Lung carcinoma	Amplification
N-*myc*	Neuroblastoma, lung carcinoma	Amplification
PDGFR	Chronic myelomonocytic leukemia	Translocation
PDGFR	Gastrointestinal stromal tumors	Point mutation
PI3K	Breast carcinoma	Point mutation
	Ovarian, gastric, lung carcinoma	Amplification
PML/RARα	Acute promyelocytic leukemia	Translocation
B-*raf*	Melanoma, colon carcinoma	Point mutation
*ras*H	Thyroid carcinoma	Point mutation
*ras*K	Colon, lung, pancreatic, and thyroid carcinomas	Point mutation
*ras*N	Acute myeloid and lymphocytic leukemias, thyroid carcinoma	Point mutation
ret	Multiple endocrine neoplasia types 2A and 2B	Point mutation
ret	Thyroid carcinoma	DNA rearrangement
SMO	Basal cell carcinoma	Point mutation

As discussed in Chapter 15, the *ras* genes encode guanine nucleotide-binding proteins that function in transduction of mitogenic signals from a variety of growth factor receptors. The activity of the Ras proteins is controlled by GTP or GDP binding, such that they alternate between active (GTP-bound) and inactive (GDP-bound) states (see Figure 15.35). The mutations characteristic of *ras* oncogenes have the effect of maintaining the Ras proteins constitutively in the active GTP-bound conformation. In large part, this effect is a result of nullifying the response of oncogenic Ras proteins to GAP (GTPase-activating protein), which stimulates hydrolysis of bound GTP by normal Ras. Because of the resulting decrease in their intracellular GTPase activity, the oncogenic Ras proteins remain in the active GTP-bound state and drive unregulated cell proliferation.

Point mutations are only one of the ways in which proto-oncogenes are converted to oncogenes in human tumors. Many cancer cells display abnor-

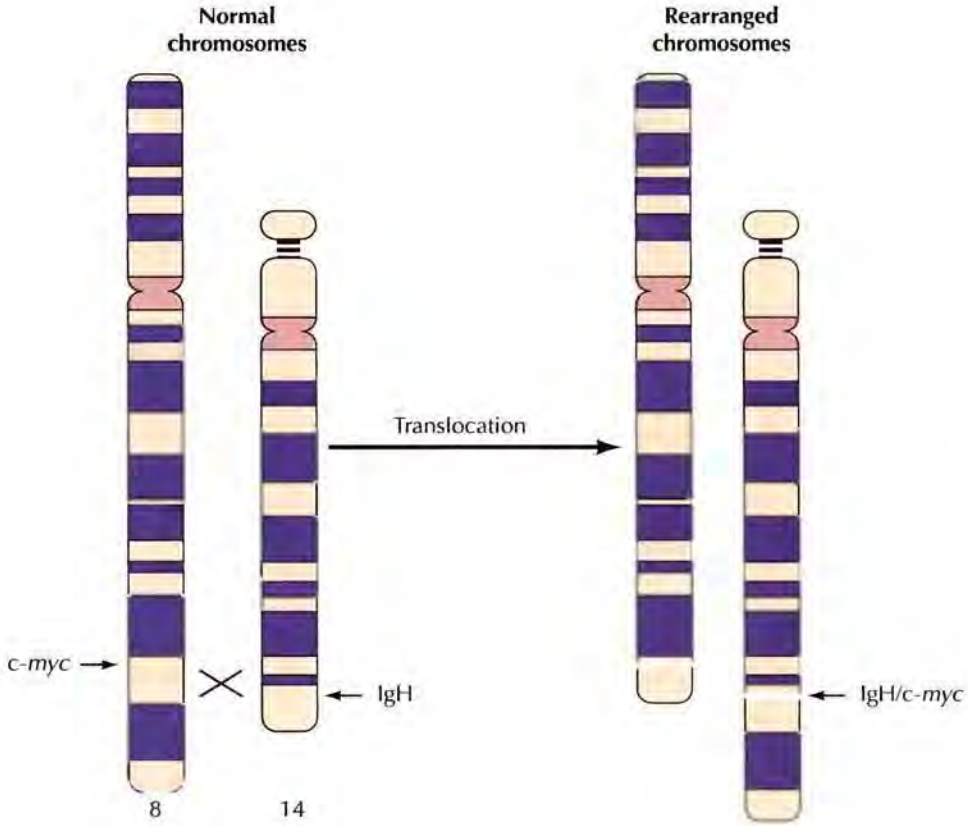


FIGURE 18.23 Translocation of *c-myc* The c-*myc* proto-oncogene is translocated from chromosome 8 to the immunoglobulin heavy-chain locus (IgH) on chromosome 14 in Burkitt's lymphomas, resulting in abnormal c-*myc* expression.

malities in chromosome structure, including translocations, duplications, and deletions. The gene rearrangements resulting from chromosome translocations frequently lead to the generation of oncogenes. In some cases, analysis of these rearrangements has implicated already known oncogenes in tumor development. In other cases, novel oncogenes have been discovered by molecular cloning and analysis of rearranged DNA sequences.

The first characterized example of oncogene activation by chromosome translocation was the involvement of the ***c-myc*** oncogene in human Burkitt's lymphomas and mouse plasmacytomas, which are malignancies of antibody-producing B lymphocytes (Figure 18.23). Both of these tumors are characterized by chromosome translocations involving the genes that encode immunoglobulins. For example, virtually all Burkitt's lymphomas have translocations of a fragment of chromosome 8 to one of the immunoglobulin gene loci, which reside on chromosomes 2 (κ light chain), 14 (heavy chain), and 22 (λ light chain). The fact that the immunoglobulin genes are actively expressed in these tumors suggested that the translocations activate a proto-oncogene from chromosome 8 by inserting it into the immunoglobulin loci. This possibility was investigated by analysis of tumor DNAs with probes for known oncogenes, leading to the finding that the c-*myc* proto-oncogene was the chromosome 8 translocation break point in Burkitt's lymphomas. These translocations inserted c-*myc* into an immunoglobulin locus, where it was expressed in an unregulated manner. Such uncontrolled expression of the c-*myc* gene, which encodes a transcription factor normally induced in response to growth factor stimulation, is sufficient to drive cell proliferation and contribute to tumor development.

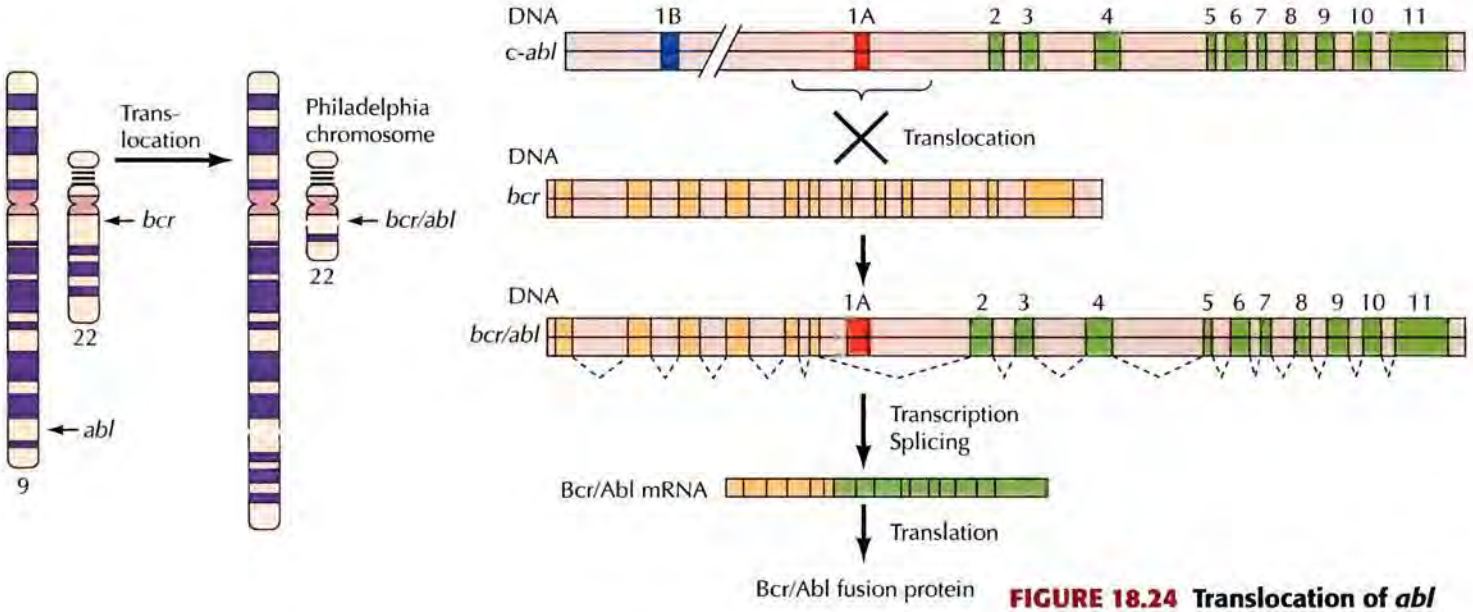


FIGURE 18.24 Translocation of *abl* The *abl* oncogene is translocated from chromosome 9 to chromosome 22, forming the Philadelphia chromosome in chronic myeloid leukemias. The *abl* proto-oncogene, which contains two alternative first exons (1A and 1B), is joined to the middle of the *bcr* gene on chromosome 22. Exon 1B is deleted as a result of the translocation. Transcription of the fused gene initiates at the *bcr* promoter and continues through *abl*. Splicing then generates a fused Bcr/Abl mRNA in which *abl* exon 1A sequences are also deleted and *bcr* sequences are joined to *abl* exon 2. The Bcr/Abl mRNA is translated to yield a recombinant Bcr/Abl fusion protein.

Translocations of other proto-oncogenes frequently result in rearrangements of coding sequences, leading to the formation of abnormal gene products. The prototype is translocation of the ***abl*** proto-oncogene from chromosome 9 to chromosome 22 in chronic myeloid leukemia (Figure 18.24). This translocation leads to fusion of *abl* with its translocation partner, a gene called *bcr*, on chromosome 22. The result is production of a Bcr/Abl fusion protein in which the normal amino terminus of the Abl proto-oncogene protein has been replaced by Bcr amino acid sequences. The fusion of Bcr sequences results in unregulated activity of the Abl protein-tyrosine kinase, leading to cell transformation.

A distinct mechanism by which oncogenes are activated in human tumors is gene amplification, which results in elevated gene expression. DNA amplification (see Figure 6.50) is common in tumor cells, occurring more than a thousand times more frequently than in normal cells, and amplification of oncogenes may play a role in the progression of many tumors to more rapid growth and increasing malignancy. Indeed, novel oncogenes have been identified by molecular cloning and characterization of DNA sequences that are amplified in tumors.

A prominent example of oncogene amplification is the involvement of the **N-*myc*** gene, which is related to c-*myc*, in neuroblastoma (a childhood tumor of neuronal cells). Amplified copies of N-*myc* are frequently present in rapidly growing aggressive tumors, indicating that N-*myc* amplification is associated with the progression of neuroblastomas to increasing malignancy. Amplification of another oncogene, ***erb*B-2**, which encodes a receptor protein-tyrosine kinase, is similarly related to progression of breast and ovarian carcinomas.

Functions of Oncogene Products

The viral and cellular oncogenes have defined a large group of genes (about 100 in total) that can contribute to the abnormal behavior of malignant cells. As already noted, many of the proteins encoded by proto-oncogenes regulate normal cell proliferation; in these cases, the elevated expression or activity of the corresponding oncogene proteins drives the uncontrolled

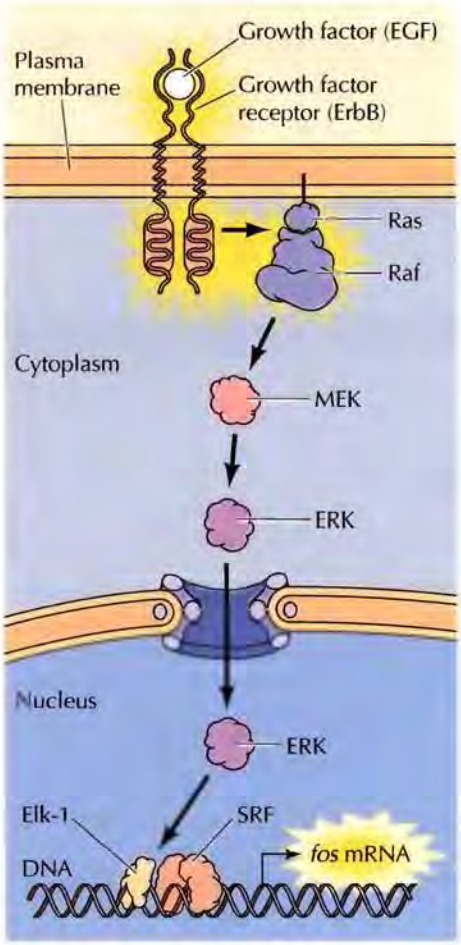


FIGURE 18.25 Oncogenes and the ERK signaling pathway Oncogene proteins act as growth factors (e.g., EGF), growth factor receptors (e.g., ErbB), and intracellular signaling molecules (Ras and Raf). Ras and Raf activate the ERK MAP kinase pathway (see Figures 15.34 and 15.37), leading to the induction of additional genes (e.g., *fos*) that encode potentially oncogenic transcriptional regulatory proteins. Proteins with known oncogenic potential are highlighted with a yellow glow.

proliferation of cancer cells. Other oncogene products contribute to other aspects of the behavior of cancer cells, such as failure to undergo programmed cell death or defective differentiation.

The function of oncogene proteins in regulation of cell proliferation is illustrated by their activities in growth factor-stimulated pathways of signal transduction, such as the activation of ERK signaling downstream of receptor protein-tyrosine kinases (Figure 18.25). The oncogene proteins within this pathway include polypeptide growth factors, growth factor receptors, intracellular signaling proteins, transcription factors, and the cell cycle regulator cyclin D1.

The action of growth factors as oncogene proteins results from their abnormal expression, leading to a situation in which a tumor cell produces a growth factor to which it also responds. The result is autocrine stimulation of the growth factor-producing cell (see Figure 18.9), which drives abnormal cell proliferation and contributes to the development of a wide variety of human tumors.

A large group of oncogenes encode growth factor receptors, most of which are protein-tyrosine kinases. These receptors can be converted to oncogene proteins by alterations of their amino-terminal domains, which would normally bind extracellular growth factors. For example, the receptor for platelet-derived growth factor (PDGF) is converted to an oncogene in some human leukemias by a chromosome translocation in which the normal amino terminus of the PDGF receptor (PDGFR) is replaced by the amino terminal sequences of a transcription factor called Tel (Figure 18.26). The Tel sequences of the resulting Tel/PDGFR fusion protein dimerize in the absence of growth factor binding, resulting in constitutive activity of the intracellular kinase domain and unregulated production of a proliferative signal from the oncogene protein. Alternatively, genes that encode receptor protein-tyrosine kinases can be activated by gene amplification or by point mutations that result in unregulated kinase activity. Other oncogenes (including *src* and *abl*) encode nonreceptor protein-tyrosine kinases that are constitutively activated by deletions or mutations of regulatory sequences.

The Ras proteins play a key role in mitogenic signaling by coupling growth factor receptors to activation of the Raf protein-serine/threonine kinase, which initiates a protein kinase cascade leading to activation of the ERK MAP kinase (see Figure 15.34). As discussed earlier, the mutations that convert *ras* proto-oncogenes to oncogenes result in constitutive Ras activity, which leads to activation of the ERK pathway. The *raf* gene can similarly be converted to an oncogene by deletions that result in loss of the amino-terminal regulatory domain of the Raf protein (see Figure 18.20). These deletions result in unregulated activity of the Raf protein kinase, which also leads to constitutive ERK activation. Alternatively, *raf* proto-oncogenes can be converted to oncogenes by point mutations that result in elevated Raf kinase activity.

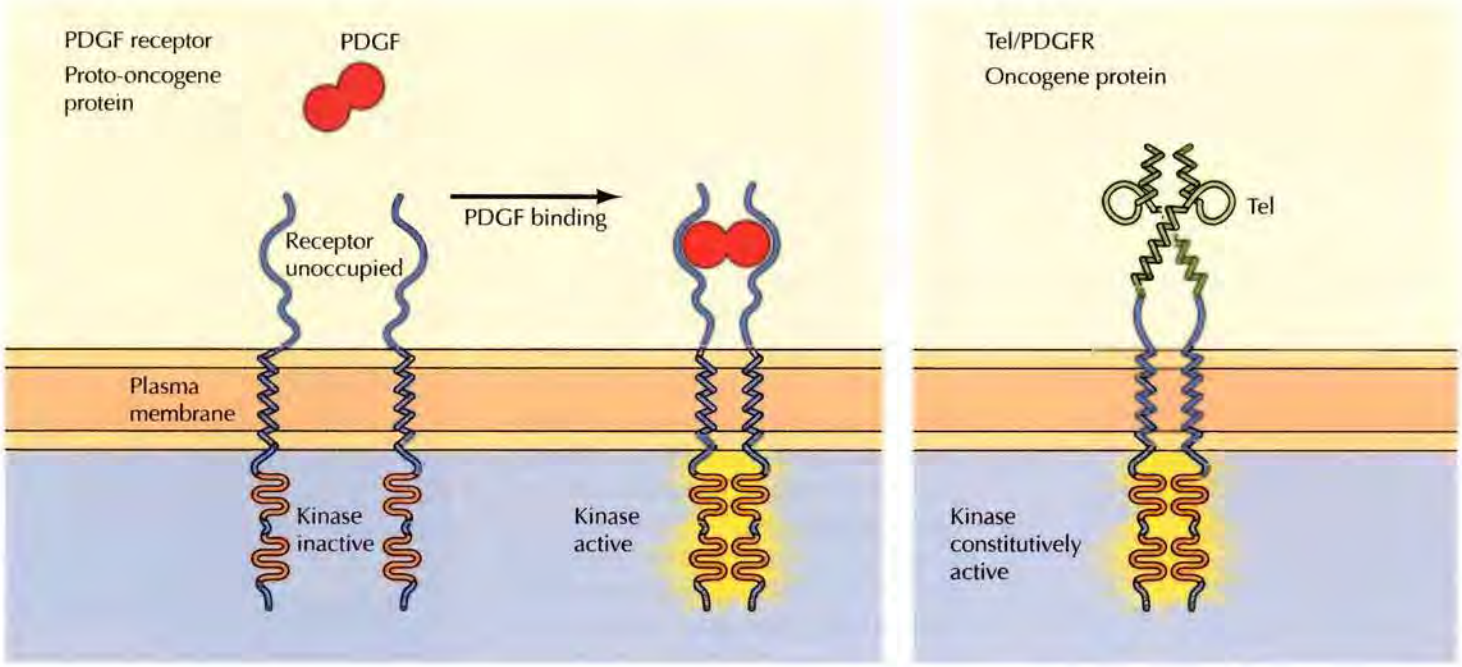


FIGURE 18.26 Mechanism of Tel/PDGFR oncogene activation The normal PDGF receptor (PDGFR) is activated by dimerization induced by PDGF binding. The Tel/PDGFR oncogene encodes a fusion protein in which the normal extracellular domain of PDGFR is replaced by the amino terminal sequences of the Tel transcription factor, which include its helix-loop-helix dimerization domain (see Figure 7.29). These sequences dimerize in the absence of PDGF, leading to constitutive activation of the oncogene protein kinase.

The ERK pathway ultimately leads to the phosphorylation of transcription factors and alterations in gene expression. As might therefore be expected, many oncogenes encode transcriptional regulatory proteins that are normally induced in response to growth factor stimulation. For example, transcription of the *fos* proto-oncogene is induced as a result of phosphorylation of Elk-1 by ERK (see Figure 18.25). **Fos** and the product of another proto-oncogene, **Jun**, are components of the AP-1 transcription factor, which activates transcription of a number of target genes, including cyclin D1, in growth factor–stimulated cells (Figure 18.27). Constitutive activity of AP-1, resulting from unregulated expression of either the Fos or Jun oncogene proteins, is sufficient to drive abnormal cell proliferation, leading to cell transformation. The Myc proteins similarly function as transcription factors regulated by mitogenic stimuli, and abnormal expression of *myc* oncogenes contributes to the development of a variety of human tumors. Other transcription factors are frequently activated as oncogenes by chromosome translocations in human leukemias and lymphomas.

The signaling pathways activated by growth factor stimulation ultimately regulate components of the cell cycle machinery that promote progression through the restriction point in G_1. The D-type cyclins are induced in response to growth factor stimulation (at least in part via activation of the AP-1 transcription factor) and play a key role in coupling growth factor signaling to cell cycle progression (see Figure 18.27). Perhaps not surprisingly, the gene encoding cyclin D1 is a proto-oncogene, which can be activated as an oncogene (called ***CCND1***) by chromosome translocation or gene amplifi-

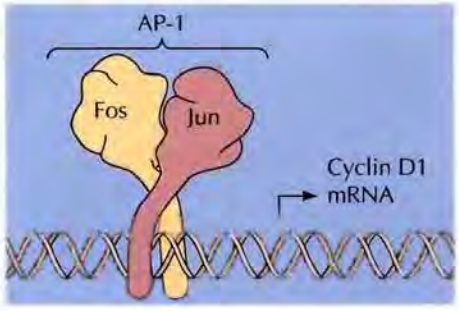


FIGURE 18.27 The AP-1 transcription factor Fos and Jun dimerize to form AP-1, which activates transcription of cyclin D1 and other growth factor–inducible genes.

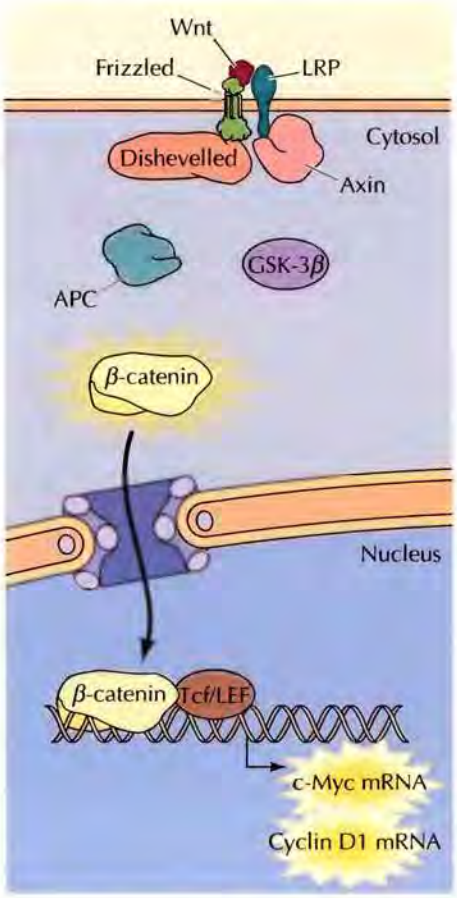


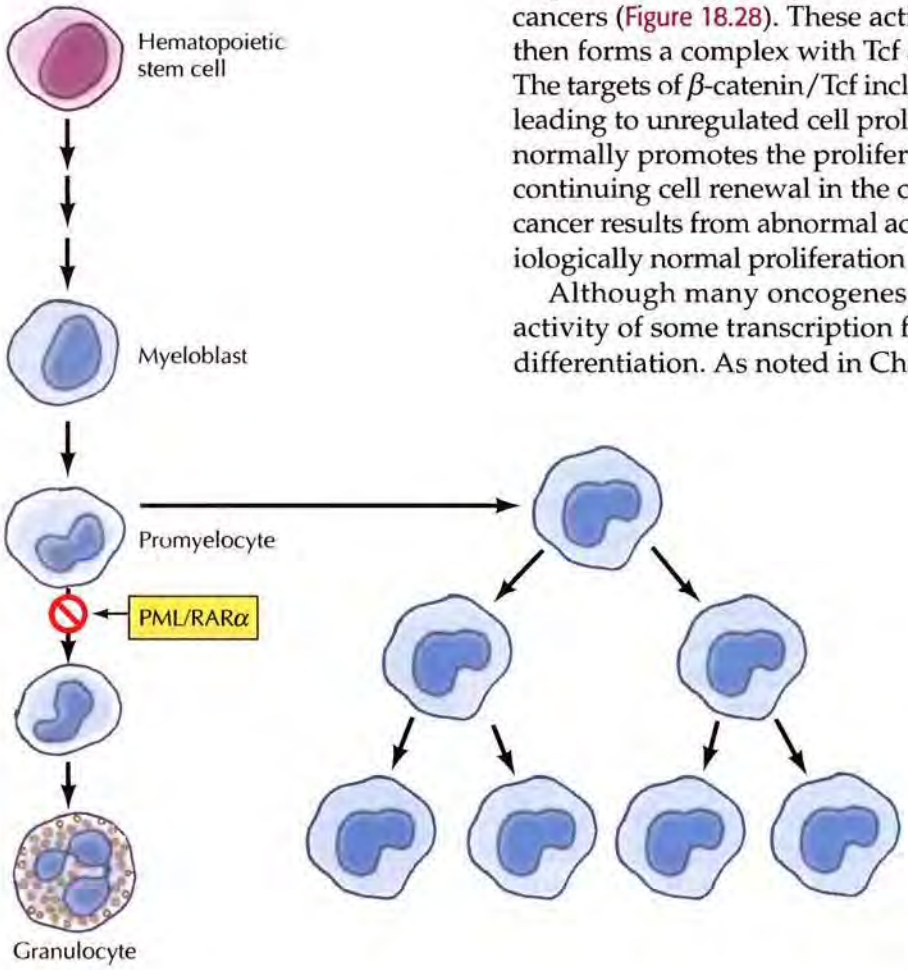
FIGURE 18.28 Oncogenic activity of the Wnt pathway Wnt polypeptides bind to Frizzled and LRP receptors, which activate Dishevelled and lead to inhibition of the axin/APC/GSK-3β complex. This results in stabilization of β-catenin, which translocates to the nucleus and forms a complex with Tcf/LEF transcription factors, leading to activation of target genes, including the genes encoding c-Myc and cyclin D1. The genes encoding both Wnt and β-catenin can act as oncogenes.

cation. These alterations lead to constitutive expression of cyclin D1, which then drives cell proliferation in the absence of normal growth factor stimulation. The catalytic partner of cyclin D1, Cdk4, is also activated as an oncogene by point mutations in melanomas.

Components of other signaling pathways discussed in Chapter 15, including G protein-coupled signaling pathways, the NF-κB pathway, and the Hedgehog, Wnt, and Notch pathways, can also act as oncogenes. For example, Wnt proteins were identified as oncogenes in mouse breast cancers, and activating mutations frequently convert the downstream target of Wnt signaling, β-catenin, to an oncogene (*CTNNB1*) in human colon cancers (Figure 18.28). These activating mutations stabilize β-catenin, which then forms a complex with Tcf and stimulates transcription of target genes. The targets of β-catenin/Tcf include the genes encoding c-Myc and cyclin D1, leading to unregulated cell proliferation. It is noteworthy that Wnt signaling normally promotes the proliferation of stem cells and their progeny during continuing cell renewal in the colon (see Figure 17.18), indicating that colon cancer results from abnormal activity of the same pathway that signals physiologically normal proliferation of colon epithelial cells.

Although many oncogenes stimulate cell proliferation, the oncogenic activity of some transcription factors instead results from inhibition of cell differentiation. As noted in Chapter 15, thyroid hormone and retinoic acid induce differentiation of a variety of cell types. These hormones diffuse through the plasma membrane and bind to intracellular receptors that act as transcriptional regulatory molecules. Mutated forms of both the thyroid hormone receptor (**ErbA**) and the retinoic acid receptor (**PML/RARα**) act as oncogene proteins in chicken erythroleukemia and human acute promyelocytic leukemia, respectively. In both cases, the mutated oncogene receptors appear to interfere with the action of their normal homologs, thereby blocking cell differentiation and maintaining the leukemic cells in an actively proliferating state (Figure 18.29). In the case of acute promyelocytic

FIGURE 18.29 Action of the PML/RARα oncogene protein The PML/RARα fusion protein blocks the differentiation of promyelocytes to granulocytes.



leukemia, high doses of retinoic acid can overcome the effect of the PML/RARα oncogene protein and induce differentiation of the leukemic cells. This biological observation has a direct clinical correlate: Patients with acute promyelocytic leukemia can be treated effectively by administration of retinoic acid, which induces differentiation and blocks continued cell proliferation.

As discussed earlier in this chapter, the failure of cancer cells to undergo programmed cell death (or apoptosis) is a critical factor in tumor development, and several oncogenes encode proteins that act to promote cell survival (Figure 18.30). The survival of most animal cells is dependent on growth factor stimulation, so those oncogenes that encode growth factors, growth factor receptors, and signaling proteins such as Ras act not only to stimulate cell proliferation but also to prevent cell death. As discussed in Chapter 16, the PI 3-kinase/Akt signaling pathway plays a key role in preventing apoptosis of many growth factor–dependent cells, and the genes encoding **PI 3-kinase** and **Akt** act as oncogenes in both retroviruses and

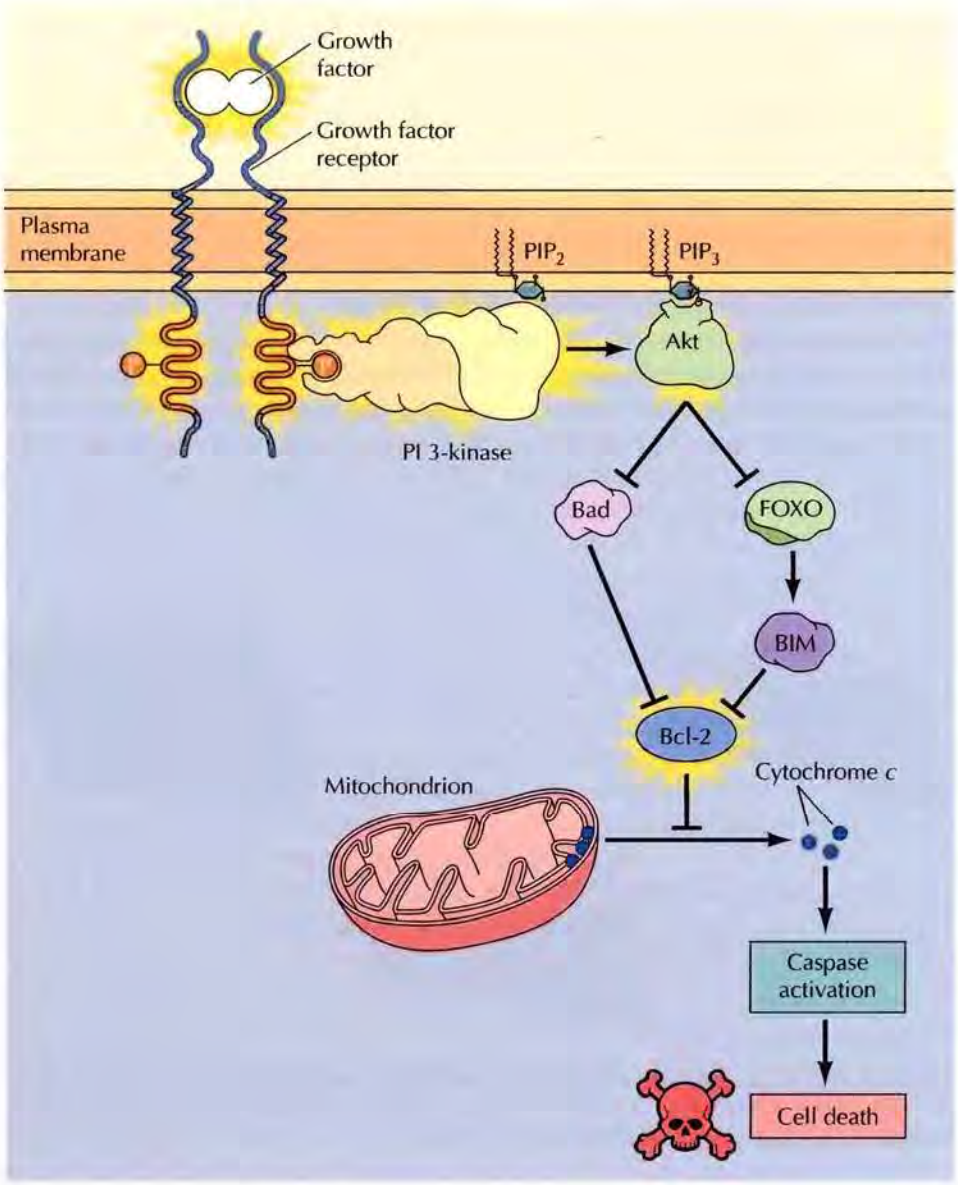


FIGURE 18.30 Oncogenes and cell survival The oncogene proteins that signal cell survival include growth factors, growth factor receptors, PI 3-kinase, Akt, and Bcl-2. The targets of Akt include the proapoptotic Bcl-2 family member Bad and the FOXO transcription factor, which stimulates transcription of another proapoptotic Bcl-2 family member, Bim. Phosphorylation by Akt inhibits both Bad and FOXO, promoting cell survival. The antiapoptotic protein Bcl-2 also functions as an oncogene by promoting cell survival and inhibiting the release of cytochrome *c* from mitochondria. Proteins with oncogenic potential are highlighted with a yellow glow.

human tumors. The downstream targets of PI 3-kinase/Akt signaling include a proapoptotic member of the Bcl-2 family, Bad, which is inactivated as a result of phosphorylation by Akt, as well as the FOXO transcription factor, which regulates expression of the proapoptotic Bcl-2 family member, Bim. In addition, it is notable that **Bcl-2** itself was first discovered as the product of an oncogene in human lymphomas. The *bcl*-2 oncogene is generated by a chromosome translocation that results in elevated expression of Bcl-2, which blocks apoptosis and maintains cell survival under conditions that normally induce cell death. The identification of *bcl*-2 as an oncogene not only provided the first demonstration of the significance of programmed cell death in the development of cancer but also led to the discovery of the role of Bcl-2 and related genes as central regulators of apoptosis in organisms ranging from *C. elegans* to humans.

Tumor Suppressor Genes

The activation of cellular oncogenes represents only one of two distinct types of genetic alterations involved in tumor development; the other is inactivation of tumor suppressor genes. Oncogenes drive abnormal cell proliferation as a consequence of genetic alterations that either increase gene expression or lead to uncontrolled activity of the oncogene-encoded proteins. **Tumor suppressor genes** represent the opposite side of cell growth control, normally acting to inhibit cell proliferation and tumor development. In many tumors, these genes are lost or inactivated, thereby removing negative regulators of cell proliferation and contributing to the abnormal proliferation of tumor cells.

Identification of Tumor Suppressor Genes

The first insight into the activity of tumor suppressor genes came from somatic cell hybridization experiments initiated by Henry Harris and his colleagues in 1969. The fusion of normal cells with tumor cells yielded hybrid cells containing chromosomes from both parents (Figure 18.31). In

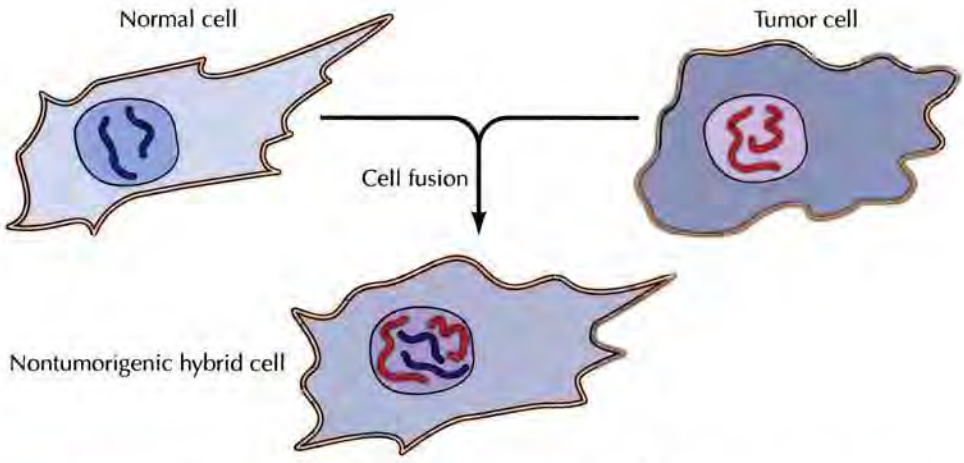


FIGURE 18.31 Suppression of tumorigenicity by cell fusion Fusion of tumor cells with normal cells yields hybrids that contain chromosomes from both parents. Such hybrids are usually nontumorigenic.

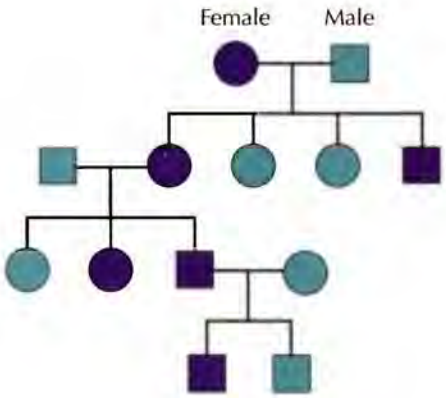


FIGURE 18.32 Inheritance of retinoblastoma Susceptibility to retinoblastoma is transmitted to approximately 50% of offspring. Affected and normal individuals are indicated by purple and green symbols, respectively.

most cases, such hybrid cells were not capable of forming tumors in animals. Therefore it appeared that genes derived from the normal cell parent acted to inhibit (or suppress) tumor development. Definition of these genes at the molecular level came, however, from a different approach—the analysis of rare inherited forms of human cancer.

The first tumor suppressor gene was identified by studies of retinoblastoma, a rare childhood eye tumor. Provided that the disease is detected early, retinoblastoma can be successfully treated and many patients survive to have families. Consequently, it was recognized that some cases of retinoblastoma are inherited. In these cases, approximately 50% of the children of an affected parent develop retinoblastoma, consistent with Mendelian transmission of a single dominant gene that confers susceptibility to tumor development (Figure 18.32).

Although susceptibility to retinoblastoma is transmitted as a dominant trait, inheritance of the susceptibility gene is not sufficient to transform a normal retinal cell into a tumor cell. All retinal cells in a patient inherit the susceptibility gene, but only a small fraction of these cells give rise to tumors. Thus tumor development requires additional events beyond inheritance of tumor susceptibility. In 1971 Alfred Knudson proposed that the development of retinoblastoma requires two mutations, which are now known to correspond to the loss of both of the functional copies of the tumor susceptibility gene (the ***Rb*** tumor suppressor gene) that would be present on homologous chromosomes of a normal diploid cell (Figure 18.33). In inherited retinoblastoma, one defective copy of *Rb* is genetically transmitted. The loss of this single *Rb* copy is not by itself sufficient to trigger tumor development, but retinoblastoma almost always develops in these individuals as a result of a second somatic mutation leading to the loss of the remaining normal *Rb* allele. Noninherited retinoblastoma, in contrast, is rare, since its development requires two independent somatic mutations to inactivate both normal copies of *Rb* in the same cell.

The functional nature of the *Rb* gene as a negative regulator of tumorigenesis was initially indicated by observations of chromosome morphology. Visible deletions of chromosome 13q14 were found in some retinoblastomas, suggesting that loss (rather than activation) of the *Rb* gene led to tumor development (Figure 18.34). Gene-mapping studies further indicated that tumor development resulted from loss of normal *Rb* alleles in the tumor cells, consistent with the function of *Rb* as a tumor suppressor gene. Isolation of the *Rb* gene as a molecular clone in 1986 then firmly established

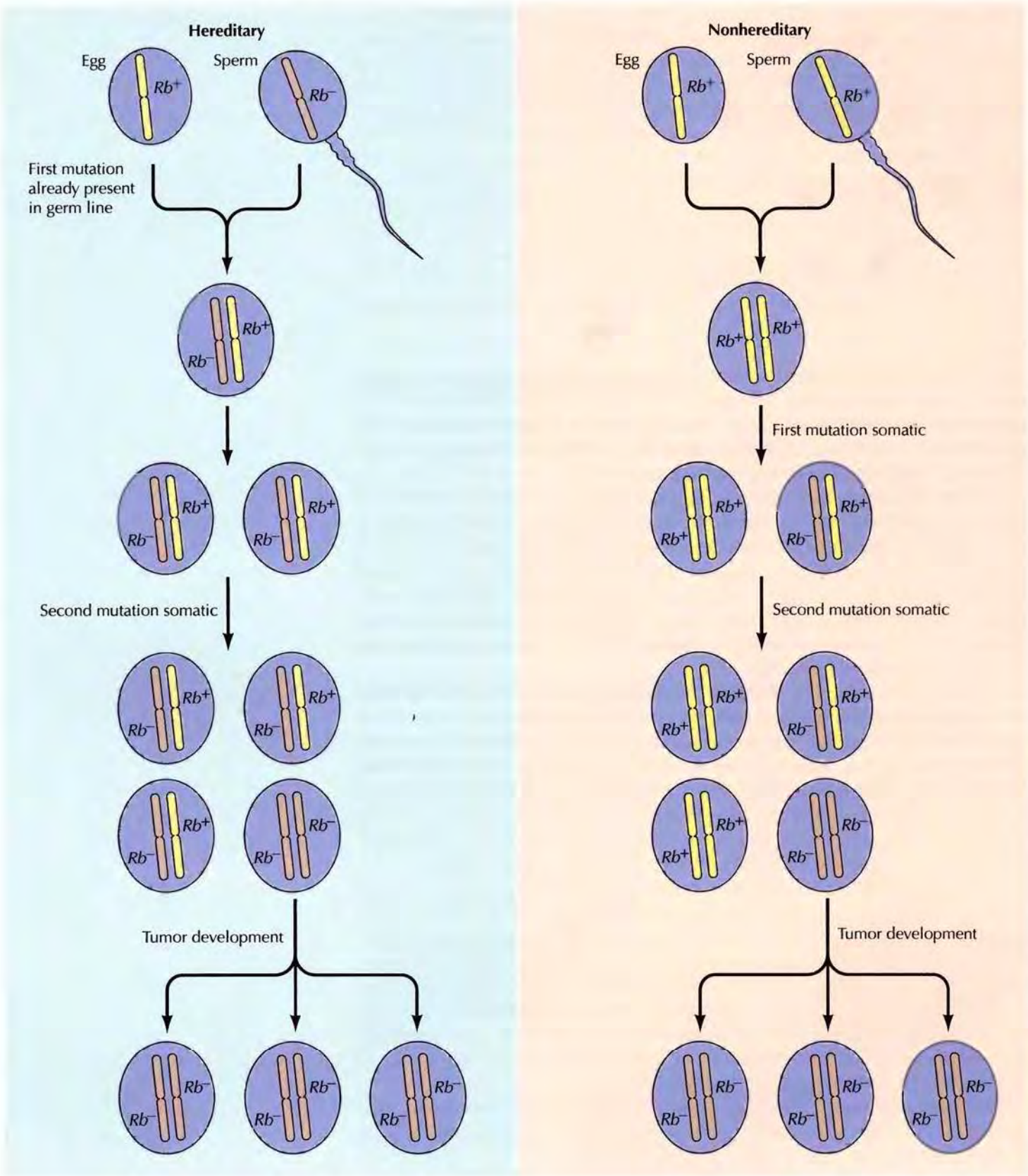


FIGURE 18.33 Mutations of *Rb* during retinoblastoma development
In hereditary retinoblastoma, a defective copy of the *Rb* gene (Rb^-) is inherited from the affected parent. A second somatic mutation, which inactivates the single normal Rb^+ copy in a retinal cell, then leads to the development of retinoblastoma. In nonhereditary cases, two normal Rb^+ genes are inherited, and retinoblastoma develops only if two somatic mutations inactivate both copies of *Rb* in the same cell.

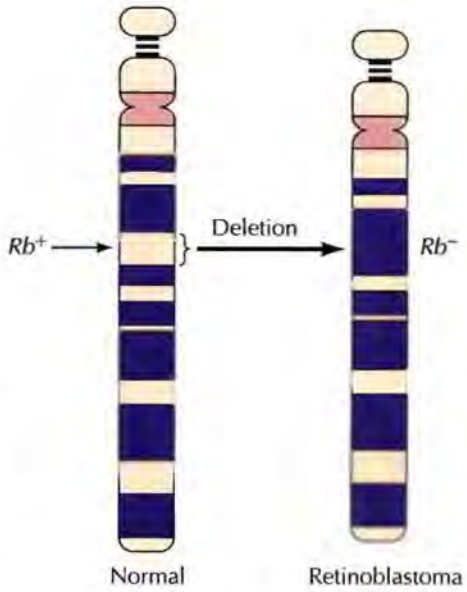


FIGURE 18.34 *Rb* deletions in retinoblastoma Many retinoblastomas have deletions of the chromosomal locus (13q14) that contains the *Rb* gene.

that *Rb* is consistently lost or mutated in retinoblastomas. Gene transfer experiments also demonstrated that introduction of a normal *Rb* gene into retinoblastoma cells reverses their tumorigenicity, providing direct evidence for the activity of *Rb* as a tumor suppressor.

Although *Rb* was identified in a rare childhood cancer, it is also involved in more common tumors of adults. In particular, studies of the cloned gene have established that *Rb* is lost or inactivated in many bladder, breast, and lung carcinomas. The significance of the *Rb* tumor suppressor gene thus extends beyond retinoblastoma, and mutations of the *Rb* gene contribute to development of a substantial fraction of human cancers. In addition, as noted earlier in this chapter, the Rb protein is a key target for the oncogene proteins of several DNA tumor viruses, including SV40, adenoviruses, and human papillomaviruses, which bind to Rb and inhibit its activity (Figure 18.35). Transformation by these viruses thus results, at least in part, from inactivation of Rb at the protein level rather than from mutational inactivation of the *Rb* gene.

Characterization of *Rb* as a tumor suppressor gene served as the prototype for the identification of additional tumor suppressor genes that contribute to the development of many different human cancers (Table 18.5). Some of these genes were identified as the causes of rare inherited cancers, playing a role similar to that of *Rb* in hereditary retinoblastoma. Other tumor suppressor genes have been identified as genes that are frequently deleted or mutated in common noninherited cancers of adults, such as colon carcinoma. In either case, it appears that most tumor suppressor genes are involved in the development of both inherited and noninherited forms of cancer. Indeed, mutations of some tumor suppressor genes appear to be the most common molecular alterations leading to human tumor development.

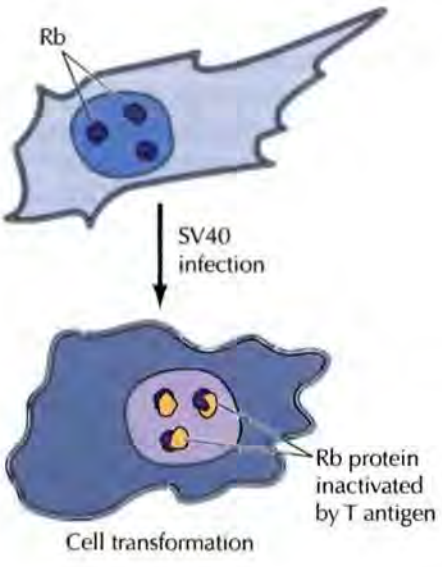


FIGURE 18.35 Interaction of Rb with oncogene proteins of DNA tumor viruses The oncogene proteins of several DNA tumor viruses (e.g., SV40 T antigen) induce transformation by binding to and inactivating Rb protein.

TABLE 18.5 Representative Tumor Suppressor Genes

Gene	Type of cancer
APC	Colon/rectum carcinoma
BRCA1	Breast and ovarian carcinomas
BRCA2	Breast carcinoma
INK4	Melanoma, lung carcinoma, brain tumors, leukemias, lymphomas
NF1	Neurofibrosarcoma
NF2	Meningioma
p53	Brain tumors; breast, colon/rectum, esophageal, liver, and lung carcinomas; sarcomas; leukemias and lymphomas
PTCH	Basal cell carcinoma
PTEN	Brain tumors; melanoma; prostate, endometrial, kidney, and lung carcinomas
Rb	Retinoblastoma; sarcomas; bladder, breast, and lung carcinomas
Smad2	Colon/rectum carcinoma
Smad4	Colon/rectum carcinoma, pancreatic carcinoma
TβRII	Colon/rectum carcinoma, gastric carcinoma
VHL	Renal cell carcinoma
WT1	Wilms' tumor

The second tumor suppressor gene to have been identified is ***p53***, which is frequently inactivated in a wide variety of human cancers, including leukemias, lymphomas, sarcomas, brain tumors, and carcinomas of many tissues, including breast, colon, and lung. In total, mutations of *p53* play a role in about 50% of all cancers, making it the most common target of genetic alterations in human malignancies. It is also of interest that inherited mutations of *p53* are responsible for genetic transmission of a rare hereditary cancer syndrome in which affected individuals develop any of several different types of cancer. In addition, the p53 protein (like Rb) is a target for the oncogene proteins of SV40, adenoviruses, and human papillomaviruses.

■ ***p53* was initially thought to be an oncogene because mutated *p53* genes found in many cancer cells induced transformation in gene transfer assays. Subsequent studies revealed that *p53* was actually a tumor suppressor and that the mutant *p53* genes found in many tumors acted as dominant negatives that induced transformation by interfering with normal *p53* function.**

Like *p53*, the *INK4* and *PTEN* tumor suppressor genes are very frequently mutated in several common cancers, including lung cancer, prostate cancer, and melanoma. Other tumor suppressor genes (including *APC*, *TβRII*, *Smad2*, and *Smad4*) are frequently inactivated in colon cancers. In addition to being involved in noninherited cases of this common adult cancer, inherited mutations of the *APC* gene are responsible for a rare hereditary form of colon cancer, called familial adenomatous polyposis. Individuals with this condition develop hundreds of benign colon adenomas (polyps), some of which inevitably progress to malignancy. Inherited mutations of two other tumor suppressor genes, *BRCA1* and *BRCA2*, are responsible for hereditary cases of breast cancer, which account for about 5% of total breast cancer incidence. Additional tumor suppressor genes have been implicated in the development of brain tumors, pancreatic cancers, and basal cell skin carcinomas, as well as in several rare inherited cancer syndromes, such as Wilms' tumor.

Functions of Tumor Suppressor Gene Products

In contrast to proto-oncogene and oncogene proteins, the proteins encoded by most tumor suppressor genes inhibit cell proliferation or survival. Inacti-

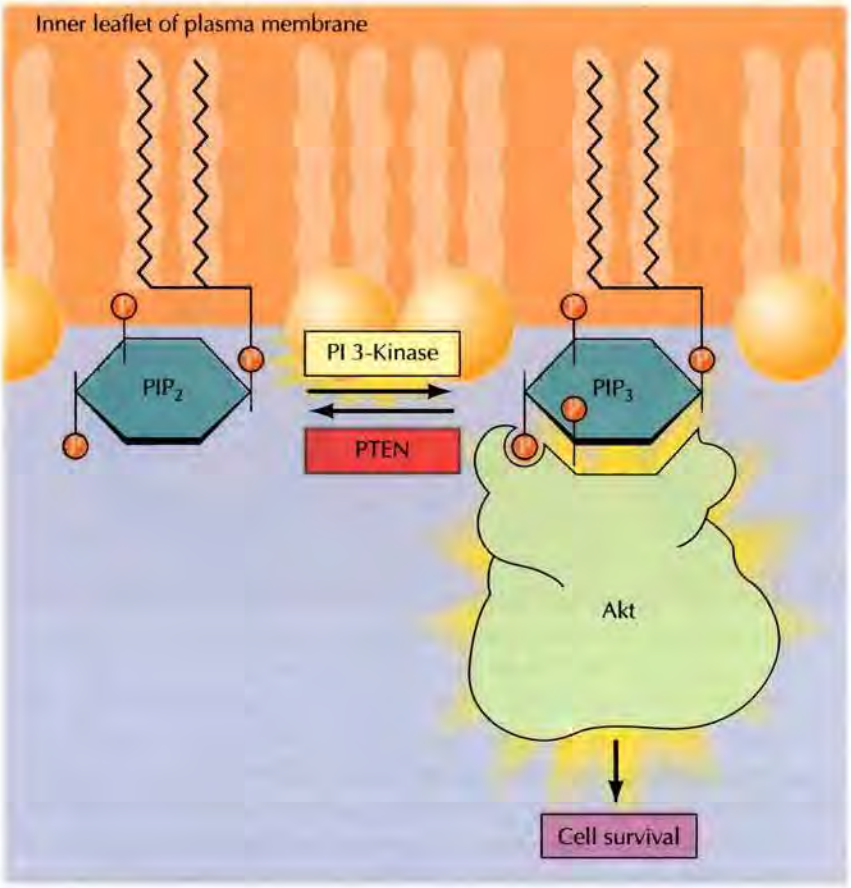


FIGURE 18.36 Suppression of cell survival by PTEN The tumor suppressor protein PTEN is a lipid phosphatase that dephosphorylates PIP_3 at the 3 position of inositol, yielding PIP_2. PTEN thus counters the action of the oncogene proteins PI 3-kinase and Akt, which promote cell survival.

vation of tumor suppressor genes therefore leads to tumor development by eliminating negative regulatory proteins. In many cases, tumor suppressor proteins inhibit the same cell regulatory pathways that are stimulated by the products of oncogenes.

The protein encoded by the ***PTEN*** tumor suppressor gene is an interesting example of antagonism between oncogene and tumor suppressor gene products (Figure 18.36). The PTEN protein is a lipid phosphatase that dephosphorylates the 3 position of phosphatidylinositides, such as phosphatidylinositol 3,4,5-bisphosphate (PIP_3). By dephosphorylating PIP_3, PTEN antagonizes the activities of PI 3-kinase and Akt, both of which can act as oncogenes by promoting cell survival. Conversely, inactivation or loss of the PTEN tumor suppressor protein can contribute to tumor development as a result of increased levels of PIP_3, activation of Akt, and inhibition of programmed cell death.

Proteins encoded by both oncogenes and tumor suppressor genes also function in the Hedgehog signaling pathway (see Figure 15.43). The receptor Smoothened is an oncogene in basal cell carcinomas, whereas Patched (the negative regulator of Smoothened) is a tumor suppressor gene. In addition, the Gli proteins (the mammalian homologs of the *Drosophila* Ci transcription factor activated by Smoothened) were first identified as the products of an amplified oncogene.

Several tumor suppressor genes encode transcriptional regulatory proteins. A good example is provided by the product of *WT1*, which is frequently inactivated in Wilms' tumors (a childhood kidney tumor). The WT1 protein is a repressor that appears to suppress transcription of a number of

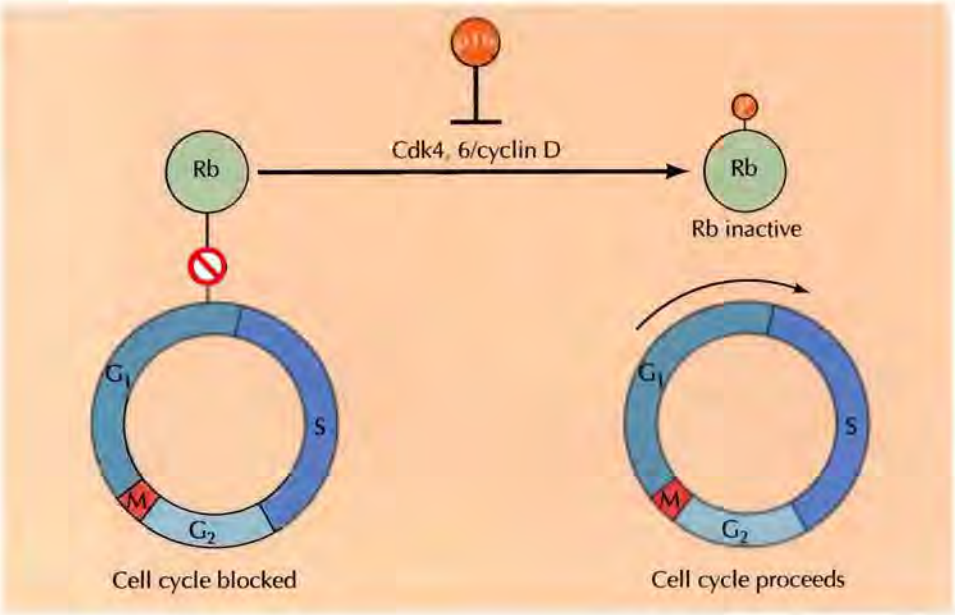


FIGURE 18.37 Inhibition of cell cycle progression by Rb and p16 Rb inhibits progression past the restriction point in G_1. Cdk4, 6/cyclin D complexes promote passage through the restriction point by phosphorylating and inactivating Rb. The activity of Cdk4, 6/cyclin D is inhibited by p16. Rb and p16 are tumor suppressors, whereas cyclin D1 and Cdk4 are oncogenes.

growth factor–inducible genes. One of the targets of WT1 is thought to be the gene that encodes insulin-like growth factor II, which is overexpressed in Wilms' tumors and may contribute to tumor development by acting as an autocrine growth factor. Inactivation of WT1 may thus lead to abnormal growth factor expression, which in turn drives tumor cell proliferation. Two other tumor suppressor genes, *Smad2* and *Smad4*, encode transcription factors that are activated by TGF-β signaling and lead to inhibition of cell proliferation (see Figure 15.41). Consistent with the activity of the TGF-β pathway in inhibiting cell proliferation, the TGF-β receptor is also encoded by a tumor suppressor gene (*TβRII*).

The products of the *Rb* and *INK4* tumor suppressor genes regulate cell cycle progression at the same point as that affected by cyclin D1 and Cdk4, both of which can act as oncogenes (Figure 18.37). Rb inhibits passage through the restriction point in G_1 by repressing transcription of a number of genes involved in cell cycle progression and DNA synthesis (see Figure 16.18). In normal cells, passage through the restriction point is regulated by Cdk4, 6/cyclin D complexes, which phosphorylate and inactivate Rb. Mutational inactivation of *Rb* in tumors thus removes a key negative regulator of cell cycle progression. The *INK4* tumor suppressor gene, which encodes the Cdk inhibitor p16, also regulates passage through the restriction point. As discussed in Chapter 16, p16 inhibits Cdk4, 6/cyclin D activity. Inactivation of *INK4* therefore leads to elevated activity of Cdk4, 6/cyclin D complexes, resulting in uncontrolled phosphorylation of Rb.

The *p53* gene product regulates both cell cycle progression and apoptosis. DNA damage leads to rapid induction of p53, which activates transcription of both proapoptotic and cell cycle inhibitory genes (Figure 18.38). The effects of p53 on apoptosis are mediated in part by activating the transcription of proapoptotic members of the Bcl-2 family (PUMA and Noxa) that induce programmed cell death. Unrepaired DNA damage normally induces apoptosis of mammalian cells, a response that is presumably advantageous to the organism because it eliminates cells carrying potentially deleterious mutations (e.g., cells that might develop into cancer cells).

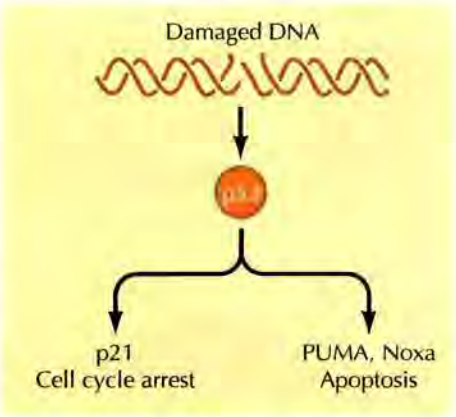


FIGURE 18.38 Action of p53 Wild-type p53 is required for both cell cycle arrest and apoptosis induced by DNA damage. Cell cycle arrest is mediated by induction of the Cdk inhibitor p21 and apoptosis by induction of the proapoptotic Bcl-2 family members PUMA and Noxa.

Cells lacking p53 fail to undergo apoptosis in response to agents that damage DNA, including radiation and many of the drugs used in cancer chemotherapy. This failure to undergo apoptosis in response to DNA damage contributes to the resistance of many tumors to chemotherapy. In addition, loss of p53 appears to interfere with apoptosis induced by other stimuli, such as growth factor deprivation and oxygen deprivation. These effects of p53 on cell survival are thought to account for the high frequency of *p53* mutations in human cancers.

In addition to inducing apoptosis, p53 blocks cell cycle progression in response to DNA damage by inducing the Cdk inhibitor p21 (see Figure 16.21). The p21 protein blocks cell cycle progression by acting as a general inhibitor of Cdk/cyclin complexes, and the resulting cell cycle arrest presumably allows time for damaged DNA to be repaired before it is replicated. Loss of p53 prevents this damage-induced cell cycle arrest, leading to increased mutation frequencies and a general instability of the cell genome. Such genetic instability is a common property of cancer cells, and it may contribute to further alterations in oncogenes and tumor suppressor genes during tumor progression.

Although their function remains to be fully understood, the products of the *BRCA1* and *BRCA2* genes, which are responsible for some inherited breast and ovarian cancers, also appear to be involved in checkpoint control of cell cycle progression and repair of double-strand breaks in DNA (see Figure 6.27). *BRCA1* and *BRCA2* thus function as **stability genes**, which act to maintain the integrity of the genome. Mutations in genes of this type lead to the development of cancer not as a result of direct effects on cell proliferation or survival but because their inactivation leads to a high frequency of mutations in oncogenes or tumor suppressor genes. Other stability genes whose loss contributes to the development of human cancers include the *ATM* gene, which acts at the DNA damage checkpoint (see Figure 16.20), the mismatch repair genes that are defective in some inherited colorectal cancers (see Chapter 6, Molecular Medicine), and the nucleotide excision repair genes that are mutated in xeroderma pigmentosum (see Figure 6.23).

Roles of Oncogenes and Tumor Suppressor Genes in Tumor Development

As discussed earlier, the development of cancer is a multistep process in which normal cells gradually progress to malignancy. Mutations resulting in both the activation of oncogenes and the inactivation of tumor suppressor genes are critical steps in tumor initiation and progression. Accumulated damage to multiple genes eventually results in the increased proliferation, survival, invasiveness, and metastatic potential that are characteristic of cancer cells.

The role of multiple genetic defects is best understood in the case of colon carcinomas, which have been studied extensively by Bert Vogelstein and his colleagues. These tumors frequently involve mutations of oncogenes or tumor suppressor genes with four distinct activities: 1) *ras* or *raf* oncogenes affecting the ERK pathway, 2) tumor suppressor or oncogene

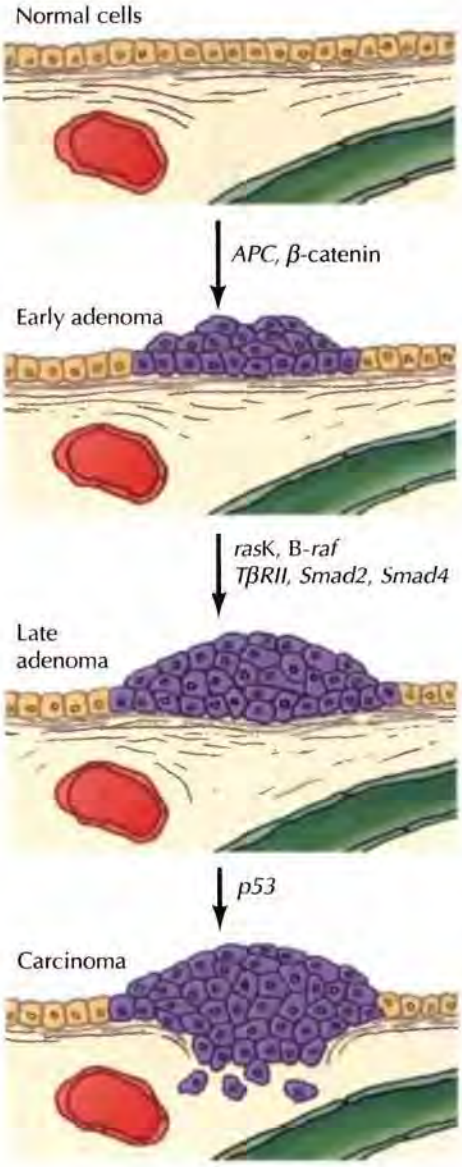


FIGURE 18.39 Genetic alterations in colon carcinomas Inactivation of APC (a component of the Wnt signaling pathway) is an early event in tumor development, giving rise to a proliferative cell population. In some cases, mutations of the gene encoding β-catenin (which is downstream of APC in Wnt signaling) occur instead. Development of adenomas is then associated with mutations in *ras*K or B-*raf* genes (which stimulate ERK signaling) and with genes encoding components of the TGF-β pathway (TGF-β receptor, Smad2 or Smad4). Mutations of *p53* are associated with later stages of tumor progression.

components of the Wnt pathway, 3) tumor suppressor proteins involved in TGF-β signaling, and 4) *p53*. Lesions representing multiple stages of colon cancer development are regularly obtained as surgical specimens, so it has been possible to correlate these genetic alterations with discrete stages of tumor progression (Figure 18.39).

These studies indicate that inactivation of *APC* (a component of the Wnt signaling pathway; see Figure 18.28) is an early event in tumor development. Genetic transmission of mutant *APC* genes in patients with familial adenomatous polyposis results in abnormal colon cell proliferation, leading to the outgrowth of multiple adenomas in the colons of affected patients. Mutations of *APC* also occur frequently in patients with noninherited colon carcinomas and are generally detected at early stages of the disease process. In some cases, activation of Wnt signaling results from mutations in the gene encoding β-catenin (which is downstream of APC in the Wnt pathway), rather than from mutations in *APC*. Wnt signaling is a major pathway responsible for proliferation of colon epithelial stem cells, and it appears that the constitutive unregulated activity of this pathway initiates the development of colon cancers.

Mutations of *ras*K genes then appear to occur, and *ras*K oncogenes are frequently present in colon adenomas. Some colon tumors have mutations leading to activation of a member of the *raf* oncogene family (B-*raf*), rather than mutations of *ras*K. Since Raf is immediately downstream of Ras, activation of either the *ras*K or B-*raf* oncogene leads to stimulation of ERK signaling in the tumor cells.

Almost all colon cancers have mutations affecting the TFG-β signaling pathway, which inhibits cell proliferation by inducing the synthesis of p15, a member of the Ink4 family of Cdk inhibitors. Mutations affecting TGF-β signaling also appear to occur relatively early in the development of colon cancers, and are frequently found in adenomas. In some tumors, mutations inactivate the *Tβ RII* tumor suppressor gene, which encodes the TGF-β receptor. In other cases, mutations inactivate tumor suppressor genes encoding the Smad2 or Smad4 transcription factors, which are targets of TFG-β signaling. Finally, the *p53* tumor suppressor gene is inactivated, usually at a later stage of tumor progression.

Accumulated damage to multiple oncogenes and tumor suppressor genes, affecting distinct pathways that regulate cell proliferation and survival, thus appears to be responsible for the multistage development of colon cancer. Accumulated damage to both oncogenes and tumor suppressor genes affecting distinct pathways of cell regulation similarly appears to be responsible for the development of other types of cancer, including breast and lung carcinomas. The progressive loss of growth control that is characteristic of cancer cells is thus thought to be the end result of abnormalities in the products of multiple genes that normally regulate cell proliferation and survival.

Molecular Approaches to Cancer Treatment

A great deal has been learned about the molecular defects responsible for the development of human cancers. However, cancer is more than a topic of scientific interest. It is a dread disease that claims the lives of nearly one out of every four Americans. Translating our growing understanding of cancer into practical improvements in cancer prevention and treatment therefore represents a major challenge for current and future research. Fortunately, advances in elucidating the molecular biology of cancer have begun to contribute to the development of new approaches to its prevention and treatment, which promise to ultimately yield major advances in dealing with this disease.

Prevention and Early Detection

The most effective way to deal with cancer would be to prevent development of the disease. A second-best, but still effective, alternative would be to reliably detect early premalignant stages of tumor development that can be easily treated. Many cancers can be cured by localized treatments, such as surgery or radiation, if they are detected before they spread throughout the body by metastasis. For example, early premalignant stages of colon cancer (adenomas) are usually completely curable by relatively minor surgical procedures (Figure 18.40). The cure rate for early carcinomas that remain localized to their site of origin is also high, about 90%. However, survival rates drop to about 50% for patients whose cancers have spread to adjacent tissues and lymph nodes, and to less than 10% for patients with metastatic colon cancer. Early detection of cancer can thus be a critical determinant of the outcome of the disease.

The major application of molecular biology to prevention and early detection may lie in the identification of individuals with inherited susceptibilities to cancer development. Such inherited cancer susceptibilities can result from mutations in tumor suppressor genes, in at least two oncogenes (*ret* and *cdk4*), and in stability genes, such as the mismatch repair genes responsible for development of hereditary nonpolyposis colon cancer (see Molecular Medicine in Chapter 6). Mutations in these genes can be detected by genetic testing, allowing the identification of high-risk individuals before disease develops.

In addition to contributing to family planning decisions, careful monitoring of high-risk individuals may allow early detection and more effective treatment of some types of cancer. For example, colon adenomas can be detected by colonoscopy and removed prior to the development of malignancy. Patients with familial adenomatous polyposis (resulting from inherited mutations of the *APC* tumor suppressor gene) typically develop hundreds of adenomas within the first 20 years of life, so the colons of these patients are usually removed before the inevitable progression of some of these polyps to malignancy. However, patients with hereditary nonpolyposis colon cancer develop a smaller number of polyps later in life and may therefore benefit from routine colonoscopy and drugs, such as nonsteroidal anti-inflammatory drugs (NSAIDs), that inhibit colon cancer development.

The direct inheritance of cancers resulting from defects in known genes is a rare event, constituting about 5% of total cancer incidence. The most common inherited cancer susceptibility is hereditary nonpolyposis colon cancer, which accounts for about 15% of colon cancers and 1 to 2% of all cancers in the United States. Mutations in the *BRCA1* and *BRCA2* tumor suppressor genes are also relatively common, accounting for approximately

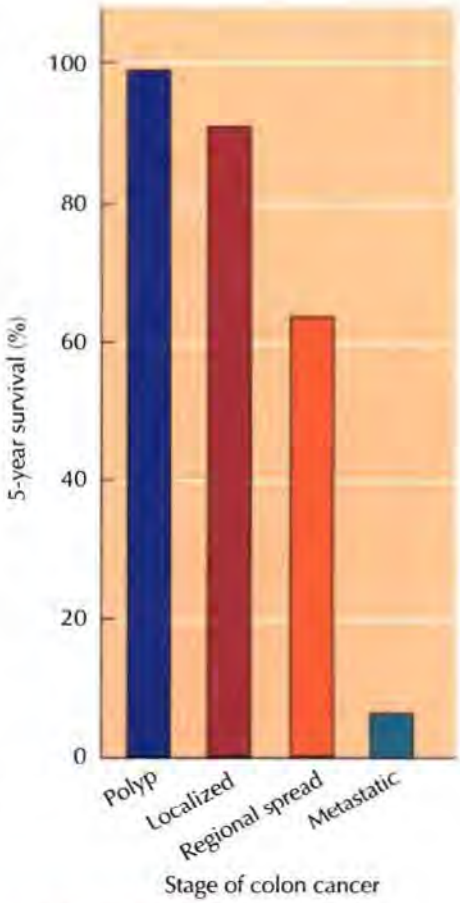


FIGURE 18.40 Survival rates of patients with colon carcinoma Five-year survival rates are shown for patients diagnosed with adenomas (polyps), with carcinoma still localized to its site of origin, with carcinoma that has spread regionally to adjacent tissues and lymph nodes, and with metastatic carcinoma.

5% of all breast cancers. However, still-unidentified genes conferring cancer susceptibility may contribute to the development of a larger fraction of common adult malignancies, such as breast, colon, and lung cancers. The continuing isolation of cancer susceptibility genes is thus an important undertaking with clear practical implications. Individuals with such inherited susceptibility genes could be appropriately advised to avoid exposure to relevant carcinogens (e.g., tobacco smoke in the case of lung cancer) and carefully monitored to detect tumors at early stages that are more readily treated. The reliable identification of susceptible individuals, if it were followed by appropriate preventive and early detection measures, might ultimately make a significant impact on cancer mortality.

Molecular Diagnosis

Molecular analysis of the oncogenes and tumor suppressor genes involved in particular types of tumors has the potential of providing information that is useful in the diagnosis of cancer and in monitoring the effects of treatment. Indeed, several applications of such molecular diagnosis have already been put into clinical practice. In some cases, mutations in oncogenes have provided useful molecular markers for monitoring the course of disease during treatment. The translocation of *abl* in chronic myeloid leukemia is a good example. As previously discussed, this translocation results in the fusion of *abl* with the *bcr* gene, leading to expression of the Bcr/Abl oncogene protein (see Figure 18.24). The polymerase chain reaction (see Figure 4.23) provides a sensitive method of detecting the recombinant *bcr/abl* oncogene in leukemic cells and is therefore used to monitor the response of patients to treatment.

In other cases, the detection of mutations in specific oncogenes or tumor suppressor genes may provide information pertinent to choosing between different therapeutic options. For example, amplification of N-*myc* in neuroblastomas and *erb*B-2 in breast and ovarian carcinomas predicts rapid disease progression. Therefore, it might be appropriate to treat patients with such amplified oncogenes more aggressively. As discussed in the following section, mutations in specific oncogenes, such as *erb*B, may also dictate the response of tumors to oncogene-targeted drugs.

In addition to analysis of individual genes, important diagnostic information may be gleaned from global analysis of gene expression in cancers. The use of DNA microarrays allows the expression of tens of thousands of genes to be analyzed simultaneously (see Figure 4.27), so it is possible to develop a molecular classification of cancers by comparing the gene expression profiles of different tumors. Studies of this type suggest that gene expression profiling can distinguish between otherwise similar tumors and provide information that predicts clinical outcome and response to treatment. Characterization of tumors by gene expression profiling may thus become a useful tool in cancer diagnosis.

Treatment

The most critical question, however, is whether the discovery of oncogenes and tumor suppressor genes will allow the development of new drugs that act selectively against cancer cells. Most of the drugs currently used in cancer treatment either damage DNA or inhibit DNA replication. Consequently, these drugs are toxic not only to cancer cells but also to normal cells, particularly those normal cells that are continually replaced by the division of stem cells (e.g., hematopoietic cells, epithelial cells of the gastrointestinal tract, and hair follicle cells). The action of anticancer drugs

against these normal cell populations accounts for most of the toxicity associated with these drugs and limits their effective use in cancer treatment.

One promising new approach to cancer therapy is the use of drugs that inhibit tumor growth by interfering with angiogenesis (blood vessel formation) or disrupting tumor blood vessels, rather than acting directly against cancer cells. As noted earlier in this chapter, the formation of new blood vessels is needed to supply the oxygen and nutrients required for tumor growth. Promoting angiogenesis is thus critical to tumor development, and tumor cells secrete a number of growth factors, including VEGF, that stimulate the proliferation of capillary endothelial cells, resulting in the outgrowth of new capillaries into the tumor (see Figure 17.14). The importance of angiogenesis was first recognized by Judah Folkman in 1971, and continuing research by Folkman and his colleagues has led to the development of new drugs that inhibit angiogenesis by blocking the proliferation of endothelial cells. Because these drugs act specifically to inhibit the formation of new blood vessels, they are less toxic to normal cells than standard anticancer agents. Angiogenesis inhibitors have shown promising results in animal tests and are currently being tested in clinical trials to evaluate their effectiveness against human cancers. In 2004, positive clinical results led the US Food and Drug Administration (FDA) to approve the use of the first angiogenesis inhibitor, a monoclonal antibody against VEGF, for treatment of colon cancer.

An alternative strategy for achieving more selective cancer treatment is the development of drugs targeted specifically against the oncogenes that drive tumor growth. Unfortunately, from the standpoint of cancer treatment, oncogenes are not unique to tumor cells. Since proto-oncogenes play important roles in normal cells, general inhibitors of oncogene expression or function are likely to act against normal cells as well as tumor cells. The exploitation of oncogenes as targets for anticancer drugs is therefore not a straightforward proposition, but several promising advances indicate that it is possible to develop selective oncogene targeted therapies (Table 18.6).

The first therapeutic regimen targeted against a specific oncogene is used for the treatment of acute promyelocytic leukemia. This leukemia is characterized by a chromosome translocation in which the gene that encodes the retinoic acid receptor (*RARα*) is fused to another gene (*PML*) to form the *PML/RARα* oncogene. The PML/RARα protein is thought to function as a transcriptional repressor that blocks cell differentiation. These leukemic cells, however, differentiate in response to treatment with high doses of

TABLE 18.6 Oncogene Targeted Therapies Approved for Clinical Use

Drug	Oncogene	Tumor Types
Retinoic acid	*PML/RARα*	Acute promyelocytic leukemia
Herceptin	*erb*B-2	Breast cancer
Erbitux	*erb*B	Colorectal cancer
STI-571	*abl*	Chronic myelogenous leukemia
	kit	Gastrointestinal stromal tumors
	PDGFR	Gastrointestinal stromal tumors, chronic myelomonocytic leukemia, hypereosinophilic syndrome, dermatofibrosarcoma protuberans
Gefitinib	*erb*B	Lung cancer

retinoic acid, which binds to and inactivates the PML/RARα oncogene protein. Such treatment with retinoic acid results in remission of the leukemia in most patients, although this favorable response is temporary and patients eventually relapse. However, combined treatment with retinoic acid and standard chemotherapeutic agents significantly reduces the incidence of relapse, so the use of retinoic acid is of substantial benefit in the treatment of acute promyelocytic leukemia. The therapeutic activity of retinoic acid was observed prior to identification of the *PML/RARα* oncogene, so its effectiveness against leukemic cells expressing this oncogene protein was discovered by chance rather than by rational drug design. Nonetheless, the use of retinoic acid for treatment of acute promyelocytic leukemia provides the first example of a clinically useful drug targeted against an oncogene protein.

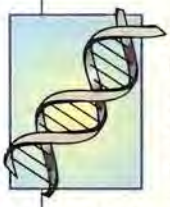
Herceptin, a monoclonal antibody against the ErbB-2 oncogene protein, was the first drug developed against a specific oncogene to achieve FDA approval for clinical use in cancer treatment. The ErbB-2 protein is overexpressed in about 30% of breast cancers as a result of amplification of the *erb*B-2 gene. It was first found that an antibody against the extracellular domain of ErbB-2 (a receptor protein-tyrosine kinase) inhibited the proliferation of tumor cells in which ErbB-2 was overexpressed. These results led to the development and clinical testing of Herceptin, which was found to significantly reduce tumor growth and prolong patient survival in clinical trials involving over 600 women with metastatic breast cancers that overexpressed the ErbB-2 protein. Based on these results, Herceptin was approved by the FDA in 1998 for treatment of metastatic breast cancers that express elevated levels of ErbB-2. Erbitux, a monoclonal antibody against the EGF receptor (the ErbB oncogene protein), was also approved by the FDA in 2004 for use in treatment of advanced colorectal cancer.

The therapeutic use of monoclonal antibodies is limited to extracellular targets, such as growth factors or cell surface receptors. A more widely applicable area of drug development is the identification of small-molecule inhibitors of oncogene proteins, including the protein kinases that play key roles in signaling the proliferation and survival of cancer cells. The pioneering advance in this area was the development of a selective inhibitor of the Bcr/Abl protein-tyrosine kinase, which is generated by the Philadelphia chromosome translocation in chronic myeloid leukemia (see Figure 18.24). Brian Druker and his colleagues developed a potent and specific inhibitor of the Bcr/Abl protein kinase and showed that this compound (called STI-571 or Gleevec) effectively blocks proliferation of chronic myeloid leukemia cells. Based on these results, a clinical trial of STI-571 was initiated in 1998. The responses to STI-571 were remarkable, and the drug had minimal side effects. The striking success of STI-571 in these clinical studies served as the basis for its rapid approval by the FDA in 2001 for use in the treatment of chronic myeloid leukemia. Although some patients relapse and develop resistance against the drug, STI-571 is unquestionably a highly effective therapy for this leukemia. Interestingly, resistance to STI-571 most often results from mutations of the Bcr/Abl protein kinase domain that prevent STI-571 binding. By analysis of these resistant mutants, it has been possible to design new inhibitors, which are currently being tested in clinical trials to determine their effectiveness against leukemias that have become resistant to STI-571.

STI-571 is also a potent inhibitor of the PDGF receptor and Kit protein-tyrosine kinases, and it has proven to be an effective therapy for tumors in

MOLECULAR MEDICINE

STI-571: Cancer Treatment Targeted against the bcr/abl Oncogene



The Disease

Chronic myeloid leukemia (CML) accounts for approximately 20% of leukemias in adults. In 2005, it was estimated that about 4600 cases of CML would be diagnosed in the United States, and that there would be about 850 deaths from this disease.

CML originates from the hematopoietic stem cell of the bone marrow. It is a slowly progressing disease, which is clinically divided into two stages: chronic phase and blast crisis. The chronic phase of CML can persist for years and is associated with minimal symptoms. Eventually, however, patients progress to an acute life-threatening stage of the disease known as blast crisis. Blast crisis is characterized by the accumulation of large numbers of rapidly proliferating leukemic cells, called blasts. Patients in blast crisis are treated with standard chemotherapeutic drugs, which may induce remission to the chronic phase of the disease. Chemotherapy may also be used during the chronic phase of CML but does not usually succeed in eliminating the leukemic cells. CML can also be treated by transplantation of bone marrow stem cells, which can be curative for about half of patients in the chronic phase of the disease.

Molecular and Cellular Basis

Activation of the *abl* oncogene by translocation from its normal locus on chromosome 9 to chromosome 22 is a highly reproducible event in CML, occurring in about 95% of these leukemias. This translocation results in formation of a fusion between *abl* and the *bcr* gene on chromosome 22, yielding the *bcr/abl* oncogene. The oncogene expresses a Bcr/Abl fusion protein in which Bcr amino acid sequences replace the first exon of Abl (see Figure 18.24). The Bcr/Abl protein is a constitutively active protein-tyrosine kinase, which leads to leukemia by activating a variety of downstream signaling pathways.

Because activation of the *bcr/abl* oncogene is such a reproducible event in the development of CML, it appeared to be a good candidate against which to develop a selective tyrosine kinase inhibitor that might be of clinical use. These studies led to the development of STI-571 (or Gleevec) as the first therapeutic drug successfully designed as a selective inhibitor of an oncogene protein.

Prevention and Treatment

The development of STI-571 started with the identification of 2-phenylaminopyrimidine as a non-specific inhibitor of protein kinases. A series of related compounds were then synthesized and optimized for activity against different targets, including the Abl tyrosine kinase. Among many compounds screened in these investigations, STI-571 was found to be a potent and specific inhibitor of Abl and two other protein-tyrosine kinases: the platelet-derived growth factor receptor and c-Kit. Further studies demonstrated that STI-571 specifically inhibited the proliferation of cells transformed by *bcr/abl* oncogenes, including cells from CML patients in culture. In addition, STI-571 prevented tumor formation by *bcr/abl*-transformed cells in mice. In contrast, normal cells or cells transformed by other oncogenes were not affected by STI-571, demonstrating its specificity against the Bcr/Abl tyrosine kinase.

Based on these results, an initial phase I clinical study of STI-571 was initiated in June, 1998. The results were strikingly positive. Of 54 chronic phase patients treated with STI-571, 53 responded to the drug. Moreover, responses to STI-571 were seen in more than half of treated blast crisis patients. The success of these initial studies prompted expanded Phase II studies involving over 1000 patients. These studies confirmed the promising results of the phase I study, with 95% of chronic phase patients and approximately 50% of blast crisis patients responding to STI-571. Moreover, in contrast to conventional chemotherapeutic drugs, STI-571 had minimal side effects. These clinical studies clearly demonstrated that STI-571 is a highly effective therapy for CML, and led to accelerated FDA approval of STI-571 in May 2001—a milestone in the translation of basic science to clinical practice.

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Crystal structure of the catalytic domain of Abl complexed with a derivative of STI-571. (From T. Schindler, W. Bornmann, P. Pellicena, W. T. Miller, B. Clarkson and J. Kuriyan. 2000. *Science* 289: 1938.)

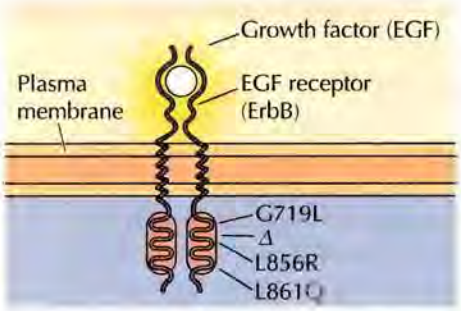

FIGURE 18.41 EGF receptor mutations associated with sensitivity to gefitinib Lung cancers that respond to gefitinib have activating mutations within the EGF receptor kinase domain. These mutations include point mutations (G719L, L856R, and L861Q) and small deletions (Δ).

which the genes encoding these protein kinases are mutationally activated oncogenes. *Kit* is activated as an oncogene by point mutations resulting in constitutive protein kinase activity in approximately 90% of gastrointestinal stromal tumors, which are tumors of stromal connective tissue of the stomach and small intestine. Many of the gastrointestinal stromal tumors that do not have activating mutations of Kit instead have activating mutations of the PDGF receptor. Consequently, gastrointestinal stromal tumors are highly responsive to STI-571. In addition, STI-571 is active against three other types of tumors in which the PDGF receptor is activated as an oncogene, including chronic myelomonocytic leukemia in which it is activated by fusion with the Tel transcription factor (see Figure 18.26).

A small molecule inhibitor of the EGF receptor (gefitinib) has recently shown striking activity against a subset of lung cancers in which the EGF receptor is activated by point mutations. It is noteworthy that the rationale for treatment of lung cancers with gefitinib was the fact that EGF receptors are overexpressed in most lung cancers, rather than the fact that they can be mutationally activated as oncogenes. In contrast to the general effectiveness of STI-571 against chronic myeloid leukemia, clinical studies indicated that gefitinib was effective in only about 10% of lung cancer patients, although the responses in these patients were quite dramatic. A major advance was thus made in 2004 when two groups of researchers found that the subset of lung cancers that responded to gefitinib were those in which point mutations resulted in constitutive activation of the EGF receptor tyrosine kinase (Figure 18.41). These results indicated that inhibition of the EGF receptor was an effective treatment for tumors in which it had been mutated to act as an oncogene but not for tumors expressing the normal protein.

The finding that gefitinib is active against only those lung cancers with mutationally activated EGF receptors supports the hypothesis that tumors with activated oncogenes are particularly susceptible to inhibitors of those oncogenes. This is also consistent with the activity of STI-571 against tumors with mutationally activated Bcr/Abl, Kit, and PDGF receptor oncogene proteins, as well as with a variety of experiments in animal model systems. The sensitivity of tumors to inhibition of activated oncogenes has been referred to as **oncogene addiction**. It is thought that an activated oncogene becomes a major driving force in the tumor cell, such that other signaling pathways in the tumor cell become secondary in importance. Consequently, the proliferation and survival of a tumor cell may become dependent on continuing activity of the oncogene, whereas normal cells have alternative signaling pathways that can compensate if any one pathway is blocked.

The examples of STI-571 and gefitinib clearly suggest that the continuing exploitation of oncogenes as targets for drug development has the potential of leading to a new generation of drugs that act selectively against cancer cells. Indeed, a wide variety of drugs targeted against both oncogene proteins (including B-Raf, PI-3 kinase, and Akt) and downstream components of oncogenic signaling pathways (such as MEK and mTOR) are being tested to evaluate their potential in the treatment of human cancer. The apparent dependence of cancer cells on mutationally activated oncogenes further offers the promise that the use of oncogene targeted drugs combined with genetic analysis of the tumors of individual patients may lead to major advances in cancer treatment. Although the eventual impact of molecular biology on the treatment of cancer remains to be seen, it is clear that the rational design of drugs targeted against specific oncogene proteins will play an important role.

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

SUMMARY

THE DEVELOPMENT AND CAUSES OF CANCER

Types of Cancer: Cancer can result from the abnormal proliferation of any type of cell. The most important distinction for the patient is between benign tumors, which remain confined to their site of origin, and malignant tumors, which can invade normal tissues and spread throughout the body.

cancer, tumor, benign tumor, malignant tumor, metastasis, carcinoma, sarcoma, leukemia, lymphoma

The Development of Cancer: Tumors develop from single altered cells that begin to proliferate abnormally. Additional mutations lead to the selection of cells with progressively increasing capacities for proliferation, survival, invasion, and metastasis.

tumor initiation, tumor progression, adenoma, polyp

Causes of Cancer: Radiation and many chemical carcinogens act by damaging DNA and inducing mutations. Other chemical carcinogens contribute to the development of cancer by stimulating cell proliferation. Viruses also cause cancer in both humans and other species.

carcinogen, tumor promoter

Properties of Cancer Cells: The uncontrolled proliferation of cancer cells is reflected in reduced requirements for extracellular growth factors and lack of inhibition by cell-cell contact. Many cancer cells are also defective in differentiation, consistent with their continued proliferation *in vivo*. The characteristic failure of cancer cells to undergo apoptosis also contributes substantially to tumor development.

density-dependent inhibition, autocrine growth stimulation, contact inhibition, angiogenesis, programmed cell death, apoptosis

Transformation of Cells in Culture: The development of *in vitro* assays for cell transformation has allowed the conversion of normal cells into tumor cells to be studied in cell culture.

cell transformation

TUMOR VIRUSES

Hepatitis B and C Viruses: The hepatitis B and C viruses cause liver cancer in humans.

tumor virus, hepatitis B virus, hepatitis C virus

SV40 and Polyomavirus: Although neither SV40 nor polyomavirus causes human cancer, they are important models for studying the molecular biology of cell transformation. SV40 T antigen induces transformation by interacting with the cellular Rb and p53 tumor suppressor proteins.

simian virus 40 (SV40), polyomavirus

Papillomaviruses: Papillomaviruses induce tumors in a variety of animals, including cervical carcinoma in humans. Like SV40 T antigen, the transforming proteins of papillomaviruses interact with Rb and p53.

papillomavirus

Adenoviruses: The adenoviruses do not cause naturally occurring cancers in either humans or other species but are important models in cancer research. Their transforming proteins also interact with Rb and p53.

adenovirus

Herpesviruses: The herpesviruses, which are among the most complex animal viruses, cause cancer in several species, including humans.

herpesvirus, Kaposi's sarcoma-associated herpesvirus, Epstein-Barr virus

Retroviruses: Retroviruses cause cancer in humans and a variety of other animals. Some retroviruses contain specific genes responsible for inducing cell transformation, and studies of these highly oncogenic retroviruses have led to the characterization of both viral and cellular oncogenes.

retrovirus, Rous sarcoma virus (RSV)

KEY TERMS

KEY TERMS

oncogene, *src*, *ras*, *raf*

proto-oncogene

***c-myc*, *abl*, N-*myc*, *erb*B-2**

Fos, Jun, *CCND1*, ErbA, PML/RARα, PI 3-kinase, Akt, Bcl-2

tumor suppressor gene, *Rb*, *p53*

stability gene

SUMMARY

ONCOGENES

Retroviral Oncogenes: The first oncogene to be identified was the *src* gene of RSV. Subsequent studies have identified more than two dozen distinct oncogenes in different retroviruses.

Proto-Oncogenes: Retroviral oncogenes originated from closely related genes of normal cells, called proto-oncogenes. The oncogenes are abnormally expressed or mutated forms of the corresponding proto-oncogenes.

Oncogenes in Human Cancer: A variety of oncogenes are activated by point mutations, DNA rearrangements, and gene amplification in human cancers. Some of these human tumor oncogenes, such as the *ras* genes, are cellular homologs of oncogenes that were first described in retroviruses.

Functions of Oncogene Products: Many oncogene proteins function as elements of signaling pathways that stimulate cell proliferation. The genes that encode cyclin D1 and Cdk4 can also act as oncogenes by stimulating cell cycle progression. Other oncogene proteins interfere with cell differentiation, and oncogenes encoding PI 3-kinase, Akt, and Bcl-2 inhibit apoptosis.

TUMOR SUPPRESSOR GENES

Identification of Tumor Suppressor Genes: In contrast to oncogenes, tumor suppressor genes inhibit tumor development. The prototype tumor suppressor gene, *Rb*, was identified by studies of inheritance of retinoblastoma. Loss or mutational inactivation of *Rb* and other tumor suppressor genes, including *p53*, contributes to the development of a wide variety of human cancers.

Functions of Tumor Suppressor Gene Products: The proteins encoded by most tumor suppressor genes act as inhibitors of cell proliferation or survival. The Rb, INK4, and p53 proteins are negative regulators of cell cycle progression. In addition, p53 is required for apoptosis induced by DNA damage and other stimuli, so its inactivation contributes to enhanced tumor cell survival. Some genes, such as *BRCA1* and *BRCA2*, act to maintain genomic stability rather than directly influencing cell proliferation.

Roles of Oncogenes and Tumor Suppressor Genes in Tumor Development: Mutations in both oncogenes and tumor suppressor genes contribute to the progressive development of human cancers. Accumulated damage to multiple such genes results in the abnormalities of cell proliferation, differentiation, and survival that characterize the cancer cell.

MOLECULAR APPROACHES TO CANCER TREATMENT

Prevention and Early Detection: Many cancers can be cured if they are detected at early stages of tumor development. Genetic testing to identify individuals with inherited cancer susceptibilities may allow early detection and more effective treatment of high-risk patients.

SUMMARY

Molecular Diagnosis: Detection of mutations in oncogenes and tumor suppressor genes may be useful in diagnosis and in monitoring response to treatment. Global analysis of gene expression may distinguish subclasses of cancers with differing clinical prognosis or response to treatment.

Treatment: The development of drugs targeted against specific oncogenes is beginning to lead to the discovery of new therapeutic agents that act selectively against cancer cells.

KEY TERMS

oncogene addiction

Questions

1. How does a benign tumor differ from a malignant tumor?

2. What is the role of clonal selection in the development of cancer?

3. How do estrogens increase the risk of cancer?

4. How does autocrine growth stimulation contribute to the progression of tumors?

5. What properties of cancer cells give them the ability to metastasize?

6. You have constructed a mutant SV40 T antigen that fails to induce cell transformation because it no longer binds Rb. Would this mutant T antigen induce transformation if introduced into cells together with a papillomavirus cDNA that encodes E6? How about one that encodes E7?

7. Why do AIDS patients have a high incidence of some types of cancer?

8. How can a proto-oncogene be converted to an oncogene without a change or mutation in its coding sequence? Explain two ways by which this can occur.

9. What effect would overexpression of the INK4 tumor suppressor have on tumor cells that express inactive Rb?

10. Which would you expect to be more sensitive to treatment with radiation – tumors with wild-type *p53* genes or tumors with mutated *p53* genes?

11. What is the mode of action of STI-571? How do some tumors develop resistance to this drug?

12. What is 'oncogene addiction' and why is this concept important for selecting molecular targets for cancer therapy?

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Answers to Questions

CHAPTER 1

1. That organic molecules, including several amino acids, can be formed spontaneously from a mixture of reducing gases.
2. RNA is uniquely capable of both serving as a template and catalyzing the chemical reactions required for its own replication.
3. Mitochondria and chloroplasts contain their own DNA and ribosomes, are similar in size to bacteria, and divide like bacteria. The ribosomal proteins and RNAs of these organelles are also more closely related to those of bacteria than to those encoded by eukaryotic nuclear genomes.
4. Because O_2 became abundant in Earth's atmosphere as a result of photosynthesis.
5. The volume of a sphere is given by the formula $4/3\pi r^3$, where r = the radius. Since the volume of the cells is proportional to the cube of their radii, the relative volume of a macrophage compared to *S. aureus* is the ratio of the cubes of their radii: $25^3/0.5^3$ = 125,000.
6. Yeast.
7. Mice.
8. The refractive index of air is 1.0; the refractive index of oil is approximately 1.4. Since resolution = $0.61/\eta \sin \alpha$ (η is the refractive index), viewing a specimen through air rather than through oil changes the limit of resolution from about 0.2 μm to about 0.3 μm.
9. Resolution = $0.61\lambda/NA$. Taking λ = 0.5 μm for visible light, the objective with an NA=1.3 will resolve objects about 0.23 µm apart, while the objective of NA =1.1 will resolve objects about 0.27 µm apart. Magnification of the images can be obtained by other means (such as projecting on a screen or using an ocular lens with higher magnification). Thus the better choice would be the objective with 60× magnification and the numerical aperture of 1.3.
10. GFP-tagged proteins can be expressed in live cells, so they can be seen without the need to fix and kill the cells for staining with a fluorescent antibody. Movement of GFP-tagged proteins can therefore be followed in living cells.
11. Velocity centrifugation separates organelles on the basis of size and shape, whereas equilibrium centrifugation separates them on the basis of density (independent of size and shape).
12. Serum contains growth factors that are required to stimulate division of most animal cells in culture.
13. Primary cell cultures are cells grown directly from an organism or tissue. Immortal cell lines have the ability to proliferate indefinitely in culture.
14. These cells retain the ability to differentiate into all of the different cell types of adult organisms, so they offer the potential of being used in transplantation therapies for treatment of a variety of diseases.

CHAPTER 2

1. Water is a polar molecule whose hydrogens and oxygen can form hydrogen bonds with each other (to make water a liquid at most of Earth's temperatures), as well as with polar organic molecules and inorganic ions. In contrast, nonpolar molecules or parts of molecules cannot interact with water, forcing them to associate with each other to form important structures like cell and organelle membranes.
2. Glycogen is a polymer of glucose residues connected primarily by $\alpha(1\rightarrow4)$ glycosidic bonds, and it has occasional $\alpha(1\rightarrow6)$ bonds that produce branches. Cellulose is a straight, unbranched polymer with $\beta(1\rightarrow4)$ glycosidic bonds between the glucose residues.

3. Fats accumulate in fat droplets of cells and are an efficient form of energy storage. Phospholipids function as the major components of cell membranes.
4. Nucleotides function as carriers of chemical energy (e.g., ATP) and as intracellular signaling molecules (e.g., cyclic AMP).
5. No. The information in nucleic acids is conveyed by the sequence of its bases. If the bases were removed, we would be left with a linear sugar phosphate chain with little informational content.
6. Christian Anfinsen and his colleagues showed that denatured ribonuclease can spontaneously refold into the active enzyme in the absence of other cellular constituents. This showed that the information required for folding was contained in the primary sequence of ribonuclease.
7. The side chain of cysteine contains a sulfhydryl group that can form covalent disulfide bonds with another cysteine residue, stabilizing the structure of cell surface and secreted proteins.
8. Cholesterol can be incorporated into membranes, where it regulates membrane fluidity. Additionally, cholesterol is used in the synthesis of steroid hormones.
9. The α-helical structure allows the CO and NH groups of peptide bonds to form hydrogen bonds with each other, thereby neutralizing their polar character. β-barrels are also capable of traversing lipid bilayers.
10. d. Alanine is the only amino acid on the list with a hydrophobic side chain capable of interacting with the fatty acids of membrane lipids.
11. In the first dimension, proteins are separated in a pH gradient according to their overall charge. In the second dimension, the proteins are separated on the basis of their mass.
12. In one approach, an organelle is purified by subcellular fractionation and the proteins contained within it are identified by mass spectrometry. Another approach, which has been used in the global analysis of yeast proteins, is to express a large number of proteins fused to GFP. The subcellular location of the fusion proteins can then be determined by fluorescence microscopy.

CHAPTER 3

1. Aspartate is an acidic amino acid that interacts with basic amino acids in the substrates of trypsin. Substitution of aspartate with lysine (a basic amino acid) would interfere with substrate binding and catalysis.
2. The side chain of histidine can be either uncharged or positively charged at physiological pH, thus allowing it to be used for exchange of hydrogen ions.
3. Enzyme E1 is probably regulated by feedback inhibition.
4. An energetically unfavorable reaction can be coupled to an energetically favorable reaction with a high negative free energy change (often the hydrolysis of ATP), such that the combined reaction is energetically favorable.
5. The reaction catalyzed by phosphofructokinase is fructose-6-phosphate + ATP $\leftrightarrow$ fructose-1,6-bisphosphate + ADP. The standard free energy change can be calculated as the sum of the standard free energy change of fructose-1,6-bisphosphate formation from fructose-6-phosphate and phosphate ($\Delta G^{o\prime} = +4$ kcal/mol) and the standard free energy change of ATP hydrolysis (–7.3 kcal/mol), giving a standard free energy change of –3.3 kcal/mol for the coupled reaction.
6. Substituting the given values into the equation:

$$\Delta G = \Delta G^{o} + RT \ln \frac{[B][C]}{[A]}$$

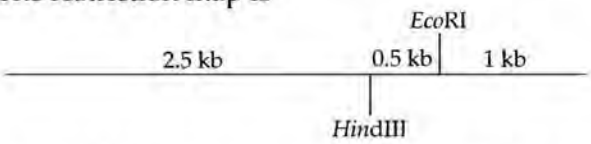
gives the following result: $\Delta G = -1.93$ kcal/mol. The reaction will therefore proceed from left to right, with A being converted to B plus C within the cell.

7. Phosphofructokinase is inhibited by high concentrations of ATP. Thus, glycolysis will be inhibited in response to an increased cellular concentration of ATP.
8. Under anaerobic conditions, glucose is metabolized only through glycolysis, with a net production of 2 ATP molecules per glucose molecule. Under aerobic conditions, glucose is completely oxidized to produce 36 – 38 molecules of ATP.
9. Under anaerobic conditions the NADH produced during glycolysis is used to reduce pyruvate to ethanol or lactate, thereby regenerating NAD^+.
10. Since lipids are more reduced than fatty acids, their oxidation yields much more energy per molecule.
11. In the light reactions, energy derived from the absorption of light by chlorophylls is used to split water to $2H^+ + 1/2\ O_2 + 2e^-$. The electrons enter an electron transport chain that results in synthesis of ATP and NADPH. In the dark reactions, ATP and NADPH drive the synthesis of glucose from CO_2 and H_2O.
12. Some reactions in glycolysis involve a large decrease in free energy and are not easily reversible. Gluconeogenesis bypasses these reactions by other reactions that are driven by the expenditure of ATP and NADH.

CHAPTER 4

1. You would breed flies with mutations in the two different genes. If mutations in the two genes are frequently inherited together, the genes would be linked and reside on the same chromosome.
2. Half the DNA will be intermediate density and half will be light.
3. Addition or deletion of one or two nucleotides would alter the reading frame of the gene, resulting in a completely different amino acid sequence of the encoded protein.

   Addition or deletion of three nucleotides adds or removes only one amino acid, often yielding a protein that functions normally.
4. To clone this piece of human DNA, a yeast artificial chromosome would need telomeres and a centromere in order to replicate as a linear chromosome-like molecule, as well as an origin of replication and a cleavage site for *Eco*RI.
5. UGC also encodes cysteine, so this mutation would have no effect on enzyme function. UGA, however, is a stop codon, so this mutation would lead to a truncated protein that would be inactive.
6. The restriction map is



7. The haploid sperm contains a single starting copy of the DNA sequence. Each cycle of PCR amplifies the starting material twofold, so 10 cycles yields 2^{10}, or approximately a thousand, copies. Amplification for 30 cycles yields 2^{30}, or more than a billion, copies.
8. From Table 4.3, cosmids carry inserts of 30–45 kb. From Table 4.2, the recognition sequence of *Bam*HI is 6 bp long, which occurs with a random frequency of once every 4000 base pairs of DNA ($4^6 = 4096$). Dividing 30,000–45,000 by 4000, we expect about 10 *Bam*HI fragments in a cosmid insert.
9. The human genome is approximately 3×10^6 kb, and the size of an insert for a BAC vector is 120–300 kb (Table 4.3). Dividing the size of the genome by the size of an insert indicates that a minimum of 10,000 to 25,000 BAC clones would be needed to cover the genome.
10. Influenza virus replicates via RNA-directed RNA synthesis, so actinomycin D (which inhibits DNA-directed RNA synthesis) will not affect influenza virus replication.
11. A selectable marker, such as drug resistance, that allows you to select stably transfected cells.
12. Proteins are dissolved in a solution containing the negatively charged detergent SDS. Each protein binds many molecules of SDS, which imparts a net negative charge to the protein. The proteins can then be separated by gel electrophoresis according to their size.

CHAPTER 5

1. Organisms with larger genomes than their complexity have larger amounts of noncoding DNA.
2. Sharp and coworkers made hybrids between adenovirus hexon mRNA and a single strand of viral DNA. Upon observation under an electron microscope, they saw that the hexon mRNA hybridized with separate regions of viral DNA and the intervening unhybridized DNA formed loops. The loops of single stranded DNA corresponded to introns that were spliced out of the long primary transcript.
3. Through alternative splicing more than one protein can be synthesized from the same gene, thereby increasing the diversity of proteins expressed from a limited number of genes.
4. Because of their distinct base composition, simple sequence repeats (e.g., ACAAACT) differ in density from the average AT/GC ratio of bulk nuclear DNA. Sequences containing tandem copies of these repeats will separate upon CsCl density-gradient centrifugation into satellite bands that are distinct from the main band of genomic DNA.
5. A centromere of *S. cerevisiae* is small (125 base pairs) and presumably binds a limited number of kinetochore proteins to form a single microtubule-binding site. Centromeres of animal chromosomes are much larger, with repeated sequences that form more extensive kinetochores and provide attachment sites for multiple microtubules.
6. Cutting the plasmid with a restriction endonuclease creates a linear chromosome with two free ends but no telomeres. The plasmid genes are quickly lost because of the instability of the chromosome ends. To test this, you could attach telomere sequences to the ends of the linearized plasmid and see if this allows the plasmid to be stably inherited.
7. The answer can be obtained by dividing the length of the human genome (3×10^6 kb) by the number of genes, after subtracting the length of DNA encompassed by the genes themselves (calculated as the number of genes multiplied by the average length of a gene [30 kb]). Thus the average distance between genes is about 100 kb.
8. One molecule of histone H1 binds genomic DNA approximately every 200 bp, so the number of histone H1 molecules bound to the yeast genome can be determined by dividing the size of the yeast genome (12 Mb) by 200 bp to yield a value of 60,000.

9. From Table 5.1, we can calculate the average length of a human intron by dividing the total length of intron sequence by the number of introns in an average human gene: approximately 3400 bp.
10. From Table 5.1, the exon sequence of an average human gene totals 2500 bp. This would be the expected average length of complete cDNA inserts in your library.
11. The International Human Genome Sequencing Consortium sequenced BAC clones that had already been mapped to distinct regions of human chromosomes. Celera Genomics used a whole-genome shotgun approach in which genomic DNA fragments were sequenced at random and overlaps between fragments were then used to reassemble a complete genome sequence.
12. Unlike protein coding sequences, regulatory sequences are short, poorly defined sequences that occur frequently by chance in large genomes, making it difficult to identify functional regulatory sequences. Approaches used to identify functional regulatory sequences include looking for clusters of regulatory elements, for sequences that are conserved in evolution, and for sequences that are present in coordinately regulated genes.
13. Single nucleotide polymorphisms (SNPs) are differences in single base pairs between the genomes of individuals. By studying SNPs, we may be able to identify specific genes associated with susceptibility to different diseases.

CHAPTER 6

1. At high temperature the mutant would have defects in replacing RNA primers with DNA in Okazaki fragments and in filling gaps in DNA following excision repair.
2. Topoisomerase I cuts one strand of the DNA double helix and allows it to swivel around the other strand to relieve twist tension. Topoisomerase II cuts both strands of the DNA double helix and can pass another DNA molecule through the cut to untangle intertwined molecules (e.g., during mitosis).
3. From Table 5.6, the size of the yeast genome is 12 Mb (12 million base pairs), while Okazaki fragments are approximately 1–3 kb (1,000–3,000 base pairs) in length. We can calculate the number of Okazaki fragments synthesized during replication by dividing the size of the genome by the length of Okazaki fragments, giving an answer of 4,000–12,000.
4. DNA polymerases are unable to initiate DNA synthesis *de novo:* they can only add nucleotides to a preexisting primer strand. Primase initiates *de novo* and synthesizes short RNA primers, which can then be extended by a DNA polymerase.
5. 3′ to 5′ exonuclease activity is required for the excision of mismatched bases in newly synthesized DNA during proofreading. A mutant *E. coli* with a DNA polymerase III lacking this activity would have a high frequency of mutations each time the DNA is replicated.
6. You would insert different DNA sequences into plasmids that lack origins of replication and determine whether the plasmids are able to transform mutant yeast that require a plasmid gene for their growth and division. Only plasmids with a functional origin will yield a high frequency of transformed yeast colonies.
7. DNA polymerases are unable to replicate the extreme 5′ ends of linear DNA molecules; yeast cells have evolved telomerase to maintain the ends of their linear chromosomes. Since the *E. coli* genome is a circular DNA molecule and has no ends, special mechanisms to replicate the ends of linear DNA are not required.
8. Double strand breaks can be repaired by recombinational repair, in which the missing portion of one chromosome can be recovered from a homologous chromosome. Single stranded breaks can be repaired by excision repair, since an undamaged complementary strand is available for use as template to direct synthesis of the excised portion of the damaged strand.
9. The high frequency of skin cancer results from DNA damage induced by solar UV irradiation, which is subject to repair by the nucleotide-excision repair system. The lack of elevated incidence of other cancers may suggest that similar types of damage are not frequent in internal organs, and that most cancers of these organs result from other types of mutations (e.g., the incorporation of mismatched bases during DNA replication).
10. The cellular processes that might be affected by these drugs include maintenance of telomeres by telomerase and the transposition of retrotransposons.
11. Instead of using methylation of parental strands, the mismatch repair system in humans uses single-strand breaks to identify newly replicated DNA. Thus, a homolog of *Mut H* is not required in human cells.
12. The mouse would be immunodeficient, lacking both B and T lymphocytes, as a result of being unable to rearrange its immunoglobulin and T cell receptor genes.

CHAPTER 7

1. The DNA sequence is radiolabeled at one end. It is then incubated with the protein and subjected to partial digestion with DNase. The site at which the protein binds will be protected from digestion, so no

labeled pieces of that length will be seen after electrophoresis of the digested DNA.

2. Sigma factors bind to sequences upstream of the transcription start site, bringing RNA polymerase specifically to the promoter region to initiate transcription.
3. The most common type of termination involves transcription of a GC-rich inverted repeat that forms a stable stem-loop structure by complementary base pairing. Formation of this structure disrupts association of the mRNA with the DNA and terminates transcription.
4. Since the diploid contains a functional repressor, it will bind the operator of the wild-type gene, in *trans*, and it will be regulated normally. The temperature-sensitive gene's operator cannot bind the repressor, so it will be expressed constitutively. β-galactosidase will therefore be produced at the permissive but not the nonpermissive temperature.
5. The promoter containing the TATA box can be transcribed *in vitro* in the presence of either TBP or TFIID. However, the Inr promoter requires TFIID, since the Inr sequence is recognized by TAFs rather than by TBP.
6. The activity of enhancers depends neither on their distance nor their orientation with respect to the transcription start site. Promoters are defined as being near the transcription start site.
7. The sequence element would be a potential binding site for a tissue-specific repressor.
8. They divide chromosomes into individual domains of chromatin structure that can be euchromatin or heterochromatin but cannot spread beyond an insulator. Also, they prevent an enhancer in one domain from acting on a promoter in the next domain.
9. One of the two X chromosomes is inactivated early in female development (or in XXY males). An RNA called *Xist* is produced by the *Xist* gene on one of the two X chromosomes, and binds to most of the genes on that chromosome. *Xist* RNA recruits proteins that induce chromatin condensation and conversion of most of the inactive X to heterochromatin.
10. The specific high-affinity binding of Sp1 to the GC box DNA sequence is exploited for its biochemical purification by DNA-affinity chromatography. To demonstrate that the purified protein is Sp1, you could carry out in vitro transcription assays.
11. The anti-Sm antiserum would bind to the snRNPs and prevent them from binding splice sites, thereby inhibiting splicing.
12. Noncoding RNAs can repress the transcription of a target gene via the RITS complex and they can induce the degradation of a target mRNA via the RISC complex. Thus noncoding RNAs can regulate both the synthesis and degradation of target transcripts.
13. Splicing factors direct snRNPs to the correct splice sites by binding to specific sequences in the pre-mRNA.
14. Apo-B100 is synthesized in the liver by translation of the unedited mRNA. The shorter Apo-B48 is synthesized in the intestine by translation of an edited mRNA in which the editing reaction has generated a stop codon.
15. Nonsense-mediated mRNA decay is a quality control process by which mRNAs that lack complete open-reading frames are degraded. This process eliminates defective mRNAs that would have given rise to abnormal truncated proteins.

CHAPTER 8

1. Many tRNAs are able to recognize more than one codon because the third base of their anticodon can "wobble" or form hydrogen bonds in nonstandard ways with bases other than the usual complementary pairs.
2. A Shine-Dalgarno sequence is needed.
3. Ribosomes depleted of most of their proteins can still synthesize polypeptides, whereas treatment with RNase completely abolishes translational activity. In addition, high-resolution structural analysis of the 50S ribosomal subunit showed that the site at which the peptidyl transferase reaction occurs is composed of rRNA, not ribosomal protein.
4. Polyadenylation is an important translational regulatory mechanism in early development. Its inhibition would block the translation of many oocyte mRNAs following fertilization.
5. Chaperones are proteins that aid the proper folding of other proteins. Elevated temperatures can denature proteins. Heat-shock proteins are chaperones that aid in the refolding of these denatured proteins, thereby restoring protein function.
6. Phospholipase treatment would release a GPI-anchored protein but not a transmembrane protein, from the cell surface.
7. The degradation of cyclin B, which allows cells to exit mitosis, requires a specific 9-amino-acid sequence called the destruction box. Mutations in this sequence prevent ubiquitination and degradation of cyclin B and block exit from mitosis.
8. No. In addition to targeting proteins for degradation, ubiquitination of a protein can have other functions, including targeting proteins for endocytosis and serving as part of the histone code.

9. miRNAs can target specific mRNAs via the RISC complex, leading to either mRNA degradation or inhibition of translation.
10. Sequences in the 3′ UTRs can regulate mRNA stability, localization, and translation by serving as binding sites for regulatory factors and miRNAs.
11. Since *T. aquaticus* grows at high temperatures, its rRNA is more stable than *E. coli* rRNA and better able to withstand the vigorous protein extraction procedures used in Noller's experiments.
12. Signal sequences that target proteins to specific cellular compartments are often removed by the action of signal peptidases. Additionally, many proteins are synthesized as larger precursor proteins and are proteolytically processed to yield the mature protein.
13. A "decoding center" in the small ribosomal subunit recognizes correct codon-anticodon base pairs and discriminates against mismatches. Insertion of a correct aminoacyl tRNA into the A site induces a conformational change that causes the hydrolysis of GTP bound to eEF-1α and the release of the elongation factor bound to GDP.
14. PDI allows rapid exchanges between paired disulfides, yielding a pattern of disulfide bonds that is compatible with the stably folded conformation of a protein. Inhibiting PDI with the siRNA will interfere with the correct folding of RNase, so you will detect much lower RNase activity in the medium of cells expressing the siRNA.

CHAPTER 9

1. Prokaryotic mRNAs are translated as they are transcribed. Separation of the site of transcription from the site of translation in eukaryotes allows regulation of mRNAs by posttranscriptional processes, such as alternative splicing, polyadenylation, and regulated transport to the cytoplasm. In addition, transcription can be regulated by modulating the nuclear localization of transcription factors.
2. Lamins form a filamentous network that supports and stabilizes the nuclear envelope. Lamins also provide binding sites for chromatin to attach to the inside of the nuclear envelope. In addition, many proteins that participate in transcription, DNA replication, and chromatin modification interact with lamins.
3. The 15 kd but not the 100 kd protein will be able to enter the nucleus, since proteins smaller than approximately 20 kd can pass freely through the nuclear pore complex.
4. The distribution of Ran/GTP across the nuclear envelope determines the directionality of nuclear transport.
5. One example is NF-κB. In unstimulated cells, IκB binds to NF-κB, and the complex cannot be imported to the nucleus. Upon stimulation, IκB is phosphorylated and degraded by ubiquitin-mediated proteolysis. This exposes a nuclear localization signal on NF-κB, allowing it to enter the nucleus and activate transcription of target genes.
6. The transcription factor could no longer be phosphorylated at these sites, so it would be constitutively imported to the nucleus and activate target gene expression.
7. Inactivating the nuclear export signal would result in retention of the protein in the nucleus.
8. You would label newly replicated DNA with bromodeoxyuridine, which is incorporated into DNA in place of thymidine. The bromodeoxyuridine-labeled DNA can be localized with antibodies against bromodeoxyuridine and fluorescence microscopy.
9. Kalderon and colleagues fused the T antigen amino acid sequence 126 to 132 to normally cytoplasmic proteins, β-galactosidase and pyruvate kinase. These fusion proteins accumulated in the nucleus.
10. Nuclear speckles contain concentrated populations of snRNPs and are thought to be storage sites for splicing components.
11. Small nucleolar RNAs (snoRNAs) localize to the nucleolus where they participate in the cleavage and modification of rRNAs.
12. Exportin-t is required for the export of tRNAs from the nucleus, so inhibiting exportin-t function would prevent tRNA export and lead to inhibition of translation.

CHAPTER 10

1. Palade and coworkers labeled pancreatic acinar cells with a pulse of radioactive amino acids, which were incorporated into proteins. Autoradiography showed that the labeled proteins were first detected in the rough ER. After a short "chase" with nonradioactive amino acids, the labeled proteins had moved to the Golgi apparatus and after longer periods, the labeled proteins were found in secretory vesicles and then outside of cells.
2. When an mRNA encoding a secreted protein is translated *in vitro* on free ribosomes, a larger protein results than when the same mRNA is translated in the presence of microsomes from the rough ER. In the latter case, the signal sequence is cleaved off by a signal peptidase in the rough ER vesicles.
3. Cotranslational translocation involves binding of the nascent polypeptide to the signal recognition particle and translocation through a translocon driven by the process of protein synthesis. Posttranslational

translocation targets a polypeptide to the ER after synthesis is complete and does not require SRP. A Sec62/63 complex recognizes a polypeptide to be incorporated, and inserts it into a translocon. The polypeptide is pulled through into the ER lumen by the chaperone BiP.

4. Proteins bound for the Golgi apparatus are unable to enter the ER in a Sec61 mutant and therefore remain in the cytosol.

5. Carbohydrate groups are added within the lumen of the ER and Golgi apparatus, both of which are topologically equivalent to the exterior of the cell.

6. Mutation of the KDEL sequence would inhibit the return of this ER-resident protein from the Golgi to the ER, so it would be secreted from the cell. Inactivation of the KDEL receptor protein would result in secretion of all KDEL-containing ER proteins.

7. Initially, a signal sequence targets the nascent lysosomal polypeptide to the rough ER. As it enters the ER, an *N*-linked oligosaccharide is added to the protein. After it moves to the Golgi, the three mannoses that are normally removed are not, but are instead converted to mannose-6-phosphate. These mannose-6-phosphates are recognized by a receptor in the *trans* Golgi network, which directs the transport of these proteins to lysosomes.

   The normally cytosolic protein lacks a signal sequence and does not enter the ER. Therefore addition of a lysosome-targeting signal would have no effect. In contrast, such an addition would direct a normally secreted protein to lysosomes from the Golgi apparatus.

8. In the absence of mannose-6-phosphate formation in the Golgi apparatus, lysosomal proteins would be secreted.

9. Glycolipids and sphingomyelin are produced by addition of sugars or phosphorylcholine to ceramide on the cytosolic and lumenal surfaces, respectively, of the Golgi apparatus. Glucosylceramide is then flipped to the lumenal surface. After vesicular transport and fusion with the plasma membrane, these lipids are located on the outer half of the plasma membrane.

10. A cell-free transport system was used by Rothman and his colleagues to demonstrate vesicular transport of proteins between Golgi cisternae.

11. The diagnosis is Gaucher disease. Since this disease involves a mutation in the lysosomal hydrolase glucocerebrosidase, you might suggest enzyme replacement therapy.

12. The enzymes destined for lysosomes are acid hydrolases that are not active at the neutral pH of the cytoplasm, ER, or Golgi apparatus. The acid hydrolases are activated by the acid pH of lysosomes, which is maintained by lysosomal proton pumps.

13. The strong interaction between the coiled-coil domains of v-SNAREs and t-SNAREs places the two membranes nearly in direct contact. This produces membrane instability and causes the membranes to fuse.

14. Rab proteins are kept in the GDP bound form in the cytosol by association with GDP dissociation inhibitors (GDIs). At membranes, the GDIs are removed by GDI displacement factors and membrane-localized guanine nucleotide exchange factors stimulate the exchange of GTP for GDP.

CHAPTER 11

1. The surface area of the mitochondrial inner membrane is high due to the formation of cristae. Also, the protein content of the membrane is unusually high (>70 %) and includes many enzymes and electron carriers.

2. Mitochondrial tRNAs are able to do an extreme form of wobble in which U in the anticodon can pair with any of the four bases in the third codon position of mRNA, allowing four codons to be recognized by a single tRNA.

3. The proton gradient of 1 pH unit corresponds to a tenfold higher $[H^+]_o$. By substituting the above values in the equation

$$\Delta G = RT \ln [H^+]_i/[H^+]_o$$

   We get an answer of approximately –1.4 kcal/mol

4. Hsp70 proteins in the cytosol maintain newly synthesized mitochondrial proteins in an unfolded state so they can be inserted into the TOM complex and be imported as an unfolded chain. Mitochondrial matrix Hsp70 chaperones bind to the polypeptide chain as it emerges from the TIM complex and use ATP hydrolysis to pull the polypeptide into the matrix. In some cases, an Hsp60 chaperonin complex binds the polypeptide and facilitates folding into its proper tertiary structure.

5. The first positively charged import signal leads to import into the mitochondrial matrix, where this presequence is removed; the second, hydrophobic sequence targets the polypeptide to Oxa1, a translocase in the inner membrane that passes cytochrome *b2* into the intermembrane space.

6. Coenzyme Q accepts a pair of electrons from either complex I or complex II and passes them to complex III. Coenzyme Q also binds two protons from the mitochondrial matrix, carries them across the inner membrane, and releases them to the intermembrane space, contributing to the generation of a proton gra-

dient. Cytochrome *c* is a small protein in the intermembrane space that picks up a pair of electrons from complex III and transfers them to complex IV.

7. F_0 spans the inner membrane of mitochondria and the thylakoid membrane of chloroplasts. F_1 is the stalk and knob that extends from F_0 into the matrix of mitochondria and into the stroma of chloroplasts. The F_0 complex provides a channel through which protons can flow back toward the matrix or stroma. Mechanical coupling by the stalk drives a rotation within the F_1 complex that catalyzes the synthesis of ATP.

8. In contrast to mitochondria, there is no electric potential across the chloroplast membrane. Therefore the charge of transit peptides does not contribute to protein translocation.

9. Two high-energy electrons are required to split each molecule of H_2O, so 24 high-energy electrons are required for the synthesis of each molecule of glucose. The passage of these electrons through the two photosystems generates 12 molecules of NADPH and between 12 and 18 molecules of ATP, depending on the stoichiometry of proton pumping at the cytochrome *bf* complex. Since 18 molecules of ATP are required for the Calvin cycle, the synthesis of glucose may require additional ATPs produced by cyclic electron flow.

10. Three out of four carbon atoms converted to glycolate are returned to chloroplasts and re-enter the Calvin cycle.

11. Most peroxisomal proteins are synthesized on free ribosomes in the cytosol and targeted to peroxisomes by either a Ser-Lys-Leu signal sequence on their carboxy terminus or a nine-amino acid sequence on their amino terminus. These are recognized by receptors and imported through transporters in the peroxisomal membrane.

12. While the thylakoid membrane is impermeable to protons, it is freely permeable to other ions, which can neutralize the voltage component of the proton gradient.

CHAPTER 12

1. The asymmetrical actin monomers associate in a head-to-tail fashion to form actin filaments. Since all the monomers are oriented in the same direction, the filament has a distinct polarity. The polarity of actin filaments defines the direction of myosin movement. If actin filaments were not polar, the unidirectional movement of myosin that results in the sliding of actin and myosin filaments could not take place.

2. Treadmilling is a dynamic behavior of actin filaments (or microtubules) in which they maintain a near-constant length by adding ATP-actins (or GTP-tubulins) to the plus end and dissociating an equal number of ADP-actins or GDP-tubulins from the minus end. During this steady-state behavior, subunits hydrolyze their nucleoside triphosphates after assembly, flux through the filament, and exit from the minus end. Treadmilling occurs at a monomer concentration between the critical concentration for the plus end and the critical concentration for the minus end.

3. Cytochalasin binds to the plus ends of actin filaments and blocks their elongation, so it would lead to depolymerization of treadmilling filaments. Phalloidin binds to actin filaments and inhibits their depolymerization, so it would cause filaments to stop treadmilling but remain present and grow longer.

4. ADF/cofilin binds to actin filaments and increases the dissociation of actin monomers. Profilin binds actin/ADP and stimulates the exchange of ADP for ATP, forming actin/ATP monomers that can join growing filaments. The Arp2/3 complex initiates the formation of branches.

5. The I band and the H zone shorten during contraction. The A band doesn't shorten because it is occupied by thick myosin filaments.

6. The contraction of smooth muscle cells is regulated by the phosphorylation of the myosin regulatory light chain by myosin light-chain kinase, which in turn is regulated by association with the calcium-binding protein calmodulin. An increase in the cytosolic concentration of calcium leads to the binding of calmodulin to myosin light-chain kinase.

7. Intermediate filaments are not required for the growth of cells in culture, so siRNA against vimentin would have no effect.

8. Dimers of cytoskeletal intermediate filaments assemble in a staggered antiparallel manner to form tetramers, which can then assemble end-to-end to form protofilaments. Since they are assembled from antiparallel tetramers, intermediate filaments do not have distinct ends and are apolar.

9. The *in vitro* movement of microtubules required ATP and was inhibited by the nonhydrolyzable ATP analog AMP-PNP. Importantly, organelles remained attached to microtubules in the presence of AMP-PNP, suggesting that the motor proteins responsible for organelle movement might also remain attached.

10. By linking the microtubule doublets in cilia together, nexin converts the sliding of individual microtubules to a bending motion that leads to the beating of cilia. If it were eliminated, the microtubules would simply slide past one another.

11. Colcemid inhibits the polymerization of microtubules and would inhibit the transport of secretory vesicles along microtubules.

12. γ-tubulin plays a key role in the formation of microtubule organizing centers. γ-tubulin associates with other proteins to form the γ-tubulin ring complex, which functions as a seed for the nucleation of new microtubules.

CHAPTER 13

1. At high temperatures cholesterol's ring structure inhibits the movement of phospholipids in the bilayer, thereby reducing membrane fluidity and increasing stability. At low temperatures cholesterol interferes with fatty acid chain interactions and maintains membrane fluidity.
2. Peripheral membrane proteins can be removed from a membrane by a high-salt wash or by solutions of extreme pH that do not disrupt the phospholipid bilayer. Integral membrane proteins can only be extracted from membranes by detergents that disrupt the phospholipid bilayer.
3. Frye and Edidin fused mouse and human cells and examined the distribution of membrane proteins after staining with anti-mouse and anti-human antibodies labeled with different fluorescent dyes. Immediately after fusion, the proteins were located in different halves of the fused cell surface, but after a brief incubation at 37°C the proteins were intermixed. This demonstrated that the proteins could diffuse laterally in a fluid membrane. If incubated at 2°C, the proteins remained separated because the membrane is not fluid at this temperature.
4. Lipid rafts are discrete membrane domains that are enriched in cholesterol and sphingolipids. Lipid rafts are believed to play important roles in cell movement, endocytosis, and cell signaling.
5. The glycocalyx protects the cell surface and is involved in cell-cell interactions.
6. Given $C_o/C_i = 10$, the K^+ equilibrium potential calculated from the Nernst equation is about –59 mV. The actual resting membrane potential differs from the K^+ equilibrium potential because resting squid axons are more permeable to K^+ than to other ions.
7. The opening of nicotinic acetylcholine receptors is required for membrane depolarization in muscle cells. Thus curare blocks the contraction of muscle cells in response to acetylcholine.
8. The K^+ channel contains a selectivity filter lined with carbonyl oxygen atoms. The pore is just wide enough to allow passage of dehydrated K^+ ions from which all water molecules have been displaced as a result of association with the carbonyl oxygen atoms. Hydrated Na^+ ions are too small to interact with the carbonyl oxygen atoms and remain associated with water molecules. This complex is too large to pass through the channel pore.
9. The uptake of glucose against its concentration gradient is coupled to the transport of Na^+ ions in the energetically favorable direction.
10. The *mdr* gene encodes an ABC transporter that is frequently overexpressed in cancer cells. The transporter can recognize a variety of drugs and pump them out of the cell, conferring resistance to chemotherapeutic drugs.
11. The addition of excess unlabeled LDL reduced the binding of labeled LDL to the surface of normal cells. This indicated that the labeled and unlabeled LDL were competing for a limited number of specific binding sites on the surface of normal cells.
12. Two types of mutations in the LDL receptor resulting in the inability to take up LDL were identified in FH patients. Cells from most FH patients failed to bind LDL, demonstrating that a specific receptor was required for LDL uptake. Other mutant receptors bound LDL, but failed to cluster in coated pits, demonstrating the role of coated pits in receptor-mediated endocytosis.

CHAPTER 14

1. Gram-positive bacteria have a single membrane—the plasma membrane—surrounded by a thick cell wall. Gram-negative bacteria have both a plasma membrane and an outer membrane, separated by a thin cell wall.
2. The rigid plant cell wall prevents cell swelling and allows the buildup of turgor pressure.
3. Hemicelluloses are branched polysaccharides that hydrogen bond to the surface of cellulose microfibrils. This interaction stabilizes the cellulose microfibrils into tough fibers.
4. The correct localization of these glucose transporters is necessary for the polarized function of intestinal epithelial cells in transferring glucose from the intestinal lumen to the blood supply. Tight junctions prevent the diffusion of these transporters between domains of the plasma membrane, as well as sealing the spaces between cells of the epithelium.
5. Hydroxylysine stabilizes the collagen triple helix by forming hydrogen bonds between the polypeptide chains. Inhibition of lysyl hydroxylase would therefore decrease the stability of collagen fibrils.
6. Fibril-forming collagens are synthesized as soluble precursors known as procollagens. Procollagens have nonhelical segments on their ends, which prevent the assembly of fibrils. Only after procollagen is secreted outside the cell are the non-helical segments removed and the fibrils assembled.
7. GAGs contain acidic sugar residues, which are modified by the addition of sulfate groups. This imparts

a high negative charge to GAGs, so they bind positively charged ions and trap water molecules to form hydrated gels.

8. The peptide most likely disrupts tight junctions, thereby allowing free diffusion of the transporter between the apical and basolateral domains of the plasma membrane.

9. Since E-cadherins mediate selective adhesion of epithelial cells, overexpression of a dominant negative version would disrupt interactions between neighboring cells.

10. The cytoplasmic domain of $\alpha_6\beta_4$ integrin is required for its interaction with the intermediate filament cytoskeleton through plectin. The mutation would therefore disrupt hemidesmosome formation and lead to decreased cell-matrix interaction.

11. Electrical synapses are specialized gap junctions found in nerve cells. They allow the rapid passage of ions between cells, thereby conducting a nerve impulse.

12. Gap junctions and plasmodesmata are similar in that they both provide channels between the cytoplasms of adjacent cells. They are likely to be analogous rather than homologous structures in animals and plants, since their structures are extremely different.

CHAPTER 15

1. In paracrine signaling, molecules released by one or a few cells affect nearby cells. In endocrine signaling, hormones are carried throughout the body to act on any target cell that has a receptor for that hormone.

2. Hydrophobic molecules like steroid hormones can diffuse through the plasma membrane and bind to cytosolic or nuclear receptors. Hydrophilic molecules (like peptide hormones) cannot cross the plasma membrane, so they act by binding to receptors on the cell surface.

3. Aspirin inhibits the enzyme cyclooxygenase, which catalyzes the first step in the synthesis of prostaglandin and thromboxanes from arachidonic acid.

4. Inhibition of cAMP phosphodiesterase would result in elevated levels of cAMP, which would stimulate cell proliferation.

5. The recombinant molecule would function as an epinephrine receptor coupled to G_i. Epinephrine would therefore inhibit adenylyl cyclase, lowering intracellular cAMP levels. Acetylcholine would have no effect, since it would not bind to the recombinant receptor.

6. PDGF monomers would not induce receptor dimerization. Since this is the first critical step in signaling from receptor protein-tyrosine kinases, they would be unable to stimulate the PDGF receptor.

7. The truncated receptor would act as a dominant-negative mutant because it would dimerize with the normal receptor. The dimers would be inactive because they would be unable to cross-phosphorylate.

8. Protein phosphatase 1 dephosphorylates serine residues that are phosphorylated by protein kinase A. Cyclic AMP-inducible genes are activated by CREB, which is phosphorylated by protein kinase A, so overexpression of protein phosphatase 1 would inhibit their induction. However, protein phosphatase 1 would not affect the activity of cAMP-gated ligand channels, since these channels are opened directly by cAMP binding rather than by protein phosphorylation.

9. Akt phosphorylates the protein kinase GSK-3β which regulates the translation factor eIF-2B. In addition, Akt regulates the mTOR/raptor protein kinase, which regulates translation by phosphorylating S6 kinase and the eIF-4E binding protein 4E-BP1.

10. Sos is required for the activation of Ras, so siRNA against Sos would inhibit induction of the immediate early gene.

11. JAKs phosphorylate and activate STATs in response to cytokine signaling. Expression of a dominant-negative JAK would inhibit STAT activation and block gene induction.

12. The mutation most likely alters a ubiquitination site on β-catenin, leading to its stabilization and nuclear accumulation. In the nucleus, β-catenin forms a complex with Tcf/LEF transcription factors and induces gene expression.

CHAPTER 16

1. G_0 and G_1 cells both have the same amount of DNA ($2n$) and both are metabolically active. G_0 cells differ from G_1 cells in that they are in a quiescent state and do not proliferate unless stimulated to re-enter the cell cycle.

2. The percentage of cells in a specific stage of the cell cycle can be used to calculate the duration of that phase. Since the cell takes 30 hours to complete all stages and 53.3% of the cells have a DNA content of $2n$ (G_1 cells), the duration of G_1 is 16 hours (53.3% of 30 hours). Cells in S phase have DNA contents between $2n$ and $4n$: these correspond to 30% of the cells, so the duration of S is 9 hours. 16.7% of the cells have DNA contents of $4n$, corresponding to cells in G_2 and M phases. Since 3.3% of the cells are in M phase, 13.4% are in G_2. The duration of G_2 is therefore 4 hours and that of M is 1 hour.

3. Cell cycle arrest at the G_1 and S phase checkpoints allows time for damaged DNA to be repaired before it is replicated. Arrest at the G_2 checkpoint allows time for DNA breaks or other damage to be repaired

before mitosis occurs, preventing the damaged DNA from being passed on to daughter cells.

4. Cdk's are regulated by four different mechanisms: association with cyclins, activating phosphorylations, inhibitory phosphorylations, and association with Cdk inhibitors.

5. One daughter cell would receive two copies of the misaligned chromosome; the other daughter cell would receive none.

6. The Cdk7/cyclin H complex is a Cdk-activating kinase and is also required for initiation of transcription by RNA polymerase II. Thus inhibiting Cdk7 would lead to cell cycle arrest and the inhibition of transcription.

7. In a normal cell, overexpression of p16 would inhibit cell cycle progression at the restriction point in G_1. Because Rb is the principal target of Cdk4, 6/cyclin D complexes, a tumor cell lacking functional Rb would be unaffected by p16 overexpression.

8. The phosphorylation of nuclear lamins is required for nuclear lamina breakdown. Expression of mutant lamins lacking a phosphorylation site for Cdk1 would prevent breakdown of the nuclear envelope.

9. Cdk1/cyclin B phosphorylates several structural proteins directly to alter their properties and initiate mitosis. Among them are condensins, nuclear lamins, Golgi matrix proteins, and microtubule-associated proteins. In addition, Cdk1/cyclin B activates other protein kinases.

10. Anaphase would initiate normally. However, Cdk1/cyclin B would remain active, so re-formation of nuclei, chromosome decondensation, and cytokinesis would not occur.

11. The anaphase-promoting complex degrades securin, leading to activation of the protease separase. Separase then degrades cohesins, breaking the link between sister chromatids.

12. Oocytes of these mice would fail to arrest at metaphase II.

13. Binding of the first sperm to its receptor releases calcium within the egg, probably by cleavage of PIP_2 to release IP_3, which opens IP_3-gated Ca^{2+} channels in the ER. This calcium induces exocytosis of secretory granules whose contents alter the extracellular coat of the egg to block entry of additional sperm.

CHAPTER 17

1. Apoptotic cells are efficiently removed from tissues by phagocytosis, whereas cells that die by acute injury release their contents into the extracellular space and cause inflammation.

2. Caspases are synthesized as long inactive precursors that are activated in complexes (e.g., the apoptosome) or converted to active enzymes by proteolytic cleavage. In addition, cells contain IAPs that associate with caspases and inhibit their activity.

3. The cleavage of lamins by caspases is required for nuclear fragmentation during apoptosis. The mutated lamins will not be cleaved by caspases, so their expression will block nuclear fragmentation.

4. Proapoptotic multidomain members of the Bcl-2 family induce apoptosis by promoting the release of cytochrome *c* from mitochondria, which leads to caspase activation. The activity of the proapoptotic multidomain proteins is regulated by antiapoptic and BH3-only members of the Bcl-2 family.

5. Activation of p53 in response to DNA damage leads to the expression of its target genes, which include the Cdk inhibitor p21 and the BH3-only Bcl-2 family members PUMA and Noxa. p21 induces cell cycle arrest and the BH3-only Bcl-2 family members induce apoptosis.

6. The mutant Bad would no longer be maintained in an inactive state by 14-3-3 protein, so it will act to induce apoptosis.

7. 14-3-3 proteins sequester proapoptotic proteins, such as Bad and FOXO transcription factors, in an inactive state. Cells expressing siRNA against 14-3-3 proteins will therefore have an increased rate of apoptosis.

8. Caspase-8 is the initiator caspase downstream of TNF receptors, so cells with inactive caspase-8 will not undergo apoptosis upon treatment with TNF. Thus TNF therapy would not be effective for this patient.

9. Ced-3 is the only caspase in *C. elegans*. Mutating it leads to the survival of all the cells that would normally die by apoptosis during development, and RNAi against Ced-3 would have the same effect.

10. The polypeptide will lead to the release of the proapoptotic proteins cytochrome *c*, Smac/Diablo, and Omi/Htr2 from the mitochondria and induce apoptosis in treated cells.

11. These tissues contain stem cells that retain the ability to proliferate and replace differentiated cells.

12. The critical characteristic of stem cells is their capacity for self-renewal. They divide to produce one daughter cell that remains a stem cell and one that divides and differentiates.

13. Embryonic stem cells are easier to isolate and culture and are capable of giving rise to all of the differentiated cell types in an adult organism.

CHAPTER 18

1. Benign tumors remain confined to their original location, whereas malignant tumors can invade surrounding normal tissue and metastasize to other parts of the body.

2. As a tumor progresses, mutations occur within cells of the tumor population. Some of these mutations confer a selective advantage to the cells in which they occur and allow them to outgrow other cells in the tumor. This process is called clonal selection, and it leads to the continuing development of more rapidly growing and increasingly malignant tumors.

3. Estrogens act as tumor promoters by stimulating the proliferation of estrogen responsive cells, such as breast and endometrial cells.

4. Autocrine growth stimulation is a positive feedback system in which tumor cells produce growth factors that stimulate their own proliferation.

5. Cancer cells are less adhesive that normal cells and are not as strictly regulated by cell-cell and cell-matrix interactions. In addition, cancer cells secrete proteases that degrade components of the extracellular matrix, facilitating invasion of adjacent tissues. Finally, cancer cells secrete angiogenic factors that promote the formation of new blood vessels that supply tumors with oxygen and nutrients and facilitate metastasis by allowing the cancer cells easy access to the circulatory system.

6. E6 interacts with p53 but not with Rb, so it would not induce transformation in combination with the mutant T antigen. However, E7 binds and inactivates Rb, so it will induce transformation in conjunction with the mutant T antigen.

7. AIDS patients are immunosuppressed and are therefore susceptible to infection by oncogenic viruses.

8. A proto-oncogene may be expressed at abnormal levels or in abnormal cell types. These changes in expression can convert a proto-oncogene to an oncogene even though a structurally normal protein is produced. A proto-oncogene can be activated in this manner either by a translocation that puts it under the control of an active promoter or by gene amplification.

9. INK4 encodes the Cdk inhibitor p16, which inhibits Cdk4, 6/cyclin D complexes. Since Rb is the critical target of Cdk4, 6/cyclin D, overexpression of p16 would not affect the proliferation of cells with inactivated Rb.

10. p53 is required for cell cycle arrest and apoptosis in response to DNA damage induced by ionizing radiation. Thus tumors with wild-type p53 genes will be more sensitive to radiation.

11. STI-571 is a specific inhibitor of the Bcr/Abl protein-tyrosine kinase expressed in chronic myeloid leukemia cells. Inhibition of Bcr/Abl blocks proliferation of these tumor cells. Resistance to STI-571 is most often caused by mutations in Bcr/Abl that prevent binding of the drug.

12. It is believed that the proliferation and survival of tumor cells become dependent on activated oncogenes, with other signaling pathways becoming secondary in importance. This dependence of tumor cells on activated oncogenes has been termed oncogene addiction. It suggests that drugs against an activated oncogene would selectively target tumor cells, while normal cells would be able to compensate by using alternative signaling pathways.

Glossary

α-actinin An actin-binding protein that crosslinks actin filaments into contractile bundles.

α helix A coiled secondary structure of a polypeptide chain formed by hydrogen bonding between amino acids separated by four residues.

ABC transporters A large family of membrane transport proteins characterized by a highly conserved ATP binding domain.

abl A proto-oncogene that encodes a protein-tyrosine kinase and is activated by chromosome translocation in chronic myeloid leukemia.

abscisic acid A plant hormone.

actin An abundant 43-kd protein that polymerizes to form cytoskeletal filaments.

actin-binding proteins Proteins that bind actin and regulate the assembly, disassembly, and organization of actin filaments.

actin bundle Actin filaments that are crosslinked into closely packed arrays.

actin-bundling proteins Proteins that crosslink actin filaments into bundles.

actin network Actin filaments that are crosslinked into loose three-dimensional meshworks.

action potential Nerve impulses that travel along axons.

activation energy The energy required to raise a molecule to its transition state to undergo a chemical reaction.

activation-induced deaminase (AID) An enzyme expressed in B lymphocytes that deaminates cytosine in DNA to form uracil in the variable regions of immunoglobulin genes. AID is required for both class switch recombination and somatic hypermutation.

active site The region of an enzyme that binds substrates and catalyzes an enzymatic reaction.

active transport The transport of molecules in an energetically unfavorable direction across a membrane coupled to the hydrolysis ofATP or other source of energy.

adaptin A protein that binds to membrane receptors and mediates the formation of clathrin-coated vesicles.

adenine A purine that base-pairs with either thymine or uracil.

adenoma A benign tumor arising from glandular epithelium.

adenovirus A widely-studied DNA tumor virus.

adenylyl cyclase An enzyme that catalyzes the formation of cyclic AMP from ATP.

ADF/cofilin A family of actin-binding proteins that disassemble actin filaments.

adherens junction A region of cell-cell adhesion at which the actin cytoskeleton is anchored to the plasma membrane.

adhesion belt A beltlike structure around epithelial cells in which a contractile bundle of actin filaments is linked to the plasma membrane.

Akt A protein-serine/threonine kinase that is activated by PIP_3 and plays a key role in signaling cell survival.

allele One copy of a gene.

allosteric regulation The regulation of enzymes by small molecules that bind to a site distinct from the active site, changing the conformation and catalytic activity of the enzyme.

alternative splicing The generation of different mRNAs by varying the pattern of pre-mRNA splicing.

amino acid Monomeric building blocks of proteins, consisting of a carbon atom bound to a carboxyl group, an amino group, a hydrogen atom, and a distinctive side chain.

aminoacyl tRNA synthetase An enzyme that joins a specific amino acid to a tRNA molecule carrying the correct anticodon sequence.

amphipathic A molecule that has both hydrophobic and hydrophilic regions.

amyloplast A plastid that stores starch.

anaphase The phase of mitosis during which sister chromatids separate and move to opposite poles of the spindle.

anaphase A The movement of daughter chromosomes toward the spindle poles during mitosis.

anaphase B The separation of the spindle poles during mitosis.

anaphase-promoting complex A ubiquitin ligase that triggers progression from metaphase to anaphase by signaling the degradation of cyclin B and cohesins.

angiogenesis The formation of new blood vessels.

ankyrin A protein that binds spectrin and links the actin cytoskeleton to the plasma membrane.

antibody A protein produced by B lymphocytes that binds to a foreign molecule.

anticodon The nucleotide sequence of transfer RNA that forms complementary base pairs with a codon sequence on messenger RNA.

antigen A molecule against which an antibody is directed.

antiport The transport of two molecules in opposite directions across a membrane.

antisense nucleic acids Nucleic acids (either RNA or DNA) that are complementary to an mRNA of interest and are used to block gene expression.

AP endonuclease A DNA repair enzyme that cleaves next to apyrimidinic or apurinic sites in DNA.

apical domain The exposed free surface of a polarized epithelial cell.

apoptosis An active process of programmed cell death, characterized by cleavage of chromosomal DNA, chromatin condensation, and fragmentation of both the nucleus and the cell.

apoptosome A protein complex in which caspase-9 is activated to initiate apoptosis following the release of cytochrome *c* from mitochondria.

Arabidopsis thaliana A small flowering plant used as a model for plant molecular biology and development.

archaebacteria One of two major groups of prokaryotes; many species of archaebacteria live in extreme conditions similar to those prevalent on primitive Earth.

ARF A GTP-binding protein required for vesicle budding from the *trans*-Golgi network.

armadillo protein family A family of proteins, including β-catenin, that link cadherins to the cytoskeleton at stable cell-cell junctions.

ARP 2/3 complex A protein complex that binds to actin filaments and initiates the formation of branches.

astral microtubules Microtubules of the mitotic spindle that extend to the cell periphery.

ATM A protein kinase that recognizes damaged DNA and leads to cell cycle arrest.

ATP (adenosine 5′-triphosphate) An adenine-containing nucleoside triphosphate that serves as a store of free energy in the cell.

ATP synthase A membrane spanning protein complex that couples the energetically favorable transport of protons across a membrane to the synthesis of ATP.

ATR A protein kinase related to ATM that leads to cell cycle arrest in response to DNA damage.

Aurora kinase A protein kinase family involved in mitotic spindle formation, kinetochore function, and cytokinesis.

autocrine growth stimulation Stimulation of cell proliferation as a result of growth factor production by a responsive cell.

autocrine signaling A type of cell signaling in which a cell produces a growth factor to which it also responds.

autonomously replicating sequence (ARS) An origin of DNA replication in yeast.

autophagosome A vesicle containing internal organelles enclosed by fragments of the endoplasmic reticulum membrane that fuses with lysosomes.

autophagy The degradation of cytoplasmic proteins and organelles by their enclosure in vesicles from the endoplasmic reticulum that fuse with lysosomes.

autophosphorylation A reaction in which a protein kinase catalyzes its own phosphorylation.

autoradiography The detection of radioisotopically labeled molecules by exposure to X-ray film.

auxin A plant hormone that controls many aspects of plant development.

axonemal dynein The type of dynein found in cilia and flagella.

axoneme The fundamental structure of cilia and flagella composed of a central pair of microtubules surrounded by nine microtubule doublets.

β-arrestin A regulatory protein that terminates signaling from G protein-coupled receptors, as well as stimulating other downstream signaling pathways.

β-barrel A transmembrane domain formed by the folding of β sheets into a barrel-like structure.

β sheet A sheetlike secondary structure of a polypeptide chain, formed by hydrogen bonding between amino acids located in different regions of the polypeptide.

bacterial artificial chromosome (BAC) A type of vector used for cloning large fragments of DNA in bacteria.

bacteriophage A bacterial virus.

baculovirus A virus commonly used as an expression vector for production of eukaryotic proteins in insect cells.

barrier element See insulator.

basal body A structure similar to a centriole that initiates the growth of axonemal microtubules and anchors cilia and flagella to the surface of the cell.

basal lamina A sheetlike extracellular matrix that supports epithelial cells and surrounds muscle cells, adipose cells, and peripheral nerves.

base-excision repair A mechanism of DNA repair in which single damaged bases are removed from a DNA molecule.

basement membrane See basal lamina.

basolateral domain The surface region of a polarized epithelial cell that is in contact with adjacent cells or the extracellular matrix.

Bcl-2 A member of a family of proteins that regulate programmed cell death.

benign tumor A tumor that remains confined to its site of origin.

bioinformatics The use of computational methods to analyze large amounts of biological data, such as genome sequences.

bone marrow transplantation A clinical procedure in which transplantation of bone marrow stem cells is used in the treatment of cancer and diseases of the hematopoietic system.

brassinosteroid A plant steroid hormone.

bright-field microscopy The simplest form of light microscopy in which light passes directly through a cell.

brush border The surface of a cell (e.g., an intestinal epithelial cell) containing a layer of microvilli.

cadherins A group of cell adhesion molecules that form stable cell-cell junctions at adherens junctions and desmosomes.

Caenorhabditis elegans A nematode used as a simple multicellular model for development.

callus An undifferentiated mass of plant cells in culture.

calmodulin A calcium-binding protein.

Calvin cycle A series of reactions by which six molecules of CO_2 are converted into glucose.

CaM kinase A member of a family of protein kinases that are activated by the binding of Ca^{2+}/calmodulin.

cAMP-dependent protein kinase See protein kinase A.

cAMP phosphodiesterase An enzyme that degrades cyclic AMP.

cAMP-response element (CRE) A regulatory sequence that mediates the transcriptional response of target genes to cAMP.

cancer A malignant tumor.

carbohydrate A molecule with the formula $(CH_2O)_n$. Carbohydrates include both simple sugars and polysaccharides.

carcinogen A cancer-inducing agent.

carcinoma A cancer of epithelial cells.

cardiolipin A phospholipid containing four hydrocarbon chains.

carrier proteins Proteins that selectively bind and transport small molecules across a membrane.

caspases A family of proteases that bring about programmed cell death.

catalase An enzyme that decomposes hydrogen peroxide.

catenin A group of cytoplasmic proteins (including α-catenin and β-catenin) that link actin filaments to cadherins at adherens junctions.

caveolae Small invaginations of the plasma membrane that may be involved in endocytosis.

caveolin A protein that interacts with lipid rafts and forms caveolae.

CCND1 The gene encoding cyclin D1, which is an oncogene in a variety of human cancers.

Cdc42 A member of the Rho subfamily of small GTP-binding proteins.

Cdk1 A protein-serine/threonine kinase that is a key regulator of mitosis in eukaryotic cells.

Cdk inhibitor (CKI) A family of proteins that bind Cdks and inhibit their activity.

Cdks Cyclin dependent protein kinases that control the cell cycle of eukaryotes.

cDNA library A collection of recombinant cDNA clones.

cell adhesion molecules Transmembrane proteins that mediate cell-cell interactions.

cell cortex The actin network underlying the plasma membrane.

cell cycle checkpoints Regulatory mechanisms that prevent entry into the next phase of the cell cycle until the events of the preceding phase have been completed.

cell lines Cells that can proliferate indefinitely in culture.

cell plate A membrane-enclosed disclike structure that forms new cell walls during cytokinesis of higher plants.

cell transformation The conversion of normal cells to tumor cells in culture.

cell wall A rigid, porous structure forming an external layer that provides structural support to bacteria, fungi, and plant cells.

cellulose The principal structural component of the plant cell wall, a linear polymer of glucose residues linked by $\beta(1\rightarrow4)$ glycosidic bonds.

cellulose microfibrils Fibers in plant cell walls that are formed by the association of several dozen parallel chains of cellulose.

cellulose synthase An enzyme that catalyzes the synthesis of cellulose.

central dogma The concept that genetic information flows from DNA to RNA to proteins.

centriole A cylindrical structure consisting of nine triplets of microtubules in the centrosomes of most animal cells.

centromere A specialized chromosomal region that connects sister chromatids and attaches them to the mitotic spindle.

centrosome The microtubule-organizing center in animal cells.

cGMP phosphodiesterase An enzyme that degrades cGMP.

channel proteins Proteins that form pores through a membrane.

chaperone A protein that facilitates the correct folding or assembly of other proteins.

chaperonin A family of heat-shock proteins within which protein folding takes place.

checkpoint kinase (CHK1 and CHK2) A protein kinase that brings about cell cycle arrest in response to damaged DNA. CHK1 and CHK2 are activated by the ATM and ATR protein kinases.

chemiosmotic coupling The generation of ATP from energy stored in a proton gradient across a membrane.

chiasmata Sites of recombination that link homologous chromosomes during meiosis.

chitin A polymer of *N*-acetylglucosamine residues that is the principal component of fungal cell walls.

chlorophyll The major photosynthetic pigment of plant cells.

chloroplast The organelle responsible for photosynthesis in the cells of plants and green algae.

cholesterol A lipid consisting of four hydrocarbon rings. Cholesterol is a major constituent of animal cell plasma membranes and the precursor of steroid hormones.

chromatin The fibrous complex of eukaryotic DNA and histone proteins. See histones, nucleosome, and chromatosome.

chromatin immunoprecipitation A method for determining regions of DNA that bind transcription factors within a cell.

chromatosome A chromatin subunit consisting of 166 base pairs of DNA wrapped around a histone core and held in place by a linker histone.

chromoplast A plastid that contains carotenoids.

chromosomal microtubules Microtubules of the mitotic spindle that attach to the ends of condensed chromosomes.

chromosomes The carriers of genes, consisting of long DNA molecules and associated proteins.

cilium A microtubule-based projection of the plasma membrane that moves a cell through fluid or fluid over a cell.

***cis*-acting control element** A regulatory DNA sequence that serves as a protein binding site and controls the transcription of adjacent genes.

***cis*-Golgi network** The region of the Golgi apparatus at which proteins enter from the endoplasmic reticulum.

citric acid cycle A series of reactions in which acetyl CoA is oxidized to CO_2. The central pathway of oxidative metabolism.

class switch recombination A type of region specific recombination responsible for the association of rearranged immunoglobulin V(D)J regions with different heavy chain constant regions.

clathrin A protein that coats the cytoplasmic surface of cell membranes and assembles into basketlike lattices that drive vesicle budding.

clathrin-coated pit A specialized region of the plasma membrane that contains receptors for macromolecules to be taken up by endocytosis.

clathrin-coated vesicle A transport vesicle coated with clathrin.

***c*-myc** A proto-oncogene that encodes a transcription factor and is frequently activated by chromosome translocation or gene amplification in human tumors.

coactivator A protein that interacts with a transcription factor to stimulate transcription.

codon The basic unit of the genetic code; one of the 64 nucleotide triplets that code for an amino acid or stop sequence.

coenzyme A (CoA) A coenzyme that functions as a carrier of acyl groups in metabolic reactions.

coenzyme Q A small lipid-soluble molecule that carries electrons between protein complexes in the mitochondrial electron transport chain.

coenzymes Low-molecular-weight organic molecules that work together with enzymes to catalyze biological reactions.

cohesins A complex of proteins that maintain the connection between sister chromatids.

colcemid A drug that inhibits the polymerization of microtubules.

colchicine A drug that inhibits the polymerization of microtubules.

collagen The major structural protein of the extracellular matrix.

collagen fibrils Fibrils formed by the assembly of collagen molecules in a regularly staggered array.

collenchyma Plant cells characterized by thick cell walls; they provide structural support to the plant.

complementary DNA (cDNA) A DNA molecule that is complementary to an mRNA molecule, synthesized *in vitro* by reverse transcriptase.

condensin A protein complex that drives metaphase chromosome condensation.

confocal microscopy A form of microscopy in which fluorescence microscopy is combined with electronic image analysis to obtain images with increased contrast and detail.

connexin A member of a family of transmembrane proteins that form gap junctions.

connexon A cylinder formed by six connexins in the plasma membrane.

contact inhibition The inhibition of movement or proliferation of normal cells that results from cell-cell contact.

contractile bundles Bundles of actin filaments that interact with myosin II and are capable of contraction.

contractile ring A structure of actin and myosin II that forms beneath the plasma membrane during mitosis and mediates cytokinesis.

COP I and **COP II** The two proteins other than clathrin that coat transport vesicles (COP indicates coat protein).

COP-coated vesicle Transport vesicles coated with COP I or COP II.

corepressor A protein that associates with repressors to inhibit gene expression, often by modifying chromatin structure.

corticosteroids Steroid hormones produced by the adrenal gland.

cosmid A vector that contains bacteriophage λ sequences, antibiotic resistance sequences, and an origin of replication. It can accomodate large DNA inserts of up to 45 kb.

CREB Cyclic AMP response element-binding protein. A transcription factor that is activated by cAMP-dependent protein kinase.

crista A fold in the inner mitochondrial membrane extending into the matrix.

crosstalk A regulatory mechanism in which one signaling pathway controls the activity of another.

cyanobacteria The largest and most complex prokaryotes in which photosynthesis is believed to have evolved.

cyclic AMP (cAMP) Adenosine monophosphate in which the phosphate group is covalently bound to both the 3′ and 5′ carbon atoms, forming a cyclic structure; an important second messenger in the response of cells to a variety of hormones.

cyclic electron flow An electron transport pathway associated with photosystem I that produces ATP without the synthesis of NADPH.

cyclic GMP (cGMP) Guanosine monophosphate in which the phosphate group is covalently bound to both the 3′ and 5′ carbon atoms, forming a cyclic structure; an important second messenger in the response of cells to a variety of hormones, and in vision.

cyclins A family of proteins that regulate the activity of Cdks and control progression through the cell cycle.

cytochalasin A drug that blocks the elongation of actin filaments.

cytochrome *bf* complex A protein complex in the thylakoid membrane that carries electrons during photosynthesis.

cytochrome *c* A mitochondrial peripheral membrane protein that carries electrons during oxidative phosphorylation.

cytochrome oxidase A protein complex in the electron transport chain that accepts electrons from cytochrome *c* and transfers them to O_2.

cytokine receptor superfamily A family of cell surface receptors that act by stimulating the activity of intracellular protein-tyrosine kinases.

cytokines Growth factors that regulate blood cells and lymphocytes.

cytokinesis Division of a cell following mitosis or meiosis.

cytokinin A plant hormone that regulates cell division.

cytoplasmic dynein The form of dynein associated with microtubules in the cytoplasm.

cytosine A pyrimidine that base-pairs with guanine.

cytoskeleton A network of protein filaments that extends throughout the cytoplasm of eukaryotic cells. It provides the structural framework of the cell and is responsible for cell movements.

cytostatic factor (CSF) A cytoplasmic factor that arrests oocyte meiosis at metaphase II.

dark reactions The series of reactions that convert carbon dioxide and water to carbohydrates during photosynthesis. See Calvin cycle.

density-dependent inhibition The cessation of the proliferation of normal cells in culture at a finite cell density.

density-gradient centrifugation A method of separating particles by centrifugation through a gradient of a dense substance, such as sucrose or cesium chloride.

deoxyribonucleic acid (DNA) The genetic material of the cell.

2′-deoxyribose The five-carbon sugar found in DNA.

desmin An intermediate filament protein expressed in muscle cells.

desmocollin A type of transmembrane cadherin that links intermediate filament cytoskeletons of adjacent cells at desmosomes.

desmoglein A type of transmembrane cadherin that links intermediate filament cytoskeletons of adjacent cells at desmosomes.

desmosome A region of contact between epithelial cells at which keratin filaments are anchored to the plasma membrane. See also hemidesmosome.

diacylglycerol A second messenger formed from the hydrolysis of PIP_2 that activates protein kinase C.

diakinesis The final stage of the prophase of meiosis I during which the chromosomes fully condense and the cell progresses to metaphase.

dideoxynucleotides Nucleotides that lack the normal 3′ hydroxyl group of deoxyribose and are used as chain-terminating nucleotides in DNA sequencing.

differential interference-contrast microscopy A type of microscopy in which variations in density or thickness between parts of the cell are converted to differences in contrast in the final image.

differential centrifugation A method used to separate the components of cells on the basis of their size and density.

diploid An organism or cell that carries two copies of each chromosome.

diplotene The stage of mieosis I during which homologous chromosomes separate along their length but remain associated at chiasmata.

DNA-affinity chromatography A method used to isolate DNA-binding proteins based on their binding to specific DNA sequences.

DNA glycosylase A DNA repair enzyme that cleaves the bond linking a purine or pyrimidine to the deoxyribose of the backbone of a DNA molecule.

DNA ligase An enzyme that seals breaks in DNA strands.

DNA microarray A glass slide or membrane filter onto which oligonucleotides or fragments of cDNAs are printed at a high density, allowing simultaneous analysis of thousands of genes by hybridization of the microarray with fluorescent probes.

DNA polymerase An enzyme catalyzing the synthesis of DNA.

DNA transposons Transposable elements that move via DNA intermediates.

dolichol phosphate A lipid molecule in the endoplasmic reticulum upon which oligosaccharides are assembled for the glycosylation of proteins.

domains Compact, globular regions of proteins that are the basic units of tertiary structure.

dominant The allele that determines the phenotype of an organism when more than one allele is present.

dominant inhibitory mutant A mutant that interferes with the function of the normal allele of the gene.

Drosophila melanogaster A species of fruit fly commonly used for studies of animal genetics and development.

dynactin A protein that acts with cytoplasmic dynein to move cargo along microtubules.

dynamic instability The alternation of microtubules between cycles of growth and shrinkage.

dynamin A membrane-associated GTPase involved in vesicle budding.

dynein A motor protein that moves along microtubules towards the minus end.

dystrophin A cytoskeletal protein of muscle cells.

E2F A family of transcription factors that regulate the expression of genes involved in cell cycle progression and DNA replication.

ecdysone An insect steroid hormone that triggers metamorphosis.

ectoderm The outer germ layer; gives rise to tissues that include the skin and nervous system.

eicosanoid A class of lipids, including prostaglandins, prostacyclins, thromboxanes, and leukotrienes, that act in autocrine and paracrine signaling.

elaioplasts Plastids that store lipids.

elastic fibers Protein fibers that are present in the extracellular matrix of connective tissues in organs that stretch and then return to their original shape.

elastin The principal component of elastic fibers.

electrical synapse Specialized assemblies of gap junctions that allow the rapid passage of ions between nerve cells.

electrochemical gradient A difference in chemical concentration and electric potential across a membrane.

electron microscopy A type of microscopy that uses an electron beam to form an image. In transmission electron microscopy, a beam of electrons is passed through a specimen stained with heavy metals. In scanning electron microscopy, electrons scattered from the surface of a specimen are analyzed to generate a three-dimensional image.

electron tomography A method used to generate three-dimensional images by computer analysis of multiple two-dimensional images obtained by electron microscopy.

electron transport chain A series of carriers through which electrons are transported from a higher to a lower energy state.

electrophoretic-mobility shift assay An assay for the binding of a protein to a specific DNA sequence.

electroporation The introduction of DNA into cells by exposure to a brief electric pulse.

Elk-1 A transcription factor that is activated by ERK phosphorylation and induces expression of immediate-early genes.

elongation factor A protein involved in the elongation phase of transcription or translation.

embryonic stem (ES) cells Stem cells cultured from early embryos.

endocrine signaling A type of cell-cell signaling in which endocrine cells secrete hormones that are carried by the circulation to distant target cells.

endocytosis The uptake of extracellular material in vesicles formed from the plasma membrane.

endoderm The inner germ layer; gives rise to internal organs.

endoplasmic reticulum (ER) An extensive network of membrane-enclosed tubules and sacs involved in protein sorting and processing as well as in lipid synthesis.

endorphin A neuropeptide that acts as a natural analgesic.

endosome A vesicular compartment involved in the sorting and transport to lysosomes of material taken up by endocytosis.

endosymbiosis A symbiotic relationship in which one cell resides within a larger cell.

enhancer A transcriptional regulatory sequence that can be located at a site distant from the promoter.

enkephalin A neuropeptide that acts as a natural analgesic.

entactin An extracellular matrix protein that interacts with laminins and type IV collagen in basal laminae.

enzymes Proteins or RNAs that catalyze biological reactions.

epidermal cells Cells forming a protective layer on the surfaces of plants and animals.

epidermal growth factor (EGF) A growth factor that stimulates cell proliferation.

epithelial cells Cells forming sheets (epithelial tissue) that cover the surface of the body and line internal organs.

Epstein-Barr virus A human herpesvirus that causes B-cell lymphomas.

equilibrium centrifugation The separation of particles on the basis of density by centrifugation to equilibrium in a gradient of a dense substance.

erbA A proto-oncogene that encodes thyroid hormone receptor.

erbB-2 A proto-oncogene encoding a receptor protein-tyrosine kinase that is frequently amplified in breast and ovarian carcinomas.

ErbA The product of the *erbA* proto-oncogene. The thyroid hormone receptor.

ERK A member of the MAP kinase family that plays a central role in growth factor-induced cell proliferation.

ERM proteins A family of proteins that link actin filaments to the plasma membranes of many kinds of cells.

erythrocytes Red blood cells.

Escherichia coli (E. coli) A species of bacteria that has been extensively used as a model system for molecular biology.

estrogen A steroid hormone produced by the ovaries.

ethylene A plant hormone responsible for fruit ripening.

etioplast An intermediate stage of chloroplast development in which chlorophyll has not been synthesized.

eubacteria One of two major groups of prokaryotes, including most common species of bacteria.

euchromatin Decondensed, transcriptionally active interphase chromatin.

eukaryotic cells Cells that have a nuclear envelope, cytoplasmic organelles, and a cytoskeleton.

excinuclease The protein complex that excises damaged DNA during nucleotide-excision repair in bacteria.

exocyst A protein complex on the plasma membrane at which exocytosis occurs.

exon A segment of a gene that contains a coding sequence.

exonuclease An enzyme that hydrolyzes DNA molecules in either the 5′ to 3′ or 3′ to 5′ direction.

exportin A karyopherin that recognizes nuclear export signals and directs transport from the nucleus to the cytosol.

expression vector A vector used to direct expression of a cloned DNA fragment in a host cell.

extracellular matrix Secreted proteins and polysaccharides that fill spaces between cells and bind cells and tissues together.

facilitated diffusion The transport of molecules across a membrane by carrier or channel proteins.

FAK (focal adhesion kinase) A nonreceptor protein-tyrosine kinase that plays a key role in integrin signaling.

fats See triacylglycerols.

fatty acids Long hydrocarbon chains usually linked to a carboxyl group (COO^-).

feedback inhibition A type of allosteric regulation in which the product of a metabolic pathway inhibits the activity of an enzyme involved in its synthesis.

feedback loop A regulatory mechanism in which a downstream element of a signaling pathway controls the activity of an upstream component of the pathway.

feedforward relay A regulatory mechanism in which one element of a signaling pathway stimulates a downstream component.

fertilization The union of a sperm and an egg.

fibroblast A cell type found in connective tissue.

fibronectin The principal adhesion protein of the extracellular matrix.

filamentous [F] actin Actin monomers polymerized into filaments.

filamin An actin-binding protein that crosslinks actin filaments into networks.

filopodium A thin projection of the plasma membrane supported by actin bundles.

fimbrin An actin-bundling protein involved in formation of cell surface projections.

flagellum A microtubule-based projection of the plasma membrane that is responsible for cell movement.

flavin adenine dinucleotide ($FADH_2$) A coenzyme that functions as an electron carrier in oxidation/reduction reactions.

flippase A protein that catalyzes the translocation of lipids across the membrane of the endoplasmic reticulum.

flow cytometer An instrument that measures the fluorescence intensity of individual cells.

fluid mosaic model A model of membrane structure in which proteins are inserted in a fluid phospholipid bilayer.

fluid-phase endocytosis The nonselective uptake of extracellular fluids during endocytosis.

fluorescence-activated cell sorter An instrument that sorts individual cells on the basis of their fluorescence intensity.

fluorescence *in situ* hybridization (FISH) A method used to localize genes on chromosomes or RNAs within cells using fluorescent probes.

fluorescence microscopy Type of microscopy in which molecules are detected based on the emission of flourescent light.

fluorescence recovery after photobleaching (FRAP) A method used to study the movement of proteins within living cells.

fluorescence resonance energy transfer (FRET) A method used to study protein interactions within living cells.

focal adhesion A site of attachment of cells to the extracellular matrix at which integrins are linked to bundles of actin filaments.

focal complex A small cluster of integrins binding to the extracellular matrix that initiates the formation of a focal adhesion.

fodrin Nonerythroid spectrin.

footprinting A method used to identify the sites at which proteins bind to DNA.

formin An actin-binding protein that nucleates and polymerizes actin filaments.

Fos A transcription factor, encoded by a proto-oncogene, that is induced in response to growth factor stimulation.

freeze fracture Method of electron microscopy in which specimens are frozen in liquid nitrogen and then fractured to split the lipid bilayer, revealing the interior faces of cell membranes.

γ-tubulin ring complex A protein complex that nucleates the formation of microtubules.

G protein A family of cell signaling proteins regulated by guanine nucleotide binding.

G protein-coupled receptor A receptor characterized by seven membrane-spanning α helices. Ligand binding causes a conformational change that activates a G protein.

G_0 A quiescent state in which cells remain metabolically active but do not proliferate.

G_1 cyclins (Clns) Yeast cyclins that control passage through START.

G_1 phase The phase of the cell cycle between the end of mitosis and the begining of DNA synthesis.

G_2 phase The phase of the cell cycle between the end of S phase and the begining of mitosis.

gap junction A plasma membrane channel forming a direct cytoplasmic connection between adjacent cells.

gel electrophoresis A method in which molecules are separated based on their migration in an electric field.

gene A segment of DNA that encodes a polypeptide chain or an RNA molecule.

gene amplification An increase in the number of copies of a gene resulting from the repeated replication of a region of DNA.

gene family A group of related genes that have arisen by duplication of a common ancestor.

gene transfer The introduction of foreign DNA into a cell.

general transcription factors Transcription factors that are part of the general transcription machinery.

genetic code The correspondence between nucleotide triplets and amino acids in proteins.

genomic imprinting The regulation of genes whose expression depends on whether they are maternally or paternally inherited, apparently controlled by DNA methylation.

genomic library A collection of recombinant DNA clones that collectively contain the genome of an organism.

genomics The systematic analysis of entire cell genomes.

genotype The genetic composition of an organism.

gibberellin A plant hormone.

Gibbs free energy (*G*) The thermodynamic function that combines the effects of enthalpy and entropy to predict the energetically favorable direction of a chemical reaction.

globular[G] actin Monomers of actin that have not been assembled into filaments.

glucocorticoid A steroid produced by the adrenal gland that acts to stimulate production of glucose.

gluconeogenesis The synthesis of glucose.

glycerol phospholipids Phospholipids consisting of two fatty acids bound to a glycerol molecule.

glycocalyx A carbohydrate coat covering the cell surface.

glycogen A polymer of glucose residues that is the principal storage form of carbohydrates in animals.

glycolipid A lipid consisting of two hydrocarbon chains linked to a polar head group containing carbohydrates.

glycolysis The anaerobic breakdown of glucose.

glycoprotein A protein linked to oligosaccharides.

glycosaminoglycan (GAG) A gel-forming polysaccharide of the extracellular matrix.

glycosidase An enzyme that removes sugar residues from its substrate.

glycosidic bond The bond formed between sugar residues in oligosaccharides or polysaccharides.

glycosylation The addition of carbohydrates to proteins.

glycosylphosphatidylinositol (GPI) anchor Glycolipids containing phosphatidylinositol that anchor proteins to the external face of the plasma membrane.

glycosyltransferase An enzyme that adds sugar residues to its substrate.

glyoxylate cycle The conversion of fatty acids to carbohydrates in plants.

glyoxysome Peroxisomes in which the reactions of the glyoxylate cycle take place.

Golgi apparatus A cytoplasmic organelle involved in the processing and sorting of proteins and lipids. In plant cells, it is also the site of the synthesis of cell wall polysaccharides.

Golgi complex see Golgi apparatus.

Golgi stack The compartments of the Golgi apparatus within which most metabolic activities take place.

granulocytes Blood cells that are involved in inflammatory reactions.

green fluorescent protein (GFP) A protein from jellyfish that is commonly used as a marker for fluorescence microscopy.

growth factors Polypeptides that control animal cell growth and differentiation.

GTPase-activating proteins Proteins that stimulate GTP hydrolysis by the small GTP-binding proteins.

guanine A purine that base-pairs with cytosine.

guanine nucleotide exchange factor A protein that acts on small GTP-binding proteins to stimulate the exchange of bound GDP for GTP.

guanylyl cyclase An enzyme that catalyzes the formation of cyclic GMP from GTP.

guidance complex A protein complex that directs proteins to chloroplasts.

haploid An organism or cell that has one copy of each chromosome.

hard keratin A keratin used for production of structures such as hair, nails, and horns.

heat-shock proteins A highly conserved group of chaperone proteins expressed in cells exposed to elevated temperatures or other forms of environmental stress.

Hedgehog A secreted signaling molecule that stimulates a pathway regulating cell fate during embryonic development.

helicase An enzyme that catalyzes the unwinding of DNA.

helix-loop-helix A transcription factor DNA-binding domain formed by the dimerization of two polypeptide chains. The dimerization domains of these proteins consist of two helical regions separated by a loop.

helix-turn-helix A transcription factor DNA-binding domain in which three or four helical regions contact DNA.

hemicellulose A polysaccharide that crosslinks cellulose microfibrils in plant cell walls.

hemidesmosome A region of contact between cells and the extracellular matrix at which keratin filaments are attached to integrin.

hepatitis B viruses A family of DNA viruses that infect liver cells and can lead to the development of liver cancer.

hepatitis C viruses A family of RNA viruses that infect liver cells and can lead to the development of liver cancer.

herpesviruses A family of DNA viruses, some members of which induce cancer.

heterochromatin Condensed, transcriptionally inactive chromatin.

heterophilic interaction An interaction between two different types of cell adhesion molecules.

heterotrimeric G protein A guanine nucleotide-binding protein consisting of three subunits.

high-energy bonds Chemical bonds that release a large amount of free energy when they are hydrolyzed.

histone acetylation The modification of histones by the addition of acetyl groups to specific lysine residues.

histone code Combinations of specific histone modifications that are thought to regulate the transcriptional activity of chromatin.

histones Proteins that package DNA in eukaryotic chromosomes.

HMGN proteins Nonhistone chromosomal proteins associated with decondensed transcriptionally active chromatin.

Holliday junction The central intermediate in recombination, consisting of a crossed-strand structure formed by homologous base pairing between strands of two DNA molecules.

Holliday model A molecular model of genetic recombination involving the formation of heteroduplex regions.

homeobox Conserved DNA sequences of 180 base pairs that encode homeodomains.

homeodomain A type of DNA binding domain found in transcription factors that regulate gene expression during embryonic development.

homologous recombination Recombination between segments of DNA with homologous nucleotide sequences.

homophilic interaction An interaction between cell adhesion molecules of the same type.

hormones Signaling molecules produced by endocrine glands that act on cells at distant body sites.

hydrophilic Soluble in water.

hydrophobic Not soluble in water.

IAP *I*nhibitor of *ap*optosis. A member of a family of proteins that inhibit apoptosis by interacting with caspases.

IκB An inhibitory subunit of NF-κB transcription factors.

immediate-early genes A family of genes whose transcription is rapidly induced in response to growth factor stimulation.

immunoblotting A method that uses antibodies to detect proteins separated by SDS-polyacrylamide gel electrophoresis.

immunoglobulin See antibody.

immunoglobulin (Ig) superfamily A family of cell adhesion molecules containing structural domains similar to immunoglobulins.

immunoprecipitation The use of antibodies to isolate proteins.

importin A karyopherin that recognizes nuclear localization signals and directs nuclear import.

induced fit A model of enzyme action in which the configurations of both the enzyme and the substrate are altered by substrate binding.

initiation factor A protein that functions in the initiation stage of translation.

***in situ* hybridization** The use of radioactive or flourescent probes to detect RNA or DNA sequences in chromosomes or intact cells.

***in vitro* mutagenesis** The introduction of mutations into cloned DNA *in vitro*.

***in vitro* translation** Protein synthesis in a cell-free extract.

inositol 1,4,5-trisphosphate (IP_3) A second messenger, formed from the hydrolysis of PIP_2, that signals the release of calcium ions from the endoplasmic reticulum.

insulator A sequence that divides chromatin into independent domains and prevents an enhancer from acting on a promoter in a separate domain.

integral membrane proteins Proteins embedded within the lipid bilayer of cell membranes.

integrin A transmembrane protein that mediates the adhesion of cells to the extracellular matrix.

intermediate filament A cytoskeletal filament about 10 nm in diameter that provides mechanical strength to cells in tissues. See also keratins and neurofilaments.

interphase The period of the cell cycle between mitoses that includes G_1, S, and G_2 phases.

intracellular signal transduction A chain of reactions that transmits chemical signals from the cell surface to their intracellular targets.

intron A noncoding sequence that interrupts exons in a gene.

ion channel A protein that mediates the rapid passage of ions across a membrane by forming open pores through the phospholipid bilayer.

ion pump A protein that couples ATP hydrolysis to the transport of ions across a membrane.

JAK/STAT pathway A signaling pathway in which STAT transcription factors are activated as a result of phosphorylation by members of the JAK family of protein kinases.

Janus kinase (JAK) A family of nonreceptor protein-tyrosine kinases associated with cytokine receptors.

Jun A transcription factor, encoded by a proto-oncogene, that is activated in response to growth factor stimulation.

junctional complex A region of cell-cell contact containing a tight junction, an adherens junction, and a desmosome.

Kaposi's sarcoma-associated herpesvirus A human herpesvirus that causes Kaposi's sarcoma.

karyopherin A nuclear transport receptor.

keratin A type of intermediate filament protein of epithelial cells.

kilobase (kb) One thousand nucleotides or nucleotide base pairs.

kinesin A motor protein that moves along microtubules toward the plus end.

kinetochore A specialized structure consisting of proteins attached to a centromere that mediates the attachment and movement of chromosomes along the mitotic spindle.

kinetochore microtubules Microtubules of the mitotic spindle that attach to condensed chromosomes at their centromeres.

knockout Inactivation of a chromosomal gene by homologous recombination with a cloned mutant allele.

Krebs cycle See citric acid cycle.

lagging strand The strand of DNA synthesized opposite to the direction of movement of the replication fork by ligation of Okazaki fragments.

lamellipodium A broad, actin-based extension of the plasma membrane involved in the movement of fibroblasts.

laminin The principal adhesion protein of basal laminae.

lamins Intermediate filament proteins that form the nuclear lamina.

leading strand The strand of DNA synthesized continuously in the direction of movement of the replication fork.

leptotene The initial stage of the extended prophase of meiosis I during which homologous chromosomes pair before condensation.

leucine zipper A protein dimerization domain containing repeated leucine residues; found in many transcription factors.

leucoplast A plastid that stores energy sources in nonphotosynthetic plant tissues.

leukemia Cancer arising from the precursors of circulating blood cells.

leukotriene An eicosanoid synthesized from arachodonic acid.

ligand A molecule that binds to a receptor.

ligand-gated channels Ion channels that open in response to the binding of signaling molecules.

light reactions The reactions of photosynthesis in which solar energy drives the synthesis of ATP and NADPH.

lignin A polymer of phenolic residues that strengthens secondary cell walls.

LINEs (long interspersed elements) A family of highly repeated retrotransposons in mammalian genomes.

lipid raft A discrete plasma membrane domain formed as a cluster of cholesterol and sphingolipids.

lipids Hydrophobic molecules that function as energy storage molecules, signaling molecules, and the major components of cell membranes.

liposome A lipid vesicle used to introduce DNA into mammalian cells.

lock-and-key model A model of enzyme action in which the substrate fits precisely into the enzyme active site.

long terminal repeat (LTR) Sequences found at the ends of retroviral DNA that are direct repeats of several hundred nucleotides resulting from reverse transcriptase activity.

low-density lipoprotein (LDL) A lipoprotein particle that transports cholesterol in the circulation.

lymphocyte A blood cell that functions in the immune response. B lymphocytes produce antibodies and T lymphocytes are responsible for cell mediated immunity.

lymphoma A cancer of lymphoid cells.

lysosomal storage diseases A family of diseases characterized by the accumulation of undegraded material in the lysosomes of affected individuals.

lysosome A cytoplasmic organelle containing enzymes that break down biological polymers.

M phase The mitotic phase of the cell cycle.

macrophage A type of white blood cell specialized for phagocytosis.

macropinocytosis The uptake of fluids in large vesicles.

malignant tumor A tumor that invades normal tissue and spreads throughout the body.

mannose-6-phosphate A modified mannose residue that targets proteins to lysosomes.

MAP kinases A family of mitogen-activated protein-serine/threonine kinases that are ubiquitous regulators of cell growth and differentiation.

mass spectrometry A method for identifying compounds based on accurate determination of their mass. Mass spectrometry is commonly used for protein identification.

matrix The inner mitochondrial space.

matrix processing peptidase (MPP) The protease that cleaves presequences from proteins imported to the matrix of mitochondria.

maturation promoting factor (MPF) A complex of Cdk1 and cyclin B that promotes entry into the M phase of either mitosis or meiosis.

Mediator A complex of proteins that allows eukaryotic protein-coding genes to respond to gene-specific regulatory factors.

megabase (Mb) One million nucleotides or nucleotide base pairs.

meiosis The division of diploid cells to haploid progeny, consisting of two sequential rounds of nuclear and cellular division.

MEK *M*AP kinase/*E*RK *k*inase. A dual-specificity protein kinase that phosphorylates and activates members of the ERK family of MAP kinases.

membrane-anchored growth factors Growth factors associated with the plasma membrane that function as signaling molecules during cell-cell contact.

mesoderm The middle germ layer; gives rise to connective tissues and the hematopoietic system.

messenger RNA (mRNA) An RNA molecule that serves as a template for protein synthesis.

metal shadowing An electron microscopic technique in which the surface of a specimen is coated with a thin layer of evaporated metal.

metaphase The phase of mitosis during which the chromosomes are aligned on a metaphase plate in the center of the cell.

metastasis Spread of cancer cells through the blood or lymphatic system to other organ sites.

7-methylguanosine cap A structure consisting of GTP and methylated sugars that is added to the 5′ ends of eukaryotic mRNAs.

microfilament A cytoskeleton filament composed of actin.

microRNA (miRNA) A naturally-occurring short noncoding RNA that acts to regulate gene expression.

microsome A small vesicle formed from the endoplasmic reticulum when cells are disrupted.

microspike See filopodium.

microtubule A cytoskeletal component formed by the polymerization of tubulin into rigid, hollow rods about 25 nm in diameter.

microtubule-associated proteins (MAPs) Proteins that bind to microtubules and modify their stability.

microtubule-organizing center An anchoring point near the center of the cell from which most microtubules extend outward.

microvillus An actin-based protrusion of the plasma membrane, abundant on the surfaces of cells involved in absorption.

middle lamella A region of the plant cell wall that acts as a glue to hold adjacent cells together.

mineralocorticoids Steroid hormones produced by the adrenal gland that act on the kidney to regulate salt and water balance.

mismatch repair A repair system that removes mismatched bases from newly synthesized DNA strands.

mitochondria Cytoplasmic organelles responsible for synthesis of most of the ATP in eukaryotic cells by oxidative phosphorylation.

mitosis Nuclear division.

mitotic spindle An array of microtubules extending from the spindle poles that is responsible for separating daughter chromosomes during mitosis. See also kinetochore microtubules, polar microtubules, chromosomal microtubules, and astral microtubules.

molecular clone See recombinant molecule.

molecular cloning The insertion of a DNA fragment of interest into a DNA molecule (vector) that is capable of independent replication in a host cell.

molecular motor A protein that generates force and movement by converting chemical energy to mechanical energy.

monocistronic Messenger RNAs that encode a single polypeptide chain.

monoclonal antibody An antibody produced by a clonal line of B lymphocytes.

monocyte A type of blood cell involved in inflammatory reactions.

monosaccharides Simple sugars with the basic formula of $(CH_2O)_n$.

Mos A protein kinase that is required for progression from meiosis I to meiosis II and maintenance of metaphase II arrest in vertebrate oocytes.

mTOR A protein kinase involved in regulation of protein synthesis in response to growth factors, nutrients, and energy availability.

multi-photon excitation microscopy A form of fluorescence microscopy in which the specimen is illuminated with a wavelength of light such that excitation of the fluorescent dye requires the simultaneous absorption of two or more photons.

muscle fibers The large cells of skeletal muscle, which are formed by the fusion of many individual cells during developmenht.

mutagen A chemical that induces a high frequency of mutations.

mutation A genetic alteration.

myofibril A bundle of actin and myosin filaments in muscle cells.

myosin A protein that interacts with actin as a molecular motor.

myosin I A type of myosin that acts to transport cargo along actin filaments.

myosin II The type of myosin that produces contraction by sliding actin filaments.

myosin light–chain kinase A protein kinase that activates myosin II by phosphorylating its regulatory light chain.

Na^+-K^+ ATPase See Na^+-K^+ pump.

Na^+-K^+ pump An ion pump that transports Na^+ out of the cell and K^+ into the cell.

NADP reductase An enzyme that transfers electrons from ferrodoxin to $NADP^+$, yielding NADPH.

nebulin A protein that regulates the length of actin filaments in muscle cells.

nectin A cell adhesion molecule involved in the formation of adherens junctions.

Nernst equation The relationship between ion concentration and membrane potential.

nerve growth factor (NGF) A polypeptide growth factor that regulates the development and survival of neurons.

neurofilament A type of intermediate filament that supports the axons of nerve cells.

neurofilament (NF) proteins The major intermediate filament proteins of many types of mature nerve cells.

neurohormone Peptides that are secreted by neurons and act on distant cells.

neuron A nerve cell specialized to receive and transmit signals throughout the body.

neuropeptides Peptide signaling molecules secreted by neurons.

neurotransmitter A small, hydrophilic molecule that carries a signal from a stimulated neuron to a target cell at a synapse.

neurotrophin A member of a family of polypeptides that regulates neuron development and survival.

nexin A protein that links microtubule doublets to each other in the axoneme.

NF-κB A family of transcription factors that are activated in response to a variety of stimuli.

nicotinamide-adenine dinucleotide (NAD^+) A coenzyme that functions as an electron carrier in oxidation/reduction reactions.

nitrogen fixation The reduction of atmospheric nitrogen (N_2) to NH_3.

nitrosylation Protein modification by addition of NO groups to the side chains of cysteine residues.

N-*myc* A proto-oncogene that encodes a transcription factor and is frequently activated by amplification in neuroblastomas.

N-myristoylation The addition of myristic acid (a 14-carbon fatty acid) to the N-terminal glycine residue of a polypeptide chain.

nonreceptor protein-tyrosine kinase An intracellular protein-tyrosine kinase.

nonsense-mediated mRNA decay Degradation of mRNAs that lack complete open-reading frames.

Northern blotting A method in which mRNAs are separated by gel electrophoresis and detected by hybridization with specific probes.

Notch A transmembrane receptor in a signaling pathway that regulates cell fate as a result of cell-cell interactions during development.

nuclear envelope The barrier separating the nucleus from the cytoplasm, composed of an inner and outer membrane, a nuclear lamina, and nuclear pore complexes.

nuclear export signal An amino acid sequence that targets proteins for transport from the nucleus to the cytosol.

nuclear lamina A meshwork of lamin filaments providing structural support to the nucleus.

nuclear localization signal An amino acid sequence that targets proteins for transportation from the cytoplasm to the nucleus.

nuclear membranes Membranes forming the nuclear envelope; the outer nuclear membrane is continuous with the endoplasmic reticulum and the inner nuclear membrane is adjacent to the nuclear lamina.

nuclear pore complex A large structure forming a transport channel through the nuclear envelope.

nuclear receptor superfamily A family of transcription factors that includes the receptors for steroid hormones, thyroid hormone, retinoic acid, and vitamin D_3.

nuclear transport receptor A protein that recognizes nuclear localization signals and mediates transport across the nuclear envelope.

nucleic acid hybridization The formation of double stranded DNA and/or RNA molecules by complementary base pairing.

nucleolar organizing regions The chromosomal regions containing the genes for ribosomal RNAs.

nucleolus The nuclear site of rRNA transcription, processing, and ribosome assembly.

nucleoside A purine or pyrimidine base linked to a sugar (ribose or deoxyribose).

nucleosome The basic structural unit of chromatin consisting of DNA wrapped around a histone core.

nucleosome core particles Particles containing 146 base pairs of DNA wrapped around an octamer consisting of two molecules each of histones H2A, H2B, H3, and H4.

nucleosome remodeling factors Proteins that disrupt chromatin structure, allowing transcription factors to bind nucleosomal DNA.

nucleotide A phosphorylated nucleoside.

nucleotide excision repair A mechanism of DNA repair in which oligonucleotides containing damaged bases are removed from a DNA molecule.

nucleus The most prominent organelle of eukaryotic cells; contains the genetic material.

Okazaki fragments Short DNA fragments that are joined to form the lagging strand of DNA.

oligonucleotide A short polymer of only a few nucleotides.

oligosaccharide A short polymer of only a few sugars.

oncogene A gene capable of inducing one or more characteristics of cancer cells.

oncogene addiction The dependence of cancer cells on the continuing activity of oncogenes.

one gene-one enzyme hypothesis The hypothesis, based on analysis of nutritional mutants of *Neurospora* in the 1940s, that a gene specifies the structure of a single enzyme. The current statement of this hypothesis is that a gene specifies the structure of a single polypeptide chain.

open-reading frame A stretch of nucleotide sequence that does not contain stop codons and can encode a polypeptide.

operator A regulatory sequence of DNA that controls transcription of an operon.

operon A group of adjacent genes transcribed as a single mRNA.

origin of replication A specific DNA sequence that serves as a binding site for proteins that initiate replication.

origin recognition complex (ORC) A protein complex that initiates DNA replication at eukaryotic origins.

oxidative metabolism The use of molecular oxygen as an electron acceptor in the breakdown of organic molecules.

oxidative phosphorylation The synthesis of ATP from ADP coupled to the energetically favorable transfer of electrons to molecular oxygen as the final acceptor in an electron transport chain.

P1 artificial chromosome (PAC) A vector used for cloning large fragments of DNA in *E. coli.*

p53 A transcription factor (encoded by the *p53* tumor suppressor gene) that arrests the cell cycle in G_1 in response to damaged DNA and is required for apoptosis induced by a variety of stimuli.

pachytene The stage of meiosis I during which recombination takes place between homologous chromosomes.

palmitoylation The addition of palmitic acid (a 16-carbon fatty acid) to cysteine residues of a polypeptide chain.

papillomavirus A member of a family of DNA viruses, some of which cause cervical and other anogenital cancers in humans.

paracrine signaling Local cell-cell signaling in which a molecule released by one cell acts on a neighboring target cell.

parenchyma cell A type of plant cell responsible for most metabolic activities.

passive diffusion The diffusion of small hydrophobic molecules through a phospholipid bilayer.

passive transport The transport of molecules across a membrane in the energetically favorable direction.

patch clamp technique A method used to isolate and study the activity of single ion channels.

pectin A gel-forming polysaccharide in plant cell walls.

peptide bond The bond joining amino acids in polypeptide chains.

peptide hormone A signaling molecule composed of amino acids.

peptidoglycan The principal component of bacterial cell walls consisting of linear polysaccharide chains crosslinked by short peptides.

peptidyl prolyl isomerase An enzyme that facilitates protein folding by catalyzing the *cis-trans* isomerization of prolyl peptide bonds.

pericentriolar material The material in the centrosome that initiates microtubule assembly.

peripheral membrane proteins Proteins indirectly associated with cell membranes by protein-protein interactions.

peroxin A protein present in peroxisomes.

peroxisome A cytoplasmic organelle specialized for carrying out oxidative reactions.

phagocytosis The uptake of large particles, such as bacteria, by a cell.

phagolysosomes A lysosome that has fused with a phagosome or autophagosome.

phagosome A vacuole containing a particle taken up by phagocytosis.

phalloidin A drug that binds to actin filaments and prevents their disassembly.

phase-contrast microscopy A type of microscopy in which variations in density or thickness between parts of the cell are converted to differences in contrast in the final image.

phenotype The physical appearance of an organism.

phosphatidylcholine A glycerol phospholipid with a head group formed from choline.

phosphatidylethanolamine A glycerol phospholipid with a head group formed from ethanolamine.

phosphatidylinositide 3-kinase (PI 3-kinase) An enzyme that phosphorylates PIP_2, yielding the second messenger phosphatidylinositol 3,4,5-trisphosphate (PIP_3).

phosphatidylinositol A glycerol phospholipids with a head group formed from inositol.

phosphatidylinositol 4,5-bisphosphate (PIP_2) A minor phospholipid component of the inner leaflet of the plasma membrane. Hormones and growth factors stimulate its hydrolysis by phospholipase C, yielding the second messengers diacylglycerol and inositol trisphosphate.

phosphatidylinositol 3,4,5-triphosphate (PIP_3) A second messenger formed by phosphorylation of PIP_2.

phosphatidylserine A glycerol phospholipid with a head group formed from serine.

phosphodiester bond A bond between the 5′-phosphate of one nucleotide and the 3′-hydroxyl of another.

phospholipase C An enzyme that hydrolyzes PIP_2 to form the second messengers diacylglycerol and inositol trisphosphate.

phospholipid bilayer The basic structure of biological membranes in which the hydrophobic tails of phospholipids are buried in the interior of the membrane and their polar head groups are exposed to the aqueous solution on either side.

phospholipid transfer protein A protein that transports phospholipid molecules between cell membranes.

phospholipids The principal components of cell membranes, consisting of two hydrocarbon chains (usually fatty acids) joined to a polar head group containing phosphate.

phosphorylation The addition of a phosphate group to a molecule.

photocenter An assembly of photosynthetic pigments in the thylakoid membrane of chloroplasts.

photoreactivation A mechanism of DNA repair in which solar energy is used to split pyrimidine dimers.

photorespiration A process that metabolizes a byproduct of photosynthesis.

photosynthesis The process by which cells harness energy from sunlight and synthesize glucose from CO_2 and water.

photosynthetic pigments Molecules that capture energy from sunlight by absorbing photons.

photosystem I A protein complex in the thylakoid membrane that uses energy absorbed from sunlight to synthesize NADPH.

photosystem II A protein complex in the thylakoid membrane that uses energy absorbed from sunlight to synthesize ATP.

pinocytosis The uptake of fluids or molecules into a cell by small vesicles.

plakin A member of a family of proteins that link intermediate filaments to other cellular structures.

plant hormones A group of small molecules that coordinate the responses of plant tissues to environmental signals.

plasma membrane A phospholipid bilayer with associated proteins that surrounds the cell.

plasmalogens A family of phospholipids that have an ether bond and an ester bond.

plasmid A small, circular DNA molecule capable of independent replication in a host cell.

plasmodesma A cytoplasmic connection between adjacent plant cells formed by a continuous region of the plasma membrane.

plastids A family of plant organelles including chloroplasts, chromoplasts, leucoplasts, amyloplasts, and elaioplasts.

platelet-derived growth factor (PDGF) A growth factor released by platelets during blood clotting to stimulate the proliferation of fibroblasts.

PML/RARα An oncogene formed by translocation of the retinoic acid receptor in acute promyelocytic leukemia.

polar body A small cell formed by asymmetric cell division following meiosis of oocytes.

polar microtubules Microtubules of the mitotic spindle that overlap in the center of the cell and push the spindle poles apart.

Polo-like kinase A protein kinase involved in mitotic spindle formation, kinetochore function, and cytokinesis.

poly-A tail A tract of about 200 adenine nucleotides added to the 3′ ends of eukaryotic mRNAs.

polyadenylation The process of adding a poly-A tail to a pre-mRNA.

polycistronic Messenger RNAs that encode multiple polypeptide chains.

polymerase chain reaction (PCR) A method for amplifying a region of DNA by repeated cycles of DNA synthesis *in vitro*.

polynucleotide A polymer containing up to millions of nucleotides.

polyomavirus A widely-studied DNA tumor virus.

polyp A benign tumor projecting from an epithelial surface.

polypeptide A polymer of amino acids.

polysaccharide A polymer containing hundreds or thousands of sugars.

polysome A series of ribosomes translating a messenger RNA.

polytene chromosome A giant chromosome found in some tissues of *Drosophila* that arises from repeated replication of DNA strands that fail to separate from each other.

porin A member of a class of proteins that cross membranes as β-barrels and form channels in the outer membranes of some bacteria, mitochondria, and chloroplasts.

pre-mRNA The primary transcript, which is processed to form messenger RNA in eukaryotic cells.

pre-rRNA The primary transcript, which is cleaved to form individual ribosomal RNAs (the 28S, 18S, and 5.8S rRNAs of higher eukaryotic cells).

pre-tRNA The primary transcript, which is cleaved to form transfer RNAs.

prenylation The addition of specific types of lipids (prenyl groups) to C terminal cysteine residues of a polypeptide chain.

presequence An amino-terminal sequence that targets proteins to mitochondria.

primary cell walls The walls of growing plant cells.

primary cultures Cell cultures established from a tissue.

primary structure The sequence of amino acids in a polypeptide chain.

primase An RNA polymerase used to intiate DNA synthesis.

processed pseudogene A pseudogene that has arisen by reverse transcription of mRNA.

procollagens Soluble precursors to the fibril-forming collagens.

product A compound formed as a result of an enzymatic reaction.

profilin An actin-binding protein that stimulates the assembly of actin monomers into filaments.

progesterone A steroid hormone produced by the ovaries.

programmed cell death A normal physiological form of cell death characterized by apoptosis.

prokaryotic cells Cells lacking a nuclear envelope, cytoplasmic organelles, and a cytoskeleton (bacteria).

prometaphase A transition period between prophase and metaphase during which the microtubules of the mitotic spindle attach to the kinetochores and the chromosomes shuffle until they align in the center of the cell.

promoter A DNA sequence at which RNA polymerase binds to initiate transcription.

pronuclei Two haploid nuclei in a newly fertilized egg.

proofreading The selective removal of mismatched bases by DNA polymerase.

prophase The beginning phase of mitosis, marked by the appearance of condensed chromosomes and the development of the mitotic spindle.

proplastid A small undifferentiated organelle that can develop into different types of mature plastids.

prostacyclin An eicosanoid formed from prostaglandin H_2.

prostaglandin A family of eicosanoid lipids involved in signaling inflammation.

prosthetic groups Small molecules bound to proteins.

proteasome A large protease complex that degrades proteins tagged by ubiquitin.

protein disulfide isomerase An enzyme that catalyzes the formation and breakage of disulfide (S–S) linkages.

protein kinase An enzyme that phosphorylates proteins by transferring a phosphate group from ATP.

protein kinase A A protein kinase regulated by cyclic AMP.

protein kinase C A family of protein-serine/threonine kinases that are activated by diacylglycerol and Ca^{2+} and function in intracellular signal transduction.

protein phosphatase An enzyme that reverses the action of protein kinases by removing phosphate groups from phosphorylated amino acid residues.

protein-serine/threonine kinase A protein kinase that phosphorylates serine and threonine residues.

protein-tyrosine kinase A protein kinase that phosphorylates tyrosine residues.

protein-tyrosine phosphatase An enzyme that removes the phosphate groups from phosphotyrosine residues.

proteins Polypeptides with a unique amino acid sequence.

proteoglycan A protein linked to glycosaminoglycans.

proteolysis Degradation of polypeptide chains.

proteome All of the proteins expressed in a given cell.

proteomics Large scale analysis of cell proteins.

proto-oncogene A normal cell gene that can be converted into an oncogene.

pseudogene A nonfunctional gene copy.

pseudopodium An actin-based extension of the plasma membrane responsible for phagocytosis and amoeboid movement.

PTB domain A protein domain that binds phosphotyrosine-containing peptides.

PTEN A lipid phosphatase that dephosphorylates PIP_3 and acts as a tumor suppressor.

purine One of the types of bases present in nucleic acids. The purines are adenine and guanine.

pyrimidine One of the types of bases present in nucleic acids. The pyrimidines are cytosine, thymine, and uracil.

pyrimidine dimer A common form of DNA damage caused by UV light in which adjacent pyrimidines are joined to form a dimer.

quaternary structure The interactions between polypeptide chains in proteins consisting of more than one polypeptide.

Rab A family of small GTP-binding proteins that play key roles in vesicular transport.

Rac A small GTP-binding protein involved in regulation of the actin cytoskeleton.

Rad51 A eukaryotic protein that functions similarly to RecA in homologous recombination.

Raf A protein-serine/threonine kinase (encoded by the *raf* oncogene) that is activated by Ras and leads to activation of the ERK MAP kinase.

raf Gene encoding Raf proteins.

Ran A small GTP-binding protein involved in nuclear import and export.

Ras A family of small GTP binding proteins (encoded by the *ras* oncogenes) that couple growth factor receptors to intracellular targets, including the Raf protein-serine/threonine kinase and the ERK MAP kinase pathway.

ras Gene encoding Ras proteins.

Rb A transcriptional regulatory protein that controls cell cycle progression and is encoded by a tumor suppressor gene that was identified by the genetic analysis of retinoblastoma.

RecA A protein that promotes the exchange of strands between homologous DNA molelcules during recombination.

receptor down-regulation The loss of receptors from the cell surface as a result of their internalization by endocytosis following ligand binding.

receptor-mediated endocytosis The selective uptake of macromolecules that bind to cell surface receptors that concentrate in clathrin-coated pits.

receptor protein-tyrosine kinase Membrane-spanning protein-tyrosine kinases that are receptors for extracellular ligands.

recessive An allele that is masked by a dominant allele.

recombinant DNA library A collection of genomic or cDNA clones.

recombinant molecule A DNA insert joined to a vector.

recombination The exchange of genetic material.

recombinational repair The repair of damaged DNA by recombination with an undamaged homologous DNA molecule.

release factor A protein that recognizes stop codons and terminates translation of mRNA.

replication fork The region of DNA synthesis where the parental strands separate and two new daughter strands elongate.

repressor A regulatory molecule that blocks transcription.

reproductive cloning The use of nuclear transfer to create a cloned organism.

resolution The ability of a microscope to distinguish objects separated by small distances.

restriction endonuclease An enzyme that cleaves DNA at a specific sequence.

restriction map The locations of restriction endonuclease cleavage sites on a DNA molecule.

restriction point A regulatory point in animal cell cycles that occurs late in G_1. After this point, a cell is committed to entering S and undergoing one cell division cycle.

retinoic acid A signaling molecule synthesized from vitamin A.

retinoid A molecule related to retinoic acid.

retrotransposon A transposable element that moves via reverse transcription of an RNA intermediate.

retrovirus A virus that replicates by making a DNA copy of its RNA genome by reverse transcription.

retrovirus-like element A retrotransposon that is structurally similar to a retrovirus.

reverse genetics Analysis of gene function by introducing mutations into a cloned gene.

reverse transcriptase A DNA polymerase that uses an RNA template.

reverse transcription Synthesis of DNA from an RNA template.

Rho A family of small GTP-binding proteins involved in regulation of the cytoskeleton.

rhodopsin A G protein-coupled photoreceptor in retinal rod cells that activates transducin in response to light absorption.

ribonucleic acid (RNA) A polymer of ribonucleotides.

ribose The five-carbon sugar found in RNA.

ribosomal RNA (rRNA) The RNA component of ribosomes.

ribosomes Particles composed of RNA and proteins that are the sites of protein synthesis.

ribozyme An RNA enzyme.

RNA editing RNA processing events other than splicing that alter the protein coding sequences of mRNAs.

RNA interference (RNAi) The degradation of mRNAs by short complementary double-stranded RNA molecules.

RNA polymerase An enzyme that catalyzes the synthesis of RNA.

RNA splicing The joining of exons in a precursor RNA molecule.

RNase H An enzyme that degrades the RNA strand of RNA-DNA hybrid molecules.

RNase P A ribozyme that cleaves the 5′ end of pre-tRNAs.

RNA world An early stage of evolution based on self-replicating RNA molecules.

rough endoplasmic reticulum (ER) The region of the endoplasmic reticulum covered with ribosomes and involved in protein metabolism.

Rous sarcoma virus (RSV) An acutely transforming retrovirus in which the first oncogene was identified.

ryanodine receptors Calcium channels in muscle and nerve cells that open in response to changes in membrane potential.

S phase The phase of the cell cycle during which DNA replication occurs.

Saccharomyces cerevisiae A frequently studied budding yeast.

sarcoma A cancer of cells of connective tissue.

sarcomere The contractile unit of muscle cells composed of interacting myosin and actin filaments.

sarcoplasmic reticulum A specialized network of membranes in muscle cells that stores a high concentration of Ca^{2+}.

satellite DNA Simple-sequence repetitive DNA with a buoyant density differing from the bulk of genomic DNA.

scaffold proteins Proteins that bind to components of signaling pathways, leading to their organization in specific signaling cassettes.

scanning electron microscopy See electron microscopy.

sclerenchyma cells Plant cells characterized by thick cell walls that provide structural support to the plant.

SDS-polyacrylamide gel electrophoresis (SDS-PAGE) A commonly used method to separate proteins by gel electrophoresis on the basis of size.

second messenger A compound whose metabolism is modified as a result of a ligand-receptor interaction; it functions as a signal transducer by regulating other intracellular processes.

secondary cell wall A thick cell wall laid down between the plasma membrane and the primary cell wall of plant cells that have ceased growth.

secondary response gene A gene whose induction following growth factor stimulation of a cell requires protein synthesis.

secondary structure The regular arrangement of amino acids within localized regions of a polypeptide chain. See α helix and β sheet.

secretory pathway The movement of secreted proteins from the endoplasmic reticulum to the Golgi apparatus and then, within secretory vesicles, to the cell surface.

secretory vesicles Membrane-enclosed sacs that transport proteins from the Golgi apparatus to the cell surface.

selectins Cell adhesion molecules that recognize oligosaccharides exposed on the cell surface.

self-splicing The ability of some RNAs to catalyze the removal of their own introns.

semiconservative replication The process of DNA replication in which the two parental strands separate and serve as templates for the synthesis of new progeny strands.

serum response element (SRE) A regulatory sequence that is recognized by the serum response factor and mediates the transcriptional induction of many immediate-early genes in response to growth factor stimulation.

serum response factor (SRF) A transcription factor that binds to the serum response element.

SH2 domain A protein domain of approximately 100 amino acids that binds phosphotyrosine-containing peptides.

Shine-Dalgarno sequence The sequence prior to the initiation site that correctly aligns bacterial mRNAs on ribosomes.

signaling network The interconnected network formed by the interactions of multiple signaling pathways within a cell.

signal patch A recognition determinant formed by the three-dimensional folding of a polypeptide chain.

signal peptidase An enzyme that removes the signal sequence of a polypeptide chain by proteolysis.

signal recognition particle (SRP) A particle composed of proteins and srpRNA that binds to signal sequences and targets polypeptide chains to the endoplasmic reticulum.

signal sequence A hydrophobic sequence at the amino terminus of a polypeptide chain that targets it for secretion in bacteria or incorporation into the endoplasmic reticulum in eukaryotic cells.

simian virus 40 (SV40) A widely-studied DNA tumor virus.

simple-sequence repeats A class of repeated DNA sequences consisting of tandem arrays of thousands of copies of short sequences.

SINEs (short interspersed elements) A family of highly repeated retrotransposons in mammalian genomes.

single-stranded DNA-binding proteins Proteins that stabilize unwound DNA by binding to single-stranded regions.

site-specific recombination Recombination mediated by proteins that recognize specific DNA sequences.

sliding filament model The model of muscle contraction in which contraction results from the sliding of actin and myosin filaments relative to each other.

Smad A family of transcription factors activated by TGF-β receptors.

small GTP-binding proteins A large family of monomeric GTP-binding proteins, including the Ras, Rab, Rho, and Ran proteins.

small nuclear RNAs (snRNAs) Nuclear RNAs ranging in size from 50 to 200 bases.

small nuclear ribonucleoprotein particles (snRNPs) Complexes of snRNAs with proteins.

small nucleolar RNAs (snoRNAs) Small RNAs present in the nucleolus that function in pre-rRNA processing.

smooth endoplasmic reticulum The major site of lipid synthesis in eukaryotic cells.

SNARE hypothesis The hypothesis that vesicle fusion is mediated by pairs of transmembrane proteins (SNAREs) on the vesicle and target membranes.

soft keratin The keratins found in the cytoplasm of epithelial cells.

somatic cell nuclear transfer The basic procedure of animal cloning in which the nucleus of an adult somatic cell is transferred to an enucleated egg.

somatic hypermutation The introduction of multiple mutations within rearranged immunoglobulin variable regions to increase antibody diversity.

Southern blotting A method in which radioactive probes are used to detect specific DNA fragments that have been separated by gel electrophoresis.

spacer sequences The DNA sequences between genes.

spectrin A major actin-binding protein of the cell cortex.

sphingomyelin A phospholipid consisting of two hydrocarbon chains bound to a polar head group containing serine.

spindle assembly checkpoint A cell cycle checkpoint that monitors the alignment of chromosomes on the metaphase spindle.

spliceosomes Large complexes of snRNAs and proteins that catalyze the splicing of pre-mRNAs.

Src A nonreceptor protein-tyrosine kinase encoded by the oncogene (*src*) of Rous sarcoma virus.

src Gene encoding Src protein.

SRP receptor A protein on the membrane of the endoplasmic reticulum that binds the signal recognition particle (SRP).

srpRNA The small cytoplasmic RNA component of SRP.

stability gene A gene that acts to maintain the integrity of the genome and whose loss can lead to the development of cancer.

starch A polymer of glucose residues that is the principal storage form of carbohydrates in plants.

START A regulatory point in the yeast cell cycle that occurs late in G_1. After this point a cell is committed to entering S and undergoing one cell division cycle.

STAT proteins Transcription factors that have an SH2 domain and are activated by tyrosine phosphorylation, which promotes their translocation from the cytoplasm to the nucleus.

stem cell A cell that divides to produce daughter cells that can either differentiate or remain as stem cells.

stereocilium A specialized microvillus of auditory hair cells.

steroid hormone receptor Transcription factors that regulate gene expression in response to hormones such as estrogen and testosterone.

steroid hormones A group of hydrophobic hormones that are derivatives of cholesterol.

stress fiber A bundle of actin filaments anchored at sites of cell adhesion to the extracellular matrix.

stroma The compartment of chloroplasts that lies between the envelope and the thylakoid membrane.

stromal processing peptidase (SPP) The protease that cleaves transit peptides from proteins imported to the chloroplast stroma.

substrate A molecule acted upon by an enzyme.

symport The transport of two molecules in the same direction across a membrane.

synapse The junction between a neuron and another cell, across which information is carried by neurotransmitters.

synapsis The association of homologous chromosomes during meiosis.

synaptic vesicle A secretory vesicle that releases neurotransmitters at a synapse.

synaptomenal complex A zipperlike protein structure that forms along the length of paired homologous chromosomes during meiosis.

systems biology A new field of biology in which large-scale experimental approaches are combined with quantitative analysis and modeling to study complex biological systems.

T cell receptor A T lymphocyte surface protein that recognizes antigens expressed on the surface of other cells.

talin A protein that mediates the association of actin filaments with integrins at focal adhesions.

TATA box A regulatory DNA sequence found in the promoters of many eukaryotic genes transcribed by RNA polymerase II.

TATA-binding protein (TBP) A basal transcription factor that binds directly to the TATA box.

taxol A drug that binds to and stabilizes microtubules.

TBP-associated factors (TAFs) Polypeptides associated with TBP in the general transcription factor TFIID.

telomerase A reverse transcriptase that synthesizes telomeric repeat sequences at the ends of chromosomes from its own RNA template.

telomeres Repeats of simple-sequence DNA that maintain the ends of linear chromosomes.

telophase The final phase of mitosis, during which the nuclei re-form and chromosomes decondense.

temperature-sensitive mutant A cell expressing a protein that is functional at one temperature but not at another, whereas the normal protein is functional at both temperatures.

tertiary structure The three-dimensional folding of a polypeptide chain that gives the protein its functional form.

testosterone A steroid hormone produced by the testis.

therapeutic cloning A procedure in which nuclear transfer into oocytes could be used to produce embryonic stem cells for use in transplantation therapy.

thylakoid membrane The innermost membrane of chloroplasts that is the site of electron transport and ATP synthesis.

thymine A pyrimidine found in DNA that base-pairs with adenine.

thyroid hormone A hormone synthesized from tyrosine in the thyroid gland.

thromboxane An eicosanoid involved in blood clotting.

Tic complex The protein translocation complex of the chloroplast inner membrane.

tight junction A continuous network of protein strands around the circumference of epithelial cells, sealing the space between cells and forming a barrier between the apical and basolateral domains.

Tim complex The protein translocation complex of the mitochondrial inner membrane.

Ti plasmid A plasmid used for gene transfer in plants.

titin A large protein that acts as a spring to keep myosin filaments centered in the muscle sarcomere.

Toc complex The protein translocation complex of the chloroplast outer membrane.

Tom complex The protein translocation complex of the mitochondrial outer membrane.

topoisomerase An enzyme that catalyzes the reversible breakage and rejoining of DNA strands.

***trans*-acting factors** Transcriptional regulatory proteins.

***trans* Golgi network** The Golgi compartment within which proteins are sorted and packaged to exit the Golgi apparatus.

transcription-coupled repair The preferential repair of damage to transcribed strands of DNA.

transcription factor A protein that regulates the activity of RNA polymerase.

transcription The synthesis of an RNA molecule from a DNA template.

transcriptional activators Transcription factors that stimulate transcription.

transcytosis The sorting and transport of proteins to different domains of the plasma membrane following endocytosis.

transducin A G protein that stimulates cGMP phosphodiesterase when it is activated by rhodopsin.

transfection The introduction of a foreign gene into eukaryotic cells.

transfer RNA (tRNA) RNA molecules that function as adaptors between amino acids and mRNA during protein synthesis.

transformation The transfer of DNA between genetically distinct bacteria. See also cell transformation.

transforming growth factor β (TGF-β) A polypeptide growth factor that generally inhibits animal cell proliferation.

transgenic mouse A mouse that carries foreign genes incorporated into the germ line.

transient expression The expression of unintegrated plasmid DNAs that have been introduced into cultured cells.

transitional ER The region of the ER from which proteins exit for the Golgi apparatus.

transition state A high energy state through which substrates must pass during the course of an enzymatic reaction.

transit peptides N-terminal sequences that target proteins for import into chloroplasts.

translation The synthesis of a polypeptide chain from an mRNA template.

translesion DNA synthesis A form of repair in which specialized DNA polymerases replicate across a site of DNA damage.

translocon The membrane channel through which polypeptide chains are transported into the endoplasmic reticulum.

transmembrane proteins Integral membrane proteins that span the lipid bilayer and have portions exposed on both sides of the membrane.

transmission electron microscopy See electron microscopy.

transposable element See transposon.

transposon A DNA sequence that can move to different positions in the genome.

treadmilling A dynamic behavior of actin filaments and microtubules in which the loss of subunits from one end of the filament is balanced by their addition to the other end.

triacylglycerol Three fatty acids linked to a glycerol molecule.

tropomyosin A fibrous protein that binds actin filaments and regulates contraction by blocking the interaction of actin and myosin.

troponin A complex of proteins that binds to actin filaments and regulates skeletal muscle contraction.

tubulin A cytoskeletal protein that polymerizes to form microtubules.

tumor Any abnormal proliferation of cells.

tumor initiation The first step in tumor development, resulting from abnormal proliferation of a single cell.

tumor necrosis factor (TNF) A polypeptide growth factor that induces programmed cell death.

tumor progression The accumulation of mutations within cells of a tumor population, resulting in increasingly rapid growth and malignancy.

tumor promoter A compound that leads to tumor development by stimulating cell proliferation.

tumor suppressor gene A gene whose inactivation leads to tumor development.

tumor virus A virus capable of causing cancer in animals or humans.

turgor pressure The internal hydrostatic pressure within plant cells.

twinfilin An actin-binding protein that stimulates the assembly of actin monomers into filaments.

two-dimensional gel electrophoresis A method for separating cell proteins based on both charge and size.

ubiquinone See coenzyme Q.

ubiquitin A highly conserved protein that acts as a marker to target other cellular proteins for rapid degradation.

ultracentrifuge A centrifuge that rotates samples at high speeds.

unfolded protein response A cellular stress response in which an excess of unfolded proteins in the endoplasmic reticulum leads to general inhibition of protein synthesis, increased expression of chaperones, and increased proteasome activity.

uniport The transport of a single molecule across a membrane.

3′ untranslated region A noncoding region at the 3′ end of mRNA.

5′ untranslated region A noncoding region at the 5′ end of mRNA.

uracil A pyrimidine found in RNA that base-pairs with adenine.

vacuole A large membrane-enclosed sac in the cytoplasm of eukaryotic cells. In plant cells, vacuoles function to store nutrients and waste products, to degrade macromolecules, and to maintain turgor pressure.

vector A DNA molecule used to direct the replication of a cloned DNA fragment in a host cell.

velocity centrifugation The separation of particles based on their rates of sedimentation.

video-enhanced microscopy The combined use of video cameras with the light microscope to allow the visualization of small objects.

villin The major actin-bundling protein of intestinal microvilli.

vimentin An intermediate filament protein found in a variety of different kinds of cells.

vinblastine A drug that inhibits microtubule polymerization.

vincristine A drug that inhibits microtubule polymerization.

vinculin A protein that mediates the association of actin filaments with integrins at focal adhesions.

voltage-gated channels Ion channels that open in response to changes in electric potential.

WASP/Scar complex A protein complex that stimulates actin filament branching.

Western blotting See immunoblotting.

Wnt A secreted signaling molecule that stimulates a pathway regulating cell fate during embryonic development.

X-chromosome inactivation A dosage compensation mechanism in which most of the genes on one X chromosome are inactivated in female cells.

X-ray crystallography A method in which the diffraction pattern of X rays is used to determine the arrangement of individual atoms within a molecule.

Xenopus laevis An African clawed frog used as a model system for developmental biology.

yeast artificial chromosome (YAC) A vector that can replicate as a chromosome in yeast cells and can accomodate very large DNA inserts (hundreds of kb).

yeast two-hybrid A genetic method for detecting protein interactions in yeast cells.

yeasts The simplest unicellular eukaryotes. Yeasts are important models for studies of eukaryotic cells.

zebrafish A species of small fish used for genetic studies of vertebrate development.

zinc finger domain A type of DNA binding domain consisting of loops containing cysteine and histidine residues that bind zinc ions.

zygote A fertilized egg.

zygotene The stage of meiosis I during which homologous chromosomes become closely associated.

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THE GENETIC CODE

First position	Second position U	C	A	G	Third position
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	Stop	Stop	A
	Leu	Ser	Stop	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

THE AMINO ACIDS

Amino acid	Three-letter abbreviation	One-letter abbreviation
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

MEASUREMENTS USED IN MOLECULAR AND CELL BIOLOGY

Length	meter (m)	1 m = 39.37 inches
	millimeter (mm)	1 mm = 10^{-3} m
	micrometer (μm)	1 μm = 10^{-6} m
	nanometer (nm)	1 nm = 10^{-9} m
	Ångström (Å)	1 Å = 0.1 nm
Volume	liter (l)	1 l = 1.057 quarts
	milliliter (ml)	1 ml = 10^{-3} l
	microliter (μl)	1 μl = 10^{-6}l
Mass	kilogram (kg)	1 kg = 2.2 pounds
	gram (g)	1 g = 10^{-3} kg
	milligram (mg)	1 mg = 10^{-3} g
	microgram (μg)	1 μg = 10^{-6} g
	nanogram (ng)	1 ng = 10^{-9} g
Molecular length	kilobase (kb)	1 kb = 1000 base pairs
	megabase (Mb)	1 Mb = 10^{3} kb

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